PRECONDITIONED CONJUGATE-GRADIENT 2 (PCG2), A COMPUTER PROGRAM FOR SOLVING GROUND-WATER FLOW EQUATIONS

U.S. GEOLOGICAL SURVEY

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ABSTRACT

This report documents PCG2: a numerical code to be used with the U.S. Geological Survey modular three-dimensional, finite-difference, ground-water flow model. PCG2 uses the preconditioned conjugate-gradient method to solve the equations produced by the model for hydraulic head. Linear or nonlinear flow conditions may be simulated.

PCG2 includes two preconditioning options: modified incomplete Cholesky preconditioning, which is efficient on scalar computers; and polynomial preconditioning, which requires less computer storage and, with modifications that depend on the computer used, is most efficient on vector computers. Convergence of the solver is determined using both head-change and residual criteria. Nonlinear problems are solved using Picard iterations.

This documentation provides a description of the preconditioned conjugate-gradient method and the two preconditioners, detailed instructions for linking PCG2 to the modular model, sample data inputs, a brief description of PCG2, and a FORTRAN listing.

INTRODUCTION

Purpose and Scope

Finite-difference numerical models commonly are used to investigate ground-water flow systems. Effective use of these models requires that the matrix equations they produce be solved efficiently, that is, that a correct solution is produced using as little computer processing time as possible. Effective use of the models also requires that the amount of computer storage be minimized to allow for solution on small computers and to avoid overburdening large computers.

The purpose of this work is to document PCG2, a numerical code which uses the preconditioned conjugate-gradient method to solve the matrix equations produced by the U.S. Geological Survey modular three-dimensional, finite-difference ground-water flow model. Two preconditioning options are included which have not previously been available for use with this model, and which perform better than available solvers for many problems.

Previous Investigations

Matrix equations have been solved using direct or iterative methods. In most direct methods the matrix is factored exactly and the true solution is obtained by executing one backward and one forward substitution. In most iterative methods an initial estimate of the solution is refined iteratively using an approximately factored matrix, and successive solutions should approach the true solution. Solution convergence is assumed to have been reached when some measure of the residual and(or) the difference in results between successive iterations is less than some user-specified convergence criteria. Direct solution is straightforward, but iterative methods are less susceptible to round-off error, are more efficient for large problems, and require less computer storage (Remson, Hornberger and Molz, 1971, p. 177; Aziz and Settari, 1979, p. 261).

The preconditioned conjugate-gradient method (Concus, Golub and O'Leary, 1976) is an iterative method which can be used to solve matrix equations if the matrix is symmetric (matrix element $a_{ij} = a_{ji}$, where the first subscript is the matrix row number, and the second is the matrix column number) and positive-definite (all eigen values are positive) (see Hildebrand, 1965, p. 30 and 48 for further discussion of these terms). The matrix produced in ground-water-flow models always is symmetric and positive-definite. The preconditioned conjugate-gradient method has been the subject of considerable interest in recent years because of its efficiency and ability to solve difficult problems (Meijerink and van der Vorst, 1977). It works well, in part, because the required iteration parameters are calculated internally and need not be estimated.

Various preconditioners may be used in the preconditioned conjugate-gradient method. Among the different preconditioners there often is a direct relationship between increased efficiency and increased computer storage (Meijerink and van der Vorst, 1977). To avoid this tradeoff, the only preconditioners considered are those that produce a solver that has computer storage requirements less than or equal to the strongly implicit procedure (SIP) as programmed for the ground-water flow problem. SIP requires additional computer storage equal to four arrays with dimensions equal to the number of grid nodes (McDonald and Harbaugh, 1988, chap. 12).

The incomplete Cholesky preconditioner (ICCG) has been very popular (Meijerink and van der Vorst, 1977; Kuiper, 1981, 1987). However, alternative methods of matrix preconditioning have been developed to achieve more efficient conjugate-gradient solvers. Axelsson and Lindskog (1986) presented a preconditioner that commonly is called the modified incomplete Cholesky preconditioner (MICCG). It is similar to preconditioners presented by Dupont and others (1968), Gustafsson (1978), Wong (1979), and Ashcraft and Grimes (1988). In this paper, MICCG refers to Axelsson and Lindskog's (1986) method. Hill (in press) showed that MICCG was more efficient than ICCG in solving eight ground-water flow test cases on a scalar computer. Saad (1985) presented a least-squares polynomial preconditioner (POLCG), in which the matrix inverse is approximated by a truncated Neuman polynomial series. It is similar to preconditioners presented by Dubois and others (1979) and Johnson and others (1983). In this paper POLCG refers to Saad's (1985) method.

POLCG is not as efficient as SIP or MICCG on scalar computers (Scandrett, 1989; Hill, in press), but is more efficient on vector computers (Scandrett, 1989; Meyer and others, 1989, p. 1445). Storage requirements are less than for SIP: POLCG requires additional storage equal to three arrays with dimensions equal to the number of grid nodes.

Many preconditioners were excluded from PCG2. Modifications of ICCG presented by Meijerink and van der Vorst (1977; 1981), Gustafsson (1978; 1979), Wong (1979), and Ashcraft and Grimes (1988) apparently converge in fewer iterations than ICCG or MICCG. These were not included in PCG2 because they require that several additional arrays with dimensions equal to the number of grid nodes be added to computer storage. Additional polynomial preconditioners have been presented in several papers (Saad, 1985; Ashby, 1987; Meyer and others, 1989). Based on Saad's results, the POLCG was chosen because it appears to be at least as efficient as the other polynomial methods and it is easier to use. However, for very large grids (greater than 100,000 cells) the optimal Chebyshev polynomial preconditioner (Meyer and others, 1989) may be more efficient, and users with large grids may wish to consider this alternative polynomial preconditioner.

Watts (1981) and Hill (in press) indicate that the greatest differences in solver efficiency on scalar computers occur for three-dimensional, nonlinear problems. Thus, for these types of problems, it may be well worth the time and effort to try more than one solver. SIP generally is a good alternative to consider.

Notation

The following notation is used in this work:

Underlined capital letters indicate matrices: $\underline{A}$

Underlined lower-case letters indicate column vectors: $\underline{r}$

The element located in matrix row i and column j is designated as follows: matrix $\underline{A}$, element a_{ij}.

Exception: a single index is used to simplify notation in some cases. These are described in the text.

THREE-DIMENSIONAL GROUND-WATER FLOW MODEL

In this work, selected numerical methods are presented for solving the matrix equations that arise when the finite-difference method is used to discretize the ground-water flow equation as applied to a two-dimensional aquifer or a three-dimensional layered aquifer system. The finite-difference model is described in detail by McDonald and Harbaugh (1988), and only aspects relevant to this report are discussed here.

The finite-difference model produces a set of linear equations which can be expressed in matrix notation as:

$$\underline{A}\,\underline{x} = \underline{b} \quad , \tag{1}$$

where $\underline{A}$ is a coefficient matrix which is discussed below, $\underline{x}$ is a vector of hydraulic heads at each grid cell, and $\underline{b}$ is a vector of defined flows, terms associated with head-dependent boundary conditions, and storage terms (for

transient problems) at each grid cell. Because of the rigid structure of the finite-difference grid, and because there are six neighboring cells to each internal cell of a three-dimensional grid, $\underline{A}$ is symmetric and there may be as many as six off-diagonals in $\underline{A}$--three above the main diagonal and three below. The elements on the off-diagonals equal the negatives of the horizontal or vertical conductances between the centers of the cells which make up the finite-difference grid (fig. 1). The horizontal conductances along columns (c_n of fig. 1) equal $T_i T_{i+1} \Delta w / (T_i \Delta L_{i+1}/2 + T_{i+1} \Delta L_i/2)$, where T_i and T_{i+1} are transmissivities between two adjoining cells, Δw is the width of the two cells, and ΔL_i and ΔL_{i+1} are lengths of the two cells. Horizontal conductances along rows (r_n of fig. 1) are defined analogously. The vertical conductances (v_n of fig. 1) equal $K_z A/\Delta z$, where K_z is the vertical hydraulic conductivity, A is the area of the cell, and Δz is the vertical distance between the centers of the two cells. Each component on the main diagonal of $\underline{A}$, a_{ii}, equals:

$$a_{ii} = \sum_{\substack{j=1 \\ i \neq j}}^{N} (-a_{ij}) + w_i \ , \qquad (2)$$

where N is the total number of nodes in the grid; a_{ij} are the off-diagonal elements of row i, which are negative numbers; and w_i are the sum of the conductances associated with head-dependent boundaries and storage terms (for transient problems), which are positive numbers. Besides being symmetric, $\underline{A}$ is also positive-definite (its eigenvalues are always positive) (Varga, 1962, p. 23, 181-188; Hildebrand, 1965, p. 48), and these properties allow equation 1 to be solved using the methods presented in this work.

Nonlinearities occur if any aquifer is unconfined or if a head-dependent boundary condition is nonlinear. If any aquifer is unconfined, the horizontal conductances are a function of hydraulic head, and the main-diagonal and four of the six off-diagonals of matrix $\underline{A}$ must be updated during the solution process. If a head-dependent boundary condition is nonlinear, the boundary condition may change from being head-dependent to defined flux depending on the hydraulic head in the aquifer adjacent to the boundary, and the main diagonal of $\underline{A}$ and vector $\underline{b}$ (eq. 1) must be updated.

SOLUTION BY THE PRECONDITIONED CONJUGATE-GRADIENT METHOD

The preconditioned conjugate-gradient method for solving a set of linear equations is iterative. In iterative methods, it is assumed that the matrix $\underline{A}$ can be split into the sum of two matrices; that is $\underline{A} = \underline{M} + \underline{N}$ (Varga, 1962, p. 87-93; Remson and others, 1971, p. 177). $\underline{M}$ is called the preconditioned form of $\underline{A}$, and the goal is to define $\underline{M}$ so that it is easy to invert and resembles $\underline{A}$ as much as possible. These two criteria generally are impossible to satisfy simultaneously, and the optimal definition of $\underline{M}$ has been the focus of much research. In the preconditioned conjugate-gradient method, $\underline{M}$ must always be symmetric and positive definite. (This brief description of matrix splitting does not include important requirements that $\underline{M}$ and $\underline{N}$ must satisfy to achieve a convergent solver. Please refer to Varga (1962) for additional information).

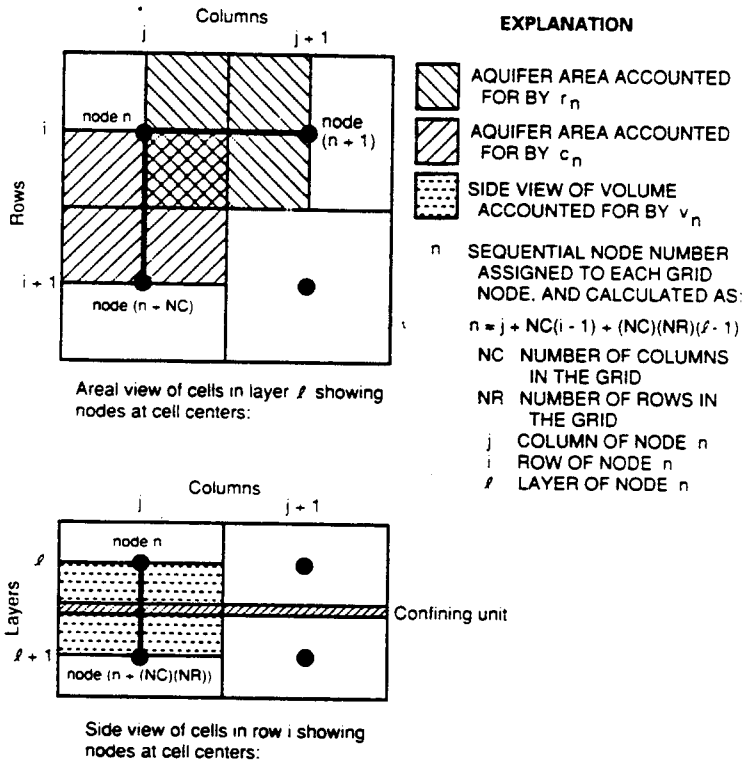


Figure 1.--Aquifer system volumes accounted for by conductances r_n, c_n, and v_n in the finite-difference method.

Once $\underline{M}$ has been defined, the basic iterative equation is developed from equation 1 and the splitting of $\underline{A}$, and can be written as:

$$\underline{M}\ \underline{x}_{k+1} = \underline{M}\ \underline{x}_k + \underline{b} - \underline{A}\ \underline{x}_k \qquad (3)$$

where k is the iteration index. Noting that $\underline{b} - \underline{A}\ \underline{x}_k$ is the residual ($\underline{r}_k$) of the original set of equations at the kth iteration, and setting $\underline{s}_k = \underline{x}_{k+1} - \underline{x}_k$ gives:

$$\underline{M}\ \underline{s}_k = \underline{r}_k \qquad (4)$$

or

$$\underline{s}_k = \underline{M}^{-1} \underline{r}_k \quad . \tag{5}$$

The new heads may then be calculated as $\underline{x}_{k+1} = \underline{x}_k + \underline{s}_k$. More generally, some function of $\underline{s}_k$ may be used to calculate $\underline{x}_{k+1}$.

Conjugate-gradient methods are second-order iterative techniques because at each iteration the new change in $\underline{x}$, which is called $\underline{p}_k$, is calculated using the change from the prior iteration, $\underline{p}_{k-1}$, in addition to the vector $\underline{s}_k$ of equation 5. Conjugate-gradient methods begin by calculating $\underline{r}_o = \underline{b} - \underline{A}\,\underline{x}_o$. The following steps are executed for each iteration, starting with k = 0:

$$\underline{s}_k = \underline{M}^{-1} \underline{r}_k \tag{6a}$$

$$\text{for } k = 0 \qquad \underline{p}_k = \underline{s}_k \tag{6b}$$

$$\text{for } k > 0 \begin{cases} \beta_k = \dfrac{\underline{s}_k^T \underline{r}_k}{\underline{s}_{k-1}^T \underline{r}_{k-1}} & (6c) \\ \underline{p}_k = \underline{s}_k + \beta_k \underline{p}_{k-1} & (6d) \end{cases}$$

$$\alpha_k = \frac{\underline{s}_k^T \underline{r}_k}{\underline{p}_k^T \underline{A}\, \underline{p}_k} \tag{6e}$$

$$\underline{x}_{k+1} = \underline{x}_k + \alpha_k \underline{p}_k \tag{6f}$$

$$\underline{r}_{k+1} = \underline{r}_k - \alpha_k \underline{A}\, \underline{p}_k \tag{6g}$$

where the superscripted T indicates the transpose of the vector. Because $\underline{r}_{k+1}$ can be calculated using the last statement, $\underline{b}$ need not be saved within the solver. Iteration parameters β_k and α_k are calculated internally such that successive updating vectors, $\underline{p}_k$, are $\underline{A}$-orthogonal to previous $\underline{p}_k$ vectors -- that is, $\underline{p}_k^T \underline{A}\, \underline{p}_\ell = 0$, $k \neq \ell$ (Hestenes and Stiefel, 1952).

Whether or not an iterative method will converge and, if so, how fast depends on the preconditioner and how $\underline{x}_k$ is updated. A discussion of convergence is beyond the scope of this report, but references are cited for the convergence properties of each solver. See Varga (1962) for the general theory of convergence of iterative methods.

Scaling of the matrix $\underline{A}$ simplifies POLCG (Dubois and others, 1979), and is accomplished as:

$$\underline{B} = \underline{S}^T \underline{A}\, \underline{S} \ , \tag{7}$$

where $\underline{B}$ is the scaled matrix, all off-diagonal entries of $\underline{S}$ are zero, and

$$s_{ii} = \frac{1}{\sqrt{a_{ii}}} \quad . \tag{8}$$

This type of scaling is called diagonal scaling, and it preserves the symmetry of the original matrix. Diagonal scaling may improve the matrix characteristics that are most important to convergence because the scaled matrix is still symmetric and positive definite, and the condition number of $\underline{A}$ (the largest eigenvalue divided by the smallest eigenvalue) is minimized (Forsythe and Strauss, 1955). However, although $\underline{A}$ is diagonally dominant because:

$$a_{ii} \geq \sum_{\substack{j=1 \\ i \neq j}}^{N} |a_{ij}| \quad , \tag{9}$$

$\underline{B}$ may not be diagonally dominant if some of the values along the diagonal of $\underline{A}$ are much smaller than others. For example, consider the following original and scaled matrices:

$$\underline{A} = \begin{bmatrix} 0.9 & -0.1 & -0.75 \\ -0.1 & 0.2 & 0.0 \\ -0.75 & 0.0 & 0.8 \end{bmatrix} \qquad \underline{B} = \begin{bmatrix} 1.0 & -0.23 & -0.88 \\ -0.23 & 1.0 & 0.0 \\ -0.88 & 0.0 & 1.0 \end{bmatrix}$$

The first row of $\underline{B}$ is no longer diagonally dominant. The lack of diagonal dominance can cause problems when using MICCG. Scaling also may cause more rounding and truncation errors because all components of the diagonal of $\underline{A}$ must be summed, as in equation 2. Without scaling, the conductance terms can be manipulated individually to reduce rounding and truncation errors (Dorn and McCracken, 1972, p. 94). Scaling was only used for POLCG.

The precision with which numbers are stored in the computer can significantly affect solver performance. For example, making the four arrays required by SIP double precision (14 to 15 significant digits on the computer used) instead of single precision (6 to 7 significant digits) can make the difference between convergence and nonconvergence for some problems (McDonald and Harbaugh, 1988, p. A-2; A.W. Harbaugh, U.S. Geological Survey, written commun., 1989). However, increasing the precision of arrays doubles the required computer storage space. All the arrays required by the solvers presented in this work were declared as single precision. Double-precision scalar variables were used to improve the accuracy of calculations, where possible. The only double-precision array in the model is, then, the array used to store calculated heads (McDonald and Harbaugh, 1988, p. A-2). In PCG2, problems caused by the limited precision of the solver arrays are most prevalent for POLCG (Hill, in press). If the limited precision of the solver arrays is suspected as the cause of convergence problems, arrays V, SS, and P may be converted to double precision by doubling their allocated storage in PCG2AL, and declaring them as double precision in PCG2AP and SPCG2E.

Nonlinear problems are solved by Picard iterations, in which the matrix $\underline{A}$ and vector $\underline{b}$ are periodically updated between iterations using the newly calculated heads. For nonlinear problems, convergence using the conjugate-gradient solvers was found to be most efficient if several iterations (called inner iterations in this report) were accomplished between Picard iteration updates (Kuiper, 1981 and 1987). This allows the solver to use equations 6c and 6d to calculate several orthogonal $\underline{p}_k$ vectors before updating $\underline{A}$ and $\underline{b}$, and thus take advantage of the orthogonality of the conjugate-gradient method.

The total number of iterations equals the sum of the inner iterations for all updates of $\underline{A}$ and $\underline{b}$. For any one $\underline{A}$ and $\underline{b}$, the inner iterations continue until one of the following occurs: (1) the user-defined maximum number of inner iterations (ITER1 of the input file) are executed; or (2) the final convergence criteria are met. Outer iterations continue until the final convergence criteria are met on the first inner iteration after an update. The total number of iterations required is minimized by adjusting ITER1 and re-executing the problem. For most problems, the optimal value of ITER1 ranges from 3 to 10.

In the absence of round-off error, only inner iterations would be required for linear problems. However, in practice, round-off error may adversely affect the residual calculated by the conjugate-gradient method (eq. 6g) when more than 50 iterations are required. Recalculating the residual as $\underline{r} = \underline{b} - \underline{A}\,\underline{x}$ occasionally by limiting the number of inner iterations to less than 50 in linear problems alleviates the error. This can be accomplished using ITER1 of the input file.

A flowchart which displays the steps discussed above is presented in the section "Documentation of PCG2" of this report.

MICCG

In modified incomplete Cholesky preconditioning, $\underline{M} = \underline{U}^T \underline{D}\,\underline{U}$, where $\underline{U}$ is an upper triangular matrix with nonzero values along the main diagonal and at off-diagonal locations where $\underline{A}$ has nonzero values. $\underline{D}$ is a positive diagonal matrix with $d_{ii} = 1/u_{ii}$. When $\underline{A}$ is structured as in the finite-difference model with natural ordering of the nodes and there are more than two columns in the grid, the off-diagonal components of $\underline{U}$ equal the off-diagonal components of $\underline{A}$. That is, $u_{ij} = a_{ij}$, for $j > i$. As an example, figure 2 shows $\underline{U}$ for a problem with 2 rows, 3 columns and 2 layers. To more clearly indicate the physical quantities involved, the variables r_n, c_n, and v_n, which are depicted in figure 1 and were described earlier in this paper, are used. To be consistent, the same subscript, n, is used for the diagonal of matrix $\underline{U}$, so that u_{ii} now becomes u_n, where $n = i$.

Calculation of the u_{ii} is explained by executing the matrix multiplication for the simple problem shown in figure 2. In matrix $\underline{U}^T \underline{D}\,\underline{U}$ (fig. 3), $-r_n$, $-c_n$, and $-v_n$ appear in the same places they occupied in the $\underline{A}$ matrix. The additional off-diagonal terms occur because $\underline{U}^T \underline{D}\,\underline{U}$ is an incomplete factorization of $\underline{A}$. The u_n are defined such that the sum of the elements along a row of $\underline{U}^T \underline{D}\,\underline{U}$ equals the sum of the elements along the same row of $\underline{A}$. To accomplish this for the matrix shown in figure 3:

$$
\begin{bmatrix}
u_1 & -r_1 & 0 & -c_1 & 0 & 0 & -v_1 & 0 & 0 & 0 & 0 & 0 \\
 & u_2 & -r_2 & 0 & -c_2 & 0 & 0 & -v_2 & 0 & 0 & 0 & 0 \\
 & & u_3 & 0 & 0 & -c_3 & 0 & 0 & -v_3 & 0 & 0 & 0 \\
 & & & u_4 & -r_4 & 0 & 0 & 0 & 0 & -v_4 & 0 & 0 \\
 & & & & u_5 & -r_5 & 0 & 0 & 0 & 0 & -v_5 & 0 \\
 & & & & & u_6 & 0 & 0 & 0 & 0 & 0 & -v_6 \\
 & & & & & & u_7 & -r_7 & 0 & -c_7 & 0 & 0 \\
 & & & & & & & u_8 & -r_8 & 0 & -c_8 & 0 \\
 & & & & & & & & u_9 & 0 & 0 & -c_9 \\
 & & & & & & & & & u_{10} & -r_{10} & 0 \\
 & & & & & & & & & & u_{11} & -r_{11} \\
 & & & & & & & & & & & u_{12}
\end{bmatrix}
$$

Figure 2.--Matrix $\underline{U}$ for MICCG. The r_n, c_n, and v_n are the conductances along rows and columns and between layers, respectively.

$$
\begin{aligned}
u_1 &= a_{11} \quad , \\
u_2 &= a_{22} - \frac{r_1^2}{u_1} - \frac{r_1 c_1}{u_1} - \frac{r_1 v_1}{u_1} \quad , \\
u_3 &= a_{33} - \frac{r_2^2}{u_2} - \frac{r_2 c_2}{u_2} - \frac{r_2 v_2}{u_2} \quad , \\
&\text{etc.}
\end{aligned} \tag{10}
$$

The general algorithm can be expressed as (R.L. Cooley, written commun., 1988):

$$
u_{ii} = a_{ii} - \sum_{k=1}^{i-1} \frac{u_{ki}^2}{u_{kk}} - \alpha \left(\sum_{j=1}^{i-1} f_{ji} + \sum_{j=i+1}^{N} f_{ij} \right) \tag{11a}
$$

$$
\begin{bmatrix}
u_1 & -r_1 & 0 & -c_1 & 0 & 0 & -v_1 & 0 & 0 & 0 & 0 & 0 & 0 \\
 & u_2+\xi_1 & -r_2 & \varphi_1 & -c_2 & 0 & \theta_1 & -v_2 & 0 & 0 & 0 & 0 & 0 \\
 & & u_3+\xi_2 & 0 & \varphi_2 & -c_3 & 0 & \theta_2 & -v_3 & 0 & 0 & 0 & 0 \\
 & & & u_4+r_1 & -r_4 & 0 & \phi_1 & 0 & 0 & -v_4 & 0 & 0 & 0 \\
 & & & & u_5+\xi_4+r_2 & -r_5 & 0 & \phi_2 & 0 & \theta_4 & -v_5 & 0 & 0 \\
 & & & & & u_6+\xi_5+r_3 & 0 & 0 & \phi_3 & 0 & \theta_5 & -v_6 & 0 \\
 & & & & & & u_7+\gamma_1 & -r_7 & 0 & -c_7 & 0 & 0 & 0 \\
 & & & & & & & u_8+\xi_7+\gamma_2 & -r_8 & \varphi_7 & -c_8 & 0 & 0 \\
 & & & & & & & & u_9+\xi_8+\gamma_3 & 0 & \varphi_8 & -c_9 & 0 \\
 & & & & & & & & & u_{10}+r_7+\gamma_4 & -r_{10} & 0 & 0 \\
 & & & & & & & & & & u_{11}+\xi_{10}+r_8+\gamma_5 & -r_{11} & 0 \\
 & & & & & & & & & & & u_{12}+\xi_{11}+r_9+\gamma_6 & 0 \\
 & & & & & & & & & & & & 0
\end{bmatrix}
$$

symmetric

$$
\xi_n = \frac{r_n^2}{u_n} \qquad r_n = \frac{c_n^2}{u_n} \qquad \gamma_n = \frac{v_n^2}{u_n} \qquad \varphi_n = \frac{r_n c_n}{u_n} \qquad \theta_n = \frac{r_n v_n}{u_n} \qquad \phi_n = \frac{c_n v_n}{u_n}
$$

Figure 3.--Matrix $\underline{M} = \underline{U}^T \underline{D} \underline{U}$ for MICCG. The r_n, c_n, and v_n are the conductances along rows and columns and between layers, respectively.

where,

$$f_{ij} = \begin{cases} \sum_{k=1}^{i-1} u_{ki}\, u_{kj}/u_{kk} & a_{ij} = 0 \\ = 0 & a_{ij} \neq 0 \end{cases} , \qquad (11b)$$

and

$$u_{ij} = 0 \text{ for } j < i \ .$$

Again, the more general notation for components of $\underline{U}$ is used. Note that the f_{ji} and f_{ij} are equivalent to the ψ_n, θ_n, and ϕ_n of figure 3, and that they occur off the main diagonal of $\underline{U}^T \underline{D} \underline{U}$. The variable α is a user-defined relaxation parameter and is used to diminish the value of the f_{ij} of equation 11a. Ashcraft and Grimes (1988) found that using a relaxation parameter value of 0.97, 0.98, or 0.99 instead of 1.00 sometimes improved convergence by as much as 50 percent. However, consistently using a value of 1.00 for the relaxation parameter generally produces solutions which are at least as efficient as those attained with other commonly used solvers (Hill, in press).

The equation $\underline{U}^T \underline{D} \underline{U} \underline{s}_k = \underline{r}_k$ is solved by a two-step process. First, $\underline{U}^T \underline{v}_k = \underline{r}_k$ is solved for $\underline{v}_k$ by forward substitution, then $\underline{D} \underline{U} \underline{s}_k = \underline{v}_k$ is solved for $\underline{s}_k$ by backward substitution.

The MICCG preconditioned matrix presented in this paper is from Axelsson and Lindskog (1986), and is nearly identical to that presented by Dupont and others (1968), Gustafsson (1978, eq. 3.1, and 1979, eq. 6), Wong (1979), and Ashcraft and Grimes (1988). In the method used in this report, the overcompensation parameter used by Gustafsson (1978) to augment a_{ii} of equation 11a equals zero. Ashcraft and Grimes (1988) found that the number of iterations required to achieve convergence was insensitive to values of the overcompensation factor, so using a value of zero should not affect convergence.

Convergence properties of incomplete Cholesky with row-sums agreement are discussed by Gustafsson (1978).

POLCG

In Neuman series polynomial preconditioning, $\underline{M}^{-1}$ equals the sum of several terms of a power-series expansion for the inverse of matrix $\underline{A}$ (Dubois and others, 1979; Johnson and others, 1983; Saad, 1985), so that

$$\underline{M}^{-1} = \underline{I} + \underline{A} + \underline{A}^2 + \cdots + \underline{A}^{\ell} \ . \qquad (12)$$

Then, by weighting the terms as suggested by Johnson and others (1983) and Saad (1985), an approximate solution can be written as

$$\underline{s}_k = \underline{M}^{-1}\,\underline{r}_k = c_0\,\underline{r}_k + c_1\,\underline{A}\,\underline{r}_k + c_2\,\underline{A}^2\underline{r}_k + \underline{A}^3\underline{r}_k\ , \qquad (13)$$

when $\ell = 3$, and c_0, c_1, and c_2 are coefficients chosen to optimize convergence. For computational efficiency, equation 13 is calculated by the following series of steps:

$$\begin{aligned} \underline{z}_1 &= c_2\,\underline{r}_k + \underline{A}\,\underline{r}_k \\ \underline{z}_2 &= c_1\,\underline{r}_k + \underline{A}\,\underline{z}_1 \\ \underline{s}_k &= c_0\,\underline{r}_k + \underline{A}\,\underline{z}_2 \end{aligned} \qquad (14)$$

so that the powers of $\underline{A}$ are never formed explicity (Dubois and others, 1979, p. 259). One of the advantages of POLCG is that the steps of equation 14 are efficient on vector and parallel computers.

The coefficients can be calculated using the method described by Saad (1985, p. 869-871; 880) as: $c_0 = \frac{-15}{32}\,g^3$, $c_1 = \frac{27}{16}\,g^2$, $c_2 = \frac{-9}{4}\,g$, where g is the upper bound on the maximum eigenvalue of A, estimated as the largest sum of the absolute values of the components in any row of $\underline{A}$ (Varga, 1962, p. 17; Gerschgorin, 1931). For a scaled matrix, g is generally close to 2. Scandrett (1989) and Hill (in press) used g=2, and the number of iterations required to achieve solutions in their test cases were generally insensitive to changes in g. Using NBPOL of the input file, the user can specify that g=2 or that g is to be estimated as described above. Estimation of g uses slightly more execution time per iteration.

Convergence properties of polynomial preconditioners are discussed by Saad (1985).

Convergence Criteria

An iterative matrix solver is assumed to have converged when some measure of the residual and(or) the difference in results between successive iterations is less than user-specified convergence criteria. In PCG2, the difference between results of successive iterations is measured using both the maximum absolute value of the change in hydraulic head and the maximum absolute value of the residual for that iteration. Typical values for these error criteria are 0.01 ft and 0.01 ft^3/s, respectively.

The defined convergence criteria are too large if the global ground-water flow budget errors calculated by the modular model (McDonald and Harbaugh, 1988, p. 3-16 to 3-22) are unacceptably large. What is unacceptable depends on the problem being considered and must be determined by the user. For most ground-water flow problems, global budget errors greater than one percent are unacceptable. If unacceptably large global budget errors occur, the error criteria should be reduced.

The defined convergence criteria are too small if the accuracy achieved by the solver exceeds the accuracy required by the user. For example, Hill (in press) found that reducing both error criteria from 10^{-3} to 10^{-6} increased the execution time by as much as 55 percent, and, especially for POLCG, resulted in lack of convergence in some test cases. If the solver is taking more iterations than expected to achieve convergence and the calculated global budget error is smaller than required by the user, or if the solver is not converging and the convergence criteria are very small, the user can increase the convergence criteria.

INPUT INSTRUCTIONS

Input for PCG2 is read from a unit specified in the IUNIT array of the Basic Package input file. In the example provided in this report, IUNIT(13) is used, but this can easily be changed, as noted below in the section "Linking this program to the modular model". The input for PCG2 is as follows.

FOR EACH SIMULATION

PCG2AL

1.	Data:	MXITER	ITER1	NPCOND
	Format:	I10	I10	I1C

PCG2RP

2.	Data:	HCLOSE	RCLOSE	RELAX	NBPOL	IPRPCG	MUTPCG	IPCGCD
		F10.0	F10.0	F10.0	I10	I10	I10	I10

Explanation of Fields Used in Input Instructions

MXITER--is the maximum number of outer iterations -- that is, calls to the solution routine. For a linear problem MXITER should be 1, unless more than 50 inner iterations are required, when MXITER could be as large as 10. A larger number (generally less than 100) is required for a nonlinear problem.

ITER1---is the maximum number of inner iterations. For nonlinear problems, ITER1 usually ranges from 3 to 10; a value of 30 will be sufficient for most linear problems.

NPCOND--is the flag used to select the matrix preconditioning method. The following options are available.

NPCOND	PRECONDITIONING METHOD
1	Modified Incomplete Cholesky (for use on scalar computers)
2	Polynomial (for use on vector computers or to conserve computer storage)

HCLOSE--is the head change criterion for convergence, in units of length. When the maximum absolute value of the head change at all nodes during an iteration is less than or equal to HCLOSE, and the criterion for RCLOSE is satisfied (see below), iteration stops. Commonly, HCLOSE equals 0.01.

RCLOSE--is the residual criterion for convergence, in units of cubic length per time. When the maximum absolute value of the residual at all nodes during an iteration is less than or equal to RCLOSE, and the criterion for HCLOSE is satisfied (see above), iteration stops. Commonly, RCLOSE equals HCLOSE.

For nonlinear problems, convergence is achieved when the convergence criteria are satisfied on the first inner iteration.

RELAX---is the relaxation parameter used with NPCOND=1 (MICCG). Usually, RELAX=1.0, but for some problems a value of 0.99, 0.98, or 0.97 will reduce the number of iterations required for convergence. RELAX is not used if NPCOND≠1.

NBPOL---is used when NPCOND=2 to indicate whether the estimate of the upper bound on the maximum eigenvalue is 2.0, or whether the estimate will be calculated. NBPOL=2 is used to specify the value as 2.0; for any other value of NBPOL, the estimate is calculated. Convergence is generally insensitive to this parameter. NBPOL is not used if NPCOND does not equal 2.

IPRPCG--is the printout interval for PCG. If IPRPCG is equal to zero, it is changed to 999. The extreme head change and residual (positive or negative) are printed for each iteration of a time step whenever the time step is an even multiple of IPRPCG. The printout also occurs at the end of each stress period regardless of the value of IPRPCG.

MUTPCG--is a flag which controls printing from the solver. If MUTPCG≠0, printing from the solver is suppressed. If MUTPCG=1, the number of iterations is printed, but the lists of extreme head changes and residuals is suppressed. If MUTPCG=2, all printing is suppressed.

IPCGCD--is a flag which is used when NPCOND=1 to control whether the same Cholesky decomposition may be used for multiple calls to PCG2AP. IPCGCD should be zero for most applications. However, future packages might benefit from nonzero values of IPCGCD.

SAMPLE DATA INPUTS

Example data set for a linear problem:

```
        10        20        30        40        50
123456789 123456789 123456789 123456789 123456789
         1        99         2
      .001      .001        1.         2         1
```

Example data set for a nonlinear problem:

```
        10        20        30        40        50
123456789 123456789 123456789 123456789 123456789
        10         5         1
       .01       .01        1.         2         1
```

LINKING PCG2 TO THE MODULAR MODEL

The following statements must be included in the main program of the modular model. Note that IUNIT(13) can be changed to allow the PCG2 input unit number to be read from a different position of the IUNIT array. IUNIT(15) is the input number for another package which may benefit from IPCGCD≠0. The "15" can be changed if this is not applicable for the package represented by IUNIT(15).

```
C     ADD BETWEEN COMMENT STATEMENTS 4 AND 5
      IF(IUNIT(13).GT.0) CALL PCG2AL(ISUM,LENX,LCV,LCSS,LCP,LCCD,
     1    LCHCHG,LCLHCH,LCRCHG,LCLRCH,MXITER,ITER1,NCOL,NROW,NLAY,
     2    IUNIT(13),IOUT,NPCOND)

C     ADD BETWEEN COMMENT STATEMENTS 6 AND 7
      IF(IUNIT(13).GT.0) CALL PCG2RP(MXITER,ITER1,HCLOSE,RCLOSE,
     1    NPCOND,NBPOL,RELAX,IPRPCG,IUNIT(13),IOUT,MUTPCG,IPCGCD)

C     ADD BETWEEN COMMENT STATEMENTS 7C2B AND 7C2C
      ICD=0
      IF(IUNIT(13).GT.0) CALL PCG2AP(X(LCHNEW),XICD=0(LCIBOU),X(LCCR),
     1    X(LCCC),X(LCCV),X(LCHCOF),X(LCRHS),X(LCV),X(LCSS),X(LCP),
     2    X(LCCD),X(LCHCHG),X(LCLHCH),X(LCRCHG),X(LCLRCH),KITER,
     3    NITER,HCLOSE,RCLOSE,ICNVG,KSTP,KPER,IPRPCG,MXITER,ITER1,
     4    NPCOND,NBPOL,NSTP,NCOL,NROW,NLAY,NODES,RELAX,IOUT,MUTPCG,
     5    IPCGCD,STEPL,DELT,IUNIT(15),IP)
```

LAYCON=3 layers may produce a matrix which is not diagonally dominant when using the BCF package as presented in McDonald and Harbaugh (1988, p. 5-22 and 5-59). MICCG (NPCOND=1) of the PCG2 package will not function if the matrix is not diagonally dominant, and POLCG may not converge. To correct this, make the following change in module BCF1FM (M.G. McDonald, U.S. Geological Survey, written commun., 1989):

McDonald and Harbaugh (1988) version:

```
C7D-----WITH HEAD BELOW TOP ADD CORRECTION TERMS TO RHS AND HCOF.
      RHS(J,I,K)=RHS(J,I,K) + CV(J,I,K-1)*TOP(J,I,KT)
      HCOF(J,I,K)=HCOF(J,I,K) + CV(J,I,K-1)
  220 CONTINUE
```

Modified version:

```
C7D-----WITH HEAD BELOW TOP ADD CORRECTION TERMS TO RHS AND HCOF.
C7D-----MODIFIED TO PUT CORRECTION COMPLETELY ONTO RIGHT HAND SIDE
      RHS(J,I,K)=RHS(J,I,K) + CV(J,I,K-1)*(TOP(J,I,KT)-HTMP)
  220 CONTINUE
```

After making the required changes and including the FORTRAN listed in this document, compile and load the modular model as usual.

DOCUMENTATION OF PCG2

Brief Description of Modules

Three primary modules and two submodules were created for the preconditioned conjugate-gradient method. The following is a brief description of the purpose of each of these modules:

Primary modules:

PCG2AL Allocates space for the conjugate-gradient calculations.

PCG2RP Reads, stores and prints the input data.

PCG2AP Performs multiple iterations of the conjugate-gradient method and checks the convergence criteria.

Submodules:

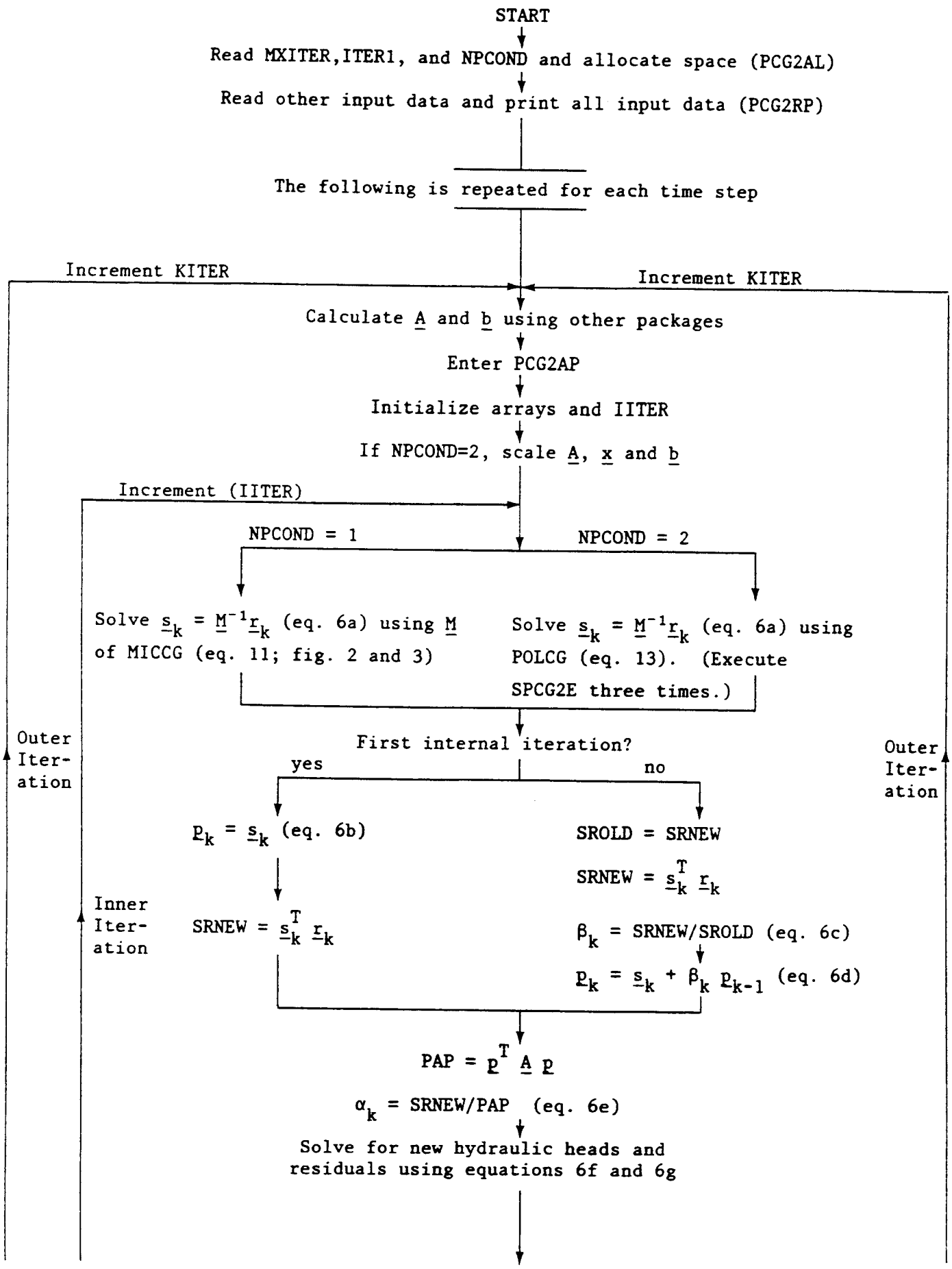
SPCG2P Called by PCG2AP to print the extreme head changes and residuals that occurred at each iteration.

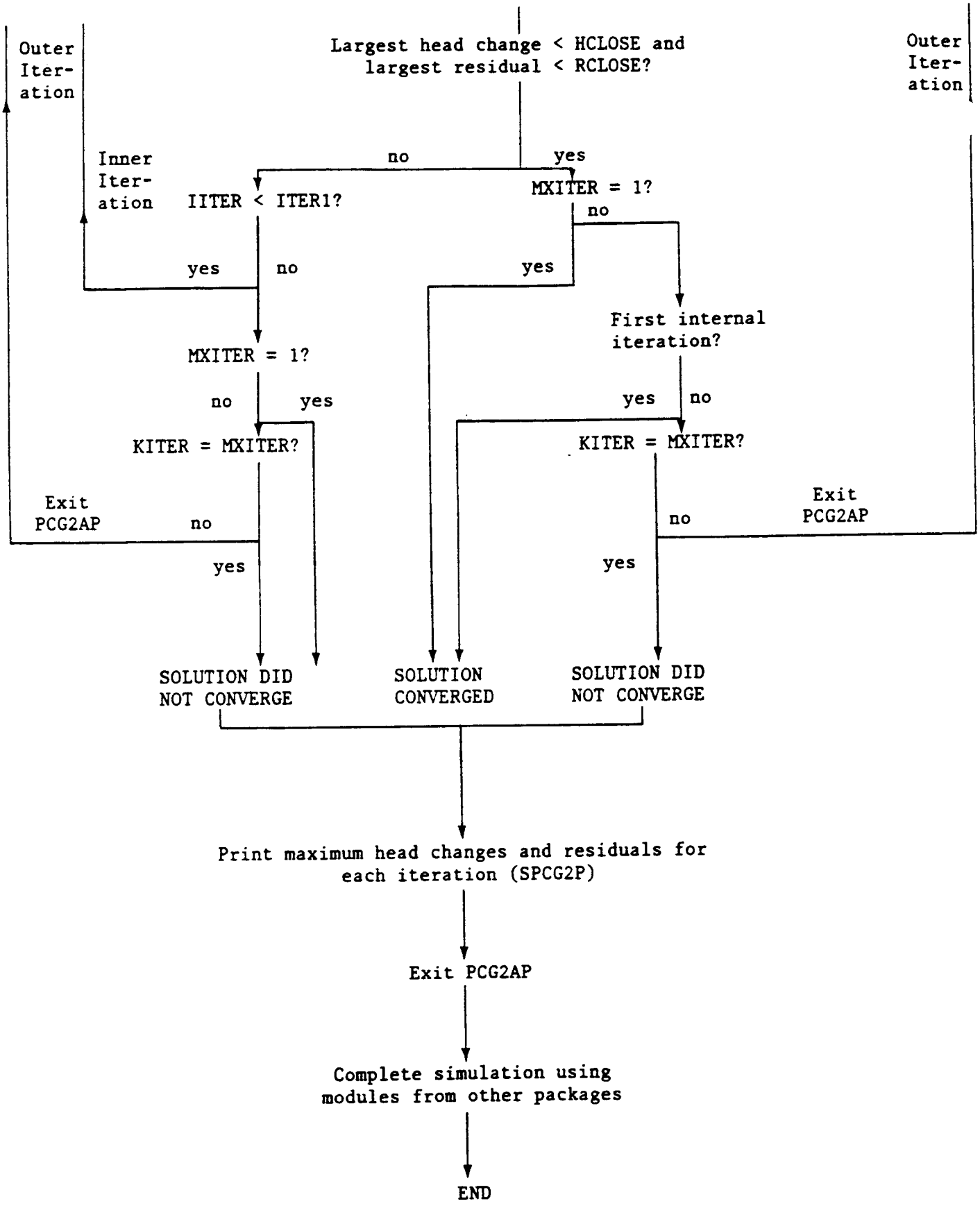
SPCG2E Called by PCG2AP to perform one matrix multiplication required by the polynomial preconditioner. This submodule is executed three times for each iteration of POLCG.

Flowchart

The flowchart for the package is shown below, and includes the functions performed by the three primary modules and the two submodules. The following variable names used in the flowchart are taken from the FORTRAN code. Some were defined in the input instructions; all are defined later in the following list of variables.

HCLOSE, IITER, ITER1, MXITER,
NPCOND, PAP, RCLOSE, SRNEW, SROLD


START
Read MXITER,ITER1, and NPCOND and allocate space (PCG2AL)
Read other input data and print all input data (PCG2RP)
The following is repeated for each time step
Increment KITER
Increment KITER
Calculate A and b using other packages
Enter PCG2AP
Initialize arrays and IITER
If NPCOND=2, scale A, x and b
Increment (IITER)
NPCOND = 1
NPCOND = 2
Solve $s_k = M^{-1} r_k$ (eq. 6a) using M of MICCG (eq. 11; fig. 2 and 3)
Solve $s_k = M^{-1} r_k$ (eq. 6a) using POLCG (eq. 13). (Execute SPCG2E three times.)
Outer Iteration
First internal iteration?
yes
no
Outer Iteration
$p_k = s_k$ (eq. 6b)
SROLD = SRNEW
SRNEW = $s_k^T r_k$
Inner Iteration
SRNEW = $s_k^T r_k$
β_k = SRNEW/SROLD (eq. 6c)
$p_k = s_k + \beta_k p_{k-1}$ (eq. 6d)
PAP = $p^T A p$
α_k = SRNEW/PAP (eq. 6e)
Solve for new hydraulic heads and residuals using equations 6f and 6g


Outer Iteration
Largest head change < HCLOSE and largest residual < RCLOSE?
Outer Iteration
Inner Iteration
no
yes
IITER < ITER1?
MXITER = 1?
no
yes
no
yes
First internal iteration?
MXITER = 1?
no
yes
yes
no
KITER = MXITER?
KITER = MXITER?
Exit PCG2AP
no
Exit PCG2AP
no
yes
yes
SOLUTION DID NOT CONVERGE
SOLUTION CONVERGED
SOLUTION DID NOT CONVERGE
Print maximum head changes and residuals for each iteration (SPCG2P)
Exit PCG2AP
Complete simulation using modules from other packages
END

Narrative for Modules

PCG2AL

Module PCG2AL reads MXITER, ITER1, and NPCOND, and allocates space in the X array for the arrays required for the solver. Arrays SS, P and V are required for both values of NPCOND; array CD is required only for NPCOND=1. Each of these four arrays are vectors dimensioned equal to the number of grid nodes. Additionally, four smaller arrays are required to store the maximum head change and residual at each iteration (HCHG and RCHG) and the cell locations where these occurred (LHCH and LRCH).

PCG2RP

Module PCG2RP reads HCLOSE, RCLOSE, RELAX, NBPOL, IPRPCG, MUTPCG, and IPCGCD, and prints all variables read for this package.

PCG2AP

Module PCG2AP performs up to ITER1 iterations of the preconditioned conjugate-gradient algorithm for solving the flow equation. To save computational time, all arrays are declared one dimensional. They are accessed by a single index which is calculated from the layer, row, and column indexes normally used to access the arrays in three dimensions.

Double precision scalar variables are used for most calculations in this module to improve the accuracy of the results. Modification of the present use of double precision may affect simulated results.

For the polynomial preconditioner, c_0 and c_1 of equation 14 are negative in the text, but are calculated as positive numbers in the FORTRAN code. The FORTRAN code is correct because, as programmed in the modular model, matrix $\underline{A}$ of equation 1 is negative definite instead of positive definite. While this poses no mathematical difficulty, it does require that the odd-numbered coefficients of equation 14 be positive instead of negative.

The steps executed by PCG2AP were outlined in the flowchart previously presented in this section.

SPCG2P

Submodule SPCG2P prints the extreme values of the head change (HCHG) and residual (RCHG) out of all cells for each iteration of a time step. The cell location (LHCH and LRCH) where the values occur also is printed.

SPCG2E

Submodule SPCG2E calculates the matrix-vector multiplication and the vector addition required by each of the three parts of equation 14. It is called three times for each polynomial iteration.

Adapting SPCG2E for Computers with Vector and Parallel Architecture

Use of the polynomial preconditioner on computers with vector and(or) parallel architecture will be most efficient if SPCG2E is modified to take advantage of the computer used. The following points should be considered when modifying SPCG2E.

The diagonal entries of $\underline{A}$ all equal -1.0 because $\underline{A}$ is scaled and is negative definite (see narrative for modeule PCG2AP). Vectors CR, CC, and CV contain the off-diagonals of $\underline{A}$ along which nonzero entries occur. Note that although array CV is dimensioned using NODES in SPCG2E and elsewhere in the model, it is originally only given storage space in X for cells in NLAY-1 layers (McDonald and Harbaugh, 1988, p. 4-25). In making changes, care must be taken to avoid overwriting elements of array HCOF, which is stored in X after CV.

Using $\underline{w}$ to represent $\underline{r}_k$, $\underline{z}_1$, or $\underline{z}_2$, and n to represent the cell number, the nth element of each vector produced by the matrix-vector multiplications of equation 14 is calculated by summing -w(n) with:

CR(n) * w(n+1)		(15a)
CR(n) * w(n)		(15b)
CC(n) * w(n+NCOL)		(15c)
CC(n) * w(n)		(15d)
CV(n) * w(n+NRC)	n≤NODES-NRC	(15e)
CV(n) * w(n)		(15f)

Equations 15a through 15f are vector-vector multiplications which can be calculated quickly on computers with vector architecture, but one problem exists. Along with rows and columns for active cells, $\underline{A}$ includes rows and columns for inactive and constant-head cells. Inactive cells are accounted for by setting appropriate entries of CR, CC, and CV to zero in the beginning of the "DO 115" loop in PCG2AP and, therefore, cause no problem. However, constant-head cells are accounted for using the IF statements in SPCG2E, and these IF statements must be eliminated to vectorize the multiplications.

One way to eliminate the IF statements in SPCG2E is to add a work vector of length NODES (allocate space in PCG2AL) and use this vector to store CR, CC, or CV temporarily while performing the multiplications of equation 15 with the work vector. In the work vector, the entries along rows associated with constant-head cells can be set to zero prior to performing the multiplications. Thus, in equation 15a, CR(n)=0 if cell n+1 is constant head; in equation 15c, CC(n)=0 if cell n+NCOL is constant head; in equation 15e, CV(n)=0 if cell n+NRC is constant head. In equations 15b, 15d, and 15f, CR(n)=0, CC(n)=0, and CV(n)=0 if cell n is constant head.

Alternatively, vectors CR, CC, and CV may be used directly in equation 15, and the values which are set to zero may be stored in a separate array and replaced once the multiplication has been completed. This would require a work vector with length equal to the number of constant-head cells in the grid. At present, the number of constant-head cells is not available when space is allocated by calling PCG2AL, and the user would have to modify the data input and the module to provide this information.

List of Variables

Variables for the entire package are listed below. Variables not listed below are used only briefly in a few calculations and their meaning can be identified from nearby lines of the code. The name of the module is listed under 'Range' for variables used by only one module.

Variable	Range	Definition
ALPHA	PCG2AP	Double-precision field containing α_k of equation 6e.
BIGH	PCG2AP	Value of head change with the largest absolute value for one iteration.
BIGR	PCG2AP	Value of residual with the largest absolute value for one iteration.
BPOLY	PCG2AP	Estimated upper bound of the maximum eigen value of $\underline{A}$; used in polynomial preconditioning.
C0, C1, C2	PCG2AP	Scalar coefficients of equations 13 and 14.
CC	Global	DIMENSION (NCOL,NROW,NLAY), conductance along columns (see fig. 1).
CD	PCG2AP	DIMENSION (NCOL,NROW,NLAY), u_{ii} of equation 11a.
CD1	PCG2AP	The first nonzero component of CD. Used to ensure that all values in CD are all ≧0 or ≦0.
CDCC, CDCR, CDCV	PCG2AP	Double-precision fields containing u_{ki}^2/u_{kk} terms of equation 11a.
CR	Global	DIMENSION (NCOL,NROW,NLAY), conductance along rows (see fig. 1).
CV	Global	DIMENSION (NCOL,NROW,NLAY), conductance between layers (see fig. 1).
DONE	Package	Double-precision field containing a one.
DZERO	PCG2AP	Double-precision field containing a zero.
FCC, FCR, FCV	PCG2AP	Double-precision fields containing f_{ij} terms of equation 11.
HCHG	Package	DIMENSION (MXITER*ITER1), extreme head change (BIGH) for each iteration.
HCHGN	PCG2AP	Double-precision field containing the head change at one cell at one iteration.
HCLOSE	Package	Closure criteria for the head change for the iterative procedure.
HCOF	Global	DIMENSION (NCOL,NROW,NLAY), coefficient of head at cell (J,I,K) in the finite-difference equation.
HNEW	Global	DIMENSION (NCOL,NROW,NLAY), most recent estimate of head in each cell. HNEW changes at each iteration.
IBOUND	Global	DIMENSION (NCOL,NROW,NLAY), status of each cell <0, constant-head cell =0, inactive cell >0, variable-head cell
ICNVG	Global	Flag set equal to zero until the iteration procedure has converged, when it is set to one.
IICNVG	PCG2AP	Inner iteration convergence flag.
IITER	PCG2AP	Inner iteration counter. Reset each time PCG2AP is called.
IN	Package	Primary unit number from which input for this package will be read.

Variable	Range	Definition
IOUT	Global	Primary unit number for all printed output. IOUT=6.
IPRPCG	Package	Frequency (in time steps) with which the extreme head changes and residuals for each iteration will be printed.
ISIZ	PCG2AL	Number of cells (nodes) in the finite-difference grid.
ISOLD	Package	Before this module allocates space, ISOLD is set equal to ISUM. After allocation, ISOLD is subtracted from ISUM to get ISP, the amount of space in the X array allocated by this module.
ISP	PCG2AL	Number of words in the X array allocated by this module.
ISUM	Global	Index number of the lowest element in the X array which has not yet been allocated. When space is allocated for an array, the size of the array is added to ISUM.
ITER1	Package	Maximum number of inner iterations.
KITER	Global	Counts the number of times PCG2AP is called.
KPER	Global	Stress period counter.
KSTP	Global	Time step counter. Reset at the start of each stress period.
LCname	Package	Location in the X array of the first element of array 'name'.
LENX	Global	Length of the X array in words. This should always be equal to the dimension of X specified in the MAIN program.
LHCH	Package	DIMENSION (3,MXITER*ITER1), Layer, row, and column of the cell containing the extreme head change (BIGH) for each iteration.
LRCH	Package	DIMENSION (3,MXITER*ITER1), Layer, row, and column of the cell containing the extreme residual (BIGR) for each iteration.
MXITER	Package	Maximum number of calls of PCG2AP.
MUTPCG	Package	Flag to control printing from the solver (see input instructions).
N	Package	Cell index.
NBPOL	Package	Used when NPCOND=2 to indicate how the value of the upper bound of the maximum eigenvalue is calculated (see input instructions).
NCD	Package	One-dimensional subscript of conductance to the adjacent cell, which is in the last column.
NCF	Package	One-dimensional subscript of conductance to the adjacent cell, which is in the next column.
NCL	Package	One-dimensional subscript of the cell index of the adjacent cell which is in the last column.
NCN	Package	One-dimensional subscript of the cell index of the adjacent cell which is in the next column.
NCOL	Global	Number of columns in the grid.
NITER	PCG2AP	Counts the total number of inner iterations that are executed.
NLAY	Global	Number of layers in the grid.
NLL	Package	One-dimensional subscript of the cell index of the adjacent cell which is in the last layer.
NLN	Package	One-dimensional subscript of the cell index of the adjacent cell which is in the next layer.

Variable	Range	Definition
NLS	Package	One-dimensional subscript of conductance to the adjacent cell which is in the next layer.
NLZ	Package	One-dimensional subscript of conductance to the adjacent cell which is in the last layer.
NODES	Global	Number of cells (nodes) in the finite-difference grid.
NORM	PCG2AP	Flag for scaling the matrix set of equations. NORM=1 and scaling is executed only when NPCOND=2.
NPCOND	Package	Preconditioner used: 1 Incomplete Cholesky with row-sums agreement 2 Polynomial
NRB	Module	One-dimensional subscript of conductance to the adjacent cell which is in the last row.
NRC	Package	Number of cells in a model layer.
NRH	Package	One-dimensional subscript of conductance to the adjacent cell which is in the next row.
NRL	Package	One-dimensional subscript of the cell index of the adjacent cell which is in the last row.
NRN	Package	One-dimensional subscript of the cell index of the adjacent cell which is in the next row.
NROW	Global	Number of rows in the grid.
NSTP	Global	Number of time steps in the current stress period.
P	PCG2AP	DIMENSION (NCOL,NROW,NLAY), $\underline{p}_k$ and $\underline{p}_{k-1}$ of equations 6d-6g.
PAP	PCG2AP	Double-precision field containing the denominator of equation 6e.
RCHG	PCG2AP	DIMENSION (MXITER*ITER1), Extreme residual (BIGR) for each iteration.
RCHGN	PCG2AP	Double-precision field containing the residual change at one cell at one iteration.
RCLOSE	Package	Closure criteria for the residual for the iterative procedure.
RELAX	Package	Relaxation parameter of equation 11a.
RHS	Global	DIMENSION (NCOL,NROW,NLAY), right-hand side of the finite-difference equation. RHS is an accumulation of terms from several different packages.
SROLD	PCG2AP	Double-precision field containing the denominator of equation 6c.
SRNEW	PCG2AP	Double-precision field containing the numnerator of equations 6c and 6e.
SS	PCG2AP	DIMENSION (NCOL,NROW,NLAY), $\underline{s}_k$ of equation 6a-6e.
V	PCG2AP	DIMENSION (NCOL,NROW,NLAY), intermediate solution when solving equation 6a, and when calculating PAP.

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FORTRAN LISTING

```
      SUBROUTINE PCG2AL(ISUM,LENX,LCV,LCSS,LCP,LCCD,LCHCHG,LCLHCH,
     1       LCRCHG,LCLRCH,MXITER,ITER1,NCOL,NROW,NLAY,IN,IOUT,NPCOND)
C
C-----VERSION 0002 01MAY1989 PCG2AL
C
C     ******************************************************************
C     ALLOCATE STORAGE IN THE X ARRAY FOR PCG ARRAYS
C     ******************************************************************
C
C        SPECIFICATIONS:
C     ------------------------------------------------------------------
C     ------------------------------------------------------------------
C
C-------PRINT A MESSAGE IDENTIFYING PCG PACKAGE
      WRITE(IOUT,1)
    1 FORMAT(1H0,'PCG2 -- CONJUGATE GRADIENT SOLUTION PACKAGE'
     1,', VERSION 2, 5/1/88')
C
C-------READ AND PRINT MXITER,ITER1 AND NPCOND
      READ(IN,2) MXITER,ITER1,NPCOND
    2 FORMAT(3I10)
      WRITE(IOUT,3) MXITER,ITER1,NPCOND
    3 FORMAT(' MAXIMUM OF',I4,' CALLS OF SOLUTION ROUTINE'/
     1       ,' MAXIMUM OF',I4,' INTERNAL ITERATIONS PER '
     2       ,'CALL TO SOLUTION ROUTINE'/
     3       ,' MATRIX PRECONDITIONING TYPE :',I5)
C
C-------ALLOCATE SPACE FOR THE PCG ARRAYS
      ISOLD=ISUM
      NRC=NROW*NCOL
      ISIZ=NRC*NLAY
      LCV=ISUM
      ISUM=ISUM+ISIZ
      LCSS=ISUM
      ISUM=ISUM+ISIZ
      LCP=ISUM
      ISUM=ISUM+ISIZ
      LCCD=ISUM
      IF(NPCOND.NE.2) ISUM=ISUM+ISIZ
      LCHCHG=ISUM
      ISUM=ISUM+MXITER*ITER1
      LCLHCH=ISUM
      ISUM=ISUM+3*MXITER*ITER1
      LCRCHG=ISUM
      ISUM=ISUM+MXITER*ITER1
      LCLRCH=ISUM
      ISUM=ISUM+3*MXITER*ITER1
C
C-------CALCULATE AND PRINT THE SPACE USED IN THE X ARRAY
      ICG=ISUM-ISOLD
      WRITE(IOUT,4) ICG
    4 FORMAT(1X,I7,' ELEMENTS IN X ARRAY ARE USED BY PCG')
      ISUM1=ISUM-1
      WRITE(IOUT,5) ISUM1,LENX
    5 FORMAT(1X,I7,' ELEMENTS OF X ARRAY USED OUT OF',I7)
```

```
      IF(ISUM1.GT.LENX) WRITE(IOUT,6)
    6 FORMAT(1X,'   ***X ARRAY MUST BE DIMENSIONED LARGER***')
C
C-------RETURN
      RETURN
      END
```

```
      SUBROUTINE PCG2RP(MXITER,ITER1,HCLOSE,RCLOSE,NPCOND,NBPOL,
     1                  RELAX,IPRPCG,IN,IOUT,MUTPCG,IPCGCD)
C
C-----VERSION 0002 01MAY1989 PCG2RP
C
C     ******************************************************************
C     READ DATA FOR PCG
C     ******************************************************************
C
C        SPECIFICATIONS:
C     ------------------------------------------------------------------
C     ------------------------------------------------------------------
C
C-------READ HCLOSE,RCLOSE,RELAX,NBPOL,IPRPCG,MUTPCG
      READ(IN,1) HCLOSE,RCLOSE,RELAX,NBPOL,IPRPCG,MUTPCG,IPCGCD
    1 FORMAT(3F10.0,4I10)
C
C-------PRINT MXITER,ITER1,NPCOND,HCLOSE,RCLOSE,RELAX,NBPOL,IPRPCG,
C-------MUTPCG, IPCGCD
      WRITE(IOUT,100)
  100 FORMAT(1H0,///57X,'SOLUTION BY THE CONJUGATE-GRADIENT METHOD'
     1/57X,43('-'))
      WRITE(IOUT,115) MXITER
  115 FORMAT(1H0,38X,'MAXIMUM NUMBER OF CALLS TO PCG ROUTINE =',I9)
      WRITE(IOUT,120) ITER1
  120 FORMAT(1H ,42X,'MAXIMUM ITERATIONS PER CALL TO PCG =',I9)
      WRITE(IOUT,122) NPCOND
  122 FORMAT(1H ,49X,'MATRIX PRECONDITIONING TYPE =',I9)
      IF(NPCOND.EQ.2) WRITE(IOUT,123)
  123 FORMAT(1H ,58X,'THE MATRIX WILL BE SCALED')
      WRITE(IOUT,124) RELAX,NBPOL
  124 FORMAT(1H ,26X,'RELAXATION FACTOR (ONLY USED WITH',
     1' PRECOND. TYPE 1) =',E15.5,/,
     2 1H ,19X,'PARAMETER OF POLYMOMIAL PRECOND.'
     3 ,' = 2 (2) OR IS CALCULATED :',I9)
      WRITE(IOUT,125) HCLOSE
  125 FORMAT(1H ,43X,'HEAD CHANGE CRITERION FOR CLOSURE =',E15.5)
      WRITE(IOUT,127) RCLOSE
  127 FORMAT(1H ,39X,'RESIDUAL CHANGE CRITERION FOR CLOSURE =',E15.5)
      IF(IPRPCG.LE.0)IPRPCG=999
      WRITE(IOUT,130) IPRPCG,MUTPCG
  130 FORMAT(1H ,30X,'PCG HEAD AND RESIDUAL CHANGE PRINTOUT INTERVAL ='
     1,I9,/,1H ,30X,'ALL PRINTING FROM THE SOLVER IS SUPPRESSED (1) ='
     2,I9)
      WRITE(IOUT,135) IPCGCD
  135 FORMAT(1H ,5X,'FOR NPCOND=1, DO (0) OR DO NOT (1) RECALC.',
     1  ' CHOL. DIAG. EACH OUTER ITER. =',I9,/)
C
      RETURN
      END
```

```
      SUBROUTINE PCG2AP(HNEW,IBOUND,CR,CC,CV,HCOF,RHS,V,SS,P,CD,
     1      HCHG,LHCH,RCHG,LRCH,KITER,NITER,HCLOSE,RCLOSE,ICNVG,
     2      KSTP,KPER,IPRPCG,MXITER,ITER1,NPCOND,NBPOL,NSTP,NCOL,NROW,
     3      NLAY,NODES,RELAX,IOUT,MUTPCG,IPCGCD,STEPL,DELT,IU,IP)
C-----VERSION 0002 01MAY1989 PCG2AP
C
C     ******************************************************************
C     SOLUTION BY THE CONJUGATE GRADIENT METHOD -
C                                          UP TO ITER1 ITERATIONS
C     ******************************************************************
C
C        SPECIFICATIONS:
C     ------------------------------------------------------------------
      PARAMETER (DZERO=0.D0,DONE=1.D0)
      DOUBLE PRECISION HNEW,HHCOF,RRHS,RES
      DOUBLE PRECISION Z,B,D,E,F,H,S,ALPHA
      DOUBLE PRECISION ZHNEW,BHNEW,DHNEW,FHNEW,HHNEW,SHNEW,HCHNEW
      DOUBLE PRECISION SRNEW,SROLD,SSCR,SSCC,SSCV,VCC,VCR,VCV
      DOUBLE PRECISION CDCC,CDCR,CDCV,CDN
      DOUBLE PRECISION PN,VN,SSN,HCHGN,RCHGN,PAP
      DOUBLE PRECISION FCC,FCR,FCV,FV
C
      DIMENSION HNEW(NODES), IBOUND(NODES), CR(NODES), CC(NODES),
     1  CV(NODES), HCOF(NODES), RHS(NODES),
     2  V(NODES), SS(NODES), P(NODES), CD(NODES), HCHG(MXITER*ITER1),
     3  LHCH(3,MXITER*ITER1), RCHG(MXITER*ITER1), LRCH(3,MXITER*ITER1)
C     ------------------------------------------------------------------
C
C-------ASSIGN VALUES TO FIELDS THAT ARE CONSTANT DURING AN ITERATION
      NRC=NROW*NCOL
C-------INITIALIZE VARIABLES USED TO CALCULATE ITERATION PARAMETERS
      SRNEW=DZERO
      BPOLY=0.
      IF(NPCOND.NE.1) RELAX=1.
      NORM=0
      IF(NPCOND.EQ.2) NORM=1
C-------INITIALIZE VARIABLE USED TO TEST FOR NEGATIVE CHOLESKY DIAGONAL
      CD1=0.
C------CLEAR PCG WORK ARRAYS.
      DO 100 N=1,NODES
      SS(N)=0.
      P(N)=0.
  100 V(N)=0.
      IF(NPCOND.EQ.1) THEN
        ITYPE=0
        IF(IPCGCD.EQ.1.AND.(IU.EQ.0.OR.IP.GT.0.OR.KPER.GT.1)) THEN
          IF(STEPL.EQ.DELT) ITYPE=1
          STEPL=DELT
        ENDIF
        IF(ITYPE.EQ.0) THEN
          DO 105 N=1,NODES
  105     CD(N)=0.
        ENDIF
      ENDIF
C
```

```
C------CALCULATE THE RESIDUAL. IF NORM=1, CALCULATE THE DIAGONALS OF
C------THE A MATRIX,AND STORE THEM IN HCOF.
      DO 115 K=1,NLAY
      DO 115 I=1,NROW
      DO 115 J=1,NCOL
C
C-------CALCULATE 1 DIMENSIONAL SUBSCRIPT OF CURRENT CELL AND
C-------SKIP CALCULATIONS IF CELL IS INACTIVE
      N=J+(I-1)*NCOL+(K-1)*NRC
      IF(IBOUND(N).EQ.0) THEN
        CC(N)=0.
        CR(N)=0.
        IF(N.LE.(NODES-NRC)) CV(N)=0.
        IF(N.GE.2) CR(N-1)=0.
        IF(N.GE.NCOL+1) CC(N-NCOL)=0.
        IF(N.LE.(NODES-NRC).AND.N.GE.NRC+1) CV(N-NRC)=0.
        HCOF(N)=0.
        RHS(N)=0.
        GO TO 115
      ENDIF
C
C-------CALCULATE 1 DIMENSIONAL SUBSCRIPTS FOR LOCATING THE 6
C-------SURROUNDING CELLS
      NRN=N+NCOL
      NRL=N-NCOL
      NCN=N+1
      NCL=N-1
      NLN=N+NRC
      NLL=N-NRC
C
C-------CALCULATE 1 DIMENSIONAL SUBSCRIPTS FOR CONDUCTANCE TO THE 6
C-------SURROUNDING CELLS.
      NCF=N
      NCD=N-1
      NRB=N-NCOL
      NRH=N
      NLS=N
      NLZ=N-NRC
C
C-----GET CONDUCTANCES TO NEIGHBORING CELLS
C-------NEIGHBOR IS 1 ROW BACK
      B=DZERO
      BHNEW=DZERO
      IF(I.NE.1) THEN
        B=CC(NRB)
        BHNEW=B*(HNEW(NRL)-HNEW(N))
      ENDIF
C
C-------NEIGHBOR IS 1 ROW AHEAD
      H=DZERO
      HHNEW=DZERO
      IF(I.NE.NROW) THEN
        H=CC(NRH)
        HHNEW=H*(HNEW(NRN)-HNEW(N))
      ENDIF
```

```
C
C-------NEIGHBOR IS 1 COLUMN BACK
      D=DZERO
      DHNEW=DZERO
      IF(J.NE.1) THEN
        D=CR(NCD)
        DHNEW=D*(HNEW(NCL)-HNEW(N))
      ENDIF
C
C-------NEIGHBOR IS 1 COLUMN AHEAD
      F=DZERO
      FHNEW=DZERO
      IF(J.NE.NCOL) THEN
        F=CR(NCF)
        FHNEW=F*(HNEW(NCN)-HNEW(N))
      ENDIF
C
C-------NEIGHBOR IS 1 LAYER BEHIND
      Z=DZERO
      ZHNEW=DZERO
      IF(K.NE.1) THEN
        Z=CV(NLZ)
        ZHNEW=Z*(HNEW(NLL)-HNEW(N))
      ENDIF
C
C-------NEIGHBOR IS 1 LAYER AHEAD
      S=DZERO
      SHNEW=DZERO
      IF(K.NE.NLAY) THEN
        S=CV(NLS)
        SHNEW=S*(HNEW(NLN)-HNEW(N))
      ENDIF
C
      IF(I.EQ.NROW) CC(N)=0.
      IF(J.EQ.NCOL) CR(N)=0.
C
C-------CALCULATE THE RESIDUAL AND STORE IT IN RHS.  TO SCALE A,
C-------CALCULATE THE DIAGONAL OF THE A MATRIX, AND STORE IT IN HCOF.
      E=-Z-B-D-F-H-S
      RRHS=RHS(N)
      HHCOF=HNEW(N)*HCOF(N)
      RHS(N)=RRHS-ZHNEW-BHNEW-DHNEW-HHCOF-FHNEW-HHNEW-SHNEW
      IF(NORM.EQ.1) HCOF(N)=HCOF(N)+E
      IF(IBOUND(N).LT.0.) RHS(N)=0.
  115 CONTINUE
C
C-------SCALE CC,CR,CV,RHS AND HNEW IF NORM=1.
      IF(NORM.EQ.1) THEN
        DO 120 K=1,NLAY
        DO 120 I=1,NROW
        DO 120 J=1,NCOL
          N=J+(I-1)*NCOL+(K-1)*NRC
          IF(IBOUND(N).EQ.0) GO TO 120
          HHCOF=SQRT(-HCOF(N))
          IF(N.LE.(NODES-NCOL).AND.CC(N).GT.0.)
```

```
     #       CC(N)=CC(N)/(HHCOF*(SQRT(-HCOF(N+NCOL))))
             IF(CR(N).GT.0.) CR(N)=CR(N)/(HHCOF*(SQRT(-HCOF(N+1))))
             IF(N.LE.(NODES-NRC).AND.CV(N).GT.0.)
     #       CV(N)=CV(N)/(HHCOF*(SQRT(-HCOF(N+NRC))))
             HNEW(N)=HNEW(N)*HHCOF
             RHS(N)=RHS(N)/HHCOF
  120    CONTINUE
         ENDIF
C
C-------CALCULATE PARAMETER B OF THE POLYNOMIAL PRECONDITIONING METHOD
         IF(NPCOND.NE.2) GO TO 152
         IF(NBPOL.EQ.2) THEN
           BPOLY=2
           GO TO 151
         ENDIF
         DO 150 K=1,NLAY
         DO 150 I=1,NROW
         DO 150 J=1,NCOL
C
         N=J+(I-1)*NCOL+(K-1)*NRC
         IF(IBOUND(N).LE.0)GO TO 150
C
         NCF=N
         NCD=N-1
         NRB=N-NCOL
         NRH=N
         NLS=N
         NLZ=N-NRC
C
         B=DZERO
         IF(I.NE.1) B=CC(NRB)
         H=DZERO
         IF(I.NE.NROW) H=CC(NRH)
         D=DZERO
         IF(J.NE.1) D=CR(NCD)
         F=DZERO
         IF(J.NE.NCOL) F=CR(NCF)
         Z=DZERO
         IF(K.NE.1) Z=CV(NLZ)
         S=DZERO
         IF(K.NE.NLAY) S=CV(NLS)
C
C-------NOTE : ABS. VAL. OF THE DIAG. OF THE SCALED A MATRIX IS 1.
         HHCOF=HCOF(N)
         IF(NORM.EQ.1) HHCOF=DONE
         T=DABS(Z)+DABS(B)+DABS(D)+ABS(HHCOF)+DABS(F)+DABS(H)+DABS(S)
         IF(T.GT.BPOLY) BPOLY=T
  150 CONTINUE
  151 CONTINUE
C
C-------CALCULATE ITERATION PARAMETERS FOR POLYNOMIAL PRECONDITIONING
C-------METHOD FOR A NEGATIVE DEFINITE MATRIX.
           C0=(15./32.)*(BPOLY**3)
           C1=(27./16.)*(BPOLY**2)
           C2=(9./4.)*BPOLY
```

```
  152 CONTINUE
C
C-------START INTERNAL ITERATIONS
      IITER=0
      IF(KITER.EQ.1) NITER=0
      ICNVG=0
      IICNVG=0
153   CONTINUE
      IITER=IITER+1
      NITER=NITER+1
C
C-------INITIALIZE VARIABLES THAT TRACK MAXIMUM HEAD CHANGE AND RESIDUAL
C-------VALUE DURING EACH ITERATIONS
      BIGH=0.
      BIGR=0.
C
C
C-------CHECK NPCOND FOR PRECONDITIONING TYPE AND EXECUTE PROPER CODE
      IF(NPCOND.EQ.2) GO TO 165
C
C-------CHOLESKY PRECONDITIONING
C
C-------STEP THROUGH CELLS TO CALCULATE THE DIAGONAL OF THE CHOLESKY
C-------MATRIX (FIRST INTERNAL ITERATION ONLY) AND THE INTERMEDIATE
C-------SOLUTION.  STORE THEM IN CD AND V, RESPECTIVELY.
      DO 155 K=1,NLAY
      DO 155 I=1,NROW
      DO 155 J=1,NCOL
C
      N=J+(I-1)*NCOL+(K-1)*NRC
      IF(IBOUND(N).LE.0)GO TO 155
C
C-------CALCULATE V
      H=DZERO
      VCC=DZERO
      IC=N-NCOL
      IF(I.NE.1) THEN
        H=CC(IC)
        IF(CD(IC).NE.0.) VCC=H*V(IC)/CD(IC)
      ENDIF
C
      F=DZERO
      VCR=DZERO
      IR=N-1
      IF(J.NE.1) THEN
        F=CR(IR)
        IF(CD(IR).NE.0.) VCR=F*V(IR)/CD(IR)
      ENDIF
C
      S=DZERO
      VCV=DZERO
      IL=N-NRC
      IF(K.NE.1) THEN
        S=CV(IL)
        IF(CD(IL).NE.0.) VCV=S*V(IL)/CD(IL)
```

```
      ENDIF
      V(N)=RHS(N)-VCR-VCC-VCV
C
C-------CALCULATE CD - FIRST INTERNAL ITERATION ONLY
      IF(IITER.EQ.1.AND.ITYPE.EQ.0) THEN
          CDCR=DZERO
          CDCC=DZERO
          CDCV=DZERO
          FCC=DZERO
          FCR=DZERO
          FCV=DZERO
          IF(IR.GT.0.AND.CD(IR).NE.0.) CDCR=(F**2)/CD(IR)
          IF(IC.GT.0.AND.CD(IC).NE.0.) CDCC=(H**2)/CD(IC)
          IF(IL.GT.0.AND.CD(IL).NE.0.) CDCV=(S**2)/CD(IL)
          IF(NPCOND.EQ.1) THEN
              IF(IR.GT.0) THEN
                  FV=CV(IR)
                  IF(K.EQ.NLAY.AND.((J+I).GT.1)) FV=DZERO
                  IF(CD(IR).NE.0.) FCR=(F/CD(IR))*(CC(IR)+FV)
              ENDIF
              IF(IC.GT.0) THEN
                  FV=CV(IC)
                  IF(K.EQ.NLAY.AND.(I.GT.1)) FV=DZERO
                  IF(CD(IC).NE.0.) FCC=(H/CD(IC))*(CR(IC)+FV)
              ENDIF
              IF(IL.GT.0) THEN
                  IF(CD(IL).NE.0.) FCV=(S/CD(IL))*(CR(IL)+CC(IL))
              ENDIF
          ENDIF
          IF(NORM.EQ.0) THEN
             B=DZERO
             H=DZERO
             D=DZERO
             F=DZERO
             Z=DZERO
             S=DZERO
             IF(I.NE.1) B=CC(IC)
             IF(I.NE.NROW) H=CC(N)
             IF(J.NE.1) D=CR(IR)
             IF(J.NE.NCOL) F=CR(N)
             IF(K.NE.1) Z=CV(IL)
             IF(K.NE.NLAY) S=CV(N)
             HHCOF=HCOF(N)-Z-B-D-F-H-S
          ENDIF
          IF(NORM.EQ.1) HHCOF=-DONE
          CD(N)=HHCOF-CDCR-CDCC-CDCV-RELAX*(FCR+FCC+FCV)
         IF(CD1.EQ.0..AND.CD(N).NE.0.) CD1=CD(N)
         IF(CD(N)*CD1.LT.0.) THEN
           WRITE(IOUT,510)
  510      FORMAT(//,'  CHOLESKY DIAGONAL LESS THAN ZERO -- '
     1        'EXECUTION TERMINATED (MATRIX NOT DIAGONALLY DOMINANT)')
           STOP
         ENDIF
       ENDIF
C
```

```
  155 CONTINUE
C
C-------STEP THROUGH EACH CELL AND SOLVE FOR S OF THE CONJUGATE
C-------GRADIENT ALGORITHM BY BACK SUBSTITUTION. STORE RESULT IN SS.
      DO 160 KK=NLAY,1,-1
      DO 160 II=NROW,1,-1
      DO 160 JJ=NCOL,1,-1
C
      N=JJ+(II-1)*NCOL+(KK-1)*NRC
      IF(IBOUND(N).LE.0)GO TO 160
C
      NC=N+1
      NR=N+NCOL
      NL=N+NRC
C
C-------BACK SUBSTITUTE, STORING RESULT IN ARRAY SS
      SSCR=DZERO
      SSCC=DZERO
      SSCV=DZERO
      IF(JJ.NE.NCOL) SSCR=CR(N)*SS(NC)/CD(N)
      IF(II.NE.NROW) SSCC=CC(N)*SS(NR)/CD(N)
      IF(KK.NE.NLAY) SSCV=CV(N)*SS(NL)/CD(N)
      VN=V(N)/CD(N)
      SS(N)=VN-SSCR-SSCC-SSCV
  160 CONTINUE
C-------SKIP OVER OTHER PRECONDITIONING TYPES
      GO TO 199
  165 CONTINUE
C
C-------POLYNOMIAL PRECONDITIONING
      DO 170 N=1,NODES
      V(N)=RHS(N)
  170 CONTINUE
      CALL SPCG2E(IBOUND,RHS,HCOF,CR,CC,CV,V,SS,C2,NORM,NCOL,NROW,
     1            NLAY,NODES)
      CALL SPCG2E(IBOUND,RHS,HCOF,CR,CC,CV,SS,V,C1,NORM,NCOL,NROW,
     1            NLAY,NODES)
      CALL SPCG2E(IBOUND,RHS,HCOF,CR,CC,CV,V,SS,C0,NORM,NCOL,NROW,
     1            NLAY,NODES)
  199 CONTINUE
C
C-------CALCULATE P OF THE CONJUGATE GRADIENT ALGORITHM
      SROLD=SRNEW
      SRNEW=DZERO
      DO 200 N=1,NODES
        IF(IBOUND(N).LE.0)GO TO 200
        SRNEW=SRNEW+SS(N)*RHS(N)
  200 CONTINUE
C
      IF(IITER.EQ.1) THEN
         DO 205 N=1,NODES
  205    P(N)=SS(N)
      ELSE
         DO 210 N=1,NODES
  210    P(N)=SS(N)+(SRNEW/SROLD)*P(N)
```

```
      ENDIF
C
C-------CALCULATE ALPHA OF THE CONJUGATE GRADIENT ROUTINE.
C-------FOR THE DENOMINATOR OF ALPHA, MULTIPLY THE MATRIX A BY THE
C-------VECTOR P, AND STORE IN V; THEN MULTIPLY P BY V.  STORE IN PAP.
      PAP=DZERO
      DO 290 K=1,NLAY
      DO 290 I=1,NROW
      DO 290 J=1,NCOL
C
      N=J+(I-1)*NCOL+(K-1)*NRC
      V(N)=0.
      IF(IBOUND(N).LE.0)GO TO 290
C
      NRN=N+NCOL
      NRL=N-NCOL
      NCN=N+1
      NCL=N-1
      NLN=N+NRC
      NLL=N-NRC
C
      NCF=N
      NCD=NCL
      NRB=NRL
      NRH=N
      NLS=N
      NLZ=NLL
C
      B=DZERO
      IF(I.NE.1) B=CC(NRB)
      H=DZERO
      IF(I.NE.NROW)H=CC(NRH)
      D=DZERO
      IF(J.NE.1) D=CR(NCD)
      F=DZERO
      IF(J.NE.NCOL) F=CR(NCF)
      Z=DZERO
      IF(K.NE.1) Z=CV(NLZ)
      S=DZERO
      IF(K.NE.NLAY) S=CV(NLS)
C
      IF(NORM.EQ.0) PN=P(N)
      IF(NORM.EQ.1) PN=DZERO
      BHNEW=DZERO
      HHNEW=DZERO
      DHNEW=DZERO
      FHNEW=DZERO
      ZHNEW=DZERO
      SHNEW=DZERO
      IF(NRL.GT.0) BHNEW=B*(P(NRL)-PN)
      IF(NRN.LE.NODES) HHNEW=H*(P(NRN)-PN)
      IF(NCL.GT.0) DHNEW=D*(P(NCL)-PN)
      IF(NCN.LE.NODES) FHNEW=F*(P(NCN)-PN)
      IF(NLL.GT.0) ZHNEW=Z*(P(NLL)-PN)
      IF(NLN.LE.NODES) SHNEW=S*(P(NLN)-PN)
```

```
C
C-------CALCULATE THE PRODUCT OF MATRIX A AND VECTOR P AND STORE
C-------RESULT IN V.
      PN=HCOF(N)*P(N)
      IF(NORM.EQ.1) PN=-P(N)
      VN=ZHNEW+BHNEW+DHNEW+PN+FHNEW+HHNEW+SHNEW
      V(N)=VN
      PAP=PAP+P(N)*VN
  290 CONTINUE
C
C-------CALCULATE ALPHA
      ALPHA=SRNEW/PAP
C
C-------CALCULATE NEW HEADS AND RESIDUALS, AND SAVE THE LARGEST
C-------CHANGE IN HEAD AND THE LARGEST VALUE OF THE RESIDUAL.
      DO 300 K=1,NLAY
      DO 300 I=1,NROW
      DO 300 J=1,NCOL
C
      N=J+(I-1)*NCOL+(K-1)*NRC
      IF(IBOUND(N).LE.0) GO TO 300
C
C-------HEAD
      HCHGN=ALPHA*P(N)
      IF(DABS(HCHGN).GT.ABS(BIGH)) THEN
          BIGH=HCHGN
          IH=I
          JH=J
          KH=K
          NH=N
      ENDIF
      HNEW(N)=HNEW(N)+HCHGN
C
C--------RESIDUAL (V IS THE PRODUCT OF MATRIX A AND VECTOR P)
      RCHGN=-ALPHA*V(N)
      RHS(N)=RHS(N)+RCHGN
      IF(ABS(RHS(N)).GT.ABS(BIGR)) THEN
          BIGR=RHS(N)
          IR=I
          JR=J
          KR=K
          NR=N
      ENDIF
  300 CONTINUE
C
C-------UNSCALE LARGEST CHANGE IN HEAD AND LARGEST RESIDUAL, AND
C-------CHECK THE CONVERGENCE CRITERION
       IF(NORM.EQ.1) THEN
          BIGH=BIGH/SQRT(-HCOF(NH))
          BIGR=BIGR*SQRT(-HCOF(NR))
       ENDIF
       IF(MXITER.EQ.1) THEN
          IF(ABS(BIGH).LE.HCLOSE.AND.ABS(BIGR).LE.RCLOSE) ICNVG=1
       ELSE
          IF(IITER.EQ.1.AND.
```

```
     1       ABS(BIGH).LE.HCLOSE.AND.ABS(BIGR).LE.RCLOSE) ICNVG=1
        ENDIF
        IF(ABS(BIGH).LE.HCLOSE.AND.ABS(BIGR).LE.RCLOSE) IICNVG=1
C
C-------STORE THE LARGEST UNSCALED HEAD CHANGE AND RESIDUAL VALUE
C-------(THIS ITERATION) AND THEIR LOCATIONS.
      II=NITER
      HCHG(II)=BIGH
      LHCH(1,II)=KH
      LHCH(2,II)=IH
      LHCH(3,II)=JH
C
      RCHG(II)=BIGR
      LRCH(1,II)=KR
      LRCH(2,II)=IR
      LRCH(3,II)=JR
C
C-------GO TO NEXT INTERNAL ITERATION IF CONVERGENCE HAS NOT BEEN
C-------REACHED AND IITER IS LESS THAN ITER1
      IF(MXITER.EQ.1) THEN
        IF(ICNVG.EQ.0.AND.IITER.LT.ITER1) GO TO 153
      ELSE
        IF(IICNVG.EQ.0.AND.IITER.LT.ITER1) GO TO 153
      ENDIF
C
C-------UNSCALE CR,CC,CV AND HNEW
      IF(NORM.EQ.1) THEN
      DO 310 N=1,NODES
      IF(IBOUND(N).EQ.0) GO TO 310
      HHCOF=SQRT(-HCOF(N))
      IF(N.LE.(NODES-NCOL).AND.CC(N).GT.0.)
     #   CC(N)=CC(N)*(HHCOF*(SQRT(-HCOF(N+NCOL))))
      IF(N.LE.(NODES-1).AND.CR(N).GT.0.)
     #   CR(N)=CR(N)*(HHCOF*(SQRT(-HCOF(N+1))))
      IF(N.LE.(NODES-NRC).AND.CV(N).GT.0.)
     #   CV(N)=CV(N)*(HHCOF*(SQRT(-HCOF(N+NRC))))
      HNEW(N)=HNEW(N)/HHCOF
  310 CONTINUE
      ENDIF
C
C-------IF END OF TIME STEP, PRINT # OF ITERATIONS THIS STEP
      IF(ICNVG.EQ.0 .AND. KITER.NE.MXITER) GO TO 600
      IF(MUTPCG.GT.1) GO TO 600
      IF(KSTP.EQ.1) WRITE(IOUT,500)
  500 FORMAT(1H0)
      WRITE(IOUT,501) KITER,KSTP,KPER,NITER
  501 FORMAT(1X,I5,' CALLS TO PCG ROUTINE FOR TIME STEP',I4,
     1' IN STRESS PERIOD',I3,/1X,I5,' TOTAL ITERATIONS')
      IF(MUTPCG.EQ.1) GO TO 600
C
C-------PRINT HEAD CHANGE EACH ITERATION IF PRINTOUT INTERVAL IS REACHED
      IF(ICNVG.EQ.0 .OR. KSTP.EQ.NSTP .OR. MOD(KSTP,IPRPCG).EQ.0)
     1      CALL SPCG2P(HCHG,LHCH,RCHG,LRCH,IITER,KITER,KSTP,KPER,
     2                  ITER1,NITER,MXITER,IOUT,NPCOND,BPOLY)
C
```

```
C-------RETURN
600    RETURN
C
       END
```

```
      SUBROUTINE SPCG2P(HCHG,LHCH,RCHG,LRCH,IITER,KITER,KSTP,KPER,ITER1,
     1                  NITER,MXITER,IOUT,NPCOND,BPOLY)
C
C
C-----VERSION 0001 01MAY1988 SPCG2P
C     ******************************************************************
C     PRINT MAXIMUM HEAD CHANGE AND RESIDUAL VALUE FOR EACH ITERATION
C                          DURING A TIME STEP
C     ******************************************************************
C
C        SPECIFICATIONS:
C     ------------------------------------------------------------------
C
      DIMENSION HCHG(MXITER*ITER1), LHCH(3,MXITER*ITER1)
      DIMENSION RCHG(MXITER*ITER1), LRCH(3,MXITER*ITER1)
C     ------------------------------------------------------------------
C
      IF(NPCOND.EQ.2) WRITE(IOUT,2) BPOLY
    2 FORMAT(1H0,'B OF THE POLYNOMIAL PRECONDITIONING METHOD: ',E12.4)
      WRITE(IOUT,5)
    5 FORMAT(1H0,'MAXIMUM HEAD CHANGE FOR EACH ITERATION:'/
     1    1H0,4('    HEAD CHANGE  LAYER,ROW,COL')/1X,120('-'))
      WRITE (IOUT,10) (HCHG(J),(LHCH(I,J),I=1,3),J=1,NITER)
      WRITE(IOUT,11)
      WRITE(IOUT,15)
   15 FORMAT(1H0,'MAXIMUM RESIDUAL FOR EACH ITERATION:'/
     1    1H0,4('     RESIDUAL    LAYER,ROW,COL')/1X,120('-'))
      WRITE (IOUT,10) (RCHG(J),(LRCH(I,J),I=1,3),J=1,NITER)
      WRITE(IOUT,11)
C
      RETURN
C
   10 FORMAT((1X,4(4X,G12.4,' (',I3,',',I3,',',I3,')')))
   11 FORMAT(1H0)
C
      END
```

```
      SUBROUTINE SPCG2E(IBOUND,RHS,HCOF,CR,CC,CV,VIN,VOUT,C,NORM,NCOL,
     1                  NROW,NLAY,NODES)
C
C
C-----VERSION 0001 01MAY1989 SPCG2E
C     ******************************************************************
C           MATRIX MULTIPLICATIONS FOR POLYNOMIAL PRECONDITIONING
C     ******************************************************************
C
C        SPECIFICATIONS:
C     ------------------------------------------------------------------
C
      DOUBLE PRECISION VN,CRHS,Z,B,D,F,H,S,ZV,BV,DV,FV,HV,SV,DZERO
      DIMENSION IBOUND(NODES),CR(NODES),CC(NODES),CV(NODES),
     1  RHS(NODES),VIN(NODES),VOUT(NODES),HCOF(NODES)
C     ------------------------------------------------------------------
C
      DZERO=0.
      NRC=NROW*NCOL
      DO 290 K=1,NLAY
      DO 290 I=1,NROW
      DO 290 J=1,NCOL
C
      N=J+(I-1)*NCOL+(K-1)*NRC
      VOUT(N)=0.
      IF(IBOUND(N).LE.0)GO TO 290
C
      NRN=N+NCOL
      NRL=N-NCOL
      NCN=N+1
      NCL=N-1
      NLN=N+NRC
      NLL=N-NRC
C
      NCF=N
      NCD=NCL
      NRB=NRL
      NRH=N
      NLS=N
      NLZ=NLL
C
      B=DZERO
      BV=DZERO
      IF(I.NE.1.AND.IBOUND(NRL).GE.0) THEN
        B=CC(NRB)
        BV=B*VIN(NRL)
      ENDIF
      H=DZERO
      HV=DZERO
      IF(I.NE.NROW.AND.IBOUND(NRN).GE.0) THEN
        H=CC(NRH)
        HV=H*VIN(NRN)
      ENDIF
      D=DZERO
      DV=DZERO
```

```
      IF(J.NE.1.AND.IBOUND(NCL).GE.0) THEN
        D=CR(NCD)
        DV=D*VIN(NCL)
      ENDIF
      F=DZERO
      FV=DZERO
      IF(J.NE.NCOL.AND.IBOUND(NCN).GE.0) THEN
        F=CR(NCF)
        FV=F*VIN(NCN)
      ENDIF
      Z=DZERO
      ZV=DZERO
      IF(K.NE.1.AND.IBOUND(NLL).GE.0) THEN
        Z=CV(NLZ)
        ZV=Z*VIN(NLL)
      ENDIF
      S=DZERO
      SV=DZERO
      IF(K.NE.NLAY.AND.IBOUND(NLN).GE.0) THEN
        S=CV(NLS)
        SV=S*VIN(NLN)
      ENDIF
C
C-------CALCULATE THE PRODUCT OF MATRIX A AND VECTOR VIN AND STORE
C------ RESULT IN VOUT
      VN=HCOF(N)*VIN(N)
      IF(NORM.EQ.1) VN=-VIN(N)
      CRHS=C*RHS(N)
      VOUT(N)=CRHS+ZV+BV+DV+VN+FV+HV+SV
  290 CONTINUE
      RETURN
      END
```

A METHOD OF CONVERTING NO-FLOW CELLS TO VARIABLE-HEAD CELLS FOR THE U.S. GEOLOGICAL SURVEY MODULAR FINITE-DIFFERENCE GROUND-WATER FLOW MODEL

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CONVERSION FACTORS

Multiply	By	To obtain
foot	0.3048	meter
foot per day	0.3048	meter per day
square foot	0.09290	square meter
cubic foot	0.02832	cubic meter
cubic foot per day	0.02832	cubic meter per day
gallon per minute	3.785	liter per minute

A METHOD OF CONVERTING NO-FLOW CELLS TO VARIABLE-HEAD CELLS IN THE U.S. GEOLOGICAL SURVEY MODULAR FINITE-DIFFERENCE GROUND-WATER FLOW MODEL

By

Michael G. McDonald, Arlen W. Harbaugh, Brennon R. Orr,
and Daniel J. Ackerman

ABSTRACT

The U.S. Geological Survey Modular Ground-Water Flow Model, commonly referred to as MODFLOW, simulates ground-water flow through porous media using the finite-difference method. The region being modeled is divided into a grid of cells, and each cell is defined to be either no-flow, variable-head, or constant-head. The model calculates a value for head at all variable-head cells whereas head at constant-head cells is specified by the user. Cells are designated as no-flow cells if they contain impermeable material or are unsaturated, and accordingly the flow of water is not simulated in such cells.

As originally published, MODFLOW could simulate the desaturation of variable-head model cells, which resulted in their conversion to no-flow cells, but could not simulate the resaturation of cells. That is, a no-flow cell could not be converted to variable head. However, such conversion is desirable in many situations. For example, one might wish to simulate pumping that desaturates some cells followed by the recovery of water levels after pumping is stopped. An option that allows cells to convert from no-flow to variable-head has been added to the model. In this option, a cell is converted to variable head based on the head at neighboring cells. The option is written in FORTRAN 77 and is fully compatible with the existing model. This report documents the new option, including a description of the concepts, detailed input instructions, and a listing of the code. Example problems illustrate the practical applications of the option. Although solution of the modified flow equations can be difficult for the model solvers, the example problems show that it is possible to solve a variety of complex problems.

INTRODUCTION

The U.S. Geological Survey developed a three-dimensional finite-difference ground-water flow model (McDonald and Harbaugh, 1988) commonly known as MODFLOW. In the model, the continuous derivatives in the ground-water flow equation are replaced with finite-difference approximations at points called nodes. Surrounding each node is a cell in which the hydraulic properties, such as hydraulic conductivity and storage coefficient, are defined. The result is a set of N equations containing N values of unknown head, where N is the number of nodes. The time derivative is approximated by the backward difference method (McDonald and Harbaugh, 1988, p. 2-16). The program solves for unknown head at each node at the end of each time interval.

The finite-difference flow equation for a model cell (i,j,k), where i, j, and k are row, column, and layer grid indexes respectively (McDonald and Harbaugh, 1988, p. 2-26), is:

$$
\begin{aligned}
&CV_{i,j,k-\frac{1}{2}}\, h^{m}_{i,j,k-1} + CC_{i-\frac{1}{2},j,k}\, h^{m}_{i-1,j,k} + CR_{i,j-\frac{1}{2},k}\, h^{m}_{i,j-1,k} \\
&+ (-CV_{i,j,k-\frac{1}{2}} - CC_{i-\frac{1}{2},j,k} - CR_{i,j-\frac{1}{2},k} \\
&\quad - CR_{i,j+\frac{1}{2},k} - CC_{i+\frac{1}{2},j,k} - CV_{i,j,k+\frac{1}{2}} + HCOF_{i,j,k})h^{m}_{i,j,k} \\
&+ CR_{i,j+\frac{1}{2},k}\, h^{m}_{i,j+1,k} + CC_{i+\frac{1}{2},j,k}\, h^{m}_{i+1,j,k} + CV_{i,j,k+\frac{1}{2}}\, h^{m}_{i,j,k+1} \\
&\qquad = RHS_{i,j,k}. \qquad (1)
\end{aligned}
$$

Hydraulic head is designated by h. The equation is written for each time step in a simulation, and the m superscript designates the time step number. The HCOF and RHS coefficients incorporate storage and external stress terms. The CR, CC, and CV coefficients are conductances. Conductance is a combination of several parameters from Darcy's law describing one-dimensional, steady-state flow occurring in a volume of porous material. Darcy's law states that flow through a block of porous material is the

product of conductance and the head difference along the flow path. The terms in equation (1) involving conductance specify the flow between the node for which the equation is being written and its six neighboring nodes.

A model cell can be one of three types: constant-head, variable-head, and no-flow. A variable-head cell is one in which the head varies and for which an equation is formulated. A constant-head cell is one for which, because the head is known and constant, an equation in terms of the head is not needed. However, the flow equation for a variable-head cell adjacent to a constant-head cell contains terms representing flow to or from the constant-head cell. A no-flow cell is one that represents a portion of the region in which there can be no movement of water. Such cells either contain impermeable material or are unsaturated. At no-flow cells, no equation is formulated, and the flow equation for a variable-head cell adjacent to a no-flow cell does not contain terms representing flow from the no-flow cell. The user initially defines a type for each cell by assigning values to the model array, IBOUND.

As part of the simulation of unconfined aquifers and aquifers that can convert between confined and unconfined, MODFLOW can change a variable-head cell to a no-flow cell. The conductance of unconfined cells and cells that can convert between confined and unconfined depends on the saturated thickness, and MODFLOW calculates the saturated thickness for these cells throughout the simulation. If the saturated thickness becomes zero, MODFLOW converts the cell to no flow (McDonald and Harbaugh, 1988, p. 5-10). This conversion is termed "drying" the cell.

The original MODFLOW does not allow a cell to be converted from no flow to variable head, which is termed "wetting" a cell. A new option to allow wetting of cells has been added to MODFLOW. Based on head in surrounding cells, the new model option will attempt to wet cells that are dry. The user can specify the cells for which wetting is attempted.

There are numerous situations for which the wetting capability is useful. An example is the recovery of water levels when wells are turned off. Another example is the mounding that can occur when irrigation is

applied to the surface above an aquifer; the mound could rise into a previously dry model layer. The wetting capability also is useful in situations where cells incorrectly go dry (convert to no flow) as part of the iterative solution process. During the solution process, heads may temporarily decrease to less than their final values, which may cause cells to go dry. The new wetting capability allows such cells to convert back to wet.

MODFLOW was designed in a modular fashion so that it would be easy to modify (McDonald and Harbaugh, 1988, p. 1-2). The program code was divided into modules, and the modules were grouped by function into packages. All modules dealing with internal aquifer flow, which includes flow between cells and flow into storage, are grouped into a package called the Block-Centered Flow (BCF) Package. To allow for future revision, each package was given a version number. The BCF Package documented by McDonald and Harbaugh (1988) is version 1 and is designated BCF1. The new wetting option was implemented by creating a replacement package for BCF1. The new package is BCF version 2, or BCF2. BCF2 is similar to BCF1; many of the modules are unchanged. All of the previous functionality is provided in addition to the new wetting capability. BCF2 is written in the FORTRAN 77 programming language (American National Standards Institute, 1978).

The remainder of this report describes the implementation of the BCF2 Package and how to use it. Several example problems are provided to illustrate its use. This report is intended to be used as a supplement to McDonald and Harbaugh (1988).

CONCEPTUALIZATION AND IMPLEMENTATION

The BCF2 Package is identical to the BCF1 Package except for the wetting and drying of cells. In particular, the BCF2 Package calculates vertical and horizontal conductance, storage terms, and vertical flow through a partially saturated zone exactly as done by the BCF1 Package (McDonald and Harbaugh, 1988, Ch. 5). Accordingly, this report discusses only the wetting and drying of cells.

For cells that are unconfined or can potentially become unconfined, the BCF1 Package calculates transmissivity as the product of hydraulic conductivity and saturated thickness. In equation form, transmissivity, TR, is calculated as (McDonald and Harbaugh, 1988, p. 5-9):

$$TR = (TOP - BOT)\,HY \quad \text{for } h \geq TOP \tag{2a}$$

$$TR = (h - BOT)\,HY \quad \text{for } TOP > h > BOT \tag{2b}$$

$$TR = 0 \quad \text{for } h \leq BOT \tag{2c}$$

where

HY is the hydraulic conductivity of a cell,
TOP is the elevation of the top of the cell, and
BOT is the elevation of the bottom of the cell.

The transmissivity values for cells are then used to calculate conductances (CR and CC) between cells as required for equation (1). Note that if horizontal anisotropy exists, hydraulic conductivity, and thus transmissivity, will be different in the row and column directions. Although transmissivity varies with saturated thickness, vertical conductance is a constant until a cell becomes dry, at which point vertical conductance is changed to zero.

As defined by equation (2), the value of head is part of the calculation of transmissivity. The head is unknown, however, until equation (1) is solved. To resolve this conflict, McDonald and Harbaugh (1988) take advantage of the fact that iterative solvers are used to solve the equations for head at each cell. Iterative solvers successively improve an approximation of the correct solution until the approximation is sufficiently accurate. Each approximation is called an iteration, and the solver is said to converge toward the correct answer. When calculating transmissivity according to equation (2) for a new iteration, the head from the previous

iteration is used. As the solver converges toward an acceptable solution, the head from the previous iteration will be nearly equal to the head for the new iteration. Thus, if all works well, the transmissivity will be accurate by the time convergence is reached.

In BCF1, a cell becomes dry whenever transmissivity as defined by equation (2c) is zero. That is, a cell becomes dry when head is below the bottom elevation. When a cell becomes dry, IBOUND is set to 0 (which indicates no flow), all conductances to the cell are set to zero, and head is set to a large value so it will serve as a visual indicator of the conversion in printouts. The conversion is irreversible in BCF1. BCF2 is identical to BCF1, except that cells can be wetted.

In BCF2, a dry cell can become wet if the head from the previous iteration in a neighboring cell is greater than or equal to the turn-on threshold. The turn-on threshold is

TURNON = BOT + THRESH

where

THRESH is a user-specified constant called the wetting threshold.

The user has two options to select which neighboring cells are checked to see if the turn-on threshold has been reached. One option is to check the cell immediately below the dry cell and the four horizontally adjacent cells. The second option is to check only the cell immediately below the dry cell. This option is useful when there are relatively large head differences between adjacent horizontal cells, which means that head in a neighboring horizontal cell is a poor indicator of when a cell should become wet.

Only variable-head cells either immediately below or horizontally adjacent to a dry cell can cause the cell to become wet. That is, if the neighboring cell is either no flow or constant head, then that cell is not checked to see if the turn-on threshold has been reached. The reason for

excluding constant-head cells from causing a dry cell to become wet is to avoid unnecessarily repeating the evaluations of whether a dry cell should become wet. If the head in a constant-head cell is high enough to cause a neighboring cell to be wet, then this neighboring cell should be wet during the entire simulation. The user can make this determination prior to running the model and can define the cell to be either wet or dry as appropriate.

When a cell is wetted, IBOUND for the cell is set to 1 (which indicates a variable-head cell), vertical conductances are set to their original values, and head at the cell is set either to

$$h = BOT + WETFCT\ (hn - BOT) \qquad (3a)$$

or

$$h = BOT + WETFCT\ (THRESH) \qquad (3b)$$

where

hn is the head at the neighboring cell that causes the cell to wet, and WETFCT is a user-specified constant called the wetting factor.

The user must select between equations (3a) and (3b); the rationale for making a selection will be discussed later in this report.

Although the head that is assigned to a cell that becomes wet might exceed the wetting threshold for a neighboring cell, a neighboring cell cannot become wet as a result of a cell that has become wet in the same iteration. If this were allowed, a single wetted cell might cause the entire model to become wet in one iteration.

Once a cell becomes wet, transmissivity and horizontal conductances are calculated as they are for any variable-head cell, which is the same as done in the BCF1 Package. The head in a wetted cell will then be recalculated in

subsequent iterations and time steps according to equation (1). A wetted cell can become dry again.

In transient simulations, the head from the previous time step, HOLD, is a part of the storage calculation (McDonald and Harbaugh, 1988, p. 5-24). For cells that become wet after becoming dry in an earlier time step, the value of HOLD that will produce the correct flow to storage is the bottom elevation. As a fundamental part of the finite-difference method for simulating the time derivative in the ground-water flow equation, MODFLOW begins each time step by setting HOLD equal to head from the end of the previous time step (McDonald and Harbaugh, 1988, pp. 2-16 and 3-2). Because the value for head at dry cells is set to a user-specified value when a cell becomes dry, the BCF2 Package must set HOLD equal to the bottom elevation at the beginning of each time step for dry cells that have the potential to become wet. To do this, a new module named BCF2AD has been added to the BCF Package.

The user should be aware of the possibility that non-unique solutions can result from the method used to wet cells. Consider a steady-state problem in which two simulations are made--the first simulation has starting heads below the final values and the second has starting heads above the final values. Heads in the first simulation may rise above the bottom of some dry cells and yet remain below the turn-on threshold. These cells would remain dry. In the second simulation, head may stay above the bottom in these same cells, so they would stay wet. Thus, there would be more variable-head cells in the second simulation even though both simulations satisfy the prescribed conditions for wetting and drying cells.

PROBLEMS WITH SOLVER CONVERGENCE

The method of wetting and drying cells can cause problems with the convergence of the iterative solvers used in MODFLOW. Convergence problems can occur in MODFLOW even without the wetting capability, but problems are more likely to occur when the wetting capability is used. Symptoms of a problem are slow convergence or divergence combined with the frequent

wetting and drying of the same cells. It is normal for the same cell to convert between wet and dry several times during the convergence process, but frequent conversions are an indication of problems. The user can detect this situation by examining the model printout; a message is printed each time a cell converts.

The convergence problems result from several factors. The primary factor is that equations (1) and (2) together form a nonlinear set of equations, but they are being solved through the use of solvers designed to solve linear equations. In particular, if the head from the previous iteration is poor, then the determination of which cells are wet and dry and the transmissivity of the wet cells will be incorrect. The overall change to the system of simultaneous equations that occurs when many cells change between wet and dry can be another cause of instability. The solvers perform differently depending on how many equations are being solved and on the values for conductances in each equation; the wetting and drying process changes both of these.

Another cause for instability is inaccuracy in the determination of when cells should become wet. While it is logical that the head at neighboring cells is a good qualitative indicator of when to wet a cell, there is no guarantee that a cell that is wetted by the described methodology should actually be wet. That is, the decision to make a cell wet should be viewed as a tentative decision. Subsequent iterations will show if the decision is correct. If head in a wetted cell drops below its bottom elevation, the cell will become dry again. It is possible for a cell to repeatedly cycle between wet and dry.

A poor estimate of initial head at wetted cells can also cause instability. Although the initial head estimate does not impact the final answer provided that convergence is reached, poor estimates can slow or prevent convergence by causing incorrect heads to be calculated at nearby cells.

The user of BCF2 has a number of ways to control solution instability.

1. The user specifies individual cells for which the wetting capability is active (see parameter WETDRY in "Input Instructions" section), which makes it possible to avoid the possibility of wetting dry cells that the user knows should never become wet.

2. The user specifies the wetting threshold for each wettable cell and which neighboring cells are checked to see if the wetting threshold has been reached (see parameter WETDRY in "Input Instructions" section).

3. The choice between using equation 3a or 3b to define the initial estimate of head at wetted cells is specified by the user.

4. The wetting factor used in 3a and 3b is specified by the user.

5. In steady-state simulations, initial conditions can be adjusted to improve stability.

For cells that can become wet, the most influential factors for controlling instability are the wetting threshold and the choice of which neighboring cells are checked to determine if the wetting threshold has been reached. The higher the threshold, the more stable the solution. Unfortunately, the wetting threshold can have a significant impact on the accuracy of the solution; the higher the wetting threshold, the less accurate the solution because cells that should become wet might stay dry. Thus the user may have to make a tradeoff between accuracy and stability.

The choice between equation (3a) and (3b) to define the initial head at wetted cells can impact solution stability. Equation (3a) is thought to be the most realistic because it varies the estimate of the head at a wetted cell according to the head at the cell that causes the cell to become wet. Equation (3b) bases its estimate of initial head on the wetting threshold. Equation (3a) can promote instability, however, when the iterative solvers are generally calculating head changes that are higher than optimum. In

that situation, equation (3b) can have a stabilizing effect. Both equations also include the wetting factor, WETFCT, which can be used to increase or decrease the estimate of head at a wetted cell.

Adjustments of the wetting threshold, the choice between equations (3a) and (3b), and wetting factor must be made on a trial and error basis. After adjusting any of these, the user should examine the output to see if a stable solution has been obtained and if acceptable determinations of which cells are wet and dry have been made.

For steady-state simulations, benefit can be gained by making an effort to set initial conditions as close as possible to final conditions. Of course the final heads are not known prior to running the simulation because the purpose of running a simulation is to solve for heads; however, it is often possible to make a good estimate of final heads. By using good estimates of initial head and the boundary array (IBOUND), which indicates which cells initially are wet and dry, conversions of cells between wet and dry can be minimized. Note that for transient simulations, there is no flexibility to adjust initial conditions; initial conditions must be set to accurately reflect conditions at the start of the simulation.

Up to this point, it has been implied that the wetting of cells is tested for at the start of every iteration. Although performing this test every iteration would appear to promote the most rapid solution, solution speed can sometimes be improved by limiting the iterations at which the wetting can occur. The reason is that the wetting of cells sometimes produces erroneous head changes in neighboring cells during the succeeding iteration, which would cause erroneous conversions of those cells. These erroneous conversions can be prevented by waiting a few iterations until heads have had a chance to adjust before testing for additional conversions. Accordingly, the ability to specify when wetting can be attempted has been incorporated into BCF2. In particular, the user can specify the iteration interval, IWETIT, for attempting to wet cells. That is, every IWETIT iterations starting from the beginning of each time step, BCF2 attempts to wet cells. At iterations for which no wetting is allowed, cells may still convert to dry. When setting IWETIT greater than one, there is some risk

that cells may be prevented from correctly converting to wet. If the solution for a time step is obtained in less than IWETIT iterations, then there will be no check during that time step to see if cells should convert to wet. The potential for this problem is greatest in transient simulations, which frequently require only a few iterations for a time step.

The last way for controlling oscillation is through the choice of the solver and adjustments of that solver. Three methods of iterative solution are available for MODFLOW. The Slice-Successive Overrelaxation (SOR) method (McDonald and Harbaugh, 1988, Ch. 13), the Strongly Implicit Procedure (SIP) method (McDonald and Harbaugh, 1988, Ch. 12), and the the Preconditioned Conjugate-Gradient (PCG) method (Kuiper, 1987, and Hill, 1990). SIP and PCG have been used successfully with BCF2. PCG appears to be the best. The SOR method is generally not practical for use with BCF2. Each solver implementation has input parameters that can affect the convergence efficiency. The reader should refer to the documentation of the solvers for specific information on the ways to adjust them.

Users of the PCG solvers should be aware that both of these solvers make use of a dual-iteration methodology, which can be useful in controlling oscillation of cells between wet and dry. Hill (1990) describes the iteration methodology using the terms "inner" and "outer" iterations. An outer iteration is a model iteration as described by McDonald and Harbaugh (1988, p. 2-20). For each outer iteration, the PCG solver can perform multiple iterations, called inner iterations. During successive inner iterations, PCG improves the calculated head assuming that all parameters in the flow equation are constant. In particular, cells cannot change between wet and dry during inner iterations. Thus, PCG's inner iterations serve a similar purpose to that of the IWETIT iteration interval, and generally there is no need to use an IWETIT value other than 1 when using PCG. Note that IWETIT applies to outer iterations in PCG; the BCF2 Package has no control over inner iterations. Note also that the BCF2 iteration interval controls only the wetting of cells and does not have an impact on the other head-dependent formulations such as transmissivity and the resulting conductances within the BCF2 Package or any other model package. No head-dependent formulations occur in any packages during PCG inner iterations.

In an unstable problem in which many cells are erratically converting between wet and dry, the solvers can have numerical problems that cause them to abort. In particular, small clusters of cells isolated from the rest of the cells by no-flow cells can cause division by zero in the solvers. In addition, the PCG2 solver (Hill, 1990) may calculate a Cholesky matrix that is not diagonally dominant, which will cause the solver to abort. In some cases setting the value of PCG2 parameter RELAX in the range of 0.97 to 0.99 prevents zero divide and non-diagonally dominant matrix errors.

APPLICABILITY AND LIMITATIONS

The BCF2 Package makes it possible to simulate situations in which a water table rises into unsaturated model layers. A typical application of this capability is in the simulation of the recovery of over-stressed aquifers either through artificial recharge or the reduction of stress. The need for the wetting capability increases as the need for vertical detail of head distribution increases. High-accuracy simulations of vertical head gradients are required for such purposes as tracking the flow of contaminants, especially when significant vertical heterogeneity exists or when stresses partially penetrate an aquifer. To obtain accuracy of vertical head requires many model layers, which implies that each layer will be thin. The thinner the model layer, the more likely it is that cells will require conversions between wet and dry.

Another benefit of the wetting capability is improved simulation of declining water tables. Although the BCF1 Package can simulate the drying of cells when water levels decrease, there is a risk of drying cells that should not become dry. This can happen when an iterative solver overshoots the correct value during an iteration, which is not unusual for iterative solvers. The wetting capability of BCF2 allows such cells to become wet again in later iterations.

Although the BCF2 Package has the ability to simulate a variety of situations, there are limitations to its use. The primary limitation has already been discussed; there may be instabilities in the solution process

that prevent solution or cause solution to require excessive computational time. To minimize solution problems, the user can choose from the SIP and PCG solvers, can adjust the solvers, and can adjust the data that controls the wetting process. All of the adjustments must be done through trial and error. Once a solver produces an answer, the user must examine the results to determine if the answer is acceptable. The budget error should be examined as well as which cells are wet and dry. Although the wetting threshold might have to be higher than one would like, the answers produced are probably more accurate than would otherwise be produced without the wetting capability.

A second limitation of BCF2 results from the method of drying cells. All stresses are removed when a cell becomes dry even though it might be more appropriate to move the stress to another location. For example, if a cell that contains a well goes dry, the pumpage will be eliminated. If the well owner would likely deepen the well rather than stop pumping water, a more appropriate action would be to move the pumpage to a deeper cell. Further, the removal of stresses when cells go dry can cause cells to repeatedly convert between wet and dry. In the example of a drying cell that contains a well, the elimination of the pumpage might then allow the cell to become wet again; this cycle could repeat indefinitely. Note that the Recharge (RCH) Package (McDonald and Harbaugh, 1988, Ch. 7) has an option to deal with this problem; it can allow recharge to pass through dry cells to the highest constant-head or variable-head cell. Likewise, this capability of the Recharge Package could be used to simulate wells whose pumpage is automatically moved to the next lower layer when a layer goes dry. Some situations may require new methods for simulating stresses; the modular design of MODFLOW makes it possible to easily add new capabilities.

A similar problem can result in transient simulations, which involve flow from storage. If a cell dries, the flow from storage at that time step is lost. Large volumes of water could be lost, causing an inaccurate simulation. For example, it is possible to specify large time steps such that the head in a cell drops from the top of the cell to the bottom in one time step. In this situation, all of the unconfined storage, which is generally a large quantity, would be lost. The solution to this problem is

to use time steps that are small enough to allow most of a cell's water to be taken from storage before the cell converts to dry. However, small time steps can cause additional problems. Small time steps result in large rates of flow from storage when a cell becomes wet because flow from storage is dependent on the inverse of time-step length. The flow to storage can result in head in the wetted cell being only slightly above the bottom of the cell. The small saturated thickness results in a low transmissivity that can restrict the flow entering the cell from adjacent cells enough to cause the cell to convert to dry again. Also, the large flow to storage caused by wetting a cell can cause the solvers to over-compensate and reduce head enough to dry the cell. Thus, small time steps can result in repeated conversions between wet and dry. If small time steps are required, a larger wetting threshold may be necessary to avoid the oscillation between wet and dry.

A third limitation is the way vertical conductance converts when a cell changes between dry and wet. When dry, conductance is zero; when wet, conductance is a constant. If a confining layer exists between layers as assumed by McDonald and Harbaugh (1988, p. 5-30), this makes sense. That is, if the confining bed is the primary restriction to vertical flow, then the lowering of the water table in the aquifer above the confining unit will not change the vertical conductance very much because the confining unit is not changed by the aquifer's desaturation. When there is no confining layer as would be the situation when a single aquifer is being simulated by multiple model layers, the changing water level would have a significant impact on vertical conductance. A more accurate representation would be to vary vertical conductance with saturated thickness. To reduce the error, more layers can be used.

Another limitation also involves vertical flow. Vertical flow between two cells is limited when the lower cell is not confined (McDonald and Harbaugh, 1988, p. 5-19). This is appropriate for simulating perched conditions that can occur when there is recharge to the upper cell and there is a low-conductivity material between the two cells or the upper cell has lower hydraulic conductivity than the lower cell. A stress in the lower cell can cause the head to drop below the top without drying the upper cell.

When vertical flow is limited, vertical flow is calculated as saturated thickness of the upper cell times the vertical conductance. Thus, flow is not dependent on the head in the lower cell. A problem can occur if flow is limited between two cells that are simulating the same aquifer. That is, it is possible for a perched water table to form in the model when heads are declining even though there is no low conductivity material between the cells that could cause the perched condition. Although this would not happen if model heads were declining smoothly from the top down, it can happen if heads in a lower cell drop below its top during the iterative convergence process while the cell above is still wet. Once the perched condition is established, it can stay even though it does not make hydrologic sense because the vertical flow limitation prevents head from dropping as low as it should drop. Thus the user should examine model heads to see if there are cells having incorrectly perched conditions.

The final limitation involves the difficulty of preparing data for the BCF2 Package. The user is faced with a major effort to prepare the input data for a typical multi-layer problem that requires the wetting capability. Errors are a likelihood given the amount of data, and errors probably will be difficult to trace. Inconsistent top and bottom elevations and other incorrect data can cause convergence problems in addition to the convergence problems that are inherent in the model formulations. When faced with a model that will not converge, it is difficult to determine whether there is a fundamental stability problem or bad data somewhere in the input files. It is often helpful to make a series of increasingly complex models rather than starting off with the "full-blown" model. For example, a complex water-table aquifer system might be simulated initially as a single unconfined aquifer even though the final model will simulate the aquifer with a sequence of model layers, some of which will use the wetting capability. This makes it possible to get the basic model operational and obtain some knowledge of the values for aquifer parameters before dealing with the problems of the wetting and drying of cells.

INPUT INSTRUCTIONS

Most of the data required by the BCF2 Package are the same as for the BCF1 Package. These data consist primarily of arrays that are used to calculate conductance and storage terms for each layer. The required arrays can include transmissivity, hydraulic conductivity, specific yield, confined storage coefficient, vertical leakance, aquifer bottom elevation, and aquifer top elevation. The specific arrays required depend on the options that are used as indicated by the layer-type code, LAYCON. The reader is referred to McDonald and Harbaugh (1988, p. 5-30) for a detailed description of these data.

A small amount of data is required for the wetting capability in addition to that required by the BCF1 Package. The additional data are a flag indicating if the wetting capability is to be used, the wetting threshold, the wetting factor, and the iteration interval for attempting to wet cells. The only additional array is WETDRY, which must be specified for each wettable layer. If the wetting capability is enabled, all layers for which transmissivity varies with saturated thickness, which is indicated by a LAYCON value of 1 or 3, are wettable.

Because BCF2 is designed as a replacement for BCF1, input is read using the same unit number, which is specified in IUNIT(1). Refer to McDonald and Harbaugh (1988, p. 3-25) for a description of the use of unit numbers in MODFLOW. The data must be in the specified order. The data are read once at the start of the simulation.

BCF2AL

1. Data: ISS IBCFCB HDRY IWDFLG WETFCT IWETIT IHDWET
   Format: I10 I10 F10.0 I10 F10.0 I10 I10

2. Data: LAYCON(NLAY) (Maximum of 80 layers)
   Format: 40I2
   (If there are 40 or fewer layers, use one record; otherwise, use two records.)

3. Data: TRPY(NLAY)
   Module: U1DREL

4. Data: DELR(NCOL)
   Module: U1DREL

5. Data: DELC(NROW)
   Module: U1DREL

A subset of the following two-dimensional arrays is used to describe each layer. The arrays needed for each layer depend on the layer-type code (LAYCON), whether the simulation is transient (ISS = 0) or steady state (ISS not 0), and if the wetting capability is active (IWDFLG not 0). If an array is not needed, it must be omitted. In no situation will all arrays be required. The required arrays (items 6-13) for layer 1 are read first; then the arrays for layer 2, etc.

IF THE SIMULATION IS TRANSIENT (ISS = 0)

6. Data: sf1(NCOL,NROW)
   Module: U2DREL

IF THE LAYER-TYPE CODE (LAYCON) is 0 or 2

7. Data: Tran(NCOL,NROW)
   Module: U2DREL

IF THE LAYER-TYPE CODE (LAYCON) IS 1 OR 3

8. Data: HY(NCOL,NROW)
   Module: U2DREL

9. Data: BOT(NCOL,NROW)
   Module: U2DREL

IF NOT THE BOTTOM LAYER

10. Data: Vcont(NCOL,NROW)
    Module: U2DREL

IF THE SIMULATION IS TRANSIENT (ISS = 0) AND THE LAYER-TYPE CODE (LAYCON) IS 2 OR 3

11. Data: sf2(NCOL,NROW)
    Module: U2DREL

IF THE LAYER-TYPE CODE (LAYCON) IS 2 OR 3

12. Data: TOP(NCOL,NROW)
    Module: U2DREL

IF THE LAYER-TYPE CODE (LAYCON) IS 1 OR 3 AND THE WETTING CAPABILITY IS ACTIVE (IWDFLG IS NOT 0)

13. Data: WETDRY(NCOL,NROW)
    Module: U2DREL

Explanation of Parameters Used in Input Instructions

ISS is the steady-state flag

If ISS is not 0, the simulation is steady state.

If ISS = 0, the simulation is transient.

IBCFCB is a flag and a unit number.

If IBCFCB > 0, it is the unit number on which cell-by-cell flow terms will be recorded whenever ICBCFL (see Output Control in McDonald and Harbaugh [1988, p. 4-15]) is set; the terms which are saved will include cell-by-cell storage terms, cell-by-cell constant-head flows, and cell-by-cell flow between adjacent cells.

If IBCFCB = 0, cell-by-cell flow terms will not be printed or recorded.

If IBCFCB < 0, flow for each constant-head cell will be printed in the listing file whenever ICBCFL is set; cell-by-cell storage terms and cell-by-cell flow between adjacent cells will not be recorded or printed.

HDRY is the head that is assigned to cells that are converted to dry during a simulation. Although this value plays no role in the model calculations, it is useful as an indicator when looking at the resulting heads that are output from the model. HDRY is thus similar to HNOFLO in the Basic Package, which is the value assigned to cells that are no-flow cells at the start of a model simulation.

IWDFLG is a flag that determines if the wetting capability is active.

If IWDFLG = 0, the wetting capability is inactive.

If IWDFLG is not 0, the wetting capability is active.

WETFCT is a factor that is included in the calculation of the head that is initially established at a cell when it is converted from dry to wet. (See IHDWET.)

IWETIT is the iteration interval for attempting to wet cells. Wetting is attempted every IWETIT iterations. If using the PCG solver (Hill, 1990), this applies to outer iterations, not inner iterations. If IWETIT is 0, it is changed to 1.

IHDWET is a flag that determines which equation is used to calculate the initial head at cells that become wet:

If IHDWET = 0, equation (3a) is used:

$$h = BOT + WETFCT\ (hn - BOT),$$

If IHDWET is not 0, equation (3b) is used:

$$h = BOT + WETFCT\ (THRESH).$$

LAYCON is the layer-type table. Each element holds the code for the respective layer. Read one value for each layer. There is a limit of 80 layers. If there are 40 or fewer layers, use one record; otherwise, use two records. Leave unused elements in a record blank.

0 - confined -- Transmissivity and storage coefficient of the layer are constant for the entire simulation.

1 - unconfined -- Transmissivity of the layer varies. It is calculated from the saturated thickness and hydraulic conductivity. The storage coefficient is constant. This type code is valid only for layer 1.

2 - confined/unconfined -- Transmissivity of the layer is constant. The storage coefficient may alternate between confined and unconfined values. Vertical flow from above is limited if the layer desaturates.

3 - confined/unconfined -- Transmissivity of the layer varies. It is calculated from the saturated thickness and hydraulic conductivity. The storage coefficient may alternate between confined and unconfined values. Vertical flow from above is limited if the aquifer desaturates.

TRPY is a one-dimensional array containing an anisotropy factor for each layer. It is the ratio of transmissivity or hydraulic conductivity (whichever is being used) along a column to transmissivity or hydraulic conductivity along a row. Set to 1.0 for isotropic conditions. This is a single array with one value per layer. Do not read an array for each layer; include only one array control record for the entire array. If the value is the same for all layers, the entire array can be specified by setting LOCAT to 0 and setting CNSTNT to the value that applies to all layers.

DELR is the cell width along rows. Read one value for each of the NCOL columns. This is a single array with one value for each column.

DELC is the cell width along columns. Read one value for each of the NROW rows. This is a single array with one value for each row.

sf1 is the primary storage coefficient. Read only for a transient simulation (steady-state flag, ISS, is 0). For LAYCON equal to 1, sf1 will always be specific yield, while for LAYCON equal to 2 or 3, sf1 will always be confined storage coefficient. For LAYCON equal to 0, sf1 would normally be confined storage coefficient; however, a LAYCON value of 0 can also be used to simulate water-table conditions where drawdowns are expected to remain everywhere a small fraction of the saturated thickness, and where there is no layer above, or flow from above is negligible. In this case, specific yield values would be entered for sf1.

Tran is the transmissivity along rows. Tran is multiplied by TRPY to obtain transmissivity along columns. Read only for layers where LAYCON is 0 or 2.

HY is the hydraulic conductivity along rows. HY is multiplied by TRPY to obtain hydraulic conductivity along columns. Read only for layers where LAYCON is 1 or 3.

BOT is the elevation of the aquifer bottom. Read only for layers where LAYCON is 1 or 3.

Vcont is the vertical hydraulic conductivity divided by the thickness from a layer to the layer below. The value for a cell is the hydraulic conductivity divided by thickness for the material between the node in that cell and the node in the cell below. Because there is not a layer beneath the bottom layer, Vcont cannot be specified for the bottom layer.

sf2 is the secondary storage coefficient. Read only for layers where LAYCON is 2 or 3 and only if the simulation is transient (steady-state flag, ISS, is 0). The secondary storage coefficient is always specific yield.

TOP is the elevation of the aquifer top. Read only for layers where LAYCON is 2 or 3.

WETDRY is a combination of the wetting threshold and a flag to indicate which neighboring cells can cause a cell to become wet. If WETDRY < 0, only the cell below a dry cell can cause the cell to become wet. If WETDRY > 0, the cell below a dry cell and the four horizontally adjacent cells can cause a cell to become wet. If WETDRY is 0, the cell cannot be wetted. The absolute value of WETDRY is the wetting threshold. When the sum of BOT and the absolute value of WETDRY at a dry cell is equaled or exceeded by the head at an adjacent cell, the cell is wetted. Read only if LAYCON is 1 or 3 and IWDFLG is not 0.

EXAMPLE PROBLEMS

Example problems were posed to illustrate the practical applications of the BCF2 option. Although solution of the modified flow equations can be difficult for the model solvers, the example problems show that it is possible to solve a variety of complex problems.

Problem 1 -- Simulation of a Two-Layer Aquifer System in which the Top Layer Converts Between Wet and Dry

In an aquifer system where two aquifers are separated by a confining bed, large pumpage withdrawals from the bottom aquifer can desaturate parts of the upper aquifer. If pumpage is discontinued, resaturation of the upper aquifer can occur. This problem demonstrates the capability of the BCF2 Package to successfully simulate this common hydrologic situation which is difficult or impossible to simulate with the original BCF1 Package. If both aquifers were simulated by one model layer using BCF1, the effect of the confining bed on vertical flow could not be simulated. Also, the transmissivity would not be correctly calculated as a function of saturated thickness. The changes in hydraulic conductivity between the confining bed and the aquifers could not be distinctly represented. If each aquifer were simulated by a separate model layer using BCF1, there would be solution difficulties. When simulating declining heads, too many cells might convert to dry; when simulating rising heads, dry cells could not convert to wet.

Conceptual Model

A hypothetical aquifer system consists of two aquifers separated by a confining unit (fig. 1). No-flow boundaries surround the system on all sides, except that the lower aquifer discharges to a stream along the right side of the area. Recharge from precipitation is applied evenly over the entire area. The stream penetrates the lower aquifer; in the region above the stream, the upper aquifer and confining unit are missing. Under natural conditions, recharge flows through the system to the stream. Under

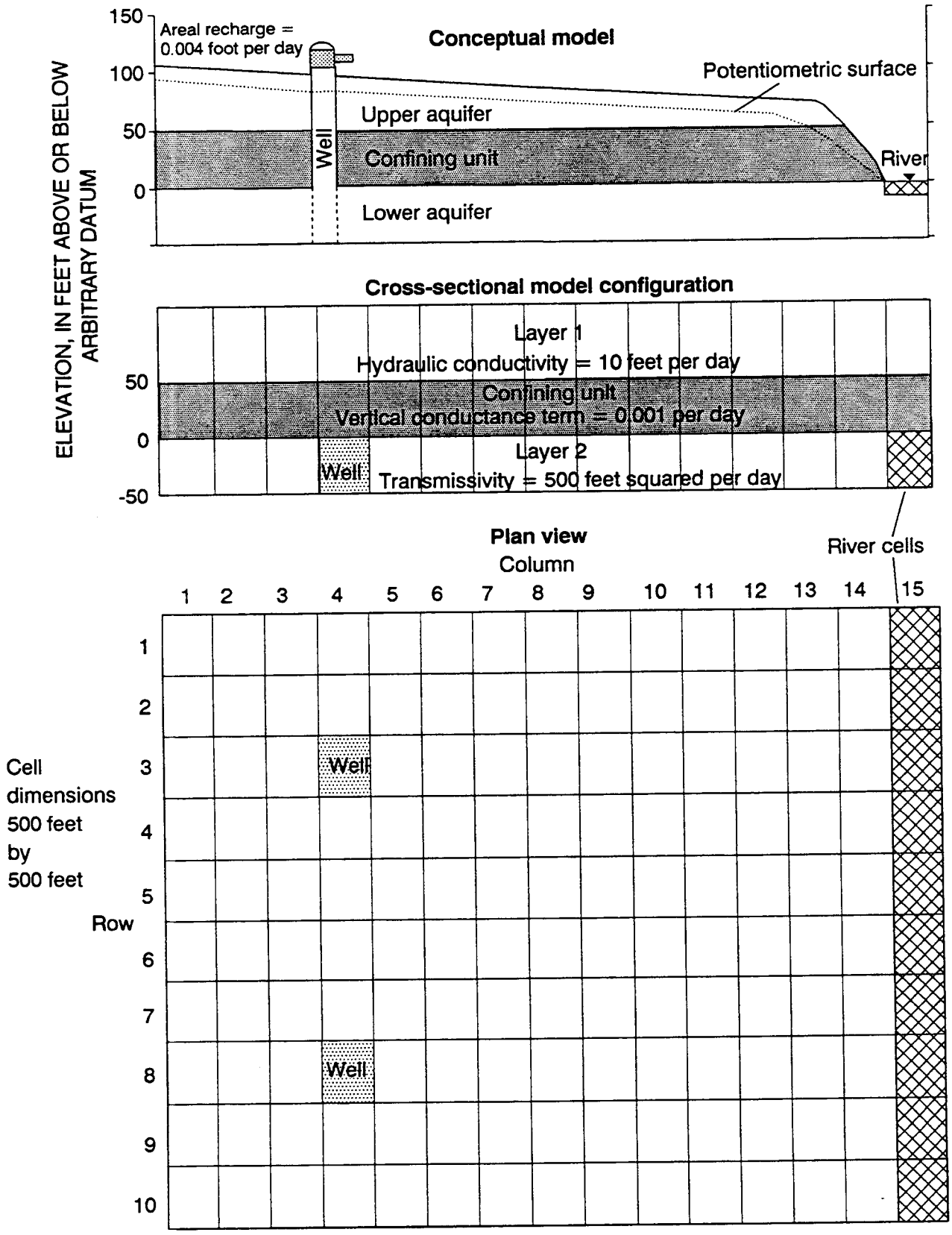


Figure 1.--Hydrogeology, model grid, and model boundary conditions for Problem 1.

stressed conditions, two wells withdraw water from the lower aquifer. If enough water is pumped, cells in the upper aquifer will desaturate. Removal of the stresses will then cause the desaturated areas to resaturate.

Modeling Approach

The model consists of two layers--one for each aquifer. Because horizontal flow in the confining bed is small compared to horizontal flow in the aquifers and storage is not a factor in steady-state simulations, the confining bed is not treated as a separate layer. The effect of the confining bed is incorporated in the value for vertical hydraulic conductivity divided by thickness between aquifer layers (McDonald and Harbaugh, 1988, p. 5-16). Note that if storage in the confining bed were significant, transient simulations would require that the confining layer be simulated using one or more layers. A uniform horizontal grid of 10 rows and 15 columns is used. Aquifer parameters are specified as shown in Figure 1.

Simulation Results

Two steady-state solutions were obtained to simulate natural conditions and pumping conditions. The two solutions are designed to demonstrate the ability of the BCF2 Package to handle a broad range of possibilities for cells converting between wet and dry in the top aquifer. When solving for natural conditions, the top aquifer initially is specified as being entirely dry and many cells must convert to wet. When solving for pumping conditions, the top aquifer is initially specified to be under natural conditions and many cells must convert to dry.

The steady-state solutions were obtained through a single simulation consisting of two stress periods. The first stress period simulates natural conditions and the second period simulates the addition of pumping wells. The simulation is declared to be steady state, so no storage values are specified and each stress period requires only a single time step to produce a steady-state result. The PCG2 Package is used to solve the flow equations

for the simulations. The complete output from the model simulation is provided in Table 1.

In the process of simulating this problem, several trial simulations were made using different values for the WETDRY parameter. The absolute value of the WETDRY parameter is the wetting threshold, and the sign of the WETDRY parameter indicates which neighboring cells can cause a cell to become wet. Determination of the WETDRY parameter often requires considerable effort. The user may have to make multiple test runs trying different values in different areas of the model. On the right side of the model, the WETDRY parameter (table 1) is negative in order to cause a cell to become wet only when head in the layer below exceeds the wetting threshold. This was done to avoid incorrectly converting dry cells to wet because of the large head differences between adjacent horizontal cells. For example, the simulation of natural conditions (Stress Period 1) shows cells in column 14 of layer 1 being dry, which is reasonable based on the head below these cells. That is, the head in column 14 of layer 2 is over 20 feet below the bottom of layer 1. However, the head in column 13 of layer 1 is 21 feet above the bottom of the aquifer, which means that, if head in adjacent horizontal cells is allowed to wet cells, column 14 would convert to wet. Thus, it is not readily apparent whether column 14 should be wet or dry. The trial simulations showed that, when horizontal wetting is allowed, column 14 repeatedly oscillates between wet and dry, indicating that column 14 should be dry. If horizontal wetting is used, oscillation between wet and dry can be prevented by raising the wetting threshold, but this also can prevent some cells that should be partly saturated from converting to wet.

On the left side of the model, horizontal head changes between adjacent cells generally are small, so head in the neighboring horizontal cells is a good indicator of whether or not a dry cell should become wet. Therefore, positive WETDRY parameters are used in most of this area to allow wetting to occur either from the cell below or from horizontally adjacent cells. Near the well, the horizontal head gradients under pumping conditions also are relatively large; consequently, a negative WETDRY parameter was used at the cells above the well. This prevents these cells from incorrectly becoming

Table 1.--Model output for problem 1

```
                U.S. GEOLOGICAL SURVEY MODULAR FINITE-DIFFERENCE GROUND-WATER MODEL

Valley aquifer with 2 sand layers separated by silt.  Stress period 1 is natural conditions.  Stress period 2 adds wells.
   2 LAYERS        10 ROWS         15 COLUMNS
  2 STRESS PERIOD(S) IN SIMULATION
MODEL TIME UNIT IS DAYS

I/O UNITS:
ELEMENT OF IUNIT:  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
        I/O UNIT: 11 12  0 14  0  0  0 18  0  0  0 20 19  0  0  0  0  0  0  0  0  0  0  0

BAS1 -- BASIC MODEL PACKAGE, VERSION 1, 9/1/87 INPUT READ FROM UNIT  5
ARRAYS RHS AND BUFF WILL SHARE MEMORY.
START HEAD WILL NOT BE SAVED -- DRAWDOWN CANNOT BE CALCULATED
    2583 ELEMENTS IN X ARRAY ARE USED BY BAS
    2583 ELEMENTS OF X ARRAY USED OUT OF  650000

BCF2 -- BLOCK-CENTERED FLOW PACKAGE, VERSION 2, 7/1/91 INPUT READ FROM UNIT 11
STEADY-STATE SIMULATION
CELL-BY-CELL FLOWS WILL BE RECORDED ON UNIT 30
HEAD AT CELLS THAT CONVERT TO DRY=   777.77
WETTING CAPABILITY IS ACTIVE
WETTING FACTOR=   1.00000     WETTING ITERATION INTERVAL=   1
FLAG THAT SPECIFIES THE EQUATION TO USE FOR HEAD AT WETTED CELLS=   0
     LAYER  AQUIFER TYPE
     ------------------
       1         1
       2         0
    602 ELEMENTS IN X ARRAY ARE USED BY BCF
   3185 ELEMENTS OF X ARRAY USED OUT OF  650000
```

Table 1.--Model output for problem 1--Continued

```
WEL1 -- WELL PACKAGE, VERSION 1, 9/1/87 INPUT READ FROM 12
MAXIMUM OF     2 WELLS
CELL-BY-CELL FLOWS WILL BE RECORDED ON UNIT 30
       8 ELEMENTS IN X ARRAY ARE USED FOR WELLS
    3193 ELEMENTS OF X ARRAY USED OUT OF  650000

RCH1 -- RECHARGE PACKAGE, VERSION 1, 9/1/87 INPUT READ FROM UNIT 18
OPTION 3 -- RECHARGE TO HIGHEST ACTIVE NODE IN EACH VERTICAL COLUMN
CELL-BY-CELL FLOW TERMS WILL BE RECORDED ON UNIT 30
     150 ELEMENTS OF X ARRAY USED FOR RECHARGE
    3343 ELEMENTS OF X ARRAY USED OUT OF  650000

RIV1 -- RIVER PACKAGE, VERSION 1, 9/1/87 INPUT READ FROM UNIT 14
MAXIMUM OF    10 RIVER NODES
CELL-BY-CELL FLOWS WILL BE RECORDED ON UNIT 30
      60 ELEMENTS IN X ARRAY ARE USED FOR RIVERS
    3403 ELEMENTS OF X ARRAY USED OUT OF  650000

PCG2 -- CONJUGATE GRADIENT SOLUTION PACKAGE, VERSION 2, 5/1/88
MAXIMUM OF  40 CALLS OF SOLUTION ROUTINE
MAXIMUM OF  20 INTERNAL ITERATIONS PER CALL TO SOLUTION ROUTINE
MATRIX PRECONDITIONING TYPE :    1
   7600 ELEMENTS IN X ARRAY ARE USED BY PCG
  11003 ELEMENTS OF X ARRAY USED OUT OF 650000


Valley aquifer with 2 sand layers separated by silt.  Stress period 1 is natural conditions.  Stress period 2 adds wells.

                                                    BOUNDARY ARRAY =            0 FOR LAYER  1
                                                    BOUNDARY ARRAY =            1 FOR LAYER  2


AQUIFER HEAD WILL BE SET TO  999.99     AT ALL NO-FLOW NODES (IBOUND=0).
```

Table 1.--Model output for problem 1--Continued

```
                                    INITIAL HEAD =  0.0000000     FOR LAYER  1

                                    INITIAL HEAD =  0.0000000     FOR LAYER  2

HEAD PRINT FORMAT IS FORMAT NUMBER  -4    DRAWDOWN PRINT FORMAT IS FORMAT NUMBER  -4

HEADS WILL BE SAVED ON UNIT 50    DRAWDOWNS WILL BE SAVED ON UNIT 50

OUTPUT CONTROL IS SPECIFIED EVERY TIME STEP

                          COLUMN TO ROW ANISOTROPY =  1.000000

                                              DELR =  500.0000

                                              DELC =  500.0000

                            HYD. COND. ALONG ROWS =  10.00000     FOR LAYER  1

                                            BOTTOM =  50.00000    FOR LAYER  1

                          VERT HYD COND /THICKNESS =  0.9999999E-03 FOR LAYER  1
```

Table 1.--Model output for problem 1--Continued

WETDRY PARAMETER FOR LAYER 1 WILL BE READ ON UNIT 11 USING FORMAT: (15f4.0)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	2.	2.	2.	2.	2.	2.	2.	2.	-2.	-2.	-2.	-2.	-2.	-2.	-2.
2	2.	2.	2.	2.	2.	2.	2.	2.	-2.	-2.	-2.	-2.	-2.	-2.	-2.
3	2.	2.	2.	-2.	2.	2.	2.	2.	-2.	-2.	-2.	-2.	-2.	-2.	-2.
4	2.	2.	2.	2.	2.	2.	2.	2.	-2.	-2.	-2.	-2.	-2.	-2.	-2.
5	2.	2.	2.	2.	2.	2.	2.	2.	-2.	-2.	-2.	-2.	-2.	-2.	-2.
6	2.	2.	2.	2.	2.	2.	2.	2.	-2.	-2.	-2.	-2.	-2.	-2.	-2.
7	2.	2.	2.	2.	2.	2.	2.	2.	-2.	-2.	-2.	-2.	-2.	-2.	-2.
8	2.	2.	2.	-2.	2.	2.	2.	2.	-2.	-2.	-2.	-2.	-2.	-2.	-2.
9	2.	2.	2.	2.	2.	2.	2.	2.	-2.	-2.	-2.	-2.	-2.	-2.	-2.
10	2.	2.	2.	2.	2.	2.	2.	2.	-2.	-2.	-2.	-2.	-2.	-2.	-2.

TRANSMIS. ALONG ROWS = 500.0000 FOR LAYER 2

Table 1.--Model output for problem 1--Continued

```
                    SOLUTION BY THE CONJUGATE-GRADIENT METHOD
                    -----------------------------------------

                    MAXIMUM NUMBER OF CALLS TO PCG ROUTINE =      40
                       MAXIMUM ITERATIONS PER CALL TO PCG =      20
                             MATRIX PRECONDITIONING TYPE =       1
           RELAXATION FACTOR (ONLY USED WITH PRECOND. TYPE 1) =   0.10000E+01
      PARAMETER OF POLYMOMIAL PRECOND. = 2 (2) OR IS CALCULATED :      2
                             HEAD CHANGE CRITERION FOR CLOSURE =   0.10000E-02
                          RESIDUAL CHANGE CRITERION FOR CLOSURE =   0.10000E+04
                 PCG HEAD AND RESIDUAL CHANGE PRINTOUT INTERVAL =       1
                 ALL PRINTING FROM THE SOLVER IS SUPPRESSED (1) =       1
FOR NPCOND=1, DO (0) OR DO NOT (1) RECALC. CHOL. DIAG. EACH OUTER ITER. =       0


                  STRESS PERIOD NO.    1, LENGTH =   1.000000
                  ----------------------------------------

                  NUMBER OF TIME STEPS =     1

                   MULTIPLIER FOR DELT =     1.000

                 INITIAL TIME STEP SIZE =   1.000000

0 WELLS

                                       RECHARGE =  0.4000000E-02
```

Table 1.--Model output for problem 1--Continued

10 RIVER REACHES

LAYER	ROW	COL	STAGE	CONDUCTANCE	BOTTOM ELEVATION	RIVER REACH
2	1	15	0.0000	0.1000E+05	-5.000	1
2	2	15	0.0000	0.1000E+05	-5.000	2
2	3	15	0.0000	0.1000E+05	-5.000	3
2	4	15	0.0000	0.1000E+05	-5.000	4
2	5	15	0.0000	0.1000E+05	-5.000	5
2	6	15	0.0000	0.1000E+05	-5.000	6
2	7	15	0.0000	0.1000E+05	-5.000	7
2	8	15	0.0000	0.1000E+05	-5.000	8
2	9	15	0.0000	0.1000E+05	-5.000	9
2	10	15	0.0000	0.1000E+05	-5.000	10

CELL CONVERSIONS FOR ITERATION= 2 LAYER= 1 TIME STEP= 1 STRESS PERIOD= 1 (ROW,COL)

WET(1, 1)	WET(1, 2)	WET(1, 3)	WET(1, 4)	WET(1, 5)	WET(1, 6)	WET(1, 7)	WET(1, 8)
WET(1, 9)	WET(1, 10)	WET(1, 11)	WET(1, 12)	WET(1, 13)	WET(2, 1)	WET(2, 2)	WET(2, 3)
WET(2, 4)	WET(2, 5)	WET(2, 6)	WET(2, 7)	WET(2, 8)	WET(2, 9)	WET(2, 10)	WET(2, 11)
WET(2, 12)	WET(2, 13)	WET(3, 1)	WET(3, 2)	WET(3, 3)	WET(3, 4)	WET(3, 5)	WET(3, 6)
WET(3, 7)	WET(3, 8)	WET(3, 9)	WET(3, 10)	WET(3, 11)	WET(3, 12)	WET(3, 13)	WET(4, 1)
WET(4, 2)	WET(4, 3)	WET(4, 4)	WET(4, 5)	WET(4, 6)	WET(4, 7)	WET(4, 8)	WET(4, 9)
WET(4, 10)	WET(4, 11)	WET(4, 12)	WET(4, 13)	WET(5, 1)	WET(5, 2)	WET(5, 3)	WET(5, 4)
WET(5, 5)	WET(5, 6)	WET(5, 7)	WET(5, 8)	WET(5, 9)	WET(5, 10)	WET(5, 11)	WET(5, 12)
WET(5, 13)	WET(6, 1)	WET(6, 2)	WET(6, 3)	WET(6, 4)	WET(6, 5)	WET(6, 6)	WET(6, 7)
WET(6, 8)	WET(6, 9)	WET(6, 10)	WET(6, 11)	WET(6, 12)	WET(6, 13)	WET(7, 1)	WET(7, 2)
WET(7, 3)	WET(7, 4)	WET(7, 5)	WET(7, 6)	WET(7, 7)	WET(7, 8)	WET(7, 9)	WET(7, 10)
WET(7, 11)	WET(7, 12)	WET(7, 13)	WET(8, 1)	WET(8, 2)	WET(8, 3)	WET(8, 4)	WET(8, 5)
WET(8, 6)	WET(8, 7)	WET(8, 8)	WET(8, 9)	WET(8, 10)	WET(8, 11)	WET(8, 12)	WET(8, 13)
WET(9, 1)	WET(9, 2)	WET(9, 3)	WET(9, 4)	WET(9, 5)	WET(9, 6)	WET(9, 7)	WET(9, 8)
WET(9, 9)	WET(9, 10)	WET(9, 11)	WET(9, 12)	WET(9, 13)	WET(10, 1)	WET(10, 2)	WET(10, 3)
WET(10, 4)	WET(10, 5)	WET(10, 6)	WET(10, 7)	WET(10, 8)	WET(10, 9)	WET(10, 10)	WET(10, 11)
WET(10, 12)	WET(10, 13)						

Table 1.--Model output for problem 1--Continued

```
   8 CALLS TO PCG ROUTINE FOR TIME STEP   1 IN STRESS PERIOD  1
  95 TOTAL ITERATIONS

HEAD/DRAWDOWN PRINTOUT FLAG = 1     TOTAL BUDGET PRINTOUT FLAG = 1     CELL-BY-CELL FLOW TERM FLAG = 0

OUTPUT FLAGS FOR ALL LAYERS ARE THE SAME:
  HEAD     DRAWDOWN  HEAD  DRAWDOWN
PRINTOUT  PRINTOUT  SAVE    SAVE
---------------------------------
   1         0       0       0


                  HEAD IN LAYER  1 AT END OF TIME STEP  1 IN STRESS PERIOD  1
         ------------------------------------------------------------------

         1      2      3      4      5      6      7      8      9     10     11     12     13     14     15
.....................................................................................................................

  1   138.94 138.23 136.79 134.61 131.65 127.87 123.19 117.53 110.77 102.77  93.33  82.39  71.06 999.99 999.99

  2   138.94 138.23 136.79 134.61 131.65 127.87 123.19 117.53 110.78 102.77  93.33  82.39  71.06 999.99 999.99

  3   138.94 138.23 136.79 134.61 131.65 127.87 123.19 117.53 110.78 102.77  93.33  82.39  71.06 999.99 999.99

  4   138.94 138.23 136.79 134.61 131.65 127.87 123.19 117.53 110.78 102.77  93.33  82.39  71.06 999.99 999.99

  5   138.94 138.23 136.79 134.61 131.65 127.87 123.19 117.53 110.78 102.77  93.33  82.39  71.06 999.99 999.99

  6   138.94 138.23 136.79 134.61 131.65 127.87 123.19 117.53 110.78 102.77  93.33  82.39  71.06 999.99 999.99

  7   138.94 138.23 136.79 134.61 131.65 127.87 123.19 117.53 110.78 102.77  93.33  82.39  71.06 999.99 999.99
```

Table 1.--Model output for problem 1--Continued

8	138.94	138.23	136.79	134.61	131.65	127.87	123.19	117.53	110.78	102.77	93.33	82.39	71.06	999.99	999.99
9	138.94	138.23	136.79	134.61	131.65	127.87	123.19	117.53	110.78	102.77	93.33	82.39	71.06	999.99	999.99
10	138.94	138.23	136.79	134.61	131.65	127.87	123.19	117.53	110.78	102.77	93.33	82.39	71.06	999.99	999.99

HEAD IN LAYER 2 AT END OF TIME STEP 1 IN STRESS PERIOD 1

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	137.46	136.72	135.24	132.97	129.89	125.92	120.98	114.93	107.58	98.63	87.60	73.72	55.50	29.50	1.50
2	137.46	136.72	135.24	132.97	129.89	125.92	120.98	114.93	107.58	98.63	87.60	73.72	55.50	29.50	1.50
3	137.46	136.72	135.24	132.97	129.89	125.92	120.98	114.93	107.58	98.63	87.60	73.72	55.50	29.50	1.50
4	137.46	136.72	135.24	132.97	129.89	125.92	120.98	114.93	107.58	98.63	87.60	73.72	55.50	29.50	1.50
5	137.46	136.72	135.24	132.97	129.89	125.92	120.98	114.93	107.58	98.63	87.60	73.72	55.50	29.50	1.50
6	137.46	136.72	135.24	132.97	129.89	125.92	120.98	114.93	107.58	98.63	87.60	73.72	55.50	29.50	1.50
7	137.46	136.72	135.24	132.97	129.89	125.92	120.98	114.93	107.58	98.63	87.60	73.72	55.50	29.50	1.50
8	137.46	136.72	135.24	132.97	129.89	125.92	120.98	114.93	107.58	98.63	87.60	73.72	55.50	29.50	1.50
9	137.46	136.72	135.24	132.97	129.89	125.92	120.98	114.93	107.58	98.63	87.60	73.72	55.50	29.50	1.50
10	137.46	136.72	135.24	132.97	129.89	125.92	120.98	114.93	107.58	98.63	87.60	73.72	55.50	29.50	1.50

Table 1.--Model output for problem 1--Continued

VOLUMETRIC BUDGET FOR ENTIRE MODEL AT END OF TIME STEP 1 IN STRESS PERIOD 1

CUMULATIVE VOLUMES	L**3	RATES FOR THIS TIME STEP	L**3/T
IN:		IN:	
STORAGE =	0.00000	STORAGE =	0.00000
CONSTANT HEAD =	0.00000	CONSTANT HEAD =	0.00000
WELLS =	0.00000	WELLS =	0.00000
RECHARGE =	0.15000E+06	RECHARGE =	0.15000E+06
RIVER LEAKAGE =	0.00000	RIVER LEAKAGE =	0.00000
TOTAL IN =	0.15000E+06	TOTAL IN =	0.15000E+06
OUT:		OUT:	
STORAGE =	0.00000	STORAGE =	0.00000
CONSTANT HEAD =	0.00000	CONSTANT HEAD =	0.00000
WELLS =	0.00000	WELLS =	0.00000
RECHARGE =	0.00000	RECHARGE =	0.00000
RIVER LEAKAGE =	0.15000E+06	RIVER LEAKAGE =	0.15000E+06
TOTAL OUT =	0.15000E+06	TOTAL OUT =	0.15000E+06
IN - OUT =	-1.8437	IN - OUT =	-1.8437
PERCENT DISCREPANCY =	0.00	PERCENT DISCREPANCY =	0.00

Table 1.--Model output for problem 1--Continued

TIME SUMMARY AT END OF TIME STEP 1 IN STRESS PERIOD 1

	SECONDS	MINUTES	HOURS	DAYS	YEARS
TIME STEP LENGTH	86400.0	1440.00	24.0000	1.00000	0.273785E-02
STRESS PERIOD TIME	86400.0	1440.00	24.0000	1.00000	0.273785E-02
TOTAL SIMULATION TIME	86400.0	1440.00	24.0000	1.00000	0.273785E-02

wet. It is also possible to use a larger positive wetting threshold to prevent these cells from incorrectly becoming wet.

Validation of Results

There is no known analytical solution for this problem. Several simulations using the BCF1 Package were made to show that the BCF2 results are valid. To do these, initial heads were set so that all cells were active, preventing the need to wet cells. In steady-state simulations, problems occurred because of the drying of too many cells. Therefore, a transient simulation was used to avoid the likelihood for cells to dry unnecessarily. With initial heads of 140 feet and all cells active, the natural and stressed conditions could be simulated in two successive stress periods in much the same way as done in the BCF2 simulation. Each stress period lasted 90,000 days, had 20 time steps, and a time step multiplier of 1.1. The specific yield for layer 1 was 0.2. By the end of the last time step, heads were essentially at steady state.

There were differences between the results from the BCF2 simulation and the transient BCF1 simulation. Cells in column 10 of layer 1 at the end of stress period 2 were dry in the BCF2 simulation and wet in the BCF1 simulation. The result is that heads in the BCF1 simulation are as much as 1 foot lower than heads in the BCF2 simulation. Analysis shows that the differences between the BCF1 and BCF2 simulations are not caused by errors in the BCF2 code. In fact, a transient BCF2 simulation was made that reproduced the BCF1 results exactly. Rather, the differences are caused by the non-unique solutions that can result from the method of solving the nonlinear equations as described previously. The cells in column 10 went dry in the BCF2 simulation because of overshoot in the iterative convergence process. Once they went dry, heads did not go high enough to cause these cells to rewet. In the steady-state BCF2 simulation, the head in column 10 of layer 2 was about 0.8 foot above the bottom of layer 1; the wetting threshold was 2 feet. The effects of flow from storage minimized the overshoot in the BCF1 transient simulation, allowing the cells in column 10 to stay wet. These simulations show that, with the specified wetting

criteria, a range of solutions is possible, depending on numerical subtleties of the solution process. The range of uncertainty can be minimized by using a smaller wetting threshold, but this is at the risk of causing instability in the solution process.

Problem 2 -- Simulation of a Water-Table Mound in Resulting from Local Recharge

Localized recharge to a water-table aquifer results in formation of a ground-water mound. For example, a ground-water mound may form in response to recharge from infiltration ponds commonly used to artificially replenish aquifers or to remove contamination by filtration through a soil column. If the aquifer has low vertical hydraulic conductivity or contains interspersed zones of low hydraulic conductivity, it may be necessary to simulate the aquifer using multiple model layers in which the mound crosses more than one layer. The BCF1 Package could not be used for such a simulation because cells cannot change from no-flow to variable head, and the simulated mound could not build up into previously dry layers. The following hypothetical problem was posed to evaluate the capability of the BCF2 Package to simulate mounding of the water table through multiple model layers.

Conceptual Model

The conceptual model consists of a rectangular, unconfined aquifer overlain by a thick unsaturated zone (fig. 2). The horizontal hydraulic conductivity is 5 feet per day and vertical hydraulic conductivity is 0.25 feet per day. A leaking pond recharges the aquifer, resulting in the formation of a ground-water mound. The pond covers approximately 6 acres and pond leakage is 12,500 cubic feet per day (65 gallons per minute). The specific yield is 20 percent. The water table is flat prior to the creation of the recharge pond. The flat water table is the result of a uniform constant-head boundary that surrounds the aquifer.

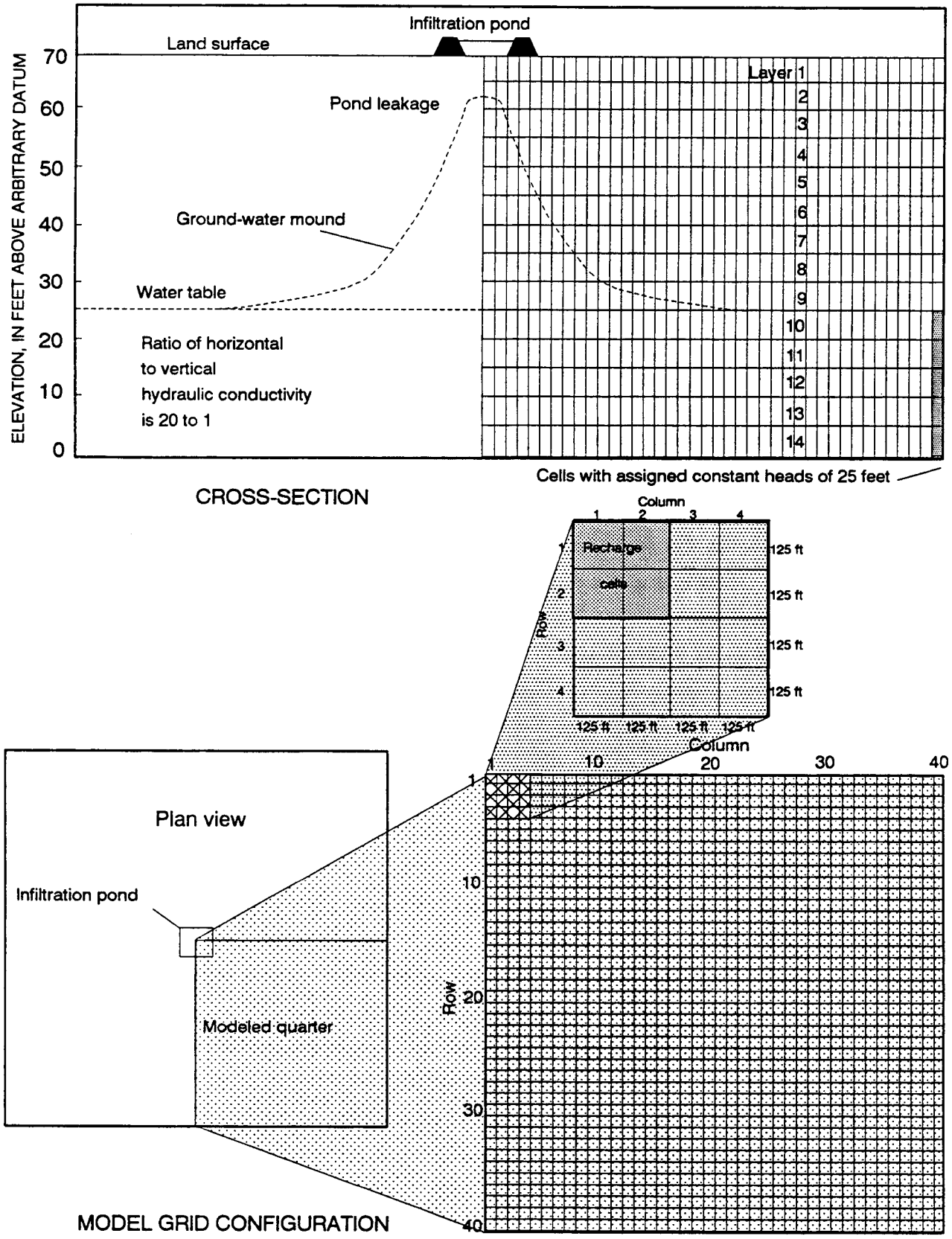


Figure 2.--Hydrogeology, model grid, and model boundary conditions for problem 2.

Modeling Approach

Because of the symmetry, heads are identical in each quadrant of the aquifer, and there is no flow between quadrants; therefore, only one quarter of the system needs to be simulated. The problem is simulated using a grid of 40 rows, 40 columns, and 14 layers (fig. 2). A uniform horizontal grid spacing of 125 feet is used, and each layer is 5 feet thick. The pond is in the upper left corner of the grid. The boundaries along row 1 and column 1 are no-flow as a result of the symmetry. A constant-head boundary of 25 feet is specified along row 40 and column 40 for layers 10-14; a no-flow boundary is assigned along row 40 and column 40 for layers 1-9. Without the recharge from the pond, layers 1-9 are dry, and the head in all the cells of layers 10-14 is 25 feet.

Recharge from the pond is applied to the horizontal area encompassed by rows 1 through 2 and columns 1 through 2. Recharge option 3 (McDonald and Harbaugh, 1988, p. 7-2) is used so that recharge will penetrate through inactive cells down to the water table. The recharge rate of 0.05 foot per day simulates leakage of 3,125 cubic feet per day (16.2 gallons per minute) through one quarter of the pond bottom, a simulated area of 62,500 square feet.

Simulation Results

Steady-state and transient simulations were made to demonstrate the capability of the BCF2 Package to simulate the formation of the mound. Solution of the flow equations was attempted using both the PCG2 solver and the SIP solver. Also, several combinations of the BCF2 parameters that affect stability of the wetting process were used. Wetting and solver parameters are shown in table 2 for comparison.

Table 2. -- Wetting and solver input data for Problem 2 simulations

Parameter	Simulation 2A	Simulation 2B	Simulation 2C
WETDRY (feet)	-0.5	1.5	-0.5
Equation used for calculating head at a cell that is wetted	3a	3a	3b
Wetting factor (WETFCT)	0.5	0.5	1.0
Wetting iteration interval (IWETIT)	1	2	11
Solver	PCG2	PCG2	SIP
Head-change closure criterion (feet)	0.001	0.001	0.00001
PCG residual closure criterion (cubic feet)	1000.	1000.	N/A
PCG maximum inner iterations	20	20	N/A
SIP WSEED	N/A	N/A	0.002

For the steady-state problem, the least computational time was required when using the PCG2 solver. Reasonable solutions to the ground-water mounding problem were obtained in two steady-state PCG2 simulations. In the first simulation, Simulation 2A, dry cells were converted to wet by comparison of the wetting threshold to head only in underlying cells, which is indicated by a negative WETDRY parameter. The wetting-iteration interval was 1 and the WETDRY parameter was -0.5 foot, which means that the wetting threshold is 10 percent of the thickness of a cell. Solution was achieved in 11 outer iterations and 179 total iterations. Figure 3 shows results from Simulation 2A.

In the second PCG2 simulation, Simulation 2B, wetting of cells was based on comparison to heads both in horizontally adjacent and underlying cells (WETDRY parameter is positive). A wetting-iteration interval of 2 and a WETDRY parameter of 1.5 feet were used in order to prevent continued oscillation between wet and dry for some cells. Solution was obtained in 23 outer iterations and 420 total iterations.

The use of horizontal wetting resulted in additional iterations for the second simulation and the need to change wetting parameters. Due to the

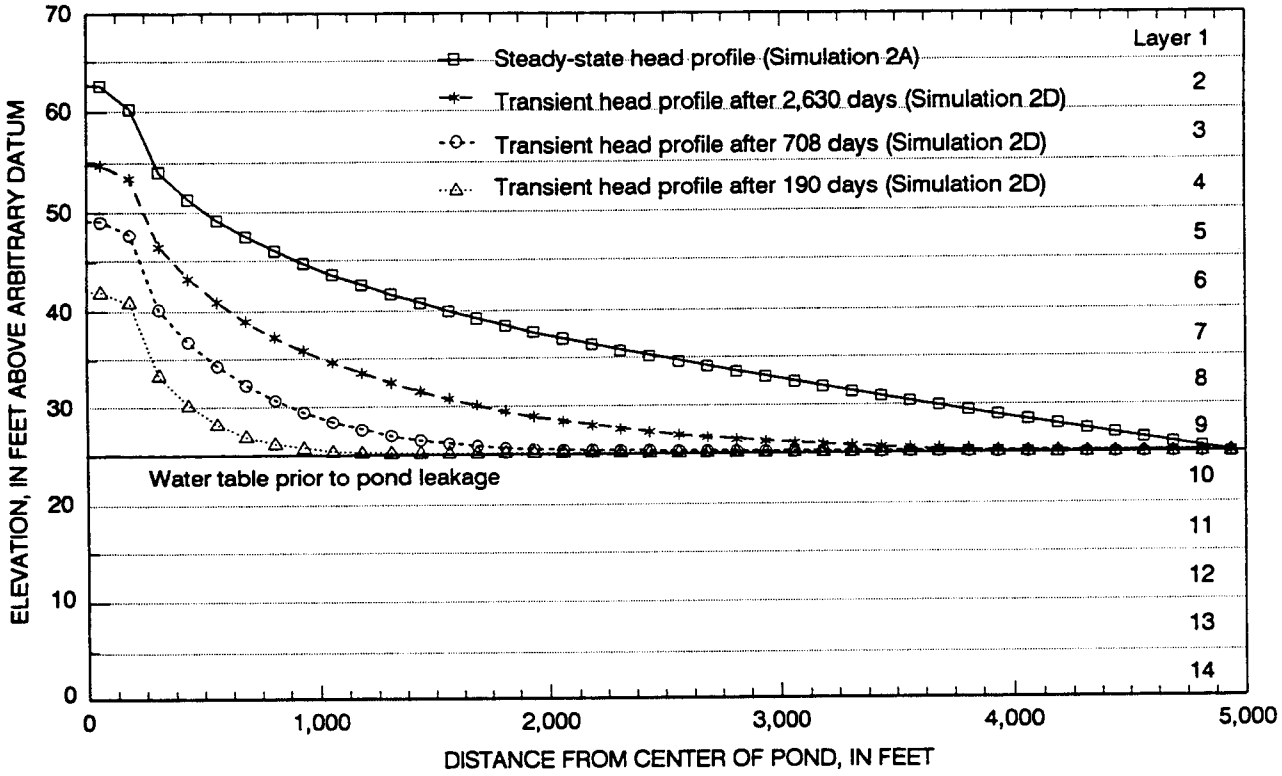


Figure 3.--Simulated head profiles beneath a leaking pond after 190 days, 708 days, 2,630 days, and at steady state conditions.

steepness of the head gradient and the grid discretization, the head difference between adjacent horizontal cells is generally much larger than the head difference between adjacent vertical cells along the mound. For example, the cell at layer 4, row 3, column 4 is supposed to be dry even though the head in the horizontally adjacent cell in column 3 is 1.4 feet above the bottom of the layer. The vertical head difference between cells in this part of the model is much less; the difference between the head at the cell in layer 4, row 3, column 3 and the cell below is only 0.05 foot. Thus, the neighboring cell to the right is repeatedly and incorrectly converted to wet during the solution process if horizontal wetting is used with a wetting threshold of 0.5 foot. The larger wetting threshold and wetting-iteration interval used in Simulation 2B allow convergence to occur, but only after many iterations. In this simulation, head in adjacent vertical cells is the best indicator of when a dry cell should become wet.

Both simulations gave similar results with minor differences in which cells were wet. For example, Simulation 2A has one wet cell in layer two while Simulation 2B has three wet cells in layer 2. However there were 244 less wet cells for Simulation 2B. This is because the higher wetting threshold of Simulation 2B prevented some cells from wetting that would have wetted with a lower threshold. Head differences between the two solutions generally were 0.05 foot or less with the exception of head at a few cells near the areas where the two simulations differed in which cells were wet. Head differences of about 0.1 foot were noted in these cells. The volumetric budget error for both solutions was zero.

A SIP solution to the steady-state ground-water mounding problem, Simulation 2C, was also obtained. In this simulation, cells were activated by comparison only to head in underlying cells. Wetting and SIP solver parameters were adjusted to minimize the required iterations (table 2). The wetting iteration interval was experimentally set at 11 iterations. The wetting threshold and wetting factor were set at 0.5 foot and 1.0, respectively. The solution met the head-change criterion for closure in 827 iterations, resulting in a budget error of 0.04 percent. This simulation took much more computer time than either PCG2 simulation. There were 43 less wet cells than in Simulation 2A. Heads matched Simulation 2A within 0.05 foot.

A steady-state SIP solution in which cells were activated by comparison of the wetting threshold to heads in horizontally adjacent and underlying cells could not be obtained despite experimental variation of WSEED and the wetting iteration interval. Chaotic wetting and drying of cells caused the simulation to terminate with divide-by-zero errors due to isolated groups of active cells.

The formation of the ground-water mound was simulated over time with a transient simulation, Simulation 2D. The transient simulation was run for one stress period with a length of 500,000 days. The stress period was divided into 50 time steps with a time-step multiplier of 1.3. The first time step was 0.3 days, and the last time step was 115,000 days. Transient simulations used a specific yield of 20 percent and a confined storage coefficient of 0.001. The PCG2 solver was used and cells were activated by comparison of the wetting threshold to heads in underlying cells. The head-change criterion for closure was 0.001 foot and the residual-change criterion was 10,000 cubic feet, the wetting threshold was 0.5 foot, the wetting factor was 0.5, and the wetting-iteration interval was 1. Figure 3 shows simulated water-table heads along row 1 at several times during the transient simulation.

Steady-state conditions were reached at the end of the transient simulation as indicated by storage flow terms being zero. Simulation 2D had 43 less wet cells at the end than did Simulation 2A. Heads at the end of Simulation 2D were within 0.02 foot of the heads in Simulation 2A. The volumetric budget error at the end of the stress period was 0.02 percent.

Validation of Results

The ground-water mounding problem is nonlinear, precluding comparison to an analytical solution. The steady-state solution from Simulation 2A was compared to two BCF1 simulations to show that the solution was reasonable.

The mounding problem theoretically can be simulated by the BCF1 Package in steady state if initial heads are above the final heads. That is, BCF1

could cause cells above the top of the mound to dry up provided that, as described earlier, heads did not temporarily drop too low during the convergence process. Several unsuccessful steady-state simulations were made to try to reproduce the BCF2 results. For these simulations, initial heads were specified as 65 feet for layers 1 through 9 and 25 feet for layers 10 through 14. The result was that too many cells went dry. In early iterations, heads dropped over the entire region so that all cells dried up in layers 1 through 5 even at the center of the mound. In later iterations, heads at the center of the mound rose again, but there was no way to rewet the dried cells. No amount of adjustment using SIP or PCG2 could avoid this problem.

Another approach at using BCF1 to simulate this problem at steady state was successful. A transient simulation was used so that the mound could form slowly over time. This produced the desired result of allowing heads to drop faster at the fringes of the mound than at the center. The steady-state solution was approximated by running the transient simulation for a 900,000-day stress period.

The successful BCF1 simulation, Simulation 2E, consisted of one stress period divided into 20 time steps with a time-step multiplier of 1.5. All cells in layers 2 through 14 initially were active and fully saturated; starting heads of 65 feet and 25 feet were assigned to all cells in layers 1-9 and 10-14, respectively. Cells were permitted to drain to a steady-state distribution of heads using the same recharge, grid configuration, and boundary conditions as used in the BCF2 simulations. The PCG2 solver was used in the simulation. PCG2 parameters included a head-change criterion for closure of 0.001 foot, residual closure criterion of 1,000 cubic feet per day and a maximum of 20 inner iterations for each outer iteration. In the last time step, essentially no water was moving in or out of storage. The overall volumetric budget error was -0.01 percent.

Comparison of Simulations 2A and 2E indicates that the steady-state heads match within 0.12 foot. Heads in Simulation 2E are slightly lower than in Simulation 2A because Simulation 2E has 159 more wet cells than Simulation 2A. For example, there are four wet cells in layer 2 in

Simulation 2E and one wet cell in Simulation 2A. This reflects the fact that the BCF1 solution was obtained by draining the system rather than raising heads. In Simulation 2A, heads below those dry cells that were wet in Simulation 2E are above the bottom of the dry cells, but not by more than the wetting threshold. Thus, the cells do not become wet. Both solutions are reasonable, but the differences point out the lack of uniqueness that results from the method of solving the nonlinear equations.

A steady-state BCF1 simulation with only one layer, Simulation 2F, was also run to help evaluate whether the steady-state BCF2 solution was reasonable. With the exception of the layering configuration and associated anisotropy, all other input parameters were the same as in the BCF2 simulations. The one-layer steady-state solution was by the SIP solver.

Figure 4 compares the head from Simulations 2F, 2E, and 2A. For Simulations 2A and 2E, head is the water-table head (head in the highest variable-head cell at a horizontal location). Heads in Simulation 2F are similar to but lower than water-table heads in the multi-layer simulations (Simulations 2A and 2E) because vertical resistance to flow is not being simulated in Simulation 2F. The vertically averaged head from all the cells at row 1, column 1 in Simulation 2A is 0.71 feet greater than the head in Simulation 2F. These results indicate that the BCF2 simulations provide a reasonable solution to the nonlinear, layered, ground-water mounding problem.

The steady-state and transient simulations of Problem 2 show that the BCF2 Package can successfully simulate the multiple-layer formation of a ground-water mound in a water-table aquifer. In many ways, Problem 2 simulates a very simple aquifer system compared to typical real-world systems. For example, boundary conditions would normally be more complex with the regional water table resulting from areal recharge flowing to natural discharge points. Also, additional stresses would normally be part of the system. However, the simplicity of the aquifer system does not reduce the complexity of the simulation regarding the wetting of cells. That is, this simulation is a difficult test of the cell-wetting capability of the BCF2 Package. In fact, typical simulations using the wetting

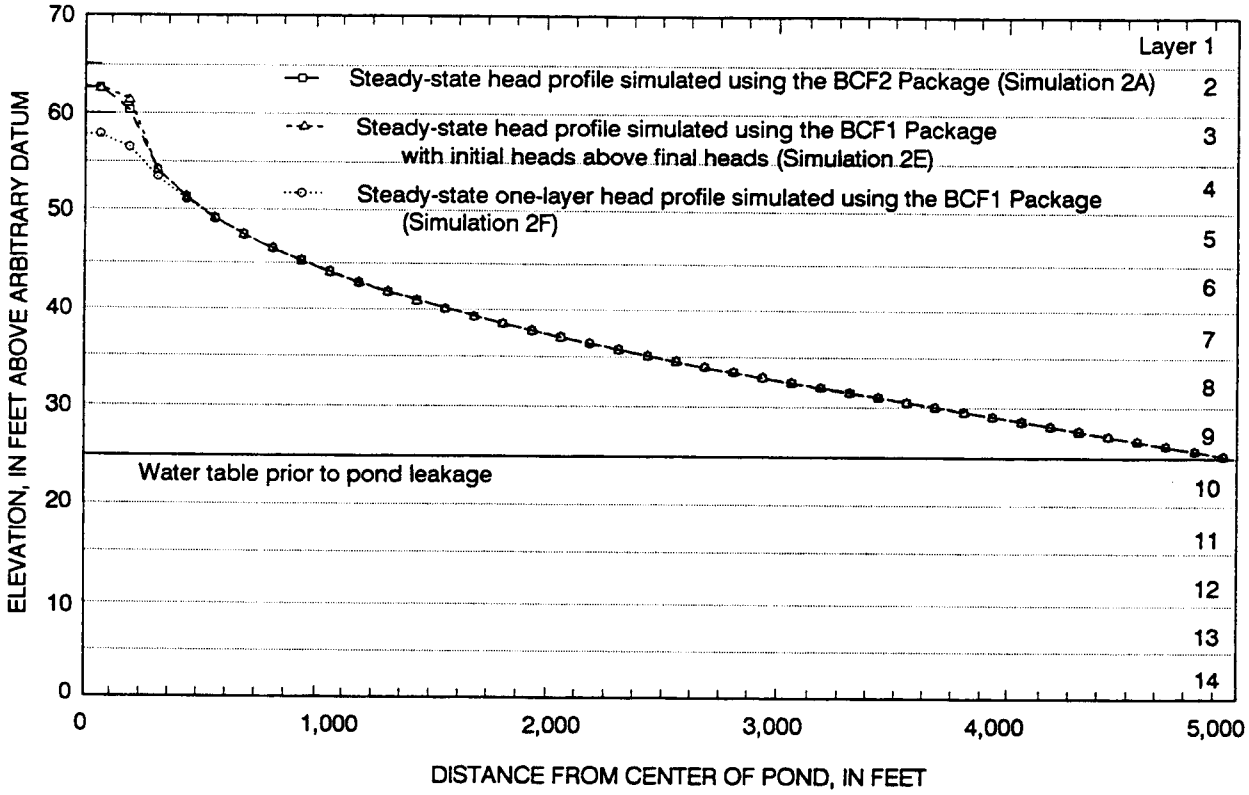


Figure 4.--Steady-state head profiles simulated using the BCF2 Package and BCF1 Package [one BCF1 simulation with initial heads higher than final heads and a one-layer BCF1 simulation].

capability probably will not involve a water table crossing so many model layers. For many purposes, this system could be simulated realistically with less layers. The large number of layers was used in order to test the BCF2 Package. With the water table crossing eight layers, there is a high possibility of repeated wetting and drying of cells. Yet stable solutions were obtained after moderate effort was used in selecting the various parameters that affect solution stability.

Problem 3 -- Simulation of a Perched Water Table

Contrasts in vertical hydraulic conductivity within the unsaturated zone can provide a mechanism for the formation of perched ground-water tables. The formation of these bodies in a previously unsaturated zone is difficult to simulate with the BCF1 Package because cells cannot change from no-flow to variable-head conditions. A problem was posed to evaluate the capability of the BCF2 Package to simulate formation of a perched water table. This problem has practical application in the simulation of recharge going to a deeper aquifer system.

Conceptual Model

The conceptual model is rectangular and consists of three geohydrologic units. The upper and lower units have higher hydraulic conductivities than the middle unit (fig. 5). There is a regional water table in which the head is below the bottom of unit 2. Natural recharge occurs over the entire area at a rate of 0.001 foot per day. This recharge can percolate through the two upper units without the formation of a water table above unit 2 because the vertical hydraulic conductivity of this middle unit is 0.002 foot per day.

Recharge at a rate of 0.01 foot per day from a pond covering 6 acres causes a perched ground-water body to form in the top two units. The total pond leakage is about 2,360 cubic feet per day (12 gallons per minute). The perched water table spreads out over an area much larger than the area

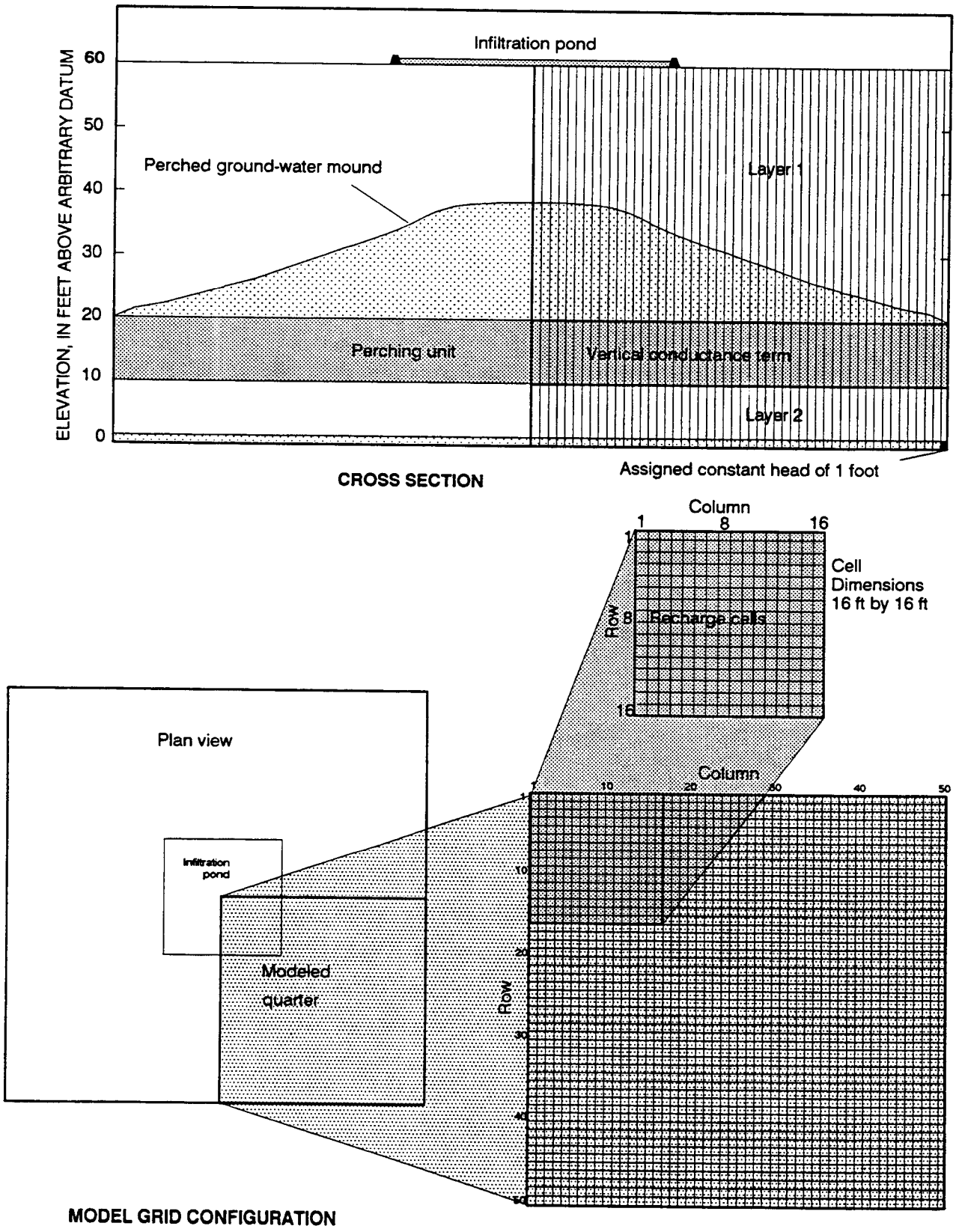


Figure 5.--Hydrogeology, model grid, and model boundary conditions for problem 3.

covered by the pond. This has an impact on the distribution of recharge to the lower unit.

Modeling Approach

Because of the rectangular symmetry of the system, there is no flow between quadrants. Therefore, only one quarter of the system must be simulated. The problem is simulated using a grid of 50 rows, 50 columns, and 2 units. A uniform grid spacing of 16 feet is used. The recharge pond is in the upper left corner of the grid; the quarter of the pond that is simulated occupies a square area that is 16 rows long and 16 columns wide. The boundaries along row 1 and along column 1 are no-flow boundaries as a result of the symmetry.

Model layer 1 simulates geohydrologic unit 1 and is assigned a hydraulic conductivity of 5 feet per day. The bottom of layer 1 is at an elevation of 20 feet.

Geohydrologic unit 3 is simulated as model layer 2. This layer is simulated as a confined/unconfined layer with constant transmissivity (layer-type 2, McDonald and Harbaugh, p. 5-34). The top and bottom of layer 2 are set at 10 and 0 feet, respectively. Because the head in this unit is always below the top of the unit, the flow from above is limited as described by McDonald and Harbaugh (p. 5-19). Thus, there is no direct hydraulic connection between the perched layer and the lower layer, but the perched heads have a direct impact on the recharge into the lower layer. The purpose of the simulation is to show that the distribution of recharge from the perched system can be estimated by using the BCF2 Package to simulate the perched water table. All cells in layer 2 are assigned a constant head of 1 foot because there is no need to simulate heads in this layer for the purpose of estimating recharge.

Geohydrologic layer 2 is not simulated as a separate model layer because it is assumed that horizontal flow and storage affects are negligible. This layer is represented by the value for VCONT (vertical

hydraulic conductivity divided by thickness) between model layers 1 and 2. Assuming the vertical hydraulic conductivity is 0.002 foot per day and the thickness is 10 feet, VCONT is 0.0002 per day.

In areas not covered by the pond, recharge is applied areally at a rate of 0.001 foot per day to simulate natural recharge. Recharge option 3 (McDonald and Harbaugh, 1988, p. 7-2) is used so that recharge will penetrate through inactive cells to the water table. A recharge rate of 0.01 foot per day is applied to the area covered by the pond.

Simulation Results

Steady-state and transient simulations were made to demonstrate the capability of the BCF2 Package to simulate the formation of a perched water table. Solutions of the flow equation were obtained using the SIP solver and the PCG2 solver. Both solvers gave good solutions, but SIP was the most efficient. The SIP results are presented in the rest of this discussion.

The steady-state simulation produced the long-term head distribution resulting from the pond recharge. Initial head in layer 1 under the pond was set at 21 feet. All other cells in layer 1 initially were specified as no-flow cells. Wetting and SIP solver parameters were adjusted to obtain a solution. The wetting-iteration interval, WETDRY parameter, and wetting factor were set at 2 iterations, 1.0 foot, and 0.5, respectively. The WETDRY parameter was positive to indicate that horizontally adjacent cells could cause dry cells to become wet. This was the only way for cells in layer 1 to become wet because heads in layer 2 were always below the bottom of layer 1. The head-change criterion for closure was set at 0.001 foot. The adjusted WSEED value was 0.0015.

Steady-state heads along row 1 in layer 1 ranged from 29.92 feet in cell 1,1,1 to 20.78 feet in cell 1,1,40 (fig. 6). The volumetric budget error was 0.03 percent, and the solution met the head-change criterion for closure in 101 iterations.

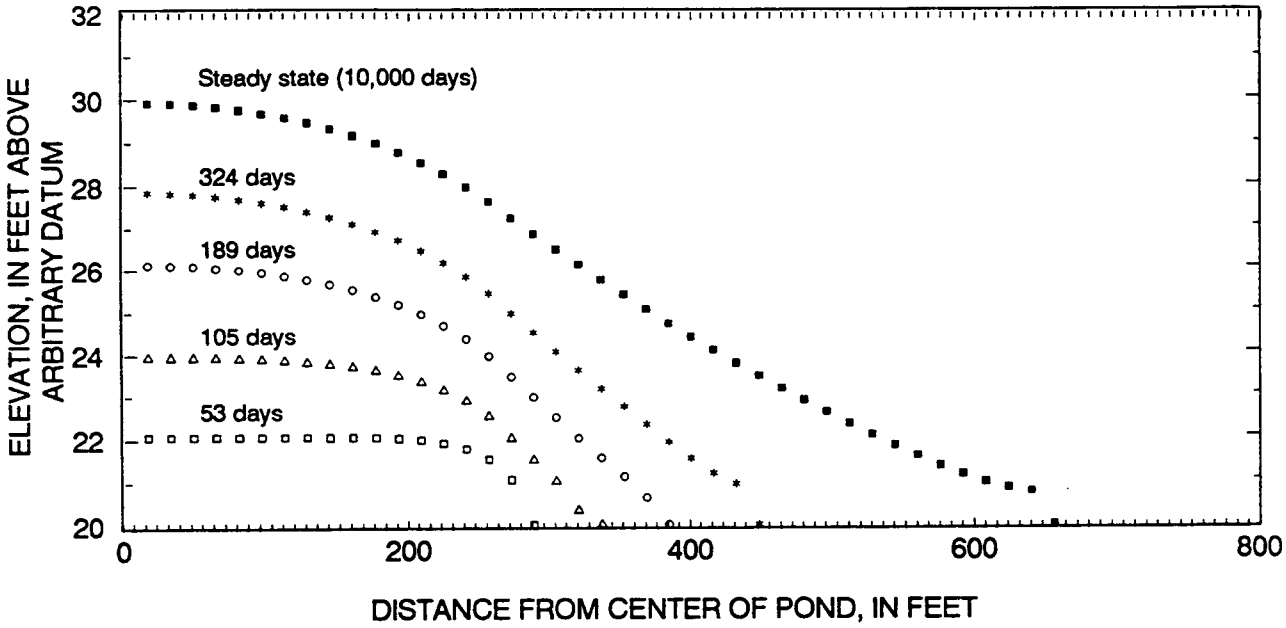


Figure 6.--Simulated perched water-table profiles after 53 days, 105 days 189 days, 324 days, and at steady-state conditions.

The formation of the perched water table was simulated over time in a transient simulation. The transient simulation was run for one stress period with a length of 10,000 days. The stress period was divided into 60 time steps with a time-step multiplier of 1.1. The specific yield was 20 percent. The wetting-iteration interval and wetting factor were experimentally set at 1 iteration and 0.1, respectively. The WETDRY parameter was 1.3 feet in the square area defined by rows 1-18 and columns 1-18, and everywhere else WETDRY was 1 foot. Initially, WETDRY was set to 1 foot everywhere, but this resulted in instability in early time. This may have been caused by the steeper head gradients that occur in early time as the mound began to form. Also, as described earlier, small time steps can contribute to instability due to the resulting large rates of flow from storage. The adjusted WSEED value was 0.0001. Initial heads of 20.01 feet were set at the cells below the pond. All other cells in layer 1 initially were no flow.

Transient heads along row 1 of layer 1 are shown in figure 6 at times of 53 days, 105 days, 189 days, and 324 days. The volumetric budget error was zero. After 10,000 days, heads essentially were at steady state because flow into storage was zero. Heads at the end of the transient simulation matched the heads from the steady-state simulation within 0.02 foot.

Validation of Results

The perched water-table problem is nonlinear, precluding comparison to an analytical solution. Comparison of the steady-state and transient solutions indicated that the BCF2 simulations reasonably solved the problem. The steady-state BCF2 solution also was compared to a transient BCF1 simulation to determine whether the solution was reasonable.

The BCF1 transient solution to the perched water-table problem was simulated by permitting cells to drain to a steady-state configuration. All cells in layer 1 initially were active with a starting head of 30 feet. SIP solver parameters were the same as those used in the BCF2 solutions. The simulation was run for 10,000 days, using 100 time steps and a time-step

multiplier of 1.1. The volumetric budget error at the end of the 10,000 days was zero.

The steady-state heads were very similar in the two simulations, but the BCF1 steady-state heads were as much as 0.6 foot lower than the BCF2 steady-state heads near the edges of the perched water table. In the BCF1 simulation, the heads were draining from high initial values while in the the BCF2 simulation, heads rose from low initial values. Thus, some cells remained wet in the BCF1 simulation that did not become wet in the BCF2 simulation because the wetting threshold was not exceeded. The larger extent of the water table in the BCF1 simulation caused the slightly lower heads.

MODULE DOCUMENTATION

The design of the BCF2 Package closely follows the design of the BCF1 Package. Several of the BCF2 modules are identical to BCF1 modules, and the replaced modules are very similar in function and structure to the previous modules. There is one new module, BCF2AD. The unchanged modules have not been given new names. That is, the unchanged modules retain the revision level "1" in their names. Accordingly, the BCF2 Package consists of the following modules: BCF2AL, BCF2RP, BCF2AD, BCF2FM, BCF1BD, SBCF2N, SBCF2H, SBCF1C, SBCF1B, SBCF1F. This report contains documentation for all of the new modules. Refer to McDonald and Harbaugh (1988, Ch. 5) for documentation of unchanged modules.

In the model program, each package is assigned a file unit on which to read input data. The file unit for the BCF2 Package is assumed to be the same input unit that is used by the BCF1 Package, which is the first element in the IUNIT array [IUNIT(1)]. Because the BCF2 Package contains all of the functionality of the BCF1 Package, there is no need to retain the BCF1 Package as a separate package in the model program.

The MAIN program must be modified to call the BCF2 Package. There are 5 calls to BCF2 Package modules -- BCF2AL, BCF2RP, BCF2AD, BCF2FM, and BCF1BD. The call to module BCF1BD should not be changed from what it is in the original MAIN program (McDonald and Harbaugh, 1988, p. 3-32). Make the following changes to the MAIN program:

Replace the BCF1AL call statement with the following:

```
      IF(IUNIT(1).GT.0) CALL BCF2AL(ISUM,LENX,LCSC1,LCHY,
     1     LCBOT,LCTOP,LCSC2,LCTRPY,IUNIT(1),ISS,
     2     NCOL,NROW,NLAY,IOUT,IBCFCB,LCWETD,IWDFLG,LCCVWD,
     3     WETFCT,IWETIT,IHDWET,HDRY)
```

Replace the BCF1RP call statement with the following:

```
      IF(IUNIT(1).GT.0) CALL BCF2RP(X(LCIBOU),X(LCHNEW),X(LCSC1),
     1          X(LCHY),X(LCCR),X(LCCC),X(LCCV),X(LCDELR),
     2     X(LCDELC),X(LCBOT),X(LCTOP),X(LCSC2),X(LCTRPY),IUNIT(1),
     3     ISS,NCOL,NROW,NLAY,NODES,IOUT,X(LCWETD),IWDFLG,X(LCCVWD))
```

Replace the BCF1FM call statement with the following:

```
      IF(IUNIT(1).GT.0) CALL BCF2FM(X(LCHCOF),X(LCRHS),X(LCHOLD),
     1          X(LCSC1),X(LCHNEW),X(LCIBOU),X(LCCR),X(LCCC),X(LCCV),
     2          X(LCHY),X(LCTRPY),X(LCBOT),X(LCTOP),X(LCSC2),
     3          X(LCDELR),X(LCDELC),DELT,ISS,KKITER,KKSTP,KKPER,NCOL,
     4          NROW,NLAY,IOUT,X(LCWETD),IWDFLG,X(LCCVWD),WETFCT,
     5          IWETIT,IHDWET,HDRY)
```

Add a new call statement for the BCF2AD module immediately following the statement that calls module BAS1AD:

```
      IF(IUNIT(1).GT.0) CALL BCF2AD(X(LCIBOU),X(LCHOLD),X(LCBOT),
     1            X(LCWETD),IWDFLG,ISS,NCOL,NROW,NLAY)
```

Narrative for Module BCF2AL

This module allocates space for data arrays for the BCF2 Package.

1. Print a message identifying the package

2. Read and print the steady-state flag ISS and the cell-by-cell flow-term unit and flag (IBCFCB). Cell-by-cell flow terms for the BCF Package are flow to the right, flow forward, flow down, inflow from storage, and flow from constant heads. Read and print HDRY, which is the value for head that is assigned at cells that convert to dry (no flow). Also read and print the flag that determines if the wetting capability is active (IWDFLG), and read the wetting parameters WETFCT (wetting factor), IWETIT (wetting iteration interval), and IHDWET (flag that determines which equation is used to calculate the initial estimate of head at a wetted cell). If wetting is active, print the wetting parameters.

3. Stop the simulation if there are more than 80 layers.

4. Read the layer-type codes (LAYCON) and print the title for the table of layer-type codes.

   0 = confined
   1 = unconfined
   2 = confined/unconfined but transmissivity is constant
   3 = confined/unconfined with transmissivity dependent on head

5. Loop through layers printing the layer-type codes and counting the layers that require TOP and BOT arrays. Initialize the counters KT and KB in which the numbers of layers needing the TOP and BOT arrays are accumulated.

   (a) Only the top layer can be unconfined. If a layer other than the top layer is unconfined (layer type 1), print an error message and STOP.

   (b) If the layer type is one or three, add one to KB, which counts the number BOT arrays.

   (c) If the layer type is two or three, add one to KT, which counts the number of TOP arrays.

6. Compute the number of cells in the entire grid and in one layer.

7. Allocate space for the following arrays:

   SC1 Primary storage capacity (only if transient);
   SC2 Secondary storage capacity (only if transient);
   TRPY Horizontal anisotropy factor;
   BOT Bottom elevation of cells;

HY Hydraulic conductivity;
TOP Top elevation of cells;
WETD Wetting threshold (only when wetting is active); and
CVWD Initial values of vertical conductance (only when wetting is active).

8. Print the amount of space used by the BCF Package.

9. RETURN.

ISS is the steady-state flag. If it is set (ISS not 0), the simulation is steady state (storage is not considered).

IBCFCB is a flag and a unit number for saving cell-by-cell flow data.

If IBCFCB > 0, it is the unit number on which cell-by-cell flow terms will be recorded whenever ICBCFL is set.

If IBCFCB = 0, cell-by-cell flow terms will not be printed or recorded.

If IBCFCB < 0, flow from constant-head cells will be printed whenever ICBCFL is set.

HDRY is the head that is assigned to cells that convert to dry.

LAYCON is a layer-type code (one for each layer).

0 - confined
1 - unconfined
2 - confined/unconfined but transmissivity is constant
3 - confined/unconfined

TOP is the elevation of the top of cells (one value for each cell in a layer).

BOT is the elevation of the bottom of cells (one value for each cell in a layer).

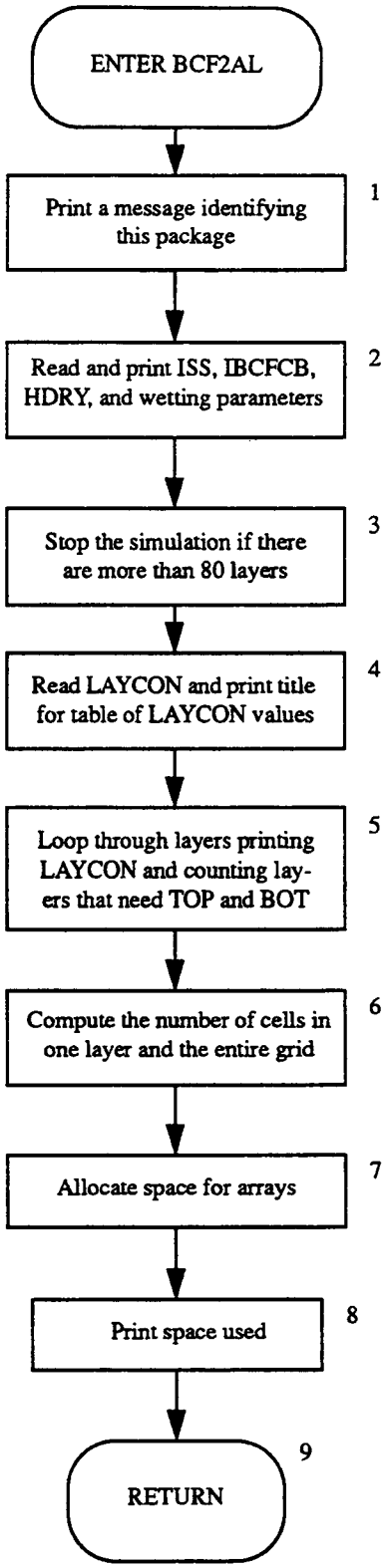


Figure 7. -- Flow chart for module BCF2AL.

```
      SUBROUTINE BCF2AL(ISUM,LENX,LCSC1,LCHY,LCBOT,LCTOP,LCSC2,LCTRPY,
     1  IN,ISS,NCOL,NROW,NLAY,IOUT,IBCFCB,LCWETD,IWDFLG,LCCVWD,
     2  WETFCT,IWETIT,IHDWET,HDRY)
C
C-----VERSION 1435 14MAY1991 BCF2AL
C     ******************************************************************
C     ALLOCATE ARRAY STORAGE FOR BLOCK-CENTERED FLOW PACKAGE, VERSION 2
C     ******************************************************************
C
C        SPECIFICATIONS:
C     ------------------------------------------------------------------
      COMMON /FLWCOM/LAYCON(80)
C     ------------------------------------------------------------------
C
C1------IDENTIFY PACKAGE
      WRITE(IOUT,1)IN
    1 FORMAT(1H0,'BCF2 -- BLOCK-CENTERED FLOW PACKAGE, VERSION 2',
     1', 7/1/91',' INPUT READ FROM UNIT',I3)
C
C2------READ AND PRINT ISS (STEADY-STATE FLAG), IBCFCB (FLAG FOR
C2------PRINTING OR UNIT# FOR RECORDING CELL-BY-CELL FLOW TERMS),
C2------HDRY (HEAD AT CELLS THAT CONVERT TO DRY), AND WETTING PARAMETERS
      READ(IN,2) ISS,IBCFCB,HDRY,IWDFLG,WETFCT,IWETIT,IHDWET
    2 FORMAT(2I10,F10.0,I10,F10.0,2I10)
      IF(ISS.EQ.0) WRITE(IOUT,3)
    3 FORMAT(1X,'TRANSIENT SIMULATION')
      IF(ISS.NE.0) WRITE(IOUT,4)
    4 FORMAT(1X,'STEADY-STATE SIMULATION')
      IF(IBCFCB.GT.0) WRITE(IOUT,9) IBCFCB
    9 FORMAT(1X,'CELL-BY-CELL FLOWS WILL BE RECORDED ON UNIT',I3)
IF(IBCFCB.LT.0) WRITE(IOUT,88)
   88 FORMAT(1X,'CONSTANT HEAD CELL-BY-CELL FLOWS WILL BE PRINTED')
      WRITE(IOUT,89) HDRY
   89 FORMAT(1X,'HEAD AT CELLS THAT CONVERT TO DRY=',G13.5)
      IF(IWDFLG.NE.0) GO TO 35
      WRITE(IOUT,31)
   31 FORMAT(1X,'WETTING CAPABILITY IS NOT ACTIVE')
      GO TO 39
C
   35 WRITE(IOUT,36)
   36 FORMAT(1X,'WETTING CAPABILITY IS ACTIVE')
      IF(IWETIT.LE.0) IWETIT=1
      WRITE(IOUT,37)WETFCT,IWETIT
   37 FORMAT(1X,'WETTING FACTOR=',F10.5,
     1     '     WETTING ITERATION INTERVAL=',I4)
      WRITE(IOUT,38)IHDWET
   38 FORMAT(1X,'FLAG THAT SPECIFIES THE EQUATION TO USE FOR HEAD AT WET
     1TED CELLS=',I4)
C
C3------STOP THE SIMULATION IF THERE ARE MORE THAN 80 LAYERS
   39 IF(NLAY.LE.80) GO TO 50
      WRITE(IOUT,11)
   11 FORMAT(1H0,'YOU HAVE SPECIFIED MORE THAN 80 MODEL LAYERS'/1X,
     1  'SPACE IS RESERVED FOR A MAXIMUM OF 80 LAYERS IN ARRAY LAYCON')
```

```
      STOP
C
C4------READ LAYCON & PRINT TITLE FOR LAYCON TABLE
   50 READ(IN,51) (LAYCON(I),I=1,NLAY)
   51 FORMAT(40I2)
      WRITE(IOUT,52)
   52 FORMAT(1X,5X,'LAYER  AQUIFER TYPE',/1X,5X,19('-'))
C
C5------LOOP THROUGH LAYERS PRINTING THE LAYER-TYPE CODE AND COUNTING
C5------LAYERS THAT NEED TOP & BOT ARRAYS
      NBOT=0
      NTOP=0
      DO 100 I=1,NLAY
      L=LAYCON(I)
      WRITE(IOUT,7) I,L
    7 FORMAT(1X,I9,I10)
C
C5A-----ONLY THE TOP LAYER CAN BE UNCONFINED(LAYCON=1).
      IF(L.NE.1 .OR. I.EQ.1) GO TO 70
      WRITE(IOUT,8)
    8 FORMAT(1H0,'AQUIFER TYPE 1 IS ONLY ALLOWED IN TOP LAYER')
      STOP
C
C5B-----LAYER TYPES 1 AND 3 NEED A BOTTOM. ADD 1 TO KB.
   70 IF(L.EQ.1 .OR. L.EQ.3) NBOT=NBOT+1
C
C5C-----LAYER TYPES 2 AND 3 NEED A TOP. ADD 1 TO KT.
      IF(L.EQ.2 .OR. L.EQ.3) NTOP=NTOP+1
  100 CONTINUE
C
C
C
C6------COMPUTE THE NUMBER OF CELLS IN THE ENTIRE GRID AND IN ONE LAYER
      NRC=NROW*NCOL
      ISIZ=NRC*NLAY
C
C7------ALLOCATE SPACE FOR ARRAYS.
      ISOLD=ISUM
      LCSC1=ISUM
      IF(ISS.EQ.0) ISUM=ISUM+ISIZ
      LCSC2=ISUM
      IF(ISS.EQ.0) ISUM=ISUM+NRC*NTOP
      LCTRPY=ISUM
      ISUM=ISUM+NLAY
      LCBOT=ISUM
      ISUM=ISUM+NRC*NBOT
      LCHY=ISUM
      ISUM=ISUM+NRC*NBOT
      LCTOP=ISUM
      ISUM=ISUM+NRC*NTOP
      LCWETD=ISUM
      IF(IWDFLG.NE.0)ISUM=ISUM+NRC*NBOT
      LCCVWD=ISUM
      IF(IWDFLG.NE.0)ISUM=ISUM+NRC*(NLAY-1)
```

```
C
C8------PRINT THE AMOUNT OF SPACE USED BY THE BCF PACKAGE.
      ISP=ISUM-ISOLD
      WRITE(IOUT,101) ISP
  101 FORMAT(1X,I8,' ELEMENTS IN X ARRAY ARE USED BY BCF')
      ISUM1=ISUM-1
      WRITE(IOUT,102) ISUM1,LENX
  102 FORMAT(1X,I8,' ELEMENTS OF X ARRAY USED OUT OF',I8)
      IF(ISUM1.GT.LENX) WRITE(IOUT,103)
  103 FORMAT(1X,'   ***X ARRAY MUST BE DIMENSIONED LARGER***')
C
C9------RETURN
      RETURN
      END
```

List of Variables for Module BCF2AL

Variable	Range	Definition
HDRY	Package	When a cell converts to dry, HNEW is set equal to HDRY.
I	Module	Index
IBCFCB	Package	Flag and a unit number. > 0, unit number on which cell-by-cell flow terms will be recorded whenever ICBCFL is set. = 0, cell-by-cell flow terms will not be printed or recorded. < 0, flow from each constant-head cell will be printed whenever ICBCFL is set.
IHDWET	Package	Flag indicating which equation to use for calculating the head at a cell that has just converted from dry to wet.
IN	Package	Primary unit number from which input for this package will be read.
IOUT	Global	Primary unit number for all printed output.
ISIZ	Module	Number of cells in the grid.
ISOLD	Module	Value of ISUM upon entry to this module.
ISP	Module	Number of elements in the X array allocated by this module.
ISS	Package	Steady-state flag. = 0, simulation is transient not 0, simulation is steady state.
ISUM	Global	Element number of the lowest element in the X array that has not yet been allocated. When space is allocated in the X array, the size of the allocation is added to ISUM.
ISUM1	Module	ISUM-1
IWDFLG	Package	Flag indicating if wetting capability is active.
IWETIT	Package	Iteration interval for attempting to wet cells.
L	Module	Temporary equivalent of LAYCON(I).
LAYCON	Package	DIMENSION (80) Layer-type code: 0 -- Layer is strictly confined. 1 -- Layer is strictly unconfined. 2 -- Layer is convertible between confined and unconfined (saturated thickness is constant). 3 -- Layer is convertible between confined and unconfined (saturated thickness varies).
LCBOT	Package	Location in the X array of the first element of array BOT.
LCCVWD	Package	Location in the X array of the first element of array CVWD.
LCHY	Package	Location in the X array of the first element of array HY.
LCSC1	Package	Location in the X array of the first element of array SC1.
LCSC2	Package	Location in the X array of the first element of array SC2.
LCTOP	Package	Location in the X array of the first element of array TOP.
LCTRPY	Package	Location in the X array of the first element of array TRPY.
LCWETD	Package	Location in the X array of the first element of array WETDRY.

LENX	Global	Number of elements in the X array. This should always be equal to the dimension of X specified in the MAIN program.
NBOT	Module	Counter for the number of layers that need elevation of the bottom and conductance divided by thickness arrays.
NCOL	Global	Number of columns in the grid.
NLAY	Global	Number of layers in the grid.
NRC	Module	Number of cells in a layer.
NROW	Global	Number of rows in the grid.
NTOP	Module	Counter for the number of layers that need elevation of the top and (if transient) secondary storage capacity arrays.
WETFCT	Package	A factor that is included in the calculation of the head that is established at a cell that has just converted from dry to wet.

Narrative for Module BCF2RP

This module reads transmissivity along rows, hydraulic conductivity along rows, storage coefficients, vertical conductance, elevation of top of cells, elevation of bottom of cells, and wetting threshold. It also calls SBCF2N to calculate parameters that are constant throughout a simulation.

1. Call utility module U1DREL to read TRPY, DELR, and DELC which have one value for each layer, column, and row, respectively. TRPY is the ratio of transmissivity (or hydraulic conductivity) along columns to transmissivity (or hydraulic conductivity) along rows for each layer.

2. For each layer, call utility module U2DREL to read the properties of the porous medium. The data requirements for each layer are determined by the layer-type code.

   (a) Find the one-dimensional array subscripts for the first element of a layer within the multi-layer arrays.

   (b) If the simulation is transient (ISS = 0), read the primary storage coefficient.

   (c) For constant transmissivity layers (LAYCON = 0 or 2), read the transmissivity.

   (d) For variable transmissivity layers (LAYCON = 1 or 3), read the hydraulic conductivity and elevation of the bottom of cells.

   (e) Read vertical hydraulic conductivity divided by thickness (VCONT). For a given layer, VCONT represents the material between the nodes in that layer and the nodes in the next lower layer. Therefore, no VCONT array is read for the lowest layer in the grid.

   (f) If the simulation is transient and the layer-type code is two or three, read the secondary storage coefficient (specific yield).

   (g) Read the elevation of the top of cells if the layer-type code is two or three.

   (h) Read the wetting threshold if the layer-type code is one or three.

3. Call SBCF2N to calculate conductance and storage terms that are constant during the simulation. Also, check that branch conductances agree with boundaries specified in the IBOUND array. Save vertical conductance if the wetting capability is active.

4. RETURN.

DELR is the grid spacing in the row direction.

DELC is the grid spacing in the column direction.

TRPY is the ratio of transmissivity in the column direction to transmissivity in the row direction.

LAYCON is a layer-type code (one for each layer).

0 - confined
1 - unconfined
2 - confined/unconfined but transmissivity is constant
3 - confined/unconfined

VCONT is vertical hydraulic conductivity divided by distance between the node in one layer and the node in the layer below (one value for each cell in a layer).

WETDRY is the wetting threshold (one value for each cell in a layer).

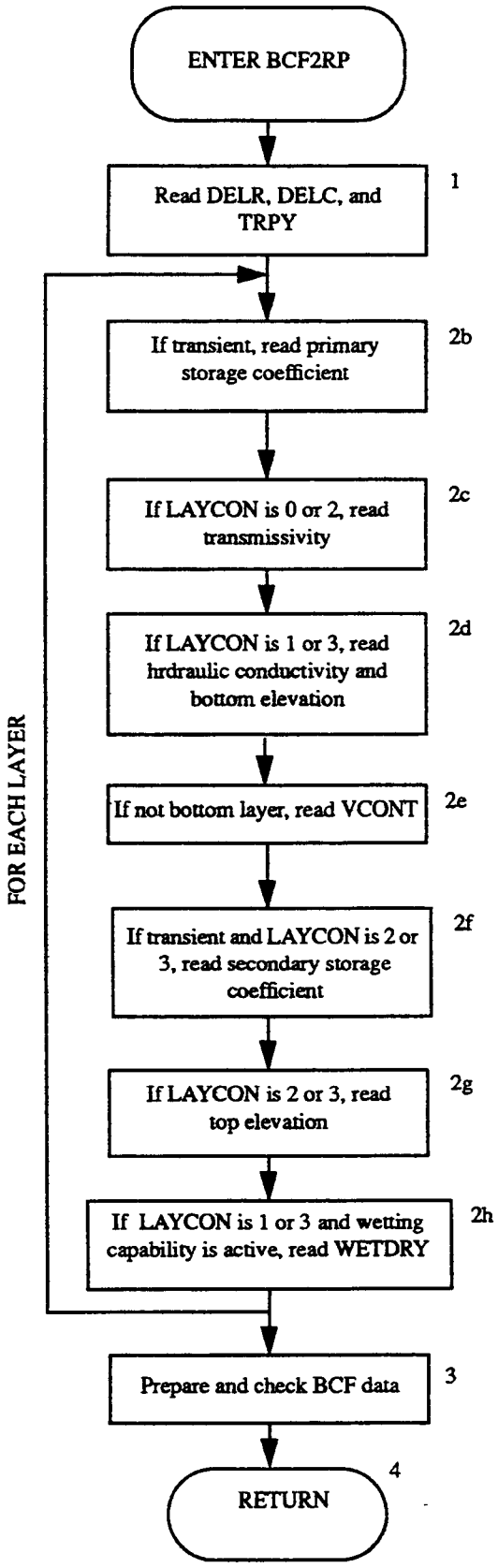


Figure 8. -- Flow chart for module BCF2RP.

```
      SUBROUTINE BCF2RP(IBOUND,HNEW,SC1,HY,CR,CC,CV,DELR,DELC,BOT,TOP,
     1 SC2,TRPY,IN,ISS,NCOL,NROW,NLAY,NODES,IOUT,WETDRY,IWDFLG,CVWD)
C
C-----VERSION 1275 6JUNE1991 BCF2RP
C     ******************************************************************
C     READ AND INITIALIZE DATA FOR BLOCK-CENTERED FLOW PACKAGE,
C     VERSION 2
C     ******************************************************************
C
C        SPECIFICATIONS:
C     ------------------------------------------------------------------
      CHARACTER*4 ANAME
      DOUBLE PRECISION HNEW
C
      DIMENSION HNEW(NODES),SC1(NODES),HY(NODES),CR(NODES),CC(NODES),
     1          CV(NODES),ANAME(6,11),DELR(NCOL),DELC(NROW),BOT(NODES),
     1          TOP(NODES),SC2(NODES),TRPY(NLAY),IBOUND(NODES),
     1          WETDRY(NODES),CVWD(NODES)
C
      COMMON /FLWCOM/LAYCON(80)
C
      DATA ANAME(1,1),ANAME(2,1),ANAME(3,1),ANAME(4,1),ANAME(5,1),
     1  ANAME(6,1) /'    ','PRIM','ARY ','STOR','AGE ','COEF'/
      DATA ANAME(1,2),ANAME(2,2),ANAME(3,2),ANAME(4,2),ANAME(5,2),
     1  ANAME(6,2) /'    ','TRAN','SMIS','. AL','ONG ','ROWS'/
      DATA ANAME(1,3),ANAME(2,3),ANAME(3,3),ANAME(4,3),ANAME(5,3),
     1  ANAME(6,3) /'   H','YD. ','COND','. AL','ONG ','ROWS'/
      DATA ANAME(1,4),ANAME(2,4),ANAME(3,4),ANAME(4,4),ANAME(5,4),
     1  ANAME(6,4) /'VERT',' HYD',' CON','D /T','HICK','NESS'/
      DATA ANAME(1,5),ANAME(2,5),ANAME(3,5),ANAME(4,5),ANAME(5,5),
     1  ANAME(6,5) /'    ','    ','    ','    ','  BO','TTOM'/
      DATA ANAME(1,6),ANAME(2,6),ANAME(3,6),ANAME(4,6),ANAME(5,6),
     1  ANAME(6,6) /'    ','    ','    ','    ','    ',' TOP'/
      DATA ANAME(1,7),ANAME(2,7),ANAME(3,7),ANAME(4,7),ANAME(5,7),
     1  ANAME(6,7) /'  SE','COND','ARY ','STOR','AGE ','COEF'/
      DATA ANAME(1,8),ANAME(2,8),ANAME(3,8),ANAME(4,8),ANAME(5,8),
     1  ANAME(6,8) /'COLU','MN T','O RO','W AN','ISOT','ROPY'/
      DATA ANAME(1,9),ANAME(2,9),ANAME(3,9),ANAME(4,9),ANAME(5,9),
     1  ANAME(6,9) /'    ','    ','    ','    ','    ','DELR'/
      DATA ANAME(1,10),ANAME(2,10),ANAME(3,10),ANAME(4,10),ANAME(5,10),
     1  ANAME(6,10) /'    ','    ','    ','    ','    ','DELC'/
      DATA ANAME(1,11),ANAME(2,11),ANAME(3,11),ANAME(4,11),ANAME(5,11),
     1  ANAME(6,11) /'    ','    ','WETD','RY P','ARAM','ETER'/
C     ------------------------------------------------------------------
C
C1------CALCULATE NUMBER OF NODES IN A LAYER AND READ TRPY,DELR,DELC
      NIJ=NCOL*NROW
C
      CALL U1DREL(TRPY,ANAME(1,8),NLAY,IN,IOUT)
      CALL U1DREL(DELR,ANAME(1,9),NCOL,IN,IOUT)
      CALL U1DREL(DELC,ANAME(1,10),NROW,IN,IOUT)
C
C2------READ ALL PARAMETERS FOR EACH LAYER
      KT=0
```

```
      KB=0
      DO 200 K=1,NLAY
      KK=K
C
C2A-----FIND ADDRESS OF EACH LAYER IN THREE DIMENSION ARRAYS.
      IF(LAYCON(K).EQ.1 .OR. LAYCON(K).EQ.3) KB=KB+1
      IF(LAYCON(K).EQ.2 .OR. LAYCON(K).EQ.3) KT=KT+1
      LOC=1+(K-1)*NIJ
      LOCB=1+(KB-1)*NIJ
      LOCT=1+(KT-1)*NIJ
C
C2B-----READ PRIMARY STORAGE COEFFICIENT INTO ARRAY SC1 IF TRANSIENT
      IF(ISS.EQ.0)CALL U2DREL(SC1(LOC),ANAME(1,1),NROW,NCOL,KK,IN,IOUT)
C
C2C-----READ TRANSMISSIVITY INTO ARRAY CC IF LAYER TYPE IS 0 OR 2
      IF(LAYCON(K).EQ.3 .OR. LAYCON(K).EQ.1) GO TO 100
      CALL U2DREL(CC(LOC),ANAME(1,2),NROW,NCOL,KK,IN,IOUT)
      GO TO 110
C
C2D-----READ HYDRAULIC CONDUCTIVITY(HY) AND BOTTOM ELEVATION(BOT)
C2D-----IF LAYER TYPE IS 1 OR 3
  100 CALL U2DREL(HY(LOCB),ANAME(1,3),NROW,NCOL,KK,IN,IOUT)
      CALL U2DREL(BOT(LOCB),ANAME(1,5),NROW,NCOL,KK,IN,IOUT)
C
C2E-----READ VERTICAL HYCOND/THICK INTO ARRAY CV IF NOT BOTTOM LAYER
C2E----- READ AS HYCOND/THICKNESS -- CONVERTED TO CONDUCTANCE LATER
  110 IF(K.EQ.NLAY) GO TO 120
      CALL U2DREL(CV(LOC),ANAME(1,4),NROW,NCOL,KK,IN,IOUT)
C
C2F-----READ SECONDARY STORAGE COEFFICIENT INTO ARRAY SC2 IF TRANSIENT
C2F-----AND LAYER TYPE IS 2 OR 3
  120 IF(LAYCON(K).NE.3 .AND. LAYCON(K).NE.2) GO TO 130
      IF(ISS.EQ.0)CALL U2DREL(SC2(LOCT),ANAME(1,7),NROW,NCOL,KK,IN,IOUT)
C
C2G-----READ TOP ELEVATION(TOP) IF LAYER TYPE IS 2 OR 3
      CALL U2DREL(TOP(LOCT),ANAME(1,6),NROW,NCOL,KK,IN,IOUT)
C
C2H-----READ WETDRY CODES IF LAYER TYPE IS 1 OR 3 AND WETTING
C2H-----CAPABILITY HAS BEEN INVOKED (IWDFLG NOT 0)
  130 IF(LAYCON(K).NE.3.AND.LAYCON(K).NE.1)GO TO 200
      IF(IWDFLG.EQ.0)GO TO 200
      CALL U2DREL(WETDRY(LOCB),ANAME(1,11),NROW,NCOL,KK,IN,IOUT)
  200 CONTINUE
C
C3------PREPARE AND CHECK BCF DATA
      CALL SBCF2N(HNEW,IBOUND,SC1,SC2,CR,CC,CV,HY,TRPY,DELR,DELC,ISS,
     1          NCOL,NROW,NLAY,IOUT,WETDRY,IWDFLG,CVWD)
C
C4------RETURN
      RETURN
      END
```

List of Variables for Module BCF2RP

Variable	Range	Definition
ANAME	Module	DIMENSION (6,11), Labels for printout of input arrays.
BOT	Package	DIMENSION (NODES), Elevation of the aquifer bottom. Although BOT is dimensioned to the size of the grid, space exists only for layers in which saturated thickness is calculated.
CC	Global	DIMENSION (NODES), Conductance in the column direction.
CR	Global	DIMENSION (NODES), Conductance in the row direction.
CV	Global	DIMENSION (NODES), Conductance in the vertical direction. Although CV is dimensioned to the size of the grid, space exists only for NLAY-1 layers.
CVWD	Package	DIMENSION (NODES), Conductance in the vertical direction. When a wet cell converts to dry, CV is set equal to 0. When a dry cell converts to wet, CV is set equal to CVWD. Although CVWD is dimensioned to the size of the grid, space exists only for NLAY-1 layers.
DELC	Global	DIMENSION (NROW), Cell dimension in the column direction. DELC(I) contains the width of row I.
DELR	Global	DIMENSION (NCOL), Cell dimension in the row direction. DELR(J) contains the width of column J.
HNEW	Global	DIMENSION (NODES), Most recent estimate of head in each cell.
HY	Package	DIMENSION (NODES), Hydraulic conductivity in the row direction for each cell. Although HY is dimensioned to the size of the grid, space exists only for layers in which saturated thickness is calculated.
IBOUND	Global	DIMENSION (NODES), Status of each cell: < 0, constant-head cell = 0, no-flow cell > 0, variable-head cell
IN	Package	Primary unit number from which input for this package will be read.
IOUT	Global	Primary unit number for all printed output.
ISS	Package	Steady-state flag. = 0, simulation is transient not 0, simulation is steady state.
IWDFLG	Package	Flag indicating if wetting capability is active.
K	Module	Index for layers.
KB	Module	Index for layer within BOT, HY, and WETDRY arrays.
KK	Module	Temporary variable set equal to K. KK is used as an actual argument in subroutine calls to avoid using the DO loop variable K as an argument, which causes problems with some compilers.
KT	Module	Index for layer within TOP and (if transient) SC2 arrays.
LAYCON	Package	DIMENSION (80) Layer-type code: 0 -- Layer is strictly confined. 1 -- Layer is strictly unconfined. 2 -- Layer is convertible between confined and unconfined (transmissivity is constant). 3 -- Layer is convertible between confined and unconfined (transmissivity varies).

LOC	Module	Pointer used to point to different layers in conductance arrays.
LOCB	Module	Pointer used to point to different layers in BOT, HY, and WETDRY arrays.
LOCT	Module	Pointer used to point to different layers in TOP and SC2 arrays.
NCOL	Global	Number of columns in the grid.
NIJ	Module	Number of cells in a layer.
NLAY	Global	Number of layers in the grid.
NODES	Global	Number of cells (nodes) in the finite-difference grid.
NROW	Global	Number of rows in the grid.
SC1	Package	DIMENSION (NODES), Primary storage capacity.
SC2	Package	DIMENSION (NODES), Secondary storage capacity. Although SC2 is dimensioned to the size of the grid, space exists only for layers that can convert between confined and unconfined.
TOP	Package	DIMENSION (NODES), Elevation of the aquifer top. Although TOP is dimensioned to the size of the grid, space exists only for layers that can convert between confined and unconfined.
TRPY	Package	DIMENSION (NLAY), Ratio of transmissivity (or hydraulic conductivity) in the column direction to transmissivity (or hydraulic conductivity) in the row direction.
WETDRY	Package	DIMENSION (NODES), Wetting threshold. Although WETDRY is dimensioned to the size of the grid, space exists only for layers that have the potential for converting from dry to wet.

Narrative for Module BCF2AD

At the beginning of each time step, this module sets head from the previous time step (HOLD) equal to the bottom elevation (BOT) for any cells that are dry and have the potential for converting to wet. This is necessary in order to compute correct storage terms for these cells.

1. Return if steady-state simulation or if the wetting capability is inactive.

2. Loop through all layers to check for dry cells that can become wet.

   (a) Keep track of the layer index for wettable layers (layers with layer-type code 1 or 3). Loop through the cells in wettable layers and skip non-wettable layers.

   (b) Skip cells that are currently wet (IBOUND not 0) or not wettable (WETDRY = 0).

   (c) Set HOLD = BOT.

3. RETURN.

LAYCON is a layer-type code (one for each layer).

0 - confined
1 - unconfined
2 - confined/unconfined but transmissivity is constant
3 - confined/unconfined

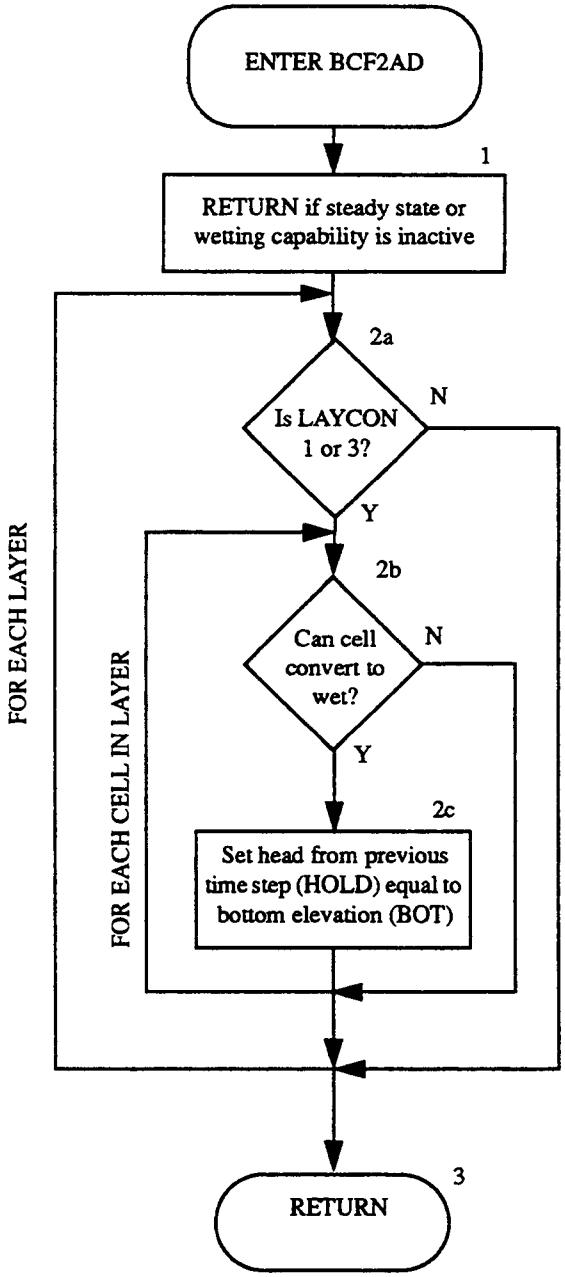


Figure 9. -- Flow chart for module BCF2AD.

```
      SUBROUTINE BCF2AD(IBOUND,HOLD,BOT,WETDRY,IWDFLG,ISS,
     1                  NCOL,NROW,NLAY)
C
C-----VERSION 1434 14MAY1991 BCF2AD
C     ******************************************************************
C     SET HOLD TO BOT WHENEVER A WETTABLE CELL IS DRY
C     ******************************************************************
C
C        SPECIFICATIONS:
C     ------------------------------------------------------------------
C
      DIMENSION IBOUND(NCOL,NROW,NLAY),HOLD(NCOL,NROW,NLAY),
     1          BOT(NCOL,NROW,NLAY),WETDRY(NCOL,NROW,NLAY)
C
      COMMON /FLWCOM/LAYCON(80)
C     ------------------------------------------------------------------
C
C1------RETURN IF STEADY STATE OR IF NOT USING WETTING CAPABILITY
      IF(IWDFLG.EQ.0 .OR. ISS.NE.0) RETURN
C
C2------LOOP THROUGH ALL LAYERS TO SET HOLD=BOT IF A WETTABLE CELL IS DRY
      KB=0
      DO 100 K=1,NLAY
C
C2A-----SKIP LAYERS THAT CANNOT CONVERT BETWEEN WET AND DRY
      IF(LAYCON(K).NE.3 .AND. LAYCON(K).NE.1) GO TO 100
      KB=KB+1
      DO 90 I=1,NROW
      DO 90 J=1,NCOL
C
C2B-----SKIP CELLS THAT ARE CURRENTLY WET OR ARE NOT WETTABLE
      IF(IBOUND(J,I,K).NE.0) GO TO 90
      IF(WETDRY(J,I,KB).EQ.0.) GO TO 90
C
C2C-----SET HOLD=BOT
      HOLD(J,I,K)=BOT(J,I,KB)
   90 CONTINUE
  100 CONTINUE
C
C3-----RETURN
      RETURN
      END
```

List of Variables for Module BCF2AD

BOT	Package	DIMENSION (NCOL,NROW,NLAY), Elevation of the aquifer bottom. Although BOT is dimensioned to the size of the grid, space exists only for layers for which saturated thickness is calculated.
HOLD	Global	DIMENSION (NCOL,NROW,NLAY), Head at the start of the current time step.
I	Module	Index for rows.
IBOUND	Global	DIMENSION (NCOL,NROW,NLAY), Status of each cell: < 0, constant-head cell = 0, no-flow cell > 0, variable-head cell
ISS	Package	Steady-state flag. = 0, simulation is transient not 0, simulation is steady state.
IWDFLG	Package	Flag indicating if wetting capability is active.
J	Module	Index for columns.
K	Module	Index for layers.
KB	Module	Index for layer within WETDRY array.
LAYCON	Package	DIMENSION (80) Layer-type code: 0 -- Layer is strictly confined. 1 -- Layer is strictly unconfined. 2 -- Layer is convertible between confined and unconfined (saturated thickness is constant). 3 -- Layer is convertible between confined and unconfined (saturated thickness varies).
NCOL	Global	Number of columns in the grid.
NLAY	Global	Number of layers in the grid.
NROW	Global	Number of rows in the grid.
WETDRY	Package	DIMENSION (NCOL,NROW,NLAY), Wetting threshold. Although WETDRY is dimensioned to the size of the grid, space exists only for layers that have the potential for converting from dry to wet.

Narrative for Module BCF2FM

This module calculates branch conductances if they are not constant throughout the simulation, adds storage terms to the HCOF and RHS accumulators, and adds terms to RHS that correct for overestimation of flow down into partially saturated cells.

1. For each layer in which transmissivity varies with head (LAYCON = 1 or 3), call module SBCF2H to calculate branch conductances.

2. If the simulation is transient, calculate storage terms for each layer. Steps 3-5 comprise the storage calculations. If the simulation is steady state, go to Step 6.

3. Determine if the layer is convertible between confined and unconfined.

4. If the layer is not convertible (LAYCON = 0 or 1), use primary storage capacity to calculate storage terms and add them to the right hand side (RHS) and h-coefficient (HCOF) terms.

5. If the layer is convertible (LAYCON = 2 or 3), then use head at the beginning of the time step (HOLD) to determine the storage factor at the beginning of the time step (SOLD). Use the latest estimate of head at the end of the time step (HNEW) to determine the storage factor at the end of the time step (SNEW). Note that the term "storage factor" as used in BCF2FM refers to storage capacity divided by time step length. Use SOLD and SNEW to calculate storage terms and add them to RHS and HCOF.

6. For each layer, determine if correction terms are needed for vertical flow down into a partially saturated layer (Steps 7-8).

7. If the layer is partially saturated and there is flow from above, calculate the vertical flow correction terms and add them to RHS (program comment C7D). This vertical flow correction is incorporated jointly in RHS and HCOF in the original model (McDonald and Harbaugh, 1988, p. 5-23 and 5-59). In BCF2, the correction is being made through RHS alone in order to avoid the possibility of producing a matrix that is not diagonally dominant. A non-diagonally dominant matrix causes problems with the Preconditioned Conjugate-Gradient Package, version 2 (Hill, 1990, p. 15). Implementing the correction through RHS alone is essentially equivalent to the former method. The method using both RHS and HCOF is an implicit representation of the correction while the method using only RHS is an explicit representation.

8. If this layer is not the bottom layer and the layer below is partially saturated, calculate the correction terms and add them to RHS.

9. RETURN.

LAYCON is a layer-type code (one for each layer).

0 - confined
1 - unconfined
2 - confined/unconfined but transmissivity is constant
3 - confined/unconfined

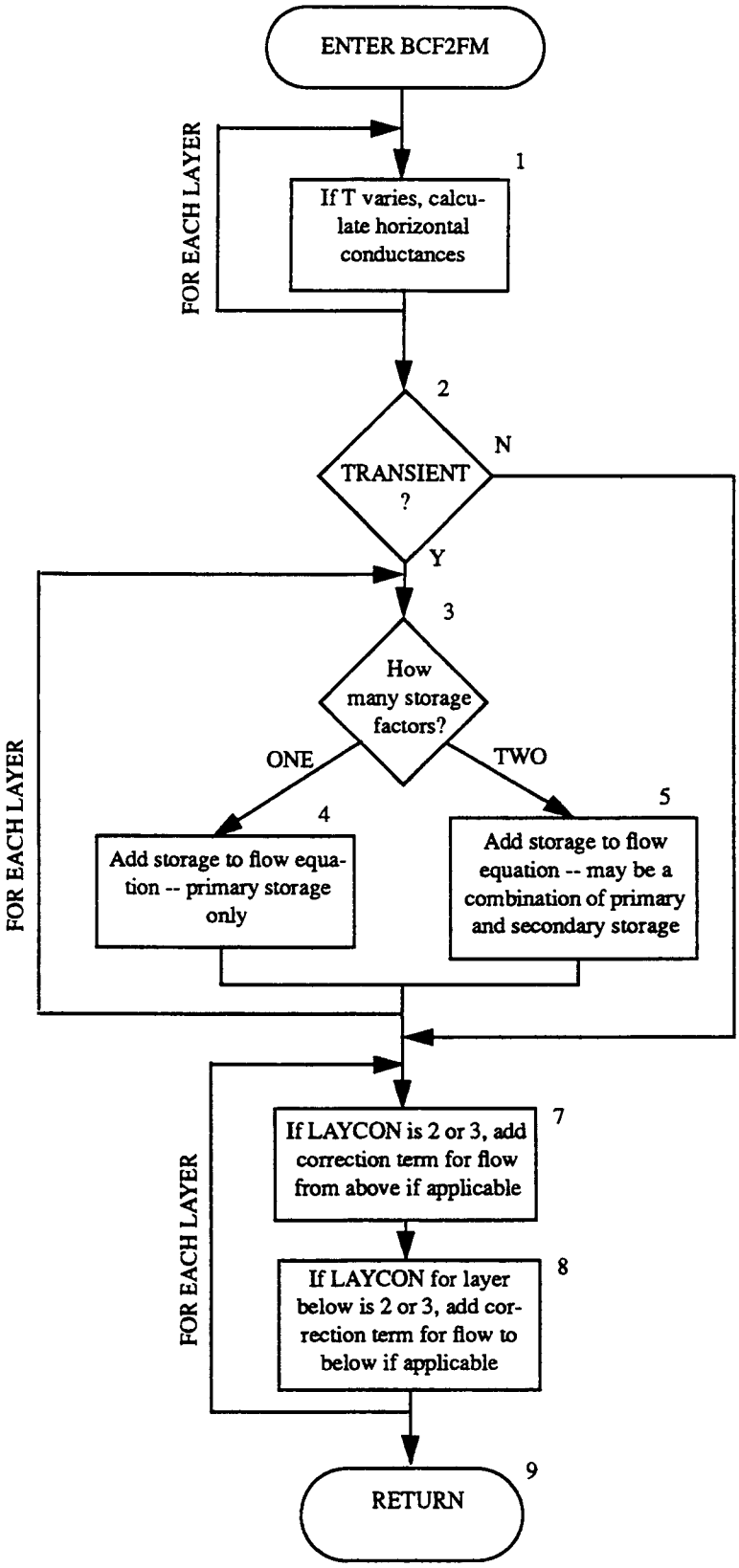


Figure 10. -- Flow chart for module BCF2FM.

```
      SUBROUTINE BCF2FM(HCOF,RHS,HOLD,SC1,HNEW,IBOUND,CR,CC,CV,HY,TRPY,
     1                  BOT,TOP,SC2,DELR,DELC,DELT,ISS,KITER,KSTP,KPER,
     2                  NCOL,NROW,NLAY,IOUT,WETDRY,IWDFLG,CVWD,
     3                  WETFCT,IWETIT,IHDWET,HDRY)
C-----VERSION 1104 5MAY1991 BCF2FM
C     ******************************************************************
C     ADD LEAKAGE CORRECTION AND STORAGE TO HCOF AND RHS, AND CALCULATE
C     CONDUCTANCE AS REQUIRED, VERSION 2
C     ******************************************************************
C
C        SPECIFICATIONS:
C     ------------------------------------------------------------------
      DOUBLE PRECISION HNEW
C
      DIMENSION HCOF(NCOL,NROW,NLAY),RHS(NCOL,NROW,NLAY),
     1    HOLD(NCOL,NROW,NLAY),SC1(NCOL,NROW,NLAY),HNEW(NCOL,NROW,NLAY),
     2    IBOUND(NCOL,NROW,NLAY),CR(NCOL,NROW,NLAY),
     3    CC(NCOL,NROW,NLAY),CV(NCOL,NROW,NLAY),HY(NCOL,NROW,NLAY),
     4    TRPY(NLAY),BOT(NCOL,NROW,NLAY),TOP(NCOL,NROW,NLAY),DELR(NCOL),
     5    DELC(NROW),SC2(NCOL,NROW,NLAY),WETDRY(NCOL,NROW,NLAY),
     6    CVWD(NCOL,NROW,NLAY)
C
      COMMON /FLWCOM/LAYCON(80)
C     ------------------------------------------------------------------
      KB=0
      KT=0
C
C1------FOR EACH LAYER: IF T VARIES CALCULATE HORIZONTAL CONDUCTANCES
      DO 100 K=1,NLAY
      KK=K
      IF(LAYCON(K).EQ.3 .OR. LAYCON(K).EQ.2) KT=KT+1
C
C1A-----IF LAYER TYPE IS NOT 1 OR 3 THEN SKIP THIS LAYER.
      IF(LAYCON(K).NE.3 .AND. LAYCON(K).NE.1) GO TO 100
      KB=KB+1
C
C1B-----FOR LAYER TYPES 1 & 3 CALL SBCF2H TO CALCULATE
C1B-----HORIZONTAL CONDUCTANCES.
      CALL SBCF2H(HNEW,IBOUND,CR,CC,CV,HY,TRPY,DELR,DELC,BOT,TOP,
     1   KK,KB,KT,KITER,KSTP,KPER,NCOL,NROW,NLAY,IOUT,WETDRY,IWDFLG,
     2   CVWD,WETFCT,IWETIT,IHDWET,HDRY)
  100 CONTINUE
C
C2------IF THE SIMULATION IS TRANSIENT ADD STORAGE TO HCOF AND RHS
      IF(ISS.NE.0) GO TO 201
      TLED=1./DELT
      KT=0
      DO 200 K=1,NLAY
C
C3------SEE IF THIS LAYER IS CONVERTIBLE OR NON-CONVERTIBLE.
      IF(LAYCON(K).EQ.3 .OR. LAYCON(K).EQ.2) GO TO 150
```

```
C4------NON-CONVERTIBLE LAYER, SO USE PRIMARY STORAGE
      DO 140 I=1,NROW
      DO 140 J=1,NCOL
      IF(IBOUND(J,I,K).LE.0) GO TO 140
      RHO=SC1(J,I,K)*TLED
      HCOF(J,I,K)=HCOF(J,I,K)-RHO
      RHS(J,I,K)=RHS(J,I,K)-RHO*HOLD(J,I,K)
  140 CONTINUE
      GO TO 200
C
C5------A CONVERTIBLE LAYER, SO CHECK OLD AND NEW HEADS TO DETERMINE
C5------WHEN TO USE PRIMARY AND SECONDARY STORAGE
  150 KT=KT+1
      DO 180 I=1,NROW
      DO 180 J=1,NCOL
C
C5A-----IF THE CELL IS EXTERNAL THEN SKIP IT.
      IF(IBOUND(J,I,K).LE.0) GO TO 180
      TP=TOP(J,I,KT)
      RHO2=SC2(J,I,KT)*TLED
      RHO1=SC1(J,I,K)*TLED
C
C5B-----FIND STORAGE FACTOR AT START OF TIME STEP.
      SOLD=RHO2
      IF(HOLD(J,I,K).GT.TP) SOLD=RHO1
C
C5C-----FIND STORAGE FACTOR AT END OF TIME STEP.
      HTMP=HNEW(J,I,K)
      SNEW=RHO2
      IF(HTMP.GT.TP) SNEW=RHO1
C
C5D-----ADD STORAGE TERMS TO RHS AND HCOF.
      HCOF(J,I,K)=HCOF(J,I,K)-SNEW
      RHS(J,I,K)=RHS(J,I,K) - SOLD*(HOLD(J,I,K)-TP) - SNEW*TP
C
  180 CONTINUE
C
  200 CONTINUE
C
C6------FOR EACH LAYER DETERMINE IF CORRECTION TERMS ARE NEEDED FOR
C6------FLOW DOWN INTO PARTIALLY SATURATED LAYERS.
  201 KT=0
      DO 300 K=1,NLAY
C
C7------SEE IF CORRECTION IS NEEDED FOR LEAKAGE FROM ABOVE.
      IF(LAYCON(K).NE.3 .AND. LAYCON(K).NE.2) GO TO 250
      KT=KT+1
      IF(K.EQ.1) GO TO 250
C
C7A-----FOR EACH CELL MAKE THE CORRECTION IF NEEDED.
      DO 220 I=1,NROW
      DO 220 J=1,NCOL
C
```

```
C7B-----IF THE CELL IS EXTERNAL(IBOUND<=0) THEN SKIP IT.
      IF(IBOUND(J,I,K).LE.0) GO TO 220
      HTMP=HNEW(J,I,K)
C
C7C-----IF HEAD IS ABOVE TOP THEN CORRECTION NOT NEEDED
      IF(HTMP.GE.TOP(J,I,KT)) GO TO 220
C
C7D-----WITH HEAD BELOW TOP ADD CORRECTION TERMS TO RHS.
      RHS(J,I,K)=RHS(J,I,K) + CV(J,I,K-1)*(TOP(J,I,KT)-HTMP)
  220 CONTINUE
C
C8------SEE IF THIS LAYER MAY NEED CORRECTION FOR LEAKAGE TO BELOW.
  250 IF(K.EQ.NLAY) GO TO 300
      IF(LAYCON(K+1).NE.3 .AND. LAYCON(K+1).NE.2) GO TO 300
      KTT=KT+1
C
C8A-----FOR EACH CELL MAKE THE CORRECTION IF NEEDED.
      DO 280 I=1,NROW
      DO 280 J=1,NCOL
C
C8B-----IF CELL IS EXTERNAL (IBOUND<=0) THEN SKIP IT.
      IF(IBOUND(J,I,K).LE.0) GO TO 280
C
C8C-----IF HEAD IN THE LOWER CELL IS LESS THAN TOP ADD CORRECTION
C8C-----TERM TO RHS.
      HTMP=HNEW(J,I,K+1)
      IF(HTMP.LT.TOP(J,I,KTT)) RHS(J,I,K)=RHS(J,I,K)
     1                         - CV(J,I,K)*(TOP(J,I,KTT)-HTMP)
  280 CONTINUE
  300 CONTINUE
C
C9------RETURN
      RETURN
      END
```

List of Variables for Module BCF2FM

Variable	Range	Definition
BOT	Package	DIMENSION (NCOL,NROW,NLAY), Elevation of the aquifer bottom. Although BOT is dimensioned to the size of the grid, space exists only for layers for which saturated thickness is calculated.
CC	Global	DIMENSION (NCOL,NROW,NLAY), Conductance in the column direction.
CR	Global	DIMENSION (NCOL,NROW,NLAY), Conductance in the row direction.
CV	Global	DIMENSION (NCOL,NROW,NLAY), Conductance in the vertical direction. Although CV is dimensioned to the size of the grid, space exists only for NLAY-1 layers.
CVWD	Package	DIMENSION (NCOL,NROW,NLAY), Conductance in the vertical direction. When a wet cell converts to dry, CV is set equal to 0. When a dry cell converts to wet, CV is set equal to CVWD. Although CVWD is dimensioned to the size of the grid, space exists only for NLAY-1 layers.
DELC	Global	DIMENSION (NROW), Cell dimension in the column direction. DELC(I) contains the width of row I.
DELR	Global	DIMENSION (NCOL), Cell dimension in the row direction. DELR(J) contains the width of column J.
DELT	GLOBAL	Length of the current time step.
HCOF	Global	DIMENSION (NCOL,NROW,NLAY), Coefficient of head in the finite-difference equation.
HDRY	Package	When a cell converts to dry, HNEW is set equal to HDRY.
HNEW	Global	DIMENSION (NCOL,NROW,NLAY), Most recent estimate of head in each cell.
HOLD	Global	DIMENSION (NCOL,NROW,NLAY), Head at the start of the current time step.
HTMP	Module	Temporary single precision equivalent of HNEW(J,I,K).
HY	Package	DIMENSION (NCOL,NROW,NLAY), Hydraulic conductivity in the row direction for each cell. Although HY is dimensioned to the size of the grid, space exists only for layers in which saturated thickness is calculated.
I	Module	Index for rows.
IBOUND	Global	DIMENSION (NCOL,NROW,NLAY), Status of each cell: < 0, constant-head cell = 0, no-flow cell > 0, variable-head cell
IHDWET	Package	Flag indicating which equation to use for calculating the head at a cell that has just converted from dry to wet.
IOUT	Global	Primary unit number for all printed output.
ISS	Package	Steady-state flag. = 0, simulation is transient not 0, simulation is steady state.
IWDFLG	Package	Flag indicating if wetting capability is active.
IWETIT	Package	Iteration interval for attempting to wet cells.
J	Module	Index for columns.
K	Module	Index for layers.

KB	Module	Index for layer within BOT, HY, and WETDRY arrays.
KITER	Global	Iteration counter. KITER=1 at the start of each time step.
KK	Module	Temporary variable set equal to K. KK is used as an actual argument in subroutine calls to avoid using the DO loop variable K as an argument, which causes problems with some compilers.
KPER	Global	Stress period counter.
KSTP	Global	Time step counter. KSTP=1 at the start of each stress period.
KT	Module	Index for layer within TOP and (if transient) SC2 arrays.
KTT	Module	Index to TOP array of layer immediately below layer K.
LAYCON	Package	DIMENSION (80) Layer-type code: 0 -- Layer is strictly confined. 1 -- Layer is strictly unconfined. 2 -- Layer is convertible between confined and unconfined (transmissivity is constant). 3 -- Layer is convertible between confined and unconfined (transmissivity varies).
NCOL	Global	Number of columns in the grid.
NLAY	Global	Number of layers in the grid.
NROW	Global	Number of rows in the grid.
RHO	Module	Storage capacity divided by time step length for strictly confined or strictly unconfined layers.
RHO1	Module	Confined storage capacity divided by time step length for convertible layers.
RHO2	Module	Unconfined storage capacity divided by time step length for convertible layers.
RHS	Global	DIMENSION (NCOL,NROW,NLAY), Right-hand side of finite-difference equation.
SC1	Package	DIMENSION (NCOL,NROW,NLAY), Primary storage capacity.
SC2	Package	DIMENSION (NCOL,NROW,NLAY), Secondary storage capacity. Although SC2 is dimensioned to the size of the grid, space exists only for cells that can convert between confined and unconfined.
SNEW	Module	Storage capacity divided by time step length at the end of the time step for convertible layers.
SOLD	Module	Storage capacity divided by time step length at the start of the time step for convertible layers.
TLED	Module	1/DELT.
TOP	Package	DIMENSION (NCOL,NROW,NLAY), Elevation of the aquifer top. Although TOP is dimensioned to the size of the grid, space exists only for cells that can convert between confined and unconfined.
TP	Module	Temporary equivalent of TOP(J,I,KT).
TRPY	Package	DIMENSION (NLAY), Ratio of transmissivity (or hydraulic conductivity) in the column direction to transmissivity (or hydraulic conductivity) in the row direction.
WETDRY	Package	DIMENSION (NCOL,NROW,NLAY), Wetting threshold. Although WETDRY is dimensioned to the size of the grid, space exists only for layers that have the potential for converting from dry to wet.

WETFCT	Package	A factor that is included in the calculation of the head that is established at a cell that has just converted from dry to wet.

Narrative for Module SBCF2N

This module insures that the transmissive properties of each cell agree with the codes specified in the boundary array (IBOUND). In addition, this module calculates parameters that remain constant throughout a simulation: horizontal-branch conductance in layers where transmissivity is constant, vertical conductance, and storage capacity for layers that cannot convert between confined and unconfined.

The array IBOUND indicates the status of every cell in the grid as shown:

IBOUND Value	Cell Status
zero	inactive
positive	variable head
negative	constant head

Module SBCF2N sets vertical conductance (CV) and transmissivity equal to zero for cells in which IBOUND is 0, and it sets IBOUND equal to 0 at cells in which all transmissive parameters are 0. IBOUND values are read by the BAS1RP module. Module SBCF2N is called by module BCF2RP, and SBCF2N calls module SBCF1C to calculate conductance for layers that have constant transmissivity.

1. Multiply vertical leakance (VCONT) by cell area to obtain vertical conductance.

2. If the wetting capability is activated, save vertical conductance (CV) as variable CVWD. At a dry cell, CV will be set to 0; at a wet cell, CV will be set to CVWD.

3. If IBOUND is 0, set vertical conductance and transmissivity to 0.

4. Insure that each active cell (variable head or constant head) has at least one non-zero transmissive parameter. Convert any cell that does not have at least one non-zero transmissive parameter to no flow. The specific tests for non-zero transmissive parameters depends on the layer-type code and if the wetting capability is active. Loop through each layer to determine the layer type.

   (a) The layer has constant transmissivity. Loop through each cell in the layer looking for cells in which transmissive parameters are all zero; transmissive parameters are horizontal transmissivity and vertical conductance. If all of these are 0 for an active cell, set IBOUND to 0, set HNEW to 888.88, and print a message.

   (b) The layer has variable transmissivity. Loop through each cell in the layer looking for cells in which transmissive parameters are all zero; transmissive parameters are horizontal hydraulic conductivity and vertical conductance.

(1) If the wetting capability is active for a cell, check to see if CVWD above or below is non zero. The test for non-zero CVWD is performed even if the cell is initially dry because the cell could convert to wet at a later time, and this would be senseless if all transmissive terms were zero.

(2) For active cells in which the wetting capability is not active, check to see if CV above or below is non zero.

(3) For cells in which CV or CVWD above and below are zero, check to see if hydraulic conductivity is zero.

(4) Hydraulic conductivity and vertical conductance above and below are all zero, so convert the cell to no flow. Set IBOUND to 0, set HNEW to 888.88, and print a message. Also, if the wetting capability is active, set WETDRY to 0 so that the cell is never allowed to become wet.

5. Call module SBCF1C to calculate conductance for layers that have constant transmissivity.

6. If a transient simulation, loop through each layer and calculate storage capacity as the product of storage coefficient and cell area.

   (a) Calculate primary storage capacity for all layers.

   (b) For layers that can convert between confined and unconfined, calculate secondary storage capacity.

7. RETURN.

LAYCON is a layer-type code (one for each layer).

0 - confined
1 - unconfined
2 - confined/unconfined but transmissivity is constant
3 - confined/unconfined

VCONT is vertical hydraulic conductivity divided by the thickness between nodes in a layer and the layer below.

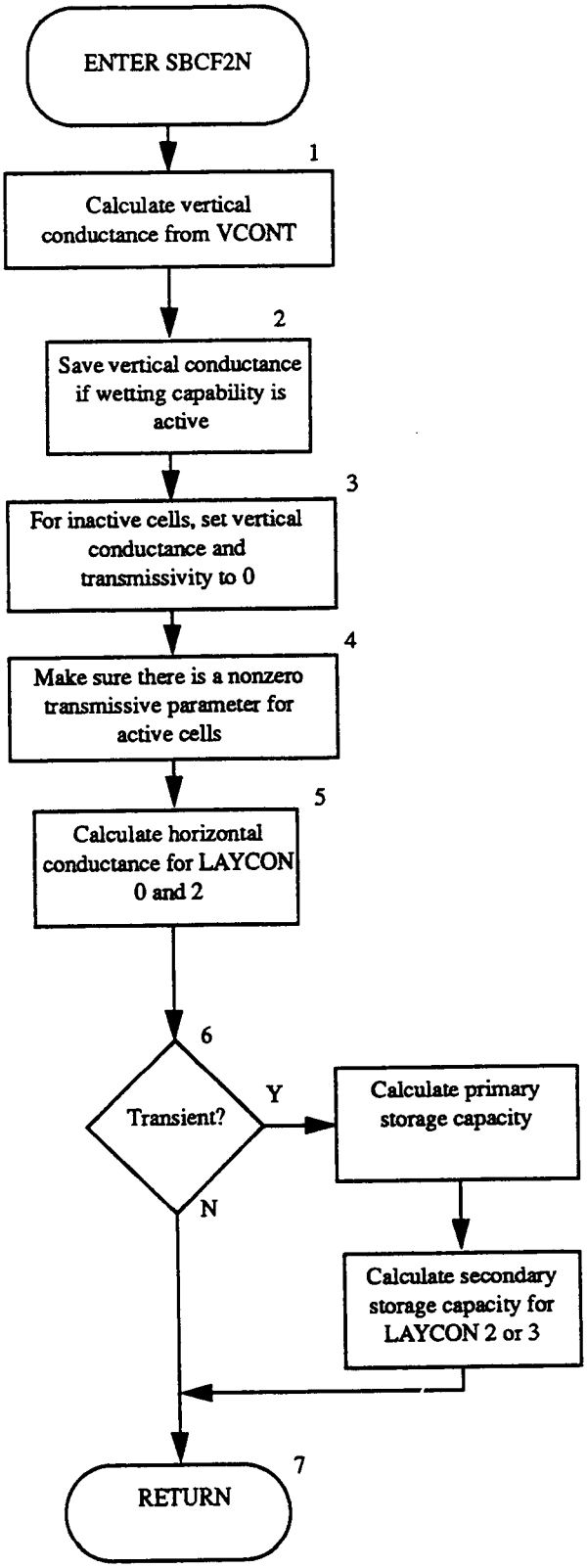


Figure 11. -- Flow chart for module SBCF2N.

```
      SUBROUTINE SBCF2N(HNEW,IBOUND,SC1,SC2,CR,CC,CV,HY,TRPY,DELR,DELC,
     1    ISS,NCOL,NROW,NLAY,IOUT,WETDRY,IWDFLG,CVWD)
C
C-----VERSION 1107 5MAY1991 SBCF2N
C
C     ******************************************************************
C     INITIALIZE AND CHECK BCF DATA, VERSION 2
C     ******************************************************************
C
C        SPECIFICATIONS:
C     ------------------------------------------------------------------
C
      DOUBLE PRECISION HNEW,HCNV
C
      DIMENSION HNEW(NCOL,NROW,NLAY),IBOUND(NCOL,NROW,NLAY)
     1    ,SC1(NCOL,NROW,NLAY),CR(NCOL,NROW,NLAY)
     2    ,CC(NCOL,NROW,NLAY),CV(NCOL,NROW,NLAY)
     3    ,HY(NCOL,NROW,NLAY),TRPY(NLAY),DELR(NCOL),DELC(NROW)
     4    ,SC2(NCOL,NROW,NLAY),WETDRY(NCOL,NROW,NLAY)
     5    ,CVWD(NCOL,NROW,NLAY)
C
      COMMON /FLWCOM/LAYCON(80)
C     ------------------------------------------------------------------
C
C1------MULTIPLY VERTICAL LEAKANCE BY AREA TO MAKE CONDUCTANCE
      IF(NLAY.EQ.1) GO TO 20
      K1=NLAY-1
      DO 10 K=1,K1
      DO 10 I=1,NROW
      DO 10 J=1,NCOL
      CV(J,I,K)=CV(J,I,K)*DELR(J)*DELC(I)
   10 CONTINUE
C
C2------IF WETTING CAPABILITY IS ACTIVATED, SAVE CV IN CVWD FOR USE WHEN
C2------WETTING CELLS.
      IF(IWDFLG.EQ.0) GO TO 20
      DO 15 K=1,K1
      DO 15 I=1,NROW
      DO 15 J=1,NCOL
      CVWD(J,I,K)=CV(J,I,K)
   15 CONTINUE
C
C3------IF IBOUND=0, SET CV=0 AND CC=0.
   20 DO 30 K=1,NLAY
      DO 30 I=1,NROW
      DO 30 J=1,NCOL
      IF(IBOUND(J,I,K).NE.0) GO TO 30
      IF(K.NE.NLAY) CV(J,I,K)=0.
      IF(K.NE.1) CV(J,I,K-1)=0.
      CC(J,I,K)=0.
   30 CONTINUE
C
C4------INSURE THAT EACH ACTIVE CELL HAS AT LEAST ONE NON-ZERO
```

```
C4------TRANSMISSIVE PARAMETER.
      HCNV=888.88
      KB=0
      DO 60 K=1,NLAY
      IF(LAYCON(K).EQ.1 .OR. LAYCON(K).EQ.3) GO TO 50
C
C4A-----WHEN LAYER TYPE IS 0 OR 2, TRANSMISSIVITY OR CV MUST BE NONZERO
      DO 45 I=1,NROW
      DO 45 J=1,NCOL
      IF(IBOUND(J,I,K).EQ.0) GO TO 45
      IF(CC(J,I,K).NE.0.) GO TO 45
      IF(K.EQ.NLAY) GO TO 41
      IF(CV(J,I,K).NE.0.) GO TO 45
   41 IF(K.EQ.1) GO TO 42
      IF(CV(J,I,K-1).NE.0.) GO TO 45
   42 IBOUND(J,I,K)=0
      HNEW(J,I,K)=HCNV
      WRITE(IOUT,43) K,I,J
   43 FORMAT(1X,'NODE (LAYER,ROW,COL)',3I4,
     1      ' ELIMINATED BECAUSE ALL CONDUCTANCES TO NODE ARE 0')
   45 CONTINUE
      GO TO 60
C
C4B-----WHEN LAYER TYPE IS 1 OR 3, HY OR CV MUST BE NONZERO
   50 KB=KB+1
      DO 59 I=1,NROW
      DO 59 J=1,NCOL
C
C4B1----IF WETTING CAPABILITY IS ACTIVE, CHECK CVWD
      IF(IWDFLG.EQ.0) GO TO 55
      IF(WETDRY(J,I,KB).EQ.0.) GO TO 55
      IF(K.EQ.NLAY) GO TO 51
      IF(CVWD(J,I,K).NE.0.) GO TO 59
   51 IF(K.EQ.1) GO TO 57
      IF(CV(J,I,K-1).NE.0.) GO TO 59
C
C4B2----WETTING CAPABILITY IS INACTIVE, SO CHECK CV AT ACTIVE CELLS
   55 IF(IBOUND(J,I,K).EQ.0) GO TO 59
      IF(K.EQ.NLAY) GO TO 56
      IF(CV(J,I,K).NE.0.) GO TO 59
   56 IF(K.EQ.1) GO TO 57
      IF(CV(J,I,K-1).NE.0.) GO TO 59
C
C4B3----CHECK HYDRAULIC CONDUCTIVITY
   57 IF(HY(J,I,KB).NE.0.) GO TO 59
C
C4B4----HY AND CV ARE ALL 0, SO CONVERT CELL TO NO FLOW
      IBOUND(J,I,K)=0
      HNEW(J,I,K)=HCNV
      IF(IWDFLG.NE.0) WETDRY(J,I,KB)=0.
      WRITE(IOUT,43) K,I,J
   59 CONTINUE
   60 CONTINUE
C
```

```
C5------CALCULATE HOR. CONDUCTANCE(CR AND CC) FOR CONSTANT T LAYERS
      DO 70 K=1,NLAY
      KK=K
      IF(LAYCON(K).EQ.3 .OR. LAYCON(K).EQ.1) GO TO 70
      CALL SBCF1C(CR,CC,TRPY,DELR,DELC,KK,NCOL,NROW,NLAY)
   70 CONTINUE
C
C6------IF TRANSIENT, LOOP THROUGH LAYERS AND CALCULATE STORAGE CAPACITY
      IF(ISS.NE.0) GO TO 100
      KT=0
      DO 90 K=1,NLAY
C
C6A-----MULTIPLY PRIMARY STORAGE COEFFICIENT BY DELR & DELC TO GET
C6A-----PRIMARY STORAGE CAPACITY
      DO 80 I=1,NROW
      DO 80 J=1,NCOL
      SC1(J,I,K)=SC1(J,I,K)*DELR(J)*DELC(I)
   80 CONTINUE
C
C6B-----IF LAYER IS CONF/UNCONF MULTIPLY SECONDARY STORAGE COEFFICIENT
C6B-----BY DELR AND DELC TO GET SECONDARY STORAGE CAPACITY(SC2).
      IF(LAYCON(K).NE.3 .AND. LAYCON(K).NE.2) GO TO 90
      KT=KT+1
      DO 85 I=1,NROW
      DO 85 J=1,NCOL
      SC2(J,I,KT)=SC2(J,I,KT)*DELR(J)*DELC(I)
   85 CONTINUE
   90 CONTINUE
C
C7------RETURN
  100 RETURN
      END
```

List of Variables for Module SBCF2N

Variable	Range	Definition
CC	Global	DIMENSION (NCOL,NROW,NLAY), Conductance in the column direction.
CR	Global	DIMENSION (NCOL,NROW,NLAY), Conductance in the row direction.
CV	Global	DIMENSION (NCOL,NROW,NLAY), Conductance in the vertical direction. Although CV is dimensioned to the size of the grid, space exists only for NLAY-1 layers.
CVWD	Package	DIMENSION (NCOL,NROW,NLAY), Conductance in the vertical direction. When a wet cell converts to dry, CV is set equal to 0. When a dry cell converts to wet, CV is set equal to CVWD. Although CVWD is dimensioned to the size of the grid, space exists only for NLAY-1 layers.
DELC	Global	DIMENSION (NROW), Cell dimension in the column direction. DELC(I) contains the width of row I.
DELR	Global	DIMENSION (NCOL), Cell dimension in the row direction. DELR(J) contains the width of column J.
HCNV	Module	Value of head used to indicate that a cell has been converted to no flow because all conductances to that cell are 0.
HNEW	Global	DIMENSION (NCOL,NROW,NLAY), Most recent estimate of head in each cell.
HY	Package	DIMENSION (NCOL,NROW,NLAY), Hydraulic conductivity in the row direction for each cell. Although HY is dimensioned to the size of the grid, space exists only for layers in which saturated thickness is calculated.
I	Module	Index for rows.
IBOUND	Global	DIMENSION (NCOL,NROW,NLAY), Status of each cell: < 0, constant-head cell = 0, no-flow cell > 0, variable-head cell
IOUT	Global	Primary unit number for all printed output.
ISS	Package	Steady-state flag. = 0, simulation is transient not 0, simulation is steady state.
IWDFLG	Package	Flag indicating if wetting capability is active.
J	Module	Index for columns.
K	Module	Index for layers.
K1	Module	NLAY-1
KB	Module	Index for layer within HY and WETDRY arrays.
KK	Module	Temporary variable set equal to K. KK is used as an actual argument in subroutine calls to avoid using the DO loop variable K as an argument, which causes problems for some compilers.
KT	Module	Index for layer within SC2 array.

LAYCON	Package	DIMENSION (80) Layer-type code: 0 -- Layer is strictly confined. 1 -- Layer is strictly unconfined. 2 -- Layer is convertible between confined and unconfined (transmissivity is constant). 3 -- Layer is convertible between confined and unconfined (transmissivity varies).
NCOL	Global	Number of columns in the grid.
NLAY	Global	Number of layers in the grid.
NROW	Global	Number of rows in the grid.
SC1	Package	DIMENSION (NCOL,NROW,NLAY), Primary storage capacity.
SC2	Package	DIMENSION (NCOL,NROW,NLAY), Secondary storage capacity. Although SC2 is dimensioned to the size of the grid, space exists only for cells that can convert between confined and unconfined.
TRPY	Package	DIMENSION (NLAY), Ratio of transmissivity (or hydraulic conductivity) in the column direction to transmissivity (or hydraulic conductivity) in the row direction.
WETDRY	Package	DIMENSION (NCOL,NROW,NLAY), Wetting threshold. Although WETDRY is dimensioned to the size of the grid, space exists only for layers that have the potential for converting from dry to wet.

Narrative for Module SBCF2H

Module SBCF2H calculates the horizontal-branch conductances for a layer in which the transmissivity is a function of head. It does this by calculating transmissivity for each cell as saturated thickness times hydraulic conductivity. Then it calls module SBCF1C to calculate the conductances from transmissivity of cells. As part of saturated thickness calculations, module SBCF2H determines if dry cells should become wet or if wet cells should become dry. Module BCF2FM calls SBCF2H for each type 1 or type 3 layer at each iteration.

1. Loop through each cell in the layer, and calculate transmissivity at each active cell. Before starting, initialize several variables. NCNVRT keeps track of the number of cells that have converted between wet and dry since the last line was printed telling which cells have converted. A line of output may list up to 8 cell conversions. IHDCNV controls the printing of the cell conversion heading; this heading is printed the first time a line of cell conversions is printed. ITFLG is 0 if the attempt to wet cells should be made this iteration, which means that the wetting capability is active and the current iteration is an even multiple of the wetting iteration interval (IWETIT).

2. If the cell is active, go to the saturated thickness calculation.

3. When the cell is inactive, check for the possibility of wetting. A cell has the potential for converting to wet only if ITFLG is zero. Also, the wetting threshold must be non zero. If these conditions are not met, go to the code that sets transmissivity to zero. If these conditions are met, calculate the turn-on threshold, TURNON, and continue the process of determining whether the cell should be converted to wet.

   (a) Check the cell below the inactive cell to see if it is variable head and if the wetting threshold has been reached. If so, go to the code that converts the cell to wet. Note that a constant head cell cannot cause another cell to become wet.

   (b) If the wetting threshold is negative, do not make any further attempts to wet the cell; the cell remains dry, and transmissivity will be set to zero. If the wetting threshold is positive, check the four horizontal neighbors to see if the turn-on threshold is reached. Neighboring cells that have converted to wet during the current iteration are not allowed to cause a cell to convert to wet. This is implemented by temporarily setting IBOUND to 30000 in Step 4 when a cell becomes wet. That is, neighboring cells having IBOUND of 30000 are not checked to see if the turn-on threshold is reached. If the turn-on threshold is reached, go to the code that converts the cell to wet.

   (c) The cell will stay dry. Set transmissivity to 0, and go to the end of the cell loop.

4. The cell becomes wet. Set initial head according to the flag IHDWET, set vertical conductance to CVWD, and set IBOUND to 30000. IBOUND will later be set to 1, but it is momentarily set to a high value as a flag to indicate cells that have been converted to wet this iteration. The flag is used in Step 3b to prevent a newly converted cell from causing other cells to convert. This could otherwise cause a chain reaction.

   (a) Save the information about which cell has converted to wet so that it can be printed. When eight conversions between wet and dry have been accumulated, print a line listing those cells.

5. Calculate saturated thickness. If LAYCON is 3, saturated thickness has a maximum value of TOP-BOT.

6. Check to see if saturated thickness is less than 0.

   (a) Saturated thickness is greater than 0; calculate transmissivity as saturated thickness times hydraulic conductivity. Store transmissivity temporarily in CC. Go to the end of the cell loop.

   (b) Saturated thickness is less than or equal to 0; convert the cell to no flow and save information flagging which cell has converted to dry. When eight conversions between wet and dry have been accumulated, print a line listing those cells. If the cell is not constant head, set IBOUND, CC (transmissivity), and CV to 0. If the cell is constant head, stop the simulation.

7. Transmissivity has been calculated for all cells. If there is a partially filled line of output of cell conversions, print the line.

8. Convert IBOUND values of 30000 to 1.

9. Call module SBCF1C to calculate conductance from transmissivity.

10. RETURN.

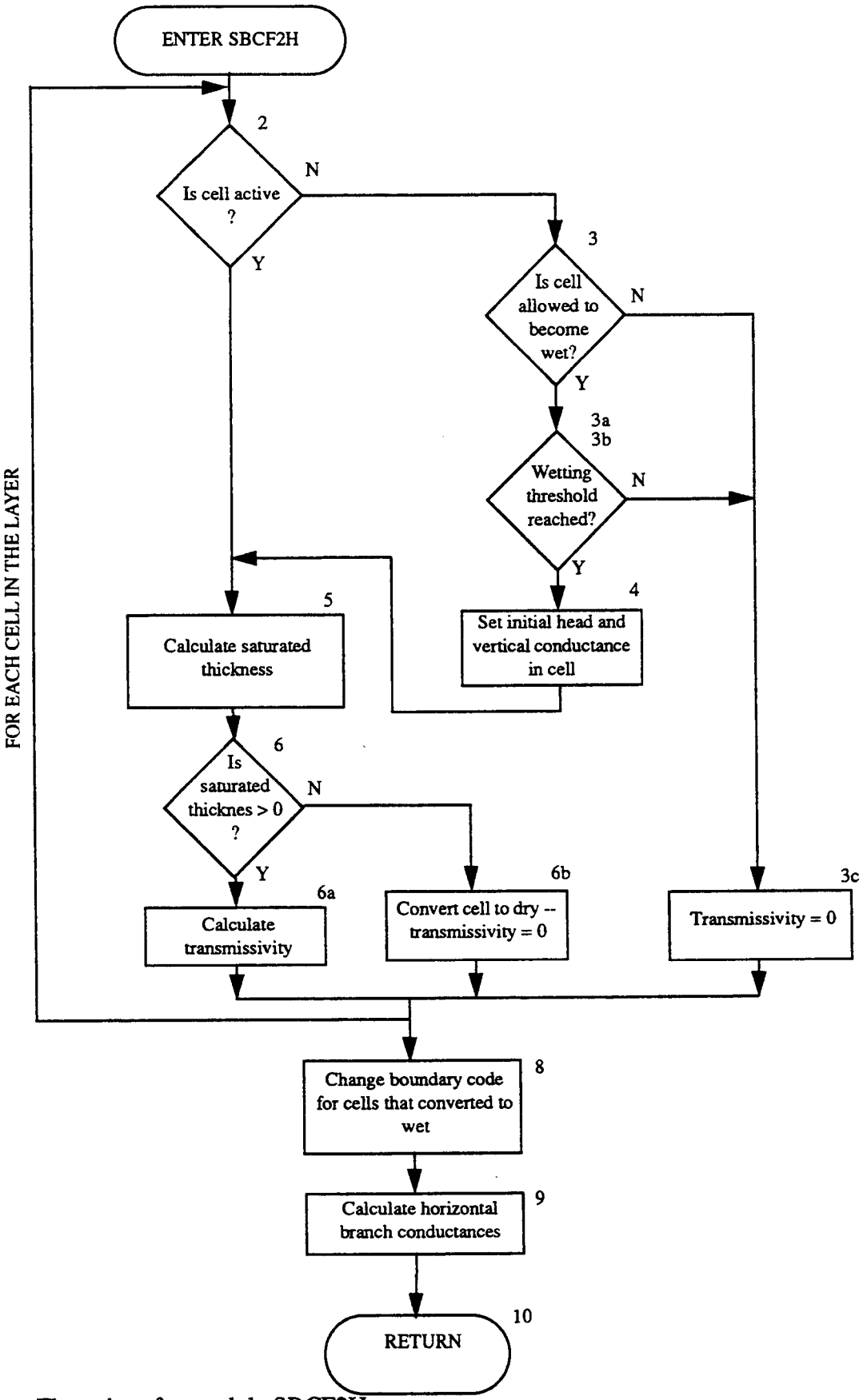


Figure 12. -- Flow chart for module SBCF2H.

```
      SUBROUTINE SBCF2H(HNEW,IBOUND,CR,CC,CV,HY,TRPY,DELR,DELC
     1,BOT,TOP,K,KB,KT,KITER,KSTP,KPER,NCOL,NROW,NLAY,IOUT
     2,WETDRY,IWDFLG,CVWD,WETFCT,IWETIT,IHDWET,HDRY)
C-----VERSION 1345 23MAY1991 SBCF2H
C
C     ******************************************************************
C     COMPUTE CONDUCTANCE FOR ONE LAYER FROM SATURATED THICKNESS AND
C     HYDRAULIC CONDUCTIVITY, VERSION 2
C     ******************************************************************
C
C      SPECIFICATIONS:
C     ------------------------------------------------------------------
      DOUBLE PRECISION HNEW
C
      DIMENSION HNEW(NCOL,NROW,NLAY),IBOUND(NCOL,NROW,NLAY)
     1,CR(NCOL,NROW,NLAY), CC(NCOL,NROW,NLAY), CV(NCOL,NROW,NLAY)
     2,HY(NCOL,NROW,NLAY), TRPY(NLAY), DELR(NCOL), DELC(NROW)
     3,BOT(NCOL,NROW,NLAY),TOP(NCOL,NROW,NLAY),WETDRY(NCOL,NROW,NLAY)
     4,CVWD(NCOL,NROW,NLAY)
      CHARACTER*4 ACNVRT
      DIMENSION ICNVRT(8),JCNVRT(8),ACNVRT(8)
C
      COMMON /FLWCOM/LAYCON(80)
C     ------------------------------------------------------------------
C
C1------LOOP THROUGH EACH CELL IN LAYER AND CALCULATE TRANSMISSIVITY AT
C1------EACH ACTIVE CELL.
      NCNVRT=0
      IHDCNV=0
      ITFLG=1
      IF(IWDFLG.NE.0) ITFLG=MOD(KITER,IWETIT)
      DO 200 I=1,NROW
      DO 200 J=1,NCOL
C
C2------IF CELL IS ACTIVE, THEN SKIP TO CODE THAT CALCULATES SATURATED
C2------THICKNESS.
      IF(IBOUND(J,I,K).NE.0) GO TO 20
C
C3------DETERMINE IF THE CELL CAN CONVERT BETWEEN CONFINED AND
C3------UNCONFINED.  IF NOT, SKIP TO CODE THAT SETS TRANSMISSIVITY TO 0.
      IF(ITFLG.NE.0) GO TO 6
      IF(WETDRY(J,I,KB).EQ.0.0)GO TO 6
      WD=WETDRY(J,I,KB)
      IF(WD.LT.0.) WD=-WD
      TURNON=BOT(J,I,KB)+WD
C
C3A-----CHECK HEAD IN CELL BELOW TO SEE IF WETTING THRESHOLD HAS BEEN
C3A-----REACHED.
      IF(K.EQ.NLAY)GO TO 2
      HTMP=HNEW(J,I,K+1)
      IF(IBOUND(J,I,K+1).GT.0.AND.HTMP.GE.TURNON)GO TO 9
C
C3B-----CHECK HEAD IN ADJACENT HORIZONTAL CELLS TO SEE IF WETTING
```

```
C3B-----THRESHOLD HAS BEEN REACHED.
    2 IF(WETDRY(J,I,KB).LT.0.) GO TO 6
      IF(J.EQ.1)GO TO 3
      HTMP=HNEW(J-1,I,K)
      IF(IBOUND(J-1,I,K).GT.0.AND.IBOUND(J-1,I,K).NE.30000.AND.
     1                           HTMP.GE.TURNON)GO TO 9
    3 IF(J.EQ.NCOL)GO TO 4
      HTMP=HNEW(J+1,I,K)
      IF(IBOUND(J+1,I,K).GT.0.AND.HTMP.GE.TURNON)GO TO 9
    4 IF(I.EQ.1)GO TO 5
      HTMP=HNEW(J,I-1,K)
      IF(IBOUND(J,I-1,K).GT.0.AND.IBOUND(J,I-1,K).NE.30000.AND.
     1                            HTMP.GE.TURNON)GO TO 9
    5 IF(I.EQ.NROW)GO TO 6
      HTMP=HNEW(J,I+1,K)
      IF(IBOUND(J,I+1,K).GT.0.AND.HTMP.GE.TURNON)GO TO 9
C
C3C-----CELL IS DRY AND STAYS DRY.  SET TRANSMISSIVITY TO 0 AND SKIP
C3C-----TO THE NEXT CELL.
    6 CC(J,I,K)=0.
      GO TO 200
C
C4------CELL BECOMES WET.  SET INITIAL HEAD AND VERTICAL CONDUCTANCE.
    9 IF(IHDWET.NE.0) HNEW(J,I,K)=BOT(J,I,KB)+WETFCT*WD
      IF(IHDWET.EQ.0) HNEW(J,I,K)=BOT(J,I,KB)+WETFCT*(HTMP-BOT(J,I,KB))
      IF(K.EQ.NLAY) GO TO 12
      IF(IBOUND(J,I,K+1).NE.0) CV(J,I,K)= CVWD(J,I,K)
   12 IF(K.EQ.1) GO TO 14
      IF(IBOUND(J,I,K-1).NE.0) CV(J,I,K-1)= CVWD(J,I,K-1)
   14 IBOUND(J,I,K)=30000
C
C4A-----PRINT MESSAGE SAYING CELL HAS BEEN CONVERTED TO WET.
      NCNVRT=NCNVRT+1
      ICNVRT(NCNVRT)=I
      JCNVRT(NCNVRT)=J
      ACNVRT(NCNVRT)=' WET'
      IF(NCNVRT.LT.8) GO TO 20
         IF(IHDCNV.EQ.0) WRITE(IOUT,17) KITER,K,KSTP,KPER
   17    FORMAT(1H0,'CELL CONVERSIONS FOR ITERATION=',I3,'  LAYER=',
     1    I2,'  TIME STEP=',I3,'  STRESS PERIOD=',I3,'      (ROW,COL)')
         IHDCNV=1
         WRITE(IOUT,18) (ACNVRT(L),ICNVRT(L),JCNVRT(L),L=1,NCNVRT)
   18    FORMAT(1X,8(A4,'(',I3,',',I3,')   '))
         NCNVRT=0
C
C5------CALCULATE SATURATED THICKNESS.
   20 HD=HNEW(J,I,K)
      IF(LAYCON(K).EQ.1) GO TO 50
      IF(HD.GT.TOP(J,I,KT)) HD=TOP(J,I,KT)
   50 THCK=HD-BOT(J,I,KB)
C
C6------CHECK TO SEE IF SATURATED THICKNESS IS GREATER THAN ZERO.
      IF(THCK.LE.0.) GO TO 100
C
```

```
C6A-----IF SATURATED THICKNESS>0 THEN TRANSMISSIVITY IS HYDRAULIC
C6A-----CONDUCTIVITY TIMES SATURATED THICKNESS.
      CC(J,I,K)=THCK*HY(J,I,KB)
      GO TO 200
C
C6B-----WHEN SATURATED THICKNESS < 0, PRINT A MESSAGE AND SET
C6B-----TRANSMISSIVITY, IBOUND, AND VERTICAL CONDUCTANCE =0
  100 NCNVRT=NCNVRT+1
      ICNVRT(NCNVRT)=I
      JCNVRT(NCNVRT)=J
      ACNVRT(NCNVRT)=' DRY'
      IF(NCNVRT.LT.8) GO TO 150
         IF(IHDCNV.EQ.0) WRITE(IOUT,17) KITER,K,KSTP,KPER
         IHDCNV=1
         WRITE(IOUT,18) (ACNVRT(L),ICNVRT(L),JCNVRT(L),L=1,NCNVRT)
         NCNVRT=0
  150 HNEW(J,I,K)=HDRY
      CC(J,I,K)=0.
      IF(IBOUND(J,I,K).GE.0) GO TO 160
         WRITE(IOUT,151)
  151    FORMAT(1H0,'CONSTANT-HEAD CELL WENT DRY -- SIMULATION ABORTED')
         WRITE(IOUT,152) K,I,J,KITER,KSTP,KPER
  152    FORMAT(1X,'LAYER=',I2,'   ROW=',I3,'   COLUMN=',I3,
     1     '   ITERATION=',I3,'   TIME STEP=',I3,'   STRESS PERIOD=',I3)
         STOP
  160 IBOUND(J,I,K)=0
      IF(K.LT.NLAY) CV(J,I,K)=0.
      IF(K.GT.1) CV(J,I,K-1)=0.
  200 CONTINUE
C
C7------PRINT ANY REMAINING CELL CONVERSIONS NOT YET PRINTED
      IF(NCNVRT.EQ.0) GO TO 203
         IF(IHDCNV.EQ.0) WRITE(IOUT,17) KITER,K,KSTP,KPER
         IHDCNV=1
         WRITE(IOUT,18) (ACNVRT(L),ICNVRT(L),JCNVRT(L),L=1,NCNVRT)
         NCNVRT=0
C
C8------CHANGE IBOUND VALUE FOR CELLS THAT CONVERTED TO WET THIS
C8------ITERATION FROM 30000 to 1.
  203 IF(IWDFLG.EQ.0) GO TO 210
      DO 205 I=1,NROW
      DO 205 J=1,NCOL
      IF(IBOUND(J,I,K).EQ.30000) IBOUND(J,I,K)=1
  205 CONTINUE
C
C9------COMPUTE HORIZONTAL BRANCH CONDUCTANCES FROM TRANSMISSIVITY.
  210 CALL SBCF1C(CR,CC,TRPY,DELR,DELC,K,NCOL,NROW,NLAY)
C
C10-----RETURN
      RETURN
      END
```

List of Variables for Module SBCF2H

Variable	Range	Definition
ACNVRT	Module	DIMENSION (8), Contains "DRY" or "WET" for up to eight cells that have converted between wet and dry.
BOT	Package	DIMENSION (NCOL,NROW,NLAY), Elevation of the aquifer bottom. Although BOT is dimensioned to the size of the grid, space exists onlfor cells for which saturated thickness is calculated.
CC	Global	DIMENSION (NCOL,NROW,NLAY), Conductance in the column direction. This array is also used to temporarily hold saturated thickness.
CR	Global	DIMENSION (NCOL,NROW,NLAY), Conductance in the row direction.
CV	Global	DIMENSION (NCOL,NROW,NLAY), Conductance in the vertical direction. Although CV is dimensioned to the size of the grid, space exists only for NLAY-1 layers.
CVWD	Package	DIMENSION (NCOL,NROW,NLAY), Conductance in the vertical direction. When a wet cell converts to dry, CV is set equal to 0. When a dry cell converts to wet, CV is set equal to CVWD. Although CVWD is dimensioned to the size of the grid, space exists only for NLAY-1 layers.
DELC	Global	DIMENSION (NROW), Cell dimension in the column direction. DELC(I) contains the width of row I.
DELR	Global	DIMENSION (NCOL), Cell dimension in the row direction. DELR(J) contains the width of column J.
HD	Module	Single precision equivalent of HNEW(J,I,K).
HDRY	Package	When a cell converts to dry, HNEW is set equal to HDRY.
HNEW	Global	DIMENSION (NCOL,NROW,NLAY), Most recent estimate of head in each cell.
HTMP	Module	Temporary single precision equivalent of HNEW.
I	Module	Index for rows.
IBOUND	Global	DIMENSION (NCOL,NROW,NLAY), Status of each cell: < 0, constant-head cell = 0, no-flow cell > 0, variable-head cell
ICNVRT	Module	DIMENSION (8), Row index for up to eight cells that have converted between wet and dry.
IHDCNV	Module	Flag indicating if any cell conversion information has been printed.
IHDWET	Package	Flag indicating which equation to use for calculating the head at a cell that has just converted from dry to wet.
IOUT	Global	Primary unit number for all printed output.
ITFLG	Module	Flag indicating if the current iteration is one in which cells are allowed to convert from dry to wet.
IWDFLG	Package	Flag indicating if wetting capability is active.
IWETIT	Package	Iteration interval for attempting to wet cells.
J	Module	Index for columns.
JCNVRT	Module	DIMENSION (8), Column index for up to eight cells that have converted between wet and dry.
K	Module	Index for layers.

KB	Module	Index for layer within BOT, HY, and WETDRY arrays.
KITER	Global	Iteration counter. KITER=1 at the start of each time step.
KPER	Global	Stress period counter.
KSTP	Global	Time step counter. KSTP=1 at the start of each stress period.
KT	Module	Index for layer within TOP array.
L	Module	Index for printing cell conversion information.
LAYCON	Package	DIMENSION (80) Layer-type code: 0 -- Layer is strictly confined. 1 -- Layer is strictly unconfined. 2 -- Layer is convertible between confined and unconfined (transmissivity is constant). 3 -- Layer is convertible between confined and unconfined (transmissivity varies).
NCNVRT	Module	Counter for number of cells that have converted between wet and dry since a line of conversion information has been printed.
NCOL	Global	Number of columns in the grid.
NLAY	Global	Number of layers in the grid.
NROW	Global	Number of rows in the grid.
THCK	Module	Saturated thickness.
TOP	Package	DIMENSION (NCOL,NROW,NLAY), Elevation of the aquifer top. Although TOP is dimensioned to the size of the grid, space exists only for cells that can convert between confined and unconfined.
TRPY	Package	DIMENSION (NLAY), Ratio of transmissivity (or hydraulic conductivity) in the column direction to transmissivity (or hydraulic conductivity) in the row direction.
TURNON	Module	Turn-on threshold.
WD	Module	The absolute value of WETDRY(J,I,KB).
WETDRY	Package	DIMENSION (NCOL,NROW,NLAY), Wetting threshold. Although WETDRY is dimensioned to the size of the grid, space exists only for layers that have the potential for converting from dry to wet.
WETFCT	Package	A factor that is included in the calculation of the head that is established at a cell that has just converted from dry to wet.

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Finite-Difference Interblock Transmissivity for Unconfined Aquifers and for Aquifers having Smoothly Varying Transmissivity

by Daniel J. Goode and Charles A. Appel

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CONVERSION FACTORS AND ABBREVIATIONS

Multiply	by	To Obtain
meter (m)	3.281	foot (ft)
meter squared per day (m^2/day)	10.76	foot squared per day (ft^2/day)
meter cubed per day (m^3/day)	35.32	foot cubed per day (ft^3/day)
meter cubed per hour (m^3/hour)	35.32	foot cubed per hour (ft^3/hour)

Finite-Difference Interblock Transmissivity for Unconfined Aquifers and for Aquifers having Smoothly Varying Transmissivity

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ABSTRACT

More accurate alternatives to the widely used harmonic mean interblock transmissivity are proposed for block-centered finite-difference models of ground-water flow in unconfined aquifers and in aquifers having smoothly varying transmissivity. The harmonic mean is the exact interblock transmissivity for steady-state one-dimensional flow with no recharge if the transmissivity is assumed to be spatially uniform over each finite-difference block, changing abruptly at the block interface. However, the harmonic mean may be inferior to other means if transmissivity varies in a continuous or smooth manner between nodes. Alternative interblock transmissivity functions are analytically derived for the case of steady-state one-dimensional flow with no recharge. The second author has previously derived the exact interblock transmissivity, the logarithmic mean, for one-dimensional flow when transmissivity is a linear function of distance in the direction of flow. We show that the logarithmic mean transmissivity is also exact for uniform flow parallel to the direction of changing transmissivity in a two- or three-dimensional model, regardless of grid orientation relative to the flow vector. For the case of horizontal flow in a homogeneous unconfined or water-table aquifer with a horizontal bottom and with areally distributed recharge, the exact interblock transmissivity is the unweighted arithmetic mean of transmissivity at the nodes. This mean also exhibits no grid-orientation effect for unidirectional flow in a two-dimensional model. For horizontal flow in an unconfined aquifer with no recharge where hydraulic conductivity is a linear function of distance in the direction of flow the exact interblock transmissivity is the product of the arithmetic mean saturated thickness and the logarithmic mean hydraulic conductivity. For several hypothetical two- and three-dimensional cases with smoothly varying transmissivity or hydraulic conductivity, the harmonic mean is shown to yield the least accurate solution to the flow equation of the alternatives considered. Application of the alternative interblock transmissivities to a regional aquifer system model indicates that the changes in computed heads and fluxes are typically small, relative to model calibration error. For this example, the use of alternative interblock transmissivities resulted in an increase in computational

effort of less than 3 percent. Numerical algorithms to compute alternative interblock transmissivity functions in a modular three-dimensional flow model are presented and documented.

INTRODUCTION

Appel (1976) examined the computation of interblock transmissivity for block-centered finite-difference models of ground-water flow and derived the exact one-dimensional interblock transmissivity for the case of transmissivity varying linearly between nodes with no recharge, and steady-state flow. He pointed out that significant errors in the solution of the ground-water flow equation can result from the use of the harmonic or arithmetic mean for some variations of block size and transmissivity. Although the harmonic mean is based on the somewhat unnatural assumption that transmissivity is spatially uniform over each finite-difference block, and changes discontinuously at the block boundaries, it remains the default function for interblock transmissivity in general ground-water flow models (for example, Konikow and Bredehoeft, 1978; McDonald and Harbaugh, 1988).

In addition to Appel (1976), a few other authors have noted the sensitivity of model results to the choice of interblock transmissivity. Anderson and others (1984) found that the computed flux into a drain in an unconfined aquifer significantly descreased when the harmonic mean was used. Goode (1990) showed that the interblock transmissivity of Appel (1976) yielded more accurate potentiometric heads than the harmonic mean in a simulation of two-dimensional flow in a confined aquifer with a linear change in transmissivity in both directions. Haverkamp and others (1977) and Schnabel and Richie (1984) evaluated several methods for computing interblock hydraulic conductivity in one-dimensional unsaturated flow models, in which the hydraulic conductivity is a nonlinear function of head. Schnabel and Richie (1984) noted that the harmonic mean gave unrealistic results for infiltration of a sharp wetting front and recommended integration of the unsaturated conductivity function to determine interblock conductance.

Visual inspection of practically any sediment outcrop reveals lithologic heterogeneity even at a small scale. As found in a variety of studies, such lithologic heterogeneity results in heterogeneity of the hydraulic conductivity. Following are some references that describe hydraulic conductivity data for closely spaced samples and thus reflect small scale variability. Gelhar (1986) shows a plot (his figure 2) of the logarithm of permeability (from laboratory analyses of cores) as compared with location of cores taken at about 0.3 meter intervals along a 60 meter section in a vertical borehole in the Mt. Simon Sandstone aquifer in central Illinois. Smith (1981) reported on the variation of hydraulic conductivity measurements made on two 100 point line samples collected at 0.3 meter

intervals. One line of samples was parallel to the bedding, and the other line was perpendicular to the bedding, of the Quadra Sand--a stratified unconsolidated sand deposit exposed in a series of cliffs near Vancouver, Canada. Measured hydraulic conductivity ranged from roughly 0.01 to 0.1 m/s and was not strongly correlated over spatial separations greater than 1 m. Goggin and others (1992) reported on thousands of permeability measurements made on outcrops and cores of eolian (wind blown) sandstones to estimate small-scale variations using a mechanical field permeameter. More than 2,800 measurements were made at an average vertical spacing of about 0.013 meter over a continuous core made up mostly of the Page Sandstone. Measurements also were made 25 meters away on the surface of the Page Sandstone outcrop. About 450 measurements were taken at an average vertical spacing of 0.025 meters and about 100 measurements were taken at an average horizontal spacing of 0.25 meters at the outcrop. Vertical transects were sampled from segments of the Nugget Sandstone and the Tensleep Sandstone at an average spacing of about 0.3 meters. These data show large ranges of permeability but relatively small coefficients of variation within a single stratification type. Weber (1986) describes two orders of magnitude range in permeability measurements made on several cores where the sample points are a few centimeters apart.

There is a growing body of literature (see review by Gomez-Hernandez and Gorelick, 1989) on approaches to determine "effective" or "best" ground-water model parameter values taking into consideration the apparent spatial variability of the parameters at different scales. That effort is still in the developmental stage. Although it is not our intent to summarize, or add to, that body of work, we do need to refer to it briefly to make clear the distinction between certain terms which could be confused. On the basis of numerical experiments, Warren and Price (1961) concluded that (p. 165) "The most probable behavior of a heterogeneous system with single-phase flow approaches that of a homogeneous system having a permeability equal to the geometric mean of the individual permeabilities." Those experiments did not recognize any spatial trends or structure to the hydraulic conductivity. It is as if the hydraulic conductivity of a block is independent of the conductivity of its neighbors. From his review of his own work and that of others, Gelhar (1986) observed (p. 138S) " . . . the stochastic results for the two-dimensional isotropic case suggest a simple rule of thumb for evaluating effective large-scale properties; that is, the [logarithm of the] large-scale transmissivity of an aquifer is obtained by averaging the logarithms of the local transmissivities that are measured." Note that this is equivalent to stating that the effective large-scale transmissivity is the geometric mean of the measured transmissivities. From their numerical experimentation, Gomez-Hernandez and Gorelick

(1989) concluded that the use of the geometric mean as an effective hydraulic conductivity seemed to be appropriate away from a pumping well but "the geometric mean failed to reproduce the expected value of simulated heads near the well locations."

Considering the degree of spatial variation observed at points centimeters or meters apart one might ask if it is overkill to represent the spatial variation between numerical block centers, separated commonly by distances of 100's or 1,000's of meters, by a function any more complex than the stairstep variation associated with the weighted harmonic mean interblock transmissivity. All of the above-mentioned references indicate that hydraulic conductivity can vary at the smallest scales for which it is even practical to measure this property. Does the conceptual model of a linear variation in hydraulic conductivity run the risk of seeming to represent the aquifer more accurately than the field data justify? In this context, is the use of the harmonic mean justified over even simpler means, such as the arithmetic mean? For the present work, we leave the overall conceptualization of spatial heterogeneity to the modeler, but suggest improved numerical approximations for a few specific deterministic cases. At worst, these methods may offer no significant improvement in real-world accuracy. However, the theoretical considerations may be sufficient to encourage consideration of alternative interblock transmissivities that have essentially the same computational cost as the harmonic mean.

In this report, one-dimensional solutions of the steady-state horizontal groundwater flow equation are interpreted to yield exact effective transmissivity between two points for different cases of deterministic spatial heterogeneity. The block-centered finite-difference solution of the steady-state two-dimensional flow equation is reviewed, highlighting computation of interblock transmissivity. Numerical solutions of the groundwater flow equation using alternative interblock transmissivities are compared for several simple hypothetical cases. Finally, model simulations of a large scale aquifer system are repeated with alternative interblock transmissivity functions to illustrate the practical effects of their use. The appendix includes subroutines for the modular three-dimensional groundwater flow model (MODFLOW) of McDonald and Harbaugh (1988) to compute alternative horizontal interblock transmissivity functions.

(PAGE 7 FOLLOWS)

THEORY

Governing Equation

The governing equation for steady-state horizontal ground-water flow may be written (Bear, 1979):

$$\nabla \mathbf{T} \nabla h + W = 0 \quad , \tag{1}$$

where h [L] is potentiometric head, **T** [L^2T^{-1}] is the transmissivity tensor, and W [LT^{-1}] is the volumetric source rate per unit area (recharge). Boundary conditions for (1) included specified head, specified flux, and mixed or leaky conditions. If the x and y coordinates are aligned with the principal axes of the transmissivity tensor then

$$\mathbf{T} = \begin{bmatrix} T_{xx} & 0 \\ 0 & T_{yy} \end{bmatrix} \quad . \tag{2}$$

In a layered three-dimensional model (for example, McDonald and Harbaugh, 1988) an equation analogous to (1) is applicable for each model layer and W is the net vertical inflow from below and above. In this case W depends on h, hence the equations for each layer are coupled through W.

Under unconfined conditions, transmissivity is often assumed to be a linear function of head:

$$\mathbf{T} = \mathbf{K} (h-b) \quad , \tag{3}$$

where **K** [LT^{-1}] is the hydraulic conductivity tensor, and b [L] is the elevation of the bottom of the aquifer or layer. In this case, the potentiometric head coincides with the water table or the top of the saturated zone, and it is free to move in time in response to hydraulic stresses. Because the head must vary spatially in order for flow to occur, transmissivity is almost everywhere spatially nonuniform for unconfined conditions.

Under confined conditions, where the top of the saturated zone is defined by the interface between the aquifer and an overlying confining unit, transmissivity is mostly assumed to be constant in time. However, transmissivity may be spatially nonuniform due

to the hydrogeologic characteristics of the aquifer. In fact, the ubiquitous heterogeneity of aquifers is the focus of much current research. For both confined and unconfined conditions, the governing equation (1) allows for nonuniform as well as anisotropic ($T_{xx} \neq T_{yy}$) transmissivity.

Effective Transmissivity for One-Dimensional Steady-State Confined Flow

For one-dimensional steady confined flow, with no leakage or recharge, the governing equation can be written:

$$\frac{d}{ds} T(s) \frac{dh}{ds} = 0 \quad , \tag{4}$$

where s is the coordinate aligned with the flow direction, and transmissivity T is a function of s. This equation is a statement of uniform flux, and is integrated once to yield the flux:

$$q = - T(s) \frac{dh}{ds} \quad , \tag{5}$$

where q [L^2T^{-1}] is the volumetric flux per unit width of aquifer, and is spatially uniform. This equation can be integrated, giving:

$$\int_{h_1}^{h_2} dh = h_2 - h_1 = - q \int_{s_1}^{s_2} \frac{ds}{T(s)} \quad , \tag{6}$$

where h_1 and h_2 [L] are potentiometric heads at s_1 and s_2, respectively. Equation (6) shows that the head difference between two points and the discharge are related by (Collins, 1961, p. 78):

$$q = \overline{T} \frac{h_1 - h_2}{\Delta s} \quad , \tag{7}$$

when $\overline{T}$ is defined from (Collins, 1961; compare to Bear, 1979, p. 177):

$$\frac{1}{\overline{T}} = \frac{1}{\Delta s}\int_{s_1}^{s_2}\frac{ds}{T(s)} \quad . \tag{8}$$

where $\Delta s = s_2 - s_1$. The inverse of equation (8), $\overline{T}$, can be considered the effective transmissivity and it is evaluated for abrupt and smooth variability of T(s) in the sequel.

Transmissivity changes abruptly

If the transmissivity changes abruptly between points 1 and 2, then (8) can be written:

$$\frac{1}{T_h} = \frac{1}{\Delta s}\left[\frac{s_b - s_1}{T_1} + \frac{s_2 - s_b}{T_2}\right] , \tag{9}$$

where T_1 and T_2 are the values of transmissivity at points 1 and 2, respectively, and s_b is the location of the discontinuous change in T. In this case, T_h is the spatially weighted harmonic mean of T_1 and T_2. If s_b is halfway between points 1 and 2, then (9) can be written:

$$T_h = \frac{2T_1T_2}{T_1 + T_2} \quad . \tag{10}$$

The harmonic mean in this form is valid as long as at least one of the transmissivities is nonzero and automatically yields zero effective transmissivity if one transmissivity is zero. When one of the transmissivities is large, relative to the other, the effective transmissivity is essentially twice the smaller value, and is insensitive to changes in the larger value.

Transmissivity linear function of distance in flow direction

If transmissivity is a linear function of s, then (8) can be integrated to yield (Butler, 1957, p. 78; Appel, 1976):

$$T_\lambda = \frac{T_2 - T_1}{\ln(T_2 / T_1)} \quad , \tag{11}$$

where T_λ is the logarithmic mean transmissivity. [Our use of the name "logarithmic mean" is due to Bird and others (1960, p. 467) who present the same expression except that the transmissivity in our case is replaced by a temperature difference between two parallel fluid streams in their case.] If the transmissivity is uniform, equation (11) cannot be used, but then from (8), $T_\lambda = T_1 = T_2$. In addition, equation (11) cannot be used if either transmissivity is zero, but in this case, the effective transmissivity is also zero.

If transmissivity is piece-wise linear across a one-dimensional flow domain, the effective transmissivity for the entire domain can be computed from equation (8) as:

$$\frac{1}{\overline{T}} = \frac{1}{\Delta s} \sum_{i=1}^{N-1} \left[(s_{i+1} - s_i) \frac{\ln\left(T_{i+1}/T_i\right)}{T_{i+1} - T_i} \right] \tag{12}$$

where N is the number of points where T is known (and assumed to vary linearly between) and Δs is the length of the domain. This is analogous to the effective hydraulic conductivity for flow across N layers of differing K.

The direction of the head gradient is parallel to the direction of flow, s, and for $s_1=0$ head is given by (after Appel, 1976):

$$h(s) = h_1 - \frac{q}{\alpha} \ln\left(1 + \frac{\alpha s}{T_1}\right) \quad . \tag{13}$$

where $\alpha \equiv \partial T/\partial s$ [LT^{-1}] is the uniform rate of change of transmissivity in the direction of flow. As with (11), equation (13) is only valid for nonzero α and T_1 as well as positive $T_1 + \alpha s_2 = T_2$.

Effective Transmissivity for One-Dimensional Steady-State Unconfined Flow

For the case of one-dimensional unconfined flow with recharge and a horizontal aquifer bottom at elevation zero, the governing equation (1) can be written:

$$\frac{d}{ds} K(s)\, h \frac{dh}{ds} = - W \quad , \tag{14}$$

where the hydraulic conductivity is a function of s and the recharge rate W is spatially uniform. Integration once yields an expression for flux:

$$q_1 + W (s - s_1) = - K(s)\, h \frac{dh}{ds} \quad , \tag{15}$$

where q_1 is here the volumetric flux per unit width at s_1. This expression will be evaluated in the sequel for the case of uniform K(s) = K with no recharge (W=0), the case of uniform K with uniform recharge W, and for the case of linearly varying K(s) with no recharge.

Homogeneous unconfined aquifer with no recharge

For the case of spatially uniform hydraulic conductivity K and no recharge (W=0), equation (15) can be integrated:

$$\int_{s_1}^{s_2} ds = - \frac{K}{q} \int_{h_1}^{h_2} h\, dh \quad , \tag{16}$$

where the subscript 1 on q has been dropped because q is uniform. Thus, the relation between the flux and the head at two points is (compare to Bear, 1979, p. 180):

$$q = \frac{K}{2\Delta s} \left(h_1^2 - h_2^2 \right) \quad . \tag{17}$$

Using the substitutions (from (3)) $T_1 = Kh_1$, $T_2 = Kh_2$, equation (17) becomes

$$q = \frac{(T_1 + T_2)}{2} \frac{(h_1 - h_2)}{\Delta s} \quad , \tag{18}$$

which shows that the effective transmissivity between points 1 and 2 (see equation (7)) is the arithmetic mean of the transmissivity at points 1 and 2:

$$T_a = \frac{T_1 + T_2}{2} \quad . \tag{19}$$

No spatial weighting occurs because the hydraulic conductivity and aquifer bottom elevation are spatially uniform between points 1 and 2. If either T_1 or T_2 is zero, then (19) cannot be used and the effective transmissivity is zero.

The head also changes only in the direction of flow, s, and for s_1=0 is given by:

$$(h(s))^2 = h_1^2 - \frac{2qs}{K} \quad . \tag{20}$$

Homogeneous unconfined aquifer with recharge

For the case of spatially uniform hydraulic conductivity K and uniform recharge W, equation (15) can be integrated to give the relation between the flux, as a function now of s, and the head at two points (compare to Bear, 1979, p. 180):

$$q(s) = q_1 + W(s - s_1) = \frac{K}{2\Delta s}\left(h_1^2 - h_2^2\right) + W\left(s - \frac{\Delta s}{2}\right) \quad . \tag{21}$$

At a point half-way between s_1 and s_2, that is s = $\Delta s/2$, the last term in (21) drops out:

$$q(s=\Delta s/2) = \frac{K}{2\Delta s}\left(h_1^2 - h_2^2\right) \quad , \tag{22}$$

which is identical to equation (17) above, except that the flux is the value midway between s_1 and s_2, instead of the uniform flux for the previous case of no recharge. Following the reasoning in the previous section, the negative of the finite-difference head gradient multiplied by the arithmetic mean transmissivity yields the flux midway between the two points for the case of uniform recharge.

The head also changes only in the direction of flow, s, and for $s_1=0$ is given by:

$$(h(s))^2 = h_1^2 - \frac{2q_1 s}{K} - \frac{Ws^2}{K} \quad . \tag{23}$$

Unconfined aquifer with hydraulic conductivity a linear function of distance in flow direction

For the case of no recharge (W=0), but spatially variable hydraulic conductivity, equation (15) can be integrated:

$$\int_{h_1}^{h_2} h \, dh = - \int_{s_1}^{s_2} \frac{q \, ds}{K(s)} \quad , \tag{24}$$

where the subscript 1 on q has been dropped because q is uniform. For the case of K(s) a linear function of s, equation (24) yields:

$$\frac{h_2^2 - h_1^2}{2} = - q\Delta s \, \frac{\ln (K_2 / K_1)}{K_2 - K_1} \quad , \tag{25}$$

where $K_1 = K(s_1)$ and $K_2 = K(s_2)$. Thus, the exact effective transmissivity is given by the product of the arithmetic mean head (that is, saturated thickness) and the logarithmic mean hydraulic conductivity:

$$T_{a\lambda} = \frac{h_1 + h_2}{2} \, \frac{K_2 - K_1}{\ln (K_2 / K_1)} \quad . \tag{26}$$

If $K_1 = K_2$, then equation (19) is used instead of (26). If either K_1 or K_2 is zero, $T_{a\lambda}$ is also zero.

The head also changes only in the direction of flow, s, and, for $s_1=0$ is given by:

$$(h(s))^2 = h_1^2 - \frac{2q}{\alpha_k} \ln\left(1 + \frac{\alpha_k s}{K_1}\right) \quad . \tag{27}$$

where $\alpha_k \equiv \partial K/\partial s$ [T^{-1}] is the uniform rate of change of hydraulic conductivity in the direction of flow.

Comparison of Effective Transmissivity Functions

The effective transmissivity functions are summarized in table 1. The logarithmic mean (Appel, 1976) is compared to the harmonic, geometric [$(T_1T_2)^{1/2}$], and arithmetic means in figures 1 and 2 in a nondimensional form (effective T over T_1 as a function of T_2 / T_1). The harmonic mean underestimates the effective T for the unconfined case with uniform hydraulic conductivity (arithmetic mean) and for confined systems with linear variability in T (logarithmic mean) because it more heavily weights the low T value. The geometric mean exhibits less error for both cases and is similar to the logarithmic mean, but still weights the lower T value excessively. For positive $T_1 \neq T_2$, the logarithmic mean is always larger than the geometric mean (Appendix A). As noted previously, the harmonic mean asymptotically approaches $2T_1$ ($T / T_1 = 2$) as T_2 approaches infinity, while the three other means continue to increase indefinitely with increasing T_2.

For the case of a water-table system with hydraulic conductivity varying linearly in space, the exact effective T depends on the relative variability in head. The effective $T_{a\lambda}$ for an unconfined aquifer with linearly varying hydraulic conductivity is shown for several ratios of h_2 / h_1 and compared to the arithmetic mean and the logarithmic mean transmissivity in figures 3 and 4. Here, the aquifer bottom elevation is assumed to be zero, so the head ratios correspond to the ratios in saturated thickness. When the ratio h_2/h_1 is greater than 1, flow is in the negative s direction (that is, from point 2 to point 1). If $h_2 / h_1 = T_2 / T_1$, then K(s) = K is uniform and the logarithmic mean transmissivity reduces to the arithmetic mean, $T_{a\lambda} = T_a$. If the transmissivities at points 1 and 2 are equal, $T_2 / T_1 = 1$, then K must be variable because the heads are not equal if flow is occuring. The effective $T_{a\lambda}$ for the case of $T_2 / T_1 = 1$ is greater than either T_1 or T_2 because the nonlinear variability in saturated thickness and the linear variability in K yield transmissivities between point 1 and 2 that are larger than the values at 1 and 2. This relation is clarified by figure 5, which shows the ratio of $T_{a\lambda} / T_1$ as a function of K_1 / K_2, for several values of h_2 / h_1.

Table 1.——Exact effective transmissivity functions for one-dimensional steady-state flow with no recharge

Flow Conditions	Spatial Variability	Function Name	Function	Notes
confined	T changes abruptly	harmonic mean	$T_h = \frac{2T_1T_2}{T_1 + T_2}$	for boundary half way between points 1 and 2
confined	T linear in s	logarithmic mean	$T_\lambda = \frac{T_2 - T_1}{\ln(T_2 / T_1)}$	$T_\lambda = 0$ if T_1 or $T_2 = 0$ $T_\lambda = T_1$ if $T_1 = T_2$
unconfined	uniform K	arithmetic mean	$T_a = \frac{T_1 + T_2}{2}$	$T_a = 0$ if T_1 or $T_2 = 0$
unconfined	K linear in s	arithmetic mean h times logarithmic mean K	$T_{a\lambda} = \frac{h_1 + h_2}{2} \frac{K_2 - K_1}{\ln(K_2 / K_1)}$	$T_{a\lambda} = 0$ if K_1 or $K_2 = 0$ $T_{a\lambda} = T_a$ if $K_1 = K_2$

Notes: s is coordinate aligned with flow direction, T>0 is transmissivity, K>0 is hydraulic conductivity, h is potentiometric head and equals the saturated thickness for bottom elevation zero. Subscript 1 or 2 indicate value of variable at points 1 or 2, respectively.

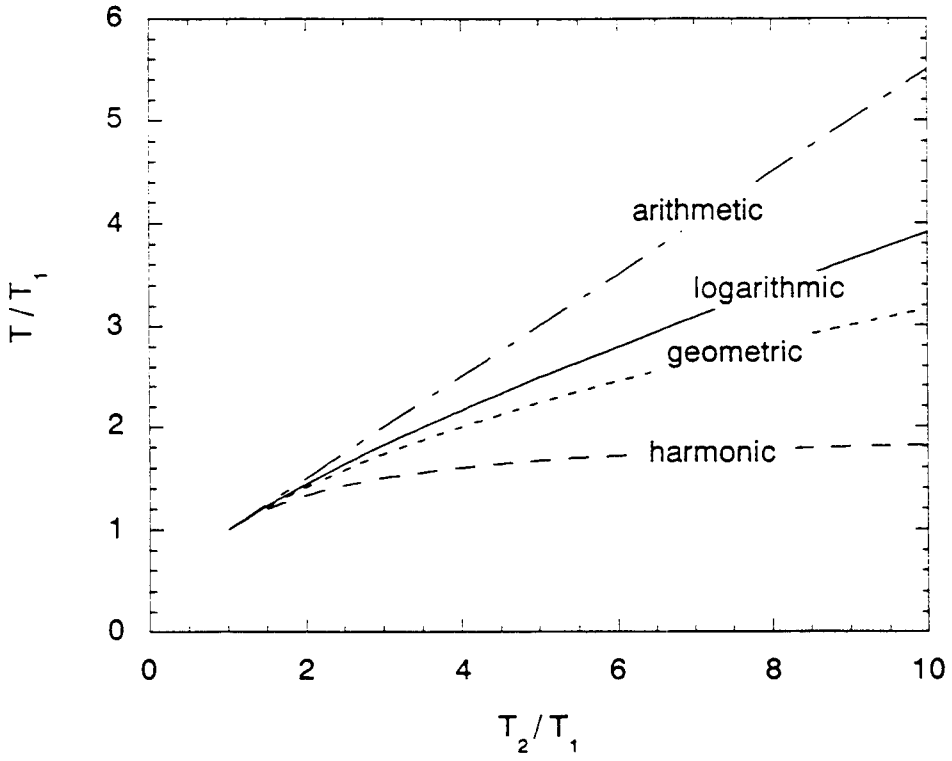


Figure 1.--Graph showing ratio of arithmetic, logarithmic, and harmonic effective transmissivities to transmissivity at point 1 (T/T_1) as a function of the ratio of transmissivity at point 2 to that at point 1 (T_2/T_1). Geometric mean transmissivity shown for comparison.

The importance of selecting the appropriate effective transmissivity depends on the magnitude of changes in transmissivity between points 1 and 2 where the head is evaluated, and on the magnitude of head changes for the unconfined case. When these changes are small then any of the functions will yield essentially the same results. Errors due to using an incorrect effective transmissivity increase with the magnitude of the changes in transmissivity, or with the magnitude of the changes in saturated thickness and hydraulic conductivity for the unconfined case.

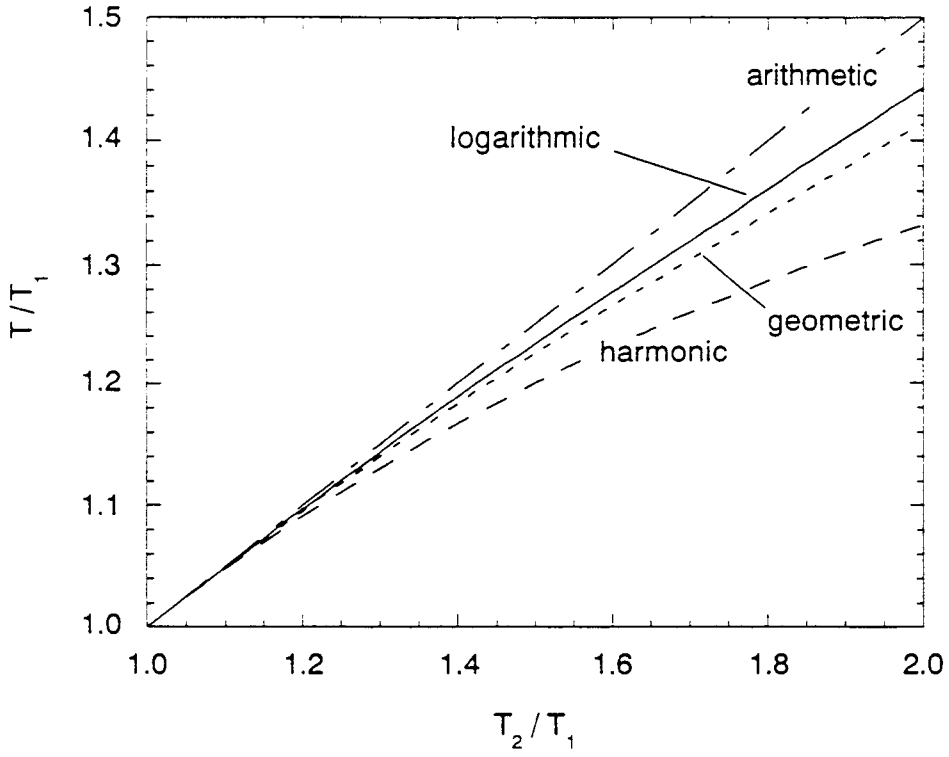


Figure 2.--Graph showing enlarged portion of Figure 1 for small values of the ratio of transmissivity at point 2 to that at point 1.

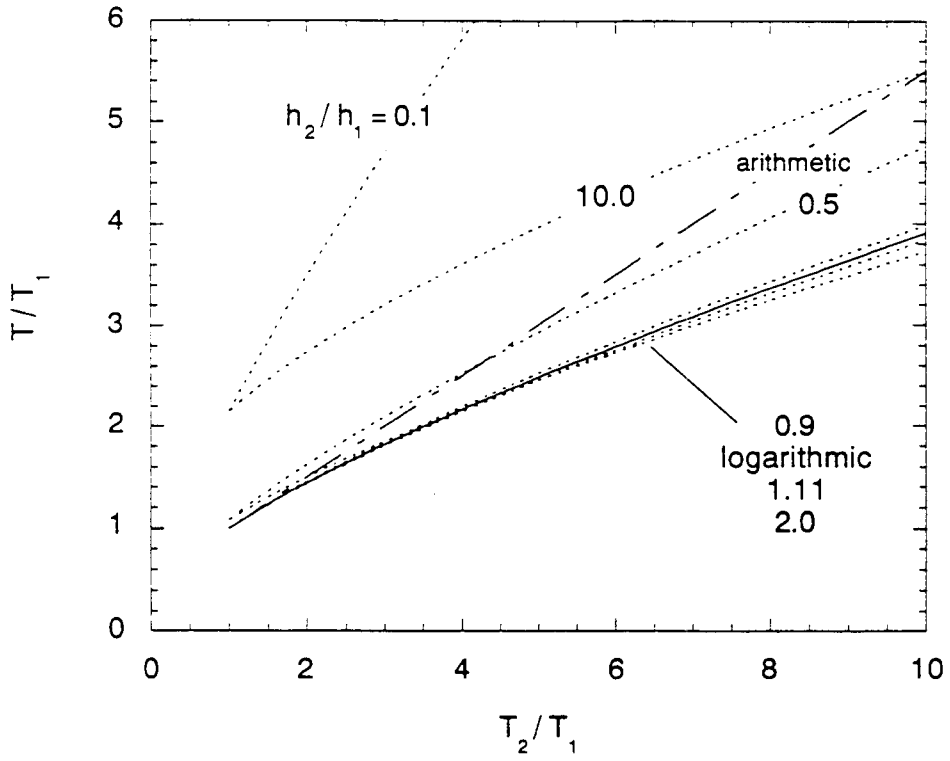


Figure 3.--Graph showing ratio of effective transmissivity as arithmetic mean saturated thickness times logarithmic mean hydraulic conductivity to the transmissivity at point 1 (T/T_1) as a function of the ratio of transmissivity at point 2 to that at point 1 (T_2/T_1). Arithmetic and logarithmic effective transmissivities shown for comparison.

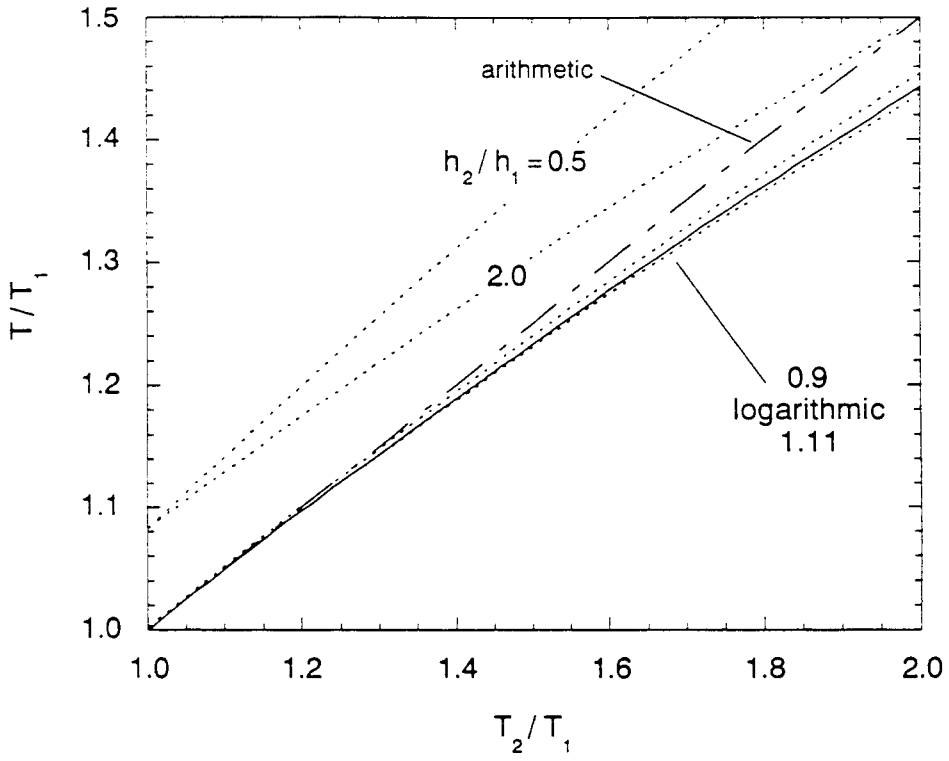


Figure 4.--Graph showing enlarged portion of Figure 3 for small values of the ratio of transmissivity at point 2 to that at point 1.

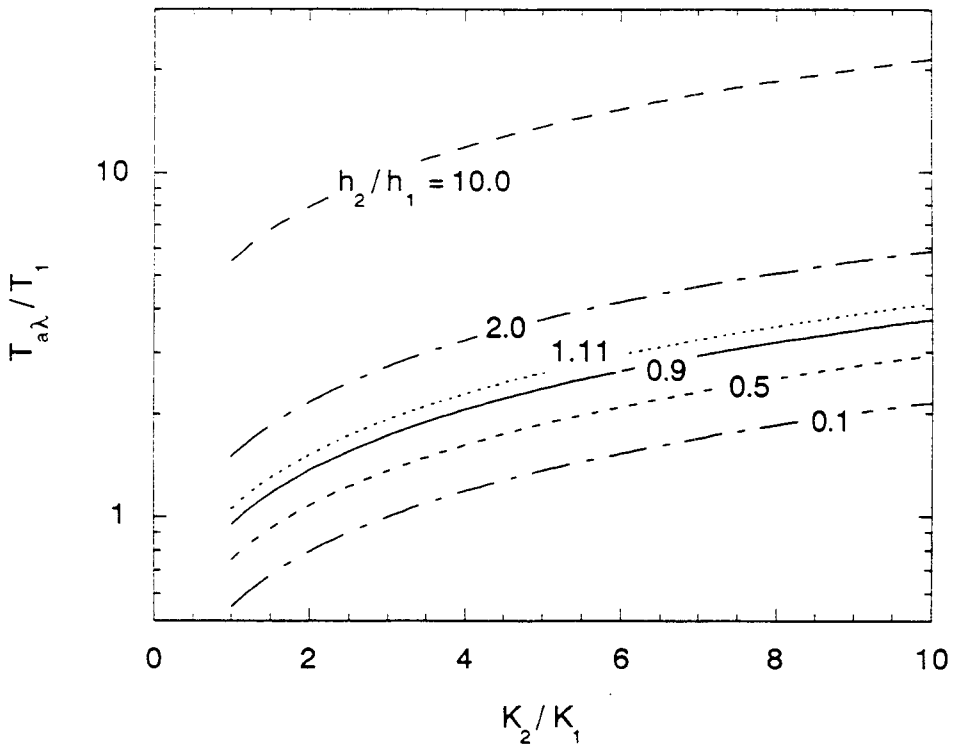


Figure 5.--Graph showing the ratio of effective transmissivity as arithmetic mean saturated thickness times logarithmic mean hydraulic conductivity to the transmissivity at point 1 (T/T_1) as a function of the ratio of hydraulic conductivity at point 2 to that at point 1 (K_2/K_1) for several values of the ratio of head at point 2 to that at point 1 (h_2/h_1).

Finite-Difference Representation of Flow Equation

The steady-state two-dimensional ground-water flow equation can be solved using a block-centered finite-difference representation. The aquifer is discretized into rectangular blocks with a node at the center of each block (fig. 6). A discrete approximation of the governing equation is obtained by solving for the heads at the nodes [after Pinder and Bredehoeft, 1968; Trescott et al., 1976; Konikow and Bredehoeft, 1978]:

$$\frac{1}{\Delta x_i}\left\{\left[T_{xx(i+1/2,j)}\frac{(h_{i+1,j}-h_{i,j})}{\Delta x_{i+1/2}}\right]-\left[T_{xx(i-1/2,j)}\frac{(h_{i,j}-h_{i-1,j})}{\Delta x_{i-1/2}}\right]\right\}$$

$$+\frac{1}{\Delta y_j}\left\{\left[T_{yy(i,j+1/2)}\frac{(h_{i,j+1}-h_{i,j})}{\Delta y_{j+1/2}}\right]-\left[T_{yy(i,j-1/2)}\frac{(h_{i,j}-h_{i,j-1})}{\Delta y_{j-1/2}}\right]\right\}$$

$$+W_{i,j}=0 \qquad (28)$$

where subscript i is the index for the discretization in the x direction and subscript j is the index for the discretization in the y direction. The term Δx_i is the x dimension of all blocks at the x location corresponding to i. The term $\Delta x_{i+1/2}$ is the distance between node i,j and node i+1,j, for all i: $\Delta x_{i+1/2} = (\Delta x_i + \Delta x_{i+1}) / 2$. The Δy terms in (28) are analogous.

The block-centered finite-difference saturated flow model of McDonald and Harbaugh (1988) (MODFLOW) solves a three-dimensional equation analogous to (28), where W includes vertical flow between layers and depends on the difference between $h_{i,j}$ and the corresponding heads in adjacent layers. The algorithms developed in the following sections have been coded in new MODFLOW subroutines for computing horizontal conductance (see Appendix B).

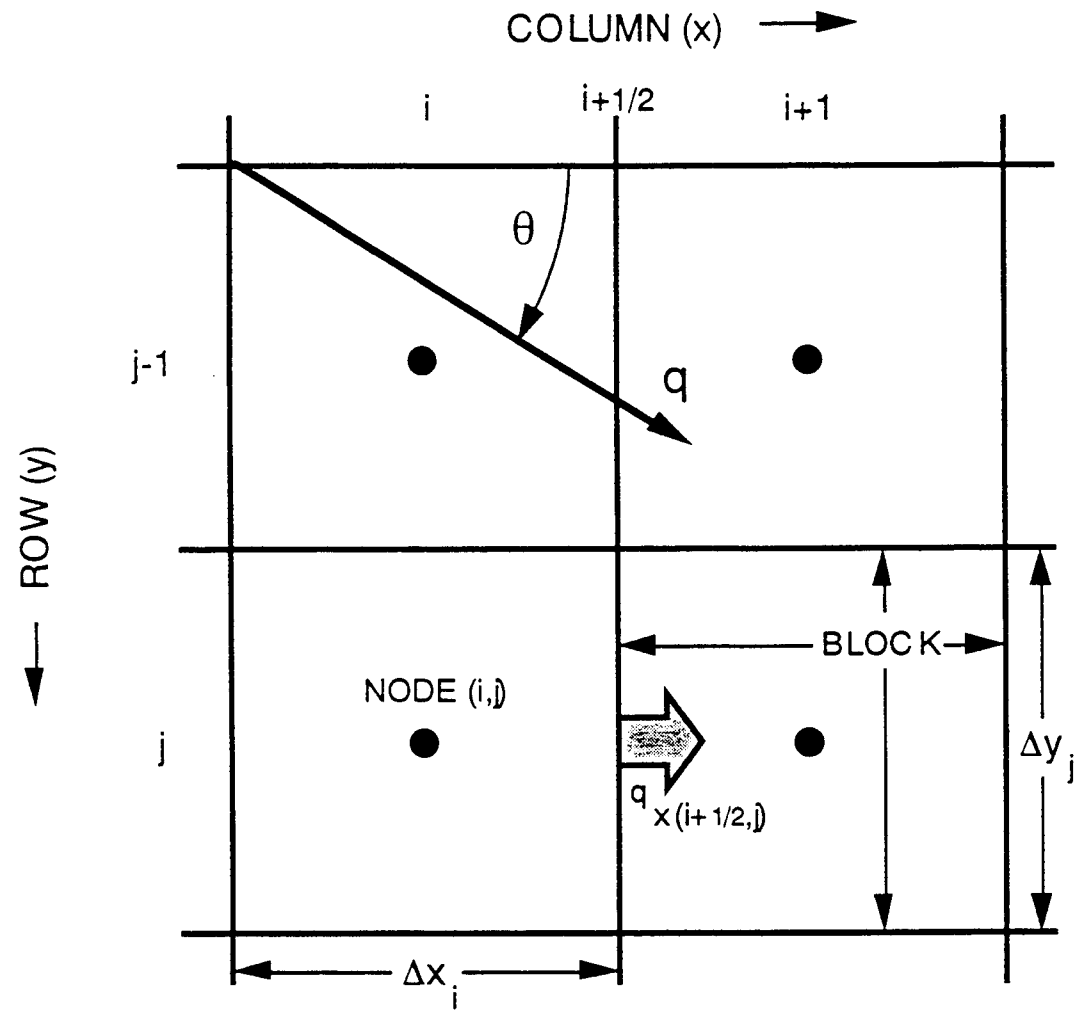


Figure 6.--Schematic of two-dimensional finite-difference discretization scheme showing flux in the x direction at (i+1/2,j) and a flux vector q oriented at angle θ to the x axis.

Interblock Transmissivity

The transmissivity terms in (28) are interblock or block interface values and are usually computed as some average of the values of T in the adjacent blocks. Two commonly used functions are the weighted arithmetic mean (Pinder and Bredehoeft, 1968)

$$T_{xx(i+1/2,j)} = \frac{\Delta x_{i+1} T_{xx(i,j)} + \Delta x_i T_{xx(i+1,j)}}{\Delta x_i + \Delta x_{i+1}} \quad , \quad T_{xx(i,j)} , T_{xx(i+1,j)} \neq 0 \quad , \qquad (29)$$

and the weighted harmonic mean (Pinder, 1970; Routt and Crawford, 1973):

$$T_{xx(i+1/2,j)} = \frac{T_{xx(i,j)} T_{xx(i+1,j)} \left(\Delta x_i + \Delta x_{i+1}\right)}{\Delta x_{i+1} T_{xx(i,j)} + \Delta x_i T_{xx(i+1,j)}} \quad ; \quad T_{xx(i,j)} + T_{xx(i+1,j)} \neq 0 \quad . \qquad (30)$$

The y components are computed analogously. As shown in the Theory section, the harmonic mean yields exact fluxes and nodal heads for one-dimensional steady-state flow with no recharge when transmissivity is uniform within each finite-difference block.

The expression for the weighted arithmetic mean (29) is the value of transmissivity at the location of the block interface obtained by linear interpolation between the nodal values. This interpolation is implied by the Taylor series expansion of Pinder and Bredehoeft (1968). In the papers of Appel (1976) and Anderson and others (1984), equation (29) is presented but with the indices on the Δx terms in the numerator switched. Their expression, which is not due to Pinder and Bredehoeft (1968), is equivalent to determining the average value of transmissivity along the line between the two nodes when the transmissivity is uniform on each block. The expression presented by Appel (1976, eq. 2) and Anderson and others (1984) has, to our knowledge, no rigorous derivation for the ground-water flow equation. Fortunately, the incorrect expression yields the same value of transmissivity when the blocks are uniform, i.e. when no spatial weighting occurs.

Other methods for determining the interblock transmissivity include using the geometric mean and using the "upstream" value only. The geometric mean, although an ad hoc method for ground-water flow, has been shown to perform satisfactorily in one-dimensional unsaturated flow simulations where hydraulic conductivity is nonlinearly dependent on head (Haverkamp et al., 1978). As shown above, the geometric mean is

similar to the logarithmic mean. Using the "upstream" value, or the value from the block with the higher head, is common in the petroleum simulation literature for the analogous finite-difference simulation of transient immiscible displacment (Crichlow, 1977), but this method has no rigorous derivation for application to steady-state ground-water flow.

The desired interblock transmissivity is that which computes the correct flux when the heads at the nodes are known, or, conversely, which yields the correct heads at the nodes when the flux is specified. Thus, the exact effective interblock transmissivity is the ratio of the exact flux divided by the finite-difference representation of the head gradient using the exact heads (for example, in the x direction):

$$T_{xx(i+1/2,j)} = \frac{-q_{x(i+1/2,j)}\,\Delta x_{i+1/2,j}}{h_{i+1,j} - h_{i,j}}\,, \qquad (31)$$

where the terms on the right hand side are known. This relation can be used, in steady-state and with no recharge, to derive the exact interblock transmissivity for cases in which the flux and heads are known analytically. The similarity between equation (31) and equation (7) above indicates that the effective transmissivity functions derived in the Theory section are suitable for interblock transmissivities in block-centered finite-difference flow models. In this use, points 1 and 2 correspond to nodes in the finite-difference grid.

Transmissivity uniform over each block

The weighted harmonic mean, equation (30), yields the exact solution for one-dimensional flow with no recharge if the transmissivity is assumed to be uniform over each finite-difference block. In addition, this mean automatically accounts for cases of uniform transmissivity and zero transmissivity, except if both transmissivities are zero. Of the alternative interblock transmissivity functions considered here (see table 1), the harmonic mean most heavily weights the lower transmissivity value. The appropriate spatial weighting depends on the grid method, the result here (equation 30) being appropriate for block-centered grids.

Transmissivity varies linearly between nodes

Appel (1976) presented a function (herein called the logarithmic mean) for computing the interblock T that is exact for steady-state one-dimensional flow with no recharge where T is a linear function of distance:

$$T_{xx(i+1/2,j)} = \frac{T_{xx(i+1,j)} - T_{xx(i,j)}}{\ln\left[\frac{T_{xx(i+1,j)}}{T_{xx(i,j)}}\right]} \quad ; \quad T_{xx(i,j)}, T_{xx(i+1,j)} \neq 0 \quad T_{xx(i,j)} \neq T_{xx(i+1,j)} \quad . \qquad (32)$$

Here, we prove that this interblock transmissivity also yields exact finite-difference solutions when the direction of flow is not aligned with the grid. That is, using the logarithmic mean for the interblock T in a two-dimensional model (or interblock K in a three-dimensional model) results in an exact solution to the flow problem if flow is uniform and aligned in the direction of changing T, regardless of the grid orientation.

Consider that the direction of flow s is at an angle θ to the finite-difference grid (fig. 6). The transformation from x and y to the s coordinate is given by:

$$s = x \cos\theta + y \sin\theta \quad . \qquad (33)$$

where it is assumed that s = 0 at x = y = 0.

The transmissivity is assumed to change in only the s direction, and to be uniform orthogonal to s. Defining the rate of change of T as a function of distance in the s direction as $\alpha = dT/ds$, the transmissivity at any point is given by:

$$T(s) = T_1 + \alpha s \quad ; \quad T(x,y) = T_1 + \alpha\left(x \cos\theta + y \sin\theta\right) \quad , \qquad (34)$$

where T_1 is the transmissivity at s=0. For this case, transmissivity is isotropic.

Assume the reference node (i,j) is located at x=y=0. The adjacent node in the x direction (i+1,j), is located at $x=\Delta x_{i+1/2}$, y=0. Using (34), the transmissivity at this node is

$$T_{xx(i+1,j)} = T_1 + \alpha \Delta x_{i+1/2} \cos\theta \quad . \qquad (35)$$

The logarithmic mean T_{xx} at (i+1/2,j) is, substituting (35) for T_2 into (32):

$$T_{xx(i+1/2,j)} = \frac{\alpha \Delta x_{i+1/2} \cos\theta}{\ln\left(1 + \frac{\alpha \Delta x_{i+1/2} \cos\theta}{T_1}\right)} \quad . \tag{36}$$

The model flux in the x direction is the negative of this interblock T multiplied by the head gradient between the nodes. The exact head at node (i+1,j) is evaluated by substituting (33) for this node into (13), yielding:

$$h_{i+1,j} = h_1 - \frac{q}{\alpha} \ln\left(1 + \frac{\alpha \Delta x_{i+1/2} \cos\theta}{T_1}\right) \quad . \tag{37}$$

The finite difference expression for the discharge per unit width at (i+1/2,j) is then

$$- T_{xx(i+1/2,j)} \frac{(h_{i+1,j} - h_{i,j})}{\Delta x_{i+1/2}} =$$

$$= \left[\frac{- \alpha \Delta x_{i+1/2} \cos\theta}{\ln\left(1 + \frac{\alpha \Delta x_{i+1/2} \cos\theta}{T_1}\right)}\right] \left[\frac{- \frac{q}{\alpha} \ln\left(1 + \frac{\alpha \Delta x_{i+1/2} \cos\theta}{T_1}\right)}{\Delta x_{i+1/2}}\right]$$

$$= q \cos\theta \quad , \tag{38}$$

which is the expected result. The results in the y direction are analogous. In addition, this proof can easily be extended to three dimensions, where T is replaced by K, the hydraulic conductivity, and q, volumetric flux per unit width, is replaced by a volumetric flux per unit area or specific discharge.

Thus, using the analytically calculated heads at the nodes and the logarithmic mean T, the flux is computed exactly. Conversely, if the flux is specified, through boundary conditions, then the heads at the nodes will be exact if logarithmic mean is used for the interblock transmissivity. This proof is also applicable to the "grid point distributed" variation (Settari and Aziz, 1972) of the finite-difference formulation for the case of uniform parallel flow because the transmissivity is assumed to vary linearly between nodes, regardless of the location of the block boundary.

Unconfined aquifer with uniform hydraulic conductivity

The analysis of the Theory Section indicates that for uniform flow under unconfined conditions, with a horizontal aquifer bottom and flow oriented in a grid direction, and with a constant hydraulic conductivity, the exact interblock transmissivity is computed as the (unweighted) artihmetic mean:

$$T_{xx(i+1/2,j)} = \frac{T_{xx(i,j)} + T_{xx(i+1,j)}}{2} \; ; \qquad T_{xx(i,j)}, T_{xx(i+1,j)} \neq 0 \tag{39}$$

It is recognized that the Dupuit approximation assumes that the change in head relative to the aquifer thickness is small. When the change in saturated thickness is small between nodes, then the differences between alternative interblock T functions is also small. However, examination of documented model applications indicate that the numerical model is likely to be applied to unconfined situations where the change in saturated thickness may be significant.

Following the proof of the previous section, we can show that use of the arithmetic mean transmissivity yields an exact solution for unidirectional unconfined flow in any direction in a two-dimensional grid, if the aquifer bottom is flat and the hydraulic conductivity is uniform. As above, head changes only in the direction of flow, s, and is given by equation (20). Thus, for flow at an angle θ to the grid, the head at node (i+1,j) is:

$$h_{i+1,j} = \left(h_1^2 - \frac{2q\Delta x_{i+1/2} \cos\theta}{K} \right)^{1/2} , \tag{40}$$

where the head at node (i,j) is $h_{i,j} = h_1$. The transmissivity at node (i,j) is Kh_1, the transmissivity at node (i+1,j) is given by (40) times K, and the arithmetic mean transmissivity $T_{xx(i+1/2,j)}$ is

$$T_{xx(i+1/2,j)} = 0.5\,K\left[h_1 + \left(h_1^2 - \frac{2q\Delta x_{i+1/2} \cos\theta}{K} \right)^{1/2} \right] , \tag{41}$$

assuming that the aquifer bottom has elevation zero. Substituting (40) and (41) into the finite-difference expression for the flux at the block interface (equation (31)) yields:

$$- T_{xx(i+1/2,j)} \frac{(h_{i+1,j} - h_{i,j})}{\Delta x_{i+1/2}} =$$

$$= \frac{- K}{2\Delta x_{i+1/2}} \left[\left(h_1^2 - \frac{2q\Delta x_{i+1/2} \cos\theta}{K} \right)^{1/2} + h_1 \right] \left[\left(h_1^2 - \frac{2q\Delta x_{i+1/2} \cos\theta}{K} \right)^{1/2} - h_1 \right]$$

$$= q \cos\theta \quad , \qquad (42)$$

which is the desired result. Thus, under the special conditions of unconfined flow, flat aquifer bottom, no recharge, and unidirectional uniform flow, use of the arithmetic mean interblock transmissivity yields the exact flow equation solution regardless of the orientation of the finite-difference grid. Furthermore, if uniform recharge is occurring, this approach will also yield the exact flux at the location halfway between the nodes. That is, if the finite-difference blocks are uniform, the method is also exact for unidirectional flow with recharge.

Unconfined aquifer with hydraulic conductivity varying linearly between nodes

The Theory Section shows that for uniform unconfined flow in a grid direction, with a horizontal aquifer bottom at elevation zero and no recharge, the exact effective transmissivity is the arithmetic mean potentiometric head times the logarithmic mean hydraulic conductivity. For a general model with variable aquifer bottom elevation, but ignoring the slope of the aquifer bottom, the finite-difference interblock transmissivity is:

$$T_{xx(i+1/2,j)} = \frac{(h_{i,j} - b_{i,j}) + (h_{i+1,j} - b_{i+1,j})}{2} K_{xx(i+1/2,j)} \quad , \qquad (43)$$

where

$$K_{xx(i+1/2,j)} = \frac{K_{xx(i+1,j)} - K_{xx(i,j)}}{\ln\left[\frac{K_{xx(i+1,j)}}{K_{xx(i,j)}}\right]} \quad ; \qquad \begin{array}{l} K_{xx(i,j)}, K_{xx(i+1,j)} \neq 0 \\ K_{xx(i,j)} \neq K_{xx(i+1,j)} \end{array} \quad . \qquad (44)$$

and where b [L] is the elevation of the aquifer bottom. This form is more general than equation (39) above and reduces to equation (39) when the hydraulic conductivity is

uniform. In addition, this form reduces to equation (32) for a confined system with uniform saturated thickness when the potentiometric head in (43) is replaced by the aquifer top elevation. For these reasons, equation (43) may be preferred for situations where one of the nodes is unconfined but the other is confined, although further investigation of this case may be warranted.

The computational effort for this interblock transmissivity is more than the previously considered methods. In effect, the effort for this case is comparable to the sum of the efforts for the arithmetic and logarithmic means, because both means are computed here. However, for most cases this increase is probably insignificant compared to the computational effort associated with solving the system of finite-difference equations. The additional computational effort associated with this interblock transmissivity could be minimized for transient problems by storing the logarithmic mean interblock hydraulic conductivity.

Following the proof of the previous section, we can show that use of the arithmetic mean transmissivity yields an exact solution for unidirectional unconfined flow in any direction in a two-dimensional grid, if the aquifer bottom is horizontal and has elevation zero, and the hydraulic conductivity varies linearly in the direction of flow. As above, the direction of the head gradient is parallel to the direction of flow, s, and head is given by equation (27). Thus, for flow at an angle θ to the grid, the head at node (i+1,j) is:

$$h_{i+1,j} = \left[h_1^2 - \frac{2q}{\alpha_k} \ln\left(1 + \frac{\alpha_k \Delta x_{i+1/2} \cos\theta}{K_1}\right)\right]^{1/2} , \qquad (45)$$

and the hydraulic conductivity at node (i,j+1) is:

$$K_{xx(i+1,j)} = K_1 + \alpha_k \Delta x_{i+1/2} \cos\theta . \qquad (46)$$

The logarithmic mean hydraulic conductivity at (i+1/2,j) is, substituting (46) and $K_1 = K_{xx(i,j)}$ into (44):

$$K_{xx(i+1/2,j)} = \frac{\alpha_k \Delta x_{i+1/2} \cos\theta}{\ln\left(1 + \frac{\alpha_k \Delta x_{i+1/2} \cos\theta}{K_1}\right)} . \qquad (47)$$

The interblock transmissivity is the arithmetic mean saturated thickness times the logarithmic mean hydraulic conductivity,

$$T_{xx(i+1/2,j)} = h_{a(i+1/2,j)} K_{xx(i+1/2,j)} \quad , \tag{48}$$

where the arithmetic mean saturated thickness is

$$h_{a(i+1/2,j)} = \frac{h_1 + h_{i+1,j}}{2} = 0.5 \left\{ h_1 + \left[h_1^2 - \frac{2q}{\alpha_k} \ln\left(1 + \frac{\alpha_k \Delta x_{i+1/2} \cos\theta}{K_1}\right)\right]^{1/2} \right\} \quad , \tag{49}$$

Substituting (45), (48) and (49) into the finite-difference expression for the flux at the block interface (equation (31)) yields,

$$- T_{xx(i+1/2,j)} \frac{(h_{i+1,j} - h_{i,j})}{\Delta x_{i+1/2}} =$$

$$- 0.5 \left\{ h_1 + \left[h_1^2 - \frac{2q}{\alpha_k} \ln\left(1 + \frac{\alpha_k \Delta x_{i+1/2} \cos\theta}{K_1}\right)\right]^{1/2} \right\} \quad \frac{\alpha_k \Delta x_{i+1/2} \cos\theta}{\ln\left(1 + \frac{\alpha_k \Delta x_{i+1/2} \cos\theta}{K_1}\right)}$$

$$\left\{ \left[h_1^2 - \frac{2q}{\alpha_k} \ln\left(1 + \frac{\alpha_k \Delta x_{i+1/2} \cos\theta}{K_1}\right)\right]^{1/2} - h_1 \right\} \Delta x_{i+1/2}^{-1}$$

$$= q \cos\theta \quad , \tag{50}$$

which is the desired result. Thus, under the special conditions of unidirectional uniform unconfined flow, where flow is parallel to the direction of changing K, and horizontal aquifer bottom, use of the arithmetic mean saturated thickness times the logarithmic mean hydraulic conductivity for interblock transmissivity yields the exact solution regardless of the orientation of the finite-difference grid.

EVALUATION OF APPROACH

In this section, the relative accuracies of the alternative interblock transmissivity functions are illustrated for two- and three-dimensional simulations. Results are first shown for several two-dimensional simulations of unidirectional flow with and without recharge. Because the exact solution for these problems is one-dimensional flow, simple analytical solutions can be used to assess model accuracy. Next, results are compared for a three-dimensional system for which the analytical solution is unavailable. In this case, coarse grid solutions are compared with solutions using a very fine grid. For all of these cases, errors are illustrated by tables of potentiometric head at certain nodes, and by tables of a simple error statistic for all computed heads and fluxes. This average error statistic is the sum of the absolute values of the difference between the model result and the exact result, divided by the sum of the absolute values of the exact result. For example, the average head error statistic is:

$$\varepsilon_h = \frac{\sum | h_{model} - h_{exact} |}{\sum | h_{exact} |} \quad . \tag{51}$$

For the cases for which the analytical solution is not available, the fine grid result is used as an estimate of the exact value. Similar statistics are computed for the applicable flux components. Finally, a previously completed three-dimensional large-scale simulation is repeated using the alternative interblock transmissivity functions. This example illustrates the practical effects of using different interblock transmissivity functions. The example simulations are summarized in table 2.

Two-Dimensional Examples with Analytical Solutions

Several simple two-dimensional examples are presented where flow is unidirectional and a one-dimensional analytical solution is available. However, the flow is oriented at an angle to the grid to activate two-dimensional aspects of the numerical solution. The numerical grid (fig. 7) is 5 rows (y) by 5 columns (x). Specified flux

Table 2.——Summary of example simulations

Case	Flow conditions	Spatial variability	Most accurate interblock T	Analytical solution
2D-C1	confined uniform flow at 30° to x axis	T linear in s	logarithmic mean	$h(s) = h_1 - \frac{q}{\alpha} \ln\left(1 + \frac{\alpha}{T_1} s\right)$
2D-C2	confined flow at 30° to x axis, with recharge	T linear in s	logarithmic mean	$h(s) = h_1 - \frac{q_1}{\alpha} \ln\left(1 + \frac{\alpha}{T_1} s\right) - \frac{W}{\alpha}\left[s - \frac{T_1}{\alpha} \ln\left(1 + \frac{\alpha}{T_1} s\right)\right]$
2D-U1	unconfined uniform flow at 30° to x axis	Uniform K	arithmetic	$(h(s))^2 = h_1^2 - \frac{2qs}{K}$
2D-U2	unconfined flow at 30° to x axis, with recharge	Uniform K	arithmetic	$(h(s))^2 = h_1^2 - \frac{2q_1 s}{K} - \frac{Ws^2}{K}$
2D-U3	unconfined uniform flow at 30° to x axis	K linear in s	arithmetic saturated thickness times logarithmic mean K	$(h(s))^2 = h_1^2 - \frac{2q}{\alpha_k} \ln\left(1 + \frac{\alpha_k s}{K_1}\right)$
3D	confined nonuniform flow, no recharge	smoothly heterogeneous	logarithmic mean	none (approximated by fine grid)
Avra	unconfined, with recharge	smoothly heterogeneous in horizontal		none

Notes: s is coordinate aligned with flow direction, T is transmissivity, K is hydraulic conductivity, h is potentiometric head, W is recharge rate per unit area, α is rate of change of transmissivity in flow direction ($\partial T/\partial s$), α_k is rate of change of hydraulic conductivity in flow direction ($\partial K/\partial s$), q is volumetric flux per unit width. Subscript 1 indicates value of variable at s=0.

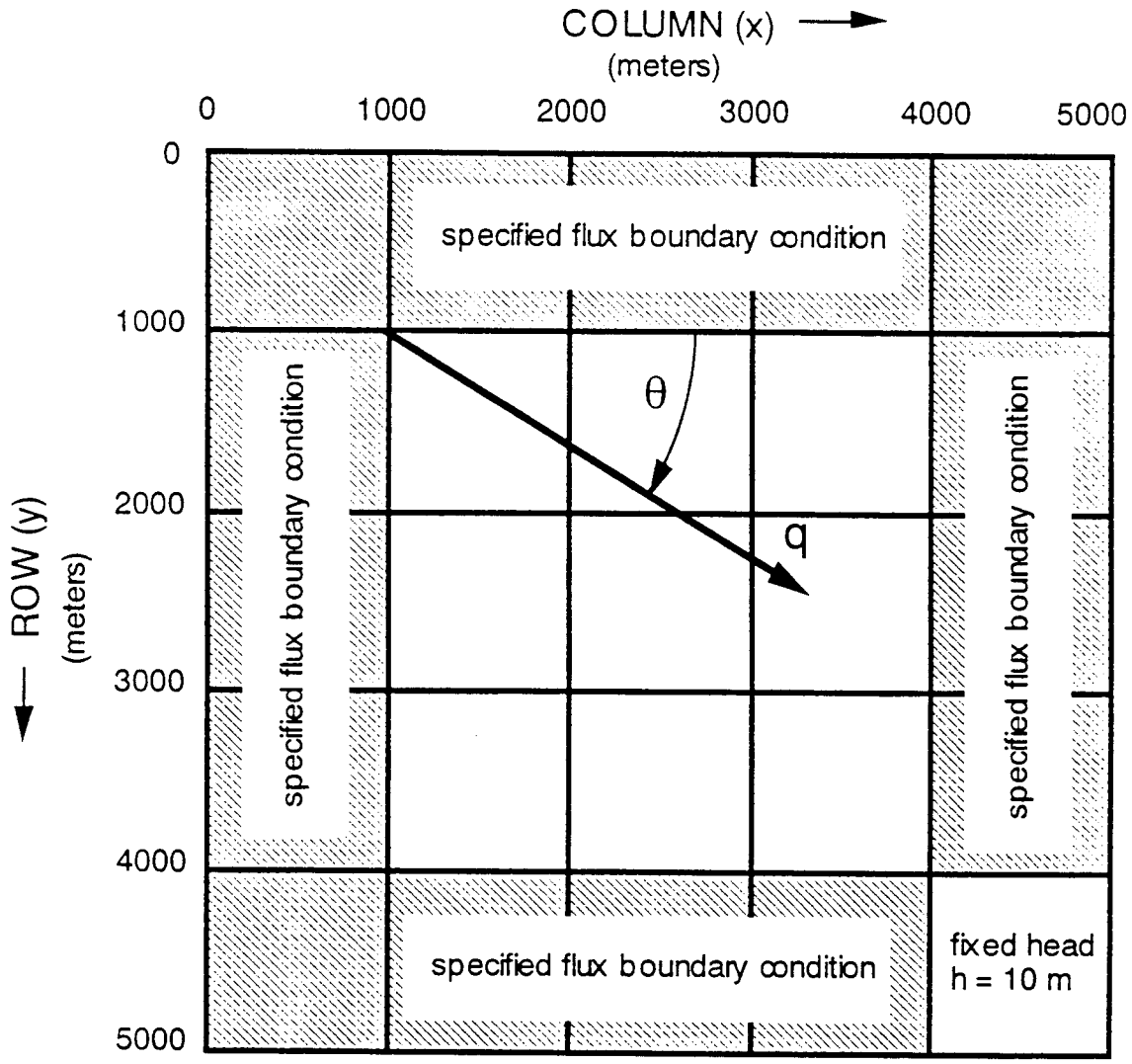


Figure 7.--Schematic of grid for two-dimensional simulations showing boundary conditions and orientation of the parallel flow vector q at angle θ to the x axis. For the simulations presented here, θ = 30 degrees.

boundary conditions are used at all boundary nodes except the node at row=5 and column 5, where the head is specified as h(5,5)=10 m. The grid spacing is $\Delta x = \Delta y = 1000$ m. For these cases, the exact solution is unidirectional flow at an angle of 30 degrees from the x axis, in the direction of increasing row and column indices.

Transmissivity a linear function of distance in flow direction, no recharge (case 2D-C1)

This case has no recharge and corresponds to the proof above showing that the linear effective mean transmissivity is exact as long as flow is unidirectional. The uniform flow rate is $q = 10^{-3}$ m^2/day, but the flux boundary conditions are nonuniform because flow is oriented at 30° to the grid (table 3). The transmissivity at node (1,1) is $T(1,1) = 10^{-2}$ m^2/day and transmissivity increases in the direction of flow (s) at a rate of $\alpha = 3 \times 10^{-5}$ m^2/day/m (table 4).

In the Numerical Methods section, we showed that the logarithmic mean transmissivity would yield an exact solution to this problem, where transmissivity is changing in the direction of flow only, and no recharge occurs. Tables 5 and 6 show that this is indeed the case, and indicate the relative errors caused by using other interblock transmissivity functions. The errors caused by using the harmonic mean is about twice the errors caused by using the arithmetic mean. The harmonic mean generally yields heads that are too high, while the arithmetic mean generally yields heads that are too low.

Table 3.——Specified volumetric flux (m^3/day) into boundary cells for cases 2D-C1, 2D-U1 and 2D-U3, flow direction 30° to grid, no recharge

	column				
row	1	2	3	4	5
1	1.366	0.500	0.500	0.500	-0.366
2	0.866				-0.866
3	0.866				-0.866
4	0.866				-0.866
5	0.366	-0.500	-0.500	-0.500	

Table 4.——Transmissivity (m^2/day) at nodes for cases 2D-C1 and 2D-C2, transmissivity a linear function of distance in flow direction 30° to grid

	column				
row	1	2	3	4	5
1	0.010	0.036	0.062	0.088	0.114
2	0.025	0.051	0.077	0.103	0.129
3	0.040	0.066	0.092	0.118	0.144
4	0.055	0.081	0.107	0.133	0.159
5	0.070	0.096	0.122	0.148	0.174

Table 5.——Head (m) in row 1 for case 2D-C1, transmissivity a linear function of distance in flow direction, no recharge [least accurate mean indicated by italics]

	column				
Interblock T mean	1	2	3	4	5
	head (m) in row 1				
harmonic	*115.5*	*62.80*	*44.33*	*32.61*	*23.95*
exact and logarithmic	105.2	62.52	44.40	32.73	24.10
arithmetic	100.8	62.35	44.43	32.79	24.18

Table 6.——Error for case 2D-C1, transmissivity a linear function of distance in flow direction, no recharge [least accurate mean indicated by italics]

	Model output component		
Interblock T mean	head	q_x	q_y
	average error statistic (dimensionless)		
harmonic	*0.0207*	*0.0106*	*0.0196*
logarithmic	0	0	0
arithmetic	0.0092	0.0049	0.0089

Transmissivity a linear function of distance in flow direction, with recharge (case 2D-C2)

This problem is identical to the previous case except that a uniform recharge rate of $W = 2 \times 10^{-7}$ m/day is applied over the entire simulated area. The flux at the outside corner of the cell centered at (1,1) is $q = 10^{-3}$ m^2/day. Again, flow is at 30 degrees from the x axis. However, for this case the flux increases linearly with distance in the flow direction due to the uniform recharge. The nonuniform specified flux boundary conditions are given in table 7. The transmissivity at node (1,1) is $T(1,1) = 10^{-2}$ m^2/day and transmissivity increases linearly with distance in the flow direction at a rate of $\alpha = 3 \times 10^{-5}$ m^2/day/m, as in the previous case (table 4).

For this case of uniform recharge, none of the interblock transmissivity functions yields the exact solution, but the logarithmic mean is the most accurate, followed by the arithmetic mean (tables 8 and 9). As with the previous case, the harmonic mean yields the least accurate solution in terms of both computed heads and fluxes.

Table 7.——Specified volumetric flux (m^3/day) into boundary cells for cases 2D-C2 and 2D-U2, flow direction 30° to grid, with recharge

	column				
row	1	2	3	4	5
1	1.4526	0.6299	0.7165	0.8031	-0.7696
2	0.9959				-1.7459
3	1.0825				-1.8325
4	1.1691				-1.9191
5	0.4624	-0.8799	-0.9665	-1.0531	

Table 8.——Head (m) in row 1 for case 2D-C2, transmissivity a linear function of distance in flow direction, with recharge [least accurate mean indicated by italics]

	column				
Interblock T mean	1	2	3	4	5
	head (m) in row 1				
exact	148.3	96.85	71.69	53.43	38.42
harmonic	*161.8*	*97.58*	*71.76*	*53.33*	*38.22*
logarithmic	149.1	96.99	71.74	53.44	38.41
arithmetic	143.6	96.66	71.71	53.49	38.50

Table 9.——Error for case 2D-C2, transmissivity a linear function of distance in flow direction, with recharge [least accurate mean indicated by italics]

Interblock T mean	Model output component: head	q_x	q_y
	average error statistic (dimensionless)		
harmonic	*0.0198*	*0.0086*	*0.0157*
logarithmic	0.0034	0.0005	0.0010
arithmetic	0.0064	0.0030	0.0057

Homogeneous unconfined aquifer, no recharge (case 2D-U1)

For this unconfined case, transmissivity is modeled as a linear function of head, and the hydraulic conductivity is assumed to be spatially uniform and equal to $K = 10^{-3}$ m/day. The uniform flow rate is $q = 10^{-3}$ m^2/day, and the aquifer bottom elevation is zero. The nonuniform flux boundary conditions are the same as case 2D-C1 (table 3).

Table 10.——Head (m) in row 1 for case 2D-U1, homogeneous unconfined aquifer, no recharge [least accurate mean indicated by italics]

Interblock T mean	column 1	2	3	4	5
	head (m) in row 1				
harmonic	*110.5*	*102.4*	*93.45*	*83.57*	*72.33*
logarithmic	106.2	97.69	88.36	77.90	65.80
exact and arithmetic	105.0	96.42	86.97	76.37	64.03

The arithmetic mean transmissivity yields the exact solution, as expected (tables 10 and 11). The harmonic mean, because it heavily weights the low transmissivity, yields the least accurate solution and generally results in heads that are too high.

Table 11.——Error for case 2D-U1, homogeneous unconfined aquifer, no recharge [least accurate mean indicated by italics]

	Model output component		
Interblock T mean	head	q_x	q_y
	average error statistic (dimensionless)		
harmonic	*0.1057*	*0.0161*	*0.0286*
logarithmic	0.0234	0.0044	0.0080
arithmetic	0	0	0

Homogeneous unconfined aquifer, with recharge (case 2D-U2)

This case is identical to the previous unconfined case except that uniform recharge occurs throughout the modeled domain. The hydraulic conductivity is spatially uniform and equal to $K = 10^{-3}$ m/day and the aquifer bottom elevation is zero. The flux at the outside corner of the cell centered at (1,1) is $q = 10^{-3}$ m^2/day, and the uniform recharge rate is $W = 2 \times 10^{-7}$ m/day. The nonuniform flux boundary conditions are the same as case 2D-C2 (table 7).

As shown in the Numerical Methods section, the use of the arithmetic mean for interblock transmissivity yields the exact solution for this problem because the grid blocks are uniform (tables 12 and 13). As with the previous example, use of the harmonic mean yields the least accurate solution and generally yields heads that are too high.

Table 12.——Head (m) in row 1 for case 2D-U2, homogeneous unconfined aquifer, with recharge [least accurate mean indicated by italics]

	column				
Interblock T mean	1	2	3	4	5
	head (m) in row 1				
harmonic	*154.7*	*147.6*	*139.1*	*128.8*	*116.4*
logarithmic	139.0	131.1	121.5	109.7	94.86
exact and arithmetic	136.0	128.0	118.1	106.0	90.65

Table 13.——Error for case 2D-U2, homogeneous uncofined aquifer, with recharge [least accurate mean indicated by italics]

	Model output component		
Interblock T mean	head	q_x	q_y
	average error statistic (dimensionless)		
harmonic	*0.2424*	*0.0275*	*0.0481*
logarithmic	0.0406	0.0077	0.0138
arithmetic	0	0	0

Unconfined aquifer with hydraulic conductivity a linear function of distance in flow direction, no recharge (case 2D-U3)

This example is similar to case 2D-U1, an unconfined aquifer with no recharge, except that here the hydraulic conductivity is assumed to be a linear function of the flow direction coordinate. The uniform flux is $q = 10^{-3}$ m^2/day and the aquifer bottom elevation is taken as zero. The hydraulic conductivity at node (1,1) is $K(1,1) = 10^{-4}$ m/day and the hydraulic conductivity increases in the direction of flow at a rate $\alpha_k = 3 \times 10^{-6}$ m/day/m (table 14).

Table 14.——Hydraulic conductivity (m/day) at nodes for case 2D-U3, hydraulic conductivity a linear function of distance in flow direction 30° to grid

	column				
row	1	2	3	4	5
1	0.0001	0.0027	0.0053	0.0079	0.0105
2	0.0016	0.0042	0.0068	0.0094	0.0120
3	0.0031	0.0057	0.0083	0.0109	0.0135
4	0.0046	0.0072	0.0098	0.0124	0.0150
5	0.0061	0.0087	0.0113	0.0139	0.0165

As shown in the Numerical Methods section, use of the arithmetic mean saturated thickness times the logarithmic mean hydraulic conductivity for the interblock transmissivity yields the exact solution (tables 15 and 16). The harmonic mean yields the least accurate results and generally yields heads that are too high. The logarithmic mean interblock transmissivity yields a more accurate solution than the arithmetic mean, indicating that, for this particular problem, incorporating the spatial variability of hydraulic conductivity is more important than accounting for the dependence of transmissivity on head.

Table 15.——Head (m) in row 1 for case 2D-U3, unconfined aquifer with hydraulic conductivity a linear function of distance in flow direction, no recharge

[least accurate mean indicated by italics]

	column				
Interblock T	1	2	3	4	5
	head (m) in row 1				
harmonic mean T	*83.96*	*35.87*	*29.10*	*24.19*	*19.91*
logarithmic mean T	61.72	36.19	29.31	24.34	20.04
arithmetic mean T	53.75	36.36	29.42	24.41	20.10
exact and arithmetic mean h times logarithmic mean K	59.19	36.15	29.28	24.31	20.04

Table 16.——Error for case 2D-U3, unconfined aquifer with hydraulic conductivity a linear function of distance in flow direction, no recharge

[least accurate mean indicated by italics]

	Model output component		
Interblock T	head	q_x	q_y
	average error statistic (dimensionless)		
harmonic mean T	*0.0479*	*0.0266*	*0.0473*
logarithmic mean T	0.0073	0.0046	0.0083
arithmetic mean T	0.0114	0.0069	0.0119
arithmetic mean h times logarithmic mean K	0	0	0

Heterogeneous Three-Dimensional Confined Aquifer (case 3D)

This case also uses a 5 row (Δy = 1000 m) by 5 column (Δx = 1000 m) grid, but additionally has 4 layers (Δz = 10 m) to illustrate performance of the alternative interblock transmissivity functions in three-dimensional flow. The steady-state boundary conditions include specified uniform flux inward (specific discharge = 10^{-4} m/day) in column 1 over all layers and rows, specified uniform head (10 m) in column 5 of layer 1, and no flow across all other model boundaries (fig. 8). These boundary conditions result in flow generally from the left (column 1) to the right with increasing upward flow towards the right. The bottom two layers have a uniform hydraulic conductivity of $K_1 = 10^{-3}$ m/day. In the top two layers (fig. 8), the first three rows, and the first two columns of rows 4 and 5, have $K = K_1$. The remaining nodes -- columns 3, 4, and 5 of rows 4 and 5 -- have $K = 10 \times K_1 = 10^{-2}$ m/day. The ratio of vertical hydraulic conductivity to horizontal hydraulic conductivity (anisotropy ratio) is 10^{-3}. Conceptually, the hydraulic conductivity is assumed to vary linearly between the nodes. The interblock vertical hydraulic conductivity, which is input explicitly, is computed with the same mean as the horizontal transmissivity.

Because an analytical solution is not available for this problem, the exact solution is estimated by a very fine grid. In this fine grid, which is designated 9x, each of the blocks of the coarse grid is replaced by 9 rows by 9 columns by 9 layers. Thus, the entire simulation area of the 9x grid is 45 rows by 45 columns by 36 layers. At this fine scale, the assumed linear variation of hydraulic conductivity between the coarse grid nodes is input explicitly.

The heads computed at selected nodes with the coarse grids are compared with the 9x results at the appropriate nodes in tables 17 and 18. Because the change in hydraulic conductivity between nodes is small for the 9x grid, the alternative interblock functions yield very similar results. The exact results for these comparisons is estimated by the average results from the three 9x simulations with the alternative interblock functions. For comparison of fluxes, the fine grid results are integrated over the area of the coarse grid block faces. At the scale of the coarse grid the alternative interblock transmissivities yields significantly different results. The average errors (table 19) are larger for this case than the two-dimensional simulations. The harmonic mean yields the least accurate solution.

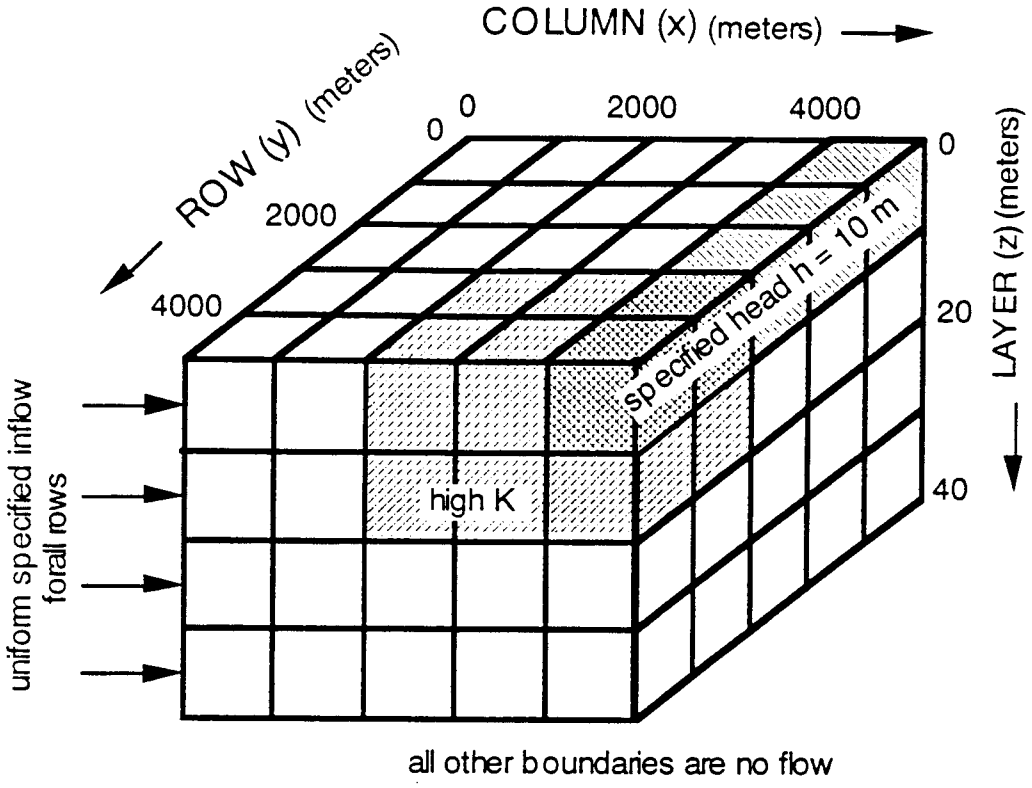


Figure 8.--Schematic of grid for three-dimensional simulation showing boundary conditions and shaded cells where node value of hydraulic conductivity is 10 times the value at other nodes.

This case shows that both the choice of interblock hydraulic conductivity and the level of spatial discretization (grid spacing) control model accuracy. When a fine grid is used, the change in hydraulic conductivity from block to block is small, hence the interblock hydraulic conductivities calculated using any method are essentially the same. Furthermore, and perhaps more importantly, solving for the potentiometric head at many additional nodes give a more accurate approximation to the curving head contours and streamlines where hydraulic conductivity is spatially variable. It is clearly desirable to have sufficiently fine spatial discretization such that the choice of interblock hydraulic conductivity is immaterial. However, when moderate or even coarse discretization are used, some improved accuracy can be obtained by choosing appropriate functions for interblock hydraulic conductivity, such as the ones presented in this report.

Table 17.——Head (m) in row 3, layer 2 for case 3D, heterogeneous confined aquifer, no recharge

	column				
Interblock T mean	1	2	3	4	5
	head (m) in row 3, layer 2				
harmonic	302.2	205.0	118.0	64.42	20.56
logarithmic	279.6	183.5	102.5	56.67	18.73
arithmetic	272.2	176.4	97.64	54.37	18.08
harmonic 9x	275.3	175.3	101.6	58.69	22.77
logarithmic 9x	274.9	175.0	101.3	58.52	22.70
arithmetic 9x	274.7	174.8	101.1	58.44	22.66

Table 18.——Head (m) in column 3, layer 3 for case 3D, heterogeneous confined aquifer, no recharge

	row				
Interblock T mean	1	2	3	4	5
	head (m) in column 3, layer 3				
harmonic	158.3	147.2	121.5	82.38	75.33
logarithmic	146.9	134.8	107.9	79.99	73.63
arithmetic	143.3	130.9	103.7	80.02	73.67
harmonic 9x	148.3	135.2	106.1	83.44	76.73
logarithmic 9x	148.1	135.0	106.0	83.42	76.72
arithmetic 9x	148.0	134.9	105.9	83.42	76.72

Table 19.——Error for case 3D, heterogeneous confined aquifer, no recharge

[least accurate mean indicated by italics]

Interblock T mean	Model output component			
	head	q_x	q_y	q_z
	average error statistic (dimensionless)			
harmonic	*0.1062*	*0.1159*	*0.1868*	*0.3053*
logarithmic	0.0237	0.0964	0.1287	0.1490
arithmetic	0.0351	0.1040	0.1384	0.1449

Simulation from Study of Avra Valley Aquifer, Arizona (case Avra)

Alternative interblock transmissivities are used to resimulate predevelopment steady-state flow in the Avra Valley aquifer to illustrate the practical significance of these methods. Hanson and others (1990) describe the hydrogeologic setting and details of the numerical simulations which are only briefly summarized here. Hanson and others (1990, p. 4) report:

> the alluvium is subdivided into lower and upper units on the basis of hydrogeologic characteristics. The lower alluvium is thousands of feet thick, consists of gravel and conglomerate along the basin margins and in the southern part of the basin, and grades into gypsiferous and anhydritic clayey silt and mudstone in the north-central part of the basin. The upper alluvium consists mainly of gravel, sand, and clayey silt, and ranges from less than [30 to about 300 m] in thickness. . . . Fan and playa environments are generally characterized by clay and silt concentrations of less than 20 percent and more than 60 percent, respectively. The interfingered-zone subregion generally contains from 20 to 60 percent clay and silt. This subregion was subdivided into two adjacent zones with 20 to 40 and 40 to 60 percent clay and silt . . .

The system was modeled using the program of McDonald and Harbaugh (1988). Two model layers were used, layer 1 representing the unconfined upper hydrogeologic layer, and layer 2 representing the confined lower hydrogeologic layer. The finite-difference grid blocks are large, 1 mile on each side. Hanson and others (1990) calibrated the steady-state model by adjusting the transmissivity of layer 2 and the hydraulic conductivity of layer 1 until a reasonable agreement with observed predevelopment (around 1940) heads was achieved. Direct recharge, streamflow infiltration, and pumping were all negligible under predevelopment conditions. All modeled inflow to the system occurs at constant head nodes and represents inflow from adjacent aquifers. Calibration yielded a (Hanson and others, 1990, p. 18)

> root-mean-square difference between measured and simulated water levels in 100 wells [of] 16 ft (5 m). . . . Most errors were negative, indicating that simulated water levels were slightly higher than measured. The largest negative errors occurred mainly in the southwestern part of the model area . . . where estimates of heads and aquifer components are less certain than elsewhere. Differences between hand-drawn contours of measured water levels and simulated head for layer 1 generally ranged from 5 to 10 ft (1.5 to 3 m) [in the] north . . .to 20 ft (6 m) [in the] south . . . with the largest difference of about 40 ft (12 m) in the south half of the valley.

For resimulation, the only modification to the predevelopment steady-state simulation of Hanson and others (1990) is the method to compute horizontal interblock transmissivity. The original model of McDonald and Harbaugh (1988) uses the harmonic mean interblock transmissivity. Here, we choose the interblock mean separately for each layer, on the basis of the character of spatial variability for that layer. Layer 1 is unconfined and has hydraulic conductivity that varies spatially due to variability in gravel content associated with stream channels (fig. 9). The hydraulic conductivity of layer 1 ranges from less than 0.2 m/day to over 30 m/day. The transmissivity of layer 1 varies spatially due to the spatially variable hydraulic conductivity and the dependence of transmissivity on saturated thickness, and hence on head. On this basis, we choose to compute the interblock transmissivity as the arithmetic mean saturated thickness times the logarithmic mean hydraulic conductivity (LAYAVG=30, input LAYCON=31). Layer 2 is confined and has spatially varying transmissivity due to the gradation of composition and interfingering of basin-margin alluvium with the lower permeability of the clay, silt and mudstone of the center of the basin (fig. 10). The transmissivity of layer 2 ranges from less than 9 m^2/day to over 400 m^2/day. Because the change in transmissivity between nodes is conceptualized

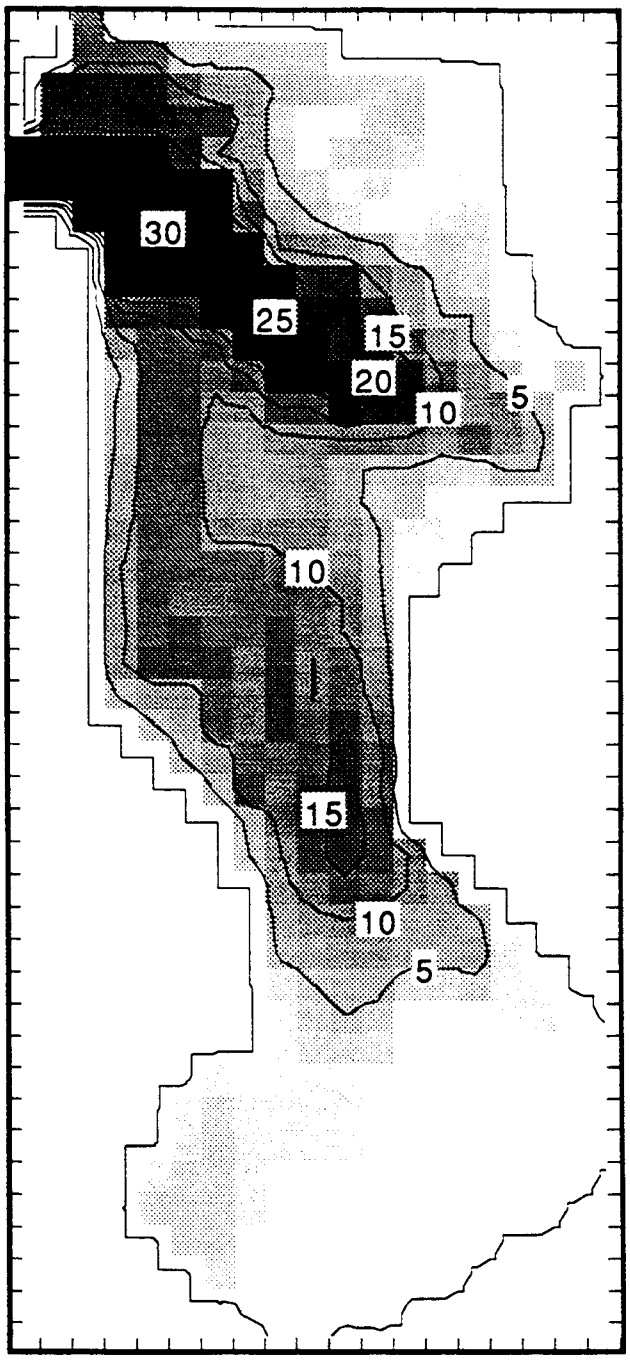


Figure 9. Contour and gray-scale map of hydraulic conductivity (in meters per day) of layer one of Avra Valley aquifer model. Model parameters from Hanson and others (1990).

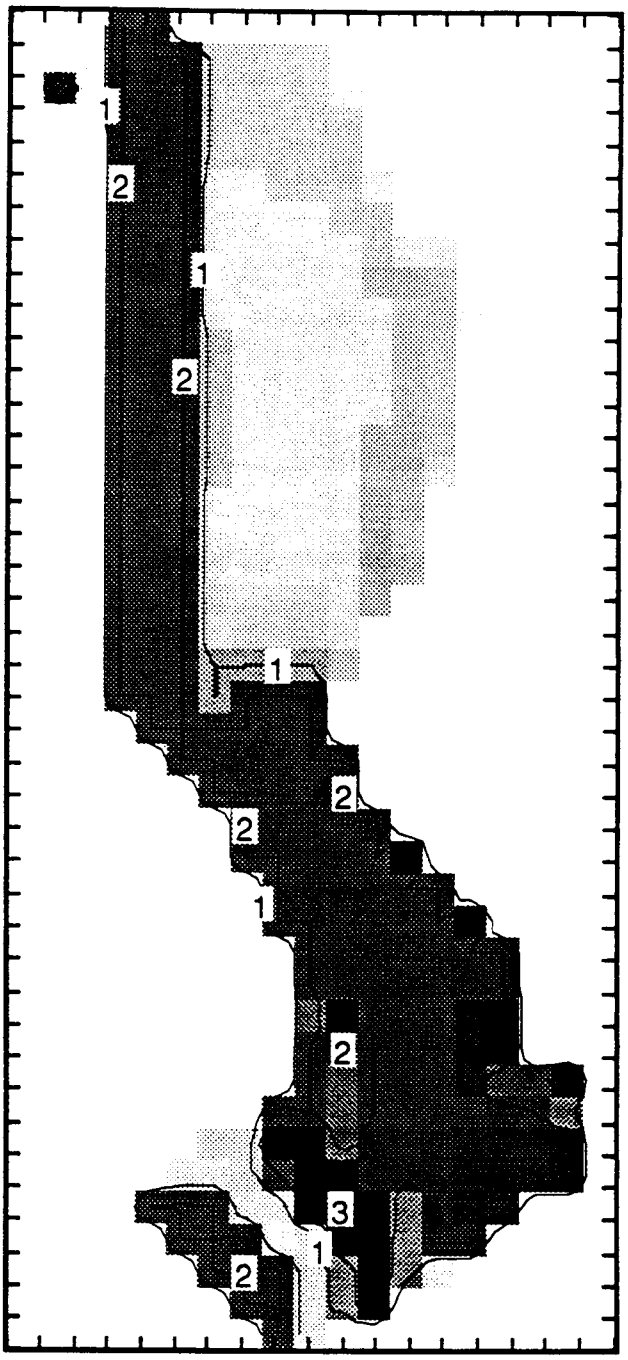


Figure 10. Contour and gray-scale map of transmissivity (in hundreds of meters squared per day) of layer two of Avra Valley aquifer model. Model parameters from Hanson and others (1990).

nodes is conceptualized as being somewhat smooth, and not abrupt, we use the logarithmic mean interblock transmissivity (LAYAVG=20, input LAYCON=20) for layer 2.

The effects of alternative interblock transmissivity functions on computed heads are relatively minor. Our resimulated heads (fig. 11) are generally higher (fig. 12), with the difference between our results and the results of Hanson and others (1990) ranging from about -1 m to about +4 m, averaging +1.2 m. Compared with the root-mean-squared calibration error of 5 m reported by Hanson and others (1990), these changes in head are small.

Because this simulation is controlled largely by specified heads at the inflow as well as outflow boundaries, simulated heads are relatively insensitive to transmissivity. However, fluxes show somewhat more significant changes because of the use of alternative interblock tranmissivity. The change in computed inflow and outflow at constant head nodes is shown in table 20. The differences in fluxes at individual nodes can be quite significant, up to 93 percent for a relatively low-flow node. However, the uncertainty inherent in computed fluxes is qualitatively greater than uncertainty in heads, because of the lack of direct measurements of aquifer discharge. Hence, these changes may still be within the bounds of uncertainty of the calibrated model. The net flow through the aquifer changes from 6.46×10^4 m^3/day to 6.74×10^4 m^3/day, an increase of about 4 percent. This result is consistent with the harmonic mean yielding lower interblock transmissivities than those used here. Overall, the interpretation of the model results are unlikely to change due to the use of the alternative interblock transmissivities.

The computational effort is slightly increased for the resimulation with alternative interblock transmissivity due to the additional operations required, particularly for the top layer with LAYAVG=30. However, this increase is relatively insignificant compared with the overall computational burden, representing an increase in total computer time of less than 3 percent for this case.

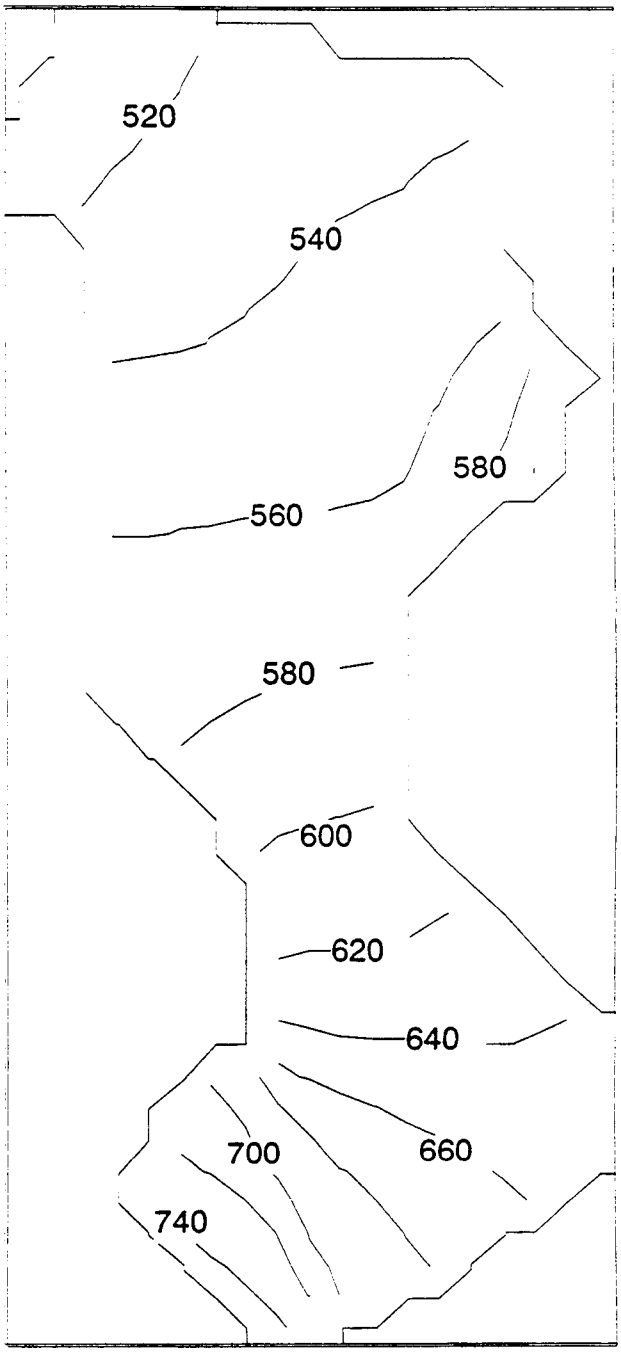


Figure 11. Contour map of new simulated head (in meters) of layer one of Avra Valley aquifer model using alternative interblock transmissivities. Contour interval is 20 meters.

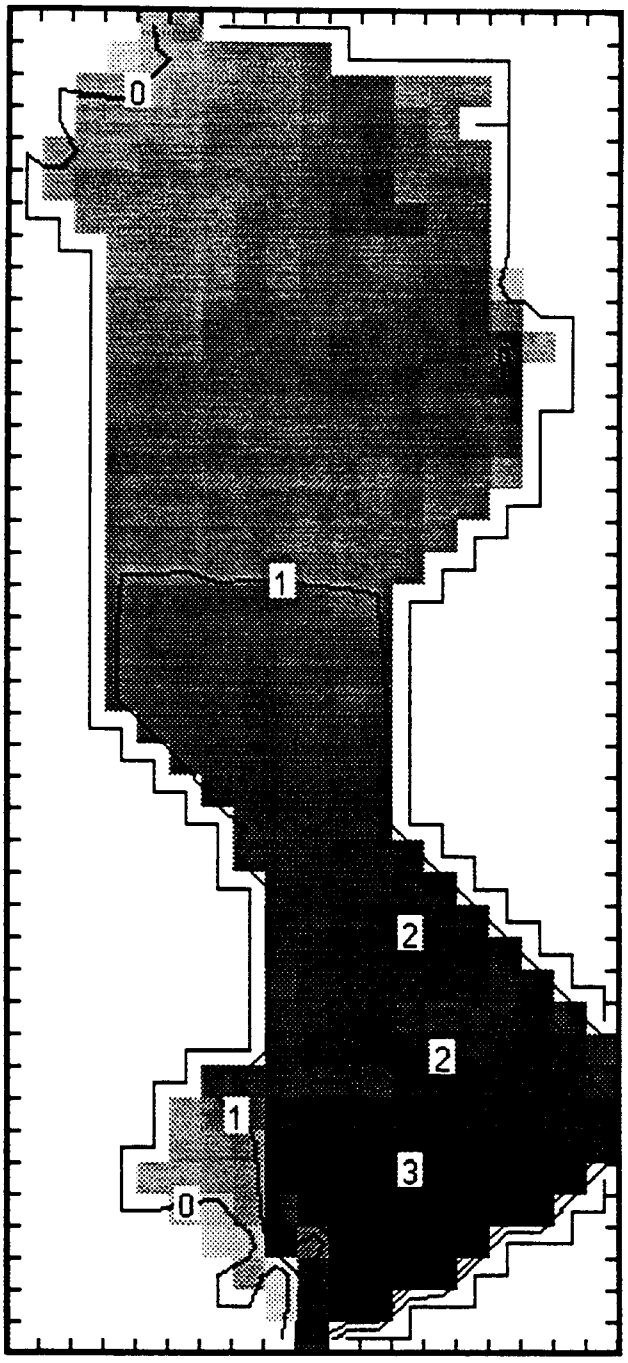


Figure 12. Contour and gray-scale map of simulated new head in layer one minus results of Hanson and others (1990) (in meters) for Avra Valley aquifer model

Table 20.——Computed fluxes at constant head boundary nodes for Avra Valley aquifer example simulation [original data from Hanson et al., 1990]

Node index			computed flux (m^3/hr)		Difference in computed fluxes	
layer	row	col	original	new	m^3/hr	percent
1	1	3	-57	-95	-38	67
1	2	3	-480	-480	0	0
1	3	2	-388	-386	1	0
1	4	2	-813	-862	-49	6
1	5	1	-642	-635	6	-1
1	6	1	-249	-256	-6	3
1	12	18	152	200	48	31
1	13	17	424	433	9	2
1	14	17	424	410	-14	-3
1	15	17	288	308	20	7
1	38	5	201	203	2	1
1	39	6	404	413	10	2
1	40	7	253	285	33	13
1	41	8	37	40	3	7
1	42	9	41	37	-4	-10
2	1	3	-12	-20	-8	65
2	2	3	-16	-27	-10	65
2	3	2	-9	-18	-9	93
2	12	17	-27	-32	-5	17
2	13	17	4	3	0	-6
2	38	5	23	24	1	3
2	39	6	119	122	3	2
2	40	7	111	111	0	0
2	41	8	117	119	2	2
2	42	9	93	103	9	10

(PAGE 55 FOLLOWS)

SUMMARY AND CONCLUSIONS

Some widely used finite-difference ground-water simulation computer programs use block-centered grids and the weighted harmonic mean of the block transmissivities for the interblock transmissivity. We have shown that the logarithmic mean transmissivity is the appropriate interblock transmissivity for one-dimensional flow in a confined aquifer having a transmissivity that varies linearly; whereas it has been known that the harmonic mean is the appropriate formulation where the transmissivity is spatially uniform in each finite-difference block but changes abruptly at the block interface. For the case of one-dimensional flow in an unconfined aquifer where the hydraulic conductivity is uniform, the appropriate interblock transmissivity is the unweighted arithmetic mean of transmissivities at the block centers; and where hydraulic conductivity varies as a linear function of distance in the direction of flow the appropriate interblock transmissivity is the arithmetic mean saturated thickness times the logarithmic mean hydraulic conductivity.

We tested the relative accuracies of different interblock transmissivities for selected hypothetical two-dimensional problems, involving confined and unconfined flow conditions, for which analytical solutions are available for comparison. We also considered a three-dimensional heterogeneous confined aquifer problem for which the results using a very fine grid (the test problem involved 100 blocks whereas the very fine grid involved 72,900 blocks) are taken to "represent" the exact unknown solution. For the two-dimensional hypothetical problems the grid block sizes used are uniform and for the hypothetical three-dimensional problem the grid block sizes in the plan view are uniform and the thickness of the blocks on the z-direction (or depth) is 1 percent of the horizontal block dimensions. We did not experiment with variable block size configurations. For the test problems selected, with uniform and linearly varying transmissivity or hydraulic conductivity, and the interblock transmissivities considered, the weighted harmonic mean formulation resulted in the least accurate computed heads. The "accuracy" criterion used for the comparisons was the sum of the absolute values for all blocks of the differences between the computed heads and "exact" heads, divided by the sum of the absolute values for all blocks of the "exact" heads. Flux component error statistics yielded the same results.

Alternative interblock transmissivities were used to resimulate predevelopment steady-state flow in the Avra Valley aquifer to illustrate the practical significance of these methods. A numerical model had been made previously for the predevelopment flow

using the harmonic mean interblock transmissivity formulation. For our resimulation the only change to the previous simulation is the method used to compute interblock transmissivity. For the unconfined (top) layer we computed the interblock transmissivity as the arithmetic mean saturated thickness times the logarithmic mean hydraulic conductivity. For the confined layer we used the logarithmic mean transmissivity formulation. The effects of the alternative interblock transmissivity functions on computed heads gave differences from the previous simulation results of from about -1 to +4 meters, averaging +1.2 meters, and an increase of net flow through the aquifer of about 4 percent. These differences in computed heads, compared to the root-mean-squared calibration error of 5 meters reported from the previous model calibration work, are small. At individual blocks the differences in the fluxes computed at constant head boundaries for the simulations varied by as much as 93 percent for a block where the flux was relatively low. We judged that the overall interpretation of the model results would likely not have changed as a result of the use of the alternative interblock transmissivities. The increased computational effort associated with the use of the alternative interblock transmissivity functions for the Avra Valley aquifer resimulation was less then 3 percent of the total computer time.

It is not an objective of this report to attempt to recommend a particular interblock transmissivity formulation for all field situations. There is little evidence to support any such universal approach. However, the results of the several hypothetical test problems that we considered suggest that for field situations where the spatial variation of the transmissivity of hydraulic conductivity is "nearly" linear and the block size dimensions areally are to be uniform, the alternative interblock transmissivity functions may give more accurate results than the weighted harmonic mean. The additional computational effort needed for the alternative transmissivity functions is nominal. One obvious limitation of the selected test problems and our comparative analysis was that the block sizes selected were uniform in the horizontal plane. Appel's (1976) results suggest that relative errors would be similar for variable spacing grids.

The block-by-block transmissivity values input to most ground-water models are estimates made by superimposing the selected model block grid and a map that characterizes the analyst's perception of the dominant large scale spatial structure of the transmissivity. It is accepted that small scale variations occur around this large scale structure. It is beyond the scope of this report to attempt to provide guidelines on how to develop such a large scale representation or to judge when such a representation is justified.

However, if such a map has been prepared and it represents the large scale variation as a continuous and smoothly varying function, the results of the test problems given here suggest that consideration of the interblock transmissivity functions other than the weighted harmonic mean may be justified.

Finally, the importance of spatial discretization in modeling flow in heterogeneous aquifers cannot be overstated. Different methods for computing interblock transmissivity yield significant differences only when the change in transmissivity from one node to the next is relatively large. In addition to making the choice of interblock transmissivity function immaterial, a fine spatial discretization (small blocks) yields a more accurate approximation to the curving head contours and streamlines where transmissivity is spatially variable. If model heads computed with alternative interblock transmissivities are significantly different, the modeler should reexamine the level of spatial discretization to ensure that a sufficiently accurate head solution is obtained.

(PAGE 59 FOLLOWS)

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APPENDIX A

PROOF THAT THE LOGARITHMIC MEAN TRANSMISSIVITY IS LARGER THAN THE GEOMETRIC MEAN TRANSMISSIVITY

We want to prove that the logarithmic mean interblock transmissivity is larger than the geometric mean transmissivity. We will restrict the cases to positive, unequal transmissivities. That is,

$$\text{Prove} \qquad \frac{T_2 - T_1}{\ln (T_2 / T_1)} > (T_1 T_2)^{1/2} \qquad \text{for } T_1, T_2 > 0 \; ; \; T_1 \neq T_2 \tag{A1}$$

It will be convenient to limit our cases to $T_2 > T_1$, but the subscripts can be interchanged so that this is always the case for two positive, unequal transmissivities. Consider the ratio of the logarithmic mean to the geometric mean, which can be rearranged to:

$$\frac{\dfrac{T_2}{T_1} - 1 \Big/ \ln \dfrac{T_2}{T_1}}{\left(\dfrac{T_2}{T_1}\right)^{1/2}} \tag{A2}$$

Defining $x = (T_2/T_1)^{1/2}$, this ratio can be written:

$$\frac{x^2 - 1 \Big/ \ln x^2}{x} = \frac{x^2 - 1}{2 x \ln x} = \frac{x - \dfrac{1}{x}}{2 \ln x} \tag{A3}$$

where it is understood that $x>1$ if $T_2>T_1$. Consider the function $F(x) = x - 1/x - 2 \ln x$ in the region $1<x$. Note that $F'(x) = (1 - 1/x)^2$. For $x>1$, $F'(x) > 0$, because the derivative is positive for all $x>1$ and the function $F(x)$ is continuous in that region, then the function $F(x)$ is monotonically increasing in that region. Note that at the "boundary" point for the region, namely $x=1$, that $F(x=1) = 0$ from which it follows that for all $x>1$, $F(x) > 0$, or $F(x) = x - 1/x - 2 \ln x > 0$; then $x - 1/x > 2 \ln x$, and, because $x>1$, the term $\ln x > 0$, so that

$$\frac{x - \dfrac{1}{x}}{2 \ln x} > 1 \quad \text{and} \quad \frac{x^2 - 1}{2 x \ln x} > 1 \tag{A4}$$

Recalling that $x=(T_2/T_1)^{1/2}$, this is equivalent to

$$\frac{T_2 - T_1}{\ln (T_2 / T_1)} > (T_1 T_2)^{1/2} \qquad \text{(A5)}$$

which is what we set out to prove.

APPENDIX B

NUMERICAL IMPLEMENTATION IN THE FLOW MODEL OF MCDONALD AND HARBAUGH (1988)

OVERVIEW

The alternative interblock transmissivity functions described in the Numerical Methods section are programmed in subroutines to be used with the modular ground-water flow model of McDonald and Harbaugh (1988). This model uses block-centered finite differences and the user inputs the horizontal hydraulic conductivity or transmissivity at each node. However the vertical interblock leakance (vertical hydraulic conductivity divided by vertical distance between nodes) is input directly. Hence, the computation of interblock conductance within the model is only applicable to the horizontal (row and column) conductances. The optional interblock transmissivity functions are implemented as a revision to the Block-Centered-Flow (BCF2) package of the modular model (McDonald and Harbaugh, 1988; McDonald and others, 1991).

Modifications include:

1. Changes to the MAIN program to identify modifications and to pass scratch array BUFF to selected BCF subroutines for possible use when LAYAVG=30. Also, common block / FLWAVG / LAYAVG(80) is added.
2. Changes to subroutine BCF2AL to interpret input LAYCON values to determine layer hydraulic characteristics (stored LAYCON) and interblock transmissivity (LAYAVG). Also, common block / FLWAVG / LAYAVG(80) is added and the program stops if any LAYAVG=30 and IAPART is zero.
3. Changes to subroutine BCF2FM to pass scratch array BUFF through for possible use when LAYAVG=30. Also, common block / FLWAVG / LAYAVG(80) is added.
4. Changes to subroutines SBCF2H and SBCF2N to call different subroutines for computing interblock conductance, depending on LAYAVG. Also, common block / FLWAVG / LAYAVG(80) is added to both subroutines.
5. Three new BCF3 subroutines are added to compute horizontal interblock conductance.

In relating the numerical algorithms in this appendix to the formula developed in the body of the report, the reader must bear in mind that the indexing used in this report is different from that used in the model and documentation of McDonald and Harbaugh (1988). In the body of this report, index i corresponds to the x coordinate and j corresponds to the y coordinate. The figures in this report illustrating two-dimensional grids have the x coordinate and i index increasing from left to right, and the y coordinate and j index increasing from top to bottom. This discretization convention has been used by numerous previous investigators (for example, Pinder and Bredehoeft, 1968; Konikow and Bredehoeft, 1978; Kipp, 1987).

The model of McDonald and Harbaugh (1988) can be applied to a problem where it is assumed that the x direction corresponds to model columns which are indexed by j. The y direction can be assumed to correspond to model rows which are indexed by i. For this conceptualization, the transmissivity in the x direction (T_{xx}) corresponds to the transmissivity in the direction of changing column index. However, in the report of McDonald and Harbaugh (1988), this transmissivity is called the "transmissivity along the rows" and is stored in the conductance term CR(j,i,k), where k is the layer (vertical) index. Likewise, the transmissivity in the y direction (T_{yy}) is called the "transmissivity along the columns" by McDonald and Harbaugh (1988) and stored in conductance array CC(i,j,k). Furthermore, the cell width in the x direction (Δx) (distance from the left side of a cell to the right side) is called the "cell width along rows" and stored in array DELR(j). Likewise, the grid spacing in the y direction (Δy) is called the "cell width along columns" and is stored in array DELC(i). These alternative discretization conventions should be kept in mind when relating the equations presented in the body of this report with the numerical algorithms presented here for the model of McDonald and Harbaugh (1988).

CHANGES TO MAIN PROGRAM

Program header **changed from**:

```
C     ******************************************************************
C     MAIN CODE FOR MODULAR MODEL --    9/1/87
C          BY MICHAEL G. MCDONALD AND ARLEN W. HARBAUGH
C-----VERSION 1638 24JUL1987 MAIN1
C-----VERSION 1323 21FEB1992 -- added BCF2, PCG2, STR1, IBS1, CHD1, and
C-----  GFD1 as documented in USGS reports
C-----                INBAS set to unit 5
C     ******************************************************************
```

to:

```
C     ******************************************************************
C     MAIN CODE FOR MODULAR MODEL --    9/1/87
C          BY MICHAEL G. MCDONALD AND ARLEN W. HARBAUGH
C-----VERSION 1638 24JUL1987 MAIN1
C-----VERSION 1323 21FEB1992 -- added BCF2, PCG2, STR1, IBS1, CHD1, and
C-----  GFD1 as documented in USGS reports
C-----BCF3 SUBSTITUTED FOR BCF2  --     9JULY1992
C-----                INBAS set to unit 5
C     ******************************************************************
```

Add common block to store interblock transmissivity flag for each layer, **changed from**:

```
C        SPECIFICATIONS:
C     ------------------------------------------------------------------
      COMMON X(350000)
      COMMON /FLWCOM/LAYCON(80)
      CHARACTER*4 HEADNG,VBNM
```

to:

```
C        SPECIFICATIONS:
C     ------------------------------------------------------------------
      COMMON X(350000)
      COMMON /FLWCOM/LAYCON(80)
      COMMON /FLWAVG/LAYAVG(80)
      CHARACTER*4 HEADNG,VBNM
```

Call to allocate space for BCF part of solution **changed from**:

```
      IF(IUNIT(1).GT.0) CALL BCF2AL(ISUM,LENX,LCSC1,LCHY,
     1     LCBOT,LCTOP,LCSC2,LCTRPY,IUNIT(1),ISS,
     2     NCOL,NROW,NLAY,IOUT,IBCFCB,LCWETD,IWDFLG,LCCVWD,
     3     WETFCT,IWETIT,IHDWET,HDRY)
```

to:

```
      IF(IUNIT(1).GT.0) CALL BCF3AL(ISUM,LENX,LCSC1,LCHY,
     1     LCBOT,LCTOP,LCSC2,LCTRPY,IUNIT(1),ISS,
     2     NCOL,NROW,NLAY,IOUT,IBCFCB,LCWETD,IWDFLG,LCCVWD,
     3     WETFCT,IWETIT,IHDWET,HDRY,LCRHS,LCBUFF)
```

Call to read BCF input **changed from**:

```
      IF(IUNIT(1).GT.0) CALL BCF2RP(X(LCIBOU),X(LCHNEW),X(LCSC1),
```

to:

```
      IF(IUNIT(1).GT.0) CALL BCF3RP(X(LCIBOU),X(LCHNEW),X(LCSC1),
```

Call to formulate BCF part of finite-difference equations **changed from**:

```
      IF(IUNIT(1).GT.0) CALL BCF2FM(X(LCHCOF),X(LCRHS),X(LCHOLD),
     1          X(LCSC1),X(LCHNEW),X(LCIBOU),X(LCCR),X(LCCC),X(LCCV),
     2          X(LCHY),X(LCTRPY),X(LCBOT),X(LCTOP),X(LCSC2),
     3          X(LCDELR),X(LCDELC),DELT,ISS,KKITER,KKSTP,KKPER,NCOL,
     4          NROW,NLAY,IOUT,X(LCWETD),IWDFLG,X(LCCVWD),WETFCT,
     5          IWETIT,IHDWET,HDRY)
```

to:

```
      IF(IUNIT(1).GT.0) CALL BCF3FM(X(LCHCOF),X(LCRHS),X(LCHOLD),
     1          X(LCSC1),X(LCHNEW),X(LCIBOU),X(LCCR),X(LCCC),X(LCCV),
     2          X(LCHY),X(LCTRPY),X(LCBOT),X(LCTOP),X(LCSC2),
     3          X(LCDELR),X(LCDELC),DELT,ISS,KKITER,KKSTP,KKPER,NCOL,
     4          NROW,NLAY,IOUT,X(LCWETD),IWDFLG,X(LCCVWD),WETFCT,
     5          IWETIT,IHDWET,HDRY,X(LCBUFF))
```

Subroutine BCF2AL changed to set LAYAVG values from input LAYCON values.

Subroutine header **changed from:**

```
      SUBROUTINE BCF2AL(ISUM,LENX,LCSC1,LCHY,LCBOT,LCTOP,LCSC2,LCTRPY,
     1  IN,ISS,NCOL,NROW,NLAY,IOUT,IBCFCB,LCWETD,IWDFLG,LCCVWD,
     2  WETFCT,IWETIT,IHDWET,HDRY)
C
C-----VERSION 1435 14MAY1991 BCF2AL
C     ******************************************************************
C     ALLOCATE ARRAY STORAGE FOR BLOCK-CENTERED FLOW PACKAGE, VERSION 2
C     ******************************************************************
C
C        SPECIFICATIONS:
C     ------------------------------------------------------------------
      COMMON /FLWCOM/LAYCON(80)
C     ------------------------------------------------------------------
C
C1------IDENTIFY PACKAGE
      WRITE(IOUT,1)IN
    1 FORMAT(1H0,'BCF2 -- BLOCK-CENTERED FLOW PACKAGE, VERSION 2',
     1', 7/1/91',' INPUT READ FROM UNIT',I3)
```

to:

```
      SUBROUTINE BCF3AL(ISUM,LENX,LCSC1,LCHY,LCBOT,LCTOP,LCSC2,LCTRPY,
     1  IN,ISS,NCOL,NROW,NLAY,IOUT,IBCFCB,LCWETD,IWDFLG,LCCVWD,
     2  WETFCT,IWETIT,IHDWET,HDRY,LCRHS,LCBUFF)
C
C-----VERSION 1436 9JULY1992 BCF3AL
C     ******************************************************************
C     ALLOCATE ARRAY STORAGE FOR BLOCK-CENTERED FLOW PACKAGE, VERSION 3
C     ******************************************************************
C
C        SPECIFICATIONS:
C     ------------------------------------------------------------------
      CHARACTER*12 AVGNAM(4)
      COMMON /FLWCOM/LAYCON(80)
      COMMON /FLWAVG/LAYAVG(80)
      DATA AVGNAM/'HARMONIC    ','ARITHMETIC  ',
     *            'LOGARITHMIC ','*UNCONFINED*'/
C     ------------------------------------------------------------------
C
C1------IDENTIFY PACKAGE
      WRITE(IOUT,1)IN
    1 FORMAT(1H0,'BCF3 -- BLOCK-CENTERED FLOW PACKAGE, VERSION 3',
     1', 7/9/92',' INPUT READ FROM UNIT',I3)
```

Reading and setting of layer codes (LAYCON) **changed from**:

```
C3------STOP THE SIMULATION IF THERE ARE MORE THAN 80 LAYERS
      IF(NLAY.LE.80) GO TO 50
      WRITE(IOUT,11)
   11 FORMAT(1H0,'YOU HAVE SPECIFIED MORE THAN 80 MODEL LAYERS'/1X,
     1  'SPACE IS RESERVED FOR A MAXIMUM OF 80 LAYERS IN ARRAY LAYCON')
      STOP
C
C4------READ LAYCON & PRINT TITLE FOR LAYCON TABLE
   50 READ(IN,51) (LAYCON(I),I=1,NLAY)
   51 FORMAT(40I2)
      WRITE(IOUT,52)
   52 FORMAT(1X,5X,'LAYER  AQUIFER TYPE',/1X,5X,19('-'))
C
C5------LOOP THROUGH LAYERS PRINTING THE LAYER-TYPE CODE AND COUNTING
C5------LAYERS THAT NEED TOP & BOT ARRAYS
      NBOT=0
      NTOP=0
      DO 100 I=1,NLAY
      L=LAYCON(I)
      WRITE(IOUT,7) I,L
    7 FORMAT(1X,I9,I10)
```

to:

```
C3------STOP THE SIMULATION IF THERE ARE MORE THAN 80 LAYERS
      IF(NLAY.LE.80) GO TO 50
      WRITE(IOUT,11)
   11 FORMAT(1H0,'YOU HAVE SPECIFIED MORE THAN 80 MODEL LAYERS'/1X,
     1  'SPACE IS RESERVED FOR A MAXIMUM OF 80 LAYERS IN ARRAY LAYCON',
     2  ' AND ARRAY LAYAVG')
      STOP
C
C4------READ LAYCON & PRINT TITLE FOR LAYCON TABLE
   50 READ(IN,51) (LAYCON(I),I=1,NLAY)
   51 FORMAT(40I2)
      WRITE(IOUT,52)
   52 FORMAT(6X,'LAYER  AQUIFER TYPE    INTERBLOCK T',/6X,37('-'))
C
C5------LOOP THROUGH LAYERS PRINTING THE LAYER-TYPE CODE AND COUNTING
C5------LAYERS THAT NEED TOP & BOT ARRAYS
      NBOT=0
      NTOP=0
      IAPART=LCRHS-LCBUFF
      DO 100 I=1,NLAY
C
C5_1----CONVERT INPUT LAYCON TO LAYAVG CODES AND ORIGINAL LAYCON
C5_1----DO NOT ALLOW LAYCON=30 OR 32
      IF(LAYCON(I).EQ.30.OR.LAYCON(I).EQ.32) LAYCON(I)=LAYCON(I)-10
      LAYAVG(I)=LAYCON(I)/10
      LAYAVG(I)=LAYAVG(I)*10
C5_2----RESET LAYCON CODE TO ORIGINAL VALUES
      LAYCON(I)=LAYCON(I)-LAYAVG(I)
C5_3----BOUND LAYAVG
      IF(LAYAVG(I).LT.0.OR.LAYAVG(I).GT.30) LAYAVG(I)=0
      L=LAYCON(I)
      LA=LAYAVG(I)
      INAM=1+LA/10
      WRITE(IOUT,7) I,L,LA,AVGNAM(INAM)
    7 FORMAT(1X,I9,2I10,'-',A12)
```

```
C5_4----CHECK IAPART=1 IF LAYAVG=30 FOR ANY LAYER
      IF(IAPART.EQ.0.AND.LAYAVG(I).EQ.30) THEN
         WRITE(IOUT,75)
   75    FORMAT(' ERROR -- IAPART MUST BE NONZERO IF LAYAVG',
     *                  '=30, STOPPING')
         STOP
      END IF
```

Subroutine BCF2RP changed to call SBCF3N.

Header **changed from**:

```
      SUBROUTINE BCF2RP(IBOUND,HNEW,SC1,HY,CR,CC,CV,DELR,DELC,BOT,TOP,
     1 SC2,TRPY,IN,ISS,NCOL,NROW,NLAY,NODES,IOUT,WETDRY,IWDFLG,CVWD)
C
C-----VERSION 1275 6JUNE1991 BCF2RP
C     ******************************************************************
C     READ AND INITIALIZE DATA FOR BLOCK-CENTERED FLOW PACKAGE,
C     VERSION 2
```

to:

```
      SUBROUTINE BCF3RP(IBOUND,HNEW,SC1,HY,CR,CC,CV,DELR,DELC,BOT,TOP,
     1 SC2,TRPY,IN,ISS,NCOL,NROW,NLAY,NODES,IOUT,WETDRY,IWDFLG,CVWD)
C
C-----VERSION 1276 9JULY1992 BCF3RP
C     ******************************************************************
C     READ AND INITIALIZE DATA FOR BLOCK-CENTERED FLOW PACKAGE,
C     VERSION 3
```

Call to SBCF2N **changed from**:

```
      CALL SBCF2N(HNEW,IBOUND,SC1,SC2,CR,CC,CV,HY,TRPY,DELR,DELC,ISS,
```

to:

```
      CALL SBCF3N(HNEW,IBOUND,SC1,SC2,CR,CC,CV,HY,TRPY,DELR,DELC,ISS,
```

Subroutine BCF2FM changed to use optional interblock transmissivity subroutines and pass BUFF for possible use.

BUFF added to dimension statement, Header **changed from**:

```
      SUBROUTINE BCF2FM(HCOF,RHS,HOLD,SC1,HNEW,IBOUND,CR,CC,CV,HY,TRPY,
     1                  BOT,TOP,SC2,DELR,DELC,DELT,ISS,KITER,KSTP,KPER,
     2                  NCOL,NROW,NLAY,IOUT,WETDRY,IWDFLG,CVWD,
     3                  WETFCT,IWETIT,IHDWET,HDRY)
C-----VERSION 1104 5MAY1991 BCF2FM
C     ******************************************************************
C     ADD LEAKAGE CORRECTION AND STORAGE TO HCOF AND RHS, AND CALCULATE
C     CONDUCTANCE AS REQUIRED, VERSION 2
C     ******************************************************************
C
C        SPECIFICATIONS:
C     ------------------------------------------------------------------
      DOUBLE PRECISION HNEW
C
      DIMENSION HCOF(NCOL,NROW,NLAY),RHS(NCOL,NROW,NLAY),
     1    HOLD(NCOL,NROW,NLAY),SC1(NCOL,NROW,NLAY),HNEW(NCOL,NROW,NLAY),
     2    IBOUND(NCOL,NROW,NLAY),CR(NCOL,NROW,NLAY),
     3    CC(NCOL,NROW,NLAY),CV(NCOL,NROW,NLAY),HY(NCOL,NROW,NLAY),
     4    TRPY(NLAY),BOT(NCOL,NROW,NLAY),TOP(NCOL,NROW,NLAY),DELR(NCOL),
     5    DELC(NROW),SC2(NCOL,NROW,NLAY),WETDRY(NCOL,NROW,NLAY),
     6    CVWD(NCOL,NROW,NLAY)
```

to:

```
      SUBROUTINE BCF3FM(HCOF,RHS,HOLD,SC1,HNEW,IBOUND,CR,CC,CV,HY,TRPY,
     1                  BOT,TOP,SC2,DELR,DELC,DELT,ISS,KITER,KSTP,KPER,
     2                  NCOL,NROW,NLAY,IOUT,WETDRY,IWDFLG,CVWD,
     3                  WETFCT,IWETIT,IHDWET,HDRY,BUFF)
C-----VERSION 1105 9JULY1992 BCF3FM
C     ******************************************************************
C     ADD LEAKAGE CORRECTION AND STORAGE TO HCOF AND RHS, AND CALCULATE
C     CONDUCTANCE AS REQUIRED, VERSION 3
C     ******************************************************************
C
C        SPECIFICATIONS:
C     ------------------------------------------------------------------
      DOUBLE PRECISION HNEW
C
      DIMENSION HCOF(NCOL,NROW,NLAY),RHS(NCOL,NROW,NLAY),
     1    HOLD(NCOL,NROW,NLAY),SC1(NCOL,NROW,NLAY),HNEW(NCOL,NROW,NLAY),
     2    IBOUND(NCOL,NROW,NLAY),CR(NCOL,NROW,NLAY),
     3    CC(NCOL,NROW,NLAY),CV(NCOL,NROW,NLAY),HY(NCOL,NROW,NLAY),
     4    TRPY(NLAY),BOT(NCOL,NROW,NLAY),TOP(NCOL,NROW,NLAY),DELR(NCOL),
     5    DELC(NROW),SC2(NCOL,NROW,NLAY),WETDRY(NCOL,NROW,NLAY),
     6    CVWD(NCOL,NROW,NLAY),BUFF(NCOL,NROW,NLAY)
```

Call to subroutine SBCF2H **changed from**:

```
C1B-----FOR LAYER TYPES 1 & 3 CALL SBCF2H TO CALCULATE
C1B-----HORIZONTAL CONDUCTANCES.
      CALL SBCF2H(HNEW,IBOUND,CR,CC,CV,HY,TRPY,DELR,DELC,BOT,TOP,
     1   KK,KB,KT,KITER,KSTP,KPER,NCOL,NROW,NLAY,IOUT,WETDRY,IWDFLG,
     2   CVWD,WETFCT,IWETIT,IHDWET,HDRY)
```

to:

```
C1B-----FOR LAYER TYPES 1 & 3 CALL SBCF3H TO CALCULATE
C1B-----HORIZONTAL CONDUCTANCES.
      CALL SBCF3H(HNEW,IBOUND,CR,CC,CV,HY,TRPY,DELR,DELC,BOT,TOP,
     1   KK,KB,KT,KITER,KSTP,KPER,NCOL,NROW,NLAY,IOUT,WETDRY,IWDFLG,
     2   CVWD,WETFCT,IWETIT,IHDWET,HDRY,BUFF)
```

Subroutine SBCF2H changed to call optional interblock transmissivity subroutines.

Header **changed from**:

```
      SUBROUTINE SBCF2H(HNEW,IBOUND,CR,CC,CV,HY,TRPY,DELR,DELC
     1,BOT,TOP,K,KB,KT,KITER,KSTP,KPER,NCOL,NROW,NLAY,IOUT
     2,WETDRY,IWDFLG,CVWD,WETFCT,IWETIT,IHDWET,HDRY)
C-----VERSION 1345 23MAY1991 SBCF2H
C
C     ******************************************************************
C     COMPUTE CONDUCTANCE FOR ONE LAYER FROM SATURATED THICKNESS AND
C     HYDRAULIC CONDUCTIVITY, VERSION 2
C     ******************************************************************
C
C      SPECIFICATIONS:
C     ------------------------------------------------------------------
      DOUBLE PRECISION HNEW
C
      DIMENSION HNEW(NCOL,NROW,NLAY),IBOUND(NCOL,NROW,NLAY)
     1,CR(NCOL,NROW,NLAY), CC(NCOL,NROW,NLAY), CV(NCOL,NROW,NLAY)
     2,HY(NCOL,NROW,NLAY), TRPY(NLAY), DELR(NCOL), DELC(NROW)
     3,BOT(NCOL,NROW,NLAY),TOP(NCOL,NROW,NLAY),WETDRY(NCOL,NROW,NLAY)
     4,CVWD(NCOL,NROW,NLAY)
      CHARACTER*4 ACNVRT
      DIMENSION ICNVRT(8),JCNVRT(8),ACNVRT(8)
C
      COMMON /FLWCOM/LAYCON(80)
```

to:

```
      SUBROUTINE SBCF3H(HNEW,IBOUND,CR,CC,CV,HY,TRPY,DELR,DELC
     1,BOT,TOP,K,KB,KT,KITER,KSTP,KPER,NCOL,NROW,NLAY,IOUT
     2,WETDRY,IWDFLG,CVWD,WETFCT,IWETIT,IHDWET,HDRY,BUFF)
C-----VERSION 1346 9JULY1992 SBCF3H
C
C     ******************************************************************
C     COMPUTE CONDUCTANCE FOR ONE LAYER FROM SATURATED THICKNESS AND
C     HYDRAULIC CONDUCTIVITY, VERSION 3
C     ******************************************************************
C
C      SPECIFICATIONS:
C     ------------------------------------------------------------------
      DOUBLE PRECISION HNEW
C
      DIMENSION HNEW(NCOL,NROW,NLAY),IBOUND(NCOL,NROW,NLAY)
     1,CR(NCOL,NROW,NLAY), CC(NCOL,NROW,NLAY), CV(NCOL,NROW,NLAY)
     2,HY(NCOL,NROW,NLAY), TRPY(NLAY), DELR(NCOL), DELC(NROW)
     3,BOT(NCOL,NROW,NLAY),TOP(NCOL,NROW,NLAY),WETDRY(NCOL,NROW,NLAY)
     4,CVWD(NCOL,NROW,NLAY),BUFF(NCOL,NROW,NLAY)
      CHARACTER*4 ACNVRT
      DIMENSION ICNVRT(8),JCNVRT(8),ACNVRT(8)
C
      COMMON /FLWCOM/LAYCON(80)
      COMMON /FLWAVG/LAYAVG(80)
```

Action for inactive nodes **changed from**:

```
C3C-----CELL IS DRY AND STAYS DRY.  SET TRANSMISSIVITY TO 0 AND SKIP
C3C-----TO THE NEXT CELL.
    6 CC(J,I,K)=0.
      GO TO 200
```

to:

```
C3C-----CELL IS DRY AND STAYS DRY.  SET TRANSMISSIVITY TO 0 AND SKIP
C3C-----TO THE NEXT CELL.
    6 CC(J,I,K)=0.
C3D-----ZERO BUFF (SATURATED THICKNESS) FOR LAYAVG=30
      IF(LAYAVG(K).EQ.30) BUFF(J,I,K)=0.
      GO TO 200
```

Temporary storage of transmissivity **changed from**:

```
C6A-----IF SATURATED THICKNESS>0 THEN TRANSMISSIVITY IS HYDRAULIC
C6A-----CONDUCTIVITY TIMES SATURATED THICKNESS.
      CC(J,I,K)=THCK*HY(J,I,KB)
      GO TO 200
```

to:

```
C6A-----IF SATURATED THICKNESS>0 THEN TRANSMISSIVITY IS HYDRAULIC
C6A-----CONDUCTIVITY TIMES SATURATED THICKNESS.
C6A1----IF LAYAVG=30, STORE K IN CC AND SAT THICKNESS IN BUFF
      IF(LAYAVG(K).EQ.30) THEN
         CC(J,I,K)=HY(J,I,KB)
         BUFF(J,I,K)=THCK
      ELSE
         CC(J,I,K)=THCK*HY(J,I,KB)
      END IF
      GO TO 200
```

Call to subroutine to compute interblock transmissivity **changed from:**

```
C9------COMPUTE HORIZONTAL BRANCH CONDUCTANCES FROM TRANSMISSIVITY.
  210 CALL SBCF1C(CR,CC,TRPY,DELR,DELC,K,NCOL,NROW,NLAY)
```

to:

```
C9------COMPUTE HORIZONTAL BRANCH CONDUCTANCES FROM TRANSMISSIVITY.
C9A-----SELECT INTERBLOCK TRANSMISSIVITY SUBROUTINE FROM LAYAVG
  210 IF(LAYAVG(K).EQ.0) THEN
         CALL SBCF1C(CR,CC,TRPY,DELR,DELC,K,NCOL,NROW,NLAY)
      ELSE IF(LAYAVG(K).EQ.10) THEN
         CALL SBCF3A(CR,CC,TRPY,DELR,DELC,K,NCOL,NROW,NLAY)
      ELSE IF(LAYAVG(K).EQ.20) THEN
         CALL SBCF3L(CR,CC,TRPY,DELR,DELC,K,NCOL,NROW,NLAY)
      ELSE
         CALL SBCF3U(CR,CC,TRPY,DELR,DELC,BUFF,K,NCOL,NROW,NLAY)
      END IF
```

Change SBCF2N to call optional interblock transmissivity subroutines when initializing for simulations with constant transmissivity.

Header **changed from:**

```
      SUBROUTINE SBCF2N(HNEW,IBOUND,SC1,SC2,CR,CC,CV,HY,TRPY,DELR,DELC,
     1    ISS,NCOL,NROW,NLAY,IOUT,WETDRY,IWDFLG,CVWD)
C
C-----VERSION 1107 5MAY1991 SBCF2N
C
C     ******************************************************************
C     INITIALIZE AND CHECK BCF DATA, VERSION 2
```

to:

```
      SUBROUTINE SBCF3N(HNEW,IBOUND,SC1,SC2,CR,CC,CV,HY,TRPY,DELR,DELC,
     1    ISS,NCOL,NROW,NLAY,IOUT,WETDRY,IWDFLG,CVWD)
C
C-----VERSION 1108 9JULY1992 SBCF3N
C
C     ******************************************************************
C     INITIALIZE AND CHECK BCF DATA, VERSION 3
```

Common block declaration changed **from:**

```
      COMMON /FLWCOM/LAYCON(80)
```

to:

```
      COMMON /FLWCOM/LAYCON(80)
      COMMON /FLWAVG/LAYAVG(80)
```

Call to subroutine to compute interblock transmissivity **changed from:**

```
C5------CALCULATE HOR. CONDUCTANCE(CR AND CC) FOR CONSTANT T LAYERS
      DO 70 K=1,NLAY
      KK=K
      IF(LAYCON(K).EQ.3 .OR. LAYCON(K).EQ.1) GO TO 70
      CALL SBCF1C(CR,CC,TRPY,DELR,DELC,KK,NCOL,NROW,NLAY)
   70 CONTINUE
```

to:

```
C5------CALCULATE HOR. CONDUCTANCE(CR AND CC) FOR CONSTANT T LAYERS
      DO 70 K=1,NLAY
      KK=K
      IF(LAYCON(K).EQ.3 .OR. LAYCON(K).EQ.1) GO TO 70
C5A-----SELECT INTERBLOCK TRANSMISSIVITY SUBROUTINE FROM LAYAVG
      IF(LAYAVG(K).EQ.0) THEN
         CALL SBCF1C(CR,CC,TRPY,DELR,DELC,KK,NCOL,NROW,NLAY)
      ELSE IF(LAYAVG(K).EQ.10) THEN
         CALL SBCF3A(CR,CC,TRPY,DELR,DELC,KK,NCOL,NROW,NLAY)
      ELSE
         CALL SBCF3L(CR,CC,TRPY,DELR,DELC,KK,NCOL,NROW,NLAY)
      END IF
   70 CONTINUE
```

NEW BCF SUBROUTINES FOR OPTIONAL INTERBLOCK TRANSMISSIVITY

```
      SUBROUTINE SBCF3A(CR,CC,TRPY,DELR,DELC,K,NCOL,NROW,NLAY)
C
C-----VERSION 1 9JULY1992 SBCF3A
C     ******************************************************************
C-------COMPUTE CONDUCTANCE USING ARITHMETIC MEAN TRANSMISSIVITY
C-------ACTIVATED BY LAYAVG=10
C     ******************************************************************
C
C      SPECIFICATIONS:
C     ------------------------------------------------------------------
C
      DIMENSION CR(NCOL,NROW,NLAY), CC(NCOL,NROW,NLAY)
     2   , TRPY(NLAY), DELR(NCOL), DELC(NROW)
C
C     ------------------------------------------------------------------
      YX=TRPY(K)
C
C1------FOR EACH CELL CALCULATE BRANCH CONDUCTANCES FROM THAT CELL
C1------TO THE ONE ON THE RIGHT AND THE ONE IN FRONT.
      DO 40 I=1,NROW
      DO 40 J=1,NCOL
      T1=CC(J,I,K)
C
C2------IF T=0 THEN SET CONDUCTANCE EQUAL TO 0. GO ON TO NEXT CELL.
      IF(T1.NE.0.) GO TO 10
      CR(J,I,K)=0.
      GO TO 40
C
C3------IF THIS IS NOT THE LAST COLUMN(RIGHTMOST) THEN CALCULATE
C3------BRANCH CONDUCTANCE IN THE ROW DIRECTION (CR) TO THE RIGHT.
   10 IF(J.EQ.NCOL) GO TO 30
      T2=CC(J+1,I,K)
C3A-----ARITHMETIC MEAN INTERBLOCK TRANSMISSIVITY
      IF(T2.EQ.0.) THEN
         CR(J,I,K)=0.
      ELSE
         CR(J,I,K)=DELC(I)*(T1+T2)/(DELR(J+1)+DELR(J))
      END IF
C
C4------IF THIS IS NOT THE LAST ROW(FRONTMOST) THEN CALCULATE
C4------BRANCH CONDUCTANCE IN THE COLUMN DIRECTION (CC) TO THE FRONT.
   30 IF(I.EQ.NROW) GO TO 40
      T2=CC(J,I+1,K)
      IF(T2.EQ.0.) THEN
         CC(J,I,K)=0.
      ELSE
         CC(J,I,K)=YX*DELR(J)*(T1+T2)/(DELC(I+1)+DELC(I))
      END IF
   40 CONTINUE
C
C5------RETURN
      RETURN
      END
```

```
      SUBROUTINE SBCF3L(CR,CC,TRPY,DELR,DELC,K,NCOL,NROW,NLAY)
C
C-----VERSION 1 9JULY1992 SBCF3L
C
C     ******************************************************************
C-------COMPUTE CONDUCTANCE USING LOGARITHMIC MEAN TRANSMISSIVITY
C-------ACTIVATED BY LAYAVG=20
C     ******************************************************************
C
C      SPECIFICATIONS:
C
C     ------------------------------------------------------------------
      DIMENSION CR(NCOL,NROW,NLAY), CC(NCOL,NROW,NLAY)
     2   , TRPY(NLAY), DELR(NCOL), DELC(NROW)
C
C     ------------------------------------------------------------------
      YX=TRPY(K)*2.
C
C1------FOR EACH CELL CALCULATE BRANCH CONDUCTANCES FROM THAT CELL
C1------TO THE ONE ON THE RIGHT AND THE ONE IN FRONT.
      DO 40 I=1,NROW
      DO 40 J=1,NCOL
      T1=CC(J,I,K)
C
C2------IF T=0 THEN SET CONDUCTANCE EQUAL TO 0. GO ON TO NEXT CELL.
      IF(T1.NE.0.) GO TO 10
      CR(J,I,K)=0.
      GO TO 40
C
C3------IF THIS IS NOT THE LAST COLUMN(RIGHTMOST) THEN CALCULATE
C3------BRANCH CONDUCTANCE IN THE ROW DIRECTION (CR) TO THE RIGHT.
   10 IF(J.EQ.NCOL) GO TO 30
      T2=CC(J+1,I,K)
      IF(T2.EQ.0.) THEN
C3A-----SET TO ZERO AND EXIT IF T2 IS ZERO
         CR(J,I,K)=0.
         GO TO 30
      END IF
C3B-----LOGARITHMIC MEAN INTERBLOCK TRANSMISSIVITY
      RATIO=T2/T1
      IF(RATIO.GT.1.005.OR.RATIO.LT.0.995) THEN
         T=(T2-T1)/ALOG(RATIO)
      ELSE
         T=0.5*(T1+T2)
      END IF
      CR(J,I,K)=2.*DELC(I)*T/(DELR(J+1)+DELR(J))
C
C4------IF THIS IS NOT THE LAST ROW(FRONTMOST) THEN CALCULATE
C4------BRANCH CONDUCTANCE IN THE COLUMN DIRECTION (CC) TO THE FRONT.
   30 IF(I.EQ.NROW) GO TO 40
      T2=CC(J,I+1,K)
      IF(T2.EQ.0.) THEN
         CC(J,I,K)=0.
         GO TO 40
      END IF
      RATIO=T2/T1
      IF(RATIO.GT.1.005.OR.RATIO.LT.0.995) THEN
         T=(T2-T1)/ALOG(RATIO)
      ELSE
         T=0.5*(T1+T2)
      END IF
      CC(J,I,K)=YX*DELR(J)*T/(DELC(I+1)+DELC(I))
   40 CONTINUE
C
C5------RETURN
      RETURN
      END
```

```
      SUBROUTINE SBCF3U(CR,CC,TRPY,DELR,DELC,BUFF,K,NCOL,NROW,NLAY)
C
C-----VERSION 1 9JULY1992 SBCF3U
C
C     ******************************************************************
C-------COMPUTE CONDUCTANCE USING ARITHMETIC MEAN SATURATED THICKNESS
C-------AND LOGARITHMIC MEAN TRANSMISSIVITY
C-------NODE HYDRAULIC CONDUCTIVITY IS IN CC,
C-------NODE SATURATED THICKNESS IS IN BUFF
C-------ACTIVATED BY LAYAVG=30
C     ******************************************************************
C
C      SPECIFICATIONS:
C     ------------------------------------------------------------------
C
      DIMENSION CR(NCOL,NROW,NLAY), CC(NCOL,NROW,NLAY)
     2   , TRPY(NLAY), DELR(NCOL), DELC(NROW)
     3   , BUFF(NCOL,NROW,NLAY)
C
C     ------------------------------------------------------------------
      YX=TRPY(K)
C
C1------FOR EACH CELL CALCULATE BRANCH CONDUCTANCES FROM THAT CELL
C1------TO THE ONE ON THE RIGHT AND THE ONE IN FRONT.
      DO 40 I=1,NROW
      DO 40 J=1,NCOL
      T1=CC(J,I,K)
C
C2------IF T=0 THEN SET CONDUCTANCE EQUAL TO 0. GO ON TO NEXT CELL.
      IF(T1.NE.0.) GO TO 10
      CR(J,I,K)=0.
      GO TO 40
C
C3------IF THIS IS NOT THE LAST COLUMN(RIGHTMOST) THEN CALCULATE
C3------BRANCH CONDUCTANCE IN THE ROW DIRECTION (CR) TO THE RIGHT.
   10 IF(J.EQ.NCOL) GO TO 30
      T2=CC(J+1,I,K)
      IF(T2.EQ.0.) THEN
C3A-----SET TO ZERO AND EXIT IF T2 IS ZERO
         CR(J,I,K)=0.
         GO TO 30
      END IF
C3B-----LOGARITHMIC MEAN HYDRAULIC CONDUCTIVITY
      RATIO=T2/T1
      IF(RATIO.GT.1.005.OR.RATIO.LT.0.995) THEN
         T=(T2-T1)/ALOG(RATIO)
      ELSE
         T=0.5*(T1+T2)
      END IF
C3C-----MULTIPLY LOGARITHMIC K BY ARITHMETIC SAT THICK
      CR(J,I,K)=DELC(I)*T*(BUFF(J,I,K)+BUFF(J+1,I,K))
     *               /(DELR(J+1)+DELR(J))
C
C4------IF THIS IS NOT THE LAST ROW(FRONTMOST) THEN CALCULATE
C4------BRANCH CONDUCTANCE IN THE COLUMN DIRECTION (CC) TO THE FRONT.
   30 IF(I.EQ.NROW) GO TO 40
      T2=CC(J,I+1,K)
      IF(T2.EQ.0.) THEN
         CC(J,I,K)=0.
         GO TO 40
      END IF
      RATIO=T2/T1
      IF(RATIO.GT.1.005.OR.RATIO.LT.0.995) THEN
         T=(T2-T1)/ALOG(RATIO)
      ELSE
         T=0.5*(T1+T2)
      END IF
      CC(J,I,K)=YX*DELR(J)*T*(BUFF(J,I,K)+BUFF(J,I+1,K))
     *            /(DELC(I+1)+DELC(I))
   40 CONTINUE
C
C5------RETURN
      RETURN
      END
```

Alternative interblock transmissivity functions are selected by layer through the value of LAYCON specified in the BCF package input. The meaning of the original LAYCON values are as follows (McDonald and Harbaugh, 1988):

LAYCON	Layer characteristics
0	confined, T and S constant in time
1	unconfined, S constant in time, T depends on h
2	confined/unconfined, T constant in time, S depends on h
3	confined/unconfined, T and S depend on h

These values are input using I2 format. To activate alternative interblock transmissivity functions, a two digit number is input using the I2 format. The first digit (tens) determines the interblock transmissivity function, and the second digit (ones) determines the layer characteristics as in the original model. These values are read from the BCF input file by subroutine BCF3AL. The factor of ten represented by the first digit is stored in a new common block array LAYAVG. For example, if the input LAYCON value is 10, 11, or 12, the value stored in LAYAVG is 10. The second digit (ones) is stored in LAYCON. The meaning of the second (ones) digit is identical to the original model of McDonald and Harbaugh (1988), shown above. The meaning of the first (tens) digit is:

LAYAVG	Interblock transmissivity
0	harmonic mean
10	arithmetic mean
20	logarithmic mean
30	arithmetic mean saturated thickness times logarithmic mean hydraulic conductivity

Thus, the tens part of the input LAYCON value is stored in array LAYAVG, and the ones part of the input LAYCON is stored in LAYCON. The stored values of LAYCON are identical to the original model (McDonald and Harbaugh, 1988), and the stored values of LAYAVG are used only to determine the method to compute interblock transmissivity.

The following table shows the input value of LAYCON used to specify the particular stored values of LAYCON (layer characteristics) and LAYAVG (interblock transmissivity):

Input LAYCON values used to specify indicated layer characteristics (stored LAYCON) and interblock transmissivity (LAYAVG)

stored	LAYAVG			
LAYCON	0	10	20	30
0	0	10	20	NA
1	1	11	21	31
2	2	12	22	NA
3	3	13	23	33

NA: The option LAYAVG=30 is not allowed unless the hydraulic conductivity is input (LAYCON=1 or 3). Input LAYCON of 30 or 32 is reset to 20 or 22, respectively.

DOCUMENTATION OF A COMPUTER PROGRAM TO SIMULATE STREAM-AQUIFER RELATIONS USING A MODULAR, FINITE-DIFFERENCE, GROUND-WATER FLOW MODEL

David E. Prudic

U.S. GEOLOGICAL SURVEY

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CONVERSION FACTORS AND ABBREVIATIONS

"Inch-pound" units of measure used in this report may be converted to metric (International System) units by using the following factors.

Multiply inch-pound unit	By	To obtain metric unit
inch (in.)	2.54	centimeter (cm)
foot (ft)	0.3048	meter (m)
foot per second (ft/s)	0.3048	meter per second (m/s)
foot per day (ft/d)	0.3048	meter per day (m/d)
square foot per second (ft^2/s)	0.09290	square meter per second (m^2/s)
square foot per day (ft^2/d)	0.09290	square meter per day (m^2/d)
cubic foot per second (ft^3/s)	0.02832	cubic meter per second (m^3/s)
cubic foot per day (ft^3/d)	0.02832	cubic meter per day (m^3/d)

ALTITUDE DATUM

Sea level: In this report, "sea level" refers to the National Geodetic Vertical Datum of 1929 (NGVD of 1929)-- a geodetic datum derived from a general adjustment of the first-order level nets of both the United States and Canada, formerly called "Sea Level Datum of 1929."

DOCUMENTATION OF A COMPUTER PROGRAM TO SIMULATE STREAM-AQUIFER RELATIONS USING A MODULAR, FINITE-DIFFERENCE, GROUND-WATER FLOW MODEL

By David E. Prudic

ABSTRACT

Computer models are widely used to simulate ground-water flow for evaluating and managing the ground-water resources of many aquifers, but few are designed to also account for surface flow in streams. A computer program was written for use in the U.S. Geological Survey modular finite-difference ground-water flow model to account for the amount of flow in streams and to simulate the interaction between surface streams and ground water. The new program is called the Streamflow-Routing Package.

The Streamflow-Routing Package is not a true surface-water flow model but rather is an accounting program that tracks the flow in one or more streams which interact with ground water. The program limits the amount of ground-water recharge to the available streamflow. It permits two or more streams to merge into one with flow in the merged stream equal to the sum of the tributary flows. The program also permits diversions from streams.

Streams are divided into segments and reaches. Each reach corresponds to individual cells in the finite-difference grid used to simulate ground-water flow. A segment consists of a group of reaches connected in downstream order. Leakage is calculated for each reach on the basis of the head difference between the stream and aquifer and a conductance term. It is subtracted or added to the amount of streamflow into the reach. The stage in each reach can be computed using the Manning formula under the assumption of a rectangular stream channel.

The amount of leakage in each reach (either into or out of the aquifer) is incorporated into the ground-water flow model by adding terms to the finite-difference equations. Recharge to the aquifer in a reach ceases when all the streamflow in upstream reaches has leaked into the aquifer and the stream is dry. A stream is permitted to flow again in downstream reaches if the head in the aquifer is above the elevation of the streambed.

Results from the program have been compared to results from two analytical solutions. One assumes time varying areal recharge to the aquifer and discharge only to a stream and the other assumes recharge to the aquifer from a change in stream stage. Results from the program reasonably duplicated the analytical solutions.

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The ground-water flow model with the Streamflow-Routing Package has an advantage over the analytical solution in simulating the interaction between aquifer and stream because it can be used to simulate complex systems that cannot be readily solved analytically.

The Streamflow-Routing Package does not include a time function for streamflow but rather streamflow entering the modeled area is assumed to be instantly available to downstream reaches during each time period. This assumption is generally reasonable because of the relatively slow rate of ground-water flow. Another assumption is that leakage between streams and aquifers is instantaneous. This assumption may not be reasonable if the streams and aquifers are separated by a thick unsaturated zone.

Documentation of the Streamflow-Routing Package includes data input instructions; flow charts, narratives, and listings of the computer program for each of four modules; and input data sets and printed results for two test problems, and one example problem.

INTRODUCTION

Computer models that simulate ground-water flow are widely used to evaluate and manage ground-water resources in many areas of the Nation. Early models were somewhat limited in the types of processes that could be simulated without making major changes in the program codes. McDonald and Harbaugh (1988) developed a computer program that simulates three-dimensional ground-water flow and includes the effects of many processes such as areal recharge, rivers, drains, evapotranspiration, and pumpage. Their program is designed to allow for additional capabilities to be easily incorporated into it. The program is divided into a main program and a series of independent subprograms called packages. A package is a group of subroutines, called modules, that deals with a single aspect of the simulation. For example, the River Package simulates leakage between rivers and aquifer. Changes to a package or an addition of a new package does not require major changes to other packages in McDonald and Harbaugh's program.

The package that simulates the effects of rivers on an aquifer only accounts for leakage to or from the rivers. It does not track the amount of flow in the rivers nor does it permit rivers to go dry during a given period of simulation. In many areas, particularly in the western United States, streams become dry as they flow across large ground-water basins. In addition, water commonly is diverted from rivers or streams through canals or ditches to irrigated fields. The effects of ephemeral streams and irrigation canals can now be simulated with a new package. The package was developed to route streamflows in one or more streams across a ground-water basin; when flow in a stream ceases, the simulation of recharge from the stream to the aquifer system also ceases.

This report documents: (1) the basic concepts of the Streamflow-Routing Package, (2) the results of three test simulations, (3) the data input instructions needed to use the program, (4) the computer program, including a narrative, flow chart, program listing and a description of variables for each module, and (5) the input data sets used for the three test simulations and selected parts of the printed results.

CONCEPTUALIZATION OF ROUTING STREAMFLOWS

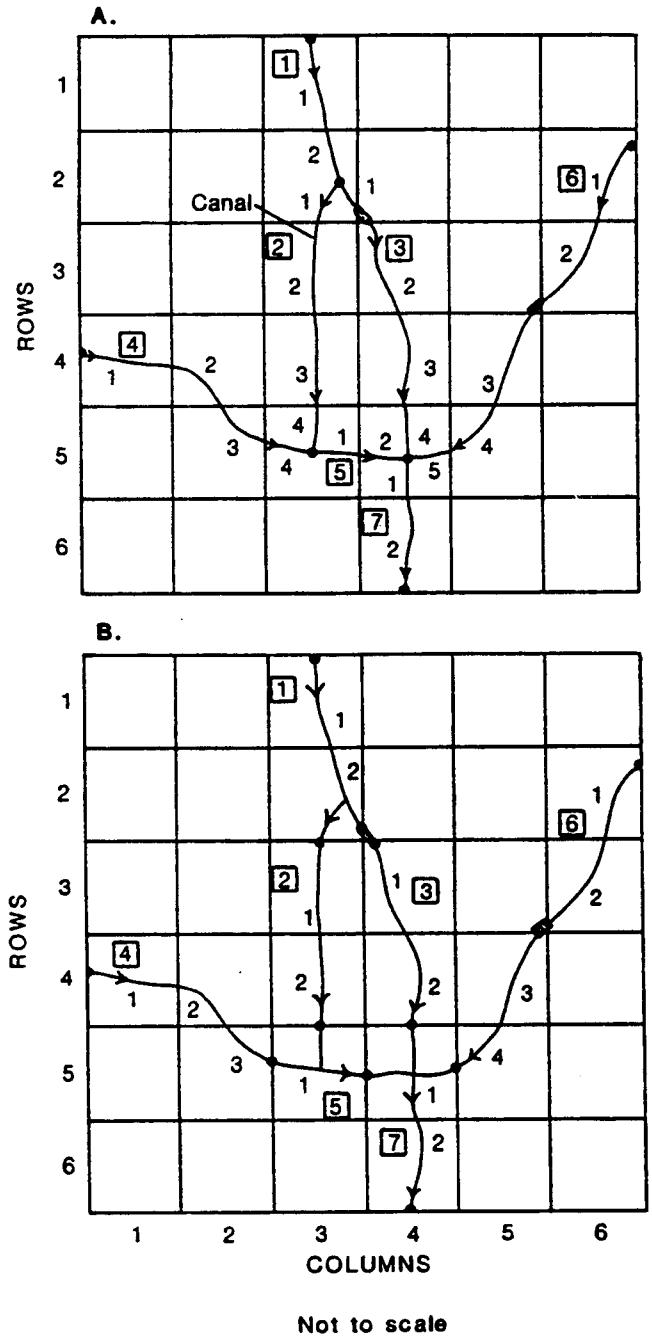
The Streamflow-Routing Package is a modification of the River Package described by McDonald and Harbaugh (1988, p. 6-1 through 6-30) and is designed to be incorporated into their program. The new package is designed to route flow through one or more rivers, streams, canals, or ditches (all of which are hereafter referred to as streams, for convenience) in addition to computing leakage between the streams and the aquifer system.

Numbering and Ordering of Streams

A brief description of the model developed by McDonald and Harbaugh (1988) is necessary prior to describing the numbering and ordering of streams. The program of McDonald and Harbaugh solves the three-dimensional ground-water flow equation using finite-difference approximations (p. 2-1 through 2-35). The finite-difference procedure requires that the aquifer be divided into blocks called cells, which have dimensions x, y, and z. The aquifer properties in each cell are assumed uniform. The unknown head in each cell is calculated at a point or node at the center of the cell. The head is calculated by iterating through the finite-difference equations for all nodes until the maximum head change in any cell between the previous iteration and the current iteration is less than a value specified by the user. Once this criterion is met, the program advances to a new time step and the process of computing heads at each node is repeated.

Streams superimposed on the aquifer are divided into reaches and segments. A segment is a stream or diversion in which streamflow from surface sources are added at the beginning of the segment or subtracted (in the case of a diversion) at the end of a segment. A reach is the part of a segment that corresponds to an individual cell in the finite-difference grid used to simulate ground-water flow in an aquifer. A segment may consist of one or more reaches. Segments are numbered sequentially from the farthest upstream segment to the last downstream segment as are reaches within each segment. Both must also be read into the program in sequential order. The use of segments in the conceptualization of routing streamflows makes the accounting of flows where streams merge to form one, or where diversions subtract flow from a stream, easier to program. Reaches, however, form the basis for the calculation of leakage between surface water and ground water.

An example of a finite-difference grid of a hypothetical aquifer connected to streams is shown in figure 1a. In this example, the locations of cells representing the aquifer are described in terms of rows and columns. Thus, cell (1,3) is the cell located in row 1, column 3. The streams are divided into segments and reaches. Segment 1 has two reaches and begins in cell (1,3) and ends in cell (2,3) where some flow is diverted into segment 2. The remaining flow from segment 1 goes to segment 3, which has 4 reaches. Segments 2 and 4 join in cell (5,3) to form segment 5. Segments 3, 5, and 6 join in cell (5,4) to form segment 7. Two small parts of streams in segments 3 and 6 are not included in the numbering scheme because they only cross the corners of cells (2,4) and (3,5).



EXPLANATION

STREAM CHANNEL OR DIVERSION—Arrow indicates direction of flow

STREAM SEGMENT—Dots indicate section of stream used to define a segment. Enclosed number is the segment number. Open number is the reach number within a segment

SECTION OF STREAM CHANNEL OR DIVERSION—Shows sections of streams or diversions not included in model simulation

Figure 1.--Example numbering systems of streams and diversions for the simulation of stream-aquifer relations with the finite-difference ground-water flow model by McDonald and Harbaugh (1988). A, one method of numbering streams and diversions. B, an alternative method of numbering streams and diversions.

A diversion from one segment can later become tributary to another segment, which is the case for segment 2 in figure 1a. Also, more than one reach in different segments can be assigned to the same model cell. For example, reach 2 in segment 5; reach 4 in segment 3; reach 5 in segment 6, and reach 1 in segment 7 are all located in cell (5,4). Leakage to or from the aquifer from all these reaches is the accumulated sum of all the leakage, however, the leakage between each reach and the cell is calculated at the cell node (located in the center of the cell). Thus, only the head at the cell node is used to calculate leakage for each reach.

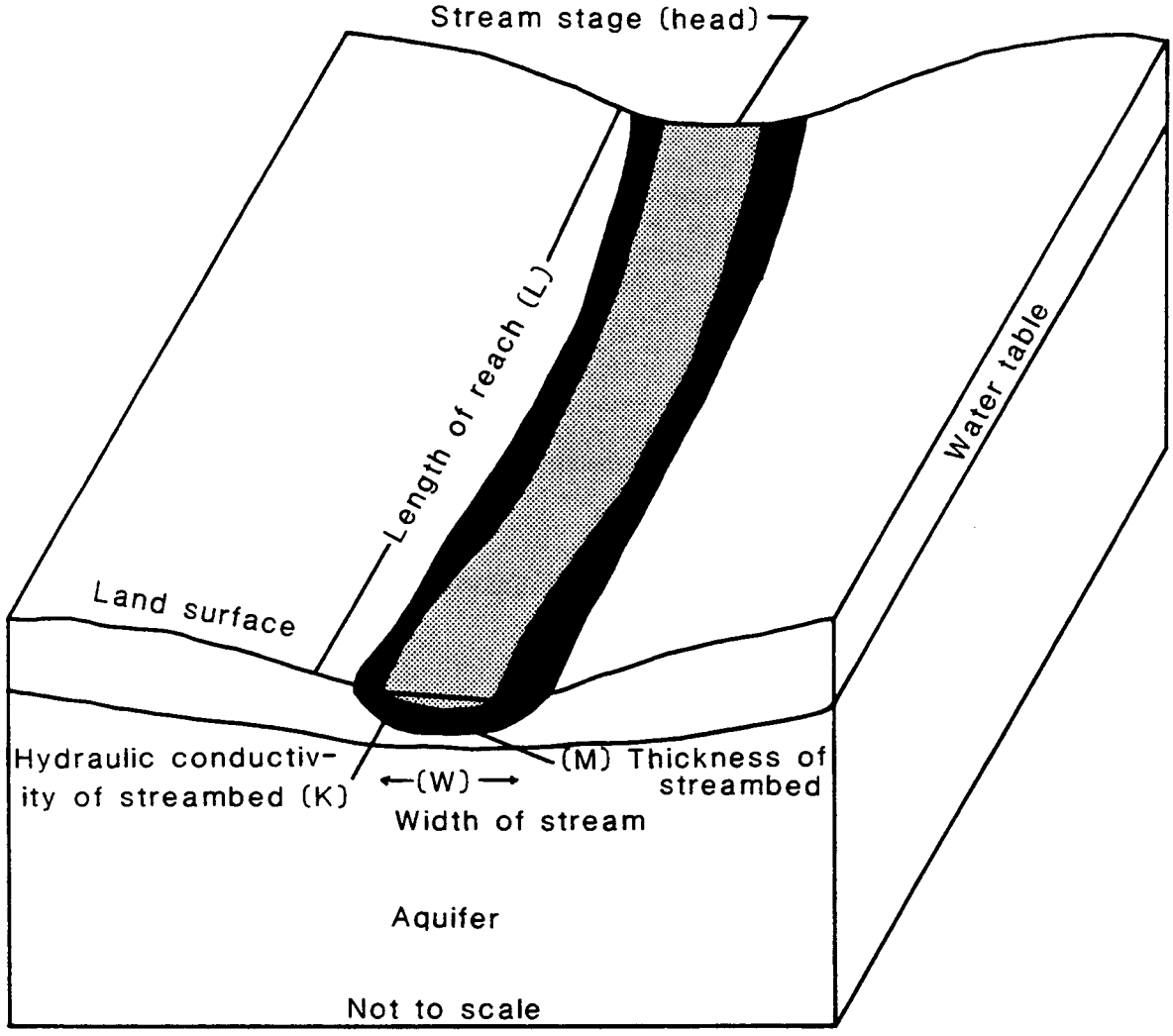
An alternative numbering scheme is shown in figure 1b whereby only one reach and segment is assigned to a cell. In the example, segment 2 (the diversion canal) begins in cell (3,3) and segment 3 begins in cell (3,4). Segment 5 has only one reach, located in cell (5,3), but it consists of parts of three streams. This scheme results in assigning fewer reaches but causes the combining of flows prior to the confluence of streams and the subtraction of flows after the diversion.

Streamflow Accounting

Streamflow is accounted for by specifying flow for the first reach in each segment that enters the modeled area, and then computing streamflow to adjacent downstream reaches in each segment as equal to inflow in the upstream reach plus or minus leakage from or to the aquifer in the upstream reach. The accounting scheme used assumes that streamflow entering the modeled layer is instantly available to downstream reaches. This assumption is generally reasonable because of the relatively slow rates of ground-water flow.

Streamflow into a segment that is formed from tributary streams is computed by adding the outflows from the last reach in each of the specified tributary segments. If a segment is a diversion, then the specified flow into the first reach of the segment is subtracted from flow in the main stream. However, if the specified flow of the diversion is greater than the flow out of the segment from which flow is to be diverted, then no flow is diverted from that segment.

Reaches in a segment are permitted to go dry whenever downward leakage to the aquifer exceeds stream inflow into the reach. The downstream reaches also go dry (streamflow into the downstream reaches is zero) and leakage into the aquifer is not permitted. Upward leakage (from the aquifer) is permitted even when there is no streamflow in a reach. This happens when the head in the aquifer exceeds the top of the streambed in a dry reach. Flow in the stream resumes and downward leakage back into the aquifer can be simulated in downstream reaches.



Conductance of streambed (CSTR)

$$CSTR = \frac{KLW}{M}$$

Figure 2.--Diagram showing part of an aquifer with a stream over it depicting how conductance of the streambed is calculated for a reach.

Computing Flow Between Stream and Aquifer

Leakage to or from a stream reach is computed by Darcy's Law as follows:

$$Q_l = CSTR\ (H_s - H_a) \qquad (1)$$

where Q_l = leakage to or from the aquifer through the streambed, (L^3/T);

H_s = head in the stream, (L);

H_a = head in aquifer side of streambed, (L); and

CSTR = conductance of the streambed (L^2/T), which is the hydraulic conductivity of the streambed times the product of the width of the stream and its length divided by the thickness of the streambed.

A sketch of the properties used in the calculation of stream leakage to or from the aquifer is shown in figure 2. The stage in the stream is for the center of the reach. The head in the aquifer side of the streambed depends upon whether the earth materials beneath the stream are saturated or not. If the materials are saturated, then the head in the aquifer side of the streambed is equal to the head in the model cell beneath the stream reach. If materials are unsaturated, then the head in the aquifer beneath the streambed is equal to the elevation of the bottom of the streambed. The relation between stream leakage and head in the model cell is shown in figure 3. For more detailed information regarding this relation, the reader is referred to McDonald and Harbaugh (1988, p. 6-1 through 6-30). If leakage through the streambed to the aquifer in a stream reach is greater than the amount of streamflow that enters the reach, then the leakage is set equal to the streamflow that enters the reach and flow out of the reach is zero. Inflow into the adjacent downstream reach is also zero.

The terms that represent leakage to or from the stream reaches are added to the finite-difference equations at the start of each iteration in the same manner as in the River Package described by McDonald and Harbaugh (1988, p. 6-12). If streamflow is zero in a reach and the head in the model cell is less than the top of the streambed, then no leakage is computed for that reach. The choice as to which terms to add to the finite-difference equations is made by comparing the most recent value of the head in the model cell to the elevation of the bottom of the streambed; or if flow in the stream reach is zero, to the top of the streambed. These comparisons are made at the start of each iteration, meaning that the most current value of head in the model cell used to calculate leakage between stream reaches and the aquifer is from the previous iteration.

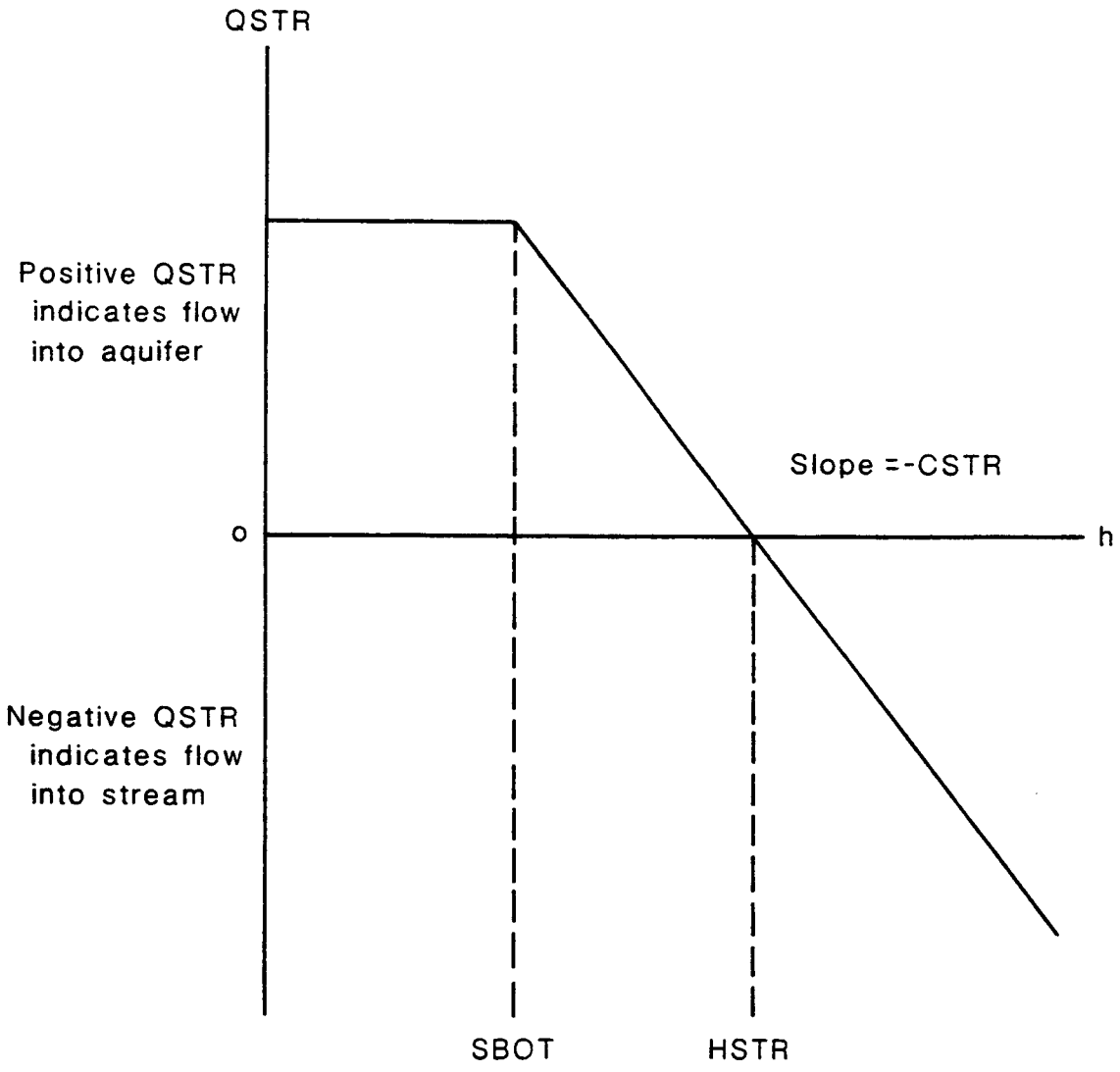


Figure 3.--Leakage, QTSR, through a streambed into an aquifer as a function of head, h, in a model cell where SBOT is the streambed bottom and HSTR is the stream stage. Slope is dependent on streambed conductance, CSTR. Modified from McDonald and Harbaugh (1988, p. 6-9).

Computing Stream Stage in Reaches

An option in the package is available that computes the stream stage in each reach. If the option is used, the stream stage is computed assuming incompressible steady flow in the stream at constant depth and using the Manning formula as described by Ozbilgin and Dickerman (1984, p. 3, eq. 1) as:

$$Q = \frac{C}{n}(A\,R^{2/3}\,S^{1/2}), \qquad (2)$$

where Q = stream discharge, (L^3/T);

n = Manning's roughness coefficient, dimensionless;

A = cross-sectional area of the stream, (L^2);

R = hydraulic radius, (L);

S = slope of the stream channel, (L/L); and

C = a constant, $(L^{1/3}/T)$, which is 1.486 for units of cubic feet per second or 1.0 for cubic meters per second.

The Manning formula, in particular, the units of (C/n), is discussed by Chow (1959, p. 98-101). The cross-sectional area and the hydraulic radius for a rectangular channel are:

$$A = wd \text{ , and} \qquad (3)$$

$$R = \frac{wd}{w + 2d} \qquad (4)$$

where d = depth of the water in the stream, (L); and

w = width of the channel, (L).

Substituting these equations into equation 2 and assuming that the depths of the streams are much less than the width, Ozbilgin and Dickerman (1984, p. 4) derived the following equation that computes stream depth:

$$d = \left[\frac{Qn}{CwS^{1/2}}\right]^{3/5} \qquad (5)$$

This is the equation used to approximate the stream stage in each reach in the Streamflow-Routing Package.

A brief discussion of Manning's roughness coefficient is presented by White (1979, p. 602-607). White (p. 605) presented estimates of Manning's roughness coefficient for a variety of stream channels. The estimates are listed in table 1.

Streamflow estimates from the Manning formula are accurate in an intermediate range of roughness coefficients (White, 1979, p. 602-607), but for deep, smooth channels and shallow, rough channels the estimates of flow may be unrealistic. Roughness coefficients can be estimated from actual measurements of stream depth and discharge. However, if field data are not available, values can be estimated from table 1.

TABLE 1.--*Experimental values of Manning's roughness coefficient* (White, 1979, p. 605)

Description of channel	Manning's roughness coefficient (n)
Excavated earth channels:	
Clean	0.022 ± 0.004
Gravelly	0.025 ± 0.005
Weedy	0.030 ± 0.005
Stony, cobbles	0.035 ± 0.010
Natural channels:	
Clean and straight	0.030 ± 0.005
Sluggish, deep pools	0.040 ± 0.010
Major rivers	0.035 ± 0.010
Floodplains:	
Pasture, farmland	0.035 ± 0.010
Light brush	0.050 ± 0.020
Heavy brush	0.075 ± 0.025
Trees	0.150 ± 0.050

The stream stage in each reach is computed prior to calculating leakage to or from the aquifer. For the first iteration, the stage for each reach is calculated using the specified inflow into a stream segment. If no streamflow is specified into a segment, the stage for all reaches in the segment will equal the top of the streambed. Leakage terms are then calculated on the basis of the computed stream stage, streambed conductance, and the initial (starting) head in each corresponding model cell. These terms are added to the finite-difference equations used to calculate a new

head for each model cell. The process continues, except in the following iterations, leakage calculated in the previous iteration is added or subtracted from streamflow prior to calculating a new stream stage, and new leakage terms are computed on the basis of the head in the corresponding model cell calculated from the previous iteration.

Assumptions and Limitations

The Streamflow-Routing Package does not include a time function for routing flows specified for the first reach in each segment that enters the modeled area to downstream reaches. The accounting scheme used assumes that streamflow entering the modeled area is instantly available in downstream reaches for each specified period in the model simulation. (The specified period is referred to as a stress period by McDonald and Harbaugh, 1988, p. 1-5). This assumption is generally valid because of the relatively slow rate of ground-water flow and because stress periods used in model simulations are commonly months or years, whereas changes in streamflow along a course of a stream are usually in days.

Stream stage for a reach can be calculated using the Manning formula and assuming a rectangular channel with a stream width much greater than depth. The assumption of a rectangular channel with width much greater than depth is generally valid but may not be appropriate for all streams. Thus, using this method to calculate a stream stage may not be valid for all cases.

The numerical scheme where information from the pervious iteration is used to determine stream stage, streamflow, and the leakage terms in the finite-difference equations can, in some cases, cause instability in the iterative solution process. Such instability may be more prominent when stream leakage is a dominant part of the simulation.

Streambed conductance values used to calculate leakage between a reach and a model cell are specified a constant for each stress period. Widths of a stream channel usually vary depending on the flow and stage in the stream which may result in varying streambed conductance values. In the package presented herein, changes in the streambed conductances due to changing streamflows must be computed separately from the model and then added at the beginning of each stress period.

Another limitation of the model is that leakage from the stream to the aquifer is assumed to be instantaneous, which is reasonable where the thickness of the unsaturated zone between a stream and aquifer is not great. The assumption may be unreasonable when the thickness of the unsaturated zone is large, because of the time lag between stream infiltration and recharge to the aquifer. In addition, the stream infiltration rate may be greater than the amount of recharge to the aquifer because some of the infiltrated water may be used to replace soil moisture and(or) be used by plants adjacent to the stream.

The Streamflow-Routing Package can be used to simulate the leakage between streams and aquifers. But the user must consider the limitations associated with the package when using it as the package may not be appropriate for all cases. The accuracy of the model results will, to a large extent, be dependent on the size of the model cells, the time

intervals used in the simulations, and the closure criteria for completing the iteration cycle in the modular model (McDonald and Harbaugh, 1988, p. 2-20 through 2-25).

RESULTS OF TEST SIMULATIONS

The package is designed primarily to simulate vertical leakage from partially penetrating streams with a distinct streambed and associated hydraulic conductivity. With some slight changes in the concept of calculating streambed conductance values, the program can be used to simulate a fully penetrating stream. This was done in comparing simulation results to two analytical solutions, which assume a fully penetrating stream. The comparisons were done to verify that the program works correctly.

For the first test, results from the Streamflow-Routing Package were compared to results from an analytical solution developed by Oakes and Wilkinson (1972). An idealized aquifer with a river flowing through the middle was chosen and is shown in figure 4. The width of the aquifer perpendicular to the river was 4,000 ft on each side, while the length parallel to the river was 13,000 ft. Assumptions used in both the analytical solution and the model simulation include:

1. The lateral boundaries of the aquifer are impermeable (no flow is allowed).
2. The rocks beneath the aquifer are impermeable.
3. The river penetrates the entire depth of the aquifer and has vertical banks.
4. The river is not separated from the aquifer by any confining material.
5. The transmissivity and storage coefficient are constant throughout the aquifer and remain constant in time.
6. The aquifer is unconfined and Darcy's Law is valid.
7. The flow of ground water is horizontal.
8. The water level in the river is constant along its length and with time.
9. The infiltration of recharge to the aquifer is instantaneous (no delay between the time precipitation infiltrates the surface until it reaches the water table).
10. The discharge from the aquifer is only to the river.

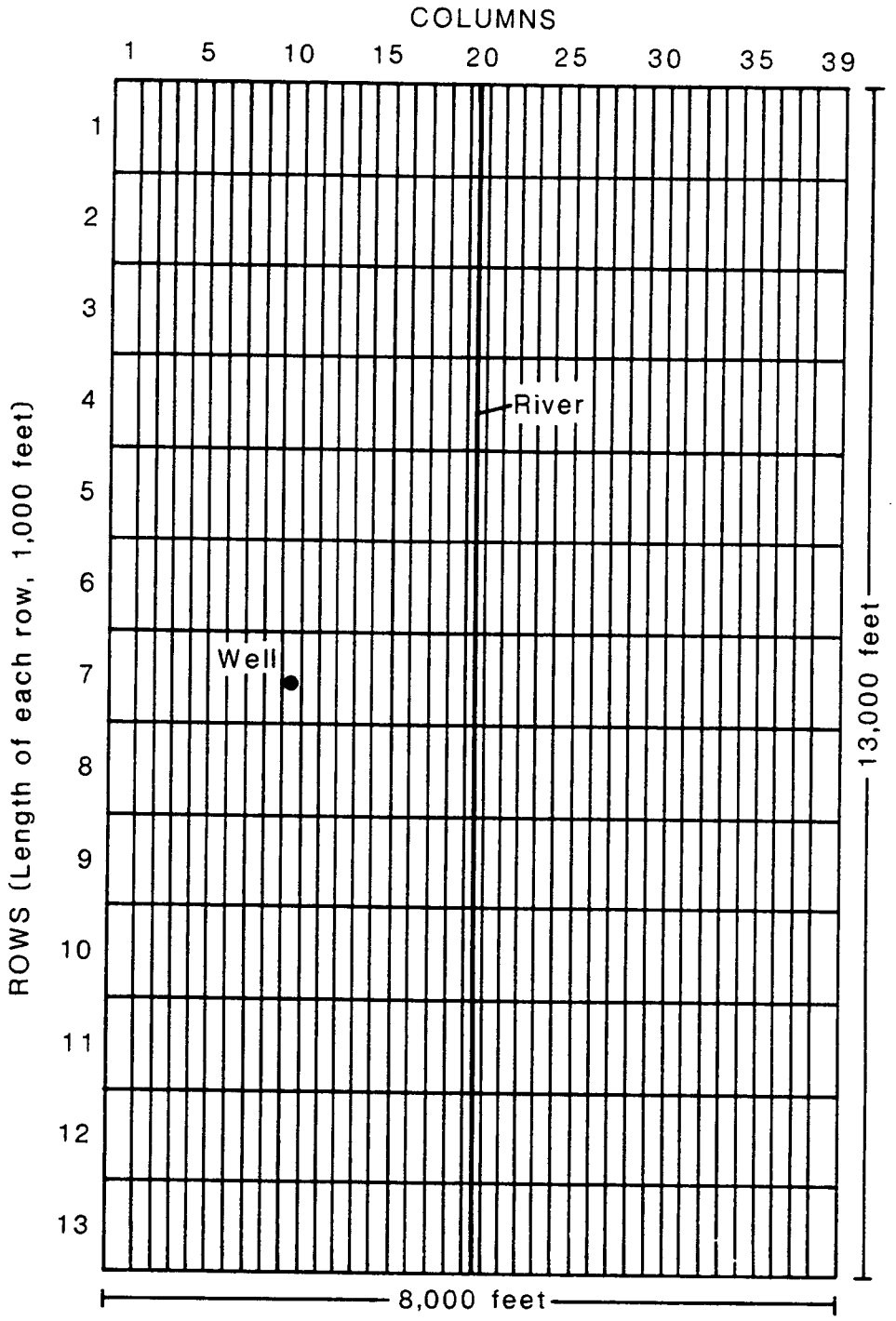


Figure 4.--Sketch showing an idealized aquifer with model grid and associated river and well.

Transmissivity of the aquifer used for both the analytical solution and in the model simulation was 3,200 ft^2/d (0.037 ft^2/s). The storage efficient was 0.20. Because the river is assumed fully penetrating and the aquifer is not separated from the river by any confining material, the streambed conductance value was assumed equal to the transmissivity of the aquifer (in this example, the width of the river is assumed equal to the depth of the aquifer) times the length of the river in each cell (1,000 ft) divided by an assumed 1-foot thickness of the riverbed. Actually, any large streambed conductance value can be used as long as the head value in the model cell (node) that corresponds to the river remains constant during the simulation. Results of varying the streambed conductance value indicates that for this problem, streambed conductances greater than 10 ft^2/d produce nearly the same results. The Streamflow-Routing Package is not really needed to simulate this condition, as the stream could have been represented using constant heads in model cells representing the stream but the simulation was done to determine if the package correctly accumulates flow from the aquifer into the stream.

Annual recharge to the aquifer was 1.5 ft and it was applied evenly over the aquifer. However, the daily recharge rate varied according to a sinusoidal distribution for the first 180 days, while no recharge was allowed for the following 180 days (fig. 5). The sinusoidal distribution was divided into 15-day intervals for the model simulation and the rate for the middle of each interval was used as a constant. The distribution used in the simulation is also shown in figure 5.

A total of six 360-day infiltration periods were used in the simulation. The first five periods were done to allow the model to reach a stable yearly cycle because the initial starting water level for each model cell was not known. Results of the model simulation from the sixth infiltration period are compared to the results from the analytical solution for an observation well 2,000 ft from the river (fig. 6). The location of the well is shown in figure 4. The same results were obtained using the River Package in the modular model (McDonald and Harbaugh, 1988). The data set used in the model simulation with the Streamflow-Routing Package and the results for the last 360-day infiltration period are listed in Appendix I.

The water level computed with the model closely matches the water level calculated from the analytical solution (fig. 6a). Streamflow computed with the model (fig. 6b) is slightly less during the period of increasing streamflow (0-100 days) and slightly more than streamflow calculated from the analytical solution immediately following peak streamflows (120-180 days).

The slight difference in flows between the simulation results and the analytical solution most likely is caused either by the size of the model cells in the direction of ground-water flow (columns) or by the length of the stress periods used to simulate the changes in recharge. Increasing the number of columns (decreasing the size of model columns) and(or) reducing the length of the stress periods probably would result in a closer approximation of the computed flow to the flow calculated with the analytical solution.

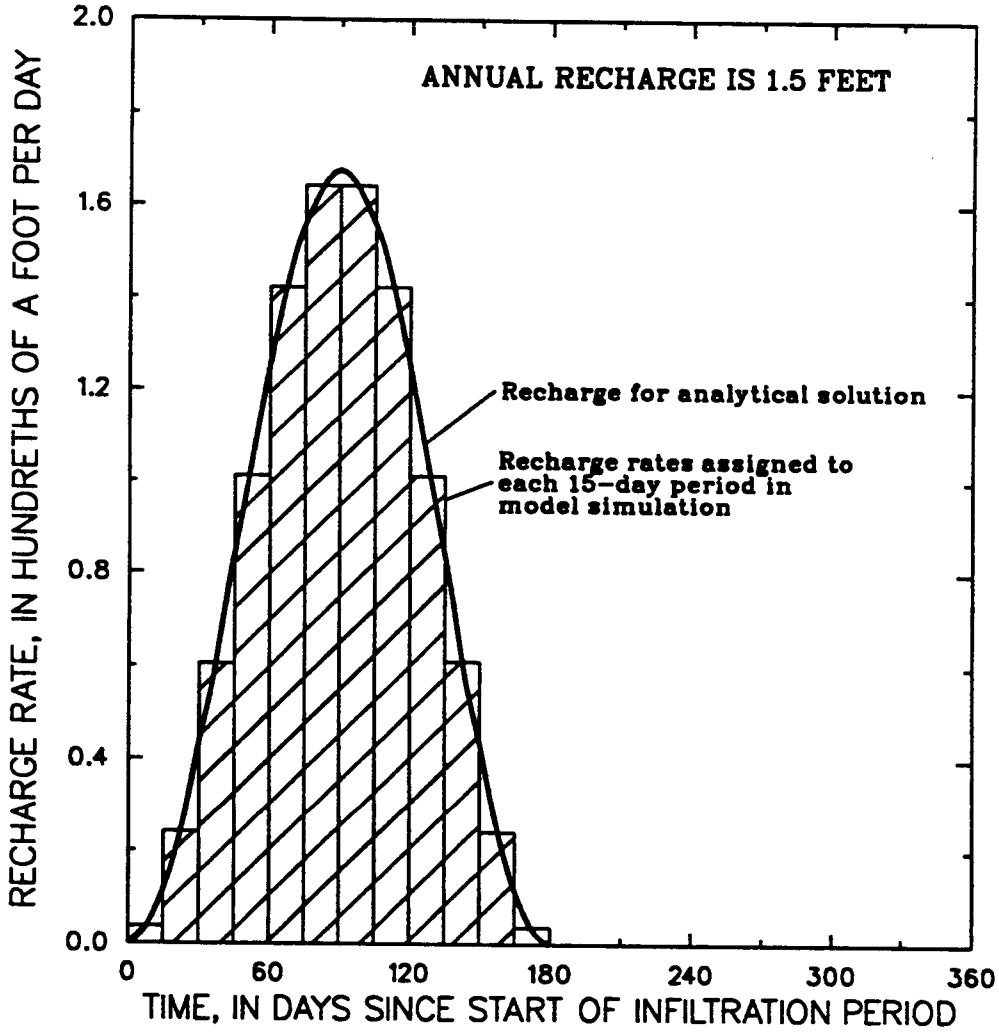


Figure 5.--Distribution of recharge used for analytical solution and how it was divided into time intervals for the simulation of test problem 1.

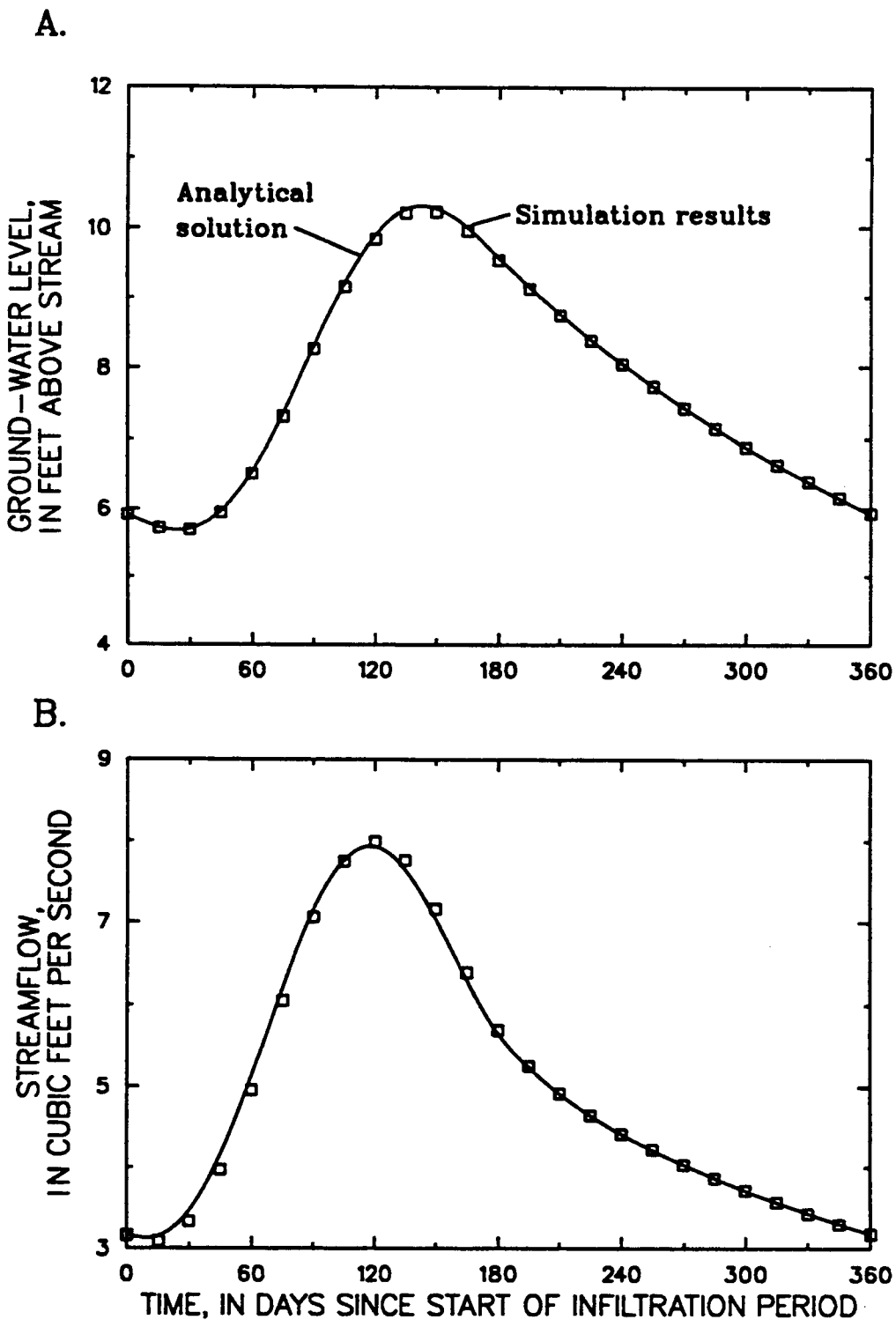


Figure 6.--Comparison of simulation results for test problem 1 to analytical solution developed by Oakes and Wilkinson (1972). A, Ground-water levels at a well 2,000 feet from river; B, Base flows for the river summed over its modeled length.

For the second test, the function that computes the head in the stream as well as simulating changes in flows to and from the aquifer was compared to an analytical solution developed by Cooper and Rorabaugh (1963). The same model grid used in the first test (fig. 4) was used in this test. The assumptions used in the previous test are the same except for assumptions 8-10, which are replaced with the following assumptions:

1. The recharge to the aquifer is only from the river as river stage increases with time.

2. The discharge from the aquifer is also only to the river as river stage decreases with time.

The aquifer properties used in this test were the same as those used in the first test. The analytical solution used from Cooper and Rorabaugh (1963, p. 355-358) is applicable for the case where the lateral boundary is at infinity (referred by Cooper and Rorabaugh as semi-infinite). The impermeable boundary assigned at 4,000 ft for this test is of sufficient distance from the river as not to interfere with the results.

A flood in the river was simulated for a 30-day period. The procedure used to calculate the distribution of streamflow for the 30-day period and for 60 days following the flood was first to calculate a distribution of river stage using equation 71 in Cooper and Rorabaugh (1963, p. 355), assuming a maximum flood stage of 4 ft above the initial river stage. The streamflow distribution shown in figure 7 then was calculated by rearranging equation 5 and solving for streamflow. The streamflow distribution was calculated from the river stage distribution, a river width of 100 ft, a roughness coefficient of 0.02377, a slope of 0.0001, and a constant (C in equation 5) of 1.486 (used when flow is in cubic feet per second). The input for this test simulation and selected parts of the printed results are included in Appendix II.

Streamflow for the first 30 days was divided into 1-day periods for simulation (fig. 7). Simulation results of computed river stage are shown in figure 8 as are the results from manually calculating river stage using equation 71 of Cooper and Rorabaugh (1963, p. 355). The simulation results are the same as the manually calculated values, indicating the equation that calculates stream stage has been correctly written in the computer program. Leakage computed between the river and aquifer in the simulation also closely approximated the analytical solution for a semi-infinite aquifer (fig. 9). Results of varying both the number of columns and the length of stress periods used to simulate the flood wave indicate that both the number of columns and the length of the time step are important in exactly duplicating the analytical solution.

The ground-water flow model with the Streamflow-Routing Package has an advantage over analytical solutions because it can be used to simulate complex systems. The example of a stream system shown in figure 1a is used to illustrate most of the features of the package. The example assumes an aquifer 6,000 ft wide by 6,000 ft long divided into six equally spaced rows and columns. The transmissivity of the aquifer is 0.08 ft^2/s (6,900 ft^2/d). Recharge to the aquifer occurs only from stream leakage as does discharge from the aquifer. Land-surface altitude varies over the modeled area as shown in figure 10.

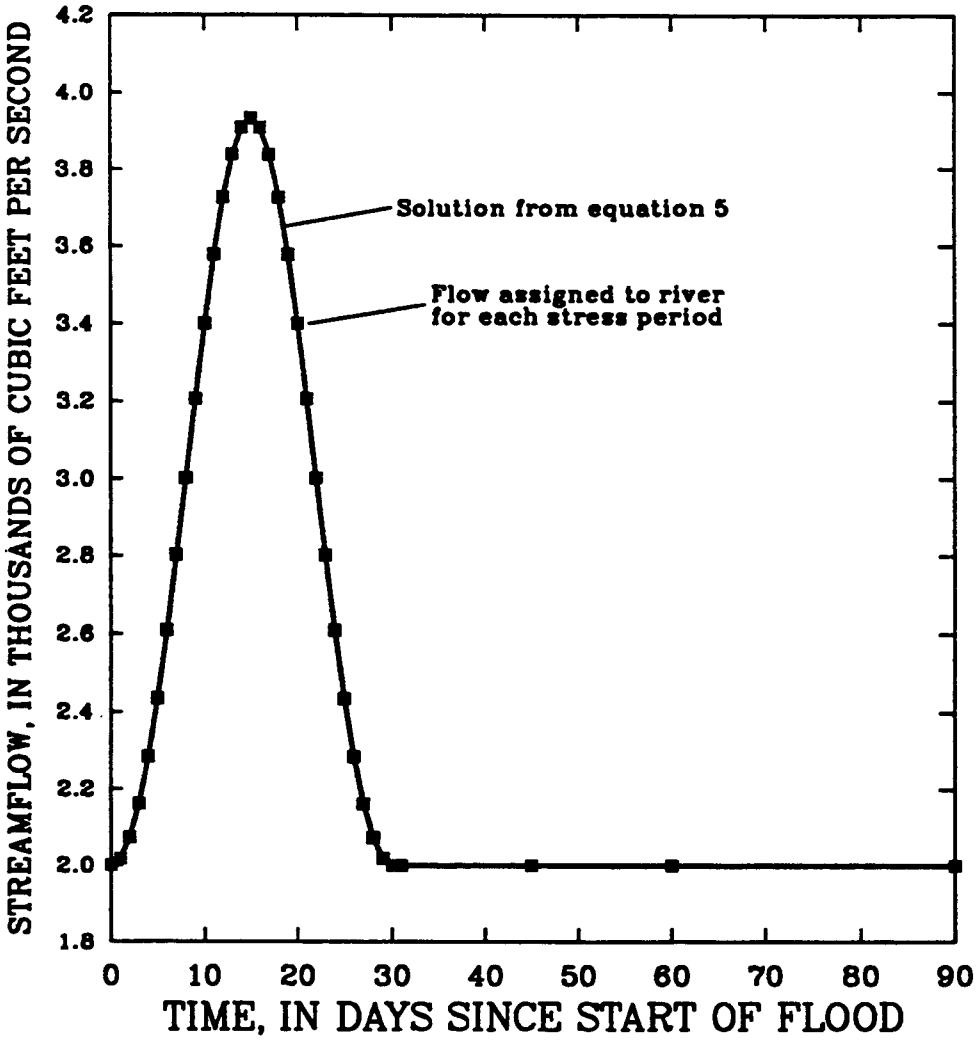


Figure 7.--Distribution of streamflow for a 30-day flood event used for the simulation of test problem 2.

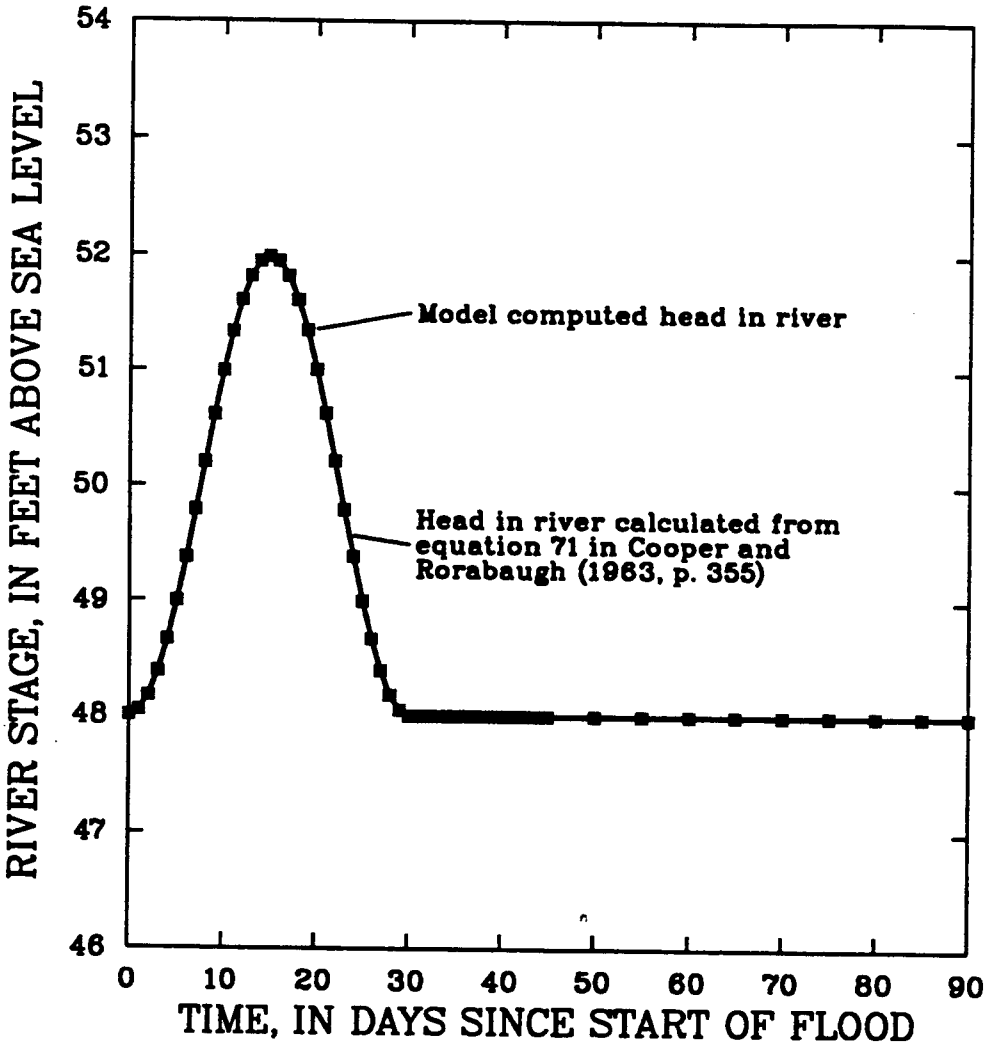
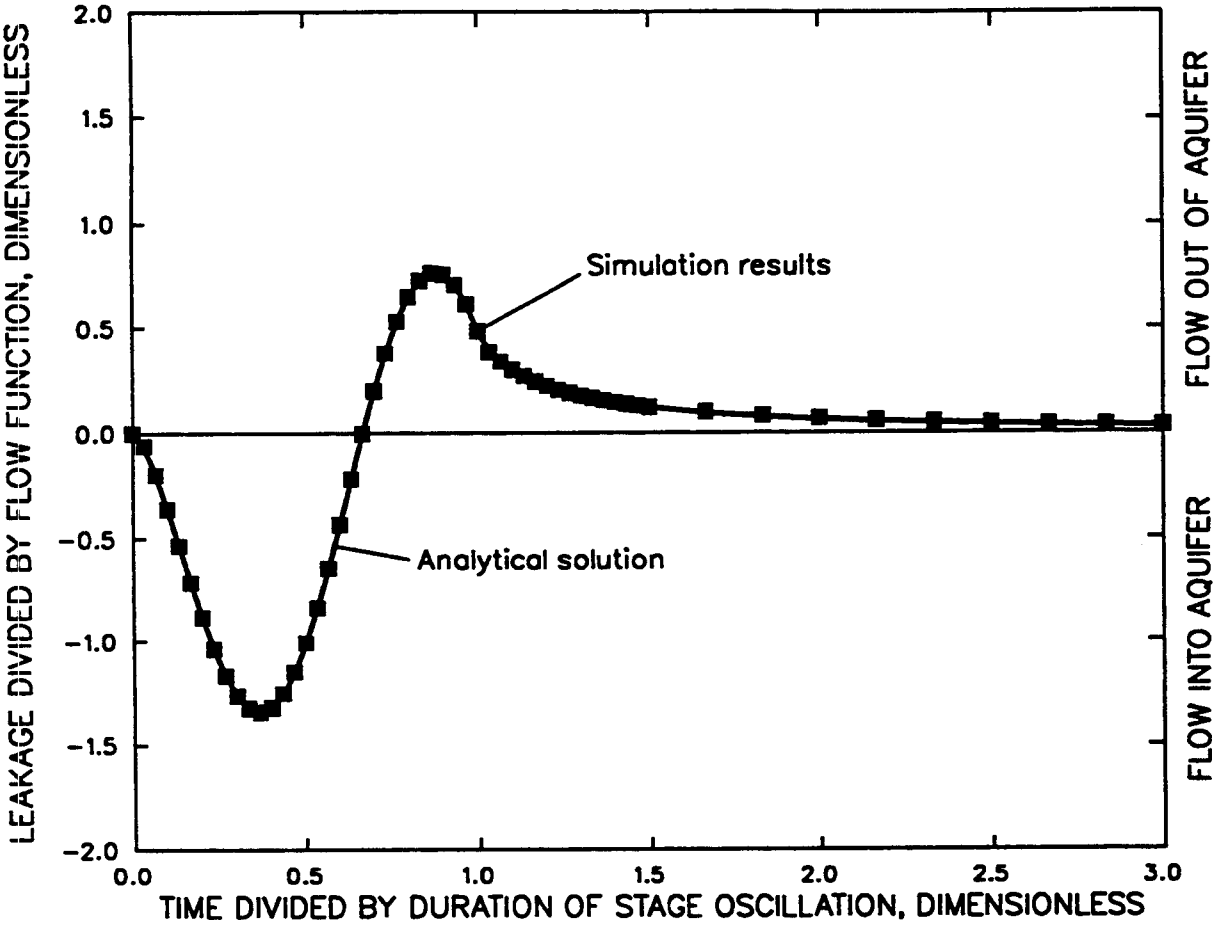


Figure 8.--Comparison of model computed river stage to stage calculated from the Manning formula assuming a rectangular stream channel (equation 5). Results indicate that equation 5 has been correctly programmed into the Streamflow-Routing Package.



EXPLANATION OF LEAKAGE AND FLOW FUNCTION

Leakage = flow between river and aquifer per unit river length, L^2 / T .

Flow function = $(h_0 / 2)\sqrt{W\ T\ St}$, L^2 / T ,

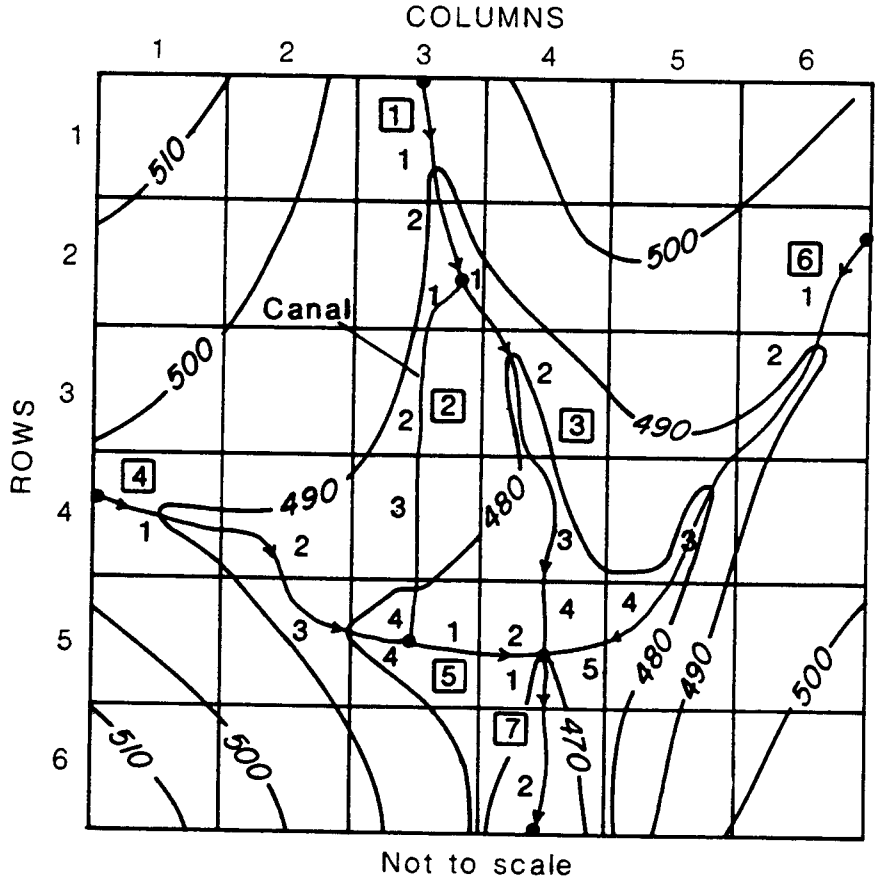
where h_0 = maximum change in river stage, L ,

T = transmissivity, L^2 / T ,

St = storage coefficient, dimensionless, and

W = frequency of river stage change, which equals two pi divided by duration of stage oscillation, T^{-1} .

Figure 9.--Comparison of model computed flows into and out of aquifer for test problem 2 to an analytical solution. The analytical solution was developed by Cooper and Rorabaugh (1963) for a semi-infinite aquifer (eqs. 81 and 82). Sign convention is opposite that used in the finite-difference ground-water flow model but was used to be consistent with Cooper and Rorabaugh.



EXPLANATION

—500— TOPOGRAPHIC CONTOUR--Shows elevation of land surface. Contour interval 10 feet. Datum is sea level

[1] / 3 STREAM CHANNEL OR DIVERSION--Arrow indicates direction of flow. Dots identify individual segments. Enclosed number is the segment number. Open number is the reach number within a segment

Figure 10.--Model grid, numbering system of streams and diversions, and topographic contours for example problem.

The example includes 7 stream segments with a total of 23 reaches. There is one diversion (segment 2) and two places where streams join (segments 2 and 4 join to make segment 5 and segments 3, 5, and 6 join to make segment 7). Stream stages are also computed for each reach and the necessary input is included in Appendix III and also table 2. The streams range in width from 5 to 10 ft. Streambed conductance values also vary depending on the length and width of each stream reach. The hydraulic conductivity of the streambed deposits is 4×10^{-4} ft/s. A steady state solution was simulated using average annual streamflows for each stream.

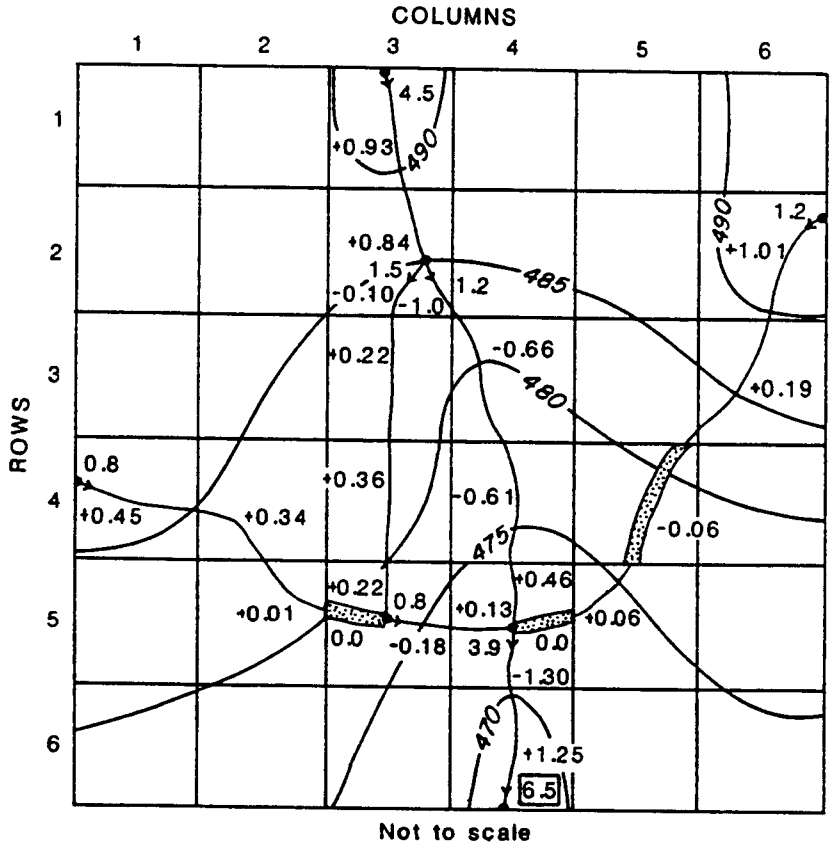
Water levels and streamflows and leakage between the stream reaches and aquifer are shown in figure 11. Total streamflow into the area was 6.5 ft^3/s with an equal amount of streamflow leaving the area. Leakage to the aquifer was 5.22 ft^3/s and leakage from the aquifer was 5.22 ft^3/s. The areas of leakage to the aquifer were generally the upstream areas while flow from the aquifer to the streams were generally in the downstream reaches. Several of the reaches went dry as leakage to the aquifer exceeded streamflows (shown in fig. 11) but downstream reaches began flowing when the head in the aquifer exceeded the top of the streambed. The data input and the simulation results are included in Appendix III.

The example illustrates that the Streamflow-Routing Package can simulate complex systems. It can also be used to simulate a variety of systems where aquifer properties change in space and where the amount and distribution of recharge and discharge can change in space and time.

IMPLEMENTATION OF STREAMFLOW-ROUTING PACKAGE IN THE GROUND-WATER FLOW MODEL

The Streamflow-Routing Package (STR1) is designed for incorporation into the ground-water flow model by McDonald and Harbaugh (1988). The package consists of four FORTRAN subroutines, referred to as modules, that carry out the following procedures: allocate memory, read and prepare data, formulate the finite-difference equations, and compute mass-balance components and print results. The modules are respectively named STR1AL, STR1RP, STR1FM, and STR1BD. The first three characters of the names identify the modules as being part of the Streamflow-Routing Package; the next character identifies the version number of the package; the last two characters identify the procedure performed in the module. The procedures used by the Streamflow-Routing Package are consistent with procedures used by existing packages in the ground-water flow model (fig. 12).

The MAIN program of the ground-water flow model must be modified to call the four modules of the Streamflow-Routing Package. The procedures on the left side of figure 12 are listed in the order each are carried out in the MAIN program of the model. Call statements to the modules of the Streamflow-Routing Package must be placed in sections of the MAIN program in which the particular procedure is being carried out for other packages. For example, the STR1AL module must be called within the section of the MAIN program in which other allocate modules (BAS1AL, RIV1AL, etc.) are called. In all sections of the MAIN program, the call to the Basic (BAS1) Package module (subroutine) must come before any other module call statements. Actually, the modules of the Streamflow-Routing Package could be used to replace the River Package in the model. The FORTRAN call statements to be added to the MAIN program are as follows:



EXPLANATION

480 — GROUND-WATER LEVEL CONTOUR--Shows elevation of ground-water level. Contour interval 5 feet. Datum is sea level.

4.5 — STREAMFLOW INTO FIRST REACH OF A SEGMENT--In cubic feet per second. Arrow indicates direction of flow. Dots identify individual segments.

6.5 — STREAMFLOW OUT OF LAST REACH OF LAST SEGMENT--In cubic feet per second. Arrow indicates direction of flow.

+0.36 — LEAKAGE BETWEEN STREAM AND AQUIFER FOR EACH REACH--In cubic feet per second. Positive value indicates flow from stream to aquifer. Negative value indicates flow from aquifer into stream.

DRY STREAM--Streamflow into reach is zero. In these reaches, leakage from stream to aquifer is zero but leakage from aquifer to stream is still allowed.

Figure 11.--Simulation results for example problem.

```
      IF (IUNIT(??).GT.0) CALL STR1AL(ISUM,LENX,LCSTRM,ICSTRM,MXSTRM,
     1            NSTREM,IUNIT(??),IOUT,ISTCB1,ISTCB2,NSS,NTRIB,
     2            NDIV,ICALC,CONST,LCTBAR,LCTRIB,LCIVAR)

      IF (IUNIT(??).GT.0) CALL STR1RP(X(LCSTRM),X(ICSTRM),NSTREM,
     1            MXSTRM,IUNIT(??),IOUT,X(LCTBAR),NDIV,NSS,
     2            NTRIB,X(LCIVAR),ICALC,IPTFLG)

      IF (IUNIT(??).GT.0) CALL STR1FM(NSTREM,X(LCSTRM),X(ICSTRM),
     1           X(LCHNEW),X(LCHCOF),X(LCRHS),X(LCIBOU),MXSTRM,
     2           NCOL,NROW,NLAY,IOUT,NSS,X(LCTBAR),NTRIB,
     3           X(LCTRIB),X(LCIVAR),ICALC,CONST)

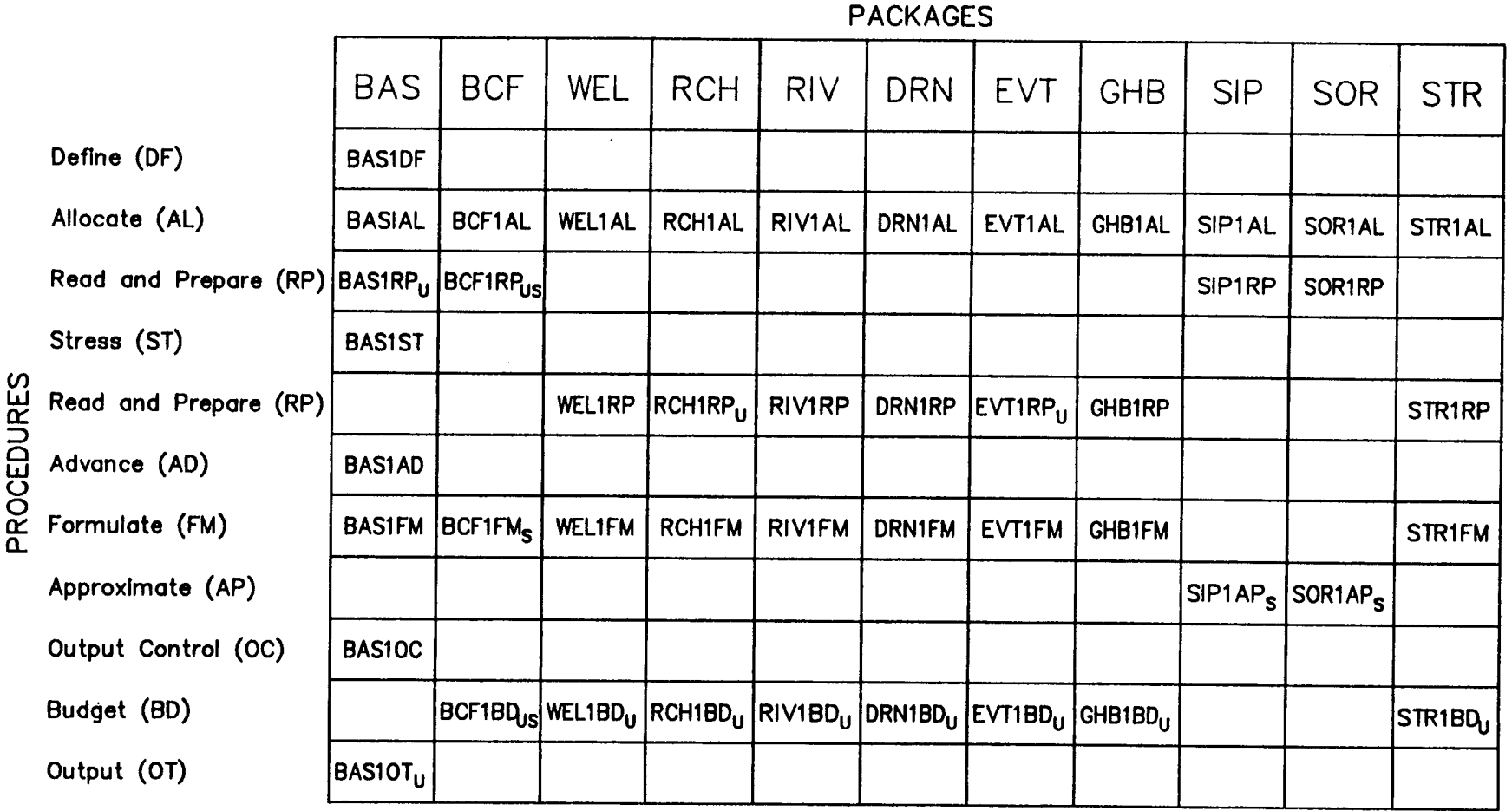
      IF (IUNIT(??).GT.0) CALL STR1BD(NSTREM,X(LCSTRM),X(ICSTRM),
     1     X(LCIBOU),MXSTRM,X(LCHNEW),NCOL,NROW,NLAY,DELT,VBVL,
     2     VBNM,MSUM,KKSTP,KKPER,ISTCB1,ISTCB2,ICBCFL,X(LCBUFF),IOUT,
     3     NTRIB,NSS,X(LCTRIB),X(LCTBAR),X(LCIVAR),ICALC,CONST,IPTFLG)
```

The user must specify a number between 13 and 24 for IUNIT(??), which is defined in the Basic Package (McDonald and Harbaugh, 1988, p. 4-9 through 4-12). Then whenever the Streamflow-Routing Package is to be used, a positive number is specified in the proper location in the Basic Package. This number corresponds to the FORTRAN unit number assigned to the Streamflow-Routing Package data set.

The first call statement is added where space is allocated to the "X" array (between comments C4 and C5 in the MAIN program). The second statement is added where information for a package is read and prepared for each stress period (between comments C7B and C7C in the MAIN program). The third statement is added where the finite-difference equations are formulated for each time step (between comments C7C2A and C7C2B in the MAIN program). The last statement is added where budget terms are calculated for each time step (between comment C7C4 and C7C5 in the MAIN program).

The four call statements are included in an example MAIN program at the end of section on module documentation. In the example, the IUNIT number is specified as 13. Thus, to activate the Streamflow-Routing Package for a simulation using the example MAIN program, a positive number must be specified to IUNIT 13 in the Basic (BAS1) Package. The number is the FORTRAN unit number assigned to the data input for the Streamflow-Routing Package.

Information needed to simulate the effects of rivers, streams, canals and ditches on aquifer systems with the Streamflow-Routing Package is in a format similar to that of the River Package described in McDonald and Harbaugh (1988, p. 6-1 through 6-36). The instructions on the ordering of the data input is explained in the following section. A sample input data set is shown in table 2, which corresponds to the example in figures 1a and 10.



Modified from McDonald and Harbaugh (1988, figure 15)

EXPLANATION

BCF1BD$_{US}$ NAME OF MODULE — Subscript, U, indicates that utility modules are utilized. Subscript, S, indicates that submodules are utilized.

Figure 12.--Primary modules of finite-difference ground-water flow model organized by procedure and package.

TABLE 2.--*Sample data set for Streamflow-Routing Package*

[From example problem; see figures 1a and 10. For description of variables in the explanation, see the following section]

Data item	Explanation	Input records								
		Column Numbers								
		1	2	3	4	5	6	7	8	
		12345678901234567890123456789012345678901234567890123456789012345678901234567890								
1	MXSTRM,NSS,NTRIB,NDIV,ICALC,CONST, ISTCB1,ISTCB2	23	7	3	1	1	1.486	-1	0	
2	ITMP,IRDFLG,IPTFLG--FIRST STRESS PERIOD	23	0	0						
3	LAY,ROW,COL,SEG,REACH,FLOW,STAGE,COND,SBOT,STOP	1	1	3	1	1	4.5	495.0	1.20	490.0	492.0
		1	2	3	1	2		490.0	0.60	485.0	487.0
		1	2	3	2	1	1.5	487.0	0.20	483.0	485.0
		1	3	3	2	2		486.0	0.40	482.0	484.0
		1	4	3	2	3		484.0	0.40	480.0	482.0
		1	5	3	2	4		480.0	0.20	476.0	478.0
		1	2	3	3	1	-1.0	486.0	0.40	481.0	483.0
		1	3	4	3	2		482.0	1.20	477.0	479.0
		1	4	4	3	3		478.0	1.20	473.0	475.0
	ONE RECORD FOR EACH REACH	1	5	4	3	4		475.0	0.60	470.0	472.0
		1	4	1	4	1	0.8	492.0	0.40	489.0	490.0
		1	4	2	4	2		488.0	0.32	485.0	486.0
		1	5	2	4	3		483.0	0.32	480.0	481.0
		1	5	3	4	4		480.0	0.20	477.0	478.0
		1	5	3	5	1	-1.0	478.0	0.20	475.0	476.0
		1	5	4	5	2		474.0	0.20	471.0	472.0
		1	2	6	6	1	1.2	495.0	0.80	491.0	493.0
		1	3	6	6	2		490.0	0.80	486.0	488.0
		1	4	5	6	3		480.0	0.80	476.0	478.0
		1	5	5	6	4		477.0	0.60	473.0	475.0
		1	5	4	6	5		474.0	0.20	470.0	472.0
		1	5	4	7	1	-1.0	472.0	0.60	467.0	469.0
		1	6	4	7	2		469.0	1.20	464.0	466.0
4	WIDTH,SLOPE,ROUGH	10.	0.007	0.030						
		10.	0.007	0.030						
		5.	0.002	0.022						
		5.	0.002	0.022						
		5.	0.002	0.022						
	ONE RECORD FOR EACH REACH	5.	0.004	0.022						
		10.	0.005	0.030						
		10.	0.005	0.030						
		10.	0.005	0.030						
		10.	0.005	0.030						

TABLE 2.--*Sample data set for Streamflow-Routing Package--Continued*

Data item	Explanation	Input records
		Column Numbers
		1 2 3 4 5 6 7 8
		12345678901234567890123456789012345678901234567890123456789012345678901234567890
4	WIDTH,SLOPE,ROUGH--CONTINUED	5. 0.004 0.022
		5. 0.004 0.022
		5. 0.004 0.022
		5. 0.004 0.022
		5. 0.005 0.022
		5. 0.005 0.022
	ONE RECORD FOR EACH REACH	5. 0.005 0.022
		5. 0.008 0.022
		5. 0.007 0.022
		5. 0.004 0.022
		5. 0.003 0.022
		10. 0.004 0.030
		10. 0.004 0.030
5	ITRIB(1),ITRIB(2),... ITRIB(NTRIB)	0 0 0
		0 0 0
		1 0 0
	ONE RECORD FOR EACH SEGMENT	0 0 0
		2 4 0
		0 0 0
		3 5 6
6	IUPSEG	0
		1
		0
	ONE RECORD FOR EACH SEGMENT	0
		0
		0
		0

Input Instructions

FOR EACH SIMULATION:

STR1AL

1. Data:	MXSTRM	NSS	NTRIB	NDIV	ICALC	CONST	ISTCB1	ISTCB2
Format:	I10	I10	I10	I10	I10	F10.0	I10	I10

FOR EACH STRESS PERIOD:

STR1RP

2. Data:	ITMP	IRDFLG	IPTFLG
Format:	I10	I10	I10

3. Data:	Layer	Row	Col	Seg	Reach	Flow	Stage	Cond	Sbot	Stop
Format:	I5	I5	I5	I5	I5	F15.0	F10.0	F10.0	F10.0	F10.0

(Item 3 normally consists of one record for each reach. Records are read in sequential order from upstream to downstream, first by segments, and then by reaches. The downstream ordering and reading of segments and reaches are important as the order determines the connection of inflows and outflows. If ITMP is negative or zero, items 3-6 are not read.)

If stream stages for each reach are to be calculated (ICALC>0), then the following data set is read in sequential order of segment and reach.

4. Data:	Width	Slope	Rough
Format:	F10.0	F10.0	F10.0

If the maximum number of tributaries (NTRIB) that can join a segment is greater than zero, then the following data set is read. One record for each segment is read in sequential order. A record is necessary even for segments that do not have tributaries. In this case a blank record or a record with all zeros is read.

5. Data:	Itrib(1)	Itrib(2)	...	Itrib(NTRIB)
Format:	I5	I5	...	I5

If diversions are specified (NDIV>0), then the following data set is read. One record is read for each segment in sequential order. For segments that are not diversions, zeros or blanks are specified for each input item.

6. Data:	Iupseg
Format:	I10

Explanation of Fields Used in Input Instructions

MXSTRM --is the maximum number of stream reaches that can be active during the simulation.

NSS --is the maximum number of segments that can be used during the simulation.

NTRIB --is the maximum number of tributary segments that can join during a simulation. Ten is the maximum number allowed as currently specified in the program.

NDIV --is a flag, which when positive, specifies that diversions from segments are to be simulated.

ICALC --is a flag, which when positive, specifies that stream stages in reaches are to be calculated.

CONST --is a constant value used in calculating stream stage in reaches. It is specified whenever ICALC is greater than zero. This constant is 1.486 for flow units of cubic feet per second and 1.0 for units of cubic meters per second. The constant must be multiplied by 86,400 when using time units of days in the simulation. (For an explanation of time units, see McDonald and Harbaugh, 1988, p. 4-10.)

ISTCB1 --is a flag and a unit number.

If ISTCB1>0, it is the unit number to which leakage between each stream reach and the corresponding model cell will be saved on disk whenever the variable ICBCFL is specified. (See McDonald and Harbaugh, 1988, p. 4-15, for details about the Output Control Package used to specify ICBCFL.)

If ISTCB1=0, leakage between each reach and corresponding model cell will not be printed nor filed on disk.

If ISTCB1<0, streamflows into and out of each reach and leakage between each reach and corresponding model cell will be printed whenever the variable ICBCFL is specified.

ISTCB2 --is a flag and unit number for an option of storing streamflow out of each reach instead of having the results printed.

If ISTCB2>0, it is the unit number where streamflow out of each reach will be saved on disk whenever the variable ICBCFL is specified.

If ISTCB2$\leq$0, streamflows out of each reach will not be saved on disk.

ITMP --is a flag and a counter.

If ITMP<0, stream data from the last stress period will be reused.

If ITMP≥0, ITMP will be the number of reaches active during the current stress period.

IRDFLG --is a flag, which when positive, suppresses printing of the input data set specified for a stress period. The input data set is printed for a stress period if the value is zero or blank.

IPTFLG --is a flag, which when positive, suppresses printing of results for a stress period. The results are printed for a stress period if the value is zero or blank and whenever the variable ICBCFL is specified.

Layer --is the layer number of the cell containing the stream reach.

Row --is the row number of the cell containing the stream reach.

Col --is the column number of the cell containing the stream reach.

Seg --is a number assigned to a group of reaches. Segments must be numbered in downstream order and are read into the program in sequential order.

Reach --is a sequential number in a segment that begins with one for the farthest upstream reach and continues in downstream order to the last reach in the segment. Reaches must be read in sequentially as the order reaches are read into the program determines the order of connection of inflows and outflows.

Flow --is the streamflow, in length cubed per time, entering a segment. This value is specified only for the first reach in each segment. The value is either a zero or a blank when the reach number (Reach) is not 1. When inflow into a segment is the sum of outflow from a specified number of tributary segments, the segment inflow values are specified as a -1. (Note: if the specified inflow to a diversion is greater than the flow in the reach from which flow is to be diverted, then no flow is diverted from the stream.)

Stage --is the stream stage, in units of length.

Cond --is the streambed hydraulic conductance, in units of length squared per time.

Sbot --is the elevation of the bottom of the streambed, in units of length.

Stop --is the elevation of the top of the streambed, in units of length.

Width --is the width of the stream channel, in units of length. It is used only when stream stage in each reach is calculated.

Slope --is the slope of the stream channel in each reach, in units of length per length. It is used only when stream stage in each reach is calculated.

Rough --is Manning's roughness coefficient for each stream reach. It is used only when stream stage in each reach is calculated.

Itrib(1)--for a segment that has tributary segments, Itrib(1) is the number of the first tributary segment. For a segment with no tributaries, Itrib(1) must be specified as zero.

Itrib(2)--for a segment that has tributary segments, Itrib(2) is the number of the second tributary segment. For a segment with no or only one tributary segment, Itrib(2) must be specified as zero.

Iupseg --for a diversion segment, Iupseg is the number of the upstream segment from which water is diverted. For a segment that is not a diversion, Iupseg must be specified as zero.

MODULE DOCUMENTATION FOR THE STREAMFLOW-ROUTING PACKAGE

The Streamflow-Routing Package (STR1) consists of four modules (subroutines), all of which are called by the MAIN program. The modules are:

STR1AL	Allocates space for arrays used in the package.
STR1RP	Reads all data needed by package and initializes reach inflow, outflow, and leakage arrays.
STR1FM	Calculates leakage to and from stream reaches and adds appropriate terms to finite-difference equations used to calculate heads in aquifer and simulate ground-water flow.
STR1BD	Calculates rates and accumulated volumes of stream leakage into and out of aquifer. Results are either saved on files or printed.

Module STR1AL

Module STR1AL allocates space for two lists (STRM and ISTRM) which will contain an entry for each stream reach. Each entry in the ISTRM list includes the location of the cell containing a reach, segment number and the reach number. Each entry in the STRM list includes the streamflow into the segment (unless outflow from tributary streams are to be added then a -1 is

entered), stream stage, conductance of the streambed, and the elevation of the bottom and the top of the streambed. If the stream stage in the reaches are to be calculated, the list also includes the width of the stream channel, the slope of the channel, and Manning's roughness coefficient. In addition, the module allocates space for three other lists (ITRBAR, ARTRIB, and IDIVAR), which will contain an entry for each segment. The ITRBAR list includes an entry for each tributary segment. The ARTRIB list is where outflow from each segment is saved and the IDIVAR list includes an entry for each segment that specifies (if any) the upstream segment number from which water is diverted.

Narrative of Module STR1AL

Module STR1AL allocates space in the X array of the modular model (McDonald and Harbaugh, 1988) so the lists needed for stream reaches and segments can be stored in the computer. A generalized flow chart of the module is shown in figure 13. A description of the steps shown in the flow chart follows:

1. Print a message identifying the package and initialize NSTREM (number of stream reaches).

2. Read and print MXSTRM (the maximum number of stream reaches), NSS (number of segments), NTRIB (maximum number of tributary segments that can join to form another segment), NDIV (flag that allows diversions from streams), ICALC (flag that specifies that stream stage for each reach will be calculated), CONST (a constant used in calculating stream stage), ISTCB1 (the unit number for saving on disk the leakage between each reach and the corresponding model cell or a flag indicating whether the flows for each reach and the leakage between the stream reach and corresponding model cell should be printed), and ISTCB2 (the unit number for saving on disk streamflow out of each reach).

3. Set LCSTRM, which is referenced to the first element in the stream list (STRM), equal to ISUM, which is currently referenced to the first unallocated element in the X array.

4. Calculate the amount of space needed for the stream list [11 values for each reach--(1) segment inflow, (2) stream stage, (3) streambed conductance, (4) streambed bottom and (5) top elevations, (6) stream width, (7) channel slope, (8) Manning's roughness coefficient, (9) streamflow out of each reach, (10) streamflow into each reach, (11) leakage through the streambed in each reach] and add it to ISUM.

5. Set ICSTRM equal to ISUM and calculate the amount of space needed for the ISTRM list (five values are needed for each reach--row, column, layer, segment number, and reach number). Add the space to ISUM.

6. Set LCTBAR equal to ISUM and calculate the amount of space needed for the ITRBAR list (NTRIB values are needed for each segment). Add the space to ISUM.

7. Set LCTRIB equal to ISUM and calculate the amount of space needed for the ARTRIB list (one value is needed for each segment). This list stores the calculated streamflows out of each segment. Add the space to ISUM.

8. Set LCIVAR equal to ISUM and calculate the amount of space needed for the IDIVAR list (one value is needed for each segment and is the number of the segment that water is to be diverted). Add the space to ISUM.

9. Print the number of elements in the X array used by the Streamflow-Routing Package.

10. RETURN to MAIN program.

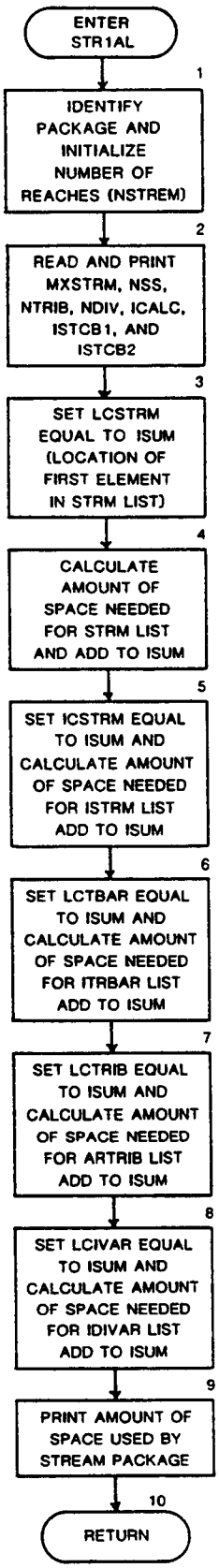


Figure 13.—Generalized flow chart of module STR1AL. Variable names are defined in section "List of variables for module STR1AL."

Program Listing for Module STR1AL

```
      SUBROUTINE STR1AL(ISUM,LENX,LCSTRM,ICSTRM,MXSTRM,NSTREM,IN,
     1                  IOUT,ISTCB1,ISTCB2,NSS,NTRIB,NDIV,ICALC,CONST,
     2                  LCTBAR,LCTRIB,LCIVAR)
C                                                                       C
C-----VERSION    1 23OCT1987 STR1AL                                     C
C     ***************************************************************C
C     ALLOCATE ARRAY STORAGE FOR STREAMS                                C
C     ***************************************************************C
C                                                                       C
C     SPECIFICATIONS:                                                   C
C     ----------------------------------------------------------------C
C     ----------------------------------------------------------------C
C                                                                       C
C1------IDENTIFY PACKAGE AND INITIALIZE NSTREM.                        C
      WRITE(IOUT,1) IN
    1 FORMAT(1H0,'STRM -- STREAM PACKAGE, VERSION 1, 10/23/87',
     1'INPUT READ FROM UNIT ',I3)
      NSTREM=0
C                                                                       C
C2------ READ MXSTRM, NSS, NTRIB, ISTCB1, AND ISTCB2.                  C
  100 READ(IN,3)MXSTRM,NSS,NTRIB,NDIV,ICALC,CONST,ISTCB1,ISTCB2
    3 FORMAT(5I10,F10.0,2I10)
      IF(MXSTRM.LT.0)MXSTRM=0
      IF(NSS.LT.0)NSS=0
      WRITE(IOUT,4)MXSTRM,NSS,NTRIB
    4 FORMAT(1H ,'MAXIMUM OF',I5,' STREAM NODES'//1X,'NUMBER OF STREAM S
     1EGMENTS IS ',I5//1X,'NUMBER OF STREAM TRIBUTARIES IS ',I5//)
      IF(NDIV.GT.0) WRITE(IOUT,5)
    5 FORMAT(1H ,'DIVERSIONS FROM STREAMS HAVE BEEN SPECIFIED')
      IF(ICALC.GT.0) WRITE(IOUT,6) CONST
    6 FORMAT(1H ,'HEADS IN STREAMS WILL BE CALCULATED USING A CONSTANT
     1OF  ',F10.4)
      IF(ISTCB1.GT.0) WRITE(IOUT,7) ISTCB1,ISTCB2
    7 FORMAT(1X,'CELL BUDGETS WILL BE SAVED ON UNITS',I3,'AND',I3)
C                                                                       C
C3------SET LCSTRM EQUAL TO ADDRESS OF FIRST UNUSED SPACE IN X.        C
  200 LCSTRM=ISUM
C                                                                       C
C4------CALCULATE AMOUNT OF SPACE NEEDED FOR STRM LIST.               C
      ISPA=11*MXSTRM
      ISUM=ISUM+ISPA
C                                                                       C
C5------CALCULATE AMOUNT OF SPACE NEEDED FOR ISTRM LIST.              C
      ICSTRM=ISUM
      ISPB=5*MXSTRM
      ISUM=ISUM+ISPB
C                                                                       C
C6------CALCULATE AMOUNT OF SPACE NEEEDED FOR ITRBAR LIST.            C
      LCTBAR=ISUM
      ISPC=NSS*NTRIB
```

```
      ISUM=ISUM+ISPC
C                                                                       C
C7------CALCULATE AMOUNT OF SPACE NEEDED FOR ARTRIB LIST.              C
      LCTRIB=ISUM
      ISPD=NSS
      ISUM=ISUM+ISPD
C                                                                       C
C8------CALCULATE AMOUNT OF SPACE NEEDED FOR IDIVAR LIST.              C
      LCIVAR=ISUM
      ISPE=NSS
      ISUM=ISUM+ISPE
      ISP=ISPA+ISPB+ISPC+ISPD+ISPE
C                                                                       C
C9------PRINT AMOUNT OF SPACE USED BY STREAM PACKAGE.                  C
      WRITE (IOUT,8)ISP
    8 FORMAT(1X,I8,' ELEMENTS IN X ARRAY ARE USED FOR STREAMS')
      ISUM1=ISUM-1
      WRITE(IOUT,9)ISUM1,LENX
    9 FORMAT(1X,I8,' ELEMENTS OF X ARRAY USED OUT OF',I7)
      IF(ISUM1.GT.LENX) WRITE(IOUT,10)
   10 FORMAT(1X,'   ***X ARRAY MUST BE DIMENSIONED LARGER***')
C                                                                       C
C10-----RETURN.                                                        C
      RETURN
      END
```

List of Variables for Module STRIAL

Variable	Range	Definition
CONST	Package	A constant used in calculating stream stage in reaches. Specified only when ICALC is greater than zero. A value of 1.486 is used for units of cubic feet per second and a value of 1.0 is used for units of cubic meters per second. The value must be multiplied by 86,400 when days are used as the time unit.
ICALC	Package	Flag. >0, stream stage in reaches will be calculated. ≤0, stream stage in reaches will not be calculated.
ICSTRM	Package	Location in X array of first element in ISTRM list.
IN	Package	Primary unit number from which input for this package will be read.
IOUT	Global	Primary unit number for all printed output.
ISTCB1	Package	Flag and unit number. >0, unit number to which leakage between each reach and corresponding model cell is saved on disk whenever the variable ICBCFL is specified in the Output Control Package (McDonald and Harbaugh, 1988, p. 4-15). =0, leakage will not be printed nor recorded. <0, leakage between each reach and corresponding model cell and reach inflow and outflow are printed whenever the variables ICBCFL and IPTFLG are specified.
ISTCB2	Package	Flag and unit number. >0, unit number to which streamflow out of each stream reach will be saved on disk whenever the variable ICBCFL is specified. ≤0, streamflow out of each reach will not be saved on disk.
ISPA	Module	Number of words in X array allocated to the STRM list by this module.

ISPB	Module	Number of words in X array allocated to the ISTRM list by this module.
ISPC	Module	Number of words in X array allocated to the ITRBAR list by this module.
ISPD	Module	Number of words in X array allocated to the ARTRIB list by this module.
ISPE	Module	Number of words in X array allocated to the IDIVAR list by this module.
ISP	Module	Total number of words in X array allocated by this module.
ISUM	Global	Index number of lowest element in X array which has not yet been allocated.
ISUM1	Module	ISUM minus one.
LCIVAR	Package	Location in X array of first element in the IDIVAR list.
LCSTRM	Package	Location in X array of first element in the STRM list.
LCTBAR	Package	Location in X array of first element in the ITRBAR list.
LCTRIB	Package	Location in X array of first element in the ARTRIB list.
LENX	Global	Length of the X array in words. This should always be equal to the dimension of the X specified in the MAIN program.
MXSTRM	Package	Maximum number of stream reaches active during the simulation.
NDIV	Package	Flag. >0, diversions from segments will be specified. ≤0, no diversions from segments will be specified.
NSS	Package	Maximum number of stream segments.
NSTREM	Package	Number of stream reaches active during current stress period.
NTRIB	Package	Maximum number of tributary segments that join one downstream segment.

Module STR1RP

Module STR1RP reads for each model cell containing a stream reach, the location of the cell, the segment and reach numbers, streamflow into the segment, stream stage, conductance of the streambed, and elevation of the bottom and top of the streambed. If the stream stage in each reach is to be calculated, then the module also reads for each reach, the width of the stream channel, the slope of the channel, and Manning's roughness coefficient. If outflow from two or more tributary stream segments is to be added as inflow to a downstream segment, then the module reads for each segment, the segments that are tributary to it. If diversions are to be subtracted from segments, then the module reads for each segment the upstream segment from which water is diverted.

Narrative for Module STR1RP

The module STR1RP reads and records data needed for each of the lists used by the Streamflow-Routing Package. A generalized flow chart of the module is shown in figure 14. A description of the steps shown in the flow chart follows:

1. Read ITMP, IRDFLG, IPTFLG. If ITMP is greater than or equal to zero, it is the number of stream reaches active during the current stress period, or when less than zero, it indicates that the stream data used in the last stress period will be used in the current stress period. If IRDFLG and IPTFLG are positive, then the input data set and the results will not be printed for the stress period.

2. Test ITMP. If ITMP is negative, print message that indicates data from the last stress period will be used and then return to MAIN program. If ITMP is greater than or equal to zero, set the number of active stream reaches (NSTREM) in the current stress period equal to ITMP.

3. Compare the number of stream reaches (NSTREM) in the current stress period to the number specified as the maximum for the simulation (MXSTRM). If NSTREM is greater than MXSTRM then STOP. If NSTREM is less than or equal to MXSTRM then print the number of stream reaches (NSTREM) used in the current stress period.

4. Check the number of stream reaches. If NSTREM is zero for the current stress period then return to MAIN program.

5. Read and print (print if IRDFLG is zero) for each stream reach: the layer, row, column, segment number, reach number, stream segment inflow, stream head, streambed conductance, and the elevation of the bottom and top of the streambed.

6. Check if ICALC is greater than zero. If ICALC is greater than zero, read and print (print if IRDFLG is zero) the channel width, channel slope, and Manning's roughness coefficient for each reach stream. If ICALC is less than or equal to zero, no data is read.

7. Initialize the ITRBAR list to zero.

8. Initialize the IDIVAR list to zero.

9. Read and print (print if IRDFLG is zero) for each segment the tributary segments. One record of NTRIB values is read for each segment even though some segments have no tributaries. The maximum number of tributary segments is 10. The number is currently set in the program but could be changed if needed (Format statements 11 and 12).

10. Check if NDIV is greater than zero. If NDIV is greater than zero, read and print (print if IRDFLG is zero) for each segment the segments in which water is diverted. A zero value is assigned to segments that do not divert water.

11. Initialize to zero for each reach, the reach inflow, outflow, and leakage to or from the aquifer system.

12. RETURN to MAIN program.

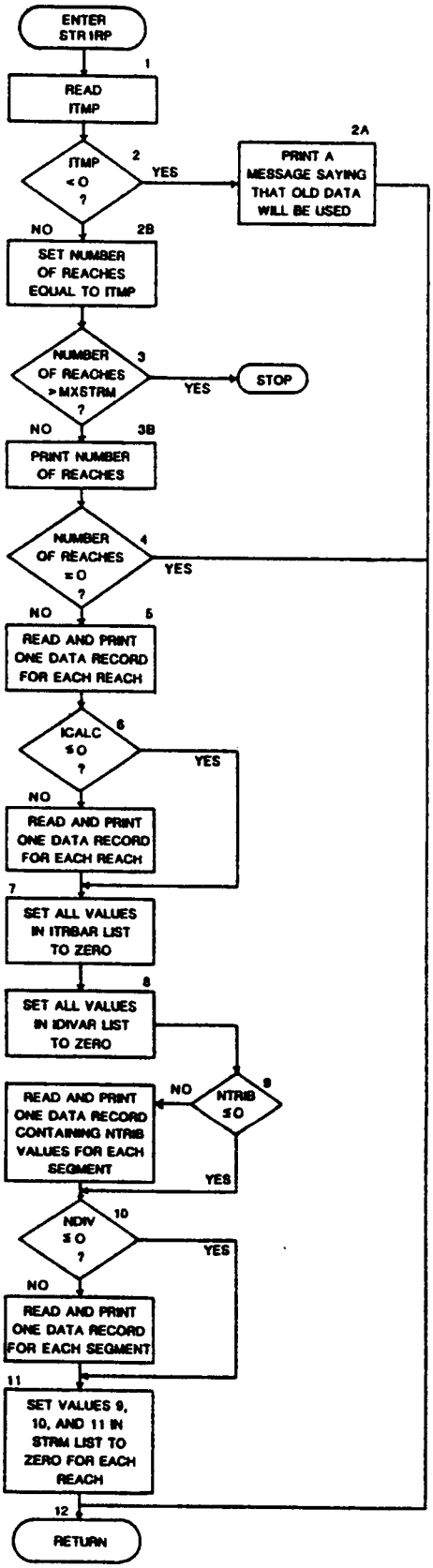


Figure 14.--Generalized flow chart of module STR1RP. Variable names are defined in section "List of variables for module STR1RP."

Program Listing for Module STR1RP

```
      SUBROUTINE STR1RP(STRM,ISTRM,NSTREM,MXSTRM,IN,IOUT,ITRBAR,NDIV,
     1                  NSS,NTRIB,IDIVAR,ICALC,IPTFLG)
C                                                                       C
C                                                                       C
C-----VERSION   1 23OCT1987 STR1RP                                      C
C     ***************************************************************C
C     READ STREAM DATA:  INCLUDES SEGMENT AND REACH NUMBERS, CELL       C
C         SEQUENCE OF SEGMENT AND REACH, FLOW INTO MODEL AT BOUNDARY,   C
C         STREAM STAGE, STREAMBED CONDUCTANCE, AND STREAMBED TOP AND    C
C         BOTTOM ELEVATIONS                                             C
C     ***************************************************************C
C                                                                       C
C     SPECIFICATIONS:                                                   C
C     ------------------------------------------------------------------C
      DIMENSION STRM(11,MXSTRM),ISTRM(5,MXSTRM),ITRBAR(NSS,NTRIB),
     1          IDIVAR(NSS)
C     ------------------------------------------------------------------C
C                                                                       C
C1A-----IF MXSTREAM IS LESS THAN 1 THEN STREAM IS INACTIVE. RETURN.     C
      IF(MXSTRM.LT.1) RETURN
C                                                                       C
C1B-----READ ITMP(NUMBER OF STREAM REACHES OR FLAG TO REUSE DATA).      C
      READ(IN,1)ITMP,IRDFLG,IPTFLG
    1 FORMAT(3I10)
C                                                                       C
C2A-----IF ITMP <0 THEN REUSE DATA FROM LAST STRESS PERIOD.             C
      IF(ITMP.GE.0)GO TO 50
      WRITE(IOUT,2)
    2 FORMAT(1H0,'REUSING STREAM NODES FROM LAST STRESS PERIOD')
      RETURN
C                                                                       C
C2B-----IF ITMP=> ZERO THEN IT IS THE NUMBER OF STREAM REACHES.         C
   50 NSTREM=ITMP
C                                                                       C
C3A-----IF NSTREM>MXSTRM THEN STOP.                                     C
      IF(NSTREM.LE.MXSTRM)GO TO 100
      WRITE(IOUT,99)NSTREM,MXSTRM
   99 FORMAT(1H0,'NSTREM(',I4,') IS GREATER THAN MXSTRM(',I4,')')
      STOP
C                                                                       C
C3B-----PRINT NUMBER OF STREAM REACHES IN THIS STRESS PERIOD.           C
  100 IF(IRDFLG.EQ.0) WRITE(IOUT,3)NSTREM
    3 FORMAT(1H0,//1X,I5,' STREAM NODES')
C                                                                       C
C4------IF THERE ARE NO STREAM CELLS THEN RETURN.                      C
      IF(NSTREM.EQ.0) RETURN
C                                                                       C
C5------READ AND PRINT DATA FOR EACH STREAM REACH.                     C
      IF(IRDFLG.EQ.0) WRITE(IOUT,4)
    4 FORMAT(1H ,3X,'LAYER   ROW    COL    SEGMENT   REACH   STREAMFLOW
```

```
     1     STREAM     STREAMBED      STREAMBED BOT  STREAMBED TOP',/27X,
     2'NUMBER   NUMBER                      STAGE   CONDUCTANCE       ELEVAT
     3ION       ELEVATION',/3X,110('-'))
      DO 250 II=1,NSTREM
      READ(IN,5)K,I,J,ISTRM(4,II),ISTRM(5,II),STRM(1,II),STRM(2,II),
     1STRM(3,II),STRM(4,II),STRM(5,II)
    5 FORMAT(5I5,F15.0,4F10.0)
      IF(IRDFLG.EQ.0) WRITE(IOUT,6)K,I,J,ISTRM(4,II),ISTRM(5,II),
     1STRM(1,II),STRM(2,II),STRM(3,II),STRM(4,II),STRM(5,II)
    6 FORMAT(1X,3X,I4,2I7,2I9,7X,G11.4,G12.4,G11.4,4X,2G13.4)
      ISTRM(1,II)=K
      ISTRM(2,II)=I
      ISTRM(3,II)=J
  250 CONTINUE
C                                                                       C
C6----READ AND PRINT DATA IF STREAM STAGE  IS CALCULATED.               C
      IF(ICALC.LE.0) GO TO 300
      IF(IRDFLG.EQ.0) WRITE(IOUT,7)
    7 FORMAT(1H0,3X,'LAYER',3X,'ROW',4X,'COL    ',' SEGMENT',3X,
     1'REACH',8X,'STREAM',13X,'STREAM',10X,'ROUGH',/27X,'NUMBER',3X,
     2 'NUMBER',8X,'WIDTH',14X,'SLOPE',10X,'COEF.',/3X,110('-'))
      DO 280 II=1,NSTREM
      READ(IN,8) STRM(6,II),STRM(7,II),STRM(8,II)
    8 FORMAT(3F10.0)
      IF(IRDFLG.EQ.0) WRITE(IOUT,9)ISTRM(1,II),ISTRM(2,II),ISTRM(3,II),
     1ISTRM(4,II),ISTRM(5,II),STRM(6,II),STRM(7,II),STRM(8,II)
    9 FORMAT(4X,I4,2I7,2I9,7X,G12.4,4X,G13.4,4X,G12.4)
  280 CONTINUE
C                                                                       C
C7------INITIALIZE ALL TRIBUTARY SEGMENTS TO ZERO.                     C
  300 DO 320 IK=1,NSS
      DO 320 JK=1,NTRIB
      ITRBAR(IK,JK)=0
  320 CONTINUE
C                                                                       C
C8-----INITIALIZE DIVERSION SEGMENT ARRAY TO ZERO.                      C
      DO 325 IK=1,NSS
      IDIVAR(IK)=0
  325 CONTINUE
C                                                                       C
C9-----READ AND PRINT TRIBUTARY SEGMENTS.                               C
      IF(NTRIB.LE.0) GO TO 343
      IF(IRDFLG.EQ.0) WRITE(IOUT,10)NTRIB
   10 FORMAT(1H0,30X,'MAXIMUM NUMBER OF TRIBUTARY STREAMS IS ',I5,//1X,
     1 20X,'STREAM SEGMENT',15X,'TRIBUTARY STREAM SEGMENT NUMBERS')
      DO 340 IK=1,NSS
      READ(IN,11) (ITRBAR(IK,JK),JK=1,NTRIB)
   11 FORMAT(10I5)
      IF(IRDFLG.EQ.0) WRITE(IOUT,12)IK,(ITRBAR(IK,JK),JK=1,NTRIB)
   12 FORMAT(20X,I5,20X,10I5)
  340 CONTINUE
C                                                                       C
C10----READ AND PRINT DIVERSION SEGMENT NUMBERS.                        C
  343 IF(NDIV.LE.0) GO TO 350
```

```
      IF(IRDFLG.EQ.0) WRITE(IOUT,13)
   13 FORMAT(1H0,10X,'DIVERSION SEGMENT NUMBER',10X,
     1       'UPSTREAM SEGMENT NUMBER')
      DO 345 IK=1,NSS
      READ(IN,14) IDIVAR(IK)
   14 FORMAT(I10)
      IF(IRDFLG.EQ.0) WRITE(IOUT,15) IK,IDIVAR(IK)
   15 FORMAT(20X,I5,28X,I5)
  345 CONTINUE
C                                                                       C
C11----SET FLOW OUT OF REACH, FLOW INTO REACH, AND FLOW THROUGH         C
C       STREAM BED TO ZERO.                                             C
  350 DO 360 II =1,NSTREM
      STRM(9,II)=0.0
      STRM(10,II)=0.0
      STRM(11,II)=0.0
  360 CONTINUE
C                                                                       C
C12------RETURN.                                                       C
      RETURN
      END
```

List of Variables for Module STR1RP

Variable	Range	Definition
I	Module	Row number of model cell containing a stream reach.
ICALC	Package	Flag. >0, stream stage in reaches will be calculated. ≤0, stream stage in reaches will not be calculated.
IDIVAR	Package	DIMENSION (NSS), For each segment: upstream segment number from which water is to be diverted.
II	Module	Index for reach.
IK	Module	Index for segment.
IN	Package	Primary unit number from which input for this package will be read.
IOUT	Global	Primary unit number for all printed output.
IPTFLG	Package	Flag to specify printing of results for each stress period. Results may be printed at selected time steps within a stress period depending on the value specified for variable ICBCFL in the Output Control Package (McDonald and Harbaugh, 1988, p. 4-15). 0, results will be printed according to the value of ICBCFL. >0, results will not be printed even if ICBCFL is specified.
IRDFLG	Module	Flag to specify printing of input data set for each stress period. 0, input data will be printed. >0, input data will not be printed.
ISTRM	Package	DIMENSION (5,MXSTRM), For each reach: layer, row, column, segment number, and reach number.

ITMP	Package	Flag or number of reaches. ≥0, number of reaches active during current stress period. <0, same reaches active last stress period will be active during the current stress period.
ITRBAR	Package	DIMENSION (NSS,NTRIB), For each segment: segment numbers of each tributary segment. A zero is assigned if no segments are tributary to a given segment.
J	Module	Column number of model cell containing a stream reach.
JK	Module	Index for tributary segments.
K	Module	Layer number of model cell containing a stream reach.
MXSTRM	Package	Maximum number of stream reaches active during the simulation.
NDIV	Package	Flag. >0, diversions from segments will be specified. ≤0, no diversions from segments will be specified.
NSS	Package	Maximum number of stream segments.
NSTREM	Package	Number of stream reaches active during the current stress period.
NTRIB	Package	Maximum number of tributary segments that join one downstream segment.
STRM	Package	DIMENSION (11,MXSTRM), For each reach: (1) segment inflow, (2) stream stage, (3) streambed conductance, (4) streambed bottom and (5) top elevations, (6) stream width, (7) channel slope, (8) Manning's roughness coefficient, (9) streamflow out of each reach, (10) streamflow into each reach, and (11) leakage through the streambed in each reach.

Module STR1FM

Module STR1FM calculates leakage from each stream reach to the corresponding model cell and subtracts the leakage from streamflow into each reach. The module then adds the appropriate terms to the finite-difference equation for each model cell that contains a stream reach. If streamflow in a reach is zero, then no terms are added to the finite-difference equation for the corresponding model cell.

Narrative for Module STR1FM

The module STR1FM routes flow in streams and adds terms representing stream leakage to the accumulators HCOF and RHS in equation 26 of McDonald and Harbaugh (1988, p. 2-26). A generalized flow chart of the module is shown in figure 15. A description of the steps shown in the flow chart follows:

1. If NSTREM is less than or equal to zero, then no stream reaches are active during the current stress period.

2. Repeat steps 3 through 20 for each reach in the ISTRM list.

3. Determine the column, row, and layer of the cell containing the reach.

4. If the cell is external to the active cells in the simulation, bypass processing on this reach and go on to the next reach.

5. Determine segment number (ISTSG) and reach number (NREACH).

6. IF NREACH equals one then set stream inflow to the reach (FLOWIN) equal to the segment inflow value.

7. If NREACH equals one and ISTSG is greater than one, set ARTRIB (ISTSG-1) equal to the outflow from the last reach in the previous segment.

8. If the segment is a diversion from an upstream segment, IDIVAR (ISTSG) is greater than zero, calculate the flow out of the upstream segment including the amount that is diverted. If more water is to be diverted than is available in the stream, set the amount of flow to be diverted to zero.

9. If flow in the first reach in a segment is negative, as specified in the input data set, then reset FLOWIN equal to zero, sum the outflow from the specified tributary segments in the ITRBAR list and add the sum to FLOWIN.

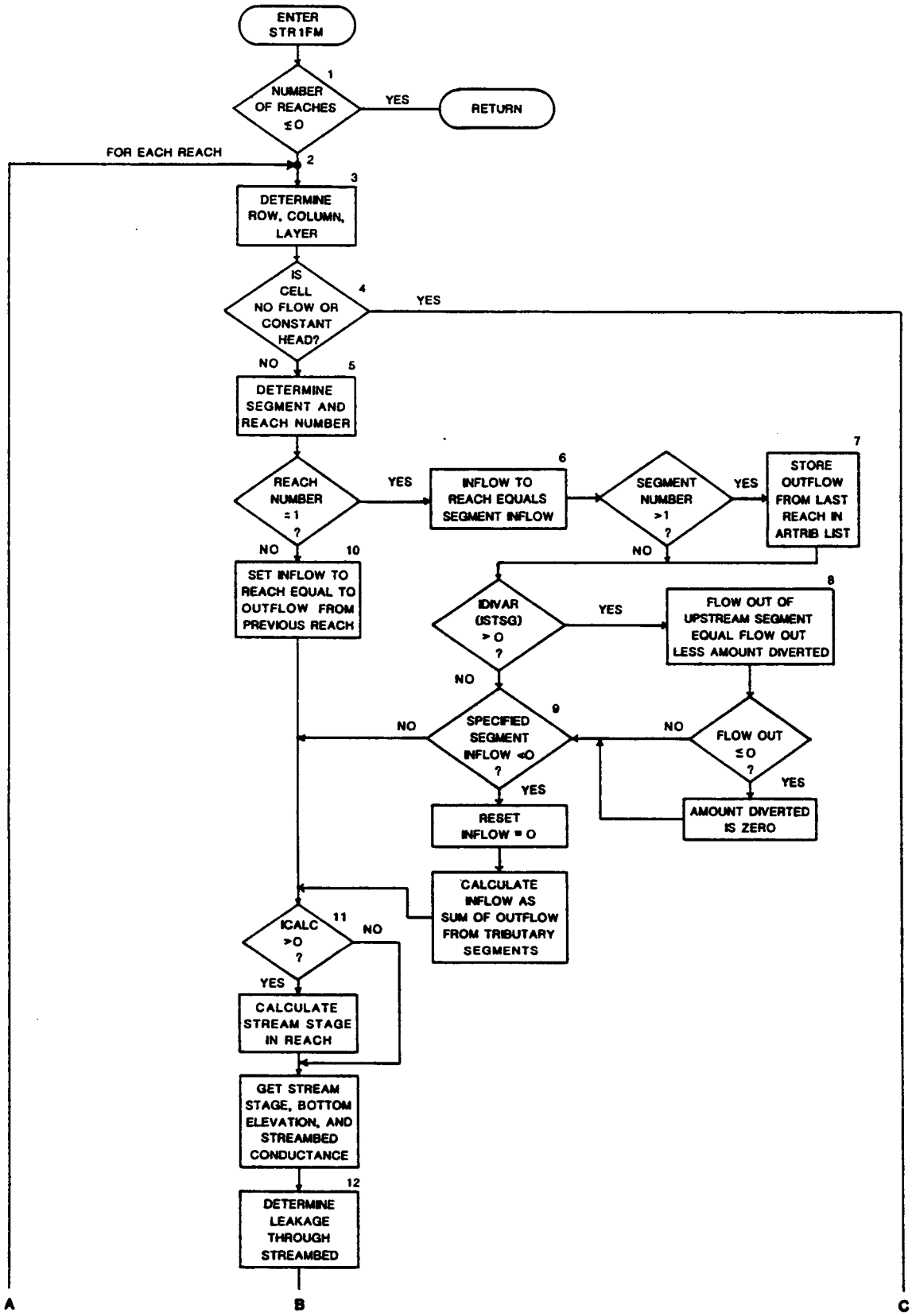


FIGURE 15.--Generalized flow chart of module STR1FM. Variable names are defined in section "List of variables for module STR1FM."

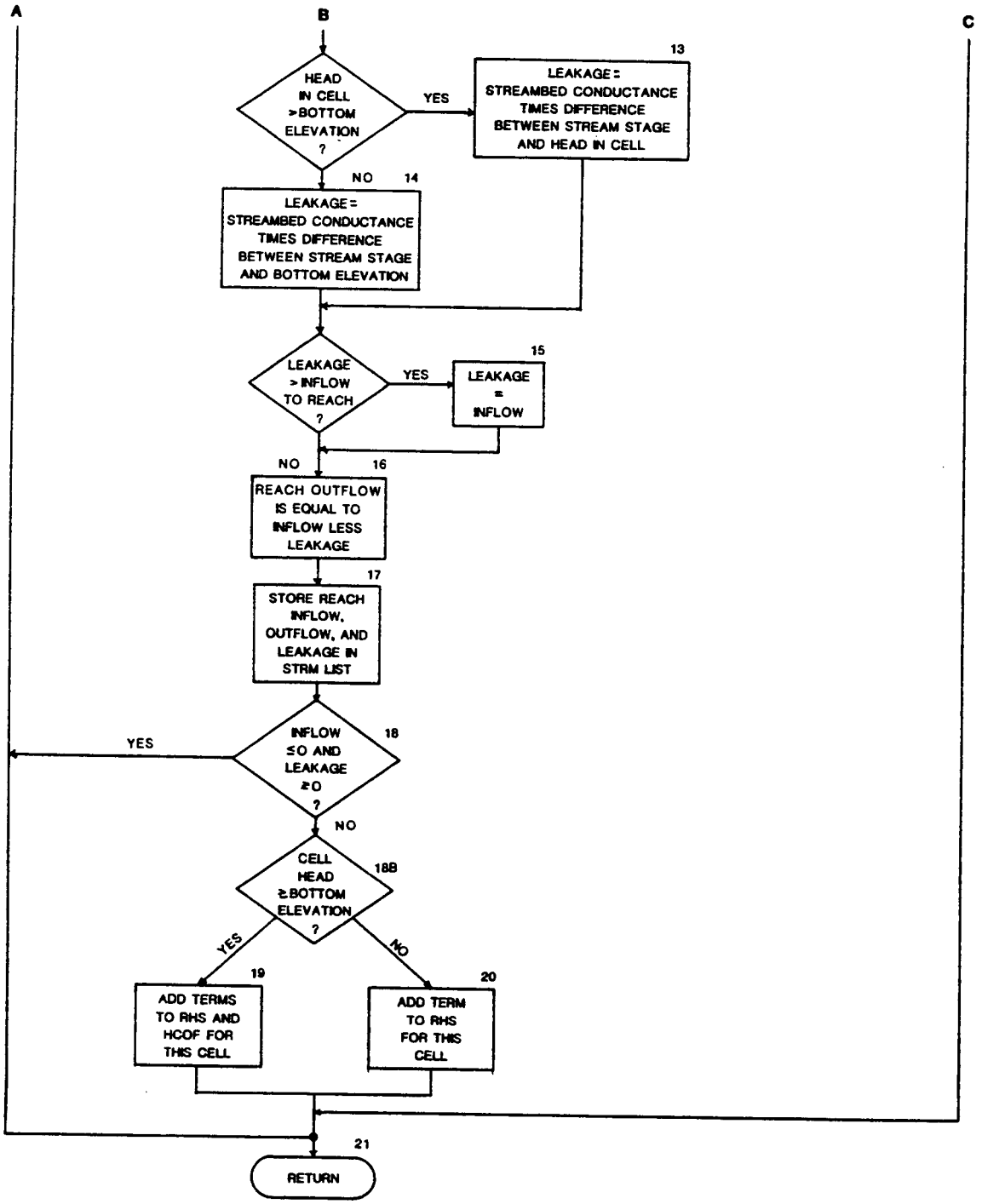


Figure 15.--(concluded)

10. If NREACH is greater than one, set the flow into the reach equal to the outflow from the adjacent upstream reach (NREACH-1).

11. If ICALC is greater than zero, then calculate stream stage by first calculating the depth then adding the value to top of streambed.

12. Determine leakage (FLOBOT) through streambed for each reach.

13. If the head in model cell (H) is greater than the elevation of bottom of streambed (SBOT), then FLOBOT is equal to streambed conductance (CSTR) times the difference between stream stage and head in model cell [CSTR*(HSTR-H)].

14. If H for the cell is less than or equal to SBOT, then FLOBOT is equal to CSTR times the difference in the elevations between HSTR and SBOT [CSTR*(HSTR-SBOT)].

15. If leakage through the streambed (FLOBOT) is greater than streamflow into the reach (FLOWIN), set FLOBOT equal to FLOWIN.

16. Streamflow out of each reach (FLOWOT) is equal to FLOWIN minus FLOBOT.

17. Store for each reach FLOWIN, FLOWOT, and FLOBOT in the STRM list.

18. If FLOWIN is less than or equal to zero and FLOBOT greater than or equal to zero return to step 3, otherwise determine which accumulators to add to the finite-difference equation.

19. If head in model cell (H) is greater than elevation of bottom of streambed (SBOT), then add the term [-CSTR*HSTR] to the accumulator RHS and the term [-CSTR] to the accumulator HCOF.

20. If head in model cell (H) is less than or equal to SBOT, then add the term [-CSTR*(HSTR-SBOT) to the accumulator RHS. Nothing is added to HCOF.

21. RETURN to MAIN program.

Program Listing for Module STR1FM

```
      SUBROUTINE STR1FM(NSTREM,STRM,ISTRM,HNEW,HCOF,RHS,IBOUND,MXSTRM,
     1              NCOL,NROW,NLAY,IOUT,NSS,ITRBAR,NTRIB,ARTRIB,
     2                   IDIVAR,ICALC,CONST)
C
C-----VERSION   1 23OCT1987 STR1FM                                      C
C     ***************************************************************C
C     ADD STREAM TERMS TO RHS AND HCOF IF FLOW OCCURS IN MODEL CELL     C
C     ***************************************************************C
C
C     SPECIFICATIONS:                                                   C
C     ---------------------------------------------------------------C
C                                                                       C
      DOUBLE PRECISION HNEW
      DIMENSION STRM(11,MXSTRM),ISTRM(5,MXSTRM),HNEW(NCOL,NROW,NLAY),
     1          HCOF(NCOL,NROW,NLAY),RHS(NCOL,NROW,NLAY),
     2          IBOUND(NCOL,NROW,NLAY),ITRBAR(NSS,NTRIB),ARTRIB(NSS),
     3          IDIVAR(NSS)
C     ---------------------------------------------------------------C
C                                                                       C
C1------IF NSTREM<=0 THERE ARE NO STREAMS. RETURN.                      C
      IF(NSTREM.LE.0)RETURN
C
C2------PROCESS EACH CELL IN THE STREAM LIST.                           C
C                                                                       C
C3------DETERMINE LAYER, ROW, COLUMN OF EACH REACH.                     C
      DO 500 L=1,NSTREM
      LL=L-1
      IL=ISTRM(1,L)
      IR=ISTRM(2,L)
      IC=ISTRM(3,L)
C
C4------DETERMINE IF CELL IS OUTSIDE OF MODEL BOUNDARIES.               C
      IF(IBOUND(IC,IR,IL).LE.0)GO TO 500
C
C5------DETERMINE SEGMENT AND REACH NUMBER.                             C
      ISTSG=ISTRM(4,L)
      NREACH=ISTRM(5,L)
C
C6------SET FLOWIN EQUAL TO SEGMENT INFLOW IF FIRST REACH.              C
      IF(NREACH.GT.1) GO TO 200
      FLOWIN=STRM(1,L)
C
C7------STORE OUTFLOW FROM PREVIOUS SEGMENT IN ARTRIB IF SEGMENT >1.    C
      IF(ISTSG.GT.1) IFLG = ISTRM(4,LL)
      IF(ISTSG.GT.1)  ARTRIB(IFLG)=STRM(9,LL)
C
C8------IF SEGMENT IS A DIVERSION, COMPUTE FLOW OUT OF UPSTREAM REACH.  C
      IF(IDIVAR(ISTSG).LE.0) GO TO 50
      NDFLG=IDIVAR(ISTSG)
      DUM=ARTRIB(NDFLG)-FLOWIN
```

```
      IF(DUM.GE.0.0) ARTRIB(NDFLG)=DUM
      IF(DUM.GE.0.0) GO TO 50
      FLOWIN=0.
   50 IF(FLOWIN.GE.0.0) GO TO 300
C                                                                       C
C9-----SUM TRIBUTARY OUTFLOW AND USE AS INFLOW INTO DOWNSTREAM SEGMENT.C
      FLOWIN =0.
      DO 100 ITRIB=1,NTRIB
      INODE=ITRBAR(ISTSG,ITRIB)
      IF(INODE.LE.0) GO TO 100
      FLOWIN=FLOWIN+ARTRIB(INODE)
  100 CONTINUE
C                                                                       C
C10-----IF REACH >1, SET INFLOW EQUAL TO OUTFLOW FROM UPSTREAM REACH.   C
  200 IF(NREACH.GT.1) FLOWIN=STRM(9,LL)
C                                                                       C
C11----COMPUTE STREAM STAGE IN REACH IF ICALC IS GREATER THAN 1.        C
  300 IF(ICALC.LE.0) GO TO 310
      XNUM=((FLOWIN+STRM(9,L))/2.0)*STRM(8,L)
      DNOM=CONST*STRM(6,L)*(SQRT(STRM(7,L)))
      DEPTH=(XNUM/DNOM)**0.6
      IF(DEPTH.LE.0.) DEPTH=0.
      STRM(2,L)=DEPTH+STRM(5,L)
  310 HSTR=STRM(2,L)
C                                                                       C
C12----DETERMINE LEAKAGE THROUGH STREAMBED.                             C
      IF(FLOWIN.LE.0.) HSTR=STRM(5,L)
      CSTR=STRM(3,L)
      SBOT=STRM(4,L)
      H=HNEW(IC,IR,IL)
      T=HSTR-SBOT
C                                                                       C
C13----COMPUTE LEAKAGE AS A FUNCTION OF STREAM STAGE AMD HEAD IN CELL.  C
      FLOBOT=CSTR*(HSTR-H)
C                                                                       C
C14----RECOMPUTE LEAKAGE IF HEAD IN CELL IS BELOW STREAMBED BOTTOM.     C
      IQFLG=0
      IF(H.GT.SBOT) GO TO 312
      IQFLG=1
      FLOBOT=CSTR*T
C                                                                       C
C15----SET LEAKAGE EQUAL TO STREAM INFLOW IF LEAKAGE MORE THAN INFLOW.  C
  312 IF(FLOBOT.LE.FLOWIN) GO TO 320
      IQFLG=1
      FLOBOT=FLOWIN
C                                                                       C
C16-----STREAMFLOW OUT EQUALS STREAMFLOW IN MINUS LEAKAGE.              C
  320 FLOWOT=FLOWIN-FLOBOT
      IF((ISTSG.GT.1).AND.(NREACH.EQ.1)) STRM(9,LL)=ARTRIB(IFLG)
C                                                                       C
C17----STORE STREAM INFLOW, OUTFLOW AND LEAKAGE FOR EACH REACH.         C
      STRM(9,L)=FLOWOT
      STRM(10,L)=FLOWIN
      STRM(11,L)=FLOBOT
```

```
C                                                                       C
C18----RETURN TO STEP 3 IF STREAM INFLOW IS LESS THAN OR EQUAL TO ZERO C
C       AND LEAKAGE IS GREATER THAN OR EQUAL TO ZERO.                   C
      IF((FLOWIN.LE.0.0).AND.(FLOBOT.GE.0.0)) GO TO 500
C                                                                       C
C19------IF HEAD > BOTTOM THEN ADD TERMS TO RHS AND HCOF.              C
      IF(IQFLG.GT.0) GO TO 400
      RHS(IC,IR,IL)=RHS(IC,IR,IL)-CSTR*HSTR
      HCOF(IC,IR,IL)=HCOF(IC,IR,IL)-CSTR
      GO TO 500
C                                                                       C
C20------IF HEAD < BOTTOM THEN ADD TERM ONLY TO RHS.                   C
  400 RHS(IC,IR,IL)=RHS(IC,IR,IL)-FLOBOT
  500 CONTINUE
C                                                                       C
C21-----RETURN.                                                         C
      RETURN
      END
```

List of Variables for Module STR1FM

Variable	Range	Definition
ARTRIB	Package	DIMENSION(NSS), For each segment: contains the streamflow out of last reach in each stream segment.
CONST	Package	A constant used in calculating stream stage in reaches. Specified only when ICALC is greater than zero. A value of 1.486 is used for units of cubic feet per second and a value of 1.0 is used for units of cubic meters per second. The value must be multiplied by 86,400 when days are used as the time unit.
CSTR	Module	Streambed hydraulic conductance.
DEPTH	Module	Depth of water in reach. The value is added to top of streambed to obtain the stream stage.
DNOM	Module	Denominator of the equation used to calculate depth of water in a reach.
DUM	Module	Amount of flow from upstream segment from which water is diverted.
FLOBOT	Module	Leakage into or out of a model cell through the streambed.
FLOWIN	Module	Streamflow into reach. If streamflow into the first reach of a segment is negative, then FLOWIN is the sum of streamflow out of the last reach in each specified tributary segment.
FLOWOT	Module	Streamflow out of reach.
H	Module	Head in model cell (HNEW).
HCOF	Global	DIMENSION (NCOL,NROW,NLAY), Coefficient of the cell (J,I,K) in the finite-difference equation.
HNEW	Global	DIMENSION (NCOL,NROW,NLAY), Most recent estimate of head in each cell. HNEW changes at each iteration.
HSTR	Module	Stream stage in reach.

IBOUND	Global	DIMENSION (NCOL,NROW,NLAY), Status of each cell. <0, constant-head cell. =0, inactive cell. >0, variable-head cell.
IC	Module	Column number of cell containing reach.
ICALC	Package	Flag. >0, stream stage in reaches will be calculated. ≤0, stream stage in reaches will not be calculated.
IDIVAR	Package	DIMENSION (NSS), For each segment: upstream segment number from which water is to be diverted.
IFLG	Module	Number of previous segment.
IL	Module	Layer number of cell containing reach.
INODE	Module	Number of a tributary segment.
IOUT	Global	Primary unit number for all printed output.
IQFLG	Module	Flag used for assigning proper terms to RHS and HCOF. 0, head in model cell is greater than streambed bottom. 1, head in model cell is less than or equal to streambed bottom.
IR	Module	Row number of cell containing reach.
ISTRM	Package	DIMENSION(5,MXSTRM), For each reach: layer, row, column, segment number, and reach number.
ISTSG	Module	Segment number of cell containing reach.
ITRBAR	Package	DIMENSION (NSS,NTRIB), For each segment: segment numbers of each tributary segment. A zero is assigned if no segments are tributary to a given segment.
ITRIB	Module	Index for the number of tributary segments.
L	Module	Index for current reach.
LL	Module	Index for previous reach.
MXSTRM	Package	Maximum number of stream reaches.

NCOL	Global	Number of columns in grid.
NDFLG	Module	Number of segments from which water is diverted.
NLAY	Global	Number of layers in grid.
NREACH	Module	Number of reach in segment.
NROW	Global	Number of rows in grid.
NSS	Package	Maximum number of segments.
NSTREM	Package	Number of stream reaches active during current stress period.
NTRIB	Package	Maximum number of tributary segments that join one downstream segment.
RHS	Global	DIMENSION (NCOL,NROW,NLAY), Right side of the finite-difference equation. RHS is an accumulation of terms from several different packages.
SBOT	Module	Elevation of streambed bottom.
STRM	Package	DIMENSION (11,MXSTRM), For each reach: (1) segment inflow, (2) stream stage, (3) streambed conductance, (4) streambed bottom and (5) top elevations, (6) stream width, (7) channel slope, (8) Manning's roughness coefficient, (9) streamflow out of each reach, (10) streamflow into each reach, and (11) leakage through the streambed in each reach.
T	Module	Difference between stream stage (HSTR) and elevation of the streambed bottom (SBOT).
XNUM	Module	Numerator of the equation used to calculate depth of water in a reach.

Module STR1BD

Module STR1BD calculates the rates and volumes of stream leakage into and out of the model cells and calculates streamflow into and out of each reach. It then saves results onto unformated disk files or prints results to a different file.

Narrative for Module STR1BD

The module STR1BD calculates rates and volumes transferred between the aquifer and streams and in addition, calculates the final stream inflow and outflow each time step for each reach. A generalized flow chart of the module is shown in figure 16. A description of the steps shown in the flow chart follows:

1. Initialize the cell-by-cell flow-term flag (IBD) and the rate accumulators (RATIN and RATOUT).

2. If no reaches are specified (NSTREM<or=0), skip down to step 25 and put zeros into the budget terms for streams.

3. Test to determine if leakage to and from the streams for each reach is to be saved on the disk. They will not be saved if either of the following conditions are true: (1) If this is not the proper time step (ICBCFL=0 in the OUTPUT package) or (2) if the term ISTCB1 is less than or equal to zero. If leakage between each stream reach and corresponding model cell is to be saved on disk, then the flag IBD is set equal to one and the buffer (BUFF) is cleared so leakage can be accumulated.

4. Repeat steps 5 through 23 for each reach in the ISTRM list accumulating leakage between each reach and corresponding model cell, and computing streamflows into and out of each reach.

5. Determine column, row, and layer of the cell containing the reach.

6. If the cell is external to active cells in the simulation, bypass processing for this reach and go to the next reach.

7. Determine the segment number (ISTSG) and reach number (NREACH).

8. If NREACH equals one, then set stream inflow to the reach (FLOWIN) equal to the segment inflow value.

9. If NREACH equals one and ISTSG is greater than one, set ARTRIB (ISTSG-1) equal to the outflow from the last reach in the previous segment.

10. If the segment is a diversion from an upstream segment, IDIVAR (ISTSG) is greater than zero, calculate the flow out of the upstream segment including the amount that is diverted. If more water is to be diverted than is available in the stream, set the amount of flow to be diverted to zero.

11. If flow in the first reach in a segment is negative, as specified in the input data set, then reset FLOWIN to zero and sum the outflow from the specified tributary segments in the ITRBAR list and add the sum to FLOWIN.

12. If NREACH is greater than one, set the flow into the reach equal to the outflow from the adjacent upstream reach (NREACH-1).

13. If ICALC is greater than zero, then calculate the stream stage by first calculating the depth then adding the value to the top of the streambed.

14. Determine the leakage (FLOBOT) through the streambed for each reach.

15. If the head in the model cell (H) is greater than the elevation of bottom of the streambed (SBOT), then FLOBOT is equal to streambed conductance (CSTR) times the difference between stream stage in reach and head in model cell [CSTR*(HSTR-H)].

16. If H in the cell is less than or equal to SBOT, then FLOBOT is equal to CSTR times the difference in elevations between HSTR and SBOT [CSTR*(HSTR-SBOT)].

17. If leakage through the streambed (FLOBOT) is greater than streamflow into the reach (FLOWIN), set FLOBOT equal to FLOWIN.

18. Set streamflow out of reach (FLOWOT) equal to FLOWIN minus FLOBOT.

19. Record FLOWIN, FLOWOT, and FLOBOT in the STRM list.

20. If the leakage values are to be saved on disk (IBD=1), add FLOBOT to the buffer (BUFF).

21. Determine if flow is into or out of model cell.

22. If FLOBOT is negative, subtract it from RATOUT.

23. If FLOBOT is positive, add it to RATIN.

24. If leakage between reach and corresponding model cell is to be saved on disk (IBD=1), call module UBUDSV to save the buffer (BUFF) onto a disk file. (Module UBUDSV is part of the program by McDonald and Harbaugh, 1988.)

25. Move RATIN and RATOUT into the VBVL array for printing by BAS1OT. Add RATIN and RATOUT multiplied by the time-step length to the volume accumulators in VBVL for printing by BAS1OT. Move the stream

budget term labels to VBNM for printing by BAS1OT. (Module BAS1OT is part of the Basic Package in McDonald and Harbaugh, 1988.)

26. Increase the budget-term counter (MSUM) by one.

27. Reset IBD to 0.

28. Determine if streamflows out of each reach are to be saved on disk. Values will not be saved if either of the following conditions are true: (1) This is not the proper time step (ICBCFL=0) or (2) the variable ISTCB2 is less than or equal to zero. If streamflows out of each reach are to be saved, set IBD=1 and initialize the buffer (BUFF) to zero.

29. If IBD=1, read into buffer the streamflow out of each reach and then call module UBUDSV to read the buffer (BUFF) onto the disk.

30. If ISTCB1 is less than zero, IPTFLG equals zero, and ICBCFL is greater than zero, print leakage and streamflows into and out of each reach.

31. RETURN to MAIN program.

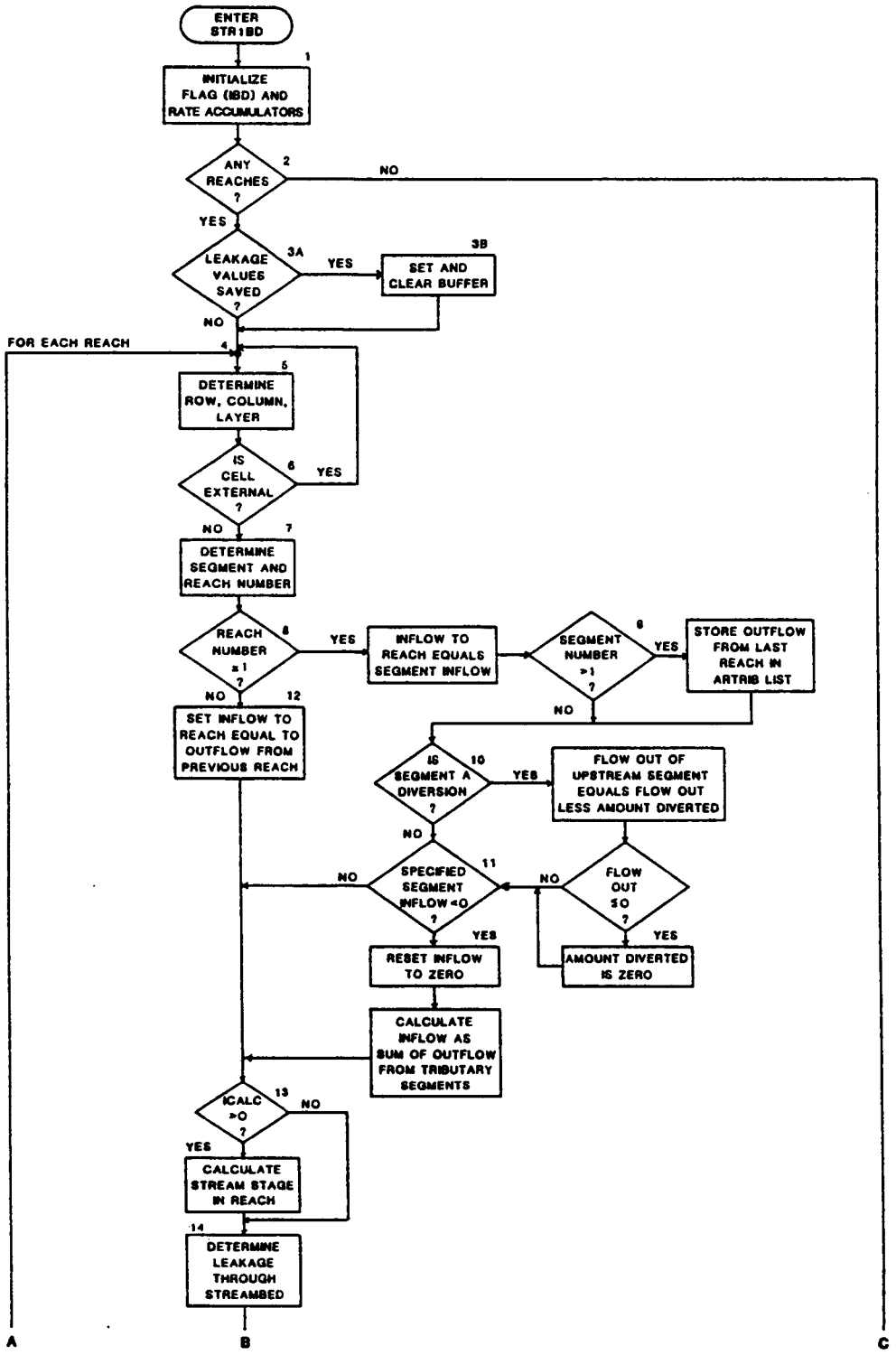


Figure 16.--Generalized flow chart of Module STR1BD. Variable names are defined in section 'List of variables for module STR1BD.'

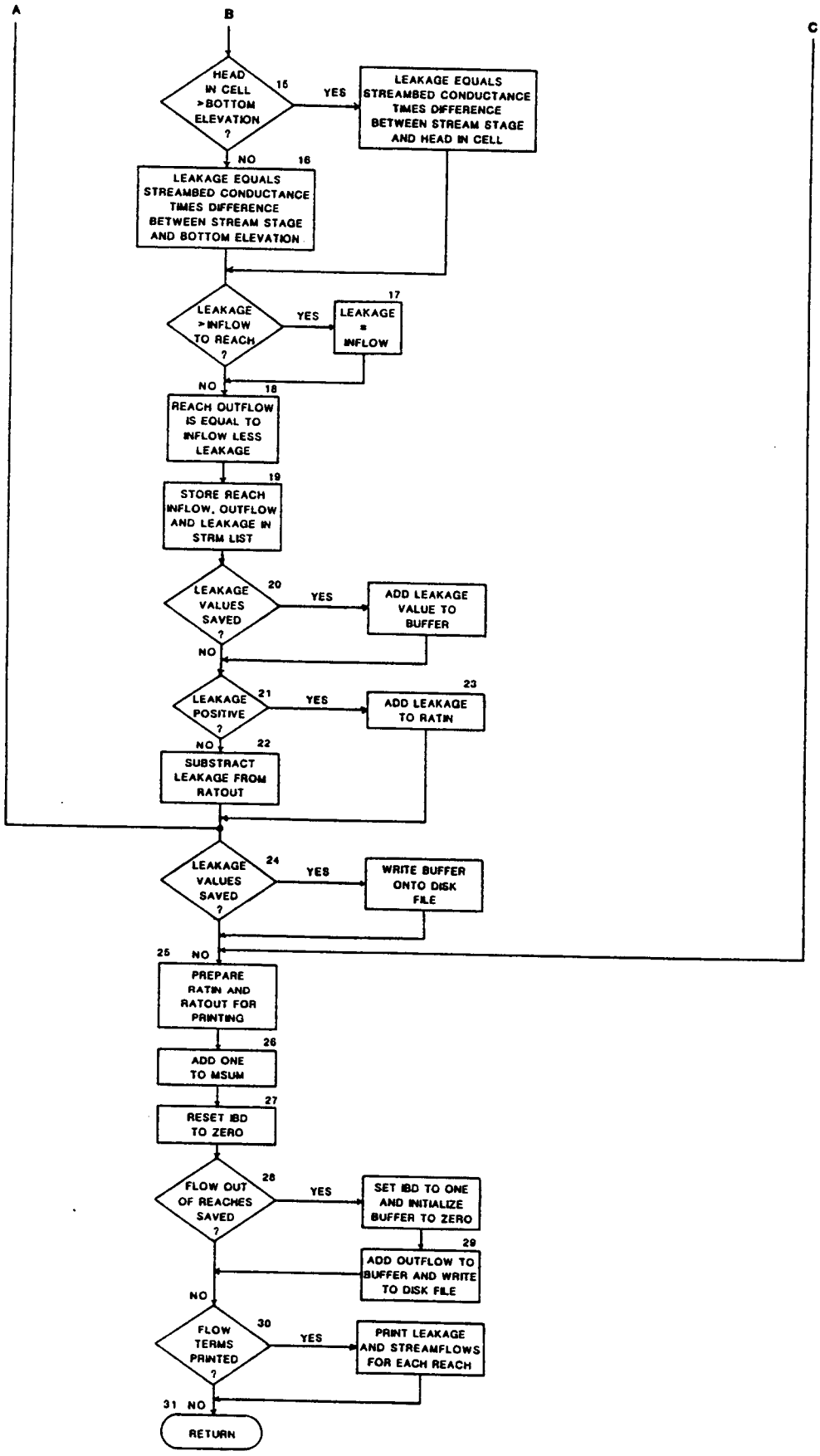


Figure 16.--(concluded)

Program Listing for Module STR1BD

```
      SUBROUTINE STR1BD(NSTREM,STRM,ISTRM,IBOUND,MXSTRM,HNEW,NCOL,NROW,
     1    NLAY,DELT,VBVL,VBNM,MSUM,KSTP,KPER,ISTCB1,ISTCB2,ICBCFL,
     2    BUFF,IOUT,NTRIB,NSS,ARTRIB,ITRBAR,IDIVAR,ICALC,CONST,IPTFLG)
C-----VERSION  1  23OCT1987 STR1BD                                      C
C     ***************************************************************C
C     CALCULATE VOLUMETRIC BUDGET FOR STREAMS                          C
C     ***************************************************************C
C                                                                      C
C     SPECIFICATIONS:                                                  C
C     -----------------------------------------------------------------C
      CHARACTER*4 VBNM,TEXT,STRTXT
      DOUBLE PRECISION HNEW
      DIMENSION STRM(11,MXSTRM),ISTRM(5,MXSTRM),IBOUND(NCOL,NROW,NLAY),
     1          HNEW(NCOL,NROW,NLAY),VBVL(4,20),VBNM(4,20),
     2          BUFF(NCOL,NROW,NLAY),ARTRIB(NSS),ITRBAR(NSS,NTRIB),
     3             IDIVAR(NSS)
      DIMENSION TEXT(4),STRTXT(4)
      DATA TEXT(1),TEXT(2),TEXT(3),TEXT(4) /'  ST','REAM',' LEA','KAGE'/
      DATA STRTXT(1),STRTXT(2),STRTXT(3),STRTXT(4) /'STRE','AM F',
     1                                              'LOW ','OUT '/
C     -----------------------------------------------------------------C
C                                                                      C
C1------SET IBD=1 IF BUDGET TERMS SHOULD BE SAVED ON DISK.             C
      IBD=0
      RATIN = 0.
      RATOUT = 0.
C                                                                      C
C2------IF NO REACHES, KEEP ZEROS IN ACCUMULATORS.                    C
      IF(NSTREM.EQ.0) GO TO 600
C                                                                      C
C3A-----TEST TO SEE IF CELL-BY-CELL TERMS ARE NEEDED.                  C
      IF((ICBCFL.EQ.0).OR.(ISTCB1.LE.0)) GO TO 10
C                                                                      C
C3B-----CELL-BY-CELL TERMS ARE NEEDED, SET IBD AND CLEAR BUFFER.       C
      IBD = 1
      DO 5 IL=1,NLAY
      DO 5 IR=1,NROW
      DO 5 IC=1,NCOL
      BUFF(IC,IR,IL)=0.
    5 CONTINUE
C                                                                      C
C4------IF THERE ARE STREAMS THEN ACCUMULATE LEAKAGE TO OR FROM THEM. C
   10 DO 500 L=1,NSTREM
      LL=L-1
C                                                                      C
C5---DETERMINE REACH LOCATION.                                         C
      IL=ISTRM(1,L)
      IR=ISTRM(2,L)
      IC=ISTRM(3,L)
C                                                                      C
C6---IF CELL IS EXTERNAL SKIP CALCULATIONS.                            C
```

```
      IF(IBOUND(IC,IR,IL).LE.0)GO TO 500
C                                                                       C
C7------DETERMINE SEGMENT AND REACH NUMBER.                            C
      ISTSG=ISTRM(4,L)
      NREACH=ISTRM(5,L)
      IF(NREACH.GT.1) GO TO 200
C                                                                       C
C8------SET FLOWIN EQUAL TO SEGMENT INFLOW IF FIRST REACH.             C
      FLOWIN=STRM(1,L)
      IF(ISTSG.GT.1) IFLG = ISTRM(4,LL)
C                                                                       C
C9------STORE OUTFLOW FROM PREVIOUS SEGMENT IN ARTRIB IF SEGMENT >1.   C
      IF(ISTSG.GT.1)  ARTRIB(IFLG)=STRM(9,LL)
C                                                                       C
C10--IF SEGMENT IS A DIVERSION, COMPUTE FLOW OUT OF UPSTREAM SEGMENT.  C
      IF(IDIVAR(ISTSG).LE.0) GO TO  50
      NDFLG=IDIVAR(ISTSG)
      DUM=ARTRIB(NDFLG)-FLOWIN
      IF(DUM.GE.0.0) ARTRIB(NDFLG)=DUM
      IF(DUM.GE.0.0) GO TO  50
      FLOWIN=0.
   50 IF(FLOWIN.GE.0.0) GO TO 300
C                                                                       C
C11--SUM TRIBUTARY OUTFLOW AND USE AS INFLOW INTO DOWNSTREAM SEGMENT.  C
      FLOWIN =0.
      DO 100 ITRIB=1,NTRIB
      INODE=ITRBAR(ISTSG,ITRIB)
      IF(INODE.LE.0) GO TO 100
      FLOWIN=FLOWIN+ARTRIB(INODE)
  100 CONTINUE
C                                                                       C
C12-----IF REACH >1, SET INFLOW EQUAL TO OUTFLOW FROM UPSTREAM REACH.  C
  200 IF(NREACH.GT.1) FLOWIN=STRM(9,LL)
C                                                                       C
C13----COMPUTE STREAM STAGE IN REACH IF ICALC > 1.                      C
  300 IF(ICALC.LE.0) GO TO 310
      XNUM=((FLOWIN+STRM(9,L))/2.0)*STRM(8,L)
      DNOM=CONST*STRM(6,L)*(SQRT(STRM(7,L)))
      DEPTH=(XNUM/DNOM)**0.6
      IF((DEPTH).LE.0) DEPTH=0.
      STRM(2,L)=DEPTH+STRM(5,L)
  310 HSTR=STRM(2,L)
C                                                                       C
C14----DETERMINE LEAKAGE THROUGH STREAMBED.                            C
      IF(FLOWIN.LE.0.0) HSTR=STRM(5,L)
      CSTR=STRM(3,L)
      SBOT=STRM(4,L)
      H=HNEW(IC,IR,IL)
      T=HSTR-SBOT
C                                                                       C
C15----COMPUTE LEAKAGE AS A FUNCTION OF STREAM STAGE AND HEAD IN CELL. C
      FLOBOT=CSTR*(HSTR-H)
C                                                                       C
C16----RECOMPUTE LEAKAGE IF HEAD IN CELL IS BELOW STREAMBED BOTTOM.    C
      IF(H.GT.SBOT) GO TO 312
```

```
      FLOBOT=CSTR*T
C                                                                       C
C17----SET LEAKAGE EQUAL TO STREAM INFLOW IF LEAKAGE MORE THAN INFLOW. C
  312 IF(FLOBOT.LE.FLOWIN) GO TO 320
      FLOBOT=FLOWIN
C                                                                       C
C18----STREAMFLOW OUT EQUALS STREAMFLOW IN MINUS LEAKAGE.               C
  320 FLOWOT=FLOWIN-FLOBOT
      IF((ISTSG.GT.1).AND.(NREACH.EQ.1)) STRM(9,LL)=ARTRIB(IFLG)
C                                                                       C
C19----STORE STREAM INFLOW, OUTFLOW AND LEAKAGE FOR EACH REACH.         C
      STRM(9,L)=FLOWOT
      STRM(10,L)=FLOWIN
      STRM(11,L)=FLOBOT
C                                                                       C
C20----IF LEAKAGE FROM STREAMS IS TO BE SAVED THEN ADD RATE TO BUFFER.  C
      IF(IBD.EQ.1) BUFF(IC,IR,IL)=BUFF(IC,IR,IL)+FLOBOT
C                                                                       C
C21----DETERMINE IF FLOW IS INTO OR OUT OF MODEL CELL.                  C
C       SKIP ESTIMATE OF LEAKAGE FROM STREAM IF LEAKAGE IS ZERO.        C
      IF(FLOBOT)494,500,496
C                                                                       C
C22-----SUBTRACT FLOW RATE FROM RATOUT IF AQUIFER DISCHARGES TO STREAM.C
  494 RATOUT=RATOUT-FLOBOT
      GO TO 500
C                                                                       C
C23-----ADD FLOW RATE TO RATIN IF STREAM DISCHARGES TO AQUIFER.         C
  496 RATIN=RATIN+FLOBOT
  500 CONTINUE
C                                                                       C
C24-----IF BUDGET TERMS WILL BE SAVED THEN WRITE TO DISK.               C
      IF(IBD.EQ.1) CALL UBUDSV(KSTP,KPER,TEXT,ISTCB1,BUFF,NCOL,NROW,
     1                         NLAY,IOUT)
C                                                                       C
C25A-----MOVE RATES INTO VBVL FOR PRINTING BY MODULE BAS_OT.            C
  600 VBVL(3,MSUM)=RATIN
      VBVL(4,MSUM)=RATOUT
C                                                                       C
C25B-----MOVE PRODUCT OF RATE AND TIME STEP INTO VBVL ACCUMULATORS.     C
      VBVL(1,MSUM)=VBVL(1,MSUM)+RATIN*DELT
      VBVL(2,MSUM)=VBVL(2,MSUM)+RATOUT*DELT
C                                                                       C
C25C-----MOVE BUDGET TERM LABELS INTO VBNM FOR PRINTING BY BAS_OT.      C
      VBNM(1,MSUM)=TEXT(1)
      VBNM(2,MSUM)=TEXT(2)
      VBNM(3,MSUM)=TEXT(3)
      VBNM(4,MSUM)=TEXT(4)
C                                                                       C
C26-----INCREASE BUDGET TERM COUNTER BY ONE.                            C
      MSUM=MSUM+1
C                                                                       C
C27-----RESET IBD COUNTER TO ZERO.                                      C
      IBD=0
C28----IF STREAM OUTFLOW FROM EACH REACH IS TO BE STORED IN ON DISK     C
C       THEN STORE OUTFLOW RATES INTO BUFFER.                           C
```

```
      IF((ICBCFL.EQ.0).OR.(ISTCB2.LE.0)) GO TO 625
      IBD = 1
      DO 605 IL=1,NLAY
      DO 605 IR=1,NROW
      DO 605 IC=1,NCOL
  605 BUFF(IC,IR,IL)=0.
C                                                                       C
C29-----SAVE STREAMFLOWS OUT OF EACH REACH ON DISK.                     C
      DO 615 L=1,NSTREM
      IC=ISTRM(3,L)
      IR=ISTRM(2,L)
      IL=ISTRM(1,L)
      IF(IBOUND(IC,IR,IL).LE.0) GO TO 615
      BUFF(IC,IR,IL)=BUFF(IC,IR,IL)+STRM(9,L)
  615 CONTINUE
      CALL UBUDSV(KSTP,KPER,STRTXT,ISTCB2,BUFF,NCOL,NROW,NLAY,IOUT)
C                                                                       C
C30-----PRINT STREAMFLOW RATES AND LEAKAGE FOR EACH REACH.              C
  625 IF((ISTCB1.GE.0).OR.(ICBCFL.LE.0)) GO TO 800
      IF(IPTFLG.GT.0) GO TO 800
      IF(ICALC.GT.0) GO TO 700
      WRITE(IOUT,650)
  650 FORMAT(1H0,12X,'LAYER',6X,'ROW',5X,'COLUMN',5X,'STREAM',4X,
     1'REACH',6X,'FLOW INTO',4X,'FLOW INTO',6X,'FLOW OUT OF'/43X,
     2       'NUMBER',3X,'NUMBER',4X,'STREAM REACH',4X,'AQUIFER',
     3       6X,'STREAM REACH'//)
      DO 690 L=1,NSTREM
      IL=ISTRM(1,L)
      IR=ISTRM(2,L)
      IC=ISTRM(3,L)
      WRITE(IOUT,675)IL,IR,IC,ISTRM(4,L),ISTRM(5,L),
     1     STRM(10,L),STRM(11,L),STRM(9,L)
  675 FORMAT(1X,5X,5I10,8X,G9.3,5X,G9.3,8X,G9.3)
  690 CONTINUE
      GO TO 800
  700 WRITE(IOUT,710)
  710 FORMAT(1H0,12X,'LAYER',6X,'ROW',5X,'COLUMN',5X,'STREAM',4X,
     1'REACH',6X,'FLOW INTO',4X,'FLOW INTO',6X,'FLOW OUT OF',5X,
     2'HEAD IN'/43X,       'NUMBER',3X,'NUMBER',4X,'STREAM REACH',
     3 4X,'AQUIFER',6X,'STREAM REACH',5X,'STREAM'//)
      DO 750 L=1,NSTREM
      IL=ISTRM(1,L)
      IR=ISTRM(2,L)
      IC=ISTRM(3,L)
      WRITE(IOUT,775)IL,IR,IC,ISTRM(4,L),ISTRM(5,L),
     1     STRM(10,L),STRM(11,L),STRM(9,L),STRM(2,L)
  775 FORMAT(1X,5X,5I10,8X,G9.3,5X,G9.3,7X,G9.3,4X,F9.2)
  750 CONTINUE
  800 CONTINUE
C                                                                       C
C31-----RETURN.                                                         C
      RETURN
      END
```

List of Variables for STR1BD

Variable	Range	Definition
ARTRIB	Package	DIMENSION (NSS), For each segment: contains the streamflow out of last reach in each stream segment.
BUFF	Global	DIMENSION (NCOL,NROW,NLAY), Buffer used to accumulate information before sending it to a disk file.
CONST	Package	A constant used in calculating stream stage in reaches. Specified only when ICALC is greater than zero. A value of 1.486 is used for units of cubic feet per second and a value of 1.0 is used for units of cubic meters per second. The constant is multiplied by 86,400 when days are used as the time unit.
CSTR	Module	Streambed hydraulic conductance.
DELT	Global	Length of current time step.
DEPTH	Module	Depth of water in reach. The value is added to top of streambed to obtain the stream stage.
DNOM	Module	Denominator of the equation used to calculate depth of water in a reach.
DUM	Module	Amount of flow from segment from which water is diverted.
FLOBOT	Module	Leakage into or out of model cell through the streambed.
FLOWIN	Module	Streamflow into reach. If, streamflow into the first reach of a segment is negative, then FLOWIN is the sum of streamflow out of the last reach in each specified tributary segment.
FLOWOT	Module	Streamflow out of a reach.
H	Module	Head in the model cell (HNEW).
HNEW	Global	DIMENSION (NCOL,NROW,NLAY), Most recent estimate of head in each cell. HNEW changes at each iteration.

HSTR	Module	Stream stage in reach.
IBD	Module	Flag. =0, flow terms for each reach will not be saved on a disk file. =1, flow terms for each reach will be saved on a disk file.
IBOUND	Global	DIMENSION (NCOL,NROW,NLAY), Status of each cell. <0, constant-head cell. =0, inactive cell. >0, variable-head cell.
ICBCFL	Global	Flag used to specify the frequency of recording or printing of results. =0, cell-by-cell flow terms will not be recorded or printed for the current time step. (not)=0, cell-by-cell flow terms will be either printed or recorded depending on the values for variables ISTCB1, ISTCB2 or IPTFLG.
IC	Module	Column number of cell containing reach.
ICALC	Package	Flag. >0, stream stage in reaches will be calculated. ≤0, stream stage in reaches will not be calculated.
IDIVAR	Package	DIMENSION (NSS), For each segment: upstream segment number from which water is to be diverted.
IFLG	Module	Number of previous segment.
IL	Module	Layer number of cell containing reach.
INODE	Module	Number of a tributary segment.
IOUT	Global	Primary unit number for all printed output. IOUT = FORTRAN unit number 6.
IPTFLG	Package	Flag to specify printing of results for each stress period. Results may be printed at selected time steps within a stress period depending on the value specified for variable ICBCFL. 0, results will be printed according to the value of ICBCFL. >0, results will not be printed even if ICBCFL is specified.

IR	Module	Row number of cell containing reach.
ISTCB1	Package	Flag and unit number. >0, unit number to which leakage between each reach and corresponding model cell is saved on disk whenever the variable ICBCFL is specified. =0, leakage will not be printed nor recorded. <0, leakage between each stream reach and corresponding model cell will be printed whenever the variables ICBCFL and IPTFLG are specified.
ISTCB2	Package	Flag and unit number. >0, unit number to which streamflow out of each reach will be saved on disk whenever the variable ICBCFL is specified. ≤0, streamflow out of each reach will not be saved on disk.
ISTRM	Package	DIMENSION (5,MXSTRM), For each reach: layer, row, column, segment number, and reach number.
ISTSG	Module	Segment number of cell containing reach.
ITRBAR	Package	DIMENSION (NSS,NTRIB), For each segment: segment numbers of each tributary segment. A zero is assigned if no segments are tributary to a given segment.
ITRIB	Module	Index for the number of tributary segments.
KPER	Global	Stress period counter.
KSTP	Global	Time step counter. Reset at start of each stress period.
L	Module	Index for current reach.
LL	Module	Index for previous reach.
MSUM	Global	Counter for budget entries and labels into VBVL and VBNM.
MXSTRM	Package	Maximum number of stream reaches.
NCOL	Global	Number of columns in grid.
NDFLG	Module	Number of segments from which water is diverted.

NLAY	Global	Number of layers in grid.
NREACH	Module	Number of reach in segment.
NROW	Global	Number of rows in grid.
NSS	Package	Maximum number of stream segments.
NSTREM	Package	Number of stream reaches active during current stress period.
NTRIB	Package	Maximum number of tributary segments that join one downstream segment.
RATIN	Module	Accumulator for the total leakage into the model cells through the streambeds of all reaches.
RATOUT	Module	Accumulator for the total leakage out of the model cells through the streambed of all reaches.
SBOT	Module	Elevation of streambed bottom.
STRM	Package	DIMENSION (11,MXSTRM), For each reach: (1) segment inflow, (2) stream stage, (3) streambed conductance, (4) streambed bottom and (5) top elevations, (6) stream width, (7) channel slope, (8) Manning's roughness coefficient, (9) streamflow out of each reach, (10) streamflow into each reach, and (11) leakage through the streambed in each reach.
STRTXT	Module	Label to be recorded with streamflows out of each reach when the data is saved on a specified unit number.
T	Module	Difference between stream stage (HSTR) and elevation of streambed bottom (SBOT).
TEXT	Module	Label to be recorded with leakage into or out of model cells through the streambed of each reach when data is saved on a specified unit number. The label is also used in the budget table.
VBNM	Global	DIMENSION (4,20), Labels for entries in volumetric budget.

VBVL	Global	DIMENSION (4,20), Entries for the volumetric budget. For flow component N, values in VBVL are: (1,N) volume into the flow field during the simulation. (2,N) volume out of the flow field during the simulation. (3,N) rate for current time step into the flow field. (4,N) rate for current time step out of the flow field.
XNUM	Module	Numerator of equation used to calculate depth of water in a reach.

Example Listing of a Modified MAIN Program

```
C     *****************************************************************
C     MAIN CODE FOR MODULAR MODEL --    9/1/87
C          BY MICHAEL G. MCDONALD AND ARLEN W. HARBAUGH
C-----VERSION 1638 24JUL1987 MAIN1
C          MODIFIED BY DAVID E. PRUDIC 23OCT1987  MAINSTREAM
C     *****************************************************************
C
C        SPECIFICATIONS:
C     -----------------------------------------------------------------
      COMMON X(200000)
      COMMON /FLWCOM/LAYCON(80)
      CHARACTER*4 HEADNG,VBNM
      DIMENSION HEADNG(32),VBNM(4,20),VBVL(4,20),IUNIT(24)
      DOUBLE PRECISION DUMMY
      EQUIVALENCE (DUMMY,X(1))
C     -----------------------------------------------------------------
C
C1------SET SIZE OF X ARRAY. REMEMBER TO REDIMENSION X.
      LENX=200000
C
C2------ASSIGN BASIC INPUT UNIT AND PRINTER UNIT.
      INBAS=5
      IOUT=6
C
C3------DEFINE PROBLEM  ROWS,COLUMNS,LAYERS,STRESS PERIODS,PACKAGES
      CALL BAS1DF(ISUM,HEADNG,NPER,ITMUNI,TOTIM,NCOL,NROW,NLAY,
     1             NODES,INBAS,IOUT,IUNIT)
C
C4------ALLOCATE SPACE IN "X" ARRAY.
      CALL BAS1AL(ISUM,LENX,LCHNEW,LCHOLD,LCIBOU,LCCR,LCCC,LCCV,
     1              LCHCOF,LCRHS,LCDELR,LCDELC,LCSTRT,LCBUFF,LCIOFL,
     2              INBAS,ISTRT,NCOL,NROW,NLAY,IOUT)
      IF(IUNIT(1).GT.0) CALL BCF1AL(ISUM,LENX,LCSC1,LCHY,
     1     LCBOT,LCTOP,LCSC2,LCTRPY,IUNIT(1),ISS,
     2     NCOL,NROW,NLAY,IOUT,IBCFCB)
      IF(IUNIT(2).GT.0) CALL WEL1AL(ISUM,LENX,LCWELL,MXWELL,NWELLS,
     1                  IUNIT(2),IOUT,IWELCB)
      IF(IUNIT(3).GT.0) CALL DRN1AL(ISUM,LENX,LCDRAI,NDRAIN,MXDRN,
     1                  IUNIT(3),IOUT,IDRNCB)
      IF(IUNIT(8).GT.0) CALL RCH1AL(ISUM,LENX,LCIRCH,LCRECH,NRCHOP,
     1            NCOL,NROW,IUNIT(8),IOUT,IRCHCB)
      IF(IUNIT(5).GT.0) CALL EVT1AL(ISUM,LENX,LCIEVT,LCEVTR,LCEXDP,
     1            LCSURF,NCOL,NROW,NEVTOP,IUNIT(5),IOUT,IEVTCB)
      IF(IUNIT(4).GT.0) CALL RIV1AL(ISUM,LENX,LCRIVR,MXRIVR,NRIVER,
     1            IUNIT(4),IOUT,IRIVCB)
C------STREAMFLOW-ROUTING PACKAGE..MODULE STR1AL
      IF(IUNIT(13).GT.0) CALL STR1AL(ISUM,LENX,LCSTRM,ICSTRM,MXSTRM,     STR1
     1                  NSTREM,IUNIT(13),IOUT,ISTCB1,ISTCB2,NSS,NTRIB,   STR1
     2                    NDIV,ICALC,CONST,LCTBAR,LCTRIB,LCIVAR)         STR1
C-------END OF MODULE STR1AL CALL STATEMENT
```

```
      IF(IUNIT(7).GT.0) CALL GHB1AL(ISUM,LENX,LCBNDS,NBOUND,MXBND,
     1            IUNIT(7),IOUT,IGHBCB)
      IF(IUNIT(9).GT.0) CALL SIP1AL(ISUM,LENX,LCEL,LCFL,LCGL,LCV,
     1          LCHDCG,LCLRCH,LCW,MXITER,NPARM,NCOL,NROW,NLAY,
     2          IUNIT(9),IOUT)
      IF(IUNIT(11).GT.0) CALL SOR1AL(ISUM,LENX,LCA,LCRES,LCHDCG,LCLRCH,
     1           LCIEQP,MXITER,NCOL,NLAY,NSLICE,MBW,IUNIT(11),IOUT)
C
C5------IF THE "X" ARRAY IS NOT BIG ENOUGH THEN STOP.
      IF(ISUM-1.GT.LENX) STOP
C
C6------READ AND PREPARE INFORMATION FOR ENTIRE SIMULATION.
      CALL BAS1RP(X(LCIBOU),X(LCHNEW),X(LCSTRT),X(LCHOLD),
     1       ISTRT,INBAS,HEADNG,NCOL,NROW,NLAY,NODES,VBVL,X(LCIOFL),
     2       IUNIT(12),IHEDFM,IDDNFM,IHEDUN,IDDNUN,IOUT)
      IF(IUNIT(1).GT.0) CALL BCF1RP(X(LCIBOU),X(LCHNEW),X(LCSC1),
     1          X(LCHY),X(LCCR),X(LCCC),X(LCCV),X(LCDELR),
     2           X(LCDELC),X(LCBOT),X(LCTOP),X(LCSC2),X(LCTRPY),
     3           IUNIT(1),ISS,NCOL,NROW,NLAY,NODES,IOUT)
      IF(IUNIT(9).GT.0) CALL SIP1RP(NPARM,MXITER,ACCL,HCLOSE,X(LCW),
     1          IUNIT(9),IPCALC,IPRSIP,IOUT)
      IF(IUNIT(11).GT.0) CALL SOR1RP(MXITER,ACCL,HCLOSE,IUNIT(11),
     1         IPRSOR,IOUT)
C
C7------SIMULATE EACH STRESS PERIOD.
      DO 300 KPER=1,NPER
      KKPER=KPER
C
C7A-----READ STRESS PERIOD TIMING INFORMATION.
      CALL BAS1ST(NSTP,DELT,TSMULT,PERTIM,KKPER,INBAS,IOUT)
C
C7B-----READ AND PREPARE INFORMATION FOR STRESS PERIOD.
      IF(IUNIT(2).GT.0) CALL WEL1RP(X(LCWELL),NWELLS,MXWELL,IUNIT(2),
     1            IOUT)
      IF(IUNIT(3).GT.0) CALL DRN1RP(X(LCDRAI),NDRAIN,MXDRN,IUNIT(3),
     1                IOUT)
      IF(IUNIT(8).GT.0) CALL RCH1RP(NRCHOP,X(LCIRCH),X(LCRECH),
     1           X(LCDELR),X(LCDELC),NROW,NCOL,IUNIT(8),IOUT)
      IF(IUNIT(5).GT.0) CALL EVT1RP(NEVTOP,X(LCIEVT),X(LCEVTR),
     1           X(LCEXDP),X(LCSURF),X(LCDELR),X(LCDELC),NCOL,NROW,
     1           IUNIT(5),IOUT)
      IF(IUNIT(4).GT.0) CALL RIV1RP(X(LCRIVR),NRIVER,MXRIVR,IUNIT(4),
     1           IOUT)
C------STREAMFLOW-ROUTING PACKAGE MODULE STR1RP
      IF(IUNIT(13).GT.0) CALL STR1RP(X(LCSTRM),X(ICSTRM),NSTREM,        STR1
     1                    MXSTRM,IUNIT(13),IOUT,X(LCTBAR),NDIV,NSS,    STR1
     1                    NTRIB,X(LCIVAR),ICALC,IPTFLG)                 STR1
C------END OF MODULE STR1RP CALL STATEMENT
      IF(IUNIT(7).GT.0) CALL GHB1RP(X(LCBNDS),NBOUND,MXBND,IUNIT(7),
     1             IOUT)
C
C7C-----SIMULATE EACH TIME STEP.
      DO 200 KSTP=1,NSTP
      KKSTP=KSTP
```

```
C
C7C1----CALCULATE TIME STEP LENGTH. SET HOLD=HNEW..
      CALL BAS1AD(DELT,TSMULT,TOTIM,PERTIM,X(LCHNEW),X(LCHOLD),KKSTP,
     1            NCOL,NROW,NLAY)
C
C7C2----ITERATIVELY FORMULATE AND SOLVE THE EQUATIONS.
      DO 100 KITER=1,MXITER
      KKITER=KITER
C
C7C2A---FORMULATE THE FINITE DIFFERENCE EQUATIONS.
      CALL BAS1FM(X(LCHCOF),X(LCRHS),NODES)
      IF(IUNIT(1).GT.0) CALL BCF1FM(X(LCHCOF),X(LCRHS),X(LCHOLD),
     1          X(LCSC1),X(LCHNEW),X(LCIBOU),X(LCCR),X(LCCC),X(LCCV),
     2          X(LCHY),X(LCTRPY),X(LCBOT),X(LCTOP),X(LCSC2),
     3          X(LCDELR),X(LCDELC),DELT,ISS,KKITER,KKSTP,KKPER,NCOL,
     4          NROW,NLAY,IOUT)
      IF(IUNIT(2).GT.0) CALL WEL1FM(NWELLS,MXWELL,X(LCRHS),X(LCWELL),
     1           X(LCIBOU),NCOL,NROW,NLAY)
      IF(IUNIT(3).GT.0) CALL DRN1FM(NDRAIN,MXDRN,X(LCDRAI),X(LCHNEW),
     1         X(LCHCOF),X(LCRHS),X(LCIBOU),NCOL,NROW,NLAY)
      IF(IUNIT(8).GT.0) CALL RCH1FM(NRCHOP,X(LCIRCH),X(LCRECH),
     1           X(LCRHS),X(LCIBOU),NCOL,NROW,NLAY)
      IF(IUNIT(5).GT.0) CALL EVT1FM(NEVTOP,X(LCIEVT),X(LCEVTR),
     1           X(LCEXDP),X(LCSURF),X(LCRHS),X(LCHCOF),X(LCIBOU),
     1           X(LCHNEW),NCOL,NROW,NLAY)
      IF(IUNIT(4).GT.0) CALL RIV1FM(NRIVER,MXRIVR,X(LCRIVR),X(LCHNEW),
     1           X(LCHCOF),X(LCRHS),X(LCIBOU),NCOL,NROW,NLAY)
C-------STREAMFLOW-ROUTING PACKAGE MODULE STR1FM
      IF(IUNIT(13).GT.0) CALL STR1FM(NSTREM,X(LCSTRM),X(ICSTRM),          STR1
     1                               X(LCHNEW),X(LCHCOF),X(LCRHS),        STR1
     2               X(LCIBOU),MXSTRM,NCOL,NROW,NLAY,IOUT,NSS,            STR1
     3              X(LCTBAR),NTRIB,X(LCTRIB),X(LCIVAR),ICALC,CONST)      STR1
C------END OF MODULE STR1FM CALL STATEMENT
      IF(IUNIT(7).GT.0) CALL GHB1FM(NBOUND,MXBND,X(LCBNDS),X(LCHCOF),
     1           X(LCRHS),X(LCIBOU),NCOL,NROW,NLAY)
C
C7C2B---MAKE ONE CUT AT AN APPROXIMATE SOLUTION.
      IF(IUNIT(9).GT.0) CALL SIP1AP(X(LCHNEW),X(LCIBOU),X(LCCR),X(LCCC),
     1     X(LCCV),X(LCHCOF),X(LCRHS),X(LCEL),X(LCFL),X(LCGL),X(LCV),
     2     X(LCW),X(LCHDCG),X(LCLRCH),NPARM,KKITER,HCLOSE,ACCL,ICNVG,
     3     KKSTP,KKPER,IPCALC,IPRSIP,MXITER,NSTP,NCOL,NROW,NLAY,NODES,
     4     IOUT)
      IF(IUNIT(11).GT.0) CALL SOR1AP(X(LCHNEW),X(LCIBOU),X(LCCR),
     1     X(LCCC),X(LCCV),X(LCHCOF),X(LCRHS),X(LCA),X(LCRES),X(LCIEQP),
     2     X(LCHDCG),X(LCLRCH),KKITER,HCLOSE,ACCL,ICNVG,KKSTP,KKPER,
     3     IPRSOR,MXITER,NSTP,NCOL,NROW,NLAY,NSLICE,MBW,IOUT)
C
C7C2C---IF CONVERGENCE CRITERION HAS BEEN MET STOP ITERATING.
      IF(ICNVG.EQ.1) GO TO 110
  100 CONTINUE
      KITER=MXITER
  110 CONTINUE
C
C7C3----DETERMINE WHICH OUTPUT IS NEEDED.
```

```
      CALL BAS1OC(NSTP,KKSTP,ICNVG,X(LCIOFL),NLAY,
     1   IBUDFL,ICBCFL,IHDDFL,IUNIT(12),IOUT)
C
C7C4----CALCULATE BUDGET TERMS. SAVE CELL-BY-CELL FLOW TERMS.
      MSUM=1
      IF(IUNIT(1).GT.0) CALL BCF1BD(VBNM,VBVL,MSUM,X(LCHNEW),
     1     X(LCIBOU),X(LCHOLD),X(LCSC1),X(LCCR),X(LCCC),X(LCCV),
     2     X(LCTOP),X(LCSC2),DELT,ISS,NCOL,NROW,NLAY,KKSTP,KKPER,
     3     IBCFCB,ICBCFL,X(LCBUFF),IOUT)
      IF(IUNIT(2).GT.0) CALL WEL1BD(NWELLS,MXWELL,VBNM,VBVL,MSUM,
     1     X(LCWELL),X(LCIBOU),DELT,NCOL,NROW,NLAY,KKSTP,KKPER,IWELCB,
     1     ICBCFL,X(LCBUFF),IOUT)
      IF(IUNIT(3).GT.0) CALL DRN1BD(NDRAIN,MXDRN,VBNM,VBVL,MSUM,
     1     X(LCDRAI),DELT,X(LCHNEW),NCOL,NROW,NLAY,X(LCIBOU),KKSTP,
     2     KKPER,IDRNCB,ICBCFL,X(LCBUFF),IOUT)
      IF(IUNIT(8).GT.0) CALL RCH1BD(NRCHOP,X(LCIRCH),X(LCRECH),
     1     X(LCIBOU),NROW,NCOL,NLAY,DELT,VBVL,VBNM,MSUM,KKSTP,KKPER,
     2     IRCHCB,ICBCFL,X(LCBUFF),IOUT)
      IF(IUNIT(5).GT.0) CALL EVT1BD(NEVTOP,X(LCIEVT),X(LCEVTR),
     1     X(LCEXDP),X(LCSURF),X(LCIBOU),X(LCHNEW),NCOL,NROW,NLAY,
     2     DELT,VBVL,VBNM,MSUM,KKSTP,KKPER,IEVTCB,ICBCFL,X(LCBUFF),IOUT)
      IF(IUNIT(4).GT.0) CALL RIV1BD(NRIVER,MXRIVR,X(LCRIVR),X(LCIBOU),
     1     X(LCHNEW),NCOL,NROW,NLAY,DELT,VBVL,VBNM,MSUM,
     2     KKSTP,KKPER,IRIVCB,ICBCFL,X(LCBUFF),IOUT)
C------STREAMFLOW-ROUTING PACKAGE MODULE STR1BD
      IF(IUNIT(13).GT.0) CALL STR1BD(NSTREM,X(LCSTRM),X(ICSTRM),        STR1
     1     X(LCIBOU),MXSTRM,X(LCHNEW),NCOL,NROW,NLAY,DELT,VBVL,         STR1
     2     VBNM,MSUM,KKSTP,KKPER,ISTCB1,ISTCB2,ICBCFL,X(LCBUFF),IOUT,   STR1
     3     NTRIB,NSS,X(LCTRIB),X(LCTBAR),X(LCIVAR),ICALC,CONST,IPTFLG)  STR1
C------END OF MODULE STR1BD CALL STATEMENT
      IF(IUNIT(7).GT.0) CALL GHB1BD(NBOUND,MXBND,VBNM,VBVL,MSUM,
     1     X(LCBNDS),DELT,X(LCHNEW),NCOL,NROW,NLAY,X(LCIBOU),KKSTP,
     2     KKPER,IGHBCB,ICBCFL,X(LCBUFF),IOUT)
C
C7C5---PRINT AND OR SAVE HEADS AND DRAWDOWNS. PRINT OVERALL BUDGET.
      CALL BAS1OT(X(LCHNEW),X(LCSTRT),ISTRT,X(LCBUFF),X(LCIOFL),
     1        MSUM,X(LCIBOU),VBNM,VBVL,KKSTP,KKPER,DELT,
     2        PERTIM,TOTIM,ITMUNI,NCOL,NROW,NLAY,ICNVG,
     3     IHDDFL,IBUDFL,IHEDFM,IHEDUN,IDDNFM,IDDNUN,IOUT)
C
C7C6----IF ITERATION FAILED TO CONVERGE THEN STOP.
      IF(ICNVG.EQ.0) STOP
  200 CONTINUE
  300 CONTINUE
C
C8------END PROGRAM
      STOP
C
      END
```

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APPENDIX I

INPUT DATA SETS AND PRINTED RESULTS FOR TEST PROBLEM 1

This problem tests the ability of the Streamflow-Routing Package to accumulate leakage from an aquifer into a stream with a constant stage. Details of the test problem and results are discussed in the section "Results of Test Simulations."

Listing of Input Data Sets for Test Problem 1

Listing of Input Data for Basic Package

Input for the Basic Package follows the column numbers. The input consists of 152 records (lines). Input for package is read from the FORTRAN unit number specified in the MAIN program.

Column Numbers

```
        1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
TEST PROBLEM1--CONSTANT STREAM STAGE AND VARIABLE RECHARGE TO AQUIFER:

         1        13        39       144         1
  7                    11 13       14 15
         0         1
         0         1
       999.
         0      55.0
 1296000.         2       1.5
Previous line is repeated 143 times.
```

Listing of Input Data for Strongly-Implicit Procedure Package

Input for the Strongly-Implicit Procedure Package follows the column numbers. The input consists of 2 records (lines). Input for package is read from FORTRAN unit 13 as specified in the data set for the Basic Package.

Column Numbers

```
        1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
       150         5
         1.    0.0001         1         0         0
```

Listing of Input Data for Block-Centered Flow Package

Input for the Block-Centered Flow Package follows the column numbers. The input data consists of 11 records (lines). Input for package is read from FORTRAN unit number 7 as specified in the data set for the Basic Package.

Column Numbers

```
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
         0         0
0
         0        1.
         7      100. (15F5.0)
  3.   2.   2.   2.   2.   2.   2.   2.   2.   2.   2.   2.   2.   2.   2.
  2.   2.   2.   2.   2.   2.   2.   2.   2.   2.   2.   2.   2.   2.   2.
  2.   2.   2.   2.   2.   2.   2.   2.   3.
         0     1000.
         0      0.20
         0  0.037000
         0        0.
```

Listing of Input Data For Output Control Package

Input for the Output Control Package follows the column numbers. The input data consists of 338 records (lines). Input for package is read from FORTRAN unit number 14 as specified in the data set for the Basic Package.

Column Numbers

```
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
         5         5         0         0
         0         0         0         0
         0         0         0         0
        -1         0         0         0
previous line repeated 238 times
         0         0         0         0
         0         0         0         0
         0         1         1         1
         1         0         0         0
sequence of previous 4 lines repeated 23 times
```



Techniques of Water-Resources Investigations
of the United States Geological Survey

Chapter A2

DOCUMENTATION OF A COMPUTER PROGRAM TO SIMULATE AQUIFER-SYSTEM COMPACTION USING THE MODULAR FINITE-DIFFERENCE GROUND-WATER FLOW MODEL

By S.A. Leake and D.E. Prudic

This chapter supersedes U.S. Geological Survey Open-File Report 88-482

Book 6
Chapter A2

U.S. DEPARTMENT OF THE INTERIOR
MANUEL LUJAN, Secretary

U.S. GEOLOGICAL SURVEY
Dallas L. Peck, Director

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PREFACE

The series of manuals on techniques describes procedures for planning and executing specialized work in water-resources investigations. The material is grouped under major subject headings called books and further subdivided into sections and chapters. Section A of Book 6 is ground-water modeling.

This report presents a computer program for simulating aquifer-system compaction resulting from ground-water storage changes in compressible beds. The program documented in this report is designed for incorporation into the modular finite-difference ground-water flow model developed by the U.S. Geological Survey. The modular model is documented in Techniques of Water-Resources Investigations Book 6, Chapter A1.

The performance of this computer program has been tested in models of both hypothetical and actual ground-water flow systems. Future applications or advancement in knowledge, equipment, or techniques may require that this program be updated. Users may inquire about the availability of updates to this computer program (Interbed-Storage Package) by writing to:

U.S. Geological Survey
375 South Euclid
Tucson, Arizona 85719

Copies of the computer program and test data sets on magnetic tape or diskette are available at cost of processing from:

U.S. Geological Survey
National Water Information System
437 National Center
Reston, VA 22092
Telephone: (703) 648-5695

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TECHNIQUES OF WATER-RESOURCES INVESTIGATIONS OF THE UNITED STATES GEOLOGICAL SURVEY

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METRIC CONVERSION FACTORS

For readers who prefer to use inch-pound units, conversion factors for the metric (International System) units in this report are listed below:

Multiply metric unit	*By*	*To obtain inch-pound unit*
meter (m)	3.281	foot (ft)
square meter (m^2)	10.76	square foot (ft^2)
cubic meter (m^3)	35.31	cubic foot (ft^3)
per meter (m^{-1})	0.3048	per foot (ft^{-1})
square meter per day (m^2/d)	10.76	square foot per day (ft^2/d)
meter per year (m/yr)	3.281	foot per year (ft/yr)

DOCUMENTATION OF A COMPUTER PROGRAM TO SIMULATE AQUIFER-SYSTEM COMPACTION USING THE MODULAR FINITE-DIFFERENCE GROUND-WATER FLOW MODEL

By S.A. Leake and D.E. Prudic

Abstract

Removal of ground water by pumping from aquifers may result in compaction of compressible fine-grained beds that are within or adjacent to the aquifers. Compaction of the sediments and resulting land subsidence may be permanent if the head declines result in vertical stresses beyond the previous maximum stress. The process of permanent compaction is not routinely included in simulations of ground-water flow. To simulate storage changes from both elastic and inelastic compaction, a computer program was written for use with the U.S. Geological Survey modular finite-difference ground-water flow model. The new program, the Interbed-Storage Package, is designed to be incorporated into this model.

In the Interbed-Storage Package, elastic compaction or expansion is assumed to be proportional to change in head. The constant of proportionality is the product of the skeletal component of elastic specific storage and the thickness of the sediments. Similarly, inelastic compaction is assumed to be proportional to decline in head. The constant of proportionality is the product of the skeletal component of inelastic specific storage and the thickness of the sediments. Storage changes are incorporated into the ground-water flow model by adding an additional term to the right-hand side of the flow equation. Within a model time step, the package appropriately apportions storage changes between elastic and inelastic components on the basis of the relation of simulated head to the previous minimum (preconsolidation) head.

Two tests were performed to verify that the package works correctly. The first test compared model-calculated storage and compaction changes to hand-calculated values for a three-dimensional simulation. Model and hand-calculated values were essentially equal. The second test was performed to compare the results of the Interbed-Storage Package with results of the one-dimensional Helm compaction model. This test problem simulated compaction in doubly draining confining beds stressed by head changes in adjacent aquifers. The Interbed-Storage Package and the Helm model computed essentially equal values of compaction.

Documentation of the Interbed-Storage Package includes data input instructions, flow charts, narratives, and listings for each of the five modules included in the package. The documentation also includes an appendix describing input instructions and a listing of a computer program for time-variant specified-head boundaries. That package was developed to reduce the amount of data input and output associated with one of the Interbed-Storage Package test problems.

Manuscript approved for publication, September 21, 1988.

Introduction

Ground water is removed from storage when water is pumped from aquifers. Jacob (1940) postulated that stored water in confined aquifers is derived from the expansion of water, the compression of the aquifer material, and compression of the clayey beds that are adjacent to and within the aquifer. He further concluded that the principal source of stored water probably is from the compression of the clayey beds. Compression or compaction of the sediments is elastic if the lowering of fluid (pore) pressures does not result in permanent rearrangement of the skeletal structure of the sediments and if water removed from storage can be replaced when fluid pressures increase. However, if the fluid pressures decrease beyond the interval where the sediments compact elastically, additional water is released from the clayey beds as the skeletal structure is rearranged and permanently compacted. This process is referred to as permanent or inelastic compaction. Water removed from storage by permanent compaction cannot be returned after pumping decreases or ceases. Thus, water released during permanent compaction can be considered a one-time source that cannot be replaced.

Concentrated pumping of large quantities of ground water (mainly from unconsolidated deposits) in many areas has resulted in the permanent compaction of fine-grained sediments and a consequent lowering of land surface (referred to as land subsidence). Land subsidence caused by pumping of ground water has been identified in seven states in the Nation (Poland, 1984, p. 6).

The process of permanent compaction is not routinely incorporated in ground-water flow simulations. Because of the need to simulate permanent compaction, particularly in areas of concentrated pumping in thick unconsolidated deposits, a subprogram was written to be incorporated into a computer program that simulates three-dimensional ground-water flow (McDonald and Harbaugh, 1988). This computer program was designed to allow for additional capabilities to be easily incorporated.

Purpose and Scope

This report documents a method for the simulation of changes in ground-water storage caused by compressible interbeds in an aquifer system. A package consisting of five subroutines or modules has been incorporated into the computer program of McDonald and Harbaugh (1988). The package is the Interbed-Storage Package. In addition to accounting for changes in storage, the package also calculates net compaction or elastic expansion of interbeds in individual model layers and sums those values to calculate a change in the land surface. The documentation includes a description of how the model computes permanent compaction as well as recoverable compaction of sediments. The documentation also includes data-input instructions and, for each module, includes a narrative, a flow chart, a program listing, and a description of variables. Also included are data-input instructions and a program listing for an additional package, the Time-Variant Specified-Head Package, which was used in a simulation to test the Interbed-Storage Package.

Interbeds

The term "interbed" is used in this report to denote a poorly permeable bed within a relatively permeable aquifer (fig. 1). Such interbeds are assumed to be (1) of significantly lower hydraulic conductivity than the surrounding sediments considered to be aquifer material yet porous and permeable enough to accept or release water in response to head changes in adjacent aquifer material, (2) of insufficient lateral extent to be considered a confining bed (or confining unit) that separates adjacent aquifers, and (3) of relatively small thickness in comparison to lateral extent. The interbeds are also assumed to consist primarily of highly compressible clay and silt beds from which water flows vertically to adjacent coarse-grained beds.

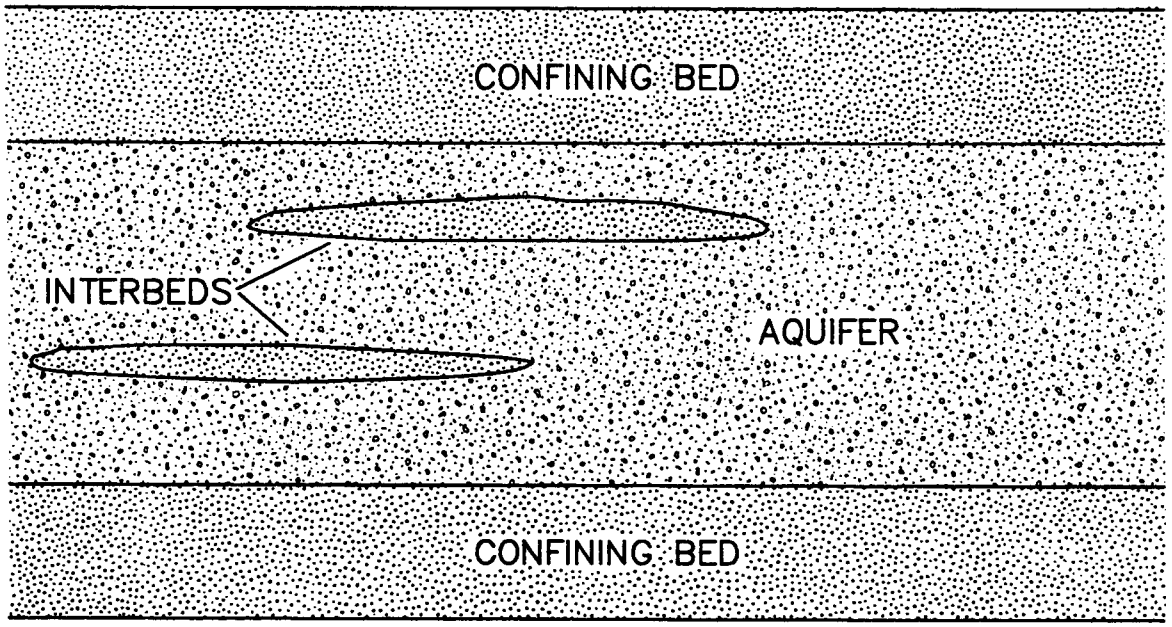


Figure 1.—Types of fine-grained beds in or adjacent to aquifers. Beds may be discontinuous interbeds or continuous confining beds.

Basic Model of Interbed Compaction

Relation of Ground-Water Head Change and Sediment Compaction

The relation between ground-water head change and sediment compaction is based on the principle of effective stress developed by Terzaghi (1925), where effective stress (p') is expressed as the difference between total stress (p), which is the total overburden load or geostatic pressure, and fluid or more pressure (u):

$$p' = p - u. \quad (1)$$

Effective stress, geostatic pressure, and fluid pressure are commonly expressed in units of force per unit area [FL^{-2}] but may also be expressed as a height of a column of water [L]. That unit may be obtained by dividing quantities of force per unit area by the unit weight of water, γ_w [FL^{-3}].

The geostatic pressure or total overburden load of the sediments and water at a given depth equals the product of the unit weight of moist sediments and the thickness of the unsaturated zone plus the product of the unit weight of saturated sediments below the water table and the thickness of the saturated sediments overlying that depth. Because of the dependency of geostatic stress on the position of the water table, a change in effective stress from a given head change generally is different in confined and unconfined aquifers. If the water table is raised or lowered in an unconfined aquifer, the geostatic pressure will change. The resulting change in effective stress in the unconfined aquifer can be expressed as (Poland and Davis, 1969, p. 195):

$$\Delta p' = -\gamma_w (1 - n + n_w) \Delta wt, \quad (2)$$

where

- Δp is change in effective stress, positive for increase and negative for decrease [FL^{-2}];
- n is porosity [dimensionless];
- n_w is moisture content above water table as a fraction of total volume [dimensionless]; and
- Δwt is change in water table, positive for raising and negative for lowering of the water table [L].

The lowering of head in a confined aquifer however does not change the geostatic pressure, assuming the overlying water table remains constant, but results in an increase of equal amount in effective stress. The increase in effective stress for a given change in head can be expressed as (Poland and Davis, 1969, p. 195):

$$\Delta p' = -\gamma_w \Delta h, \quad (3)$$

where

- Δh is change in head in a confined aquifer, positive for increase and negative for decrease in head [L].

The change in effective stress is less for an unconfined aquifer than for a confined aquifer by the relation ($1-n+n_w$). For example, if n=0.3 and n_w=0.1, the lowering of the water table increases effective stress (p') by a fraction of 0.8 of the corresponding value for an equal lowering of the head in a confined aquifer. This example assumes that the water table that overlies the confined aquifer remains constant. The maximum increase in effective stress for a confined aquifer occurs when the water table rises in an overlying unconfined aquifer while the head declines in a confined aquifer (Lofgren, 1968, p. 225).

Previous studies (Riley, 1969; Helm, 1975) have indicated that elastic compaction or expansion of sediments is proportional or nearly proportional to change in effective stress. That relation can be expressed as

$$\Delta b = \frac{\Delta p'}{\gamma_w} S_{ske} b_o, \quad (4)$$

where

- Δb is change in thickness, positive for compaction and negative for expansion [L];
- S_{ske} is skeletal component of elastic specific storage [L^{-1}]; and
- b_o is the thickness of the interbed [L].

The product of elastic skeletal specific storage and thickness is the skeletal component of elastic storage coefficient, S_{ke} [dimensionless]. For sediments in confined aquifers in which geostatic pressure is constant, the relation between change in head and change in thickness is

$$\Delta b = -\Delta h S_{ske} b_o. \quad (5)$$

Equations 4 and 5 generally are applicable to both fine- and coarse-grained sediments. Laboratory consolidation tests and field data indicate

that when compressible fine-grained sediments are stressed beyond a previous maximum stress, compaction is permanent (inelastic) and nonrecoverable (Jorgenson, 1980, p. 20; Riley, 1969). Compaction per unit increase in effective stress in the inelastic range is considerably greater than in the elastic range. When effective stress of sediments compacting in the inelastic range is reduced, the sediments again expand and compact with the elastic characteristics until effective stress increases beyond the new maximum effective stress.

Approximate inelastic compaction can be related to increase in effective stress with an expression analogous to equation 4:

$$\Delta b^* = \frac{\Delta p'}{\gamma_w} S_{skv} b_o, \quad (6)$$

where

Δb^* is approximate inelastic compaction [L] and

S_{skv} is the skeletal component of inelastic specific storage [L^{-1}].

For confined aquifers in which geostatic pressure is constant, the expression analogous to equation 5 is

$$\Delta b^* = \Delta h S_{skv} b_o. \quad (7)$$

Expressions similar to equations 4 and 6 were used by Helm (1975) in development of a one-dimensional compaction model. Meyer and Carr (1979) and Williamson and others (1989) also used analogous relations in regional subsidence models of the Houston area, Texas, and the Central Valley, California, respectively. According to laboratory consolidation tests, inelastic compaction is more nearly proportional to increase in log of effective stress (Jorgenson, 1980, p. 20-21). The validity of assuming that inelastic compaction is proportional to increase in effective stress is discussed in the section entitled "Assumptions and Limitations."

Incorporating Storage Changes into Ground-Water Flow Equation

Storage changes in a saturated aquifer system under conditions of decreasing water levels result from three primary processes: (1) the draining of pore spaces as the water table declines; (2) the compression of the aquifer skeleton caused by increasing effective stresses in the aquifer below the water table; and (3) the expansion of water caused by decreasing fluid pressures. Compression of the aquifer skeleton is elastic if it does not result in a permanent rearrangement of the individual grains. Water removed from storage by this process can be replaced when heads in the aquifer increase. Conversely, compression of the aquifer skeleton is inelastic if it results in a permanent rearrangement of the grains. Water removed from storage by inelastic compaction cannot be replaced when heads in the aquifer increase. In general, layers of compressible clayey beds are the most susceptible to inelastic compaction. Storage changes caused by elastic compression of the aquifer skeleton and expansion of the water are much smaller than those caused by the draining of the pore spaces or by inelastic compaction.

The ground-water flow model of McDonald and Harbaugh (1988) is designed to account for changes in storage caused by water-table fluctuations, elastic compression of the aquifer skeleton, and expansion of water. Their model can be used to simulate inelastic compaction but considerable effort would be required to select and manually specify the proper storage values for given periods of time. The Interbed-Storage Package adds the capability of accounting for storage changes from inelastic compaction of interbeds within an aquifer and the resultant land subsidence without having to manually change the storage value during the model simulation.

Equations that describe ground-water flow include terms that account for flow into or out of storage in response to head changes. The general form of the flow equation used by McDonald and Harbaugh (1988) is

$$\frac{\partial}{\partial x}\left(K_{xx}\frac{\partial h}{\partial x}\right)+\frac{\partial}{\partial y}\left(K_{yy}\frac{\partial h}{\partial y}\right)+\frac{\partial}{\partial z}\left(K_{zz}\frac{\partial h}{\partial z}\right)-W=S_s\frac{\partial h}{\partial t}, \quad (8)$$

where

x, y, z are cartesian coordinates, aligned along the major axes of the hydraulic conductivity tensor [L];

K_{xx}, K_{yy}, K_{zz} are principal components of the hydraulic conductivity tensor [LT^{-1}];

h is hydraulic head [L];

W is volumetric flux per unit volume of sources and (or) sinks of water [T^{-1}];

S_s is specific storage of aquifer material [L^{-1}]; and

t is time [T].

The term on the right-hand side of this equation describes the flow rate of water into storage in the aquifer per unit volume. The storage term can vary over the modeled area depending on the storage properties of the aquifer materials but the values are assumed constant in time, except for the case where a confined aquifer becomes water table or vice versa (McDonald and Harbaugh, 1988, chap. 5, p. 26). To account for changes in storage caused by compaction of fine-grained interbeds, an additional term has been added to the ground-water flow equation. Assuming that changes in head result in equal but opposite changes in effective stress, the term can be expressed as

$$q_i = S'_{sk} \frac{\partial h}{\partial t}, \qquad (9)$$

where

q_i is rate of flow per unit volume of water flowing into storage in compressible interbeds [T^{-1}] and

S'_{sk} is skeletal component of specific storage of interbeds, a function of previous maximum effective stress [L^{-1}].

Equation 9 represents a volume average flow into or out of storage per unit volume of interbeds.

The skeletal specific-storage value in equation 9 varies between an elastic and an inelastic value depending on the relation of the head in the cell to the preconsolidation head. The preconsolidation head corresponds to the previous maximum effective stress (previous lowest head). The term S'_{sk} is the elastic skeletal specific storage whenever the head in a cell is greater than the preconsolidation head. The term is the inelastic skeletal specific storage whenever the head in a cell is less than the preconsolidation head. The preconsolidation head can also change during the simulation as it is assigned the most recent lowest head value.

For an aquifer with multiple interbeds of differing specific-storage values, a single elastic and a single inelastic skeletal specific-storage value can be determined to account for changes in storage in all the interbeds. For a system with n interbeds with specific-storage values S_{s1}, S_{s2}, ... S_{sn} and with thicknesses b_1, b_2, ... b_n, a single equivalent specific-storage value, $S_{s\ system}$, is given by Jorgenson (1980, equation 69):

$$S_{s\ system} = \frac{S_{s1}b_1 + S_{s2}b_2 + \ldots S_{sn}b_n}{b_1 + b_2 + \ldots b_n}. \qquad (10)$$

Individual thicknesses and elastic skeletal specific-storage values of the interbeds can be combined by equation 10 to compute a single elastic specific-storage value for the S'_{sk} function. Similarly, the thicknesses and inelastic specific-storage values can be combined by equation 10 to compute a single inelastic specific-storage value.

Storage changes related to the compressibility of water in interbeds are not included in the q_i function. The changes are small in relation to changes resulting from inelastic skeletal compressibility and generally can be neglected without significantly affecting estimates of ground-water flow and subsidence. Storage changes from compression or expansion of water in interbeds, however, can be accounted for in the model. To do so, the product of the component of specific storage from compressibility of water, S_{sw}, and thickness of the interbeds is added to the storage coefficient read into the Block-Centered Flow Package (McDonald and Harbaugh, 1988).

The computer program of McDonald and Harbaugh (1988) requires specification of dimensionless storage-coefficient values for each layer. The value for a confined layer is the specific storage assigned to a model cell multiplied by the thickness of the cell. The storage coefficient is then multiplied by the cell area in the program to create a storage capacity. The storage-capacity value for each cell is used in the basic finite-difference equation for ground-water flow. The program described in this report also requires specification of the storage coefficient. Storage coefficients for individual layers of sediments can be combined into an equivalent value in a manner analogous to combining specific-storage values in equation 10. The expression for computing the equivalent storage coefficient, S_{system}, is given by Jorgenson (1980, equation 67) as

$$S_{system} = S_{s1}b_1 + S_{s2}b_2 + \ldots S_{sn}b_n. \qquad (11)$$

The skeletal storage coefficients (elastic and inelastic) are used to estimate elastic and inelastic components of subsidence in addition to the flow of water into and out of storage. The dimensionless storage coefficient is the constant of proportionality between compaction and head change (equations 5, 7). This proportionality is used in the calculation of storage changes and compaction in the computer program.

Regional subsidence models by Meyer and Carr (1979) and Williamson and others (1989) used approaches similar to that outlined in equation 9 to account for storage changes in com-

pressible sediments. In approximating the storage term in the finite-difference equations, they computed the storage function explicitly from the relation of head to preconsolidation head at the end of a time step in a manner analogous to

$$q_i^m = \frac{S_{sk}^m}{\Delta t}\left(h^m - h^{m-1}\right) \qquad (12)$$

$$S_{sk}^m = \begin{cases} S_{ske}, h^{m-1} > H^{m-1} \\ S_{skv}, h^{m-1} \le H^{m-1} \end{cases},$$

where

q_i^m is rate of flow per unit volume of water flowing into storage in compressible interbeds during time step m [T^{-1}];
S_{sk}^m is specific storage for time step m [L^{-1}];
Δt is length of time step [T];
h^m is head at finite-difference node at end of time step m (h^0 is starting head [L]), and
H^m is preconsolidation head at node at end of time step m (H^0 is starting preconsolidation head [L]).

In this explicit method, if the computed head is less than the preconsolidation head at the end of the step, an inelastic storage value is selected for the next time step and the preconsolidation head value is reset to the new lowest head. Similarly, if computed head at the end of a time step is greater than the preconsolidation head, an elastic storage value would be selected for the next time step.

The explicit method is relatively easy to formulate in a ground-water model program but may lead to significant errors unless time steps are small. The problem arises from the large differences between the elastic and inelastic storage values. When computed head values are in the elastic range but are declining toward the preconsolidation head, storage values are relatively small and head-decline rates may be relatively large. Consequently, the computed head values may overshoot the preconsolidation head value by a significant amount. This overshoot causes the new preconsolidation head value to be set too low and may result in incorrect calculations of water released from compressible sediments. The problem is most significant in systems with cyclical head fluctuations that result in repeated switching between the elastic and inelastic regimes. In recognition of this problem, Williamson and others (1989) made the model time steps as small as was feasible with the available computing resources.

For the Interbed-Storage Package, a more complex but more accurate method of approximating q_i is used. The method implicitly selects the appropriate specific storage at the end of a time step and apportions storage changes to either elastic, inelastic, or both components within a time step. The finite-difference expression of the implicit method is

$$q_i^m = \frac{S_{sk}^m}{\Delta t}\left(h^m - H^{m-1}\right) + \frac{S_{ske}}{\Delta t}\left(H^{m-1} - h^{m-1}\right), \qquad (13)$$

$$S_{sk}^m = \begin{cases} S_{ske}, h^m > H^{m-1} \\ S_{skv}, h^m \le H^{m-1} \end{cases}.$$

With this formulation, time steps need not be particularly small. The correct value of specific storage at the end of a time step is easily determined within the iterations of solution schemes provided in the model by McDonald and Harbaugh (1988).

Compaction during a time step can be computed by multiplying equation 13 by the length of the time step, Δt, and by interbed thickness, b_0. That rearrangement of equation 13 results in an expression for total compaction that is equivalent to the sum of expressions for elastic and inelastic compaction in equations 5 and 7.

In addition to using the preceding formulation to simulate release of water and compaction of thin interbeds, it could be used to simulate compaction of thicker confining beds (fig. 1). That simulation could be done by representing the confining layer with a number of adjacent finite-difference layers, each with the formulation for the nonlinear q_i function. The degree to which the vertical head distribution can be approximated is dependent on the number of finite-difference layers used to simulate the confining layer. Several finite-difference layers may be required to simulate the vertical head distribution. Although this approach is valid, it does not make efficient use of computer resources because the model would solve for horizontal components of flow and store horizontal-conductance values for layers in which flow is dominantly vertical.

Assumptions and Limitations

The Interbed-Storage Package does not consider changes in effective stress caused by

changes in geostatic pressure but assumes that changes in effective stress are a function only of the head changes. Permanent compaction may be overestimated for an unconfined aquifer unless estimates of inelastic storage coefficients are decreased in proportion to the amount that effective stress would be overestimated. However, the specific yield in an unconfined aquifer generally is greater than the inelastic storage coefficient unless the total thickness of the interbeds exceeds 300 meters. In this case it may be better to model the thick unconfined aquifer with more than one model layer. In a confined aquifer overlain by an unconfined aquifer, the method will overestimate effective stress if the water table declines in the unconfined aquifer or underestimate effective stress if the water table increases in the unconfined aquifer. The error in the estimate of effective stress depends on the amount of change in the geostatic pressure.

In adding the function q_i to the ground-water flow equation, the assumption is made that head changes within a model time step in aquifer material also take place throughout fine-grained interbeds. If time steps are not large enough to allow for drainage of all excess pore pressure in the interbeds, the flow from the interbeds will be unrealistically large for the time step. A measure of time required for excess pore pressure to dissipate in a doubly draining interbed following an instantaneous step load is given by Riley (1969, equation 1):

$$\tau = \frac{S_s (b_0/2)^2}{K}, \tag{14}$$

where

τ is time constant of interbed [T];
S_s is specific storage of interbed [L^{-1}];
b_0 is thickness of interbed [L], and
K is hydraulic conductivity of interbed [LT^{-1}].

The time constant τ is the time required for 93 percent of excess pore pressure to dissipate following an instantaneous change in head at the boundaries of the interbed. The variable S_s in equation 14 includes components that account for the compressibility of the sediment skeleton and of water; however, the inelastic skeletal specific storage, S_{skv}, may be substituted into the expression to obtain an estimate of the upper limit of τ. The resulting time constant will apply when the entire layer is compacting inelastically. For that situation, the component of specific storage that accounts for the compressibility of water is negligible. The upper limit of τ can be compared with the length of model time steps to evaluate the validity of the assumption that all excess pore pressure is dissipated in a time step. If this assumption is not valid, the model results may overestimate storage changes and compaction in early time and underestimate those quantities in later time.

Another assumption mentioned previously is that inelastic compaction is proportional to change in effective stress. Laboratory consolidation tests indicate that inelastic compaction of many fine-grained sediments is proportional to the increase in logarithm of effective stress. A relation of compaction to increase in logarithm of effective stress can be derived from equations 45a and 50 in Jorgenson (1980). That relation is

$$\Delta b' = b_0 C_c \frac{(\Delta \log_{10} p')}{(1+e_0)}, \tag{15}$$

where

$\Delta b'$ is inelastic compaction [L];
b_0 is initial thickness of interbeds [L];
C_c is compression index [dimensionless]; and
e_0 is initial void ratio [dimensionless].

The skeletal component of inelastic specific storage is not a constant but rather is a function of effective stress. Jorgenson (1980, equation 59) relates the two quantities by

$$S_{skv} = \frac{0.434 C_c \gamma_w}{p'(1+e_0)}. \tag{16}$$

The error in assuming that S_{skv} is a constant can be expressed as

$$\text{error} = 100 \frac{(\Delta b^* - \Delta b')}{\Delta b'}, \tag{17}$$

where

error is the percentage by which compaction will be overestimated by assuming S_{skv} is constant [dimensionless].

If S_{skv} is assumed to be a constant selected on the basis of the initial state of effective stress, p'_0, equations 6, 15, 16, and 17 can be combined to express the percentage error in computed compaction:

$$\text{error} = 100 \left(\frac{0.434\, \Delta p'}{p'_0 \Delta \log_{10} p'} - 1 \right). \tag{18}$$

This relation indicates that the percentage error in computed compaction is less than one-half of the percentage increase in effective stress (fig. 2). The analysis points out the merit of application of equation 1 to estimate initial effective stress and changes in effective stress from changes in pore pressure or head. For sediments relatively deep below the land surface, a given decline in head will result in a smaller percentage increase in effective stress than for shallower sediments.

For many ground-water flow systems, increases in effective stress are a relatively small percentage of the initial state of stress. Additionally, errors in compaction computed using equation 7 can be minimized by selecting the constant S_{skv} on the basis of an effective stress in the center of the range of stress change, rather than at the beginning. Alternatively, specific-storage values used in the package can be changed with time, if necessary, by restarting a simulation with new storage values.

Applicability of the Interbed-Storage Package

The Interbed-Storage Package is designed for simulation of ground-water flow in aquifers where permanent compaction of compressible interbeds may be a significant source of pumped water. The amount of permanent compaction of compressible interbeds depends on a number of factors including the total stress borne by the system (preconsolidation stress); variations in the amount, compressibility, and bedding of clayey or silty interbeds; and variations in the applied stress that tends to compact the interbeds. The package is also designed to estimate the amount of land subsidence caused by compaction. In general, the package would not be applied for flow systems in which water-level declines do not cause the preconsolidation stress to be exceeded, unless there is interest in calculating elastic compaction and (or) rebound.

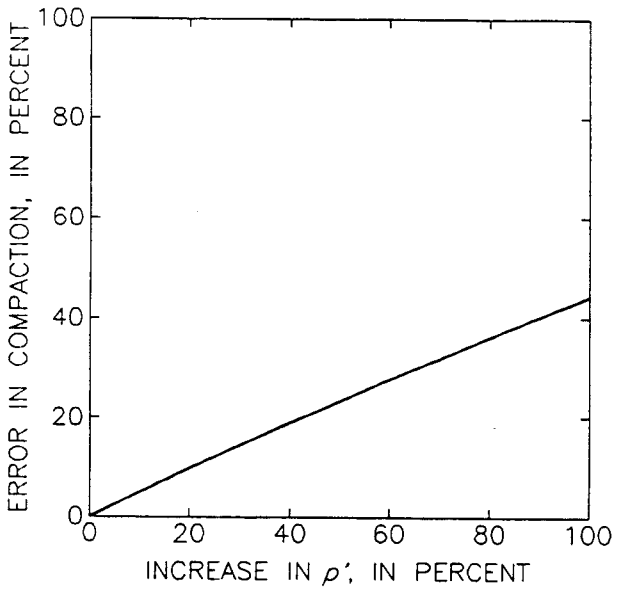


Figure 2.—Relation between increase in effective stress, p', and error in computed compaction. Error is the amount by which computed compaction will be overestimated if inelastic skeletal specific storage is assumed to be a constant selected on the basis of the initial state of effective stress.

Holzer (1981, p. 693) estimated the preconsolidation stress in four areas where permanent compaction was caused by pumping of ground water. The estimated stresses are equivalent to water-level declines of 16 to 63 m below the predevelopment water level. Bull (1975) noted that the interlayering of numerous thin-bedded compressible clayey sediments with coarse-grained sediments resulted in a system that compacted more rapidly than one with fewer but thicker beds of clayey sediments. The amount of compaction also is dependent on the type of clay minerals present. Meade (1967) noted that montmorillonite is more susceptible to compact than either illite or kaolinite and was the predominant clay mineral in the major subsiding areas in the San Joaquin Valley of California.

As ground-water reservoirs in the United States and elsewhere continue to be developed, permanent compaction of sediments will undoubtedly become an increasing problem. Poland (1981, p. 116) identified 17 areas in the United States where ground-water withdrawal has resulted in permanent compaction of sediments and accompanying land subsidence. In several alluvial basins in California and Arizona and in the coastal plain around Houston, Texas, compaction has resulted in more than 3 m of land subsidence. Subsequent investigation by Chi and Reilinger (1984, p. 155) identified more than 30 new areas in the United States where land subsidence may have been caused by ground-water pumping. Their identifications were made on the basis of repeated level surveys and ground-water pumping histories. Many of the new areas identified are along the Mississippi Valley. They also suggested that land subsidence may have occurred in additional ground-water basins but remains undetected because of a lack of repeated levelings.

Test Problems

Storage-Depletion Test Problem

A simulation was run to test the results of the Interbed-Storage Package given a specified uniform decline in head. The finite-difference grid consists of two layers with 10 rows and 12 columns of cells. The first and last column of cells in both layers are constant-head boundaries, so that the other 10 active columns of cells are a flow region subject to head and storage changes. The grid dimensions for each cell are 1,000 m along both rows and columns. Each layer has a transmissivity of 1,000 m^2/d and a storage coefficient of 0.0001. The starting head in the active area specifies a uniform gradient of 0.001 along the rows. The starting head values in the constant-head columns also specify a gradient of 0.001 along the rows over the entire grid but at a level exactly 10 m lower than the starting heads in the active area of the model. Thus, if the transient solution is allowed to run until steady-state conditions are reached, the head at each interior cell will be exactly 10 m below the corresponding starting head.

For this test problem, compressible interbeds are assumed to occur in the aquifer represented by the upper layer. The preconsolidation head is assumed to be 5 m below the starting head at each cell. The sum of the products of thickness and elastic skeletal specific storage for interbeds in the upper layer is assumed to result in an elastic storage coefficient of 0.0001 at each cell. Similarly, the inelastic storage coefficient is assumed to be 0.001 at each cell. Data sets for this test problem, referred to as the storage-depletion test problem, including input for all model packages, are given in Appendix A.

Within 1,000 days of simulation time, head values at all cells will be within about 0.001 m of the final steady-state value. The volumetric budget for the simulation is shown in table 1. The correct volume of water released by the interbeds can be computed as the product of total area excluding boundary cells ($1x10^8$ m^2), storage coefficient (0.0001 for elastic and 0.001 for inelastic), and head decline (5 m under elastic conditions and 5 m under inelastic conditions). The correct amount of water released from interbed storage therefore is $5x10^4$ m^3 for declines in the elastic range and $5x10^5$ m^3 for declines in the inelastic range. The sum of the two values is consistent with the $5.5x10^5$ m^3 release of water from interbed storage calculated by the model (table 1). For this test problem, the model computes $5.5x10^{-3}$ m of compaction. That value is consistent with the sum of elastic and inelastic compaction computed from equations 5 and 7.

Ramp-Load Test Problem

Another more complex test problem involves comparing the results of the Interbed-Storage Package with those of one-dimensional compaction model COMPAC1 (Helm, 1975, 1984). The problem is similar to the non-declining ramp-load problem given by Helm (1975, p. 470-473). Model COMPAC1 computes compaction of compress-

Table 1.—*Volumetric budget for storage-depletion test problem at end of time step 10 in stress period 3*

CUMULATIVE VOLUMES L**3	RATES FOR THIS TIME STEP L**3/T
IN:	
STORAGE = 0.20000E+06	STORAGE = 0.00000
CONSTANT HEAD = 0.59683E+08	CONSTANT HEAD = 20000.
INTERBED STORAGE = 0.55000E+06	INTERBED STORAGE = 0.00000
TOTAL IN = 0.60433E+08	TOTAL IN = 20000.
OUT:	
STORAGE = 0.25808E-01	STORAGE = 0.00000
CONSTANT HEAD = 0.60433E+08	CONSTANT HEAD = 20000.
INTERBED STORAGE = 0.43711E-01	INTERBED STORAGE = 0.00000
TOTAL OUT = 0.60433E+08	TOTAL OUT = 20000.
IN - OUT = -104.00	IN - OUT = -0.35156E-01
PERCENT DISCREPANCY = 0.00	PERCENT DISCREPANCY = 0.00

ible beds given effective stress changes at the boundary by solving a one-dimensional partial-differential equation by finite differences. The ramp-load test problem presented here involves cumulative compaction of 100 identical interbeds with the following properties:

Property	*Value*
Thickness (m)	10
Hydraulic conductivity (m/yr)	$1x10^{-3}$
Elastic specific storage (m^{-1})	$1x10^{-6}$
Inelastic specific storage (m^{-1})	$1x10^{-4}$

With these properties, the elastic and inelastic time constants are 0.025 and 2.5 years, respectively. The head at the boundaries of the interbeds is set to successively decline linearly for 180 days to 10 m below the starting value. Following the declines, the head recovers linearly for 180 days to the original value. Successive cycles of declines and recoveries are approximated with 180 1-day steps in each ramp (fig. 3*A*). The preconsolidation head throughout the interbeds is assumed to be equal to the starting head so that compaction is initially in the inelastic range.

The problem can be solved by COMPAC1 and by the Interbed-Storage Package by simulating compaction in one-half of a single doubly draining bed and multiplying those results by 200 to obtain the total compaction for all 100 interbeds. For model COMPAC1, a grid spacing of 0.2 m was selected, thus requiring 25 finite-difference cells to simulate one-half of a doubly draining bed. To apply the Interbed-Storage Package to a problem of flow and compaction in a confining bed, one approach would be to discretize the confining bed into a number of model layers. Flow in the confining bed is assumed to be one-dimensional; therefore, the problem could also be solved with a single row of finite-difference cells in one model layer. The single-row approach is less awkward than the multiple-layer approach because input to and output from the flow model generally are carried out layer by layer. A disadvantage of orienting the grid along a row is that the Interbed-Storage Package sums compaction in all layers to compute total compaction. The package is incapable of summing compaction along a row to calculate total compaction; however, the total compaction can be obtained by dividing the net cumulative volume of water released from interbed storage by the cross-sectional area of the grid. The finite-difference grid for the Interbed-Storage Package is shown in figure 4. A cross-sectional area of 1 m by 1 m was chosen so that the net volume of water from interbed storage in the volumetric budget of the model would directly relate to total compaction for half of the doubly draining bed. A cell dimension of 0.25 m was selected thereby requiring 20 finite-difference cells to represent one-half of the doubly draining bed. An additional cell was required to impose the specified-head boundary condition of cyclical ramp loading.

Computed interbed-storage values from the volumetric budget were converted to total compaction for all 100 layers for comparison with the results of model COMPAC1 (fig. 3*B*). The results of the two models are virtually the same. The input for all model packages for the ramp-load test problem is given in Appendix B.

In the ground-water model by McDonald and Harbaugh (1988), the only provision for changing specified head at boundary cells is by use of the River Package, Drain Package, or General-Head Boundary Package. Those packages allow specification of a different boundary head at each stress period. If associated river-bed conductances are large enough, the head in the aquifer will almost equal the boundary head. For the ramp-load test problem, difficulties may arise from using these packages to approximate a specified aquifer head. First, the flow from the boundary to the aquifer is computed as the product of conductance and head difference between the boundary and the aquifer. If the intent is to make that difference small, mass-balance errors result because of imprecision in computing the head difference across the boundary. Precision is lost when two nearly equal numbers are subtracted. Second, a large amount of input to and output from the model are associated with each stress period. Simulation of one complete cycle of the loading to the precision shown in figure 3*A* would require 360 stress periods. In an attempt to use the River Package to simulate one complete cycle of loading with 1-day stress periods, more than 28,000 lines of output were generated by the model.

As an alternative way of simulating a specified boundary head, a new model package was developed to simulate variable-head boundaries. The new package is named the Time-Variant Specified-Head Package. It makes use of the capability of the model to incorporate constant-head cells in a simulation. At the start of a simulation, the Time-Variant Specified-Head Package reads in the maximum number of constant-head cells that may be specified for any stress period.

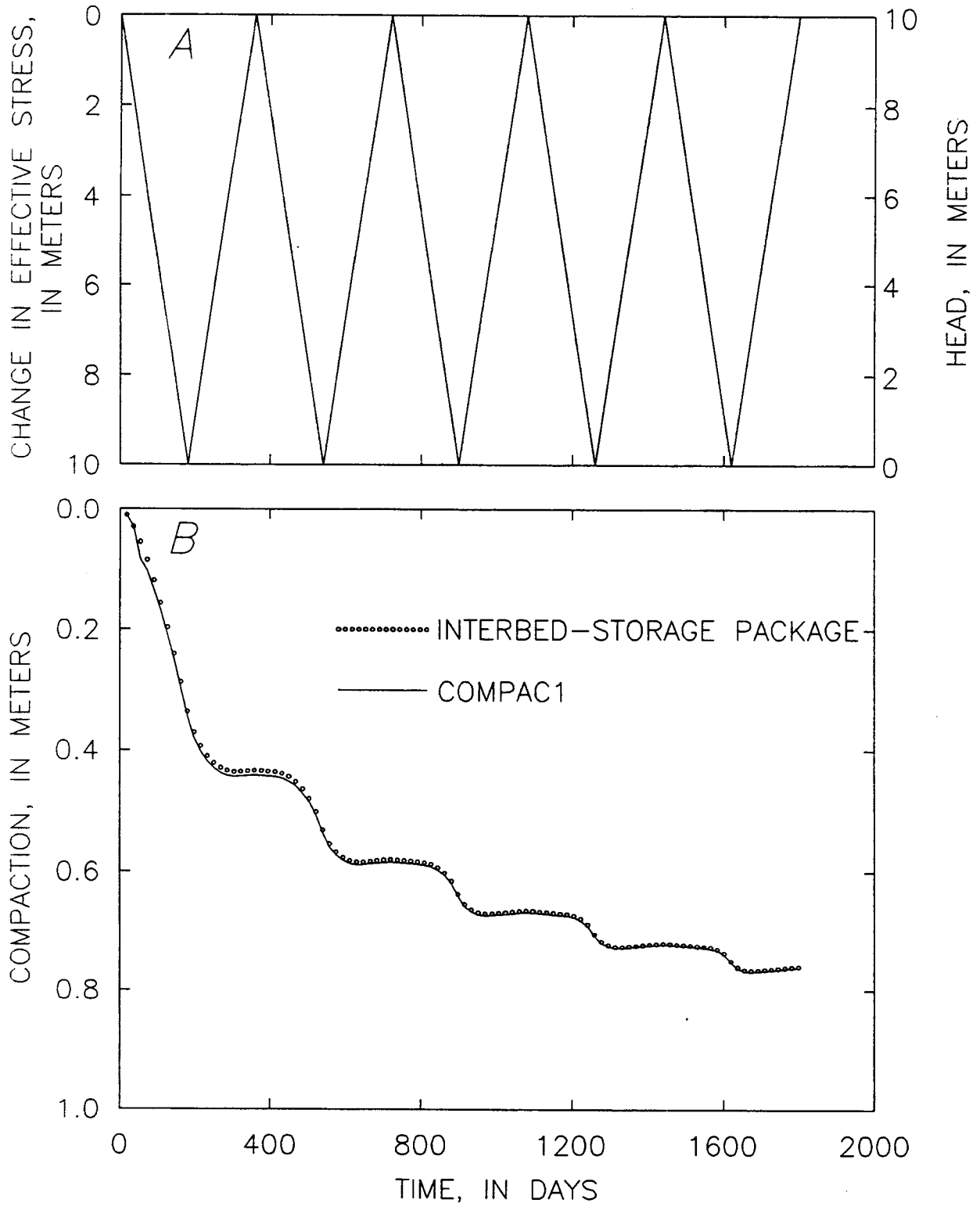


Figure 3.—Stress and compaction for the ramp-load test problem. *A*, Non-declining cyclical ramp load. *B*, Compaction computed by the Interbed-Storage Package and by model COMPAC1 (Helm, 1975, 1984).

For each stress period, the package reads the number of boundary cells at which head will be specified. For each boundary cell, the package reads layer, row, and column indices and corresponding head values at the start and end of the stress period. For each time step within the stress period, the value of each specified head is linearly interpolated from the starting and ending heads of the period and the proportion of elapsed time within the stress period to length of the stress period. The package sets the corresponding elements in the IBOUND array to negative numbers to indicate that the cells are constant-head cells. The Block-Centered Flow Package of the model (McDonald and Harbaugh, 1988) accounts for flow to or from the constant-head cells as though they were specified as such by the Block-Centered Flow Package. The data-input instructions and listings of subroutines of the Time-Variant Specified-Head Package are given in Appendix C.

Implementation of Interbed-Storage Package in the Ground-Water Model

The Interbed-Storage Package is designed for incorporation into the ground-water flow

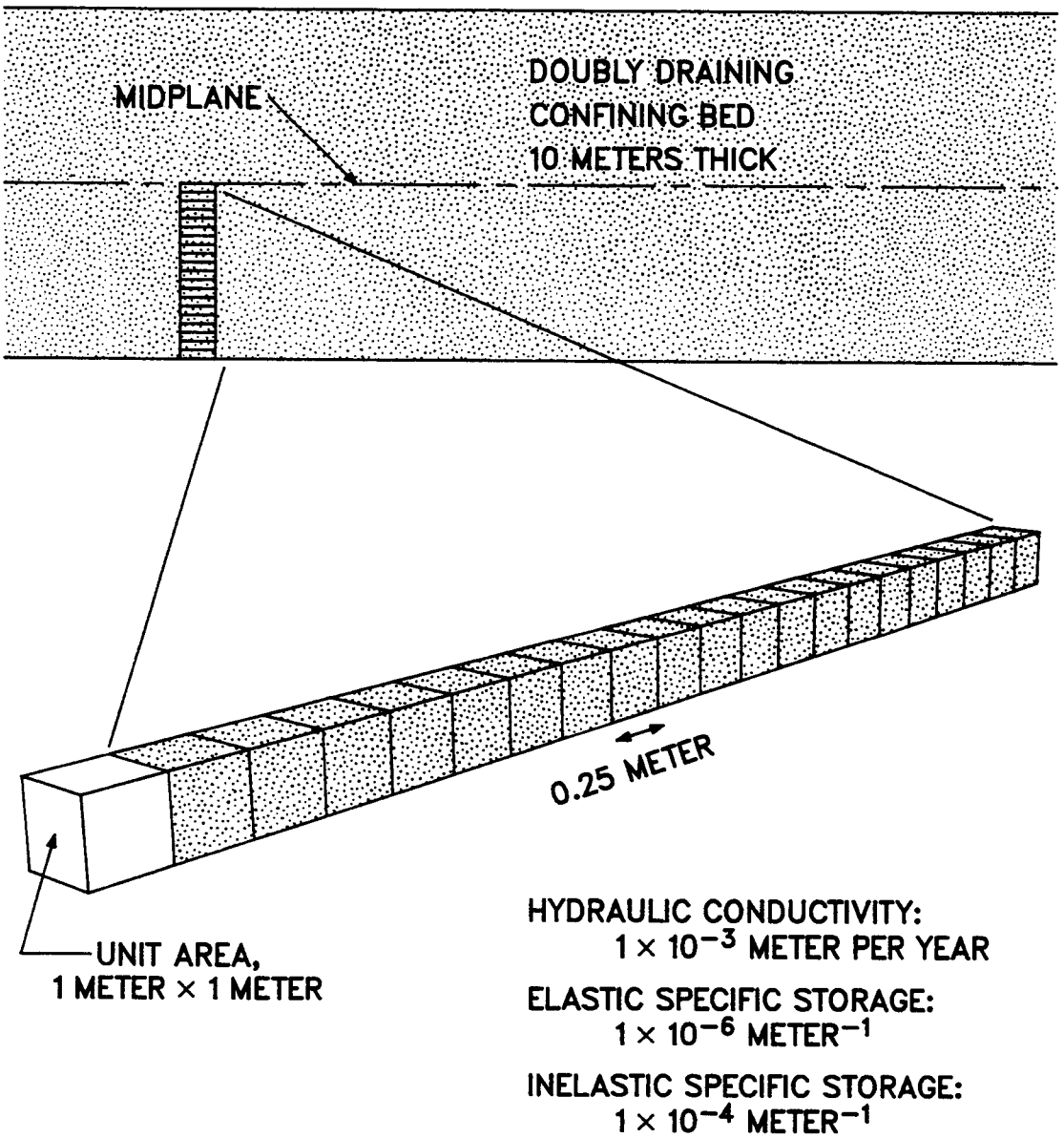


Figure 4.—Finite-difference grid used by the Interbed-Storage Package for the ramp-load test problem.

model by McDonald and Harbaugh (1988). It consists of five FORTRAN subroutines, referred to as modules, that carry out the following procedures: Allocate memory, read and prepare data, formulate finite-difference equations, compute mass-balance components, and output compaction arrays. The modules are named IBS1AL, IBS1RP, IBS1FM, IBS1BD, and IBS1OT, respectively. The first three characters of the names identify the modules as being a part of the Interbed-Storage Package. The next character identifies the version number of the package. The last two characters identify the procedure carried out by the module; for example, "AL" for allocate memory, "RP" for read and prepare data. The procedures used by the Interbed-Storage Package are consistent with procedures used by existing packages in the ground-water flow model (fig. 5). Detailed descriptions of each of the five modules are presented in a following section entitled "Module Documentation for the Interbed-Storage Package."

The main program of the ground-water model must be modified to call the five modules of the Interbed-Storage Package. The procedures on the left side of figure 5 are listed in the order that procedures are carried out in the main program of the model. Calls to modules of the Interbed-Storage Package must be placed in sections of the main program in which the particular procedure is being carried out for other packages. For instance, the IBS1RP module must be called within the section of the main program in which other read and prepare modules such as BAS1RP and BCF1RP are called. The FORTRAN calls to be added to the main program are as shown below. An example of a main program that includes the above calls to the Interbed-Storage Package modules is given in Appendix D.

The package is selected by entering a positive unit number in element 19 of the IUNIT array in the basic package input of the model (McDonald and Harbaugh, 1988, chap. 4, p. 9). If a different element of the IUNIT array is desired for selecting the package, each of the preceding calls to the five modules must be modified so that "19" is changed to the desired element number. The reference to IUNIT(19) occurs in each of the five IF statements as well as within the argument lists of calls to modules IBS1AL, IBS1RP, and IBS1OT.

The Interbed-Storage Package allows a user to select which model layers are to include calculations to account for release of water from compressible interbeds. For each layer with interbed storage, the user must enter several two-dimensional arrays that represent areal distributions of properties within the model layer. The arrays are elastic storage coefficient [dimensionless], inelastic storage coefficient [dimensionless], and initial preconsolidation head [L]. Storage coefficient for a model cell is the sum of the products of specific-storage values and thicknesses of all -compressible interbeds within the cell (equation 11).

```
      IF (IUNIT(19).GT.0) CALL IBS1AL(ISUM,LENX,LCHC,LCSCE,LCSCV,
     1    LCSUB,NCOL,NROW,NLAY,IIBSCB,IIBSOC,ISS,IUNIT(19),IOUT)

      IF(IUNIT(19).GT.0) CALL IBS1RP(X(LCDELR),X(LCDELC),X(LCHNEW),
     1  X(LCHC),X(LCSCE),X(LCSCV),X(LCSUB),NCOL,NROW,NLAY,
     2  NODES,IIBSOC,ISUBFM,ICOMFM,IHCFM,ISUBUN,ICOMUN,IHCUN,
     3  IUNIT(19),IOUT)

      IF(IUNIT(19).GT.0) CALL IBS1FM(X(LCRHS),X(LCHCOF),X(LCHNEW),
     1  X(LCHOLD),X(LCHC),X(LCSCE),X(LCSCV),X(LCIBOU),
     2  NCOL,NROW,NLAY,DELT)

      IF(IUNIT(19).GT.0) CALL IBS1BD(X(LCIBOU),X(LCHNEW),X(LCHOLD),
     1  X(LCHC),X(LCSCE),X(LCSCV),X(LCSUB),X(LCDELR),X(LCDELC),
     2  NCOL,NROW,NLAY,DELT,VBVL,VBNM,MSUM,KKSTP,KKPER,IIBSCB,
     3  ICBCFL,X(LCBUFF),IOUT)

      IF(IUNIT(19).GT.0)CALL IBS1OT(NCOL,NROW,NLAY,PERTIM,TOTIM,KKSTP,
     1  KKPER,NSTP,X(LCBUFF),X(LCSUB),X(LCHC),IIBSOC,ISUBFM,ICOMFM,
     2  IHCFM,ISUBUN,ICOMUN,IHCUN,IUNIT(19),IOUT)
```

PACKAGES

PROCEDURES	BAS	BCF	WEL	RCH	RIV	DRN	EVT	GHB	SIP	SOR	IBS
Define (DF)	BAS1DF										
Allocate (AL)	BAS1AL	BCF1AL	WEL1AL	RCH1AL	RIV1AL	DRN1AL	EVT1AL	GHB1AL	SIP1AL	SOR1AL	IBS1AL
Read & Prepare (RP)	$BAS1RP_U$	$BCF1RP_{US}$							SIP1RP	SOR1RP	$IBS1RP_U$
Stress (ST)	BAS1ST										
Read & Prepare (RP)			WEL1RP	$RCH1RP_U$	RIV1RP	DRN1RP	$EVT1RP_U$	GHB1RP			
Advance (AD)	BAS1AD										
Formulate (FM)	BAS1FM	$BCF1FM_S$	WEL1FM	RCH1FM	RIV1FM	DRN1FM	EVT1FM	GHB1FM			IBS1FM
Approximate (AP)									$SIP1AP_S$	$SOR1AP_S$	
Output Control (OC)	BAS1OC										
Budget (BD)		$BCF1BD_{US}$	$WEL1BD_U$	$RCH1BD_U$	$RIV1BD_U$	$DRN1BD_U$	$EVT1BD_U$	$GHB1BD_U$			$IBS1BD_U$
Output (OT)	$BAS1OT_U$										$IBS1OT_U$

Modified from McDonald and Harbaugh (1988, figure 15)

EXPLANATION

$BCF1BD_{US}$ NAME OF MODULE — Subscript, U, indicates that utility modules are utilized. Subscript, S, indicates that submodules are utilized

Figure 5.—Primary modules of ground-water flow model organized by procedure and package.

The finite-difference equations in the ground-water flow model describe flow rates in terms of volumetric rates [L^3T^{-1}], whereas equation 13 expresses flow into storage as a rate per unit volume [T^{-1}]. To make equation 13 match the dimensionality of the finite-difference equations, specific-storage quantities are replaced by corresponding storage-capacity values in module IBS1RP. Storage capacity is defined by McDonald and Harbaugh (1988, chap. 5, p. 25) as the product of specific storage and volume. The S_{ske} quantity in equation 13 is replaced by elastic storage capacity SCE [L^2], which is the product of S_{ske}, thickness of compacting interbeds, and area of the finite-difference cell. The inelastic storage capacity, SCV [L^2], is computed the same way using S_{skv}. The storage capacity at the end of time step m is expressed as SC^m. The resulting equation is:

$$QI^m = \frac{SC^m}{\Delta t}\left(h^m - H^{m-1}\right) + \frac{SCE}{\Delta t}\left(H^{m-1} - h^{m-1}\right), \quad (19)$$

where

QI^m is rate of flow into storage in compressible interbeds during time step m [L^3T^{-1}].

In this equation, the only unknown head quantity is h^m. The coefficient of this quantity, $SC^m/\Delta t$, must be subtracted from the corresponding element in the array HCOF of the ground-water model. The remaining terms, $-(SC^m/\Delta t)H^{m-1} + (SCE/\Delta t)(H^{m-1}-h^{m-1})$ are added to the corresponding element of the array RHS. In these expressions, the quantity SC^m is unknown because it is a function of h^m. Within each iteration that the model carries out to compute h^m, a value of SC^m is selected in module IBS1FM on the basis of the most recent estimate of h^m.

Interbed-Storage Package output consists of the rate of flow to and from interbed storage during a time step and the accumulated volume to and from interbed storage whenever a volumetric budget is printed for the entire model. Also, at the end of each stress period, the package prints out a single two-dimensional array showing the sum of the compaction of all model layers that have interbed storage. That sum is referred to as SUBSIDENCE in the model output. Optionally, for any particular model time step, the user may print and (or) write to a disk file the sum of compaction for all layers and (or) compaction for each layer that has interbed storage and (or) the current preconsolidation head for each layer that has interbed storage. Additionally, for any time step, a user may write to a disk file an unformatted array containing values of rate of flow from storage for each cell in the finite-difference grid. The disk files of compaction and subsidence are unformatted arrays written with the ground-water model utility subroutine ULASAV (McDonald and Harbaugh, 1988, chap. 14, p. 9-11). The text label written with each array is COMPACTION for individual compaction arrays, SUBSIDENCE for the array that is the sum of compaction for all layers, and CRITICAL HEAD for preconsolidation head arrays. Each of those character strings are left justified in four elements of text arrays, with four characters included in each element. Trailing blanks are included to total 16 characters in each text label. The unformatted cell-by-cell budget array is written with model utility subroutine UBUDSV (McDonald and Harbaugh, 1988, chap. 14, p. 6-8). The text string written out with that array is INTERBED STORAGE.

Input Instructions

Input for the Interbed-Storage Package is read from the unit IUNIT(19), specified in the Basic Package input (McDonald and Harbaugh, 1988, chap. 4, p. 9–11).

FOR EACH SIMULATION

IBS1AL

1. Data: IIBSCB IIBSOC
   Format: I10 I10

2. Data: IBQ(NLAY) (Maximum of 80 layers)
   Format: 40I2

   (If there are 40 or fewer layers, use one record; otherwise, use two records)

IBS1RP

The following four arrays are needed to specify the material properties and initial compaction of model layers having interbed storage as indicated in the IBQ array. The four arrays are first read for the uppermost layer with interbed storage, and then continuing downward to lower layers with interbed storage.

FOR EACH LAYER WITH IBQ CODE GREATER THAN ZERO

3. Data: HC(NCOL,NROW)
   Module: U2DREL

4. Data: Sfe(NCOL,NROW)
   Module: U2DREL

5. Data: Sfv(NCOL,NROW)
   Module: U2DREL

6. Data: COM(NCOL,NROW)
   Module: U2DREL

IF IIBSOC IS GREATER THAN ZERO

7. Data: ISUBFM ICOMFM IHCFM ISUBUN ICOMUN IHCUN
   Format: I10 I10 I10 I10 I10 I10

IBS1OT

FOR EACH TIME STEP
IF IIBSOC IS GREATER THAN ZERO

8. Data: ISUBPR ICOMPR IHCPR ISUBSV ICOMSV IHCSV
   Format: I10 I10 I10 I10 I10 I10

Explanation of Fields Used in Input Instructions

IIBSCB is a flag and unit number.

If IIBSCB > 0, it is the unit number on which cell-by-cell flow terms will be recorded whenever ICBCFL (see McDonald and Harbaugh, 1988, chap. 4, p. 14-15) is set.
If IIBSCB ≤ 0, cell-by-cell flow terms will not be recorded.

IIBSOC is a flag for selecting output control for each time step.

If IIBSOC > 0, output control will be read each time step for printing and recording subsidence (total compaction of all layers with interbed storage), and compaction and preconsolidation head for all layers with interbed storage.
If IIBSOC ≤ 0, subsidence will be printed at the end of each stress period using format (10G11.4). Compaction and preconsolidation head will not be printed and subsidence, compaction, and preconsolidation head will not be recorded.

IBQ is an indicator to specify which model layers have interbed storage.

If IBQ(K) > 0, model layer K has interbed storage.
If IBQ(K) ≤ 0, model layer K does not have interbed storage.

HC is an array specifying the preconsolidation head or preconsolidation stress in terms of head in the aquifer. Preconsolidation head is the previous minimum head value in the aquifer. For any model cells in which specified HC is greater than the corresponding value of starting head (see McDonald and Harbaugh, 1988, chap. 4, p. 9-12), value of HC will be set to that of starting head.

Sfe is an array specifying the dimensionless elastic storage factor for interbeds present in model layer. The storage factor may be estimated as the sum of the products of elastic skeletal specific storage and thickness of all interbeds in a model layer.

Sfv is an array specifying the dimensionless inelastic storage factor for interbeds present in model layer. The storage factor may be estimated as the sum of the products of inelastic skeletal specific storage and thickness of all interbeds in a model layer.

Com is an array specifying the starting compaction in each layer with interbed storage. Compaction values computed by the package are added to values in this array so that printed or stored values of compaction and land subsidence may include previous components. Values in this array do not affect calculations of storage changes or resulting compaction. For simulations in which output values are to reflect compaction and subsidence since the start of the simulation, enter zero values for all elements of this array.

ISUBFM is a code for the format in which subsidence will be printed.

ICOMFM is a code for the format in which compaction will be printed.

IHCFM is a code for the format in which preconsolidation head will be printed. Format codes have the same meaning for subsidence, compaction, and preconsolidation head and are the same as codes specified by McDonald and Harbaugh (1988, chap. 4, p. 14–15) for printing head and drawdown. A positive code selects their wrap format and a negative code selects their strip format. The absolute value of the code specifies the printout format as follows:

0 -	(10G11.4)	7 -	(20F5.0)
1 -	(11G10.3)	8 -	(20F5.1)
2 -	(9G13.6)	9 -	(20F5.2)
3 -	(15F7.1)	10 -	(20F5.3)
4 -	(15F7.2)	11 -	(20F5.4)
5 -	(15F7.3)	12 -	(10G11.4)
6 -	(15F7.4)		

ISUBUN is the unit number to which subsidence arrays will be written if they are saved on disk.

ICOMUN is the unit number to which compaction arrays will be written if they are saved on disk.

IHCUN is the unit number to which preconsolidation head arrays will be written if they are saved on disk.

ISUBPR is the output flag for subsidence printout.

If ISUBPR ≤ 0, subsidence is not printed.
If ISUBPR > 0, subsidence is printed.

ICOMPR is the output flag for compaction printout.

If ICOMPR ≤ 0, compaction is not printed.
If ICOMPR > 0, compaction is printed for each layer with interbed storage.

IHCPR is the output flag for preconsolidation head printout.

If IHCPR ≤ 0, preconsolidation head is not printed.
If IHCPR > 0, preconsolidation head is printed for each layer with interbed storage.

ISUBSV is the output flag for saving subsidence in an unformatted disk file.

If ISUBSV ≤ 0, subsidence is not saved.
If ISUBSV > 0, subsidence is saved.

ICOMSV is the output flag for saving compaction in an unformatted disk file.

If ICOMSV ≤ 0, compaction is not saved.
If ICOMSV > 0, compaction is saved for each layer with interbed storage.

IHCSV is the output flag for saving preconsolidation head in an unformatted disk file.

If IHCSV ≤ 0, preconsolidation head is not saved.
If IHCSV > 0, preconsolidation head is saved for each layer with interbed storage.

MODULE DOCUMENTATION FOR THE INTERBED-STORAGE PACKAGE

The Interbed-Storage Package (IBS1) consists of five modules, all of which are called by the main program. The modules are

IBS1AL Allocates space for data arrays.

IBS1RP Reads all data needed by package and initializes the preconsolidation-head and storage-capacity arrays. Initializes flags for output control.

IBS1FM Calculates terms to add to right-hand side (RHS) and main diagonal (HCOF).

IBS1BD Calculates flow rates and accumulated flow volumes from interbed storage. Calculates compaction and updates preconsolidation head each time step.

IBS1OT Reads flags for printing and storing subsidence, compaction, and preconsolidation head. Prints and stores arrays.

IBS1AL

Narrative for Module IBS1AL

This module allocates space in the X array (McDonald and Harbaugh, 1988, chap. 3, p. 22–23) for all arrays needed for the Interbed-Storage Package.

1. Print a message indentifying the package.
2. Check flag ISS to determine if simulation is steady-state or transient.
3. If simulation is steady-state, cancel the Interbed-Storage option and continue with the simulation.
4. Read flag for saving cell-by-cell storage changes (IIBSCB) and flag for output control of compaction, subsidence and preconsolidation head.
5. If option to save cell-by-cell storage terms is selected, print message giving unit number selected. If option to specify output control is is selected, print message.
6. Read one-dimensional array indicator (IBQ) to determine which model layers have interbed storage.
7. Print a message indicating which layers have interbed storage.
8. Allocate space for preconsolidation head array (HC), elastic storage capacity (SCE), inelastic storage capacity (SCV), and compaction (SUB). Space is allocated by computing the location of the first element of each array as the first unused element in the X array (ISUM). The locations for the arrays are stored in variables LCHC, LCSCE, LCSCV, and LCSUB.
9. Calculate and print the amount of space used by the Interbed-Storage Package. Calculate the total amount of space used in the X array. If the amount of space used exceeds the amount allocated (LENX), print a message and terminate simulation.
10. RETURN.

Flow chart for module IBS1AL

ISS is a steady-state flag.

If ISS ≠ 0, the simulation is steady-state.

If ISS = 0, the simulation is transient.

IIBSCB is a flag and a unit number.

If IIBSCB > 0, it is the unit number on which cell-by-cell flow terms will be recorded.

If IIBSCB ≤ 0, cell-by-cell terms will not be recorded.

IIBSOC is a flag.

If IIBSOC > 0, output control flags for printing and recording subsidence, compaction, and preconsolidation head will be read each time step.

If IIBSOC ≤ 0, output control flags will not be read.

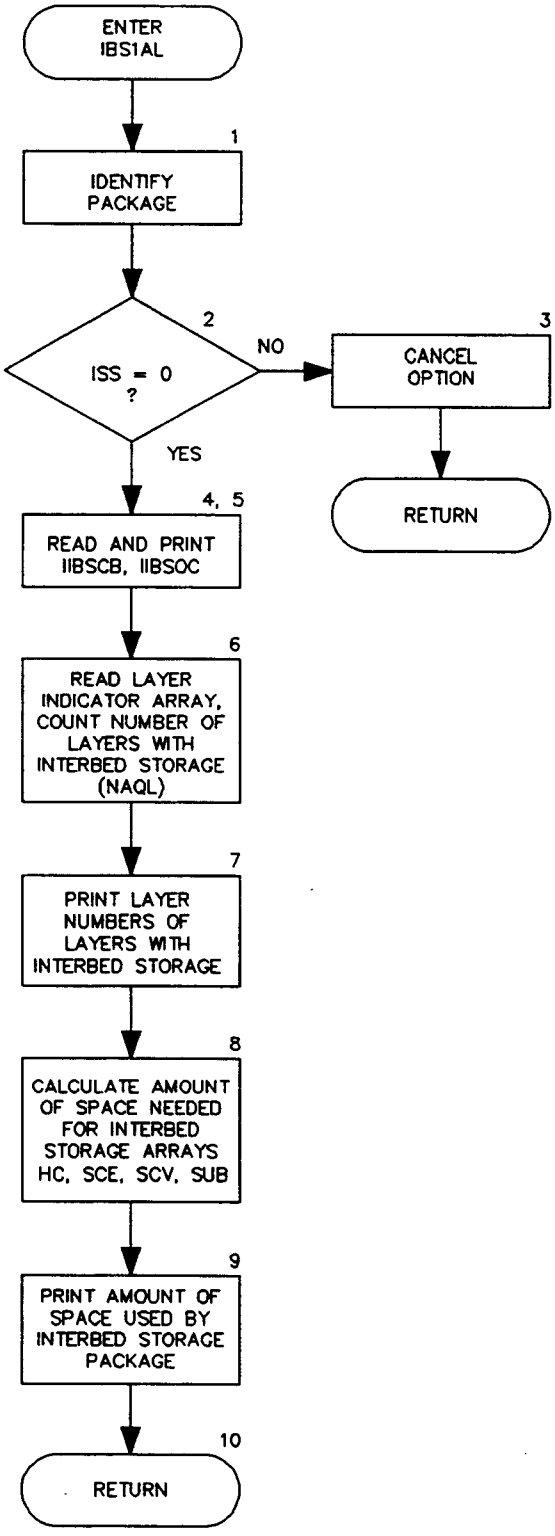
NAQL is the number of layers with interbed storage.

HC is the preconsolidation head array.

SCE is the elastic storage capacity array.

SCV is the inelastic storage capacity array.

SUB is the compaction array.



Program Listing for Module IBS1AL

```
      SUBROUTINE IBS1AL(ISUM,LENX,LCHC,LCSCE,LCSCV,LCSUB,
     1                  NCOL,NROW,NLAY,IIBSCB,IIBSOC,ISS,IN,IOUT)
C
C-----VERSION 1115 06JUN1988 IBS1AL
C     ******************************************************************
C     ALLOCATE ARRAY STORAGE FOR INTERBED STORAGE PACKAGE
C     ******************************************************************
C
C        SPECIFICATIONS:
C     ------------------------------------------------------------------
      DIMENSION IBQ1(80)
      COMMON /IBSCOM/ IBQ(80)
C     ------------------------------------------------------------------
C
C1------IDENTIFY PACKAGE.
      WRITE(IOUT,1)IN
    1 FORMAT(1H0,'IBS1 -- INTERBED STORAGE PACKAGE, VERSION 1,',
     1     ' 06/02/88',' INPUT READ FROM UNIT',I3)
C
C2------CHECK TO SEE THAT INTERBED STORAGE OPTION IS APPROPRIATE
      IF(ISS.EQ.0) GO TO 100
C
C3------IF INAPPROPRIATE PRINT A MESSAGE & CANCEL OPTION.
      WRITE(IOUT,8)
    8 FORMAT(1X,'INTERBED STORAGE INAPPROPRIATE FOR STEADY-STATE',
     1 ' PROBLEM.',/,1X,'OPTION CANCELLED, SIMULATION CONTINUING.')
      IN=0
      RETURN
C
C4------READ FLAG FOR STORING CELL-BY-CELL STORAGE CHANGES AND
C4------FLAG FOR PRINTING AND STORING COMPACTION, SUBSIDENCE, AND
C4------CRITICAL HEAD ARRAYS.
  100 READ(IN,3) IIBSCB,IIBSOC
    3 FORMAT(2I10)
C
C5------IF CELL-BY-CELL TERMS TO BE SAVED THEN PRINT UNIT NUMBER.
      IF(IIBSCB.GT.0) WRITE(IOUT,105) IIBSCB
  105 FORMAT(1X,'CELL-BY-CELL FLOW TERMS WILL BE SAVED ON UNIT',I3)
C
C5A-----IF OUTPUT CONTROL FOR PRINTING ARRAYS IS SELECTED PRINT MESSAGE.
      IF(IIBSOC.GT.0) WRITE(IOUT,106)
  106 FORMAT(1X,'OUTPUT CONTROL RECORDS FOR IBS1 PACKAGE WILL BE ',
     1 'READ EACH TIME STEP.')
C
C6------READ INDICATOR AND FIND OUT HOW MANY LAYERS HAVE INTERBED STORAGE.
      READ(IN,110) (IBQ(K),K=1,NLAY)
  110 FORMAT(40I2)
      NAQL=0
      DO 120 K=1,NLAY
      IF(IBQ(K).LE.0) GO TO 120
      NAQL=NAQL+1
```

```
      IBQ1(NAQL)=K
  120 CONTINUE
C
C7------IDENTIFY WHICH LAYERS HAVE INTERBED STORAGE.
      WRITE(IOUT,130) (IBQ1(K),K=1,NAQL)
  130 FORMAT(1X,'INTERBED STORAGE IN LAYER(S) ',80I2)
C
C8------ALLOCATE SPACE FOR THE ARRAYS HC, SCE, SCV, AND SUB.
      IRK=ISUM
      NA=NROW*NCOL*NAQL
      LCHC=ISUM
      ISUM=ISUM+NA
      LCSCE=ISUM
      ISUM=ISUM+NA
      LCSCV=ISUM
      ISUM=ISUM+NA
      LCSUB=ISUM
      ISUM=ISUM+NA
C
C9------CALCULATE & PRINT AMOUNT OF SPACE USED BY PACKAGE.
  300 IRK=ISUM-IRK
      WRITE(IOUT,4)IRK
    4 FORMAT(1X,I8,' ELEMENTS OF X ARRAY USED FOR INTERBED STORAGE')
      ISUM1=ISUM-1
      WRITE(IOUT,5)ISUM1,LENX
    5 FORMAT(1X,I8,' ELEMENTS OF X ARRAY USED OUT OF',I8)
      IF(ISUM1.GT.LENX)WRITE(IOUT,6)
    6 FORMAT(1X,'   ***X ARRAY MUST BE MADE LARGER***')
C
C10-----RETURN.
      RETURN
      END
```

List of Variables for Module IBS1AL

Variable	Range	Definition
IBQ	Package	DIMENSION (80) Layer flag for interbed storage: > 0, Layer has interbed storage. ≤ 0, Layer does not have interbed storage.
IBQ1	Module	DIMENSION (80) List of layer numbers of layers with interbed storage.
IIBSCB	Package	Flag and a unit number. > 0, it is the unit number on which cell-by-cell flow terms will be recorded whenever ICBCFL (see McDonald and Harbaugh, 1988, chap. 4, p. 14-15) is set. ≤ 0, cell-by-cell flow terms will not be recorded.

Variable	Range	Definition
IIBSOC	Package	Flag. > 0, output control will be read each time step for printing and recording subsidence, compaction and preconsolidation head. ≤ 0, subsidence will be printed at the end of each stress period.
IN	Package	Primary unit number from which input for this package will be read.
IOUT	Global	Primary unit number for all printed output. IOUT = 6.
IRK	Module	Before this module allocates space, IRK is set equal to ISUM. After allocation, IRK is subtracted from ISUM to get the amount of space in the X array allocated by this module.
ISS	Global	Flag. = 0, simulation is transient. ≠ 0, simulation is steady state.
ISUM	Global	Index number of the lowest element in the X array which has not yet been allocated. When space is allocated for an array, the size of the array is added to ISUM.
ISUM1	Module	ISUM+1
K	Module	Index
LCHC	Package	Location in the X array of the first element of array HC.
LCSCE	Package	Location in the X array of the first element of array SCE.
LCSCV	Package	Location in the X array of the first element of array SCV.
LCSUB	Package	Location in the X array of the first element of array SUB.
LENX	Global	Length of the X array in words. This should always be equal to the dimension of X specified in the MAIN program.
NA	Module	Number of cells in all layers with interbed storage.
NAQL	Module	Number of layers with interbed storage.
NCOL	Global	Number of columns in the grid.
NLAY	Global	Number of layers in the grid.
NROW	Global	Number of rows in the grid.

IBS1RP

Narrative for Module IBS1RP

This module reads storage properties and preconsolidation-head arrays that are used by the package.

1. Read preconsolidation head for all layers with interbed storage. Read elastic and inelastic storage coefficients for all layers with interbed storage. Storage properties initially read in are the product of specific storage and thickness. Read starting compaction for all layers with interbed storage.

2. Loop through all cells in the model grid with interbed storage.

3. Multiply storage coefficients by area of the cell to obtain storage capacity.

4. Make sure that preconsolidation head at each node is equal to or greater than the starting head. If preconsolidation head is less than the starting head, set it equal to the starting head value.

5. Initialize and read flags that specify when subsidence, compaction, and preconsolidation head will be printed and saved on disk. Print messages indicating which options were selected.

6. RETURN.

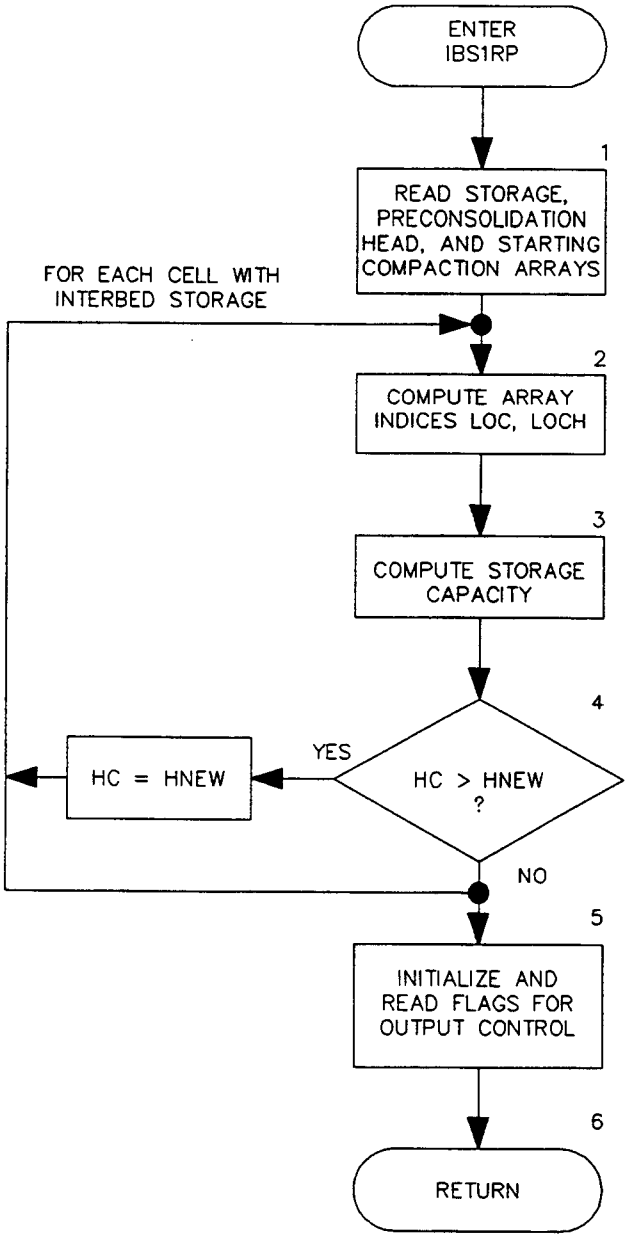
Flow chart for module IBS1RP

LOC is a pointer to elements in the Interbed-Storage Package arrays.

LOCH is a pointer to elements in the HNEW array.

HC is the preconsolidation head array.

HNEW is the array containing computed head for the end of the time step.



Program Listing for Module IBS1RP

```
      SUBROUTINE IBS1RP(DELR,DELC,HNEW,HC,SCE,SCV,SUB,NCOL,NROW,
     1                  NLAY,NODES,IIBSOC,ISUBFM,ICOMFM,IHCFM,
     2                  ISUBUN,ICOMUN,IHCUN,IN,IOUT)
C
C-----VERSION 1117 02JUN1988 IBS1RP
C     ******************************************************************
C     READ INTERBED STORAGE DATA
C     ******************************************************************
C
C        SPECIFICATIONS:
C     ------------------------------------------------------------------
      CHARACTER*4 ANAME
      DOUBLE PRECISION HNEW
      DIMENSION HNEW(NODES),HC(NODES),SCE(NODES),
     1       SCV(NODES),SUB(NODES),ANAME(6,4),
     2       DELR(NCOL),DELC(NROW)
C
      COMMON /IBSCOM/ IBQ(80)
C
      DATA ANAME(1,1),ANAME(2,1),ANAME(3,1),ANAME(4,1),ANAME(5,1),
     1  ANAME(6,1) /'   P','RECO','NSOL','IDAT','ION ','HEAD'/
      DATA ANAME(1,2),ANAME(2,2),ANAME(3,2),ANAME(4,2),ANAME(5,2),
     1  ANAME(6,2) /'ELAS','TIC ','INTE','RBED',' STO','RAGE'/
      DATA ANAME(1,3),ANAME(2,3),ANAME(3,3),ANAME(4,3),ANAME(5,3),
     1  ANAME(6,3) /' VIR','GIN ','INTE','RBED',' STO','RAGE'/
      DATA ANAME(1,4),ANAME(2,4),ANAME(3,4),ANAME(4,4),ANAME(5,4),
     1  ANAME(6,4) /'    ',' STA','RTIN','G CO','MPAC','TION'/
C     ------------------------------------------------------------------
C
C1------READ IN STORAGE AND CRITICAL HEAD ARRAYS
      NIJ=NROW*NCOL
      KQ=0
      DO 60 K=1,NLAY
      IF(IBQ(K).LE.0) GO TO 60
      KQ=KQ+1
      LOC=1+(KQ-1)*NIJ
      CALL U2DREL(HC(LOC),ANAME(1,1),NROW,NCOL,K,IN,IOUT)
      CALL U2DREL(SCE(LOC),ANAME(1,2),NROW,NCOL,K,IN,IOUT)
      CALL U2DREL(SCV(LOC),ANAME(1,3),NROW,NCOL,K,IN,IOUT)
      CALL U2DREL(SUB(LOC),ANAME(1,4),NROW,NCOL,K,IN,IOUT)
   60 CONTINUE
C
C2------LOOP THROUGH ALL CELLS WITH INTERBED STORAGE.
      KQ=0
      DO 80 K=1,NLAY
      IF(IBQ(K).LE.0) GO TO 80
      KQ=KQ+1
      NQ=(KQ-1)*NIJ
      NK=(K-1)*NIJ
      DO 70 IR=1,NROW
      NQR=NQ+(IR-1)*NCOL
```

```
      NKR=NK+(IR-1)*NCOL
      DO 70 IC=1,NCOL
      LOC=NQR+IC
      LOCH=NKR+IC
C
C3------MULTIPLY STORAGE BY AREA TO GET STORAGE CAPACITY.
      AREA=DELR(IC)*DELC(IR)
      SCE(LOC)=SCE(LOC)*AREA
      SCV(LOC)=SCV(LOC)*AREA
C
C4------MAKE SURE THAT PRECONSOLIDATION HEAD VALUES
C4------ARE CONSISTANT WITH STARTING HEADS.
      IF(HC(LOC).GT.HNEW(LOCH)) HC(LOC)=HNEW(LOCH)
   70 CONTINUE
   80 CONTINUE
C
C5------INITIALIZE AND READ OUTPUT FLAGS.
      ICOMFM=0
      ISUBFM=0
      IHCFM=0
      ICOMUN=0
      ISUBUN=0
      IHCUN=0
      IF(IIBSOC.LE.0) GO TO 200
      READ(IN,100) ISUBFM,ICOMFM,IHCFM,ISUBUN,ICOMUN,IHCUN
  100 FORMAT(6I10)
      WRITE(IOUT,110) ISUBFM,ICOMFM,IHCFM
  110 FORMAT(1HO,'    SUBSIDENCE PRINT FORMAT IS NUMBER',I4/
     1          '    COMPACTION PRINT FORMAT IS NUMBER',I4/
     2          '  CRITICAL HEAD PRINT FORMAT IS NUMBER',I4)
      IF(ISUBUN.GT.0) WRITE(IOUT,120) ISUBUN
  120 FORMAT(1HO,'    UNIT FOR SAVING SUBSIDENCE IS',I4)
      IF(ICOMUN.GT.0) WRITE(IOUT,130) ICOMUN
  130 FORMAT(1H ,'    UNIT FOR SAVING COMPACTION IS',I4)
      IF(IHCUN.GT.0)  WRITE(IOUT,140) IHCUN
  140 FORMAT(1H ,' UNIT FOR SAVING CRITICAL HEAD IS',I4)
C
C6------RETURN
  200 RETURN
      END
```

List of Variables for Module IBS1RP

Variable	Range	Definition
ANAME	Module	Labels for printout of input arrays.
AREA	Module	Area of cell.
DELC	Global	DIMENSION (NROW), Cell dimensions in the column direction. DELC(I) contains the width of row I.
DELR	Global	DIMENSION (NCOL), Cell dimensions in the row direction. DELC(J) contains the width of column J.
HC	Package	DIMENSION (NCOL,NROW,NAQL), Preconsolidation head in each cell with interbed storage. NAQL is number of layers for which IBQ > 0.
HNEW	Global	DIMENSION (NCOL,NROW,NLAY), Most recent estimate of head in each cell. HNEW changes at each iteration.
IBQ	Package	DIMENSION (80) Layer flag for interbed storage: > 0, Layer has interbed storage. ≤ 0, Layer does not have interbed storage.
IC	Module	Index for columns.
ICOMFM	Package	Code for format in which compaction will be printed.
ICOMUN	Package	Unit number on which an unformatted record containing compaction should be recorded.
IHCFM	Package	Code for format in which preconsolidation head will be printed.
IHCUN	Package	Unit number on which an unformatted record containing preconsolidation head should be recorded.
IIBSOC	Package	Flag. > 0, output control will be read each time step for printing and recording subsidence, compaction and preconsolidation head. ≤ 0, subsidence will be printed at the end of each stress period.
IN	Package	Primary unit number from which input for this package will be read.
IOUT	Global	Primary unit number for all printed output. IOUT = 6.
IR	Module	Index for rows.
SUBFM	Package	Code for format in which subsidence will be printed.

Variable	Range	Definition
ISUBUN	Package	Unit number on which an unformatted record containing subsidence should be recorded.
K	Module	Index of all model layers.
KQ	Module	Index of model layers with interbed storage
LOC	Module	Pointer to parts of SUB, HC, SCE, and SCV arrays.
LOCH	Module	Pointer to parts of HNEW array.
NCOL	Global	Number of columns in the grid.
NIJ	Module	Number of cells in a layer.
NK	Module	Number of cells preceding a particular layer in the HNEW array.
NKR	Module	Number of cells preceding a particular row in the HNEW array.
NLAY	Global	Number of layers in the grid.
NODES	Global	Number of cells (nodes) in the finite-difference grid.
NQ	Module	Number of cells preceding a particular layer in SUB, HC, SCE, and SCV arrays.
NQR	Module	Number of cells preceding a particular row in SUB, HC, SCE, and SCV arrays.
NROW	Global	Number of rows in the grid.
SCE	Package	DIMENSION (NCOL,NROW,NAQL), Elastic storage capacity of each cell with interbed storage. NAQL is number of layers for which IBQ > 0.
SCV	Package	DIMENSION (NCOL,NROW,NAQL), Inelastic storage capacity of each cell with interbed storage. NAQL is number of layers for which IBQ > 0.
SUB	Package	DIMENSION (NCOL,NROW,NAQL), Computed compaction of each cell with interbed storage. NAQL is number of layers for which IBQ > 0.

IBS1FM

Narrative for Module IBS1FM

This module adds terms representing interbed storage to the accumulators in which the terms RHS and HCOF are formulated.

1. Initialize counter for layers with interbed storage. Calculate reciprocal of time-step length to reduce number of divisions later on.

2. Find all layers with interbed storage using IBQ indicator. For layers with interbed storage, loop through cells, skipping those outside of the flow region.

3. Determine applicable storage capacities at the start and end of time step so that storage changes for cell may be apportioned between elastic and inelastic components.

4. Add contributions from interbed storage to the RHS and HCOF accumulators.

5. RETURN.

Flow chart for module IBS1FM

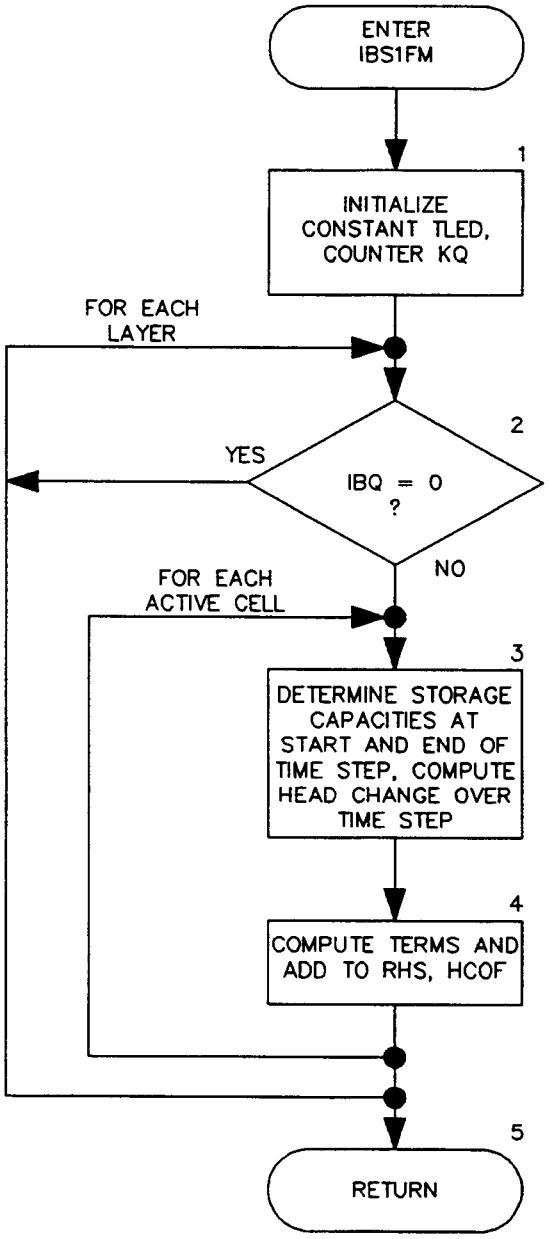
TLED is the reciprocal of the time-step length.

KQ is an index of model layers with interbed storage.

IBQ is a layer flag.
IBQ > 0 means that model layer has interbed storage.
IBQ ≤ 0 means that model layer does not have interbed storage.

RHS is an accumulator in which the right-hand side of the finite-difference equation is formulated.

HCOF is an accumulator in which the coefficient of head in the cell is formulated.



Program Listing for Module IBS1FM

```
      SUBROUTINE IBS1FM(RHS,HCOF,HNEW,HOLD,HC,SCE,SCV,
     1                  IBOUND,NCOL,NROW,NLAY,DELT)
C
C-----VERSION 1223 02JUN1988 IBS1FM
C     ******************************************************************
C        ADD INTERBED STORAGE TO RHS AND HCOF
C     ******************************************************************
C
C        SPECIFICATIONS:
C     ------------------------------------------------------------------
      DOUBLE PRECISION HNEW
      DIMENSION RHS(NCOL,NROW,NLAY),HCOF(NCOL,NROW,NLAY),
     1          IBOUND(NCOL,NROW,NLAY),HNEW(NCOL,NROW,NLAY),
     2          HOLD(NCOL,NROW,NLAY),HC(NCOL,NROW,NLAY),
     3          SCE(NCOL,NROW,NLAY),SCV(NCOL,NROW,NLAY)
C
      COMMON /IBSCOM/ IBQ(80)
C     ------------------------------------------------------------------
C
C1------INITIALIZE
       TLED=1./DELT
      KQ=0
C
C2------FIND LAYERS WITH INTERBED STORAGE
      DO 110 K=1,NLAY
      IF(IBQ(K).EQ.0) GO TO 110
      KQ=KQ+1
      DO 100 I=1,NROW
      DO 100 J=1,NCOL
      IF(IBOUND(J,I,K).LE.0) GO TO 100
C
C3------DETERMINE STORAGE CAPACITIES FOR CELL AT START AND END OF STEP
      RHO1=SCE(J,I,KQ)*TLED
      RHO2=RHO1
      HCTMP=HC(J,I,KQ)
      IF(HNEW(J,I,K).LT.HCTMP) RHO2=SCV(J,I,KQ)*TLED
C
C4------ADD APPROPRIATE TERMS TO RHS AND HCOF
      RHS(J,I,K)=RHS(J,I,K)-HCTMP*(RHO2-RHO1)-RHO1*HOLD(J,I,K)
      HCOF(J,I,K)=HCOF(J,I,K)-RHO2
  100 CONTINUE
  110 CONTINUE
C
C5------RETURN
      RETURN
      END
```

List of Variables for Module IBS1FM

Variable	Range	Definition
DELT	Global	Length of the current time step.
HC	Package	DIMENSION (NCOL,NROW,NAQL), Preconsolidation head in each cell with interbed storage. NAQL is number of layers for which IBQ > 0.
HCOF	Global	DIMENSION (NCOL,NROW,NLAY), Coefficient of head in cell (J,I,K) in the finite-difference equation.
HCTMP	Module	Temporary HC(J,I,K).
HNEW	Global	DIMENSION (NCOL,NROW,NLAY), Most recent estimate of head in each cell. HNEW changes at each iteration.
HOLD	Global	DIMENSION (NCOL,NROW,NLAY), Head at the start of the current time step.
I	Module	Index for rows.
IBOUND	Global	DIMENSION (NCOL,NROW,NLAY), Status of each cell. < 0, constant-head cell = 0, inactive cell > 0, variable-head cell
IBQ	Package	DIMENSION (80), Layer flag for interbed storage: > 0, Layer has interbed storage. ≤ 0, Layer does not have interbed storage.
J	Module	Index for columns.
K	Module	Index for layers.
KQ	Module	Index of model layers with interbed storage.
NCOL	Global	Number of columns in the grid.
NLAY	Global	Number of layers in the grid.
NROW	Global	Number of rows in the grid.
RHO1	Module	Elastic storage capacity divided by length of current time step.
RHO2	Module	Storage capacity at end of time step (elastic or inelastic) divided by length of current time step.

Variable	Range	Definition
RHS	Global	DIMENSION (NCOL,NROW,NLAY), Right hand side of finite-difference equation. RHS is an accumulation of terms from several different packages.
SCE	Package	DIMENSION (NCOL,NROW,NAQL), Elastic storage capacity of each cell with interbed storage. NAQL is number of layers for which IBQ > 0.
SCV	Package	DIMENSION (NCOL,NROW,NAQL), Inelastic storage capacity of each cell with interbed storage. NAQL is number of layers for which IBQ > 0.
TLED	Module	Reciprocal of the length of the current time step.

IBS1BD

Narrative for Module IBS1BD

This module calculates rates and volumes of water derived from interbed storage and updates the preconsolidation-head array.

1. Initialize the cell-by-cell flow term flag (IBD), and rate accumulators STOIN and STOUT.

2. Test to determine whether or not cell-by-cell flow terms are to be saved. If not, skip item 3, following.

3. Set cell-by-cell flow term flag (IBD) and clear the array (BUFF) in which they will be accumulated.

4. Loop through all cells in grid with interbed storage to perform rate and volume calculations.

5. Begin calculations of flow rates to or from interbed storage. If cell is outside of active flow region, skip it. Set temporary variables equal to previous head in cell, current head in cell, and preconsolidation head in cell.

6. Determine storage capacities at beginning and end of time step.

7. Calculate volume change in interbed storage for time step.

8. Calculate compaction associated with storage change. Accumulate compaction in array SUB.

9. If cell-by-cell flow rates are to be saved, calculate flow rate and add to BUFF.

10. Determine whether flow rate is into or out of storage, add magnitude of rate to STOIN OR STOUT.

11. If cell-by-cell flow terms are to be saved, call module UBUDSV to write the buffer (BUFF) onto disk.

12. Calculate rates from volume storage changes and time-step length. Move rates, volumes, and budget labels into arrays for printing.

13. Increment the budget term counter (MSUM).

14. Update the preconsolidation-head array. If head at end of time step is less than preconsolidation head, set preconsolidation head equal to head in cell.

15. RETURN.

STOIN is an accumulator in which all flows into the aquifer from interbed storage are added.

STOUT is an accumulator in which all flows out of the aquifer to interbed storage are added.

BUFF is an array in which values are stored as they are being gathered for printing or recording.

IBQ is a layer flag.
IBQ > 0 means that model layer has interbed storage.
IBQ ≤ 0 means that model layer does not have interbed storage.

SUB is the compaction array.

VBVL is a table of budget entries calculated by component-of-flow packages for use in calculating the volumetric budget.

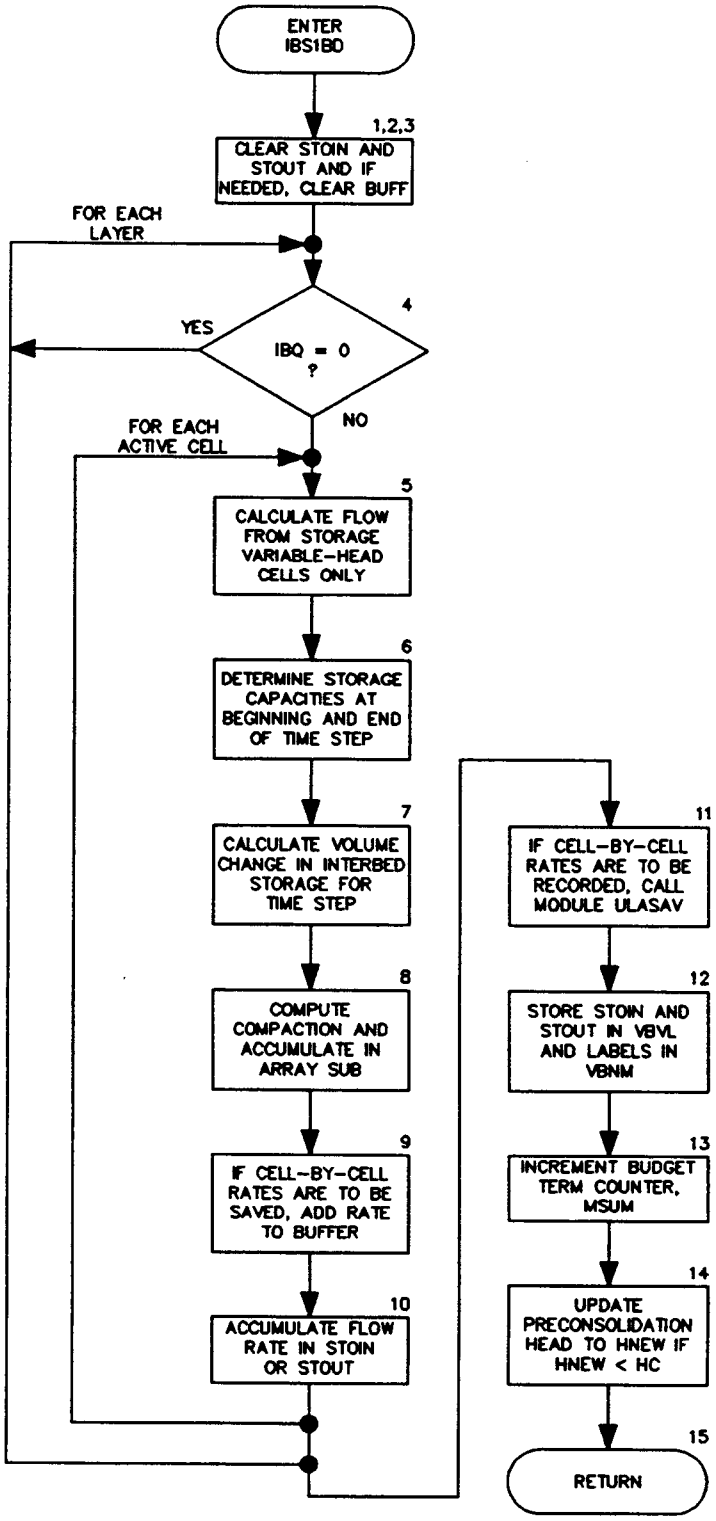
VBNM is a table of labels for budget terms.

MSUM is the counter for budget entries and labels in VBVL and VBNM.

HC is the preconsolidation head array.

HNEW is the array containing computed head for the end of the time step.

Flow chart for module IBS1BD



Program Listing for Module IBS1BD

```
      SUBROUTINE IBS1BD(IBOUND,HNEW,HOLD,HC,SCE,SCV,SUB,DELR,DELC,
     1      NCOL,NROW,NLAY,DELT,VBVL,VBNM,MSUM,KSTP,KPER,IIBSCB,
     2      ICBCFL,BUFF,IOUT)
C-----VERSION 1224 02JUN1988 IBS1BD
C     ******************************************************************
C     CALCULATE VOLUMETRIC BUDGET FOR INTERBED STORAGE
C     ******************************************************************
C
C     SPECIFICATIONS:
C     ------------------------------------------------------------------
      CHARACTER*4 TEXT,VBNM
      DOUBLE PRECISION HNEW
      DIMENSION IBOUND(NCOL,NROW,NLAY),HOLD(NCOL,NROW,NLAY),
     1          HNEW(NCOL,NROW,NLAY),HC(NCOL,NROW,NLAY),
     2          SCE(NCOL,NROW,NLAY),SCV(NCOL,NROW,NLAY),
     3          SUB(NCOL,NROW,NLAY),VBVL(4,20),VBNM(4,20),
     4          BUFF(NCOL,NROW,NLAY),DELR(NCOL),DELC(NROW)
      DIMENSION TEXT(4)
C
      COMMON /IBSCOM/ IBQ(80)
      DATA TEXT(1),TEXT(2),TEXT(3),TEXT(4) /'INTE','RBED',' STO','RAGE'/
C     ------------------------------------------------------------------
C
C1------INITIALIZE CELL-BY-CELL FLOW TERM FLAG (IBD) AND
C1------ACCUMULATORS (STOIN AND STOUT).
      IBD=0
      STOIN=0.
      STOUT=0.
C
C2------TEST TO SEE IF CELL-BY-CELL FLOW TERMS ARE NEEDED.
      IF(ICBCFL.EQ.0  .OR. IIBSCB.LE.0 ) GO TO 10
C
C3------CELL-BY-CELL FLOW TERMS ARE NEEDED SET IBD AND CLEAR BUFFER.
      IBD=1
      DO 5 IL=1,NLAY
      DO 5 IR=1,NROW
      DO 5 IC=1,NCOL
      BUFF(IC,IR,IL)=0.
    5 CONTINUE
C
C4------RUN THROUGH EVERY CELL IN THE GRID WITH INTERBED STORAGE.
   10 KQ=0
      TLED=1./DELT
      DO 110 K=1,NLAY
      IF(IBQ(K).EQ.0) GO TO 110
      KQ=KQ+1
      DO 100 I=1,NROW
      DO 100 J=1,NCOL
C
C5------CALCULATE FLOW FROM STORAGE (VARIABLE HEAD CELLS ONLY)
      IF(IBOUND(J,I,K).LE.0) GO TO 100
```

```
      HHOLD=HOLD(J,I,K)
      HHNEW=HNEW(J,I,K)
      HHC=HC(J,I,KQ)
C
C6------GET STORAGE CAPACITIES AT BEGINNING AND END OF TIME STEP.
      SBGN=SCE(J,I,KQ)
      SEND=SBGN
      IF(HHNEW.LT.HHC) SEND=SCV(J,I,KQ)
C
C7------CALCULATE VOLUME CHANGE IN INTERBED STORAGE FOR TIME STEP.
      STRG=HHC*(SEND-SBGN)+SBGN*HHOLD-SEND*HHNEW
C
C8------ACCUMULATE SUBSIDENCE ASSOCIATED WITH CHANGE IN STORAGE
      SUB(J,I,KQ)=SUB(J,I,KQ)+STRG/(DELR(J)*DELC(I))
C
C9------IF C-B-C FLOW TERMS ARE TO BE SAVED THEN ADD RATE TO BUFFER.
      IF(IBD.EQ.1) BUFF(J,I,K)=BUFF(J,I,K)+STRG*TLED
C
C10-----SEE IF FLOW IS INTO OR OUT OF STORAGE.
      IF(STRG)94,100,96
   94 STOUT=STOUT-STRG
      GO TO 100
   96 STOIN=STOIN+STRG
  100 CONTINUE
  110 CONTINUE
C
C11-----IF C-B-C FLOW TERMS WILL BE SAVED CALL UBUDSV TO RECORD THEM.
      IF(IBD.EQ.1) CALL UBUDSV(KSTP,KPER,TEXT,IIBSCB,BUFF,NCOL,NROW,
     1                         NLAY,IOUT)
C
C12-----MOVE RATES,VOLUMES & LABELS INTO ARRAYS FOR PRINTING.
  200 VBVL(3,MSUM)=STOIN*TLED
      VBVL(4,MSUM)=STOUT*TLED
      VBVL(1,MSUM)=VBVL(1,MSUM)+STOIN
      VBVL(2,MSUM)=VBVL(2,MSUM)+STOUT
      VBNM(1,MSUM)=TEXT(1)
      VBNM(2,MSUM)=TEXT(2)
      VBNM(3,MSUM)=TEXT(3)
      VBNM(4,MSUM)=TEXT(4)
C
C13-----INCREMENT BUDGET TERM COUNTER
      MSUM=MSUM+1
C
C14-----UPDATE PRECONSOLIDATION HEAD ARRAY
      KQ=0
      DO 310 K=1,NLAY
      IF(IBQ(K).LE.0) GO TO 310
      KQ=KQ+1
      DO 300 I=1,NROW
      DO 300 J=1,NCOL
      IF(IBOUND(J,I,K).LE.0) GO TO 300
      HHNEW=HNEW(J,I,K)
      IF(HHNEW.LT.HC(J,I,KQ)) HC(J,I,KQ)=HHNEW
  300 CONTINUE
```

```
  310 CONTINUE
C
C15-----RETURN
      RETURN
      END
```

List of Variables for Module IBS1BD

Variable	Range	Definition
BUFF	Global	DIMENSION (NCOL,NROW,NLAY), Buffer used to accumulate information before printing or recording it.
DELC	Global	DIMENSION (NROW), Cell dimensions in the column direction. DELC(I) contains the width of row I.
DELR	Global	DIMENSION (NCOL), Cell dimensions in the row direction. DELC(J) contains the width of column J.
DELT	Global	Length of the current time step.
HC	Package	DIMENSION (NCOL,NROW,NAQL), Preconsolidation head in each cell with interbed storage. NAQL is number of layers for which IBQ > 0.
HHC	Module	Temporary storage of HC(I,J,KQ).
HHNEW	Module	Temporary storage of HNEW(J,I,K) (single precision).
HHOLD	Module	Temporary storage of HOLD(J,I,K).
HNEW	Global	DIMENSION (NCOL,NROW,NLAY), Most recent estimate of head in each cell. HNEW changes at each iteration.
HOLD	Global	DIMENSION (NCOL,NROW,NLAY), Head at the start of the current time step.
I	Module	Index for rows.
IBD	Module	Flag. = 0, cell-by-cell flow terms for this package will not be recorded. ≠ 0, cell-by-cell flow terms for this package will be recorded.
IBOUND	Global	DIMENSION (NCOL,NROW,NLAY), Status of each cell. < 0, constant-head cell = 0, inactive cell > 0, variable-head cell

Variable	Range	Definition
IBQ	Package	DIMENSION (80), Layer flag for interbed storage: > 0, Layer has interbed storage. ≤ 0, Layer does not have interbed storage.
IC	Module	Index for columns.
ICBCFL	Global	Flag. = 0, cell-by-cell flow terms will not be recorded or printed for the current time step. ≠ 0, cell-by-cell flow terms will be recorded for the current time step.
IIBSCB	Package	Flag and a unit number. > 0, it is the unit number on which cell-by-cell flow terms will be recorded whenever ICBCFL (see McDonald and Harbaugh, 1988, chap. 4, p. 14-15) is set. ≤ 0, cell-by-cell flow terms will not be recorded.
IL	Module	Index for layers.
IOUT	Global	Primary unit number for all printed output. IOUT = 6.
IR	Module	Index for rows.
J	Module	Index for columns.
K	Module	Index for layers.
KPER	Global	Stress period counter.
KQ	Module	Index of model layers with interbed storage.
KSTP	Global	Time step counter. Reset at the start of each stress period.
MSUM	Global	Counter for budget entries and labels in VBVL and VBNM.
NCOL	Global	Number of columns in the grid.
NLAY	Global	Number of layers in the grid.
NROW	Global	Number of rows in the grid.
SBGN	Module	Storage capacity for cell at beginning of time step.
SCE	Package	DIMENSION (NCOL,NROW,NAQL), Elastic storage capacity of each cell with interbed storage. NAQL is number of layers for which IBQ > 0.

Variable	Range	Definition
SCV	Package	DIMENSION (NCOL,NROW,NAQL), Inelastic storage capacity of each cell with interbed storage. NAQL is number of layers for which IBQ > 0.
SEND	Module	Storage capacity for cell at end of time step.
STOIN	Module	Sum of decreases in interbed storage from individual cells.
STOUT	Module	Sum of increases in interbed storage from individual cells.
STRG	Module	Volume of flow into or out interbed storage for a single cell.
SUB	Package	DIMENSION (NCOL,NROW,NAQL), Computed compaction of each cell with interbed storage. NAQL is number of layers for which IBQ > 0.
TEXT	Module	Labels recorded along with cell-by-cell flow terms.
TLED	Module	Reciprocal of the length of the current time step.
VBNM	Global	DIMENSION(4,20), Labels for entries in the volumetric budget.
VBVL	Global	DIMENSION(4,20), Entries for the volumetric budget. For flow component N, the values in VBVL are: (1,N), Rate for the current time step into the flow field. (2,N), Rate for the current time step out of the flow field. (3,N), Volume into the flow field during the simulation. (4,N), Volume out of the flow field during the simulation.

IBS1OT

Narrative for Module IBS1OT

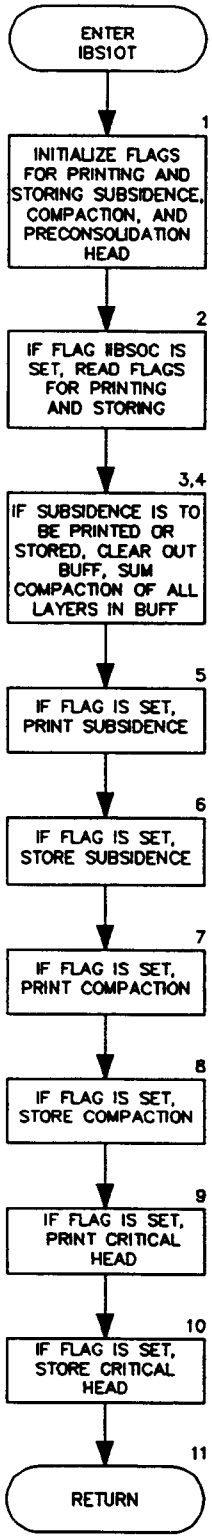
This module prints and stores subsidence, compaction, and preconsolidation head.

1. Initialize flags for printing and saving subsidence, compaction, and preconsolidation head.
2. Read flags for printing and saving subsidence, compaction, and preconsolidation head. Print flags.
3. Prepare to print and store subsidence. First, clear out buffer (BUFF).
4. Sum compaction in all layers to compute total subsidence.
5. If flag is set, print subsidence array with module ULAPRS or ULAPRW.
6. If flag is set, store subsidence array with module ULASAV.
7. If ICOMPR flag is set, print compaction array (all layers) with module ULAPRS or ULAPRW.
8. If ICOMSV flag is set, save compaction array (all layers) with module ULASAV.
9. If IHCPR flag is set, print preconsolidation-head array (all layers) with module ULAPRS or ULAPRW.
10. If IHCSV flag is set, save preconsolidation-head array (all layers) with module ULASAV.
11. RETURN.

IIBSOC is a flag.
If IIBSOC > 0, output control flags for printing and recording subsidence, compaction, and preconsolidation head will be read each time step.
If IIBSOC ≤ 0, output control flags will not be read.

BUFF is an array in which values are stored as they are being gathered for printing or recording.

Flow chart for module IBS1OT



Program Listing for Module IBS1OT

```
      SUBROUTINE IBS1OT(NCOL,NROW,NLAY,PERTIM,TOTIM,KSTP,KPER,NSTP,
     1           BUFF,SUB,HC,IIBSOC,ISUBFM,ICOMFM,IHCFM,ISUBUN,
     2           ICOMUN,IHCUN,IN,IOUT)
C-----VERSION 1227 02JUN1988 IBS1OT
C     ******************************************************************
C     PRINT AND STORE SUBSIDENCE, COMPACTION AND CRITICAL HEAD.
C     ******************************************************************
C
C     SPECIFICATIONS:
C     ------------------------------------------------------------------
      CHARACTER*4 TEXT
      DIMENSION HC(NCOL,NROW,NLAY),SUB(NCOL,NROW,NLAY),
     1          BUFF(NCOL,NROW,NLAY),TEXT(4,3)
      COMMON /IBSCOM/ IBQ(80)
      DATA TEXT(1,1),TEXT(2,1),TEXT(3,1),TEXT(4,1) /'    ',' SU',
     1     'BSID','ENCE'/,TEXT(1,2),TEXT(2,2),TEXT(3,2),TEXT(4,2)
     2     /'    ',' CO','MPAC','TION'/,TEXT(1,3),TEXT(2,3),
     3     TEXT(3,3),TEXT(4,3) /'  C','RITI','CAL ','HEAD'/
C     ------------------------------------------------------------------
C
C1------INITIALIZE FLAGS FOR PRINTING AND SAVING SUBSIDENCE, COMPACTION,
C1------AND CRITICAL HEAD
      ISUBPR=0
      ICOMPR=0
      IHCPR=0
      ISUBSV=0
      ICOMSV=0
      IHCSV=0
      IF(KSTP.EQ.NSTP) ISUBPR=1
C2------READ FLAGS FOR PRINTING AND SAVING.
      IF(IIBSOC.LE.0) GO TO 28
      READ(IN,10) ISUBPR,ICOMPR,IHCPR,ISUBSV,ICOMSV,IHCSV
   10 FORMAT(6I10)
      WRITE(IOUT,15) ISUBPR,ICOMPR,IHCPR,ISUBSV,ICOMSV,IHCSV
   15 FORMAT(1HO,'FLAGS FOR PRINTING AND STORING SUBSIDENCE, '
     1 'COMPACTION, AND CRITICAL HEAD:'/
     2 '   ISUBPR    ICOMPR    IHCPR     ISUBSV    ICOMSV    IHCSV   '/
     3 ' ------------------------------------------------------------'/
     4 I6,5I10)
C
C3------PRINT AND STORE SUBSIDENCE, FIRST, CLEAR OUT BUFF.
   28 IF(ISUBPR.LE.0.AND.ISUBSV.LE.0) GO TO 100
      DO 30 IR=1,NROW
      DO 30 IC=1,NCOL
      BUFF(IC,IR,1)=0.
   30 CONTINUE
C
C4------SUM COMPACTION IN ALL LAYERS TO GET SUBSIDENCE.
      KQ=0
      DO 50 K=1,NLAY
      IF(IBQ(K).EQ.0) GO TO 50
```

```
      KQ=KQ+1
      DO 40 I=1,NROW
      DO 40 J=1,NCOL
      BUFF(J,I,1)=BUFF(J,I,1)+SUB(J,I,KQ)
   40 CONTINUE
   50 CONTINUE
C
C5------PRINT SUBSIDENCE.
      IF(ISUBPR.LE.0) GO TO 60
      IF(ISUBFM.LT.0) CALL ULAPRS(BUFF,TEXT(1,1),KSTP,KPER,NCOL,NROW,1,
     1          -ISUBFM,IOUT)
      IF(ISUBFM.GE.0) CALL ULAPRW(BUFF,TEXT(1,1),KSTP,KPER,NCOL,NROW,1,
     1           ISUBFM,IOUT)
C
C6------STORE SUBSIDENCE.
   60 IF(ISUBSV.LE.0) GO TO 100
      CALL ULASAV(BUFF,TEXT(1,1),KSTP,KPER,PERTIM,TOTIM,NCOL,NROW,1,
     1             ISUBUN)
C
C7------PRINT COMPACTION FOR ALL LAYERS WITH INTERBED STORAGE.
  100 IF(ICOMPR.LE.0) GO TO 140
      KQ=0
      DO 130 K=1,NLAY
      IF(IBQ(K).LE.0) GO TO 130
      KQ=KQ+1
      IF(ICOMFM.LT.0) CALL ULAPRS(SUB(1,1,KQ),TEXT(1,2),KSTP,KPER,NCOL,
     1          NROW,K,-ICOMFM,IOUT)
      IF(ICOMFM.GE.0) CALL ULAPRW(SUB(1,1,KQ),TEXT(1,2),KSTP,KPER,NCOL,
     1           NROW,K,ICOMFM,IOUT)
  130 CONTINUE
C
C8------SAVE COMPACTION FOR ALL LAYERS WITH INTERBED STORAGE.
  140 IF(ICOMSV.LE.0) GO TO 200
      KQ=0
      DO 160 K=1,NLAY
      IF(IBQ(K).LE.0) GO TO 160
      KQ=KQ+1
      CALL ULASAV(SUB(1,1,KQ),TEXT(1,2),KSTP,KPER,PERTIM,TOTIM,NCOL,
     1            NROW,K,ICOMUN)
  160 CONTINUE
C
C9------PRINT CRITICAL HEAD FOR ALL LAYERS WITH INTERBED STORAGE.
  200 IF(IHCPR.LE.0) GO TO 240
      KQ=0
      DO 230 K=1,NLAY
      IF(IBQ(K).LE.0) GO TO 230
      KQ=KQ+1
      IF(IHCFM.LT.0) CALL ULAPRS(HC(1,1,KQ),TEXT(1,3),KSTP,KPER,NCOL,
     1          NROW,K,-IHCFM,IOUT)
      IF(IHCFM.GE.0) CALL ULAPRW(HC(1,1,KQ),TEXT(1,3),KSTP,KPER,NCOL,
     1           NROW,K,IHCFM,IOUT)
  230 CONTINUE
C
C10-----SAVE CRITICAL HEAD FOR ALL LAYERS WITH INTERBED STORAGE.
```

```
  240 IF(IHCSV.LE.0) GO TO 300
      KQ=0
      DO 260 K=1,NLAY
      KQ=KQ+1
      IF(IBQ(K).LE.0) GO TO 260
      CALL ULASAV(HC(1,1,KQ),TEXT(1,3),KSTP,KPER,PERTIM,TOTIM,NCOL,
     1            NROW,K,IHCUN)
  260 CONTINUE
C
C11-----RETURN
  300 RETURN
      END
```

List of Variables for Module IBS1OT

Variable	Range	Definition
BUFF	Global	DIMENSION (NCOL,NROW,NLAY), Buffer used to accumulate information before printing or recording it.
HC	Package	DIMENSION (NCOL,NROW,NAQL), Preconsolidation head in each cell with interbed storage. NAQL is number of layers for which IBQ > 0.
I	Module	Index for rows.
IBQ	Package	DIMENSION (80), Layer flag for interbed storage: > 0, Layer has interbed storage. ≤ 0, Layer does not have interbed storage.
IC	Module	Index for columns.
ICOMFM	Package	Code for format in which compaction will be printed.
ICOMPR	Module	Output flag for printing entire compaction array for current time step. ≤ 0, compaction is not printed. > 0, compaction is printed.
ICOMSV	Module	Output flag for recording entire compaction array on disk for current time step. ≤ 0, compaction is not recorded. > 0, compaction is recorded.
ICOMUN	Package	Unit number on which an unformatted record containing compaction should be recorded.
IHCFM	Package	Code for format in which preconsolidation head will be printed.

Variable	Range	Definition
IHCPR	Module	Output flag for printing entire preconsolidation-head array for current time step. ≤ 0, preconsolidation head is not printed. > 0, preconsolidation head is printed.
IHCSV	Module	Output flag for recording entire preconsolidation-head array on disk for current time step. ≤ 0, preconsolidation head is not recorded. > 0, preconsolidation head is recorded.
IHCUN	Package	Unit number on which an unformatted record containing preconsolidation head should be recorded.
IIBSOC	Package	Flag. > 0, output control will be read each time step for printing and recording subsidence, compaction, and preconsolidation head. ≤ 0, subsidence will be printed at the end of each stress period.
IN	Package	Primary unit number from which input for this package will be read.
IOUT	Global	Primary unit number for all printed output. IOUT = 6.
IR	Module	Index for rows.
ISUBFM	Package	Code for format in which subsidence will be printed.
ISUBPR	Module	Output flag for printing entire subsidence array for current time step. ≤ 0, subsidence is not printed. > 0, subsidence is printed.
ISUBSV	Module	Output flag for recording entire subsidence array on disk for current time step. ≤ 0, subsidence is not recorded. > 0, subsidence is recorded.
ISUBUN	Package	Unit number on which an unformatted record containing subsidence should be recorded.
J	Module	Index for columns.
K	Module	Index for layers.

Variable	Range	Definition
KPER	Global	Stress period counter.
KQ	Module	Index of model layers with interbed storage.
KSTP	Global	Time step counter. Reset at the start of each stress period.
MSUM	Global	Counter for budget entries and labels in VBVL and VBNM.
NCOL	Global	Number of columns in the grid.
NLAY	Global	Number of layers in the grid.
NROW	Global	Number of rows in the grid.
NSTP	Global	Number of time steps in the current stress period.
PERTIM	Global	Elapsed time during the current stress period.
SUB	Package	DIMENSION (NCOL,NROW,NAQL), Computed compaction of each cell with interbed storage. NAQL is number of layers for which IBQ > 0.
TEXT	Module	DIMENSION(4,3), Labels to be printed or recorded with array data.
TOTIM	Global	Elapsed time in the simulation.

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APPENDIXES A–D

APPENDIX A

INPUT DATA SETS FOR STORAGE-DEPLETION TEST PROBLEM

The storage-depletion test problem tests the ability of the Interbed-Storage Package to compute compaction and storage changes from a specified head decline. Details of the test problem and results are discussed in the section entitled "Test Problems."

Basic Package Input Data Set

Input for the Basic Package follows the column numbers. The input consists of 53 records, all of which are shown below and on the following page. Input for this package is read from the unit number for Basic Package input specified in the MAIN program.

```
                                        Column Numbers
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
MODULAR MODEL STORAGE-DEPLETION TEST PROBLEM--TESTS CALCULATION OF ULTIMATE
COMPACTION OF INTERBEDS
         2        10        12         3         4
 7  0  0  0  0  0  0  0 10  0  0 11  0  0  0  0  0  0 13
         0         1    IAPART,ISTRT
         5         1(20I3)
-1  1  1  1  1  1  1  1  1  1  1 -1
-1  1  1  1  1  1  1  1  1  1  1 -1
-1  1  1  1  1  1  1  1  1  1  1 -1
-1  1  1  1  1  1  1  1  1  1  1 -1
-1  1  1  1  1  1  1  1  1  1  1 -1
-1  1  1  1  1  1  1  1  1  1  1 -1
-1  1  1  1  1  1  1  1  1  1  1 -1
-1  1  1  1  1  1  1  1  1  1  1 -1
-1  1  1  1  1  1  1  1  1  1  1 -1
-1  1  1  1  1  1  1  1  1  1  1 -1
         5         1(20I3)
-1  1  1  1  1  1  1  1  1  1  1 -1
-1  1  1  1  1  1  1  1  1  1  1 -1
-1  1  1  1  1  1  1  1  1  1  1 -1
-1  1  1  1  1  1  1  1  1  1  1 -1
-1  1  1  1  1  1  1  1  1  1  1 -1
-1  1  1  1  1  1  1  1  1  1  1 -1
-1  1  1  1  1  1  1  1  1  1  1 -1
-1  1  1  1  1  1  1  1  1  1  1 -1
-1  1  1  1  1  1  1  1  1  1  1 -1
-1  1  1  1  1  1  1  1  1  1  1 -1
```

Input for the Basic Package is continued following the column numbers.

```
                                        Column Numbers
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
    444.44      HNOFLO
         5          1.(18F3.0)
 0 11 12 13 14 15 16 17 18 19 20 11
 0 11 12 13 14 15 16 17 18 19 20 11
 0 11 12 13 14 15 16 17 18 19 20 11
 0 11 12 13 14 15 16 17 18 19 20 11
 0 11 12 13 14 15 16 17 18 19 20 11
 0 11 12 13 14 15 16 17 18 19 20 11
 0 11 12 13 14 15 16 17 18 19 20 11
 0 11 12 13 14 15 16 17 18 19 20 11
 0 11 12 13 14 15 16 17 18 19 20 11
 0 11 12 13 14 15 16 17 18 19 20 11
         5          1.(18F3.0)
 0 11 12 13 14 15 16 17 18 19 20 11
 0 11 12 13 14 15 16 17 18 19 20 11
 0 11 12 13 14 15 16 17 18 19 20 11
 0 11 12 13 14 15 16 17 18 19 20 11
 0 11 12 13 14 15 16 17 18 19 20 11
 0 11 12 13 14 15 16 17 18 19 20 11
 0 11 12 13 14 15 16 17 18 19 20 11
 0 11 12 13 14 15 16 17 18 19 20 11
 0 11 12 13 14 15 16 17 18 19 20 11
 0 11 12 13 14 15 16 17 18 19 20 11
    1000.          10        1.5
    1000.          10        1.5
    1000.          10        1.5
```

Block-Centered Flow Package Input Data Set

Input for the Block-Centered Flow Package follows the column numbers. The input consists of 10 records, all of which are shown below. Input for this package is read from unit number 7 as specified in the data set for the Basic Package.

```
                                        Column Numbers
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
         0         0   ISS  IBCFCB
 0 0
         0        1.                                          TRPY
         0     1000.                                          DELR
         0     1000.                                          DELC
         0     .0001                                          sfl -layer 1
         0     1000.                                          tran -layer 1
         0  0.000001                                          VCONT -layer 1
         0     .0001                                          sfl -layer 2
         0     1000.                                          tran -layer2
```

Strongly Implicit Procedure Package Input Data Set

Input for the Strongly-Implicit Procedure Package follows the column numbers. The input consists of two records, both of which are shown below. Input for this package is read from unit number 8 as specified in the data set for the Basic Package.

```
                                        Column Numbers
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
        50         5
        1.    .00001         1         0         1
```

Interbed-Storage Package Input Data Set

Input for the Interbed-Storage Package follows the column numbers. The input consists of 16 records, all of which are shown below. Input for this package is read from unit number 13 as specified in the data set for the Basic Package.

```
                                        Column Numbers
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
        0         0
 1 0
       13        1.(12F3.0)
  5  6  7  8  9 10 11 12 13 14 15 16
  5  6  7  8  9 10 11 12 13 14 15 16
  5  6  7  8  9 10 11 12 13 14 15 16
  5  6  7  8  9 10 11 12 13 14 15 16
  5  6  7  8  9 10 11 12 13 14 15 16
  5  6  7  8  9 10 11 12 13 14 15 16
  5  6  7  8  9 10 11 12 13 14 15 16
  5  6  7  8  9 10 11 12 13 14 15 16
  5  6  7  8  9 10 11 12 13 14 15 16
  5  6  7  8  9 10 11 12 13 14 15 16
        0      .0001
        0       .001
        0        0.0
```

Output Control Package Input Data Set

Input for the Output Control Package follows the column numbers. The input consists of 61 records, 21 of which are shown below. Input for this package is read from unit number 11 as specified in the data set for the Basic Package.

Column Numbers

```
        1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
        5         5         0         0
        0         0         0         0
        0         0         0         0
        0         0         0         0
        0         0         0         0
        0         0         0         0
        0         0         0         0
        0         0         0         0
        0         0         0         0
        0         0         0         0
        0         0         0         0
        0         0         0         0
        0         0         0         0
        0         0         0         0
        0         0         0         0
        0         0         0         0
        0         0         0         0
        0         0         0         0
        0         0         0         0
        0         0         0         0
        0         1         0         0
        1         0         0         0
   (include 2 additional sets of the above 20 records)
```

APPENDIX B

INPUT DATA SETS FOR RAMP-LOAD TEST PROBLEM

The ramp-load test problem tests the ability of the Interbed-Storage Package to compute compaction consistent with a one-dimensional compaction model by Helm (1975, 1984). Details of the test problem and results are discussed in the section entitled "Ramp-Load Test Problem."

Basic Package Input Data Set

Input for the Basic Package follows the column numbers. The input consists of 109 records, 10 of which are shown below. Input for this package is read from the unit number for Basic Package input specified in the MAIN program.

```
                                  Column Numbers
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
Test of IBS1 package--Ramp loading problem for comparison with model by
Helm (1975, 1984)
         1         1        21       100         4
 07 00 00 00 00 00 00 00 08 00 00 00 00 00 00 00 00 00 12 11
         0         0
         5         1(40I2)
-1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
      999.
         0       10.
       18.        18       1.0
  (include 99 additional records identical to the above record)
```

Block-Centered Flow Package Input Data Set

Input for the Block-Centered Flow Package follows the column numbers. The input consists of eight records, all of which are shown below. Input for this package is read from unit number 7 as specified in the data set for the Basic Package.

```
                                  Column Numbers
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
         0         0
 0
         0         1
         0      0.25
         0      1.00
         0      0.00
         7    1.E-06(16F5.0)
1.E10 2.78 2.78 2.78 2.78 2.78 2.78 2.78 2.78 2.78 2.78 2.78 2.78 2.78 2.78 2.78
```

Strongly Implicit Procedure Package Input Data Set

Input for the Strongly Implicit Procedure Package follows the column numbers. The input consists of two records, both of which are shown below. Input for this package is read from unit number 8 as specified in the data set for the Basic Package.

```
                                    Column Numbers
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
        50         5
        1.      .001         1         0       999
```

Interbed-Storage Package Input Data Set

Input for the Interbed-Storage Package follows the column numbers. The input consists of eight records, all of which are shown below. Input for this package is read from unit number 12 as specified in the data set for the Basic Package.

```
                                    Column Numbers
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
         0         0
 1
         0       10.
        12    1.0E-6(40F2.0)
 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
        12    1.0E-4(40F2.0)
 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
         0        0.
```

Time-Variant Specified-Head Package Input Data Set

Input for the Time-Variant Specified-Head Package follows the column numbers. The input consists of 201 records, 41 of which are shown below. Input for this package is read from unit number 11 as specified in the data set for the Basic Package. Input instructions for the Time-Variant Specified-Head Package are given in appendix C.

```
                                   Column Numbers
        1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
        1
        1
        1         1         1      10.0      9.0
        1
        1         1         1       9.0      8.0
        1
        1         1         1       8.0      7.0
        1
        1         1         1       7.0      6.0
        1
        1         1         1       6.0      5.0
        1
        1         1         1       5.0      4.0
        1
        1         1         1       4.0      3.0
        1
        1         1         1       3.0      2.0
        1
        1         1         1       2.0      1.0
        1
        1         1         1       1.0      0.0
        1
        1         1         1       0.0      1.0
        1
        1         1         1       1.0      2.0
        1
        1         1         1       2.0      3.0
        1
        1         1         1       3.0      4.0
        1
        1         1         1       4.0      5.0
        1
        1         1         1       5.0      6.0
        1
        1         1         1       6.0      7.0
        1
        1         1         1       7.0      8.0
        1
        1         1         1       8.0      9.0
        1
        1         1         1       9.0     10.0
(include 4 additional sets of the above 40 records)
```

APPENDIX C

TIME-VARIANT SPECIFIED-HEAD PACKAGE

The Time-Variant Specified-Head Package was developed to allow constant-head cells to take on different values for each time step. It provides a means of simulating head variations with time that can be approximated with piecewise linear segments. The Time-Variant Specified-Head Package does not alter the way constant-head boundaries are formulated in the finite-difference equations of the model by McDonald and Harbaugh (1988). Because the package is not a complex modification of the flow model, documentation in this appendix is the minimum amount required to implement the package. Additional information regarding the Time-Variant Specified-Head Package is given in the section entitled "Ramp-Load Test Problem." Modifications to the main program to implement the package are given in Appendix D.

As the package is presented here, a specified number of cells are defined for each stress period. The package sets the element in the IBOUND array to a negative value for all cells where a time-variant specified-head boundary is selected. Because layer, row, and column locations for cells having a time-variant specified head are read for each stress period, the opportunity exists to specify different sets of cells for each stress period. If a cell is specified as a time-variant specified-head boundary for a stress period and omitted in the specification for a subsequent period, it remains a constant-head boundary with a head equal to the end head of the last stress period for which it was specified.

The Time-Variant Specified-Head Package includes three modules. The first module, CHD1AL, allocates space for an array that is needed for the package. The second module, CHD1RP, reads boundary-cell indices and head values at the start of each stress period and sets corresponding elements in the IBOUND array to negative values. The third module, CHD1FM, linearly interpolates boundary head for each time-variant specified-head boundary cell and sets the corresponding elements in the HNEW and HOLD arrays to the interpolated values. As defined by McDonald and Harbaugh (1988), the Formulate Procedure calculates conductances and coefficients required by the finite-difference flow equation. Module CHD1FM is considered to be part of the Formulate Procedure because its function is to calculate values for the known heads, which are components of the flow equation. The flowchart for the model (McDonald and Harbaugh, 1988, chap. 3, p. 2) shows that modules in the Formulate Procedure are called each iteration during the iterative solution process; however, module CHD1FM is called outside the iterative program loop. CHD1FM is called only once each time step because the specified heads are constant during a time step.

Listing of Input Data for Streamflow-Routing Package

Input for the Streamflow-Routing Package follows the column numbers. The input consists of 158 records (lines). Input for package is read from FORTRAN unit number 15 as specified in the data set for the Basic Package.

Column Numbers

```
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
        13         1         0         0         0         0        -1         0
        13         0         0
    1   1  20   1   1              0.01      50.0    37.037       0.0       0.0
    1   2  20   1   2                        50.0    37.037       0.0       0.0
    1   3  20   1   3                        50.0    37.037       0.0       0.0
    1   4  20   1   4                        50.0    37.037       0.0       0.0
    1   5  20   1   5                        50.0    37.037       0.0       0.0
    1   6  20   1   6                        50.0    37.037       0.0       0.0
    1   7  20   1   7                        50.0    37.037       0.0       0.0
    1   8  20   1   8                        50.0    37.037       0.0       0.0
    1   9  20   1   9                        50.0    37.037       0.0       0.0
    1  10  20   1  10                        50.0    37.037       0.0       0.0
    1  11  20   1  11                        50.0    37.037       0.0       0.0
    1  12  20   1  12                        50.0    37.037       0.0       0.0
    1  13  20   1  13                        50.0    37.037       0.0       0.0
       -1         0         0
previous line repeated 142 times
```

Listing of Input Data for Recharge Package

Input for the Recharge Package follows the column numbers. The input consists of 289 records (lines). Input for package is read from FORTRAN unit number 11 as specified in the data set for the Basic Package.

Column Numbers

```
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
         1         0
         1
         02.8935E-09
         1
         02.7778E-08
         1
         07.1181E-08
         1
         01.2095E-07
         1
         01.6435E-07
```

Listing of Input Data for Recharge Package (continued)

Column Numbers

```
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
         1
         01.8924E-07
         1
         01.8924E-07
         1
         01.6435E-07
         1
         01.2095E-07
         1
         07.1181E-08
         1
         02.7778E-08
         1
         02.8935E-09
         1
         0       0.0
         1
         0      0.00
         1
         0      0.00
         1
         0      0.00
         1
         0      0.00
         1
         0      0.00
         1
         0       0.0
         1
         0      0.00
         1
         0      0.00
         1
         0      0.00
         1
         0      0.00
         1
         0      0.00
sequence of previous 48 lines repeated 5 times
```

SELECTED PAGES FROM PRINTED RESULTS FOR TEST PROBLEM 1

```
                U.S. GEOLOGICAL SURVEY MODULAR FINITE-DIFFERENCE GROUND-WATER MODEL

TEST PROBLEM 1--CONSTANT STREAM STAGE AND VARIABLE RECHARGE TO AQUIFER:

   1 LAYERS        13 ROWS        39 COLUMNS
144 STRESS PERIOD(S) IN SIMULATION
MODEL TIME UNIT IS SECONDS

I/O UNITS:
ELEMENT OF IUNIT:  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
       I/O UNIT:  7  0  0  0  0  0  0 11 13  0  0 14 15  0  0  0  0  0  0  0  0  0  0  0

BAS1 -- BASIC MODEL PACKAGE, VERSION 1, 9/1/87 INPUT READ FROM UNIT  5
ARRAYS RHS AND BUFF WILL SHARE MEMORY.
START HEAD WILL BE SAVED
    4619 ELEMENTS IN X ARRAY ARE USED BY BAS
    4619 ELEMENTS OF X ARRAY USED OUT OF  200000

BCF1 -- BLOCK-CENTERED FLOW PACKAGE, VERSION 1, 9/1/87 INPUT READ FROM UNIT  7
TRANSIENT SIMULATION
     LAYER  AQUIFER TYPE
     ------------------
        1         0
     508 ELEMENTS IN X ARRAY ARE USED BY BCF
    5127 ELEMENTS OF X ARRAY USED OUT OF  200000

RCH1 -- RECHARGE PACKAGE, VERSION 1, 9/1/87 INPUT READ FROM UNIT 11
OPTION 1 -- RECHARGE TO TOP LAYER
     507 ELEMENTS OF X ARRAY USED FOR RECHARGE
    5634 ELEMENTS OF X ARRAY USED OUT OF  200000

STRM -- STREAM PACKAGE, VERSION 1, 10/23/87 INPUT READ FROM UNIT 15
MAXIMUM OF   13 STREAM NODES

NUMBER OF STREAM SEGMENTS IS     1

NUMBER OF STREAM TRIBUTARIES IS     0


     210 ELEMENTS IN X ARRAY ARE USED FOR STREAMS
    5844 ELEMENTS OF X ARRAY USED OUT OF 200000

SIP1 -- STRONGLY IMPLICIT PROCEDURE SOLUTION PACKAGE, VERSION 1, 9/1/87 INPUT READ FROM UNIT 13
MAXIMUM OF 150 ITERATIONS ALLOWED FOR CLOSURE
 5 ITERATION PARAMETERS
    2633 ELEMENTS IN X ARRAY ARE USED BY SIP
    8477 ELEMENTS OF X ARRAY USED OUT OF  200000
```

STRESS PERIOD NO. 121, LENGTH = 1296000.

NUMBER OF TIME STEPS = 2

MULTIPLIER FOR DELT = 1.500

INITIAL TIME STEP SIZE = 518400.0

RECHARGE = 0.2893500E-08

REUSING STREAM NODES FROM LAST STRESS PERIOD

4 ITERATIONS FOR TIME STEP 1 IN STRESS PERIOD121

HEAD/DRAWDOWN PRINTOUT FLAG = 0 TOTAL BUDGET PRINTOUT FLAG = 0 CELL-BY-CELL FLOW TERM FLAG = 0
OUTPUT FLAGS FOR ALL LAYERS ARE THE SAME:

HEAD PRINTOUT	DRAWDOWN PRINTOUT	HEAD SAVE	DRAWDOWN SAVE
0	0	0	0

4 ITERATIONS FOR TIME STEP 2 IN STRESS PERIOD121
MAXIMUM HEAD CHANGE FOR EACH ITERATION:

HEAD CHANGE	LAYER,ROW,COL	HEAD CHANGE	LAYER,ROW,COL	HEAD CHANGE	LAYER,ROW,COL	HEAD CHANGE	LAYER,ROW,COL	HEAD CHANGE	LAYER,ROW,COL
-0.1584	(1, 13, 39)	-0.2075E-01	(1, 2, 2)	-0.1028E-02	(1, 6, 4)	-0.1414E-04	(1, 13, 1)		

HEAD/DRAWDOWN PRINTOUT FLAG = 1 TOTAL BUDGET PRINTOUT FLAG = 1 CELL-BY-CELL FLOW TERM FLAG = 1
OUTPUT FLAGS FOR ALL LAYERS ARE THE SAME:

HEAD PRINTOUT	DRAWDOWN PRINTOUT	HEAD SAVE	DRAWDOWN SAVE
1	0	0	0

LAYER	ROW	COLUMN	STREAM NUMBER	REACH NUMBER	FLOW INTO STREAM REACH	FLOW INTO AQUIFER	FLOW OUT OF STREAM REACH
1	1	20	1	1	0.100E-01	-.237	0.247
1	2	20	1	2	0.247	-.237	0.484
1	3	20	1	3	0.484	-.237	0.721
1	4	20	1	4	0.721	-.237	0.958
1	5	20	1	5	0.958	-.237	1.20
1	6	20	1	6	1.20	-.237	1.43
1	7	20	1	7	1.43	-.237	1.67
1	8	20	1	8	1.67	-.237	1.91
1	9	20	1	9	1.91	-.237	2.14
1	10	20	1	10	2.14	-.237	2.38
1	11	20	1	11	2.38	-.237	2.62
1	12	20	1	12	2.62	-.237	2.85
1	13	20	1	13	2.85	-.237	3.09

HEAD IN LAYER 1 AT END OF TIME STEP 2 IN STRESS PERIOD121

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
	31	32	33	34	35	36	37	38	39						
1	58.049	57.957	57.836	57.666	57.450	57.188	56.882	56.534	56.145	55.719	55.257	54.762	54.237	53.686	53.111
	52.516	51.905	51.281	50.647	50.006	50.647	51.281	51.905	52.516	53.111	53.686	54.237	54.762	55.257	55.719
	56.145	56.534	56.882	57.188	57.450	57.666	57.836	57.957	58.049						
2	58.049	57.957	57.836	57.666	57.450	57.188	56.882	56.534	56.145	55.719	55.257	54.762	54.237	53.686	53.111
	52.516	51.905	51.281	50.647	50.006	50.647	51.281	51.905	52.516	53.111	53.686	54.237	54.762	55.257	55.719
	56.145	56.534	56.882	57.188	57.450	57.666	57.836	57.957	58.049						
3	58.049	57.957	57.836	57.666	57.450	57.188	56.882	56.534	56.145	55.719	55.257	54.762	54.237	53.686	53.111
	52.516	51.905	51.281	50.647	50.006	50.647	51.281	51.905	52.516	53.111	53.686	54.237	54.762	55.257	55.719
	56.145	56.534	56.882	57.188	57.450	57.666	57.836	57.957	58.049						
4	58.049	57.957	57.836	57.666	57.450	57.188	56.882	56.534	56.145	55.719	55.257	54.762	54.237	53.686	53.111
	52.516	51.905	51.281	50.647	50.006	50.647	51.281	51.905	52.516	53.111	53.686	54.237	54.762	55.257	55.719
	56.145	56.534	56.882	57.188	57.450	57.666	57.836	57.957	58.049						
5	58.049	57.957	57.836	57.666	57.450	57.188	56.882	56.534	56.145	55.719	55.257	54.762	54.237	53.686	53.111
	52.516	51.905	51.281	50.647	50.006	50.647	51.281	51.905	52.516	53.111	53.686	54.237	54.762	55.257	55.719
	56.145	56.534	56.882	57.188	57.450	57.666	57.836	57.957	58.049						
6	58.049	57.957	57.836	57.666	57.450	57.188	56.882	56.534	56.145	55.719	55.257	54.762	54.237	53.686	53.111
	52.516	51.905	51.281	50.647	50.006	50.647	51.281	51.905	52.516	53.111	53.686	54.237	54.762	55.257	55.719
	56.145	56.534	56.882	57.188	57.450	57.666	57.836	57.957	58.049						
7	58.049	57.957	57.836	57.666	57.450	57.188	56.882	56.534	56.145	55.719	55.257	54.762	54.237	53.686	53.111
	52.516	51.905	51.281	50.647	50.006	50.647	51.281	51.905	52.516	53.111	53.686	54.237	54.762	55.257	55.719
	56.145	56.534	56.882	57.188	57.450	57.666	57.836	57.957	58.049						
8	58.049	57.957	57.836	57.666	57.450	57.188	56.882	56.534	56.145	55.719	55.257	54.762	54.237	53.686	53.111
	52.516	51.905	51.281	50.647	50.006	50.647	51.281	51.905	52.516	53.111	53.686	54.237	54.762	55.257	55.719
	56.145	56.534	56.882	57.188	57.450	57.666	57.836	57.957	58.049						
9	58.049	57.957	57.836	57.666	57.450	57.188	56.882	56.534	56.145	55.719	55.257	54.762	54.237	53.686	53.111
	52.516	51.905	51.281	50.647	50.006	50.647	51.281	51.905	52.516	53.111	53.686	54.237	54.762	55.257	55.719
	56.145	56.534	56.882	57.188	57.450	57.666	57.836	57.957	58.049						
10	58.049	57.957	57.836	57.666	57.450	57.188	56.882	56.534	56.145	55.719	55.257	54.762	54.237	53.686	53.111
	52.516	51.905	51.281	50.647	50.006	50.647	51.281	51.905	52.516	53.111	53.686	54.237	54.762	55.257	55.719
	56.145	56.534	56.882	57.188	57.450	57.666	57.836	57.957	58.049						
11	58.049	57.957	57.836	57.666	57.450	57.188	56.882	56.534	56.145	55.719	55.257	54.762	54.237	53.686	53.111
	52.516	51.905	51.281	50.647	50.006	50.647	51.281	51.905	52.516	53.111	53.686	54.237	54.762	55.257	55.719
	56.145	56.534	56.882	57.188	57.450	57.666	57.836	57.957	58.049						
12	58.049	57.957	57.836	57.666	57.450	57.188	56.882	56.534	56.145	55.719	55.257	54.762	54.237	53.686	53.111
	52.516	51.905	51.281	50.647	50.006	50.647	51.281	51.905	52.516	53.111	53.686	54.237	54.762	55.257	55.719
	56.145	56.534	56.882	57.188	57.450	57.666	57.836	57.957	58.049						
13	58.049	57.957	57.836	57.666	57.450	57.188	56.882	56.534	56.145	55.719	55.257	54.762	54.237	53.686	53.111
	52.516	51.905	51.281	50.647	50.006	50.647	51.281	51.905	52.516	53.111	53.686	54.237	54.762	55.257	55.719
	56.145	56.534	56.882	57.188	57.450	57.666	57.836	57.957	58.049						

VOLUMETRIC BUDGET FOR ENTIRE MODEL AT END OF TIME STEP 2 IN STRESS PERIOD121

CUMULATIVE VOLUMES	L**3	RATES FOR THIS TIME STEP	L**3/T
IN:		IN:	
STORAGE =	0.41038E+09	STORAGE =	2.7859
CONSTANT HEAD =	0.00000	CONSTANT HEAD =	0.00000
RECHARGE =	0.77725E+09	RECHARGE =	0.30091
STREAM LEAKAGE =	0.00000	STREAM LEAKAGE =	0.00000
TOTAL IN =	0.11876E+10	TOTAL IN =	3.0868
OUT:		OUT:	
STORAGE =	0.41335E+09	STORAGE =	0.00000
CONSTANT HEAD =	0.00000	CONSTANT HEAD =	0.00000
RECHARGE =	0.00000	RECHARGE =	0.00000
STREAM LEAKAGE =	0.77386E+09	STREAM LEAKAGE =	3.0820
TOTAL OUT =	0.11872E+10	TOTAL OUT =	3.0820
IN - OUT =	0.42240E+06	IN - OUT =	0.48499E-02
PERCENT DISCREPANCY =	0.04	PERCENT DISCREPANCY =	0.16

TIME SUMMARY AT END OF TIME STEP 2 IN STRESS PERIOD121

	SECONDS	MINUTES	HOURS	DAYS	YEARS
TIME STEP LENGTH	777600.	12960.0	216.000	9.00000	0.246407E-01
STRESS PERIOD TIME	0.129600E+07	21600.0	360.000	15.0000	0.410678E-01
TOTAL SIMULATION TIME	0.156816E+09	0.261360E+07	43560.0	1815.00	4.96920

STRESS PERIOD NO. 134, LENGTH = 1296000.

NUMBER OF TIME STEPS = 2

MULTIPLIER FOR DELT = 1.500

INITIAL TIME STEP SIZE = 518400.0

RECHARGE = 0.0000000

REUSING STREAM NODES FROM LAST STRESS PERIOD

4 ITERATIONS FOR TIME STEP 1 IN STRESS PERIOD134

HEAD/DRAWDOWN PRINTOUT FLAG = 0 TOTAL BUDGET PRINTOUT FLAG = 0 CELL-BY-CELL FLOW TERM FLAG = 0
OUTPUT FLAGS FOR ALL LAYERS ARE THE SAME:

HEAD PRINTOUT	DRAWDOWN PRINTOUT	HEAD SAVE	DRAWDOWN SAVE
0	0	0	0

4 ITERATIONS FOR TIME STEP 2 IN STRESS PERIOD134

MAXIMUM HEAD CHANGE FOR EACH ITERATION:

HEAD CHANGE	LAYER,ROW,COL	HEAD CHANGE	LAYER,ROW,COL	HEAD CHANGE	LAYER,ROW,COL	HEAD CHANGE	LAYER,ROW,COL	HEAD CHANGE	LAYER,ROW,COL
-0.2121	(1, 13, 9)	-0.2933E-01	(1, 2, 8)	-0.1548E-02	(1, 7, 7)	-0.2070E-04	(1, 13, 1)		

HEAD/DRAWDOWN PRINTOUT FLAG = 1 TOTAL BUDGET PRINTOUT FLAG = 1 CELL-BY-CELL FLOW TERM FLAG = 1
OUTPUT FLAGS FOR ALL LAYERS ARE THE SAME:

HEAD PRINTOUT	DRAWDOWN PRINTOUT	HEAD SAVE	DRAWDOWN SAVE
1	0	0	0

LAYER	ROW	COLUMN	STREAM NUMBER	REACH NUMBER	FLOW INTO STREAM REACH	FLOW INTO AQUIFER	FLOW OUT OF STREAM REACH
1	1	20	1	1	0.100E-01	-.379	0.389
1	2	20	1	2	0.389	-.379	0.767
1	3	20	1	3	0.767	-.379	1.15
1	4	20	1	4	1.15	-.379	1.52
1	5	20	1	5	1.52	-.379	1.90
1	6	20	1	6	1.90	-.379	2.28
1	7	20	1	7	2.28	-.379	2.66
1	8	20	1	8	2.66	-.379	3.04
1	9	20	1	9	3.04	-.379	3.42
1	10	20	1	10	3.42	-.379	3.80
1	11	20	1	11	3.80	-.379	4.18
1	12	20	1	12	4.18	-.379	4.55
1	13	20	1	13	4.55	-.379	4.93

HEAD IN LAYER 1 AT END OF TIME STEP 2 IN STRESS PERIOD134

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
	31	32	33	34	35	36	37	38	39						
1	61.742	61.632	61.485	61.279	61.011	60.682	60.291	59.837	59.320	58.739	58.094	57.387	56.621	55.798	54.922
	54.000	53.039	52.047	51.034	50.010	51.034	52.047	53.039	54.000	54.922	55.798	56.621	57.387	58.094	58.739
	59.320	59.837	60.291	60.682	61.011	61.278	61.485	61.632	61.742						
2	61.742	61.632	61.485	61.279	61.011	60.682	60.291	59.837	59.320	58.739	58.094	57.387	56.621	55.798	54.922
	54.000	53.039	52.047	51.034	50.010	51.034	52.047	53.039	54.000	54.922	55.798	56.621	57.387	58.094	58.739
	59.320	59.837	60.291	60.682	61.011	61.278	61.485	61.632	61.742						
3	61.742	61.632	61.485	61.279	61.011	60.682	60.291	59.837	59.320	58.739	58.094	57.387	56.621	55.798	54.922
	54.000	53.039	52.047	51.034	50.010	51.034	52.047	53.039	54.000	54.922	55.798	56.621	57.387	58.094	58.739
	59.320	59.837	60.291	60.682	61.011	61.278	61.485	61.632	61.742						
4	61.742	61.632	61.485	61.279	61.011	60.682	60.291	59.837	59.320	58.739	58.094	57.387	56.621	55.798	54.922
	54.000	53.039	52.047	51.034	50.010	51.034	52.047	53.039	54.000	54.922	55.798	56.621	57.387	58.094	58.739
	59.320	59.837	60.291	60.682	61.011	61.278	61.485	61.632	61.742						
5	61.742	61.632	61.485	61.279	61.011	60.682	60.291	59.837	59.320	58.739	58.094	57.387	56.621	55.798	54.922
	54.000	53.039	52.047	51.034	50.010	51.034	52.047	53.039	54.000	54.922	55.798	56.621	57.387	58.094	58.739
	59.320	59.837	60.291	60.682	61.011	61.278	61.485	61.632	61.742						
6	61.742	61.632	61.485	61.278	61.011	60.682	60.291	59.837	59.320	58.739	58.094	57.387	56.621	55.798	54.922
	54.000	53.039	52.047	51.034	50.010	51.034	52.047	53.039	54.000	54.922	55.798	56.621	57.387	58.094	58.739
	59.320	59.837	60.291	60.682	61.011	61.278	61.485	61.632	61.742						
7	61.742	61.632	61.485	61.278	61.011	60.682	60.291	59.837	59.320	58.739	58.094	57.387	56.621	55.798	54.922
	54.000	53.039	52.047	51.034	50.010	51.034	52.047	53.039	54.000	54.922	55.798	56.621	57.387	58.094	58.739
	59.320	59.837	60.291	60.682	61.011	61.278	61.485	61.632	61.742						
8	61.742	61.632	61.485	61.278	61.011	60.682	60.291	59.837	59.320	58.739	58.094	57.387	56.621	55.798	54.922
	54.000	53.039	52.047	51.034	50.010	51.034	52.047	53.039	54.000	54.922	55.798	56.621	57.387	58.094	58.739
	59.320	59.837	60.291	60.682	61.011	61.278	61.485	61.632	61.742						
9	61.742	61.632	61.485	61.278	61.011	60.682	60.291	59.837	59.320	58.739	58.094	57.387	56.621	55.798	54.922
	54.000	53.039	52.047	51.034	50.010	51.034	52.047	53.039	54.000	54.922	55.798	56.621	57.387	58.094	58.739
	59.320	59.837	60.291	60.682	61.011	61.279	61.485	61.632	61.742						
10	61.742	61.632	61.485	61.278	61.011	60.682	60.291	59.837	59.320	58.739	58.094	57.387	56.621	55.798	54.922
	54.000	53.039	52.047	51.034	50.010	51.034	52.047	53.039	54.000	54.922	55.798	56.621	57.387	58.094	58.739
	59.320	59.837	60.291	60.682	61.011	61.279	61.485	61.632	61.742						
11	61.742	61.632	61.485	61.278	61.011	60.682	60.291	59.837	59.320	58.739	58.094	57.387	56.621	55.798	54.922
	54.000	53.039	52.047	51.034	50.010	51.034	52.047	53.039	54.000	54.922	55.798	56.621	57.387	58.094	58.739
	59.320	59.837	60.291	60.682	61.011	61.279	61.485	61.632	61.742						
12	61.742	61.632	61.485	61.278	61.011	60.682	60.291	59.837	59.320	58.739	58.094	57.387	56.621	55.798	54.922
	54.000	53.039	52.047	51.034	50.010	51.034	52.047	53.039	54.000	54.922	55.798	56.621	57.387	58.094	58.739
	59.320	59.837	60.291	60.682	61.011	61.279	61.485	61.632	61.742						
13	61.742	61.632	61.485	61.278	61.011	60.682	60.291	59.837	59.320	58.739	58.094	57.387	56.621	55.798	54.922
	54.000	53.039	52.047	51.034	50.010	51.034	52.047	53.039	54.000	54.922	55.798	56.621	57.387	58.094	58.739
	59.320	59.837	60.291	60.682	61.011	61.279	61.485	61.632	61.742						

VOLUMETRIC BUDGET FOR ENTIRE MODEL AT END OF TIME STEP 2 IN STRESS PERIOD134

CUMULATIVE VOLUMES	L**3	RATES FOR THIS TIME STEP	L**3/T
IN:		IN:	
STORAGE =	0.43765E+09	STORAGE =	4.9254
CONSTANT HEAD =	0.00000	CONSTANT HEAD =	0.00000
RECHARGE =	0.93224E+09	RECHARGE =	0.00000
STREAM LEAKAGE =	0.00000	STREAM LEAKAGE =	0.00000
TOTAL IN =	0.13699E+10	TOTAL IN =	4.9254
OUT:		OUT:	
STORAGE =	0.49468E+09	STORAGE =	0.00000
CONSTANT HEAD =	0.00000	CONSTANT HEAD =	0.00000
RECHARGE =	0.00000	RECHARGE =	0.00000
STREAM LEAKAGE =	0.87473E+09	STREAM LEAKAGE =	4.9224
TOTAL OUT =	0.13694E+10	TOTAL OUT =	4.9224
IN - OUT =	0.47002E+06	IN - OUT =	0.30622E-02
PERCENT DISCREPANCY =	0.03	PERCENT DISCREPANCY =	0.06

TIME SUMMARY AT END OF TIME STEP 2 IN STRESS PERIOD134

	SECONDS	MINUTES	HOURS	DAYS	YEARS
TIME STEP LENGTH	777600.	12960.0	216.000	9.00000	0.246407E-01
STRESS PERIOD TIME	0.129600E+07	21600.0	360.000	15.0000	0.410678E-01
TOTAL SIMULATION TIME	0.173664E+09	0.289440E+07	48240.0	2010.00	5.50308

STRESS PERIOD NO. 140, LENGTH = 1296000.

NUMBER OF TIME STEPS = 2

MULTIPLIER FOR DELT = 1.500

INITIAL TIME STEP SIZE = 518400.0

RECHARGE = 0.0000000

REUSING STREAM NODES FROM LAST STRESS PERIOD

4 ITERATIONS FOR TIME STEP 1 IN STRESS PERIOD140

HEAD/DRAWDOWN PRINTOUT FLAG = 0 TOTAL BUDGET PRINTOUT FLAG = 0 CELL-BY-CELL FLOW TERM FLAG = 0
OUTPUT FLAGS FOR ALL LAYERS ARE THE SAME:

HEAD PRINTOUT	DRAWDOWN PRINTOUT	HEAD SAVE	DRAWDOWN SAVE
0	0	0	0

4 ITERATIONS FOR TIME STEP 2 IN STRESS PERIOD140

MAXIMUM HEAD CHANGE FOR EACH ITERATION:

HEAD CHANGE	LAYER,ROW,COL	HEAD CHANGE	LAYER,ROW,COL	HEAD CHANGE	LAYER,ROW,COL	HEAD CHANGE	LAYER,ROW,COL	HEAD CHANGE	LAYER,ROW,COL
-0.1965	(1, 13, 39)	-0.2610E-01	(1, 2, 3)	-0.1310E-02	(1, 6, 4)	-0.1804E-04	(1, 13, 1)		

HEAD/DRAWDOWN PRINTOUT FLAG = 1 TOTAL BUDGET PRINTOUT FLAG = 1 CELL-BY-CELL FLOW TERM FLAG = 1
OUTPUT FLAGS FOR ALL LAYERS ARE THE SAME:

HEAD PRINTOUT	DRAWDOWN PRINTOUT	HEAD SAVE	DRAWDOWN SAVE
1	0	0	0

LAYER	ROW	COLUMN	STREAM NUMBER	REACH NUMBER	FLOW INTO STREAM REACH	FLOW INTO AQUIFER	FLOW OUT OF STREAM REACH
1	1	20	1	1	0.100E-01	-.285	0.295
1	2	20	1	2	0.295	-.285	0.580
1	3	20	1	3	0.580	-.285	0.864
1	4	20	1	4	0.864	-.285	1.15
1	5	20	1	5	1.15	-.285	1.43
1	6	20	1	6	1.43	-.285	1.72
1	7	20	1	7	1.72	-.285	2.00
1	8	20	1	8	2.00	-.285	2.29
1	9	20	1	9	2.29	-.285	2.57
1	10	20	1	10	2.57	-.285	2.86
1	11	20	1	11	2.86	-.285	3.14
1	12	20	1	12	3.14	-.285	3.43
1	13	20	1	13	3.43	-.285	3.71

HEAD IN LAYER 1 AT END OF TIME STEP 2 IN STRESS PERIOD140

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
	31	32	33	34	35	36	37	38	39						
1	59.621	59.514	59.373	59.175	58.923	58.616	58.257	57.847	57.388	56.882	56.331	55.740	55.111	54.448	53.756
	53.037	52.299	51.544	50.778	50.008	50.778	51.544	52.299	53.037	53.756	54.448	55.111	55.740	56.331	56.882
	57.388	57.847	58.257	58.616	58.923	59.175	59.373	59.514	59.621						
2	59.621	59.514	59.373	59.175	58.923	58.616	58.257	57.847	57.388	56.882	56.331	55.740	55.111	54.448	53.756
	53.037	52.299	51.544	50.778	50.008	50.778	51.544	52.299	53.037	53.756	54.448	55.111	55.740	56.331	56.882
	57.388	57.847	58.257	58.616	58.923	59.175	59.373	59.514	59.621						
3	59.621	59.514	59.373	59.175	58.923	58.616	58.257	57.847	57.388	56.882	56.331	55.740	55.111	54.448	53.756
	53.037	52.299	51.544	50.778	50.008	50.778	51.544	52.299	53.037	53.756	54.448	55.111	55.740	56.331	56.882
	57.388	57.847	58.257	58.616	58.923	59.175	59.373	59.514	59.621						
4	59.621	59.514	59.373	59.175	58.923	58.616	58.257	57.847	57.388	56.882	56.331	55.740	55.111	54.448	53.756
	53.037	52.299	51.544	50.778	50.008	50.778	51.544	52.299	53.037	53.756	54.448	55.111	55.740	56.331	56.882
	57.388	57.847	58.257	58.616	58.923	59.175	59.373	59.514	59.621						
5	59.621	59.514	59.373	59.175	58.923	58.616	58.257	57.847	57.388	56.882	56.331	55.740	55.111	54.448	53.756
	53.037	52.299	51.544	50.778	50.008	50.778	51.544	52.299	53.037	53.756	54.448	55.111	55.740	56.331	56.882
	57.388	57.847	58.257	58.616	58.923	59.175	59.373	59.514	59.621						
6	59.621	59.514	59.373	59.175	58.923	58.616	58.257	57.847	57.388	56.882	56.331	55.740	55.111	54.448	53.756
	53.037	52.299	51.544	50.778	50.008	50.778	51.544	52.299	53.037	53.756	54.448	55.111	55.740	56.331	56.882
	57.388	57.847	58.257	58.616	58.923	59.175	59.373	59.514	59.621						
7	59.621	59.514	59.373	59.175	58.923	58.616	58.257	57.847	57.388	56.882	56.331	55.740	55.111	54.448	53.756
	53.037	52.299	51.544	50.778	50.008	50.778	51.544	52.299	53.037	53.756	54.448	55.111	55.740	56.331	56.882
	57.388	57.847	58.257	58.616	58.923	59.175	59.373	59.514	59.621						
8	59.621	59.514	59.373	59.175	58.923	58.616	58.257	57.847	57.388	56.882	56.331	55.740	55.111	54.448	53.756
	53.037	52.299	51.544	50.778	50.008	50.778	51.544	52.299	53.037	53.756	54.448	55.111	55.740	56.331	56.882
	57.388	57.847	58.257	58.616	58.923	59.175	59.373	59.514	59.621						
9	59.621	59.514	59.373	59.175	58.923	58.616	58.257	57.847	57.388	56.882	56.331	55.740	55.111	54.448	53.756
	53.037	52.299	51.544	50.778	50.008	50.778	51.544	52.299	53.037	53.756	54.448	55.111	55.740	56.331	56.882
	57.388	57.847	58.257	58.616	58.923	59.175	59.373	59.514	59.621						
10	59.621	59.514	59.373	59.175	58.923	58.616	58.257	57.847	57.388	56.882	56.331	55.740	55.111	54.448	53.756
	53.037	52.299	51.544	50.778	50.008	50.778	51.544	52.299	53.037	53.756	54.448	55.111	55.740	56.331	56.882
	57.388	57.847	58.257	58.616	58.923	59.175	59.373	59.514	59.621						
11	59.621	59.514	59.373	59.175	58.923	58.616	58.257	57.847	57.388	56.882	56.331	55.740	55.111	54.448	53.756
	53.037	52.299	51.544	50.778	50.008	50.778	51.544	52.299	53.037	53.756	54.448	55.111	55.740	56.331	56.882
	57.388	57.847	58.257	58.616	58.923	59.175	59.373	59.514	59.621						
12	59.621	59.514	59.373	59.175	58.923	58.616	58.257	57.847	57.388	56.882	56.331	55.740	55.111	54.448	53.756
	53.037	52.299	51.544	50.778	50.008	50.778	51.544	52.299	53.037	53.756	54.448	55.111	55.740	56.331	56.882
	57.388	57.847	58.257	58.616	58.923	59.175	59.373	59.514	59.621						
13	59.621	59.514	59.373	59.175	58.923	58.616	58.257	57.847	57.388	56.882	56.331	55.740	55.111	54.448	53.756
	53.037	52.299	51.544	50.778	50.008	50.778	51.544	52.299	53.037	53.756	54.448	55.111	55.740	56.331	56.882
	57.388	57.847	58.257	58.616	58.923	59.175	59.373	59.514	59.621						

VOLUMETRIC BUDGET FOR ENTIRE MODEL AT END OF TIME STEP 2 IN STRESS PERIOD140

CUMULATIVE VOLUMES	L**3	RATES FOR THIS TIME STEP	L**3/T
IN:		IN:	
STORAGE =	0.47028E+09	STORAGE =	3.7073
CONSTANT HEAD =	0.00000	CONSTANT HEAD =	0.00000
RECHARGE =	0.93224E+09	RECHARGE =	0.00000
STREAM LEAKAGE =	0.00000	STREAM LEAKAGE =	0.00000
TOTAL IN =	0.14025E+10	TOTAL IN =	3.7073
OUT:		OUT:	
STORAGE =	0.49468E+09	STORAGE =	0.00000
CONSTANT HEAD =	0.00000	CONSTANT HEAD =	0.00000
RECHARGE =	0.00000	RECHARGE =	0.00000
STREAM LEAKAGE =	0.90735E+09	STREAM LEAKAGE =	3.7028
TOTAL OUT =	0.14020E+10	TOTAL OUT =	3.7028
IN - OUT =	0.49126E+06	IN - OUT =	0.45352E-02
PERCENT DISCREPANCY =	0.04	PERCENT DISCREPANCY =	0.12

TIME SUMMARY AT END OF TIME STEP 2 IN STRESS PERIOD140

	SECONDS	MINUTES	HOURS	DAYS	YEARS
TIME STEP LENGTH	777600.	12960.0	216.000	9.00000	0.246407E-01
STRESS PERIOD TIME	0.129600E+07	21600.0	360.000	15.0000	0.410678E-01
TOTAL SIMULATION TIME	0.181440E+09	0.302400E+07	50400.0	2100.00	5.74949

APPENDIX II

INPUT DATA SETS AND PRINTED RESULTS FOR TEST PROBLEM 2

This problem tests the ability of the Streamflow-Routing Package to calculate stream stage on the basis of flow and to simulate leakage between stream and aquifer. Details of the test problem and results are discussed in the section "Results of Test Simulations."

Listing of Input Data Sets for Test Problem 2

Listing of Input Data for Basic Package

Input for the Basic Package follows the column numbers. The input consists of 40 records (lines). Input for package is read from FORTRAN unit number specified in the MAIN program.

Column Numbers

```
        1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
TEST PROBLEM 2--COMPUTATION OF STREAM STAGE FOR A 30-DAY FLOOD: RESULTS COMPARED
TO COOPER AND RORABAUGH (1963)
         1        13        39        32         1
  7                        13       14 15
         0         1
         0         1
      999.
         0      48.0
    86400.         1       1.0
previous line repeated 29 times
  1296000.        15       1.0
  3888000.         9       1.0
```

Listing of Input Data for Strongly-Implicit Procedure Package

Input for the Strongly-Implicit Procedure Package follows the column numbers. The input consists of 2 records (lines). Input for package is read from FORTRAN unit number 13 as specified in the data set for the Basic Package.

Column Numbers

```
        1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
        150         5
        1.0    0.0001         1         0         0
```

Listing of Input Data for Block-Centered Flow Package

Input for the Block-Centered Flow Package follows the column numbers. The input consists of 11 records (lines). Input for package is read from FORTRAN unit number 7 as specified in the data set for the Basic Package.

Column Numbers

```
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
         0         0
0
         0        1.
         7      100.  (15F5.0)
   3.    2.    2.    2.    2.    2.    2.    2.    2.    2.    2.    2.    2.    2.    2.
   2.    2.    2.    2.    2.    2.    2.    2.    2.    2.    2.    2.    2.    2.    2.
   2.    2.    2.    2.    2.    2.    2.    2.    3.
         0     1000.
         0      0.20
         0  0.037000
         0        0.
```

Listing of Input Data for Output Control Package

Input for the Output Control Package follows the column numbers. The input consists of 77 records (lines). Input for package is read from FORTRAN unit number 14 as specified in the data set for the Basic Package.

Column Numbers

```
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
         5         5         0         0
         0         1         1         1
         1         0         0         0
    previous two lines repeated 29 more times
         0         1         1         1
         1         0         0         0
        -1         1         1         0
     previous line repeated 13 more times
         0         1         1         1
         1         0         0         0
        -1         1         1         0
     previous line repeated 7 more times
```

Listing of Input Data for Streamflow-Routing Package

Input for the Streamflow-Routing Package follows the column numbers. The input consists of 813 records (lines). Input for package is read from FORTRAN unit number 15 as specified in the data set for the Basic Package.

Column Numbers

```
        1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
      13         1         0         0         1     1.486        -1         0
      13         0         0
 1    1   20    1    1              2018.5      48.0    37.000      40.0      40.0
 1    2   20    1    2                          48.0    37.000      40.0      40.0
 1    3   20    1    3                          48.0    37.000      40.0      40.0
 1    4   20    1    4                          48.0    37.000      40.0      40.0
 1    5   20    1    5                          48.0    37.000      40.0      40.0
 1    6   20    1    6                          48.0    37.000      40.0      40.0
 1    7   20    1    7                          48.0    37.000      40.0      40.0
 1    8   20    1    8                          48.0    37.000      40.0      40.0
 1    9   20    1    9                          48.0    37.000      40.0      40.0
 1   10   20    1   10                          48.0    37.000      40.0      40.0
 1   11   20    1   11                          48.0    37.000      40.0      40.0
 1   12   20    1   12                          48.0    37.000      40.0      40.0
 1   13   20    1   13                          48.0    37.000      40.0      40.0
   100.    0.0001   0.02377
   100.    0.0001   0.02377
   100.    0.0001   0.02377
   100.    0.0001   0.02377
   100.    0.0001   0.02377
   100.    0.0001   0.02377
   100.    0.0001   0.02377
   100.    0.0001   0.02377
   100.    0.0001   0.02377
   100.    0.0001   0.02377
   100.    0.0001   0.02377
   100.    0.0001   0.02377
   100.    0.0001   0.02377
```

Listing of Input Data for Streamflow-Routing Package (continued)

Sequence of previous 27 lines repeated 29 times except the second line in each sequence changes as listed following the column numbers.

Column Numbers

1			2		3	4	5	6	7	8
1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	1	20	1	1		2073.1	48.0	37.000	40.0	40.0
1	1	20	1	1		2162.2	48.0	37.000	40.0	40.0
1	1	20	1	1		2283.8	48.0	37.000	40.0	40.0
1	1	20	1	1		2434.4	48.0	37.000	40.0	40.0
1	1	20	1	1		2609.0	48.0	37.000	40.0	40.0
1	1	20	1	1		2801.4	48.0	37.000	40.0	40.0
1	1	20	1	1		3003.5	48.0	37.000	40.0	40.0
1	1	20	1	1		3206.7	48.0	37.000	40.0	40.0
1	1	20	1	1		3401.3	48.0	37.000	40.0	40.0
1	1	20	1	1		3577.3	48.0	37.000	40.0	40.0
1	1	20	1	1		3725.7	48.0	37.000	40.0	40.0
1	1	20	1	1		3838.1	48.0	37.000	40.0	40.0
1	1	20	1	1		3908.3	48.0	37.000	40.0	40.0
1	1	20	1	1		3932.1	48.0	37.000	40.0	40.0
1	1	20	1	1		3908.3	48.0	37.000	40.0	40.0
1	1	20	1	1		3838.1	48.0	37.000	40.0	40.0
1	1	20	1	1		3725.7	48.0	37.000	40.0	40.0
1	1	20	1	1		3577.3	48.0	37.000	40.0	40.0
1	1	20	1	1		3401.3	48.0	37.000	40.0	40.0
1	1	20	1	1		3206.7	48.0	37.000	40.0	40.0
1	1	20	1	1		3003.5	48.0	37.000	40.0	40.0
1	1	20	1	1		2801.4	48.0	37.000	40.0	40.0
1	1	20	1	1		2609.0	48.0	37.000	40.0	40.0
1	1	20	1	1		2434.4	48.0	37.000	40.0	40.0
1	1	20	1	1		2283.8	48.0	37.000	40.0	40.0
1	1	20	1	1		2162.2	48.0	37.000	40.0	40.0
1	1	20	1	1		2073.1	48.0	37.000	40.0	40.0
1	1	20	1	1		2018.5	48.0	37.000	40.0	40.0
1	1	20	1	1		2000.0	48.0	37.000	40.0	40.0

last 2 records are as follows:

-1

-1

SELECTED PAGES FROM PRINTED RESULTS FOR TEST PROBLEM 2

```
                    U.S. GEOLOGICAL SURVEY MODULAR FINITE-DIFFERENCE GROUND-WATER MODEL

TEST PROBLEM 2--COMPUTATION OF STREAM STAGE FOR A 30-DAY FLOOD: RESULTS COMPARED TO COOPER AND RORABAUGH (1963)

   1 LAYERS        13 ROWS        39 COLUMNS
 32 STRESS PERIOD(S) IN SIMULATION
MODEL TIME UNIT IS SECONDS

I/O UNITS:
ELEMENT OF IUNIT:  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
       I/O UNIT:  7  0  0  0  0  0  0  0 13  0  0 14 15  0  0  0  0  0  0  0  0  0  0  0

BAS1 -- BASIC MODEL PACKAGE, VERSION 1, 9/1/87 INPUT READ FROM UNIT  5
ARRAYS RHS AND BUFF WILL SHARE MEMORY.
START HEAD WILL BE SAVED
    4619 ELEMENTS IN X ARRAY ARE USED BY BAS
    4619 ELEMENTS OF X ARRAY USED OUT OF  200000

BCF1 -- BLOCK-CENTERED FLOW PACKAGE, VERSION 1, 9/1/87 INPUT READ FROM UNIT  7
TRANSIENT SIMULATION
     LAYER  AQUIFER TYPE
     ------------------
        1         0
     508 ELEMENTS IN X ARRAY ARE USED BY BCF
    5127 ELEMENTS OF X ARRAY USED OUT OF  200000

STRM -- STREAM PACKAGE, VERSION 1, 10/23/87 INPUT READ FROM UNIT 15
MAXIMUM OF   13 STREAM NODES

NUMBER OF STREAM SEGMENTS IS     1

NUMBER OF STREAM TRIBUTARIES IS     0


STREAM STAGES WILL BE CALCULATED USING A CONSTANT OF      1.4860
     210 ELEMENTS IN X ARRAY ARE USED FOR STREAMS
    5337 ELEMENTS OF X ARRAY USED OUT OF 200000

SIP1 -- STRONGLY IMPLICIT PROCEDURE SOLUTION PACKAGE, VERSION 1, 9/1/87 INPUT READ FROM UNIT 13
MAXIMUM OF 150 ITERATIONS ALLOWED FOR CLOSURE
 5 ITERATION PARAMETERS
    2633 ELEMENTS IN X ARRAY ARE USED BY SIP
    7970 ELEMENTS OF X ARRAY USED OUT OF  200000
```

STRESS PERIOD NO. 11, LENGTH = 86400.00

NUMBER OF TIME STEPS = 1

MULTIPLIER FOR DELT = 1.000

INITIAL TIME STEP SIZE = 86400.00

13 STREAM NODES

LAYER	ROW	COL	SEGMENT NUMBER	REACH NUMBER	STREAMFLOW	STREAM STAGE	STREAMBED CONDUCTANCE	STREAMBED BOT ELEVATION	STREAMBED TOP ELEVATION
1	1	20	1	1	3577.	48.00	37.00	40.00	40.00
1	2	20	1	2	0.0000	48.00	37.00	40.00	40.00
1	3	20	1	3	0.0000	48.00	37.00	40.00	40.00
1	4	20	1	4	0.0000	48.00	37.00	40.00	40.00
1	5	20	1	5	0.0000	48.00	37.00	40.00	40.00
1	6	20	1	6	0.0000	48.00	37.00	40.00	40.00
1	7	20	1	7	0.0000	48.00	37.00	40.00	40.00
1	8	20	1	8	0.0000	48.00	37.00	40.00	40.00
1	9	20	1	9	0.0000	48.00	37.00	40.00	40.00
1	10	20	1	10	0.0000	48.00	37.00	40.00	40.00
1	11	20	1	11	0.0000	48.00	37.00	40.00	40.00
1	12	20	1	12	0.0000	48.00	37.00	40.00	40.00
1	13	20	1	13	0.0000	48.00	37.00	40.00	40.00

LAYER	ROW	COL	SEGMENT NUMBER	REACH NUMBER	STREAM WIDTH	STREAM SLOPE	ROUGH COEF.
1	1	20	1	1	100.0	0.1000E-03	0.2377E-01
1	2	20	1	2	100.0	0.1000E-03	0.2377E-01
1	3	20	1	3	100.0	0.1000E-03	0.2377E-01
1	4	20	1	4	100.0	0.1000E-03	0.2377E-01
1	5	20	1	5	100.0	0.1000E-03	0.2377E-01
1	6	20	1	6	100.0	0.1000E-03	0.2377E-01
1	7	20	1	7	100.0	0.1000E-03	0.2377E-01
1	8	20	1	8	100.0	0.1000E-03	0.2377E-01
1	9	20	1	9	100.0	0.1000E-03	0.2377E-01
1	10	20	1	10	100.0	0.1000E-03	0.2377E-01
1	11	20	1	11	100.0	0.1000E-03	0.2377E-01
1	12	20	1	12	100.0	0.1000E-03	0.2377E-01
1	13	20	1	13	100.0	0.1000E-03	0.2377E-01

15 ITERATIONS FOR TIME STEP 1 IN STRESS PERIOD 11

MAXIMUM HEAD CHANGE FOR EACH ITERATION:

HEAD CHANGE	LAYER,ROW,COL	HEAD CHANGE	LAYER,ROW,COL	HEAD CHANGE	LAYER,ROW,COL	HEAD CHANGE	LAYER,ROW,COL	HEAD CHANGE	LAYER,ROW,COL
-3.443	(1, 1, 20)	3.902	(1, 1, 20)	-1.044	(1, 13, 20)	1.142	(1, 13, 20)	-0.3119	(1, 13, 20)
0.2921	(1, 13, 20)	-0.6935E-01	(1, 13, 20)	0.5328E-01	(1, 13, 20)	-0.1243E-01	(1, 13, 20)	0.7865E-02	(1, 13, 20)
-0.1882E-02	(1, 13, 20)	0.9960E-03	(1, 13, 20)	-0.2458E-03	(1, 13, 20)	0.1229E-03	(1, 13, 20)	-0.3233E-04	(1, 13, 20)

HEAD/DRAWDOWN PRINTOUT FLAG = 1 TOTAL BUDGET PRINTOUT FLAG = 1 CELL-BY-CELL FLOW TERM FLAG = 1

OUTPUT FLAGS FOR ALL LAYERS ARE THE SAME:

HEAD PRINTOUT	DRAWDOWN PRINTOUT	HEAD SAVE	DRAWDOWN SAVE
1	0	0	0

LAYER	ROW	COLUMN	STREAM NUMBER	REACH NUMBER	FLOW INTO STREAM REACH	FLOW INTO AQUIFER	FLOW OUT OF STREAM REACH	HEAD IN STREAM
1	1	20	1	1	0.358E+04	0.721	0.358E+04	51.34
1	2	20	1	2	0.358E+04	0.721	0.358E+04	51.34
1	3	20	1	3	0.358E+04	0.721	0.358E+04	51.33
1	4	20	1	4	0.358E+04	0.721	0.357E+04	51.33
1	5	20	1	5	0.357E+04	0.720	0.357E+04	51.33
1	6	20	1	6	0.357E+04	0.720	0.357E+04	51.33
1	7	20	1	7	0.357E+04	0.720	0.357E+04	51.33
1	8	20	1	8	0.357E+04	0.720	0.357E+04	51.33
1	9	20	1	9	0.357E+04	0.720	0.357E+04	51.33
1	10	20	1	10	0.357E+04	0.720	0.357E+04	51.33
1	11	20	1	11	0.357E+04	0.720	0.357E+04	51.32
1	12	20	1	12	0.357E+04	0.720	0.357E+04	51.32
1	13	20	1	13	0.357E+04	0.719	0.357E+04	51.32

HEAD IN LAYER 1 AT END OF TIME STEP 1 IN STRESS PERIOD 11

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
	31	32	33	34	35	36	37	38	39						
1	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.001	48.002	48.006	48.015	48.037	48.088
	48.200	48.437	48.909	49.792	51.318	49.792	48.909	48.437	48.200	48.088	48.037	48.015	48.006	48.002	48.001
	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000						
2	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.001	48.002	48.006	48.015	48.037	48.088
	48.200	48.436	48.908	49.791	51.317	49.791	48.908	48.436	48.200	48.088	48.037	48.015	48.006	48.002	48.001
	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000						
3	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.001	48.002	48.006	48.015	48.037	48.088
	48.200	48.436	48.908	49.790	51.315	49.790	48.908	48.436	48.200	48.088	48.037	48.015	48.006	48.002	48.001
	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000						
4	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.001	48.002	48.006	48.015	48.037	48.088
	48.200	48.436	48.907	49.789	51.314	49.789	48.907	48.436	48.200	48.088	48.037	48.015	48.006	48.002	48.001
	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000						
5	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.001	48.002	48.006	48.015	48.037	48.088
	48.200	48.436	48.907	49.788	51.312	49.788	48.907	48.436	48.200	48.088	48.037	48.015	48.006	48.002	48.001
	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000						
6	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.001	48.002	48.006	48.015	48.037	48.088
	48.200	48.435	48.906	49.788	51.311	49.788	48.906	48.435	48.200	48.088	48.037	48.015	48.006	48.002	48.001
	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000						
7	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.001	48.002	48.006	48.015	48.037	48.088
	48.199	48.435	48.906	49.787	51.310	49.787	48.906	48.435	48.199	48.088	48.037	48.015	48.006	48.002	48.001
	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000						
8	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.001	48.002	48.006	48.015	48.037	48.088
	48.199	48.435	48.905	49.786	51.308	49.786	48.905	48.435	48.199	48.088	48.037	48.015	48.006	48.002	48.001
	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000						
9	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.001	48.002	48.006	48.015	48.037	48.088
	48.199	48.434	48.905	49.785	51.307	49.785	48.905	48.434	48.199	48.088	48.037	48.015	48.006	48.002	48.001
	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000						
10	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.001	48.002	48.006	48.015	48.037	48.088
	48.199	48.434	48.904	49.784	51.306	49.784	48.904	48.434	48.199	48.088	48.037	48.015	48.006	48.002	48.001
	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000						
11	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.001	48.002	48.006	48.015	48.037	48.088
	48.199	48.434	48.904	49.783	51.304	49.783	48.904	48.434	48.199	48.088	48.037	48.015	48.006	48.002	48.001
	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000						
12	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.001	48.002	48.006	48.015	48.037	48.087
	48.199	48.434	48.903	49.782	51.303	49.782	48.903	48.434	48.199	48.087	48.037	48.015	48.006	48.002	48.001
	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000						
13	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.001	48.002	48.006	48.015	48.037	48.087
	48.199	48.433	48.903	49.781	51.301	49.781	48.903	48.433	48.199	48.087	48.037	48.015	48.006	48.002	48.001
	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000	48.000						

VOLUMETRIC BUDGET FOR ENTIRE MODEL AT END OF TIME STEP 1 IN STRESS PERIOD 11

CUMULATIVE VOLUMES	L**3	RATES FOR THIS TIME STEP	L**3/T
IN:		IN:	
STORAGE =	0.00000	STORAGE =	0.00000
CONSTANT HEAD =	0.00000	CONSTANT HEAD =	0.00000
STREAM LEAKAGE =	0.53424E+07	STREAM LEAKAGE =	9.3624
TOTAL IN =	0.53424E+07	TOTAL IN =	9.3624
OUT:		OUT:	
STORAGE =	0.53371E+07	STORAGE =	9.3589
CONSTANT HEAD =	0.00000	CONSTANT HEAD =	0.00000
STREAM LEAKAGE =	0.00000	STREAM LEAKAGE =	0.00000
TOTAL OUT =	0.53371E+07	TOTAL OUT =	9.3589
IN - OUT =	5253.0	IN - OUT =	0.34771E-02
PERCENT DISCREPANCY =	0.10	PERCENT DISCREPANCY =	0.04

TIME SUMMARY AT END OF TIME STEP 1 IN STRESS PERIOD 11

	SECONDS	MINUTES	HOURS	DAYS	YEARS
TIME STEP LENGTH	86400.0	1440.00	24.0000	1.00000	0.273785E-02
STRESS PERIOD TIME	86400.0	1440.00	24.0000	1.00000	0.273785E-02
TOTAL SIMULATION TIME	950400.	15840.0	264.000	11.0000	0.301164E-01

STRESS PERIOD NO. 29, LENGTH = 86400.00

NUMBER OF TIME STEPS = 1

MULTIPLIER FOR DELT = 1.000

INITIAL TIME STEP SIZE = 86400.00

13 STREAM NODES

LAYER	ROW	COL	SEGMENT NUMBER	REACH NUMBER	STREAMFLOW	STREAM STAGE	STREAMBED CONDUCTANCE	STREAMBED BOT ELEVATION	STREAMBED TOP ELEVATION
1	1	20	1	1	2018.	48.00	37.00	40.00	40.00
1	2	20	1	2	0.0000	48.00	37.00	40.00	40.00
1	3	20	1	3	0.0000	48.00	37.00	40.00	40.00
1	4	20	1	4	0.0000	48.00	37.00	40.00	40.00
1	5	20	1	5	0.0000	48.00	37.00	40.00	40.00
1	6	20	1	6	0.0000	48.00	37.00	40.00	40.00
1	7	20	1	7	0.0000	48.00	37.00	40.00	40.00
1	8	20	1	8	0.0000	48.00	37.00	40.00	40.00
1	9	20	1	9	0.0000	48.00	37.00	40.00	40.00
1	10	20	1	10	0.0000	48.00	37.00	40.00	40.00
1	11	20	1	11	0.0000	48.00	37.00	40.00	40.00
1	12	20	1	12	0.0000	48.00	37.00	40.00	40.00
1	13	20	1	13	0.0000	48.00	37.00	40.00	40.00

LAYER	ROW	COL	SEGMENT NUMBER	REACH NUMBER	STREAM WIDTH	STREAM SLOPE	ROUGH COEF.
1	1	20	1	1	100.0	0.1000E-03	0.2377E-01
1	2	20	1	2	100.0	0.1000E-03	0.2377E-01
1	3	20	1	3	100.0	0.1000E-03	0.2377E-01
1	4	20	1	4	100.0	0.1000E-03	0.2377E-01
1	5	20	1	5	100.0	0.1000E-03	0.2377E-01
1	6	20	1	6	100.0	0.1000E-03	0.2377E-01
1	7	20	1	7	100.0	0.1000E-03	0.2377E-01
1	8	20	1	8	100.0	0.1000E-03	0.2377E-01
1	9	20	1	9	100.0	0.1000E-03	0.2377E-01
1	10	20	1	10	100.0	0.1000E-03	0.2377E-01
1	11	20	1	11	100.0	0.1000E-03	0.2377E-01
1	12	20	1	12	100.0	0.1000E-03	0.2377E-01
1	13	20	1	13	100.0	0.1000E-03	0.2377E-01

15 ITERATIONS FOR TIME STEP 1 IN STRESS PERIOD 29

MAXIMUM HEAD CHANGE FOR EACH ITERATION:

```
 HEAD CHANGE LAYER,ROW,COL HEAD CHANGE LAYER,ROW,COL HEAD CHANGE LAYER,ROW,COL HEAD CHANGE LAYER,ROW,COL HEAD CHANGE LAYER,ROW,COL
-------------------------------------------------------------------------------------------------------------------------------------
  -2.814     (  1,  1, 20)   2.807     (  1,  1, 20) -0.9299     (  1, 13, 20)  0.9100     (  1, 13, 20) -0.3413     (  1, 13, 20)
   0.2986    (  1, 13, 20) -0.9357E-01 (  1, 13, 20)  0.6964E-01 (  1, 13, 20) -0.2084E-01 (  1, 13, 20)  0.1315E-01 (  1, 13, 20)
 -0.3939E-02 (  1, 13, 20)  0.2134E-02 (  1, 13, 20) -0.6467E-03 (  1, 13, 20)  0.3104E-03 (  1, 13, 20) -0.9701E-04 (  1, 13, 20)


HEAD/DRAWDOWN PRINTOUT FLAG = 1     TOTAL BUDGET PRINTOUT FLAG = 1     CELL-BY-CELL FLOW TERM FLAG = 1

OUTPUT FLAGS FOR ALL LAYERS ARE THE SAME:
  HEAD     DRAWDOWN  HEAD  DRAWDOWN
PRINTOUT  PRINTOUT  SAVE    SAVE
---------------------------------
    1         0       0       0
```

LAYER	ROW	COLUMN	STREAM NUMBER	REACH NUMBER	FLOW INTO STREAM REACH	FLOW INTO AQUIFER	FLOW OUT OF STREAM REACH	HEAD IN STREAM
1	1	20	1	1	0.202E+04	-.329	0.202E+04	48.04
1	2	20	1	2	0.202E+04	-.329	0.202E+04	48.04
1	3	20	1	3	0.202E+04	-.329	0.202E+04	48.05
1	4	20	1	4	0.202E+04	-.329	0.202E+04	48.05
1	5	20	1	5	0.202E+04	-.329	0.202E+04	48.05
1	6	20	1	6	0.202E+04	-.329	0.202E+04	48.05
1	7	20	1	7	0.202E+04	-.329	0.202E+04	48.05
1	8	20	1	8	0.202E+04	-.329	0.202E+04	48.05
1	9	20	1	9	0.202E+04	-.329	0.202E+04	48.05
1	10	20	1	10	0.202E+04	-.329	0.202E+04	48.05
1	11	20	1	11	0.202E+04	-.328	0.202E+04	48.05
1	12	20	1	12	0.202E+04	-.328	0.202E+04	48.05
1	13	20	1	13	0.202E+04	-.327	0.202E+04	48.05

HEAD IN LAYER 1 AT END OF TIME STEP 1 IN STRESS PERIOD 29

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
	31	32	33	34	35	36	37	38	39						
1	48.000	48.000	48.001	48.002	48.004	48.007	48.013	48.025	48.045	48.078	48.133	48.218	48.342	48.510	48.717
	48.934	49.095	49.091	48.779	48.052	48.779	49.091	49.095	48.934	48.717	48.510	48.342	48.218	48.133	48.078
	48.045	48.025	48.013	48.007	48.004	48.002	48.001	48.000	48.000						
2	48.000	48.000	48.001	48.002	48.004	48.007	48.013	48.025	48.045	48.078	48.133	48.218	48.342	48.510	48.717
	48.934	49.095	49.091	48.780	48.053	48.780	49.091	49.095	48.934	48.717	48.510	48.342	48.218	48.133	48.078
	48.045	48.025	48.013	48.007	48.004	48.002	48.001	48.000	48.000						
3	48.000	48.000	48.001	48.002	48.004	48.007	48.013	48.025	48.045	48.078	48.133	48.218	48.342	48.510	48.717
	48.934	49.095	49.091	48.780	48.054	48.780	49.091	49.095	48.934	48.717	48.510	48.342	48.218	48.133	48.078
	48.045	48.025	48.013	48.007	48.004	48.002	48.001	48.000	48.000						
4	48.000	48.000	48.001	48.002	48.004	48.007	48.013	48.025	48.045	48.078	48.133	48.218	48.341	48.510	48.717
	48.933	49.095	49.091	48.781	48.055	48.781	49.091	49.095	48.933	48.717	48.510	48.341	48.218	48.133	48.078
	48.045	48.025	48.013	48.007	48.004	48.002	48.001	48.000	48.000						
5	48.000	48.000	48.001	48.002	48.004	48.007	48.013	48.025	48.045	48.078	48.133	48.218	48.341	48.510	48.716
	48.933	49.095	49.091	48.781	48.056	48.781	49.091	49.095	48.933	48.716	48.510	48.341	48.218	48.133	48.078
	48.045	48.025	48.013	48.007	48.004	48.002	48.001	48.000	48.000						
6	48.000	48.000	48.001	48.002	48.004	48.007	48.013	48.025	48.045	48.078	48.133	48.218	48.341	48.509	48.716
	48.933	49.095	49.092	48.782	48.056	48.782	49.092	49.095	48.933	48.716	48.509	48.341	48.218	48.133	48.078
	48.045	48.025	48.013	48.007	48.004	48.002	48.001	48.000	48.000						
7	48.000	48.000	48.001	48.002	48.004	48.007	48.013	48.025	48.045	48.078	48.133	48.217	48.341	48.509	48.716
	48.933	49.095	49.092	48.782	48.057	48.782	49.092	49.095	48.933	48.716	48.509	48.341	48.217	48.133	48.078
	48.045	48.025	48.013	48.007	48.004	48.002	48.001	48.000	48.000						
8	48.000	48.000	48.001	48.002	48.004	48.007	48.013	48.025	48.045	48.078	48.133	48.217	48.341	48.509	48.716
	48.933	49.095	49.092	48.782	48.058	48.782	49.092	49.095	48.933	48.716	48.509	48.341	48.217	48.133	48.078
	48.045	48.025	48.013	48.007	48.004	48.002	48.001	48.000	48.000						
9	48.000	48.000	48.001	48.002	48.004	48.007	48.013	48.025	48.044	48.078	48.133	48.217	48.341	48.509	48.716
	48.933	49.095	49.092	48.783	48.059	48.783	49.092	49.095	48.933	48.716	48.509	48.341	48.217	48.133	48.078
	48.044	48.025	48.013	48.007	48.004	48.002	48.001	48.000	48.000						
10	48.000	48.000	48.001	48.002	48.004	48.007	48.013	48.025	48.044	48.078	48.133	48.217	48.341	48.509	48.716
	48.933	49.095	49.092	48.783	48.059	48.783	49.092	49.095	48.933	48.716	48.509	48.341	48.217	48.133	48.078
	48.044	48.025	48.013	48.007	48.004	48.002	48.001	48.000	48.000						
11	48.000	48.000	48.001	48.002	48.004	48.007	48.013	48.025	48.044	48.078	48.133	48.217	48.341	48.509	48.716
	48.932	49.095	49.092	48.784	48.060	48.784	49.092	49.095	48.932	48.716	48.509	48.341	48.217	48.133	48.078
	48.044	48.025	48.013	48.007	48.004	48.002	48.001	48.000	48.000						
12	48.000	48.000	48.001	48.002	48.004	48.007	48.013	48.025	48.044	48.078	48.133	48.217	48.341	48.509	48.715
	48.932	49.095	49.093	48.784	48.061	48.784	49.093	49.095	48.932	48.715	48.509	48.341	48.217	48.133	48.078
	48.044	48.025	48.013	48.007	48.004	48.002	48.001	48.000	48.000						
13	48.000	48.000	48.001	48.002	48.004	48.007	48.013	48.025	48.044	48.078	48.133	48.217	48.341	48.509	48.715
	48.932	49.095	49.093	48.785	48.062	48.785	49.093	49.095	48.932	48.715	48.509	48.341	48.217	48.133	48.078
	48.044	48.025	48.013	48.007	48.004	48.002	48.001	48.000	48.000						

VOLUMETRIC BUDGET FOR ENTIRE MODEL AT END OF TIME STEP 1 IN STRESS PERIOD 29

CUMULATIVE VOLUMES	L**3	RATES FOR THIS TIME STEP	L**3/T
IN:		IN:	
STORAGE =	0.52264E+07	STORAGE =	5.3231
CONSTANT HEAD =	0.00000	CONSTANT HEAD =	0.00000
STREAM LEAKAGE =	0.94751E+07	STREAM LEAKAGE =	0.00000
TOTAL IN =	0.14701E+08	TOTAL IN =	5.3231
OUT:		OUT:	
STORAGE =	0.11489E+08	STORAGE =	1.0429
CONSTANT HEAD =	0.00000	CONSTANT HEAD =	0.00000
STREAM LEAKAGE =	0.32005E+07	STREAM LEAKAGE =	4.2724
TOTAL OUT =	0.14689E+08	TOTAL OUT =	5.3153
IN - OUT =	12434.	IN - OUT =	0.77477E-02
PERCENT DISCREPANCY =	0.08	PERCENT DISCREPANCY =	0.15

TIME SUMMARY AT END OF TIME STEP 1 IN STRESS PERIOD 29

	SECONDS	MINUTES	HOURS	DAYS	YEARS
TIME STEP LENGTH	86400.0	1440.00	24.0000	1.00000	0.273785E-02
STRESS PERIOD TIME	86400.0	1440.00	24.0000	1.00000	0.273785E-02
TOTAL SIMULATION TIME	0.250560E+07	41760.0	696.000	29.0000	0.793977E-01

APPENDIX III

INPUT DATA SETS AND PRINTED RESULTS FOR EXAMPLE PROBLEM

This problem is an example of the capabilities of the Streamflow-Routing Package. Details of the test problem and results are discussed in the section "Results of Test Simulations."

Listing of Input Data Sets for Example Problem

Listing of Input Data for Basic Package

Input for the Basic Package follows the column numbers. The input consists of 15 records (lines). Input for package is read from a FORTRAN unit number specified in the MAIN program.

Column Numbers

```
        1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
EXAMPLE SIMULATION OF STREAM ROUTING PACKAGE -- STEADY STATE  OCTOBER  21,1987
-- STREAM STAGE IS CALCULATED
         1         6         6         1         1
  7                        13       14 15
         0         1
         0         1
      999.
         5       1.0 (6F8.0)
  480.00  480.00  480.00  480.00  480.00  480.00
  480.00  480.00  480.00  480.00  480.00  480.00
  480.00  480.00  480.00  480.00  480.00  480.00
  480.00  480.00  480.00  480.00  480.00  480.00
  480.00  480.00  480.00  480.00  480.00  480.00
  480.00  480.00  480.00  480.00  480.00  480.00
  1296000.          3       1.5
```

Listing of Input Data for Strongly-Implicit Procedure Package

Input for the Strongly-Implicit Procedure Package follows the column numbers. The input consists of 2 records (lines). Input for package is read from FORTRAN unit number 13 as specified in the data set for the Basic Package.

Listing of Input Data for Strongly-Implicit Procedure Package (continued)

Column Numbers

```
        1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
       150         5
        1.    0.0001         1         0         0
```

Listing of Input Data for Block-Centered Flow Package

Input for the Block-Centered Flow Package follows the column numbers. The input consists of 7 records (lines). Input for package is read from FORTRAN unit number 7 as specified in the data set for the Basic Package.

Column Numbers

```
        1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
        1         0
0
        0        1.
        0     1000.
        0     1000.
        0  0.080000
        0        0.
```

Listing of Input Data for Output Control Package

Input for the Output Control Package follows the column numbers. The input consists of 7 records (lines). Input for package is read from FORTRAN unit number 14 as specified in the data set for the Basic Package.

Column Numbers

```
        1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
        5         5         0         0
        0         0         0         0
        0         0         0         0
        0         0         0         0
        0         0         0         0
        0         1         1         1
        1         0         0         0
```

Listing of Input Data for Streamflow-Routing Package

Input for the Streamflow-Routing Package follows the column numbers. The input consists of 62 records (lines). Input for package is read from FORTRAN unit number 15 as specified in the data set for the Basic Package.

Column Numbers

```
      1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
      23        7         3         1         1     1.486        -1         0
      23        0         0
 1    1    3    1    1              4.5     495.0      1.20     490.0     492.0
 1    2    3    1    2                      490.0      0.60     485.0     487.0
 1    2    3    2    1              1.5     487.0      0.20     483.0     485.0
 1    3    3    2    2                      486.0      0.40     482.0     484.0
 1    4    3    2    3                      484.0      0.40     480.0     482.0
 1    5    3    2    4                      480.0      0.20     476.0     478.0
 1    2    3    3    1             -1.0     486.0      0.40     481.0     483.0
 1    3    4    3    2                      482.0      1.20     477.0     479.0
 1    4    4    3    3                      478.0      1.20     473.0     475.0
 1    5    4    3    4                      475.0      0.60     470.0     472.0
 1    4    1    4    1              0.8     492.0      0.40     489.0     490.0
 1    4    2    4    2                      488.0      0.32     485.0     486.0
 1    5    2    4    3                      483.0      0.32     480.0     481.0
 1    5    3    4    4                      480.0      0.20     477.0     478.0
 1    5    3    5    1             -1.0     478.0      0.20     475.0     476.0
 1    5    4    5    2                      474.0      0.20     471.0     472.0
 1    2    6    6    1              1.2     495.0      0.80     491.0     493.0
 1    3    6    6    2                      490.0      0.80     486.0     488.0
 1    4    5    6    3                      480.0      0.80     476.0     478.0
 1    5    5    6    4                      477.0      0.60     473.0     475.0
 1    5    4    6    5                      474.0      0.20     470.0     472.0
 1    5    4    7    1             -1.0     472.0      0.60     467.0     469.0
 1    6    4    7    2                      469.0      1.20     464.0     466.0
    10.       0.007     0.030
    10.       0.007     0.030
     5.       0.002     0.022
     5.       0.002     0.022
     5.       0.002     0.022
     5.       0.004     0.022
    10.       0.005     0.030
    10.       0.005     0.030
    10.       0.005     0.030
    10.       0.005     0.030
     5.       0.004     0.022
     5.       0.004     0.022
     5.       0.004     0.022
     5.       0.004     0.022
     5.       0.005     0.022
     5.       0.005     0.022
```

Listing of Input Data for Streamflow-Routing Package (continued)

Column Numbers

```
         1         2         3         4         5         6         7         8
12345678901234567890123456789012345678901234567890123456789012345678901234567890
      5.     0.005     0.022
      5.     0.008     0.022
      5.     0.007     0.022
      5.     0.004     0.022
      5.     0.003     0.022
     10.     0.004     0.030
     10.     0.004     0.030
0    0  0
0    0  0
1    0  0
0    0  0
2    4  0
0    0  0
3    5  6
       0
       1
       0
       0
       0
       0
       0
```

PRINTED RESULTS FOR EXAMPLE PROBLEM

```
                U.S. GEOLOGICAL SURVEY MODULAR FINITE-DIFFERENCE GROUND-WATER MODEL

EXAMPLE SIMULATION OF STREAM ROUTING PACKAGE -- STEADY STATE OCTOBER 21, 1987  -- STREAM STAGE IS CALCULATED

   1 LAYERS          6 ROWS          6 COLUMNS

  1 STRESS PERIOD(S) IN SIMULATION

MODEL TIME UNIT IS SECONDS

I/O UNITS:

ELEMENT OF IUNIT:  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
        I/O UNIT:  7  0  0  0  0  0  0  0 13  0  0 14 15  0  0  0  0  0  0  0  0  0  0  0

BAS1 -- BASIC MODEL PACKAGE, VERSION 1, 9/1/87 INPUT READ FROM UNIT  5
ARRAYS RHS AND BUFF WILL SHARE MEMORY.
START HEAD WILL BE SAVED
     340 ELEMENTS IN X ARRAY ARE USED BY BAS
     340 ELEMENTS OF X ARRAY USED OUT OF  200000

BCF1 -- BLOCK-CENTERED FLOW PACKAGE, VERSION 1, 9/1/87 INPUT READ FROM UNIT  7
STEADY-STATE SIMULATION
     LAYER  AQUIFER TYPE
     ------------------
        1         0

       1 ELEMENTS IN X ARRAY ARE USED BY BCF
     341 ELEMENTS OF X ARRAY USED OUT OF  200000

STRM -- STREAM PACKAGE, VERSION 1, 10/23/87 INPUT READ FROM UNIT 15
MAXIMUM OF   23 STREAM NODES

NUMBER OF STREAM SEGMENTS IS     7

NUMBER OF STREAM TRIBUTARIES IS     3

DIVERSIONS FROM STREAMS HAVE BEEN SPECIFIED
STREAM STAGES WILL BE CALCULATED USING A CONSTANT OF      1.4860

     403 ELEMENTS IN X ARRAY ARE USED FOR STREAMS
     744 ELEMENTS OF X ARRAY USED OUT OF 200000

SIP1 -- STRONGLY IMPLICIT PROCEDURE SOLUTION PACKAGE, VERSION 1, 9/1/87 INPUT READ FROM UNIT 13
MAXIMUM OF 150 ITERATIONS ALLOWED FOR CLOSURE

 5 ITERATION PARAMETERS

     749 ELEMENTS IN X ARRAY ARE USED BY SIP
    1493 ELEMENTS OF X ARRAY USED OUT OF  200000
```

EXAMPLE SIMULATION OF STREAM ROUTING PACKAGE -- STEADY STATE OCTOBER 21, 1987 -- STREAM STAGE IS CALCULATED

BOUNDARY ARRAY = 1 FOR LAYER 1

AQUIFER HEAD WILL BE SET TO 999.00 AT ALL NO-FLOW NODES (IBOUND=0).

INITIAL HEAD FOR LAYER 1 WILL BE READ ON UNIT 5 USING FORMAT: (6F8.0)

	1	2	3	4	5	6
1	480.0	480.0	480.0	480.0	480.0	480.0
2	480.0	480.0	480.0	480.0	480.0	480.0
3	480.0	480.0	480.0	480.0	480.0	480.0
4	480.0	480.0	480.0	480.0	480.0	480.0
5	480.0	480.0	480.0	480.0	480.0	480.0
6	480.0	480.0	480.0	480.0	480.0	480.0

HEAD PRINT FORMAT IS FORMAT NUMBER 5 DRAWDOWN PRINT FORMAT IS FORMAT NUMBER 5

HEADS WILL BE SAVED ON UNIT 0 DRAWDOWNS WILL BE SAVED ON UNIT 0

OUTPUT CONTROL IS SPECIFIED EVERY TIME STEP

COLUMN TO ROW ANISOTROPY = 1.000000

DELR = 1000.000

DELC = 1000.000

TRANSMIS. ALONG ROWS = 0.8000000E-01 FOR LAYER 1

SOLUTION BY THE STRONGLY IMPLICIT PROCEDURE

MAXIMUM ITERATIONS ALLOWED FOR CLOSURE = 150
ACCELERATION PARAMETER = 1.0000
HEAD CHANGE CRITERION FOR CLOSURE = 0.10000E-03
SIP HEAD CHANGE PRINTOUT INTERVAL = 999
CALCULATE ITERATION PARAMETERS FROM MODEL CALCULATED WSEED

STRESS PERIOD NO. 1, LENGTH = 1296000.

NUMBER OF TIME STEPS = 3

MULTIPLIER FOR DELT = 1.500

INITIAL TIME STEP SIZE = 272842.1

23 STREAM NODES

LAYER	ROW	COL	SEGMENT NUMBER	REACH NUMBER	STREAMFLOW	STREAM STAGE	STREAMBED CONDUCTANCE	STREAMBED BOT ELEVATION	STREAMBED TOP ELEVATION
1	1	3	1	1	4.500	495.0	1.200	490.0	492.0
1	2	3	1	2	0.0000	490.0	0.6000	485.0	487.0
1	2	3	2	1	1.500	487.0	0.2000	483.0	485.0
1	3	3	2	2	0.0000	486.0	0.4000	482.0	484.0
1	4	3	2	3	0.0000	484.0	0.4000	480.0	482.0
1	5	3	2	4	0.0000	480.0	0.2000	476.0	478.0
1	2	3	3	1	-1.000	486.0	0.4000	481.0	483.0
1	3	4	3	2	0.0000	482.0	1.200	477.0	479.0
1	4	4	3	3	0.0000	478.0	1.200	473.0	475.0
1	5	4	3	4	0.0000	475.0	0.6000	470.0	472.0
1	4	1	4	1	0.8000	492.0	0.4000	489.0	490.0
1	4	2	4	2	0.0000	488.0	0.3200	485.0	486.0
1	5	2	4	3	0.0000	483.0	0.3200	480.0	481.0
1	5	3	4	4	0.0000	480.0	0.2000	477.0	478.0
1	5	3	5	1	-1.000	478.0	0.2000	475.0	476.0
1	5	4	5	2	0.0000	474.0	0.2000	471.0	472.0
1	2	6	6	1	1.200	495.0	0.8000	491.0	493.0
1	3	6	6	2	0.0000	490.0	0.8000	486.0	488.0
1	4	5	6	3	0.0000	480.0	0.8000	476.0	478.0
1	5	5	6	4	0.0000	477.0	0.6000	473.0	475.0
1	5	4	6	5	0.0000	474.0	0.2000	470.0	472.0
1	5	4	7	1	-1.000	472.0	0.6000	467.0	469.0
1	6	4	7	2	0.0000	469.0	1.200	464.0	466.0

LAYER	ROW	COL	SEGMENT NUMBER	REACH NUMBER	STREAM WIDTH	STREAM SLOPE	ROUGH COEF
1	1	3	1	1	10.00	0.7000E-02	0.3000E-01
1	2	3	1	2	10.00	0.7000E-02	0.3000E-01
1	2	3	2	1	5.000	0.2000E-02	0.2200E-01
1	3	3	2	2	5.000	0.2000E-02	0.2200E-01
1	4	3	2	3	5.000	0.2000E-02	0.2200E-01
1	5	3	2	4	5.000	0.4000E-02	0.2200E-01
1	2	3	3	1	10.00	0.5000E-02	0.3000E-01
1	3	4	3	2	10.00	0.5000E-02	0.3000E-01
1	4	4	3	3	10.00	0.5000E-02	0.3000E-01
1	5	4	3	4	10.00	0.5000E-02	0.3000E-01
1	4	1	4	1	5.000	0.4000E-02	0.2200E-01
1	4	2	4	2	5.000	0.4000E-02	0.2200E-01
1	5	2	4	3	5.000	0.4000E-02	0.2200E-01
1	5	3	4	4	5.000	0.4000E-02	0.2200E-01
1	5	3	5	1	5.000	0.5000E-02	0.2200E-01
1	5	4	5	2	5.000	0.5000E-02	0.2200E-01
1	2	6	6	1	5.000	0.5000E-02	0.2200E-01
1	3	6	6	2	5.000	0.8000E-02	0.2200E-01
1	4	5	6	3	5.000	0.7000E-02	0.2200E-01
1	5	5	6	4	5.000	0.4000E-02	0.2200E-01
1	5	4	6	5	5.000	0.3000E-02	0.2200E-01
1	5	4	7	1	10.00	0.4000E-02	0.3000E-01
1	6	4	7	2	10.00	0.4000E-02	0.3000E-01

MAXIMUM NUMBER OF TRIBUTARY STREAMS IS 3

STREAM SEGMENT	TRIBUTARY STREAM SEGMENT NUMBERS		
1	0	0	0
2	0	0	0
3	1	0	0
4	0	0	0
5	2	4	0
6	0	0	0
7	3	5	6

DIVERSION SEGMENT NUMBER	UPSTREAM SEGMENT NUMBER
1	0
2	1
3	0
4	0
5	0
6	0
7	0

```
AVERAGE SEED = 0.06853890
MINIMUM SEED = 0.06853890

   5 ITERATION PARAMETERS CALCULATED FROM AVERAGE SEED:

          0.0000000E+00  0.4883367E+00  0.7382007E+00  0.8660468E+00  0.9314611E+00

   20 ITERATIONS FOR TIME STEP   1 IN STRESS PERIOD  1

HEAD/DRAWDOWN PRINTOUT FLAG = 0     TOTAL BUDGET PRINTOUT FLAG = 0     CELL-BY-CELL FLOW TERM FLAG = 0

OUTPUT FLAGS FOR ALL LAYERS ARE THE SAME:

  HEAD     DRAWDOWN  HEAD  DRAWDOWN
PRINTOUT  PRINTOUT  SAVE    SAVE
---------------------------------
    0         0       0       0

    1 ITERATIONS FOR TIME STEP   2 IN STRESS PERIOD  1

HEAD/DRAWDOWN PRINTOUT FLAG = 0     TOTAL BUDGET PRINTOUT FLAG = 0     CELL-BY-CELL FLOW TERM FLAG = 0

OUTPUT FLAGS FOR ALL LAYERS ARE THE SAME:

  HEAD     DRAWDOWN  HEAD  DRAWDOWN
PRINTOUT  PRINTOUT  SAVE    SAVE
---------------------------------
    0         0       0       0

    1 ITERATIONS FOR TIME STEP   3 IN STRESS PERIOD  1

MAXIMUM HEAD CHANGE FOR EACH ITERATION:

 HEAD CHANGE LAYER,ROW,COL HEAD CHANGE LAYER,ROW,COL HEAD CHANGE LAYER,ROW,COL HEAD CHANGE LAYER,ROW,COL HEAD CHANGE AYER,ROW,COL
-------------------------------------------------------------------------------------------------------------------------------
  0.2274E-06 (  1,  5,  4)

HEAD/DRAWDOWN PRINTOUT FLAG = 1     TOTAL BUDGET PRINTOUT FLAG = 1     CELL-BY-CELL FLOW TERM FLAG = 1

OUTPUT FLAGS FOR ALL LAYERS ARE THE SAME:

  HEAD     DRAWDOWN  HEAD  DRAWDOWN
PRINTOUT  PRINTOUT  SAVE    SAVE
---------------------------------
    1         0       0       0
```

LAYER	ROW	COLUMN	STREAM NUMBER	REACH NUMBER	FLOW INTO STREAM REACH	FLOW INTO AQUIFER	FLOW OUT OF STREAM REACH	HEAD IN STREAM
1	1	3	1	1	4.50	0.931	3.57	492.25
1	2	3	1	2	3.57	0.841	1.23	487.18
1	2	3	2	1	1.50	-.105	1.60	485.26
1	3	3	2	2	1.60	0.222	1.38	484.25
1	4	3	2	3	1.38	0.357	1.03	482.22
1	5	3	2	4	1.03	0.221	0.805	478.15
1	2	3	3	1	1.23	-1.05	2.27	483.17
1	3	4	3	2	2.27	-.663	2.94	479.21
1	4	4	3	3	2.94	-.612	3.55	475.24
1	5	4	3	4	3.55	0.455	3.09	472.24
1	4	1	4	1	0.800	0.446	0.354	490.11
1	4	2	4	2	0.354	0.339	0.157E-01	486.06
1	5	2	4	3	0.157E-01	0.157E-01	0.000	481.01
1	5	3	4	4	0.000	0.000	0.000	478.00
1	5	3	5	1	0.805	-.182	0.987	476.14
1	5	4	5	2	0.987	0.131	0.855	472.14
1	2	6	6	1	1.20	1.01	0.187	493.12
1	3	6	6	2	0.187	0.187	0.000	488.03
1	4	5	6	3	0.000	-.587E-01	0.587E-01	478.02
1	5	5	6	4	0.587E-01	0.587E-01	0.000	475.02
1	5	4	6	5	0.000	0.000	0.000	472.00
1	5	4	7	1	3.95	-1.30	5.25	469.32
1	6	4	7	2	5.25	-1.25	6.50	466.37

HEAD IN LAYER 1 AT END OF TIME STEP 3 IN STRESS PERIOD 1

	1	2	3	4	5	6
1	487.639	488.554	491.471	488.439	488.579	490.216
2	486.725	486.551	485.779	485.266	487.082	491.854
3	485.985	485.144	483.696	479.763	482.630	485.603
4	486.087	484.345	481.326	475.750	478.073	479.984
5	482.357	480.593	477.047	471.485	474.663	476.275
6	480.392	478.426	474.294	467.409	472.084	474.179

VOLUMETRIC BUDGET FOR ENTIRE MODEL AT END OF TIME STEP 3 IN STRESS PERIOD 1

CUMULATIVE VOLUMES	L**3	RATES FOR THIS TIME STEP	L**3/T
IN:		IN:	
STORAGE =	0.00000	STORAGE =	0.00000
CONSTANT HEAD =	0.00000	CONSTANT HEAD =	0.00000
STREAM LEAKAGE =	0.67615E+07	STREAM LEAKAGE =	5.2172
TOTAL IN =	0.67615E+07	TOTAL IN =	5.2172
OUT:		OUT:	
STORAGE =	0.00000	STORAGE =	0.00000
CONSTANT HEAD =	0.00000	CONSTANT HEAD =	0.00000
STREAM LEAKAGE =	0.67619E+07	STREAM LEAKAGE =	5.2175
TOTAL OUT =	0.67619E+07	TOTAL OUT =	5.2175
IN - OUT =	-382.00	IN - OUT =	-0.29469E-03
PERCENT DISCREPANCY =	-0.01	PERCENT DISCREPANCY =	-0.01

TIME SUMMARY AT END OF TIME STEP 3 IN STRESS PERIOD 1

	SECONDS	MINUTES	HOURS	DAYS	YEARS
TIME STEP LENGTH	613894.	10231.6	170.526	7.10526	0.194531E-01
STRESS PERIOD TIME	0.129600E+07	21600.0	360.000	15.0000	0.410677E-01
TOTAL SIMULATION TIME	0.129600E+07	21600.0	360.000	15.0000	0.410677E-01

Input Instructions

Input for the Time-Variant Specified-Head Package is read from the unit in IUNIT(20) specified in the Basic Package input (see McDonald and Harbaugh, 1988, chap. 4, p. 9-11).

FOR EACH SIMULATION

CHD1AL

1. Data: MXCHD
   Format: I10

FOR EACH STRESS PERIOD

CHD1RP

2. Data: ITMP
   Format: I10

					Start	End
3. Data:	Layer	Row	Column	Head	Head
Format:	I10	I10	I10	F10.0	F10.0

(Input item 3 normally consists of one record for each constant-head boundary cell. If ITMP is zero, item 3 is not read)

Explanation of Fields Used in Input Instructions

MXCHD is the maximum number of specified-head cells to be specified each stress period.

ITMP is a flag.

If ITMP < 0, specified-head boundary data from the previous stress period will be reused and input item 3 will not be read. Reusing data from a previous stress period means that the head values at the start and end of the current stress period will be the same as they were at the start and end of the previous stress period.

If ITMP ≥ 0, it is the number of records of specified-head boundary data that will be read for the current stress period.

Layer is the layer number of the cell affected by the specified-head boundary.

Row is the row number of the cell affected by the specified-head boundary.

Column is the column number of the cell affected by the specified-head boundary.

Start head is the head value at the cell at the start of the stress period. Because the package assigns head values on the basis of linear interpolation to the end of a time step, the assigned specified-head value will never equal Start head unless Start head and End head are equal.

End head is the head value at the cell at the end of the stress period. It is the value that will be assigned to the cell for the last time step in the stress period.

Program Listing for Module CHD1AL

```
      SUBROUTINE CHD1AL(ISUM,LENX,LCCHDS,NCHDS,MXCHD,IN,IOUT)
C
C-----VERSION 0000 23SEP1987 CHD1AL
C   ******************************************************************************
C     ALLOCATE ARRAY STORAGE FOR CONSTANT-HEAD CELLS
C   ******************************************************************************
C
C     SPECIFICATIONS:
C   ------------------------------------------------------------------------------
C   ------------------------------------------------------------------------------
C
C1------IDENTIFY PACKAGE AND INITIALIZE # OF CONSTANT HEAD CELLS
      WRITE(IOUT,1)IN
    1 FORMAT(1HO,'CHD1 -- CHD PACKAGE, VERSION 1, 09/23/87',
     2' INPUT READ FROM UNIT',I3)
      NCHDS=0
C
C2------READ AND PRINT MXCHD (NUMBER OF CONSTANT-HEAD
C2------CELLS TO BE SPECIFIED EACH STRESS PERIOD)
      READ(IN,2) MXCHD
    2 FORMAT(I10)
      WRITE(IOUT,3) MXCHD
    3 FORMAT(1H ,'A TOTAL OF',I5,' CONSTANT-HEAD CELLS MUST BE',
     2 ' SPECIFIED EACH STRESS PERIOD.')
C
C3------SET LCCHDS EQUAL TO ADDRESS OF FIRST UNUSED SPACE IN X.
      LCCHDS=ISUM
C
C4------CALCULATE AMOUNT OF SPACE USED BY THE CONSTANT-HEAD LIST.
      ISP=5*MXCHD
      ISUM=ISUM+ISP
C
C5------PRINT AMOUNT OF SPACE USED BY THE CHD PACKAGE.
      WRITE(IOUT,4) ISP
    4 FORMAT(1X,I6,' ELEMENTS IN X ARRAY ARE USED FOR CONSTANT',
     1      '-HEAD CELLS')
      ISUM1=ISUM-1
      WRITE(IOUT,5) ISUM1,LENX
    5 FORMAT(1X,I6,' ELEMENTS OF X ARRAY USED OUT OF',I7)
      IF(ISUM1.GT.LENX) WRITE(IOUT,6)
    6 FORMAT(1X,'   ***X ARRAY MUST BE DIMENSIONED LARGER***')
C
C6------RETURN
      RETURN
      END
```

Program Listing for Module CHD1RP

```
      SUBROUTINE CHD1RP(CHDS,NCHDS,MXCHD,IBOUND,NCOL,NROW,NLAY,
     2  PERLEN,DELT,NSTP,TSMULT,IN,IOUT)
C
C
C-----VERSION 0000 23SEP1987 CHD1RP
C   ******************************************************************************
C     READ DATA FOR CHD
C   ******************************************************************************
C
C     SPECIFICATIONS:
C   ------------------------------------------------------------------------------
      DIMENSION CHDS(5,MXCHD),IBOUND(NCOL,NROW,NLAY)
C   ------------------------------------------------------------------------------
C
C1------READ ITMP(FLAG TO REUSE DATA.)
      READ(IN,8) ITMP
    8 FORMAT(I10)
C
C2------TEST ITMP
      IF(ITMP.GE.0) GO TO 50
C
C2A-----IF ITMP<0 THEN REUSE DATA FROM LAST STRESS PERIOD.
      WRITE(IOUT,7)
    7 FORMAT(1H0,'REUSING CONSTANT-HEAD CELLS FROM LAST STRESS',
     1      ' PERIOD')
      GO TO 260
C
C3------IF ITMP=>0 THEN IT IS THE # OF CONSTANT-HEAD CELLS.
   50 NCHDS=MXCHD
C
C4------PRINT # OF CONSTANT-HEAD CELLS THIS STRESS PERIOD.
  100 WRITE(IOUT,1) NCHDS
    1 FORMAT(1H0,//1X,I5,' CONSTANT-HEAD CELLS')
C
C5------IF THERE ARE NO CONSTANT-HEAD CELLS THEN RETURN.
      IF(NCHDS.EQ.0) GO TO 260
C
C6------READ & PRINT DATA FOR EACH CONSTANT-HEAD CELL.
      WRITE(IOUT,3)
    3 FORMAT(1H0,15X,'LAYER',5X,'ROW',5X
     1,'COL   STRT HEAD   ENDING HEAD'/1X,15X,48('-'))
      DO 250 II=1,NCHDS
      READ (IN,4) K,I,J,CHDS(4,II),CHDS(5,II)
    4 FORMAT(3I10,2F10.0)
      WRITE (IOUT,5) K,I,J,CHDS(4,II),CHDS(5,II)
    5 FORMAT(1X,15X,I4,I9,I8,G13.4,G14.4)
      CHDS(1,II)=K
      CHDS(2,II)=I
      CHDS(3,II)=J
      IF(IBOUND(J,I,K).NE.0) IBOUND(J,I,K)=-IABS(IBOUND(J,I,K))
  250 CONTINUE
```

```
C
C7------RECOMPUTE LENGTH OF PERIOD, PERLEN, A LOCAL VARIABLE IN
C7------SUBROUTINE BAS1AD
      PERLEN=DELT*FLOAT(NSTP)
      IF(TSMULT.NE.1.) PERLEN=DELT*(1.-TSMULT**NSTP)/(1.-TSMULT)
C8------RETURN
  260 RETURN
      END
```

Program Listing for Module CHD1FM

```
      SUBROUTINE CHD1FM(NCHDS,MXCHD,CHDS,IBOUND,HNEW,
     1         HOLD,PERLEN,PERTIM,DELT,NCOL,NROW,NLAY)
C
C-----VERSION 0000 23SEP1987 CHD1FM
C   ******************************************************************************
C     COMPUTE HEAD FOR TIME STEP AT EACH CONSTANT HEAD CELL
C   ******************************************************************************
C
C     SPECIFICATIONS:
C   ------------------------------------------------------------------------------
      DOUBLE PRECISION HNEW
C
      DIMENSION CHDS(5,MXCHD),IBOUND(NCOL,NROW,NLAY),
     2          HNEW(NCOL,NROW,NLAY),HOLD(NCOL,NROW,NLAY)
C   ------------------------------------------------------------------------------
C
C1------IF NCHDS<=0 THEN THERE ARE NO CONSTANT-HEAD CELLS. RETURN.
      IF(NCHDS.LE.0) RETURN
C
C2------COMPUTE PROPORTION OF STRESS PERIOD TO END OF THIS TIME STEP
      FRAC=PERTIM/PERLEN
C
C2------PROCESS EACH ENTRY IN THE CONSTANT-HEAD CELL LIST (CHDS)
      DO 100 L=1,NCHDS
C
C3------GET COLUMN, ROW AND LAYER OF CELL CONTAINING BOUNDARY
      IL=CHDS(1,L)
      IR=CHDS(2,L)
      IC=CHDS(3,L)
C
C5------COMPUTE HEAD AT CELL BY LINEAR INTERPOLATION.
      HB=CHDS(4,L)+(CHDS(5,L)-CHDS(4,L))*FRAC
C
C6------UPDATE THE APPROPRIATE HNEW VALUE
      HNEW(IC,IR,IL)=HB
      HOLD(IC,IR,IL)=HB
  100 CONTINUE
C
C7------RETURN
      RETURN
      END
```

APPENDIX D

EXAMPLE OF MAIN PROGRAM TO USE WITH INTERBED-STORAGE AND TIME-VARIANT SPECIFIED-HEAD PACKAGES

The following main program has been modified to allow use of the Interbed-Storage Package and the Time-Variant Specified-Head Package. Either one of the packages can be run independently of the other. The additions to the code specify that input data for the Interbed-Storage Package are to be read from the unit number stored in element 19 of the IUNIT array. Input data for the Time-Variant Specified-Head Package are to be read from the unit number stored in element 20 of the IUNIT array.

In the following program listing, records added to access the Interbed-Storage Package are identified with the characters "IBS" in rightmost columns. Records added to access the Time-Variant Specified-Head Package are identified with the characters "CHD" in rightmost columns. The identifiers "IBS" and "CHD" are not necessary parts of the added records and if they are included in the FORTRAN program, they must be placed within columns 73-80.

Program Listing

```
C   ******************************************************************************
C     MAIN CODE FOR MODULAR MODEL --    9/1/87
C          BY MICHAEL G. MCDONALD AND ARLEN W. HARBAUGH
C-----VERSION 1638 24JUL1987 MAIN1
C   ******************************************************************************
C
C        SPECIFICATIONS:
C   ------------------------------------------------------------------------------
      COMMON X(120000)
      COMMON /FLWCOM/LAYCON(80)
      CHARACTER*4 HEADNG,VBNM
      DIMENSION HEADNG(32),VBNM(4,20),VBVL(4,20),IUNIT(24)
      DOUBLE PRECISION DUMMY
      EQUIVALENCE (DUMMY,X(1))
C   ------------------------------------------------------------------------------
C
C1------SET SIZE OF X ARRAY. REMEMBER TO REDIMENSION X.
      LENX=120000
C
C2------ASSIGN BASIC INPUT UNIT AND PRINTER UNIT.
      INBAS=5
      IOUT=6
C
C3------DEFINE PROBLEM__ROWS,COLUMNS,LAYERS,STRESS PERIODS,PACKAGES
      CALL BAS1DF(ISUM,HEADNG,NPER,ITMUNI,TOTIM,NCOL,NROW,NLAY,
     1            NODES,INBAS,IOUT,IUNIT)
C
```

```
C4------ALLOCATE SPACE IN "X" ARRAY.
      CALL BAS1AL(ISUM,LENX,LCHNEW,LCHOLD,LCIBOU,LCCR,LCCC,LCCV,
     1              LCHCOF,LCRHS,LCDELR,LCDELC,LCSTRT,LCBUFF,LCIOFL,
     2              INBAS,ISTRT,NCOL,NROW,NLAY,IOUT)
      IF(IUNIT(1).GT.0) CALL BCF1AL(ISUM,LENX,LCSC1,LCHY,
     1     LCBOT,LCTOP,LCSC2,LCTRPY,IUNIT(1),ISS,
     2     NCOL,NROW,NLAY,IOUT,IBCFCB)
      IF(IUNIT(2).GT.0) CALL WEL1AL(ISUM,LENX,LCWELL,MXWELL,NWELLS,
     1                 IUNIT(2),IOUT,IWELCB)
      IF(IUNIT(3).GT.0) CALL DRN1AL(ISUM,LENX,LCDRAI,NDRAIN,MXDRN,
     1                 IUNIT(3),IOUT,IDRNCB)
      IF(IUNIT(8).GT.0) CALL RCH1AL(ISUM,LENX,LCIRCH,LCRECH,NRCHOP,
     1            NCOL,NROW,IUNIT(8),IOUT,IRCHCB)
      IF(IUNIT(5).GT.0) CALL EVT1AL(ISUM,LENX,LCIEVT,LCEVTR,LCEXDP,
     1            LCSURF,NCOL,NROW,NEVTOP,IUNIT(5),IOUT,IEVTCB)
      IF(IUNIT(4).GT.0) CALL RIV1AL(ISUM,LENX,LCRIVR,MXRIVR,NRIVER,
     1            IUNIT(4),IOUT,IRIVCB)
      IF(IUNIT(7).GT.0) CALL GHB1AL(ISUM,LENX,LCBNDS,NBOUND,MXBND,
     1            IUNIT(7),IOUT,IGHBCB)
      IF(IUNIT(9).GT.0) CALL SIP1AL(ISUM,LENX,LCEL,LCFL,LCGL,LCV,
     1          LCHDCG,LCLRCH,LCW,MXITER,NPARM,NCOL,NROW,NLAY,
     2          IUNIT(9),IOUT)
      IF(IUNIT(11).GT.0) CALL SOR1AL(ISUM,LENX,LCA,LCRES,LCHDCG,LCLRCH,
     1          LCIEQP,MXITER,NCOL,NLAY,NSLICE,MBW,IUNIT(11),IOUT)
      IF (IUNIT(19).GT.0) CALL IBS1AL(ISUM,LENX,LCHC,LCSCE,LCSCV,       IBS
     1          LCSUB,NCOL,NROW,NLAY,IIBSCB,IIBSOC,ISS,IUNIT(19),IOUT)  IBS
      IF(IUNIT(20).GT.0) CALL CHD1AL(ISUM,LENX,LCCHDS,NCHDS,MXCHD,      CHD
     1          IUNIT(20),IOUT)                                         CHD
C
C5------IF THE "X" ARRAY IS NOT BIG ENOUGH THEN STOP.
      IF(ISUM-1.GT.LENX) STOP
C
C6------READ AND PREPARE INFORMATION FOR ENTIRE SIMULATION.
      CALL BAS1RP(X(LCIBOU),X(LCHNEW),X(LCSTRT),X(LCHOLD),
     1       ISTRT,INBAS,HEADNG,NCOL,NROW,NLAY,NODES,VBVL,X(LCIOFL),
     2       IUNIT(12),IHEDFM,IDDNFM,IHEDUN,IDDNUN,IOUT)
      IF(IUNIT(1).GT.0) CALL BCF1RP(X(LCIBOU),X(LCHNEW),X(LCSC1),
     1          X(LCHY),X(LCCR),X(LCCC),X(LCCV),X(LCDELR),
     2           X(LCDELC),X(LCBOT),X(LCTOP),X(LCSC2),X(LCTRPY),
     3           IUNIT(1),ISS,NCOL,NROW,NLAY,NODES,IOUT)
      IF(IUNIT(9).GT.0) CALL SIP1RP(NPARM,MXITER,ACCL,HCLOSE,X(LCW),
     1          IUNIT(9),IPCALC,IPRSIP,IOUT)
      IF(IUNIT(11).GT.0) CALL SOR1RP(MXITER,ACCL,HCLOSE,IUNIT(11),
     1         IPRSOR,IOUT)
      IF(IUNIT(19).GT.0) CALL IBS1RP(X(LCDELR),X(LCDELC),X(LCHNEW),     IBS
     1      X(LCHC),X(LCSCE),X(LCSCV),X(LCSUB),NCOL,NROW,NLAY,          IBS
     2      NODES,IIBSOC,ISUBFM,ICOMFM,IHCFM,ISUBUN,ICOMUN,IHCUN,       IBS
     3      IUNIT(19),IOUT)                                             IBS
C
C7------SIMULATE EACH STRESS PERIOD.
      DO 300 KPER=1,NPER
      KKPER=KPER
C
```

```
C7A-----READ STRESS PERIOD TIMING INFORMATION.
      CALL BAS1ST(NSTP,DELT,TSMULT,PERTIM,KKPER,INBAS,IOUT)
C
C7B-----READ AND PREPARE INFORMATION FOR STRESS PERIOD.
      IF(IUNIT(2).GT.0) CALL WEL1RP(X(LCWELL),NWELLS,MXWELL,IUNIT(2),
     1             IOUT)
      IF(IUNIT(3).GT.0) CALL DRN1RP(X(LCDRAI),NDRAIN,MXDRN,IUNIT(3),
     1                  IOUT)
      IF(IUNIT(8).GT.0) CALL RCH1RP(NRCHOP,X(LCIRCH),X(LCRECH),
     1            X(LCDELR),X(LCDELC),NROW,NCOL,IUNIT(8),IOUT)
      IF(IUNIT(5).GT.0) CALL EVT1RP(NEVTOP,X(LCIEVT),X(LCEVTR),
     1            X(LCEXDP),X(LCSURF),X(LCDELR),X(LCDELC),NCOL,NROW,
     1            IUNIT(5),IOUT)
      IF(IUNIT(4).GT.0) CALL RIV1RP(X(LCRIVR),NRIVER,MXRIVR,IUNIT(4),
     1            IOUT)
      IF(IUNIT(7).GT.0) CALL GHB1RP(X(LCBNDS),NBOUND,MXBND,IUNIT(7),
     1              IOUT)
      IF(IUNIT(20).GT.0) CALL CHD1RP(X(LCCHDS),NCHDS,MXCHD,X(LCIBOU),    CHD
     1         NCOL,NROW,NLAY,PERLEN,DELT,NSTP,TSMULT,IUNIT(20),IOUT)    CHD
C
C7C-----SIMULATE EACH TIME STEP.
      DO 200 KSTP=1,NSTP
      KKSTP=KSTP
C
C7C1----CALCULATE TIME STEP LENGTH. SET HOLD=HNEW..
      CALL BAS1AD(DELT,TSMULT,TOTIM,PERTIM,X(LCHNEW),X(LCHOLD),KKSTP,
     1             NCOL,NROW,NLAY)
      IF(IUNIT(20).GT.0) CALL CHD1FM(NCHDS,MXCHD,X(LCCHDS),X(LCIBOU),    CHD
     1          X(LCHNEW),X(LCHOLD),PERLEN,PERTIM,DELT,NCOL,NROW,NLAY)   CHD
C
C7C2----ITERATIVELY FORMULATE AND SOLVE THE EQUATIONS.
      DO 100 KITER=1,MXITER
      KKITER=KITER
C
C7C2A---FORMULATE THE FINITE DIFFERENCE EQUATIONS.
      CALL BAS1FM(X(LCHCOF),X(LCRHS),NODES)
      IF(IUNIT(1).GT.0) CALL BCF1FM(X(LCHCOF),X(LCRHS),X(LCHOLD),
     1           X(LCSC1),X(LCHNEW),X(LCIBOU),X(LCCR),X(LCCC),X(LCCV),
     2           X(LCHY),X(LCTRPY),X(LCBOT),X(LCTOP),X(LCSC2),
     3           X(LCDELR),X(LCDELC),DELT,ISS,KKITER,KKSTP,KKPER,NCOL,
     4           NROW,NLAY,IOUT)
      IF(IUNIT(2).GT.0) CALL WEL1FM(NWELLS,MXWELL,X(LCRHS),X(LCWELL),
     1            X(LCIBOU),NCOL,NROW,NLAY)
      IF(IUNIT(3).GT.0) CALL DRN1FM(NDRAIN,MXDRN,X(LCDRAI),X(LCHNEW),
     1          X(LCHCOF),X(LCRHS),X(LCIBOU),NCOL,NROW,NLAY)
      IF(IUNIT(8).GT.0) CALL RCH1FM(NRCHOP,X(LCIRCH),X(LCRECH),
     1            X(LCRHS),X(LCIBOU),NCOL,NROW,NLAY)
      IF(IUNIT(5).GT.0) CALL EVT1FM(NEVTOP,X(LCIEVT),X(LCEVTR),
     1            X(LCEXDP),X(LCSURF),X(LCRHS),X(LCHCOF),X(LCIBOU),
     1            X(LCHNEW),NCOL,NROW,NLAY)
      IF(IUNIT(4).GT.0) CALL RIV1FM(NRIVER,MXRIVR,X(LCRIVR),X(LCHNEW),
     1            X(LCHCOF),X(LCRHS),X(LCIBOU),NCOL,NROW,NLAY)
      IF(IUNIT(7).GT.0) CALL GHB1FM(NBOUND,MXBND,X(LCBNDS),X(LCHCOF),
     1            X(LCRHS),X(LCIBOU),NCOL,NROW,NLAY)
```

```
      IF(IUNIT(19).GT.0) CALL IBS1FM(X(LCRHS),X(LCHCOF),X(LCHNEW),      IBS
     1       X(LCHOLD),X(LCHC),X(LCSCE),X(LCSCV),X(LCIBOU),             IBS
     2       NCOL,NROW,NLAY,DELT)                                       IBS
C
C7C2B---MAKE ONE CUT AT AN APPROXIMATE SOLUTION.
      IF(IUNIT(9).GT.0) CALL SIP1AP(X(LCHNEW),X(LCIBOU),X(LCCR),X(LCCC),
     1     X(LCCV),X(LCHCOF),X(LCRHS),X(LCEL),X(LCFL),X(LCGL),X(LCV),
     2     X(LCW),X(LCHDCG),X(LCLRCH),NPARM,KKITER,HCLOSE,ACCL,ICNVG,
     3     KKSTP,KKPER,IPCALC,IPRSIP,MXITER,NSTP,NCOL,NROW,NLAY,NODES,
     4     IOUT)
      IF(IUNIT(11).GT.0) CALL SOR1AP(X(LCHNEW),X(LCIBOU),X(LCCR),
     1     X(LCCC),X(LCCV),X(LCHCOF),X(LCRHS),X(LCA),X(LCRES),X(LCIEQP),
     2     X(LCHDCG),X(LCLRCH),KKITER,HCLOSE,ACCL,ICNVG,KKSTP,KKPER,
     3     IPRSOR,MXITER,NSTP,NCOL,NROW,NLAY,NSLICE,MBW,IOUT)
C
C7C2C---IF CONVERGENCE CRITERION HAS BEEN MET STOP ITERATING.
      IF(ICNVG.EQ.1) GO TO 110
  100 CONTINUE
      KITER=MXITER
  110 CONTINUE
C
C7C3----DETERMINE WHICH OUTPUT IS NEEDED.
      CALL BAS1OC(NSTP,KKSTP,ICNVG,X(LCIOFL),NLAY,
     1  IBUDFL,ICBCFL,IHDDFL,IUNIT(12),IOUT)
C
C7C4----CALCULATE BUDGET TERMS. SAVE CELL-BY-CELL FLOW TERMS.
      MSUM=1
      IF(IUNIT(1).GT.0) CALL BCF1BD(VBNM,VBVL,MSUM,X(LCHNEW),
     1     X(LCIBOU),X(LCHOLD),X(LCSC1),X(LCCR),X(LCCC),X(LCCV),
     2     X(LCTOP),X(LCSC2),DELT,ISS,NCOL,NROW,NLAY,KKSTP,KKPER,
     3     IBCFCB,ICBCFL,X(LCBUFF),IOUT)
      IF(IUNIT(2).GT.0) CALL WEL1BD(NWELLS,MXWELL,VBNM,VBVL,MSUM,
     1     X(LCWELL),X(LCIBOU),DELT,NCOL,NROW,NLAY,KKSTP,KKPER,IWELCB,
     1     ICBCFL,X(LCBUFF),IOUT)
      IF(IUNIT(3).GT.0) CALL DRN1BD(NDRAIN,MXDRN,VBNM,VBVL,MSUM,
     1     X(LCDRAI),DELT,X(LCHNEW),NCOL,NROW,NLAY,X(LCIBOU),KKSTP,
     2     KKPER,IDRNCB,ICBCFL,X(LCBUFF),IOUT)
      IF(IUNIT(8).GT.0) CALL RCH1BD(NRCHOP,X(LCIRCH),X(LCRECH),
     1     X(LCIBOU),NROW,NCOL,NLAY,DELT,VBVL,VBNM,MSUM,KKSTP,KKPER,
     2     IRCHCB,ICBCFL,X(LCBUFF),IOUT)
      IF(IUNIT(5).GT.0) CALL EVT1BD(NEVTOP,X(LCIEVT),X(LCEVTR),
     1     X(LCEXDP),X(LCSURF),X(LCIBOU),X(LCHNEW),NCOL,NROW,NLAY,
     2     DELT,VBVL,VBNM,MSUM,KKSTP,KKPER,IEVTCB,ICBCFL,X(LCBUFF),IOUT)
      IF(IUNIT(4).GT.0) CALL RIV1BD(NRIVER,MXRIVR,X(LCRIVR),X(LCIBOU),
     1     X(LCHNEW),NCOL,NROW,NLAY,DELT,VBVL,VBNM,MSUM,
     2     KKSTP,KKPER,IRIVCB,ICBCFL,X(LCBUFF),IOUT)
      IF(IUNIT(7).GT.0) CALL GHB1BD(NBOUND,MXBND,VBNM,VBVL,MSUM,
     1     X(LCBNDS),DELT,X(LCHNEW),NCOL,NROW,NLAY,X(LCIBOU),KKSTP,
     2     KKPER,IGHBCB,ICBCFL,X(LCBUFF),IOUT)
      IF(IUNIT(19).GT.0) CALL IBS1BD(X(LCIBOU),X(LCHNEW),X(LCHOLD),      IBS
     1      X(LCHC),X(LCSCE),X(LCSCV),X(LCSUB),X(LCDELR),X(LCDELC),      IBS
     2      NCOL,NROW,NLAY,DELT,VBVL,VBNM,MSUM,KKSTP,KKPER,IIBSCB,       IBS
     3      ICBCFL,X(LCBUFF),IOUT)                                       IBS
C
```

```
C7C5---PRINT AND OR SAVE HEADS AND DRAWDOWNS. PRINT OVERALL BUDGET.
C7C5A--FIRST PRINT AND OR SAVE SUBSIDENCE, COMPACTION,                  IBS
C7C5A-- AND CRITICAL HEAD.                                              IBS
      IF(IUNIT(19).GT.0) CALL IBS1OT(NCOL,NROW,NLAY,PERTIM,TOTIM,KKSTP, IBS
     1      KKPER,NSTP,X(LCBUFF),X(LCSUB),X(LCHC),IIBSOC,ISUBFM,ICOMFM, IBS
     2      IHCFM,ISUBUN,ICOMUN,IHCUN,IUNIT(19),IOUT)                   IBS
      CALL BAS1OT(X(LCHNEW),X(LCSTRT),ISTRT,X(LCBUFF),X(LCIOFL),
     1        MSUM,X(LCIBOU),VBNM,VBVL,KKSTP,KKPER,DELT,
     2        PERTIM,TOTIM,ITMUNI,NCOL,NROW,NLAY,ICNVG,
     3     IHDDFL,IBUDFL,IHEDFM,IHEDUN,IDDNFM,IDDNUN,IOUT)
C
C7C6----IF ITERATION FAILED TO CONVERGE THEN STOP.
      IF(ICNVG.EQ.0) STOP
  200 CONTINUE
  300 CONTINUE
C
C8------END PROGRAM
      STOP
C
      END
```

DOCUMENTATION OF A COMPUTER PROGRAM TO SIMULATE HORIZONTAL-FLOW BARRIERS USING THE U.S. GEOLOGICAL SURVEY'S MODULAR THREE-DIMENSIONAL FINITE-DIFFERENCE GROUND-WATER FLOW MODEL

By Paul A. Hsieh *and* John R. Freckleton

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PREFACE

This report presents a computer program for simulating horizontal-flow barriers in the Modular Three-Dimensional Finite-Difference Ground-Water Flow Model (MODFLOW) developed by the U.S. Geological Survey. Users of the report are encouraged to notify the originating office of any errors found in the report or in the computer program. Updates may be made occasionally to both the report and computer program. Users who wish to be added to the mailing list to receive updates, if any, may send a request to:

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DEFINITIONS OF SELECTED TERMS AND MISCELLANEOUS NOTES

Definitions

Executable program: An organized set of instructions that directs a computer in solving a particular problem or performing a designated task. This is the program that "runs" on the computer.
MAIN program: The controlling program unit at which program execution begins. It controls the order in which the primary modules are executed and it serves as a switching system for information.
Module: A collection of related source-code instructions to perform a functional task.
Package or PACKAGE: A collection of related modules (includes submodules).
Primary Module: A module that is called directly by the MAIN program.
Source code: The collection of programming language instructions, Fortran in this report, written by a programmer.
Submodule: A module that is called by other modules.
Subroutine: For this report, synonymous with module or submodule.

Notes

1. The source code used for the subroutines contained in this report is written in the Fortran computer programming language and conforms to the Fortran 66 standard (American National Standards Institute, 1966). The only exception to the Fortran 66 standard is that character strings in FORMAT statements are in single quotes rather than Hollerith constants. Most Fortran 66 compilers include this capability. The modified MAIN program in this report includes conversions by McDonald and Harbaugh (1988) to conform to the Fortran 77 standard (American National Standards Institute, 1978).
2. Rather than writing out a complete logical IF statement as it appears in the source code, the authors simply refer to it, when it is used in the text, as an IF statement or an 'IF...CALL...' statement (quotation marks included).
3. Fortran allows for a program identification/sequence alphanumeric to be placed in columns 73 through 80 of a statement field. In the source-code listing for MAIN found in the text, those columns that contain *HFB* identify code produced for this report.
4. In the text, the word 'CALL', written in capitals, refers to the Fortran keyword CALL and is sometimes used as a verb as in 'CALLed' or a plural as in 'CALLs'.

DOCUMENTATION OF A COMPUTER PROGRAM TO SIMULATE HORIZONTAL-FLOW BARRIERS USING THE U.S. GEOLOGICAL SURVEY'S MODULAR THREE-DIMENSIONAL FINITE-DIFFERENCE GROUND-WATER FLOW MODEL

By Paul A. Hsieh *and* John R. Freckleton

Abstract

As an addition to the U.S. Geological Survey's Modular Three-Dimensional Finite-Difference Ground-Water Flow Model, the Horizontal-Flow-Barrier (HFB) Package simulates thin, vertical low-permeability geologic features that impede the horizontal flow of ground water. These geologic features are approximated as a series of horizontal-flow barriers conceptually situated on the boundaries between pairs of adjacent cells in the finite-difference grid. The key assumption underlying the HFB Package is that the width of the barrier is negligibly small in comparison with the horizontal dimensions of the cells in the grid. Barrier width is not explicitly considered in the Package, but is included implicitly in a hydraulic characteristic defined as either (1) barrier transmissivity divided by barrier width, if the barrier is in a constant-transmissivity layer, or (2) barrier hydraulic conductivity divided by barrier width, if the barrier is in a variable-transmissivity layer. Furthermore, the barrier is assumed to have zero storage capacity. Its sole function is to lower the horizontal branch conductance between the two cells that it separates.

Documentation of the HFB Package includes data-input instructions, narratives, flowcharts, program listings, and variables lists for the three primary modules and one submodule in the Package.

INTRODUCTION

In some aquifers there may exist thin, vertical low-permeability geologic features, such as vertical faults, that act as barriers to the horizontal flow of water. These features can be simulated in the model developed by McDonald and Harbaugh (1988) by reducing model grid spacing appropriately or by using variable grid spacing in the region of the features to be modeled. However, reduced grid spacing may result in an excessive number of model cells, and variable grid spacing is efficient (in terms of number of model cells used for a feature) only if all the features to be modeled are parallel to either the row or column direction of the model. In addition, the abruptness with which changing hydraulic properties are modeled must be considered in order to avoid possible numerical difficulties. The Horizontal-Flow-Barrier (HFB) Package, documented in this report, allows these features to be modeled without increasing the number of model cells or using variable grid spacing; thus, the HFB Package may enhance model efficiency.

The Horizontal-Flow-Barrier (HFB) Package consists of three primary modules and one submodule. The documentation includes data-input instructions, narratives, flowcharts, program listings, and variables lists for the three primary modules and one submodule.

PROGRAM DESIGN

The HFB Package is designed to be totally compatible with the standard version of the Modular Three-Dimensional Finite-Difference Ground-Water Flow Model (McDonald and Harbaugh, 1988), which commonly is referred to as MODFLOW. The HFB Package can be incorporated into MODFLOW by (1) designating an unused and unreserved element of the IUNIT array to serve as a flag for invoking the Package, (2) inserting into the MAIN program of MODFLOW three IF statements which, when true, invoke CALLs to HFB subroutine modules, (3) adding the source codes of four HFB modules (three primary modules and one submodule) to the MODFLOW source code, and (4) for a typical computer system, compiling and linking the code into an executable program. In the present case, IUNIT(16) is chosen to be the flag for invoking the Package.

ADDITIONS TO THE MAIN PROGRAM

The three IF statements added to the MAIN program of MODFLOW are enclosed by boxes in the listing of the modified MAIN program on pages 3-6 of this report. These statements cause execution of CALLs to the three primary modules (HFB1AL, HFB1RP, and HFB1FM) of the HFB Package when the Package is invoked. The IF statement resulting in the CALL to module (subroutine) HFB1AL can be placed anywhere within the "allocate space" procedure of the MAIN program (McDonald and Harbaugh, 1988, chap. 3, p. 1-2). However, the IF statement that results in a CALL to module HFB1RP should be placed after the 'IF...CALL...' to module BCF1RP, and the IF statement that results in a CALL to module HFB1FM should be placed after the 'IF...CALL...' to module BCF1FM. In the event that the user is using an alternative package to the Block-Centered Flow Package, then the IF statements resulting in CALLs to the HFB1RP and HFB1FM modules should be placed after the statements that replace the 'IF...CALL...' to BCF1RP and BCF1FM, respectively. The relations among the primary modules, submodule, and MAIN program are shown in figure 1.

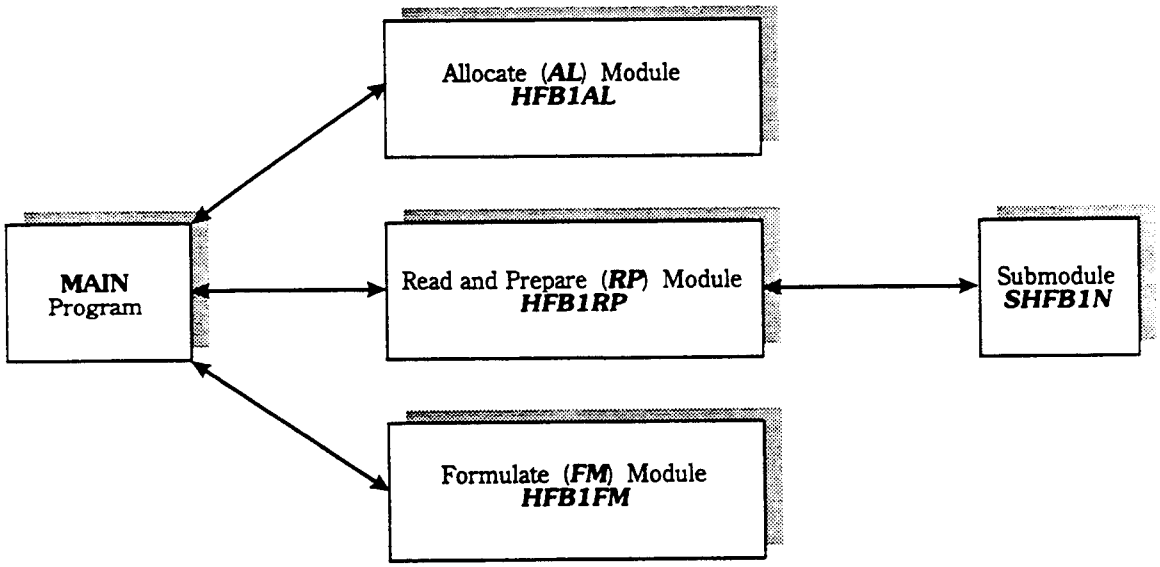


Figure 1. Relations among the primary modules, submodule, and MAIN program in the Horizontal-Flow-Barrier Package.

INVOKING THE HORIZONTAL-FLOW-BARRIER PACKAGE

As described in this report, the HFB Package is invoked by assigning a nonzero input unit number to IUNIT(16) of the Basic Package input (McDonald and Harbaugh, 1988, chap. 4, p. 9). The Package is invoked by entering a nonzero input unit number in columns 46 to 48 of data-item 4 of the Basic Package input.

PROGRAM LISTING FOR MODIFIED MAIN PROGRAM OF THE MODULAR MODEL (WITH THREE IF..CALL STATEMENTS INSERTED TO INVOKE HFB)

```
C *****************************************************************
C     MAIN CODE FOR MODULAR MODEL --    9/1/87
C          BY MICHAEL G. MCDONALD AND ARLEN W. HARBAUGH
C-----VERSION 1638 24 JUL 1987 MAIN1 MODIFIED TO INCLUDE HORIZONTAL-FLOW-
C     BARRIER (HFB) PACKAGE
C     *****************************************************************
C
C        SPECIFICATIONS:
C     -----------------------------------------------------------------
      COMMON X(30000)
      COMMON /FLWCOM/LAYCON(80)
      CHARACTER*4 HEADNG,VBNM
      DIMENSION HEADNG(32),VBNM(4,20),VBVL(4,20),IUNIT(24)
      DOUBLE PRECISION DUMMY
      EQUIVALENCE (DUMMY,X(1))
C     -----------------------------------------------------------------
C
C1------SET SIZE OF X ARRAY. REMEMBER TO REDIMENSION X.
      LENX=30000
C
C2------ASSIGN BASIC INPUT UNIT AND PRINTER UNIT.
      INBAS=1
      IOUT=6
C
C3------DEFINE PROBLEM__ROWS,COLUMNS,LAYERS,STRESS PERIODS,PACKAGES
      CALL BAS1DF(ISUM,HEADNG,NPER,ITMUNI,TOTIM,NCOL,NROW,NLAY,
     1             NODES,INBAS,IOUT,IUNIT)
C
C4------ALLOCATE SPACE IN "X" ARRAY.
      CALL BAS1AL(ISUM,LENX,LCHNEW,LCHOLD,LCIBOU,LCCR,LCCC,LCCV,
     1              LCHCOF,LCRHS,LCDELR,LCDELC,LCSTRT,LCBUFF,LCIOFL,
     2              INBAS,ISTRT,NCOL,NROW,NLAY,IOUT)
      IF(IUNIT(1).GT.0) CALL BCF1AL(ISUM,LENX,LCSC1,LCHY,
     1     LCBOT,LCTOP,LCSC2,LCTRPY,IUNIT(1),ISS,
     2     NCOL,NROW,NLAY,IOUT,IBCFCB)
      IF(IUNIT(2).GT.0) CALL WEL1AL(ISUM,LENX,LCWELL,MXWELL,NWELLS,
     1                 IUNIT(2),IOUT,IWELCB)
      IF(IUNIT(3).GT.0) CALL DRN1AL(ISUM,LENX,LCDRAI,NDRAIN,MXDRN,
     1                 IUNIT(3),IOUT,IDRNCB)
      IF(IUNIT(8).GT.0) CALL RCH1AL(ISUM,LENX,LCIRCH,LCRECH,NRCHOP,
     1            NCOL,NROW,IUNIT(8),IOUT,IRCHCB)
      IF(IUNIT(5).GT.0) CALL EVT1AL(ISUM,LENX,LCIEVT,LCEVTR,LCEXDP,
     1            LCSURF,NCOL,NROW,NEVTOP,IUNIT(5),IOUT,IEVTCB)
      IF(IUNIT(4).GT.0) CALL RIV1AL(ISUM,LENX,LCRIVR,MXRIVR,NRIVER,
     1            IUNIT(4),IOUT,IRIVCB)
      IF(IUNIT(7).GT.0) CALL GHB1AL(ISUM,LENX,LCBNDS,NBOUND,MXBND,
     1            IUNIT(7),IOUT,IGHBCB)
      IF(IUNIT(9).GT.0) CALL SIP1AL(ISUM,LENX,LCEL,LCFL,LCGL,LCV,
     1          LCHDCG,LCLRCH,LCW,MXITER,NPARM,NCOL,NROW,NLAY,
     2          IUNIT(9),IOUT)
      IF(IUNIT(11).GT.0) CALL SOR1AL(ISUM,LENX,LCA,LCRES,LCHDCG,LCLRCH,
     1           LCIEQP,MXITER,NCOL,NLAY,NSLICE,MBW,IUNIT(11),IOUT)
```

```
      IF(IUNIT(16).GT.0) CALL HFB1AL(ISUM,LENX,LCHFBR,NHFB,IUNIT(16),    *HFB*
     1           IOUT)                                                   *HFB*
C
C5------IF THE "X" ARRAY IS NOT BIG ENOUGH THEN STOP.
      IF(ISUM-1.GT.LENX) STOPC
C6------READ AND PREPARE INFORMATION FOR ENTIRE SIMULATION.
      CALL BAS1RP(X(LCIBOU),X(LCHNEW),X(LCSTRT),X(LCHOLD),
     1       ISTRT,INBAS,HEADNG,NCOL,NROW,NLAY,NODES,VBVL,X(LCIOFL),
     2       IUNIT(12),IHEDFM,IDDNFM,IHEDUN,IDDNUN,IOUT)
      IF(IUNIT(1).GT.0) CALL BCF1RP(X(LCIBOU),X(LCHNEW),X(LCSC1),
     1          X(LCHY),X(LCCR),X(LCCC),X(LCCV),X(LCDELR),
     2           X(LCDELC),X(LCBOT),X(LCTOP),X(LCSC2),X(LCTRPY),
     3           IUNIT(1),ISS,NCOL,NROW,NLAY,NODES,IOUT)
      IF(IUNIT(16).GT.0) CALL HFB1RP(X(LCCR),X(LCCC),X(LCDELR),          *HFB*
     1         X(LCDELC),X(LCHFBR),IUNIT(16),NCOL,NROW,NLAY,NODES,       *HFB*
     2         NHFB,IOUT)                                                *HFB*
      IF(IUNIT(9).GT.0) CALL SIP1RP(NPARM,MXITER,ACCL,HCLOSE,X(LCW),
     1          IUNIT(9),IPCALC,IPRSIP,IOUT)
      IF(IUNIT(11).GT.0) CALL SOR1RP(MXITER,ACCL,HCLOSE,IUNIT(11),
     1         IPRSOR,IOUT)
C
C7------SIMULATE EACH STRESS PERIOD.
      DO 300 KPER=1,NPER
      KKPER=KPER
C
C7A-----READ STRESS PERIOD TIMING INFORMATION.
      CALL BAS1ST(NSTP,DELT,TSMULT,PERTIM,KKPER,INBAS,IOUT)
C
C7B-----READ AND PREPARE INFORMATION FOR STRESS PERIOD.
      IF(IUNIT(2).GT.0) CALL WEL1RP(X(LCWELL),NWELLS,MXWELL,IUNIT(2),
     1             IOUT)
      IF(IUNIT(3).GT.0) CALL DRN1RP(X(LCDRAI),NDRAIN,MXDRN,IUNIT(3),
     1                  IOUT)
      IF(IUNIT(8).GT.0) CALL RCH1RP(NRCHOP,X(LCIRCH),X(LCRECH),
     1            X(LCDELR),X(LCDELC),NROW,NCOL,IUNIT(8),IOUT)
      IF(IUNIT(5).GT.0) CALL EVT1RP(NEVTOP,X(LCIEVT),X(LCEVTR),
     1            X(LCEXDP),X(LCSURF),X(LCDELR),X(LCDELC),NCOL,NROW,
     1            IUNIT(5),IOUT)
      IF(IUNIT(4).GT.0) CALL RIV1RP(X(LCRIVR),NRIVER,MXRIVR,IUNIT(4),
     1            IOUT)
      IF(IUNIT(7).GT.0) CALL GHB1RP(X(LCBNDS),NBOUND,MXBND,IUNIT(7),
     1              IOUT)
C
C7C-----SIMULATE EACH TIME STEP.
      DO 200 KSTP=1,NSTP
      KKSTP=KSTP
C
C7C1----CALCULATE TIME STEP LENGTH. SET HOLD=HNEW.
      CALL BAS1AD(DELT,TSMULT,TOTIM,PERTIM,X(LCHNEW),X(LCHOLD),KKSTP,
     1             NCOL,NROW,NLAY)
C
```

```
C7C2----ITERATIVELY FORMULATE AND SOLVE THE EQUATIONS.
      DO 100 KITER=1,MXITER
      KKITER=KITER
C
C7C2A---FORMULATE THE FINITE DIFFERENCE EQUATIONS.
      CALL BAS1FM(X(LCHCOF),X(LCRHS),NODES)
      IF(IUNIT(1).GT.0) CALL BCF1FM(X(LCHCOF),X(LCRHS),X(LCHOLD),
     1          X(LCSC1),X(LCHNEW),X(LCIBOU),X(LCCR),X(LCCC),X(LCCV),
     2          X(LCHY),X(LCTRPY),X(LCBOT),X(LCTOP),X(LCSC2),
     3          X(LCDELR),X(LCDELC),DELT,ISS,KKITER,KKSTP,KKPER,NCOL,
     4          NROW,NLAY,IOUT)
      IF(IUNIT(16).GT.0) CALL HFB1FM(X(LCHNEW),X(LCCR),X(LCCC),          *HFB*
     1          X(LCBOT),X(LCTOP),X(LCDELR),X(LCDELC),X(LCHFBR),         *HFB*
     2          NCOL,NROW,NLAY,NHFB)                                     *HFB*
      IF(IUNIT(2).GT.0) CALL WEL1FM(NWELLS,MXWELL,X(LCRHS),X(LCWELL),
     1           X(LCIBOU),NCOL,NROW,NLAY)
      IF(IUNIT(3).GT.0) CALL DRN1FM(NDRAIN,MXDRN,X(LCDRAI),X(LCHNEW),
     1         X(LCHCOF),X(LCRHS),X(LCIBOU),NCOL,NROW,NLAY)
      IF(IUNIT(8).GT.0) CALL RCH1FM(NRCHOP,X(LCIRCH),X(LCRECH),
     1            X(LCRHS),X(LCIBOU),NCOL,NROW,NLAY)
      IF(IUNIT(5).GT.0) CALL EVT1FM(NEVTOP,X(LCIEVT),X(LCEVTR),
     1            X(LCEXDP),X(LCSURF),X(LCRHS),X(LCHCOF),X(LCIBOU),
     1            X(LCHNEW),NCOL,NROW,NLAY)
      IF(IUNIT(4).GT.0) CALL RIV1FM(NRIVER,MXRIVR,X(LCRIVR),X(LCHNEW),
     1            X(LCHCOF),X(LCRHS),X(LCIBOU),NCOL,NROW,NLAY)
      IF(IUNIT(7).GT.0) CALL GHB1FM(NBOUND,MXBND,X(LCBNDS),X(LCHCOF),
     1            X(LCRHS),X(LCIBOU),NCOL,NROW,NLAY)
C
C7C2B---MAKE ONE CUT AT AN APPROXIMATE SOLUTION.
      IF(IUNIT(9).GT.0) CALL SIP1AP(X(LCHNEW),X(LCIBOU),X(LCCR),X(LCCC),
     1     X(LCCV),X(LCHCOF),X(LCRHS),X(LCEL),X(LCFL),X(LCGL),X(LCV),
     2     X(LCW),X(LCHDCG),X(LCLRCH),NPARM,KKITER,HCLOSE,ACCL,ICNVG,
     3     KKSTP,KKPER,IPCALC,IPRSIP,MXITER,NSTP,NCOL,NROW,NLAY,NODES,
     4     IOUT)
      IF(IUNIT(11).GT.0) CALL SOR1AP(X(LCHNEW),X(LCIBOU),X(LCCR),
     1     X(LCCC),X(LCCV),X(LCHCOF),X(LCRHS),X(LCA),X(LCRES),X(LCIEQP),
     2     X(LCHDCG),X(LCLRCH),KKITER,HCLOSE,ACCL,ICNVG,KKSTP,KKPER,
     3     IPRSOR,MXITER,NSTP,NCOL,NROW,NLAY,NSLICE,MBW,IOUT)
C
C7C2C---IF CONVERGENCE CRITERION HAS BEEN MET STOP ITERATING.
      IF(ICNVG.EQ.1) GO TO 110
  100 CONTINUE
      KITER=MXITER
  110 CONTINUE
C
C7C3----DETERMINE WHICH OUTPUT IS NEEDED.
      CALL BAS1OC(NSTP,KKSTP,ICNVG,X(LCIOFL),NLAY,
     1  IBUDFL,ICBCFL,IHDDFL,IUNIT(12),IOUT)
C
C7C4----CALCULATE BUDGET TERMS. SAVE CELL-BY-CELL FLOW TERMS.
      MSUM=1
      IF(IUNIT(1).GT.0) CALL BCF1BD(VBNM,VBVL,MSUM,X(LCHNEW),
     1     X(LCIBOU),X(LCHOLD),X(LCSC1),X(LCCR),X(LCCC),X(LCCV),
```

```
     2      X(LCTOP),X(LCSC2),DELT,ISS,NCOL,NROW,NLAY,KKSTP,KKPER,
     3      IBCFCB,ICBCFL,X(LCBUFF),IOUT)
       IF(IUNIT(2).GT.0) CALL WEL1BD(NWELLS,MXWELL,VBNM,VBVL,MSUM,
     1      X(LCWELL),X(LCIBOU),DELT,NCOL,NROW,NLAY,KKSTP,KKPER,IWELCB,
     1      ICBCFL,X(LCBUFF),IOUT)
       IF(IUNIT(3).GT.0) CALL DRN1BD(NDRAIN,MXDRN,VBNM,VBVL,MSUM,
     1      X(LCDRAI),DELT,X(LCHNEW),NCOL,NROW,NLAY,X(LCIBOU),KKSTP,
     2      KKPER,IDRNCB,ICBCFL,X(LCBUFF),IOUT)
       IF(IUNIT(8).GT.0) CALL RCH1BD(NRCHOP,X(LCIRCH),X(LCRECH),
     1      X(LCIBOU),NROW,NCOL,NLAY,DELT,VBVL,VBNM,MSUM,KKSTP,KKPER,
     2      IRCHCB,ICBCFL,X(LCBUFF),IOUT)
       IF(IUNIT(5).GT.0) CALL EVT1BD(NEVTOP,X(LCIEVT),X(LCEVTR),
     1      X(LCEXDP),X(LCSURF),X(LCIBOU),X(LCHNEW),NCOL,NROW,NLAY,
     2      DELT,VBVL,VBNM,MSUM,KKSTP,KKPER,IEVTCB,ICBCFL,X(LCBUFF),IOUT)
       IF(IUNIT(4).GT.0) CALL RIV1BD(NRIVER,MXRIVR,X(LCRIVR),X(LCIBOU),
     1      X(LCHNEW),NCOL,NROW,NLAY,DELT,VBVL,VBNM,MSUM,
     2      KKSTP,KKPER,IRIVCB,ICBCFL,X(LCBUFF),IOUT)
       IF(IUNIT(7).GT.0) CALL GHB1BD(NBOUND,MXBND,VBNM,VBVL,MSUM,
     1      X(LCBNDS),DELT,X(LCHNEW),NCOL,NROW,NLAY,X(LCIBOU),KKSTP,
     2      KKPER,IGHBCB,ICBCFL,X(LCBUFF),IOUT)
C
C7C5---PRINT AND OR SAVE HEADS AND DRAWDOWNS. PRINT OVERALL BUDGET.
       CALL BAS1OT(X(LCHNEW),X(LCSTRT),ISTRT,X(LCBUFF),X(LCIOFL),
     1         MSUM,X(LCIBOU),VBNM,VBVL,KKSTP,KKPER,DELT,
     2         PERTIM,TOTIM,ITMUNI,NCOL,NROW,NLAY,ICNVG,
     3      IHDDFL,IBUDFL,IHEDFM,IHEDUN,IDDNFM,IDDNUN,IOUT)
C
C7C6----IF ITERATION FAILED TO CONVERGE THEN STOP.
       IF(ICNVG.EQ.0) STOP
  200 CONTINUE
  300 CONTINUE
C
C8------END PROGRAM
       STOP
C
       END
```

DESCRIPTION OF HORIZONTAL-FLOW-BARRIER PACKAGE

CONCEPTUALIZATION AND IMPLEMENTATION

A plan view of a model layer containing a thin, vertical low-permeability geologic feature is shown in figure 2*A*. This geologic feature is approximated, in figure 2*B*, as a series of horizontal-flow barriers conceptually situated on the boundary between pairs of adjacent cells in the finite-difference grid. Note that each segment of the geologic feature lying between two adjacent cells is defined as a horizontal-flow barrier. Thus, geologically speaking, figure 2*A* shows a single low-permeability feature in the model layer. However, in the finite-difference grid (fig. 2*B*), this feature is treated as six horizontal-flow barriers in the model layer.

The key assumption underlying the Horizontal-Flow-Barrier Package is that the width of the barrier (that is, the shortest distance through the barrier) is negligibly small in comparison with the horizontal dimensions of the cells in the finite-difference grid. Thus, the barrier width is not explicitly considered in the Package, but it is included implicitly in a hydraulic characteristic defined as either (1) barrier transmissivity divided by barrier width, if it is in a constant-transmissivity layer, or (2) barrier hydraulic conductivity divided by barrier width, if it is in a variable-transmissivity layer. Furthermore, the barrier is assumed to have zero storage capacity. Its sole function is to lower the horizontal branch conductance between the two cells that it separates.

Shown in figure 3*A* is a horizontal-flow barrier situated between cell *i,j,k,* and cell *i,j*+1,*k*--that is, two adjacent cells in the same row in a model layer. The branch conductance in the row direction between the two cells can be determined as shown by McDonald and Harbaugh (1988, chap. 5, p. 6, eq. 38, and p. 7, fig. 25):

$$\frac{1}{CR_{i,j+1/2,k}} = \frac{1}{\frac{TR_{i,j,k}DELC_i}{\frac{DELR_j}{2}}} + \frac{1}{TDW_{i,j+1/2,k}DELC_i} + \frac{1}{\frac{TR_{i,j+1,k}DELC_i}{\frac{DELR_{j+1}}{2}}}, \quad (1)$$

where

$CR_{i,j+1/2,k}$ is the branch conductance in the row direction between nodes *i, j, k* and *i,j*+1,*k* [L^2T^{-1}];
$TR_{i,j,k}$ is the transmissivity in the row direction of cell *i,j,k* [L^2T^{-1}];
$TR_{i,j+1,k}$ is the transmissivity in the row direction of cell *i,j*+1,*k* [L^2T^{-1}];
$DELR_j$ is the grid width in the row direction of column *j* [L];
$DELR_{j+1}$ is the grid width in the row direction of column *j*+1 [L];
$DELC_i$ is the grid width in the column direction of row *i* [L]; and
$TDW_{i,j+1/2,k}$ is the barrier transmissivity divided by the width of the barrier between cell *i,j,k* and cell *i,j*+1,*k* [LT^{-1}].

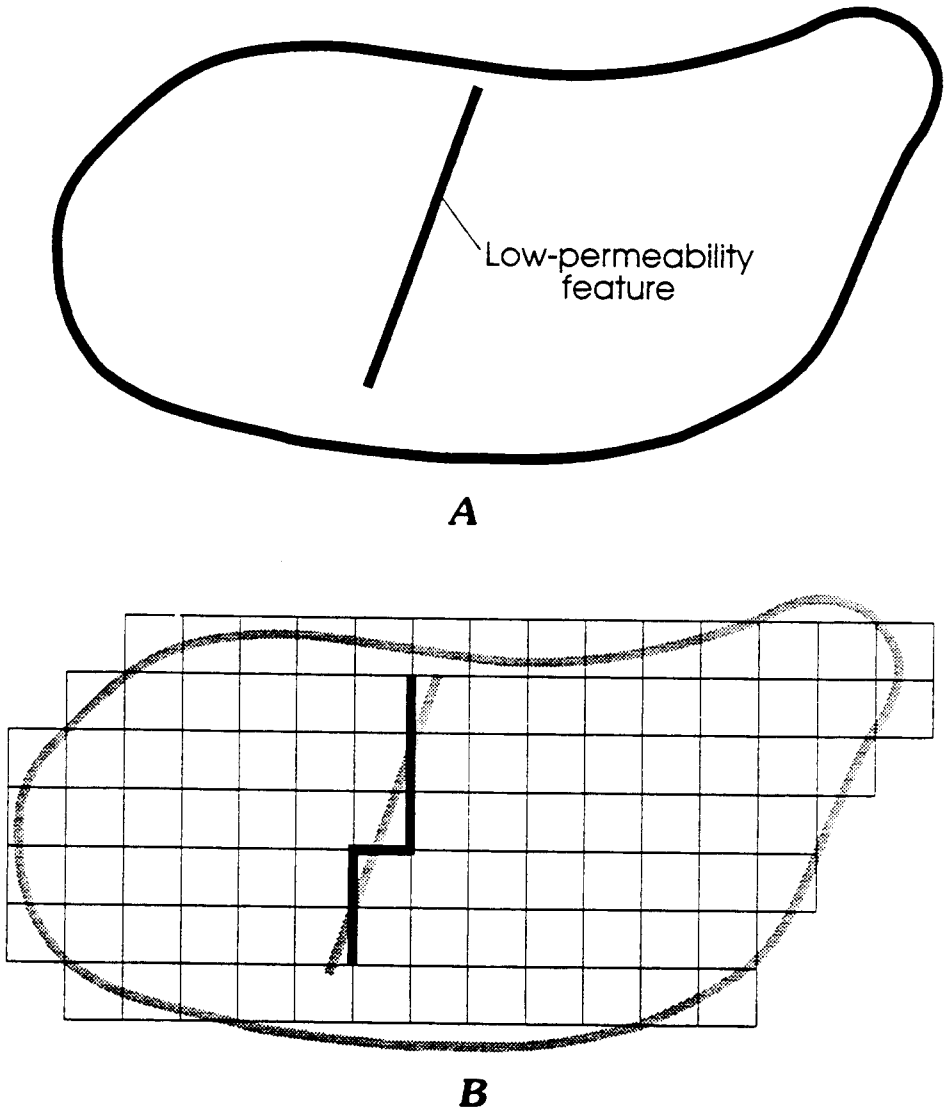


Figure 2. Schematic representation of model layer (***A***) and grid (***B***) with a low-permeability feature and its representation in the grid as a series of six horizontal-flow barriers.

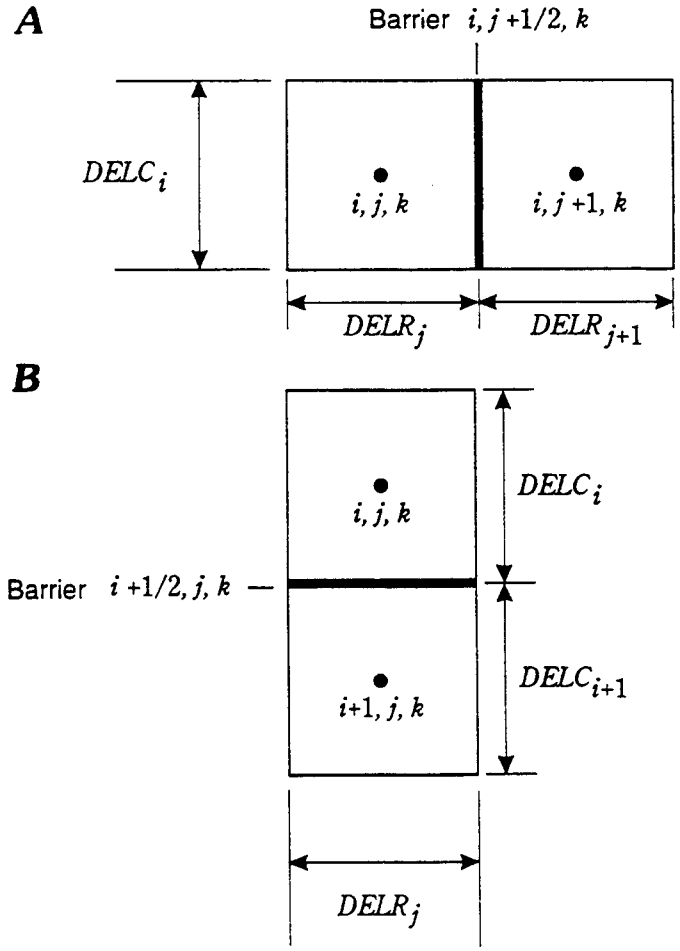


Figure 3. Schematic representation of a horizontal-flow barrier separating two adjacent model cells in the same row (***A***) and two adjacent model cells in the same column (***B***).

Combining the first and third terms on the right side of equation 1 gives

$$\frac{1}{CR_{i,j+1/2,k}} = \frac{1}{CR^{*}_{i,j+1/2,k}} + \frac{1}{TDW_{i,j+1/2,k}DELC_i}, \tag{2}$$

where $CR^{*}_{i,j+1/2,k}$ is the branch conductance if the barrier did not exist, and is given by

$$CR^{*}_{i,j+1/2,k} = 2\ DELC_i \frac{TR_{i,j,k}TR_{i,j+1,k}}{TR_{i,j,k}DELR_{j+1} + TR_{i,j+1,k}DELR_j}. \tag{3}$$

Solving for $CR_{i,j+1/2,k}$ in equation 2 yields

$$CR_{i,j+1/2,k} = \frac{CR^{*}_{i,j+1/2,k}TDW_{i,j+1/2,k}DELC_i}{CR^{*}_{i,j+1/2,k} + TDW_{i,j+1/2,k}DELC_i}. \tag{4}$$

The same process can be applied when the horizontal-flow barrier is situated between two cells on the same column in a layer (fig. 3*B*). In this case, the branch conductance in the column direction between nodes i,j,k and nodes $i+1,j,k$ is

$$CC_{i+1/2,j,k} = \frac{CC^{*}_{i+1/2,j,k} TDW_{i+1/2,j,k} DELR_j}{CC^{*}_{i+1/2,j,k} + TDW_{i+1/2,j,k} DELR_j}, \quad (5)$$

where $CC^{*}_{i+1/2,j,k}$ is the branch conductance if the barrier did not exist, and is given by

$$CC^{*}_{i+1/2,j,k} = 2\ DELR_j \frac{TC_{i,j,k} TC_{i+1,j,k}}{TC_{i,j,k} DELC_{i+1} + TC_{i+1,j,k} DELC_i}, \quad (6)$$

and

$TC_{i,j,k}$ is the transmissivity in the column direction of cell i,j,k [L^2T^{-1}]; and
$TC_{i+1,j,k}$ is the transmissivity in the column direction of cell $i+1,j,k$ [L^2T^{-1}].

In the computer program, the horizontal branch conductances are computed in two steps. First, the horizontal-flow barriers are neglected and the branch conductances CC^{*} and CR^{*} are computed by the standard method (for example, eqs. 3 and 6; see also McDonald and Harbaugh, 1988, chap. 5, p. 6-10). Next, the effects of the barriers are taken into account by means of equations 4 and 5.

In a model layer that is confined, TDW (in this case, barrier transmissivity divided by barrier width) of a barrier will be constant for the simulation. If a layer is potentially unconfined, new values of TDW must be calculated as the head fluctuates. This is done at the start of each iteration. In this case, TDW is obtained by multiplying the average saturated thickness of the two cells on either side of the barrier by the barrier hydraulic conductivity divided by the width of the barrier.

INPUT FOR HORIZONTAL-FLOW-BARRIER PACKAGE

FOR EACH SIMULATION

HFB1AL

1. Data: NHFB
   Format: I10

HFB1RP

The following input data are read one layer at a time; that is, all the data for layer 1 are read first, then all the data for layer 2, and so forth.

FOR EACH LAYER

2. Data: NBRLAY
   Format: I10

3. Data: IROW1 ICOL1 IROW2 ICOL2 HYDCHR
   Format: I10 I10 I10 I10 F10.0
   (Input item 3 consists of one record for each horizontal-flow barrier. If NBRLAY is zero, item 3 will not be read.)

EXPLANATION OF FIELDS USED IN INPUT INSTRUCTIONS

NHFB--is the total number of horizontal-flow barriers in the finite-difference grid.

NBRLAY--is the number of horizontal-flow barriers in a layer.

NOTE: Within a layer, the location of a horizontal-flow barrier is identified by the two cells on either side of the barrier. The row and column numbers of these two cells are IROW1, ICOL1, and IROW2, ICOL2, respectively. There is no requirement regarding the order in which the cells are identified.

IROW1--is the row number of the cell on one side of the horizontal-flow barrier.

ICOL1--is the column number of the cell on one side of the horizontal-flow barrier.

IROW2--is the row number of the cell on the other side of the horizontal-flow barrier.

ICOL2--is the column number of the cell on the other side of the horizontal-flow barrier.

HYDCHR--is a hydraulic characteristic of the horizontal-flow barrier. If the layer type is 0 or 2, it is the barrier transmissivity divided by the width of the horizontal-flow barrier. If the layer type is 1 or 3, it is the barrier hydraulic conductivity divided by the width of the horizontal-flow barrier.

SAMPLE INPUT FOR THE HORIZONTAL-FLOW-BARRIER PACKAGE

The following example illustrates input for the HFB Package. There are three simulated layers, as shown in figure 4. In layer 1, a low-permeability feature is represented by four horizontal-flow barriers. The hydraulic characteristic (HYDCHR) of this feature is nonuniform, as reflected by nonuniform values of HYDCHR in the input data. Layer 2 does not have any horizontal-flow barriers. In layer 3, a low-permeability feature is represented by three horizontal-flow barriers. The hydraulic characteristic of this feature is uniform, as reflected by uniform values of HYDCHR in the input data.

Data item	Explanation	Input record			
1	{NHBF} 7				
2	{NBRLAY} FOR LAYER 1 4				
3	{IROW1, ICOL1, IROW2, ICOL2, HYDCHR} 3	4	3	5	0.00025
3	{IROW1, ICOL1, IROW2, ICOL2, HYDCHR} 3	4	4	4	.00025
3	{IROW1, ICOL1, IROW2, ICOL2, HYDCHR} 4	3	4	4	.00041
3	{IROW1, ICOL1, IROW2, ICOL2, HYDCHR} 5	3	5	4	.00054
2	{NBRLAY} FOR LAYER 2 0				
2	{NBRLAY} FOR LAYER 3 3				
3	{IROW1, ICOL1, IROW2, ICOL2, HYDCHR} 2	4	3	4	.00013
3	{IROW1, ICOL1, IROW2, ICOL2, HYDCHR} 3	3	3	4	.00013
3	{IROW1, ICOL1, IROW2, ICOL2, HYDCHR} 3	3	4	3	.00013

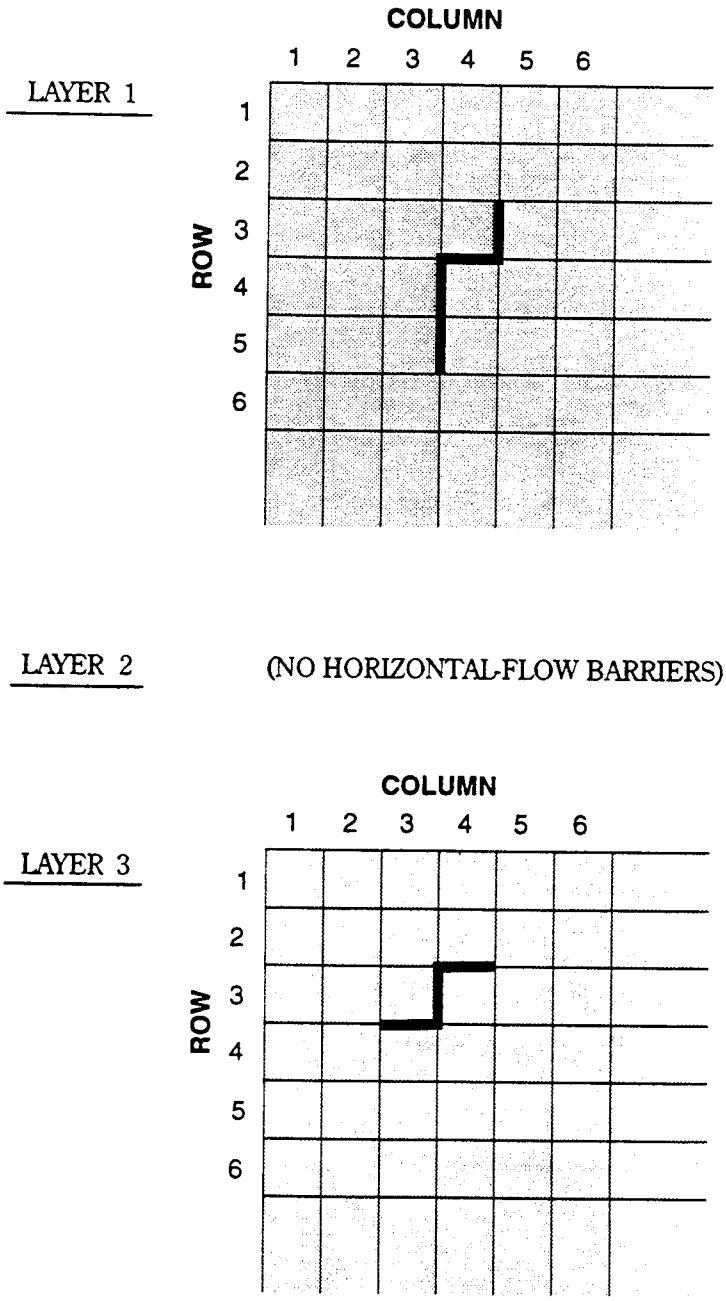


Figure 4. Schematic representation of model grid locations of horizontal-flow barriers (heavy lines) in layers 1 and 3.

MODULE DOCUMENTATION FOR THE HORIZONTAL-FLOW-BARRIER PACKAGE

The Horizontal-Flow-Barrier Package (HFB1) has three primary modules and one submodule. The primary modules HFB1AL, HFB1RP, and HFB1FM are called in the MAIN program; the submodule SHFB1N is called in the primary module HFB1RP. Following is an overview of the modules.

HFB1AL Allocates space for data array. A narrative, flowchart, program listing, and variables list for this module are included in this report on pages 13-16.

HFB1RP Reads all data needed by the package and invokes SHFB1N, which checks input data and modifies horizontal branch conductances in constant-transmissivity layers to account for horizontal-flow barriers. A narrative, flowchart, program listing, and variables list for this module are included in this report on pages 16-20.

HFB1FM Modifies horizontal branch conductances in variable-transmissivity layers to account for horizontal-flow barriers. A narrative, flowchart, program listing, and variables list for this module are included in this report on pages 21-26.

SHFB1N Checks input data and modifies horizontal branch conductances in constant-transmissivity layers to account for horizontal-flow barriers. A narrative, flowchart, program listing, and variables list for this module are included in this report on pages 27-31.

PRIMARY MODULES

HFB1AL

Narrative for module HFB1AL.--This module allocates space in the X array to store the locations of the horizontal-flow barriers and their hydraulic characteristics. The order of execution is:

1. Print a message identifying the package.

2. Read and print NHFB (number of horizontal-flow barriers in the finite-difference grid).

3. Set LCHFBR, which will point to the first element in the horizontal-flow-barrier data list (HFBR), equal to ISUM, which, in turn, currently is pointing to the first unallocated element in the X array.

4. Calculate the amount of space needed for the horizontal-flow-barrier data list (five values for each barrier--IROW1, ICOL1, IROW2, ICOL2, and hydraulic characteristic HYDCHR) and add it to ISUM.

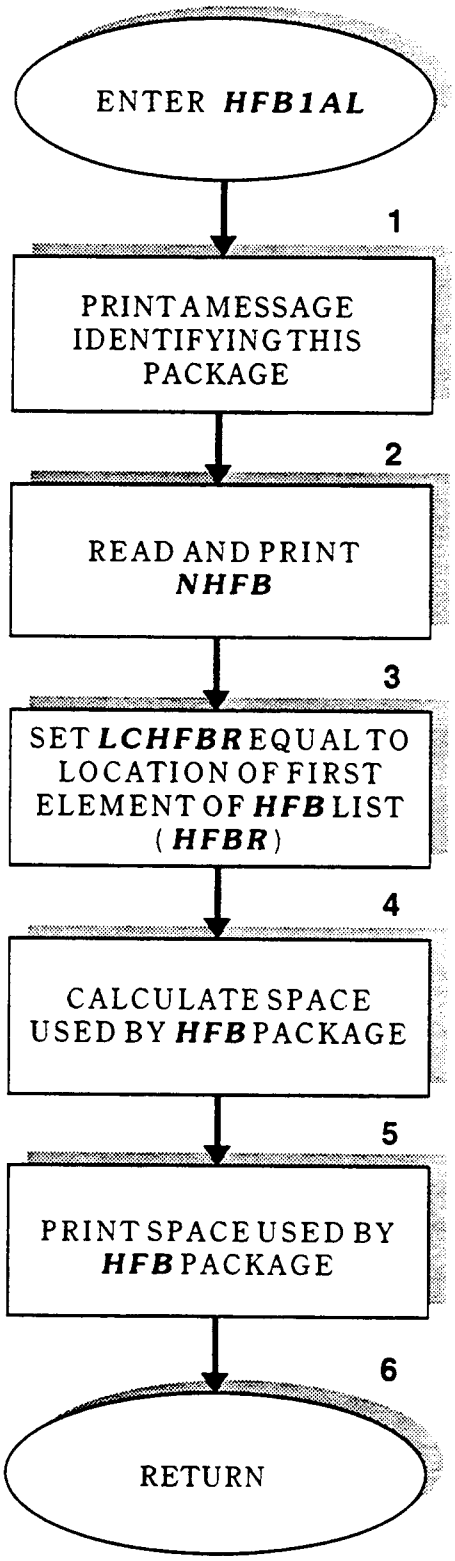
5. Print the number of elements in the X array used by the HFB Package.

6. Return.

Flowchart for module HFB1AL.--

NHFB is the number of horizontal-flow barriers in the finite-difference grid.

LCHFBR is the location in the X array of the list of horizontal-flow-barrier data (HFBR).



Program listing for module HFB1AL.--

```
      SUBROUTINE HFB1AL(ISUM,LENX,LCHFBR,NHFB,IN,IOUT)
C
C-------VERSION 0001 13JUNE1986 HFB1AL
C
C     ***************************************************************
C     ALLOCATE ARRAY STORAGE FOR HORIZONTAL-FLOW-BARRIER PACKAGE
C     ***************************************************************
C
C        SPECIFICATIONS:
C     -------------------------------------------------------------
      COMMON/HFBCOM/NBRLAY(80)
C     -------------------------------------------------------------
C
C1------IDENTIFY PACKAGE.
      WRITE(IOUT,1)IN
    1 FORMAT(1H0,'HFB1 -- HORIZONTAL-FLOW-BARRIER PACKAGE, VERSION 1',
     1', 06/13/86',' INPUT READ FROM UNIT',I3)
C
C2------READ AND PRINT NHFB (TOTAL NUMBER OF HORIZONTAL-FLOW BARRIERS).
      READ(IN,2) NHFB
    2 FORMAT(I10)
      WRITE(IOUT,3)NHFB
    3 FORMAT(1H0,'A TOTAL OF',I5,' HORIZONTAL-FLOW BARRIERS')
C
C3------SET LCHFBR EQUAL TO ADDRESS OF FIRST UNUSED SPACE IN THE X ARRAY.
      LCHFBR=ISUM
C
C4------CALCULATE AMOUNT OF SPACE USED BY HFB PACKAGE.
      ISP=5*NHFB
      ISUM=ISUM+ISP
C
C5------PRINT AMOUNT OF SPACE USED BY HFB PACKAGE.
      WRITE(IOUT,4)ISP
    4 FORMAT(1X,I6,' ELEMENTS IN X ARRAY ARE USED FOR HORIZONTAL-FLOW'
     1,' BARRIERS')
      ISUM1=ISUM-1
      WRITE(IOUT,5)ISUM1,LENX
    5 FORMAT(1X,I6,' ELEMENTS OF X ARRAY USED OUT OF',I7)
      IF(ISUM1.GT.LENX) WRITE(IOUT,6)
    6 FORMAT(1X,'   ***X ARRAY MUST BE DIMENSIONED LARGER***')
C
C6------RETURN
      RETURN
      END
```

List of variables for module HFB1AL.--

Variable	Range	Definition
IN	Package	Primary unit number from which input for this package will be read.
IOUT	Global	Primary unit number for all printed output. IOUT is set to be 6 in MAIN.
ISP	Module	Number of words in the X array allocated by this module.
ISUM	Global	Index number of the lowest element in the X array that has not yet been allocated. When space is allocated for an array, this size of the array is added to ISUM.
ISUM1	Module	ISUM-1.
LCHFBR	Package	Location in the X array of the first element of array HFBR.
LENX	Global	Length of X array in words. This always should be equal to the dimension of X specified in the MAIN program.
NHFB	Package	Number of horizontal-flow barriers in the finite-difference grid.

HFB1RP

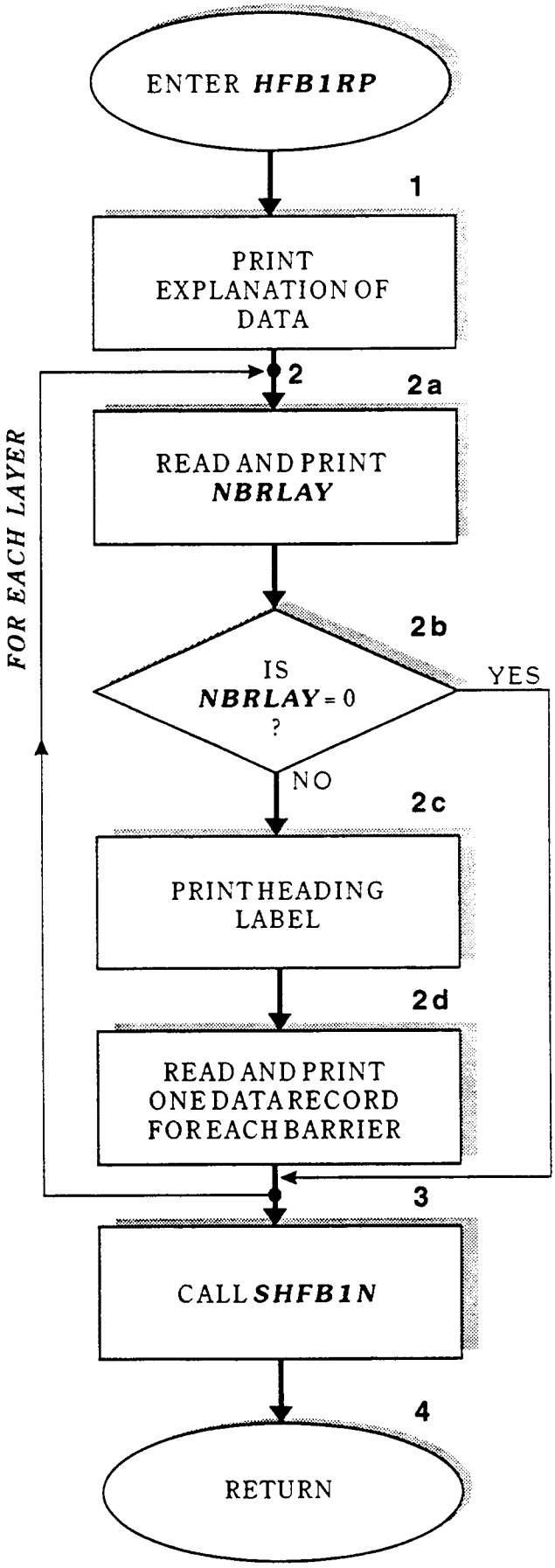
Narrative for module HFB1RP.--This module reads data to build the horizontal-flow-barrier data list one layer at a time. Within a layer, the location of a horizontal-flow barrier is identified by the two cells on either side of the barrier. The row and column numbers of the two cells are respectively IROW1, ICOL1, and IROW2, ICOL2. Thus, for each barrier, the horizontal-flow-barrier data list consists of five items: IROW1, ICOL1, IROW2, ICOL2, and HYDCHR (hydraulic characteristic of the barrier). After all data are read, the module calls the submodule SHFB1N to check the input data and to modify horizontal branch conductances in constant-transmissivity layers to account for the horizontal-flow barriers. The order of execution is:

1. Print explanation of data.
2. For each layer, read and print input data.
   (a) Read and print NBRLAY (number of horizontal-flow barriers in the layer).
   (b) If there are no horizontal-flow barriers in the layer (NBRLAY = 0), then go to the next layer.
   (c) Print heading label of horizontal-flow-barrier data list. For constant-transmissivity layer (LAYCON = 0 or 2), HYDCHR is read as the barrier transmissivity divided by the width of the barrier. For variable-transmissivity layer (LAYCON = 1 or 3), HYDCHR is read as the barrier hydraulic conductivity divided by the width of the barrier.
   (d) For each horizontal-flow barrier, read and print IROW1, ICOL1, IROW2, ICOL2, and HYDCHR.
3. Call submodule SHFB1N to check data and modify horizontal branch conductances in constant-transmissivity layers to account for horizontal-flow barriers.
4. Return.

Flowchart for module HFB1RP.--

NBRLAY is the number of horizontal-flow barriers in a layer.

SHFB1N is the submodule called to check data and modify horizontal branch conductances in constant transmissivity layers.



Program listing for module HFB1RP.--

```
      SUBROUTINE HFB1RP(CR,CC,DELR,DELC,HFBR,IN,NCOL,NROW,NLAY,NODES,
     1        NHFB,IOUT)
C
C-----VERSION 0001 13JUNE1986 HFB1RP
C
C     ******************************************************************
C     READ AND INITIALIZE DATA FOR HORIZONTAL-FLOW-BARRIER PACKAGE
C     ******************************************************************
C
C       SPECIFICATIONS:
C     ------------------------------------------------------------------
      DIMENSION CR(NODES),CC(NODES),DELR(NCOL),DELC(NROW),HFBR(5,NHFB)
C
      COMMON/HFBCOM/NBRLAY(80)
      COMMON/FLWCOM/LAYCON(80)
C     ------------------------------------------------------------------
C1------PRINT EXPLANATION OF DATA.
      WRITE(IOUT,8)
    8 FORMAT(///1X,'HORIZONTAL-FLOW BARRIERS -- LISTED BY ',
     1'LAYERS.  WITHIN EACH LAYER, THE LOCATION OF A BARRIER IS ',
     2'IDENTIFIED BY'/1X,'THE 2 CELLS ON EITHER SIDE OF THE BARRIER.  ',
     3'THE ROW AND COLUMN NUMBER OF THE TWO CELLS ARE RESPECTIVELY '/1X,
     4'IROW1, ICOL1, AND IROW2, ICOL2.')
C
C2------FOR EACH LAYER, READ AND PRINT INPUT DATA.
      II=0
      DO 100 K=1,NLAY
C
C2A-----READ AND PRINT NUMBER OF HORIZONTAL-FLOW BARRIERS IN LAYER K.
      READ(IN,1) NBRLAY(K)
    1 FORMAT(I10)
      WRITE(IOUT,2) NBRLAY(K),K
    2 FORMAT(1H0,I5,' HORIZONTAL-FLOW BARRIERS IN LAYER',I3)
C
C2B-----IF NO BARRIERS, THEN GO TO NEXT LAYER.
      IF(NBRLAY(K).EQ.0) GOTO 100
C
C2C------PRINT HEADING LABEL.
      IF (LAYCON(K).EQ.0 .OR. LAYCON(K).EQ.2) WRITE(IOUT,3)
    3 FORMAT(1X,20X,'IROW1',5X,'ICOL1',5X,'IROW2',5X,'ICOL2',3X,
     1'TRANSMIS./WIDTH',3X,'BARRIER NO.'/1X,20X,71('-'))
      IF (LAYCON(K).EQ.1 .OR. LAYCON(K).EQ.3) WRITE(IOUT,4)
    4 FORMAT(1X,20X,'IROW1',5X,'ICOL1',5X,'IROW2',5X,'ICOL2',2X,
     1'HYD. COND./WIDTH',2X,'BARRIER NO.'/1X,20X,71('-'))
C
C2D-----FOR EACH HORIZONTAL-FLOW BARRIER IN LAYER, READ AND PRINT BARRIER
C2D-----LOCATION AND HYDRAULIC CHARACTERISTIC.
      DO 90 JJ=1,NBRLAY(K)
      II=II+1
      READ(IN,5)IROW1,ICOL1,IROW2,ICOL2,HYDCHR
    5 FORMAT(4I10,F10.0)
      WRITE(IOUT,6)IROW1,ICOL1,IROW2,ICOL2,HYDCHR,II
```

```
    6 FORMAT(1X,19X,I5,3I10,5X,G12.4,I11)
      HFBR(1,II)=IROW1
      HFBR(2,II)=ICOL1
      HFBR(3,II)=IROW2
      HFBR(4,II)=ICOL2
      HFBR(5,II)=HYDCHR
   90 CONTINUE
  100 CONTINUE
C
C3------CHECK HFB DATA AND MODIFY HORIZONTAL BRANCH CONDUCTANCES FOR
C3------CONSTANT TRANSMISSIVITY LAYERS.
      CALL SHFB1N(CR,CC,DELR,DELC,HFBR,NCOL,NROW,NLAY,NHFB,IOUT)
C
C4------RETURN
      RETURN
      END
```

List of variables for module HFB1RP.--

Variable	Range	Definition
CC	Global	DIMENSION (NCOL, NROW, NLAY), Conductance in the column direction. CC(J,I,K) contains conductance between nodes (J,I,K) and (J,I+1,K).
CR	Global	DIMENSION (NCOL, NROW, NLAY), Conductance in the row direction. CR(J,I,K) contains conductance between nodes (J,I,K) and (J+1,I,K).
DELC	Global	DIMENSION (NROW), Cell dimension in the column direction. DELC(I) contains width of row I.
DELR	Global	DIMENSION (NCOL), Cell dimension in the row direction. DELR(J) contains width of column J.
HFBR	Package	DIMENSION (5,NHFB), For each horizontal-flow barrier: IROW1, ICOL1, IROW2, ICOL2, and HYDCHR.
HYDCHR	Module	Hydraulic characteristic of the horizontal-flow barrier. If LAYCON = 0 or 2, HYDCHR is the barrier transmissivity divided by the width of the barrier. If LAYCON = 1 or 3, HYDCHR is the barrier hydraulic conductivity divided by the width of the barrier.
ICOL1	Module	Column number of cell on one side of horizontal-flow barrier.
ICOL2	Module	Column number of cell on other side of horizontal-flow barrier.
II	Module	Index for horizontal-flow barriers.
IN	Package	Primary unit number from which input for this package will be read.
IOUT	Global	Primary unit number for all printed output. IOUT is set to be 6 in MAIN.
IROW1	Module	Row number of cell on one side of horizontal-flow barrier.
IROW2	Module	Row number of cell on other side of horizontal-flow barrier.
JJ	Module	Index for horizontal-flow barriers within a layer.
K	Module	Index for layer.
LAYCON	Package	DIMENSION (80) Layer type code: 0 - Layer strictly confined. 1 - Layer strictly unconfined. 2 - Layer confined/unconfined (transmissivity is constant). 3 - Layer confined/unconfined (transmissivity varies).
NBRLAY	Package	DIMENSION (80), Number of horizontal-flow barriers in each layer. NBRLAY(K) is the number of horizontal-flow barriers in layer K.
NCOL	Global	Number of columns in the grid.
NHFB	Package	Number of horizontal-flow barriers in the grid.
NLAY	Global	Number of cells (nodes) in the grid.
NROW	Global	Number of rows in the grid.

Narrative for module HFB1FM.--This module modifies the horizontal branch conductance in variable-transmissivity layers to account for the horizontal-flow barriers. Upon entry of the module, the arrays CR and CC must contain horizontal branch conductance values computed for the case of no horizontal-flow barriers in the variable-transmissivity layers. The following steps are executed:

1. For each layer, modify horizontal branch conductance if transmissivity varies.
   (a) If layer type is not 1 or 3, go to the next layer.
   (b) If zero barriers in the layer, go the next layer.
   (c) For each barrier in the layer, modify horizontal branch conductances.
      (1) Determine I1 and J1, which are the row and column numbers of the cell whose horizontal branch conductances are to be modified.
      (2) Determine I2 and J2, which are the row and column numbers of the cell that is separated from cell (J1, I1, K) by the barrier.
      (3) If I1≠I2, then horizontal branch conductance along row direction is not modified. Otherwise, go to step (4).
      (4) If CR(J1,I1,K) is zero, then go to the next barrier.
      (5) Determine the average saturated thickness between cells (J1,I1,K) and (J2,I2,K).
      (6) Modify CR(J1,I1,K) to account for barrier and go to next barrier.
      (7) Case of J1=J2. Horizontal branch conductance along column direction to be modified.
      (8) If CC(J1,I1,K) is zero, then go to the next barrier.
      (9) Determine the average saturated thickness between cells (J1,I1,K) and (J2,I2,K).
      (10) Modify CC(J1,I1,K) to account for barrier and go to next barrier.
2. Return.

Flowchart for module HFB1FM.--

LAYCON is a layer-type code (one for each layer).
- 0-confined
- 1-unconfined
- 2-confined/unconfined but transmissivity is constant
- 3-confined/unconfined

NBRLAY is the number of horizontal-flow barriers in a layer.

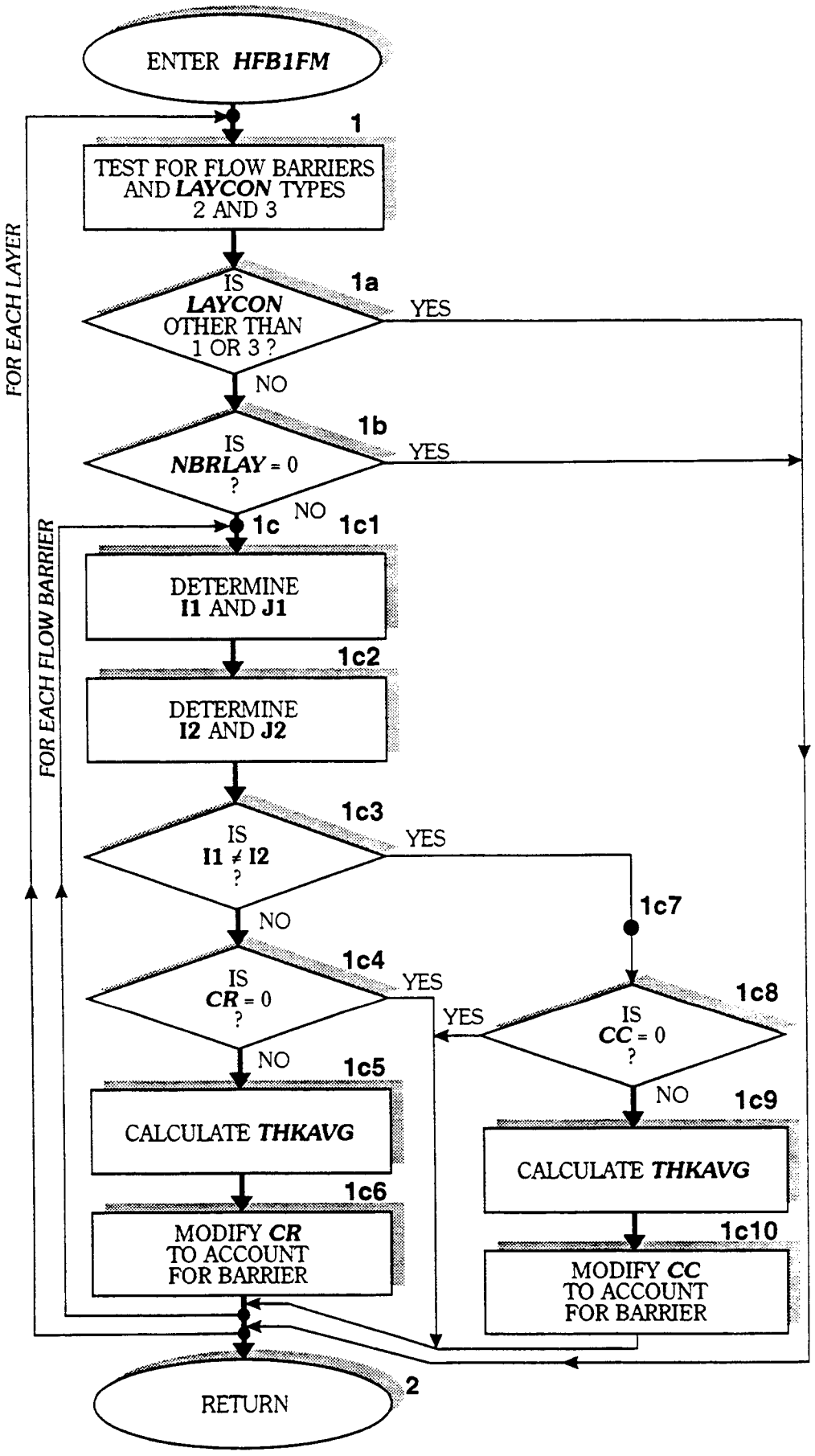
I1 and I2 are the row numbers of the two cells separated by the horizontal-flow barrier.

J1 and J2 are the column numbers of the two cells separated by the horizontal-flow barrier.

CR is the conductance in the row direction.

CC is the conductance in the column direction.

THKAVG is the average saturated thickness of the two cells separated by the horizontal-flow barrier.



Program listing for module HFB1FM.--

```
      SUBROUTINE HFB1FM(HNEW,CR,CC,BOT,TOP,DELR,DELC,HFBR,NCOL,NROW,
     1      NLAY,NHFB)
C
C-----VERSION 0001 13JUNE1986 HFB1FM
C
C     ******************************************************************
C     MODIFY HORIZONTAL BRANCH CONDUCTANCES IN VARIABLE-TRANSMISSIVITY
C     LAYERS TO ACCOUNT FOR HORIZONTAL-FLOW BARRIERS.
C     ******************************************************************
C
C        SPECIFICATIONS:
C     ------------------------------------------------------------------
      DOUBLE PRECISION HNEW
C
      DIMENSION HNEW(NCOL,NROW,NLAY),CR(NCOL,NROW,NLAY),
     1    CC(NCOL,NROW,NLAY),BOT(NCOL,NROW,NLAY),TOP(NCOL,NROW,NLAY),
     2    DELR(NCOL),DELC(NROW),HFBR(5,NHFB)
C
      COMMON/HFBCOM/NBRLAY(80)
      COMMON/FLWCOM/LAYCON(80)
C     ------------------------------------------------------------------
      KB=0
      KT=0
      II2=0
C
C1------FOR EACH LAYER: CHECK IF TRANSMISSIVITY VARIES.
      DO 100 K=1,NLAY
      IF (NBRLAY(K).EQ.0) GO TO 5
      II1=II2+1
      II2=II2+NBRLAY(K)
    5 IF (LAYCON(K).EQ.3 .OR. LAYCON(K).EQ.2) KT=KT+1
C
C1A-----IF LAYER TYPE IS NOT 1 OR 3, THEN SKIP THIS LAYER.
      IF (LAYCON(K).NE.3 .AND. LAYCON(K).NE.1) GO TO 100
      KB=KB+1
C
C1B-----IF NO BARRIER IN THE LAYER, GO TO NEXT LAYER.
      IF (NBRLAY(K).EQ.0) GO TO 100
C
C1C-----FOR EACH BARRIER IN THE LAYER, MODIFY HORIZONTAL BRANCH
C1C-----CONDUCTANCES
      DO 90 II=II1,II2
C1C1----CELL (J1,I1,K) IS THE ONE WHOSE HORIZONTAL BRANCH
C1C1----CONDUCTANCES ARE TO BE MODIFIED.
      I1=HFBR(1,II)
      J1=HFBR(2,II)
C1C2----CELL (J2,I2,K) IS THE CELL NEXT TO CELL (J1,I1,K) AND SEPARATED
C1C2----FROM IT BY THE BARRIER.
      I2=HFBR(3,II)
      J2=HFBR(4,II)
      HCDW=HFBR(5,II)
C
C1C3----IF I1=I2, THEN MODIFY HORIZONTAL BRANCH CONDUCTANCES ALONG ROW
C1C3----DIRECTION.
```

```
      IF (I1.NE.I2) GOTO 20
C
C1C4----IF CR(J1,I1,K)=0, THEN GO TO NEXT BARRIER
      IF (CR(J1,I1,K).EQ.0.) GO TO 90
C
C1C5----CALCULATE AVERAGE SATURATED THICKNESS BETWEEN CELLS (J1,I1,K)
C1C5----AND (J2,I2,K).  NOTE: NEGATIVE SATURATED THICKNESS DOES NOT
C1C5----OCCUR; OTHERWISE, CR(J1,I1,K) WOULD BE ZERO AND THE FOLLOWING
C1C5----CALCULATION FOR SATURATED THICKNESS WOULD BE SKIPPED.
      HD1=HNEW(J1,I1,K)
      HD2=HNEW(J2,I2,K)
      IF (LAYCON(K).EQ.1) GO TO 10
      IF (HD1.GT.TOP(J1,I1,KT)) HD1=TOP(J1,I1,KT)
      IF (HD2.GT.TOP(J2,I2,KT)) HD2=TOP(J2,I2,KT)
   10 THKAVG=((HD1-BOT(J1,I1,KB))+(HD2-BOT(J2,I2,KB)))/2.
C
C1C6----MODIFY CR(J1,I1,K) TO ACCOUNT FOR BARRIER.
      TDW=THKAVG*HCDW
      CR(J1,I1,K)=TDW*CR(J1,I1,K)*DELC(I1)/(TDW*DELC(I1)+CR(J1,I1,K))
      GOTO 90
C
C1C7----CASE OF J1=J2. MODIFY HORIZONTAL BRANCH CONDUCTANCES ALONG
C1C7----COLUMN DIRECTION.
   20 CONTINUE
C
C1C8----IF CC(J1,I1,K)=0, THEN GO TO NEXT BARRIER.
      IF (CC(J1,I1,K).EQ.0.) GO TO 90
C
C1C9----CALCULATE AVERAGE SATURATED THICKNESS BETWEEN CELLS (J1,I1,K)
C1C9----AND (J2,I2,K).  NEGATIVE SATURATED THICKNESS DOES NOT OCCUR
C1C9----FOR THE SAME REASON AS DESCRIBED ABOVE.
      HD1=HNEW(J1,I1,K)
      HD2=HNEW(J2,I2,K)
      IF (LAYCON(K).EQ.1) GO TO 30
      IF (HD1.GT.TOP(J1,I1,KT)) HD1=TOP(J1,I1,KT)
      IF (HD2.GT.TOP(J2,I2,KT)) HD2=TOP(J2,I2,KT)
   30 THKAVG=((HD1-BOT(J1,I1,KB))+(HD2-BOT(J2,I2,KB)))/2.
C
C1C10---MODIFY CC(J1,I1,K) TO ACCOUNT FOR BARRIER.
      TDW=THKAVG*HCDW
      CC(J1,I1,K)=TDW*CC(J1,I1,K)*DELR(J1)/(TDW*DELR(J1)+CC(J1,I1,K))
   90 CONTINUE
  100 CONTINUE
C
C2------RETURN
      RETURN
      END
```

List of variables for module HFB1FM.--

Variable	Range	Definition
BOT	Package	DIMENSION (NCOL,NROW,NBOT), Elevation of bottom of each layer. (NBOT is the number of layers for which LAYCON = 1 or 3.)
CC	Global	DIMENSION (NCOL,NROW,NLAY) Conductance in the column direction. CC(J,I,K) contains conductance between nodes (J,I,K) and (J,I+1,K).
CR	Global	DIMENSION (NCOL,NROW,NLAY), Conductance in the row direction. CR(J,I,K) contains conductance between nodes (J,I,K) and (J+1,I,K).
DELC	Global	DIMENSION (NROW), Cell dimension in the column direction. DELC(I) contains width of row I.
DELR	Global	DIMENSION (NCOL), Cell dimension in the row direction. DELR(J) contains width of column J.
HCDW	Module	Hydraulic conductivity divided by the width of the horizontal-flow barrier.
HD1	Module	Temporary label of HNEW(J1,I1,K).
HD2	Module	Temporary label of HNEW(J2,I2,K).
HFBR	Package	DIMENSION (5,NHFB), Horizontal-flow barrier list. Contains row and column numbers of the two cells on either side of the barrier and the hydraulic characteristic of the barrier.
HNEW	Global	DIMENSION (NCOL,NROW,NLAY), Most recent estimation of head in each cell. HNEW changes at each iteration.
I1	Module	Row number of cell whose horizontal branch conductances are to be modified.
I2	Module	Row number of the cell next to cell (J1,I1,K) and separated from it by the horizontal-flow barrier.
II	Module	Counter for horizontal-flow barriers.
II1	Module	Lower limit of II in DO loop.
II2	Module	Upper limit of II in DO loop.
IN	Package	Primary unit number from which input for this package will be read.
IOUT	Global	Primary unit number for all printed output. IOUT is set to be 6 in MAIN.
J1	Module	Column number of cell whose horizontal branch conductances are to be modified.
J2	Module	Column number of the cell next to cell (J1,I1,K) and separated from it by the horizontal-flow barrier.

K	Module	Index for layers.
KB	Module	Counter for layers for which bottom elevation is needed.
KT	Module	Counter for layers for which top elevation is needed.
LAYCON	Package	DIMENSION (80) Layer type code: 0 - Layer strictly confined. 1 - Layer strictly unconfined. 2 - Layer confined/unconfined (transmissivity is constant.) 3 - Layer confined/unconfined (transmissivity varies).
NBRLAY	Package	DIMENSION (80), Number of horizontal-flow barriers in each layer. NBRLAY(K) is the number of horizontal-flow barriers in layer K.
NCOL	Global	Number of columns in the grid.
NHFB	Package	Number of horizontal-flow barriers in the grid.
NLAY	Global	Number of layers in the grid.
NROW	Global	Number of rows in the grid.
THKAVG	Module	Average saturated thickness of the cells (J1,I1,K) and (J2,I2,K).
TOP	Package	DIMENSION (NCOL,NROW,NTOP), Elevation of top of layer. (NTOP is the number of layers for which LAYCON = 2 or 3.)

Submodule SHFB1N

Narrative for submodule SHFB1N.--This submodule checks the input data read by the primary module HFB1RP and modifies the horizontal branch conductances in the constant-transmissivity layers to account for horizontal-flow barriers. Upon entry of the submodule, the arrays CR and CC must contain the horizontal branch conductance values of the constant-transmissivity layers computed for the case of no horizontal-flow barriers. The following steps are executed in the submodule:

1. Initialize the error flag to zero.
2. Check input data one layer at a time.
   (a) If no barriers in layer, then go to next layer.
   (b) Check input data for each barrier in the layer.
      (1) Find row and column number of the two cells on either side of the barrier, and rearrange HFBR array so that HFBR(1,II) and HFBR(2,II) contain the row and column number of the cell whose horizontal branch conductances are to be modified to account for the presence of barrier no. II. This cell is denoted by (J1,I1,K).
      (2) Check if the two cells are on different rows. If true, then go to step (7).
      (3) Check if the two cells are on non-adjacent columns. If true, then print error message, set error flag to 1 (2b12), and go to next barrier.
      (4) If layer type is 1 or 3, then go to next barrier.
      (5) If CR(J1,I1,K) is zero, then go to next barrier.
      (6) Modify CR(J1,I1,K) to account for barrier and go to next barrier.
      (7) Check if the two cells are on different columns. If true, then print error message, set error flag to 1 (2b12), and go to next barrier.
      (8) Check if the two cells are on nonadjacent rows. If true, then print error message, set error flag to 1 (2b12), and go to next barrier.
      (9) If layer type is 1 or 3, then go to next barrier.
      (10) If CC(J1,I1,K) is zero, then go to next barrier.
      (11) Modify CC(J1,I1,K) to account for barrier and go to next barrier.
      (12) Print an error message if (3) or (7) or (8) is true.
3. If error flag is 1, stop execution.
4. Return.

Flowchart for sub-module SHFB1N.--

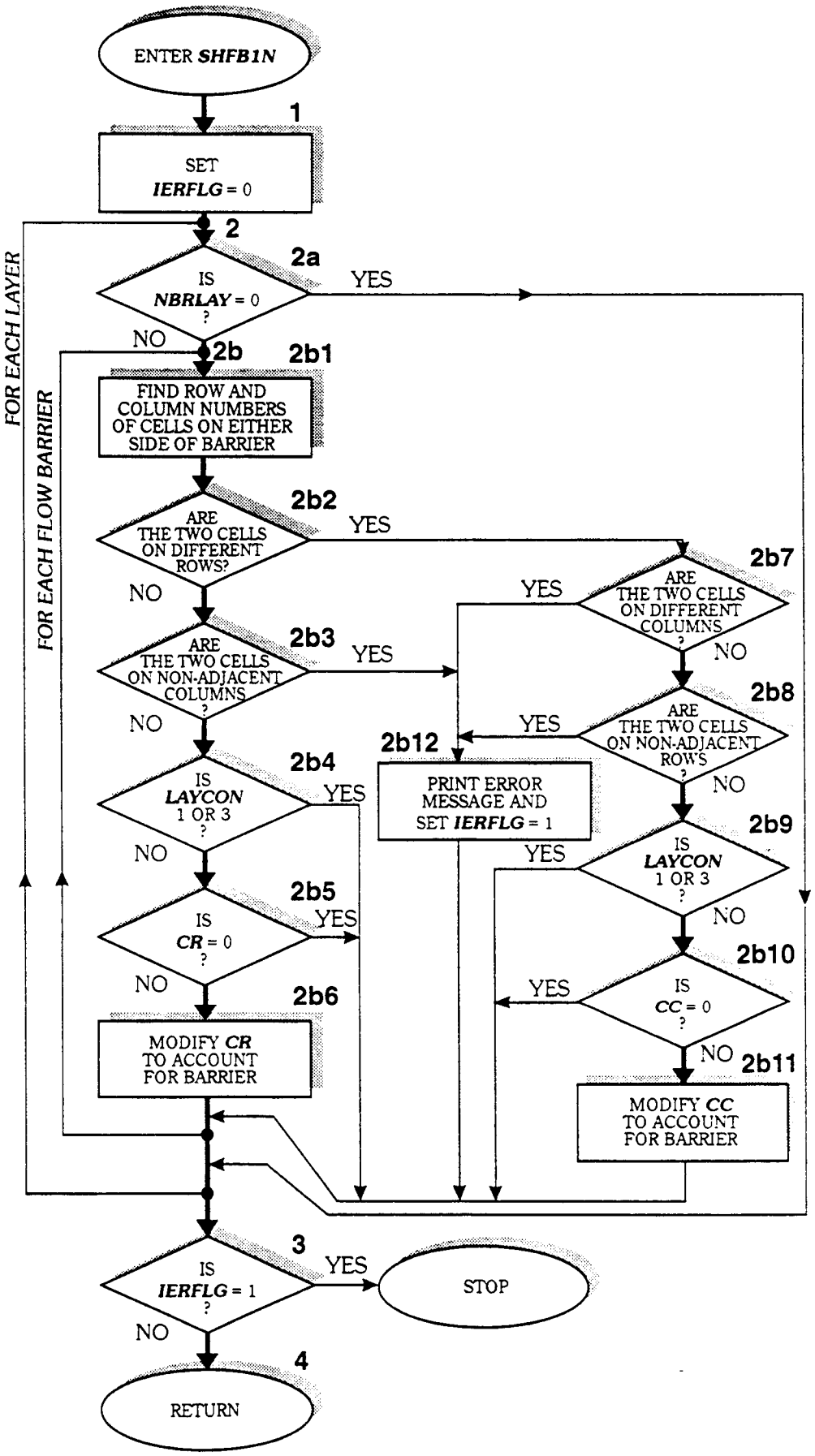
IERFLG is an error flag.
0-no error detected
1-error detected

NBRLAY is the number of horizontal-flow barriers in a layer.

LAYCON is a layer-type code (one for each layer).
0-confined
1-unconfined
2-confined/ unconfined but transmissivity is constant
3-confined/ unconfined

CR is the conductance in the row direction.

CC is the conductance in the column direction.



Program listing for submodule SHFB1N.--

```
      SUBROUTINE SHFB1N(CR,CC,DELR,DELC,HFBR,NCOL,NROW,NLAY,NHFB,IOUT)
C
C-----VERSION 0001 13JUNE1986 SHFB1N
C
C     ******************************************************************
      CHECK HFB DATA AND MODIFY HORIZONTAL CONDUCTANCES (CR AND CC)
C     FOR CONSTANT TRANSMISSIVITY LAYERS TO ACCOUNT FOR HORIZONTAL-FLOW
C     BARRIERS.
C     ******************************************************************
C
C        SPECIFICATIONS:
C     ------------------------------------------------------------------
      DIMENSION CR(NCOL,NROW,NLAY),CC(NCOL,NROW,NLAY),DELR(NCOL),
     1    DELC(NROW),HFBR(5,NHFB)
C
      COMMON/HFBCOM/NBRLAY(80)
      COMMON/FLWCOM/LAYCON(80)
C     ------------------------------------------------------------------
C
C1------INITIALIZE ERROR FLAG TO ZERO.
      IERFLG=0
C
C2------CHECK HFB DATA ONE LAYER AT A TIME.
      II=0
      DO 100 K=1,NLAY
C
C2A-----IF ZERO BARRIER IN LAYER, THEN GO TO NEXT LAYER.
      IF (NBRLAY(K).EQ.0) GOTO 100
C
C2B-----CHECK EACH BARRIER IN LAYER K.
      DO 90 JJ=1,NBRLAY(K)
      II=II+1
      TDW=HFBR(5,II)
C
C2B1----FIND ROW AND COLUMN NUMBERS OF THE TWO CELLS ON EITHER SIDE
C2B1----OF THE BARRIER AND REARRANGE HFBR ARRAY.
      I1=MIN(HFBR(1,II),HFBR(3,II))
      J1=MIN(HFBR(2,II),HFBR(4,II))
      I2=MAX(HFBR(1,II),HFBR(3,II))
      J2=MAX(HFBR(2,II),HFBR(4,II))
      HFBR(1,II)=I1
      HFBR(2,II)=J1
      HFBR(3,II)=I2
      HFBR(4,II)=J2
C
C2B2----IF I1=I2, BARRIER IS BETWEEN TWO CELLS ON THE SAME ROW.
      IF (I1.NE.I2) GOTO 10
C
C2B3----IF J2-J1=1, THE TWO CELLS ARE NEXT TO ONE ANOTHER (DATA OK).
C2B3----OTHERWISE, PRINT ERROR MESSAGE AND SET ERROR FLAG TO 1.
      IF ((J2-J1).NE.1) GOTO 80
C
C2B4----IF LAYER TYPE IS 1 OR 3, THEN GO TO NEXT BARRIER.
      IF (LAYCON(K).EQ.1 .OR. LAYCON(K).EQ.3) GOTO 90
```

```
C
C2B5-----IF CR(J1,I1,K)=0 THEN GO TO NEXT BARRIER.
      IF (CR(J1,I1,K).EQ.0.) GO TO 90
C
C2B6----MODIFY CR(J1,I1,K) TO ACCOUNT FOR BARRIER.
      CR(J1,I1,K)=TDW*CR(J1,I1,K)*DELC(I1)/(TDW*DELC(I1)+CR(J1,I1,K))
      GOTO 90
C
C2B7----IF J1=J2, BARRIER IS BETWEEN TWO CELLS ON THE SAME COLUMN.
C2B7----OTHERWISE, PRINT ERROR MESSAGE AND SET ERROR FLAG TO 1.
   10 IF (J1.NE.J2) GOTO 80
C
C2B8----IF I2-I1=1, THE TWO CELLS ARE NEXT TO ONE ANOTHER (DATA OK).
C2B8----OTHERWISE, PRINT ERROR MESSAGE AND SET ERROR FLAG TO 1.
      IF ((I2-I1).NE.1) GOTO 80
C
C2B9----IF LAYER TYPE IS 1 OR 3, THEN GO TO NEXT BARRIER.
      IF (LAYCON(K).EQ.1 .OR. LAYCON(K).EQ.3) GOTO 90
C
C2B10---IF CC(J1,I1,K)=0, THEN GO TO NEXT BARRIER.
      IF (CC(J1,I1,K).EQ.0.) GOTO 90
C
C2B11---MODIFY CC(J1,I1,K) TO ACCOUNT FOR BARRIER
      CC(J1,I1,K)=TDW*CC(J1,I1,K)*DELR(J1)/(TDW*DELR(J1)+CC(J1,I1,K))
      GOTO 90
C
C2B12---PRINT ERROR MESSAGE AND SET ERROR FLAG.
   80 WRITE (IOUT,1) II
    1 FORMAT (1X,'ERROR DETECTED IN LOCATION DATA OF BARRIER NO.',I4)
      IERFLG=1
   90 CONTINUE
  100 CONTINUE
C
C3-----HALT EXECUTION IF ERRORS ARE DETECTED.
      IF (IERFLG.EQ.1) STOP
C
C4-----RETURN
      RETURN
      END
```

List of variables for submodule SHFB1N.--

Variable	Range	Definition
CC	Global	DIMENSION (NCOL,NROW,NLAY), Conductance in the column direction. CC(J,I,K) contains conductance between nodes (J,I,K) and (J,I+1,K).
CR	Global	DIMENSION (NCOL,NROW,NLAY), Conductance in the row direction. CR(J,I,K) and (J+1,I,K).
DELC	Global	DIMENSION (NROW), Cell dimension in the column direction. DELC(I) contains width of row I.
DELR	Global	DIMENSION (NCOL), Cell dimension in the row direction. DELR(J) contains width of column J.
HFBR	Package	DIMENSION (5,NHFB), Horizontal-flow-barrier data list. Contains row and column numbers of the two cells on either side of the barrier and the hydraulic characteristics of the barrier.
I1	Submodule	Row number of cell whose horizontal branch conductances are to be modified.
I2	Submodule	Row number of the cell next to cell (J2,I2,K) and separated from it by the horizontal-flow barrier.
IERFLG	Submodule	Error flag.
II	Submodule	Counter for horizontal-flow barriers.
IN	Package	Primary unit number from which input for this package will be read.
IOUT	Global	Primary unit number for all printed output. IOUT is set to be 6 in MAIN.
J1	Submodule	Column number of cell whose horizontal branch conductances are to be modified.
J2	Submodule	Column number of the cell next to cell (J2,I2,K) and separated from it by the horizontal-flow barrier.
JJ	Submodule	Counter for horizontal-flow barriers in a layer.
K	Submodule	Index for layers.
LAYCON	Package	DIMENSION (80) Layer type code: 0 - Layer strictly confined. 1 - Layer strictly unconfined. 2 - Layer confined/unconfined (transmissivity is constant). 3 - Layer confined/unconfined (transmissivity varies).
NBRLAY	Package	DIMENSION (80), Number of horizontal-flow barriers in each layer. NBRLAY(K) is the number of horizontal-flow barriers in layer K.
NCOL	Global	Number of columns in the grid.
NHFB	Package	Number of horizontal-flow barriers in the grid.
NLAY	Global	Number of layers in the grid.
NROW	Global	Number of rows in the grid.
TDW	Submodule	Barrier transmissivity divided by barrier width.

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